

Study on a potential CBAM scope extension to other goods and transportation services

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Abstract

Using public statistics and confidential EU ETS data, product groups at risk of carbon leakage in the sectors inorganic and organic chemicals, polymers, pulp and paper, refinery products, glass, ceramics, ferro-alloys, aluminium, and non-ferrous metals were identified as a priority for further consideration in a potential CBAM scope extension. Starting from these sectors, technical feasibility was assessed, focusing on whether products and their production routes can be clearly identified, and whether specific MRV issues prevent their inclusion. Most goods in the sectors pulp and paper, glass, and ceramics proved feasible for inclusion, while non-ferrous metals and some chemicals appear less feasible due to complex value chains. Precursors like lime, dolime, magnesia, and alumina were identified as feasible, coke and refinery products need to be treated considering which chemicals would be included. Regarding scrap, methodological choices outside this study are required. No robust evidence of high carbon leakage risk exists because of EU ETS inclusion of the transport sector.

Executive Summary

English version

This "Study on a potential CBAM scope extension to other goods and transportation services. Specific Contract No TAXUD/2024/DE/311 implementing framework contract No GROW/2021/OP/0001 Lot 2" had as main objective the assessment of the feasibility of extending the scope of the CBAM goods horizontally to include other sectors currently not in scope and vertically to include precursors, based on a risk assessment of carbon leakage. The feasibility assessment included extension of transport emissions of CBAM goods and of the new CBAM goods identified in the study. The study collected and assessed evidence based on three key pillars: data collection and analysis, stakeholder consultations, and continuous alignment with the European Commission regarding assessment options.

The stakeholder consultation covered two topics, CBAM goods scope extension, and CBAM transport emissions scope extension. The consultations were implemented in three stages: scoping interviews with industry associations implemented during August and September 2024, stakeholder survey implemented from April – June 2025 and follow up interviews conducted during June – July 2025. Different methods were used for the implementation of the study, including desk research, quantitative analysis, input-output modelling, scenario analysis, surveys, and interviews.

The key findings of the assessments of the study are presented below per tasks.

Key findings for Task 1

The first task was to establish a shortlist of goods in the prioritisation of the CBAM extension for further consideration under a potential CBAM scope extension. The starting point was the current list of sectors and products regarded as being at risk of carbon leakage under the EU ETS, as stipulated in Article 30 (2b) of the CBAM Regulation. From this longlist, sectors and products that are already included in the CBAM, that relate to a service and not a good, or that are not produced in the EU were first excluded.

A three-stage filtering approach was subsequently applied to the longlist of sectors and products to establish a prioritised shortlist of goods based on public statistics and confidential EU ETS data from the Commission:

- **Stage 1:** the longlist was first reduced to the NACE 4-digit sectors most at risk of carbon leakage. Other criteria used in the Stage 1 filtering were the presence of at least one EU ETS installation in the sector and having annual EU ETS emissions of at least 25 ktCO₂e.
- **Stage 2:** within the resulting sectors from the Stage 1 filtering, goods were identified that had significant levels of trade, imports and production associated with EU ETS installations to establish a preliminary shortlist.
- **Stage 3:** the preliminarily shortlisted goods were subsequently analysed to determine whether they could be grouped at a more aggregated CN level and/or if it were more logical to include or exclude certain products based on factors not directly reflected in statistical data.

This filtering approach resulted in product groups prioritised for further consideration in a potential CBAM scope extension in the sectors pulp and paper, refinery products, glass, ceramics, ferro-alloys, aluminium, and non-ferrous metals.

Additionally, a considerable number of inorganic and organic chemicals and polymers were also identified for prioritisation. This was, however, not based on the filtering approach with

statistics. The filtering approach led to goods that were spread out over various parts of the chemical value chain, were not congruent with the processes with significant emissions in the chemical industry. Instead, a “substance-centred” value chain approach was used, i.e., focusing on those value chains with chemicals that are substances only and not mixtures or formulations of several substances. Substances that were produced at high volumes in the EU and/or leading to significant emissions were subsequently selected for prioritisation. The list of prioritised chemicals and polymers included downstream substances with significant embedded emissions from goods currently included in the CBAM, key precursors and co-products, and/or substances deemed important by the Commission for further analysis.

Key findings for Task 2

This task aimed to identify the technical feasibility of including new goods in the CBAM scope. The focus was on goods prioritised under Task 1, in particular on chemicals and polymers. Analysis included a comprehensive review of the existing legislative framework of the CBAM (the implementing act for the transitional phase) and of similar EU legislation dealing with environmental reporting and value chains. The criteria for the technical feasibility assessment included: (1) goods must be clearly identifiable (by CN code or other means, e.g. a CAS number in case of chemicals), (2) monitoring rules must be available, in particular (a) production processes and routes must be known appropriately for setting boundaries in the legislation, (b) rules for attributing emissions to co-products in multi-functional processes must be available, and (c) overly complex MRV issues need to be avoided. Furthermore, (3) the determination of default values must be feasible, and (4) the MRV system must be able to handle longer value chains. For chemicals and polymers, criterion (2) was supplemented by a methodology to calculate a “complexity measure” for the relevant value chains and production processes.

The task’s results suggest that it is technically feasible to include in the CBAM: (1) Most of the CN codes of the pulp and paper sector, (2) most of the CN codes in the glass sector, (3) several CN codes in the ceramics sector, including refractory materials, (4) in the ferro-alloys sector: ferro-silicon. In these sectors, groups of goods were found as unfit for inclusion in the CBAM where they are too diverse in nature or too complex in terms of production processes. The latter is a reason to consider the sector of non-ferrous metals as too complex for inclusion in the CBAM. Furthermore, groups of goods were identified which are usually not carried out in installations that would qualify for inclusion in the EU ETS, i.e. they would be considered as “downstream goods” and could be included in the CBAM if the inclusion of their precursors would lead to a risk of carbon leakage also for the downstream goods.

The products of the refinery sector were assessed on the one hand as individual sector, and on the other hand under the aspect of providing precursors to the organic chemicals and polymers sectors. Findings include that it would be rather difficult to include refineries only for certain products, e.g. naphtha, pyrolysis gas and reformat as precursors for the chemical industry. The whole sector was identified as difficult to perform MRV (monitoring, reporting and verification), but not to an extent that it would make inclusion in the CBAM impossible, as similar MRV is already being conducted under the EU ETS.

Regarding chemicals and polymers, a total of 118 substances were assessed. However, the assessment did not result in a straightforward result, but a more complex situation: For most substances, if they do not have their own CN code, it is possible to identify e.g. a CAS number or a systematic IUPAC name for identifying the substance. However, in some cases, this was not possible with the methodology and resources in this project. Also, a mere ranking of substances in terms of complexity of the value chain will not provide a clear case in favour of inclusion in the CBAM. While priority was given in the assessment to substances already covered in the EU ETS by product benchmarks and to other high-volume chemicals, high

volume does not always correspond with good technical feasibility. On the other hand, some simpler goods were only included in the assessment because they might be precursors to higher volume chemicals. But should the more complex good be dropped from the list for inclusion in the CBAM, also those simpler precursors could be dropped. Similarly, some goods were only included because they appear as co-products with other goods assessed and would not merit inclusion on their own. Overall, it was found that the inclusion of some chemicals and polymers would be technically feasible, while keeping it open, where a limit in complexity would have to be set.

Key findings for Task 3

The objective of Task 3 was to identify precursors of goods under the existing CBAM scope, and of the goods identified under Task 2, and whether they should be included in the CBAM. Again, the assessment focussed on technical feasibility (using a similar criteria set as under Task 2) and the relevant value chains. The following goods were identified to be possible candidates for inclusion in the CBAM as precursors to the existing scope: Lime, dolime, magnesia, alumina. A special consideration is required for coke: Technically its inclusion is feasible, but a solution must be found for the treatment of its co-products and the related emission attribution rules. Depending on which chemicals would be included in the CBAM (see Task 2), the co-products of coke (tar, light oils, crude aromatics) would become precursors for chemicals and should then be explicitly considered on their own for inclusion in the CBAM.

Refinery products were not specifically assessed under this task, although relevant as precursors for chemicals and polymers. However, the assessment was already performed in Task 2. For chemicals and polymers, no specific assessment was carried out under Task 3, as their precursors were already included under Task 2. Finally, scrap can be an important precursor for several sectors, such as iron and steel, aluminium, but also for glass (cullet), and paper (recycled pulp). However, a detailed discussion of the inclusion in the CBAM was excluded from this project, as it was considered necessary to first develop an appropriate classification of pre-/post-consumer scrap and a suitable method for attributing embedded emissions to scrap.

Key findings for Task 5

This task was devoted to developing a methodology for assessing the scale of non-EU transport emissions associated with CBAM goods. This includes an estimate of the non-EU transport emissions of goods in the T2 and T3 list into the EU. The key findings of this assessment are the following: There seem to be not enough evidence of increased risk of carbon leakage because of carbon price obligations for the transport sector in the EU ETS. In addition, the share of transport emissions in total emissions is rather low (less than 5%). This share varies considerably depending on the country of origin and mode of transport, highlighting the importance of distance and logistics patterns in understanding the carbon intensity of goods imported under the CBAM. This assessment, however, could change depending on the final list of CN codes for extension of the CBAM scope.

Key findings for Task 6

The objective of Task 6 was to assess the technical feasibility of including emissions from transport services associated with the movement of CBAM goods outside the EU, analysing data requirements, evaluating data availability for importers, and determining the extent and cost of EU authorities' ability to monitor the accuracy of reported information. The key findings of this assessment are the following: The assessment indicates that the extension to include transport emission of CBAM goods could potentially be possible from the viewpoint of the measurement, storing and sharing of transport emissions data along logistics operators, with some limitations. Moreover, it could present challenges for small transport operators in third

countries, who are usually not known even by the international logistics operators. While EU institutions have the information technologies available for reporting, the verification process of transport emissions at the level of CN code remains a challenge, as it is not a practice that transport operators are used to. CBAM verifiers would require per single trade flow to verify emissions in different transport chain elements (all those used for the movement of a CN code lote from origin to destination) rendering the verification process complex to coordinate, and probably costly, as it would also require specific competences to be defined for the CBAM verifier/certifier.

Deutsche Version

Die Studie mit dem Titel „ Study on a potential CBAM scope extension to other goods and transportation services. Specific Contract No TAXUD/2024/DE/311 implementing framework contract No GROW/2021/OP/0001 Lot 2“ hatte als Hauptziel die Bewertung der Machbarkeit einer horizontalen Erweiterung des CBAM-Warenbereichs, um weitere Sektoren einzubeziehen, die derzeit nicht erfasst sind, sowie einer vertikalen Erweiterung, um Vorprodukte aufzunehmen. Dies erfolgte auf Grundlage einer Risikobewertung von Kohlenstofflecks. Die Machbarkeitsbewertung umfasste auch die Einbeziehung von Transportemissionen der CBAM-Waren sowie der im Rahmen der Studie identifizierten neuen CBAM-Waren.

Die Studie stützte sich auf drei zentrale Säulen: Datenerhebung und -analyse, Stakeholder-Konsultationen sowie eine kontinuierliche Abstimmung mit der Europäischen Kommission zu Bewertungsoptionen.

Die Stakeholder-Konsultationen behandelten zwei Themen: die Erweiterung des CBAM-Warenbereichs und die Erweiterung des CBAM-Transportemissionsbereichs. Die Konsultationen wurden in drei Phasen durchgeführt: Scoping-Interviews mit Branchenverbänden (August–September 2024), eine Stakeholder-Umfrage (April–Juni 2025) sowie vertiefende Follow-up-Interviews (Juni–Juli 2025). Für die Durchführung der Studie kamen verschiedene Methoden zum Einsatz, darunter Desk Research, quantitative Analysen, Input-Output-Modellierung, Szenarioanalysen, Umfragen und Interviews.

Die wichtigsten Ergebnisse der Bewertungen der Studie werden im Folgenden nach Aufgabenbereichen dargestellt.

Wichtigste Ergebnisse für Aufgabe 1

Erstellung einer engeren Auswahl von Waren für die mögliche Erweiterung des CBAM

Die erste Aufgabe bestand darin, eine engere Auswahl von Waren zu erstellen, die für eine weitere Prüfung im Rahmen einer möglichen Erweiterung des Geltungsbereichs des CBAM priorisiert werden sollten. Ausgangspunkt war die aktuelle Liste der Sektoren und Produkte, die gemäß Artikel 30 (2b) der CBAM-Verordnung als gefährdet für Kohlenstofflecks im Rahmen des EU-Emissionshandels (EU ETS) gelten. Von dieser Langliste wurden zunächst Sektoren und Produkte ausgeschlossen, die bereits im CBAM enthalten sind, die sich auf eine Dienstleistung und nicht auf ein Gut beziehen oder die nicht in der EU produziert werden.

Anschließend wurde ein dreistufiger Filteransatz auf die Langliste der Sektoren und Produkte angewendet, um eine priorisierte engere Auswahl von Waren auf der Grundlage öffentlicher Statistiken und vertraulicher EU-ETS-Daten der Kommission zu erstellen:

- **Stufe 1:** Die Langliste wurde zunächst auf die NACE-4-Steller-Sektoren mit dem höchsten Risiko für Kohlenstofflecks reduziert. Weitere Kriterien für die Filterung in Stufe 1 waren das Vorhandensein mindestens einer EU-ETS-Anlage im Sektor und jährliche EU-ETS-Emissionen von mindestens 25 kt CO₂e.
- **Stufe 2:** Innerhalb der aus der Filterung der Stufe 1 resultierenden Sektoren wurden Waren identifiziert, die signifikante Handels-, Import- und Produktionsvolumina im Zusammenhang mit EU-ETS-Anlagen aufwiesen, um eine vorläufige engere Auswahl zu erstellen.
- **Stufe 3:** Die vorläufig ausgewählten Waren wurden anschließend analysiert, um zu bestimmen, ob sie auf einer aggregierteren CN-Ebene gruppiert werden konnten und/oder ob es logischer war, bestimmte Produkte auf der Grundlage von Faktoren, die nicht direkt in den statistischen Daten widerspiegelt werden, einzuschließen oder auszuschließen.

Dieser Filteransatz führte zu Produktgruppen, die für eine weitere Prüfung im Rahmen einer möglichen Erweiterung des CBAM-Geltungsbereichs in den Sektoren Zellstoff und Papier, Raffinerieprodukte, Glas, Keramik, Ferrolegierungen, Aluminium und Nichteisenmetalle priorisiert wurden.

Zusätzlich wurde eine beträchtliche Anzahl anorganischer und organischer Chemikalien sowie Polymere für die Priorisierung identifiziert. Dies basierte jedoch nicht auf dem statistischen Filteransatz. Der Filteransatz führte zu Waren, die über verschiedene Teile der chemischen Wertschöpfungskette verteilt waren und nicht mit den Prozessen übereinstimmten, die in der chemischen Industrie zu erheblichen Emissionen führen. Stattdessen wurde ein „stoffzentrierter“ Wertschöpfungskettenansatz verwendet, d. h., der Fokus lag auf den Wertschöpfungsketten mit Chemikalien, die reine Substanzen und keine Gemische oder Formulierungen mehrerer Substanzen sind. Substanzen, die in der EU in großen Mengen produziert werden und/oder zu erheblichen Emissionen führen, wurden anschließend für die Priorisierung ausgewählt. Die Liste der priorisierten Chemikalien und Polymere umfasste nachgelagerte Substanzen mit erheblichen eingebetteten Emissionen aus Waren, die derzeit im CBAM enthalten sind, wichtige Vorprodukte und Koprodukte und/oder Substanzen, die von der Kommission als wichtig für eine weitere Analyse erachtet wurden.

Wichtigste Ergebnisse für Aufgabe 2

Diese Aufgabe zielte darauf ab, die technische Machbarkeit der Aufnahme neuer Waren in den Geltungsbereich des CBAM (Carbon Border Adjustment Mechanism) zu bewerten. Der Fokus lag auf den in Aufgabe 1 priorisierten Waren, insbesondere auf Chemikalien und Polymeren. Die Analyse beinhaltete eine umfassende Überprüfung des bestehenden Rechtsrahmens des CBAM (der Durchführungsrechtsakt für die Übergangsphase) sowie ähnlicher EU-Gesetzgebung zur Umweltberichterstattung und zu Wertschöpfungsketten. Die Kriterien für die Bewertung der technischen Machbarkeit umfassten: (1) Die Waren müssen klar identifizierbar sein (durch CN-Code oder andere Mittel, z.B. eine CAS-Nummer im Falle von Chemikalien), (2) Überwachungsregeln müssen verfügbar sein, insbesondere (a) Produktionsprozesse und -wege müssen für die Festlegung von Grenzen in der Gesetzgebung hinreichend bekannt sein, (b) Regeln für die Zurechnung von Emissionen zu Koprodukten in multifunktionalen Prozessen müssen verfügbar sein und (c) übermäßig komplexe MRV-Fragen (Monitoring, Reporting, Verification) müssen vermieden werden. Darüber hinaus muss (3) die Bestimmung von Standardwerten machbar sein und (4) das MRV-System muss in der Lage sein, längere Wertschöpfungsketten zu bewältigen. Für Chemikalien und Polymere wurde Kriterium (2) durch eine Methodik zur Berechnung eines „Komplexitätsmaßes“ für die relevanten Wertschöpfungsketten und Produktionsprozesse ergänzt.

Die Ergebnisse der Aufgabe deuten darauf hin, dass es technisch machbar ist, folgende Waren in den CBAM aufzunehmen: (1) Die meisten CN-Codes des Zellstoff- und Papiersektors, (2) die meisten CN-Codes im Glassektor, (3) mehrere CN-Codes im Keramiksektor, einschließlich feuerfester Materialien, (4) im Bereich der Ferrolegierungen: Ferrosilizium. In diesen Sektoren wurden Warengruppen als ungeeignet für die Aufnahme in den CBAM eingestuft, wenn sie zu vielfältig oder zu komplex in Bezug auf die Produktionsprozesse sind. Letzteres ist ein Grund, den Sektor der Nichteisenmetalle als zu komplex für die Aufnahme in den CBAM zu betrachten. Darüber hinaus wurden Warengruppen identifiziert, die in der Regel nicht in Anlagen hergestellt werden, die für die Aufnahme in den EU-Emissionshandel (EU ETS) qualifiziert wären, d.h. sie würden als „nachgelagerte Waren“ betrachtet und könnten in den CBAM aufgenommen werden, wenn die Aufnahme ihrer Vorprodukte zu einem Risiko von Kohlenstofflecks auch für die nachgelagerten Waren führen würde.

Die Produkte des Raffineriesektors wurden einerseits als eigenständiger Sektor und andererseits unter dem Aspekt der Bereitstellung von Vorprodukten für die Sektoren organische

Chemikalien und Polymere bewertet. Die Ergebnisse zeigen, dass es eher schwierig wäre, Raffinerien nur für bestimmte Produkte, z.B. Naphtha, Pyrolysegas und Reformat als Vorprodukte für die chemische Industrie, aufzunehmen. Der gesamte Sektor wurde als schwierig für die Durchführung von MRV eingestuft, jedoch nicht in einem Maße, das die Aufnahme in den CBAM unmöglich machen würde, da ähnliche MRV-Bestimmungen bereits im Rahmen des EU-Emissionshandels durchgeführt werden.

Hinsichtlich Chemikalien und Polymeren wurden insgesamt 118 Substanzen bewertet. Die Bewertung ergab jedoch kein klares Ergebnis, sondern eine komplexere Situation: Für die meisten Substanzen ist es möglich, sie z.B. über eine CAS-Nummer oder einen systematischen IUPAC-Namen zu identifizieren, falls sie keinen eigenen CN-Code haben. In einigen Fällen war dies jedoch mit der Methodik und den Ressourcen dieses Projekts nicht möglich. Auch eine bloße Rangfolge der Substanzen nach Komplexität der Wertschöpfungskette liefert keinen klaren Fall für die Aufnahme in den CBAM. Obwohl bei der Bewertung Priorität auf Substanzen gelegt wurde, die bereits im EU-Emissionshandel durch Produktbenchmarks abgedeckt sind, und auf andere Hochvolumen-Chemikalien, entspricht ein hohes Volumen nicht immer einer guten technischen Machbarkeit. Andererseits wurden einige einfachere Waren nur in die Bewertung einbezogen, weil sie Vorprodukte für Chemikalien mit höherem Volumen sein könnten. Sollte jedoch das komplexere Gut von der Liste für die Aufnahme in den CBAM gestrichen werden, könnten auch diese einfacheren Vorprodukte gestrichen werden. Ebenso wurden einige Waren nur einbezogen, weil sie als Koprodukte mit anderen bewerteten Waren auftreten und für sich allein keine Aufnahme rechtfertigen würden. Insgesamt wurde festgestellt, dass die Aufnahme einiger Chemikalien und Polymere technisch machbar wäre, während offen bleibt, wo eine Grenze in Bezug auf die Komplexität zu ziehen ist.

Wichtigste Ergebnisse für Aufgabe 3

Ziel von Aufgabe 3 war es, Vorprodukte von Waren im bestehenden Geltungsbereich des CBAM sowie der in Aufgabe 2 identifizierten Waren zu ermitteln und zu bewerten, ob sie in den CBAM aufgenommen werden sollten. Auch hier lag der Schwerpunkt auf der technischen Machbarkeit (unter Verwendung eines ähnlichen Kriteriensatzes wie in Aufgabe 2) und den relevanten Wertschöpfungsketten. Folgende Waren wurden als mögliche Kandidaten für die Aufnahme in den CBAM als Vorprodukte des bestehenden Geltungsbereichs identifiziert: Kalk, Dolomitkalk, Magnesia, Aluminiumoxid. Eine besondere Betrachtung erfordert Koks: Technisch ist seine Aufnahme machbar, aber es muss eine Lösung für die Behandlung seiner Koprodukte und die damit verbundenen Emissionszurechnungsregeln gefunden werden. Je nachdem, welche Chemikalien in den CBAM aufgenommen werden (siehe Aufgabe 2), würden die Koprodukte von Koks (Teer, Leichtöle, Roharomaten) zu Vorprodukten für Chemikalien und sollten dann explizit auf ihre eigene Aufnahme in den CBAM geprüft werden.

Raffinerieprodukte wurden in dieser Aufgabe nicht spezifisch bewertet, obwohl sie als Vorprodukte für Chemikalien und Polymere relevant sind. Die Bewertung wurde jedoch bereits in Aufgabe 2 durchgeführt. Für Chemikalien und Polymere wurde in Aufgabe 3 keine spezifische Bewertung vorgenommen, da ihre Vorprodukte bereits in Aufgabe 2 berücksichtigt wurden. Schließlich kann Schrott ein wichtiges Vorprodukt für mehrere Sektoren sein, z. B. Eisen und Stahl, Aluminium, aber auch für Glas (Glasscherben) und Papier (Recyclingzellstoff). Eine detaillierte Diskussion über die Aufnahme in den CBAM wurde jedoch aus diesem Projekt ausgeschlossen, da es zunächst als notwendig erachtet wurde, eine angemessene Klassifizierung von Vor-/Nachverbraucherschrott und eine geeignete Methode zur Zurechnung der eingebetteten Emissionen an Schrott zu entwickeln.

Wichtigste Ergebnisse für Aufgabe 5

Diese Aufgabe widmete sich der Entwicklung einer Methodik zur Bewertung des Umfangs der nicht-EU-Transportemissionen, die mit CBAM-Waren verbunden sind. Dazu gehört eine Schätzung der nicht-EU-Transportemissionen der Waren aus den Listen T2 und T3 in die EU. Die wichtigsten Erkenntnisse dieser Bewertung sind folgende: Es gibt anscheinend nicht genug Belege für ein erhöhtes Risiko von Kohlenstofflecks aufgrund von CO₂-Preisverpflichtungen für den Verkehrssektor im EU-Emissionshandel (EU ETS). Es scheint nicht genügend Belege dafür zu geben, dass aufgrund der CO₂-Preisverpflichtungen für den Verkehrssektor im EU-ETS ein erhöhtes Risiko der Carbon Leakage besteht. Darüber hinaus ist der Anteil der Verkehrsemissionen an den Gesamtemissionen relativ gering (weniger als 5 %). Dieser Anteil variiert jedoch erheblich je nach Herkunftsland und Transportmittel, was die Bedeutung von Entfernung und Logistikmustern für das Verständnis der Kohlenstoffintensität von im Rahmen des CBAM importierten Waren unterstreicht. Diese Einschätzung könnte sich jedoch ändern, abhängig von der endgültigen Liste der KN-Codes für die Erweiterung des CBAM-Geltungsbereichs.

Wichtigste Ergebnisse für Aufgabe 6

Ziel von Aufgabe 6 war es, die technische Machbarkeit der Einbeziehung von Emissionen aus Transportdienstleistungen, die mit dem Transport von CBAM-Waren außerhalb der EU verbunden sind, zu bewerten. Dabei wurden die Datenanforderungen analysiert, die Datenverfügbarkeit für Importeure bewertet und das Ausmaß und die Kosten der Fähigkeit der EU-Behörden zur Überprüfung der Richtigkeit der gemeldeten Informationen bestimmt. Die wichtigsten Erkenntnisse dieser Bewertung sind folgende: Die Bewertung zeigt, dass eine Erweiterung zur Einbeziehung der Verkehrsemissionen von CBAM-Waren grundsätzlich möglich sein könnte – zumindest aus der Perspektive der Erfassung, Speicherung und Weitergabe von Emissionsdaten entlang der Logistikketten, wenn auch mit gewissen Einschränkungen. Darüber hinaus könnte dies Herausforderungen für kleinere Transportunternehmen in Drittländern mit sich bringen, die häufig selbst den internationalen Logistikdienstleistern nicht bekannt sind. Während die EU-Institutionen über die notwendigen Informationstechnologien für die Berichterstattung verfügen, bleibt der Verifizierungsprozess der Verkehrsemissionen auf Ebene der KN-Codes eine Herausforderung, da dies keine gängige Praxis für Transportunternehmen ist. CBAM-Prüfer müssten für jeden einzelnen Handelsfluss die Emissionen in den verschiedenen Elementen der Transportkette (alle, die für die Beförderung einer Charge eines KN-Codes vom Ursprungs- bis zum Zielort genutzt werden) verifizieren, was den Verifizierungsprozess komplex und vermutlich kostenintensiv macht. Zudem müssten spezifische Kompetenzen für den CBAM-Prüfer bzw. -Zertifizierer definiert werden.

Version Française

Cette "Study on a potential CBAM scope extension to other goods and transportation services. Specific Contract No TAXUD/2024/DE/311 implementing framework contract No GROW/2021/OP/0001 Lot 2" avait pour objectif principal d'évaluer la faisabilité d'une extension horizontale du périmètre des biens soumis au mécanisme d'ajustement carbone aux frontières (MACF) pour inclure d'autres secteurs actuellement non couverts, et verticalement pour intégrer les précurseurs, sur la base d'une évaluation des risques de fuite de carbone. L'évaluation de faisabilité incluait également l'extension des émissions de transport des biens soumis au MACF et des nouveaux biens identifiés dans l'étude.

L'étude a recueilli et analysé des données selon trois piliers clés : la collecte et l'analyse de données, les consultations des parties prenantes, et un alignement continu avec la Commission européenne concernant les options d'évaluation.

Les consultations des parties prenantes ont porté sur deux thèmes: l'extension du périmètre des biens soumis au MACF et l'extension du périmètre des émissions de transport du MACF. Elles se sont déroulées en trois phases: des entretiens d'orientation avec des associations industrielles menés en août et septembre 2024, une enquête auprès des parties prenantes réalisée d'avril à juin 2025, des entretiens de suivi conduits de juin à juillet 2025.

Différentes méthodes ont été utilisées pour mener l'étude, notamment des recherches documentaires, des analyses quantitatives, une modélisation entrées-sorties, des analyses de scénarios, des enquêtes et des entretiens.

Les principales conclusions des évaluations de l'étude sont présentées ci-dessous par tâche.

Principales conclusions de la tâche 1

La première tâche consistait à établir une liste restreinte de biens prioritaires pour une extension potentielle du champ d'application du MACF, en vue d'un examen plus approfondi. Le point de départ était la liste actuelle des secteurs et produits considérés comme exposés à un risque de fuite de carbone dans le cadre du système d'échange de quotas d'émission de l'Union européenne (SEQE-UE), conformément à l'article 30 (2b) du règlement MACF. Ont d'abord été exclus de cette liste longue les secteurs et produits déjà inclus dans le MACF, ceux relatifs à un service et non à un bien, ou ceux qui ne sont pas produits dans l'UE.

Une approche de filtrage en trois étapes a ensuite été appliquée à la liste longue des secteurs et produits afin d'établir une liste restreinte prioritaire de biens, basée sur des statistiques publiques et des données confidentielles du SEQE-UE de la Commission :

- **Étape 1 :** la liste longue a d'abord été réduite aux secteurs NACE à 4 chiffres les plus exposés au risque de fuite de carbone. D'autres critères utilisés lors du filtrage de l'étape 1 étaient la présence d'au moins une installation SEQE-UE dans le secteur et des émissions annuelles SEQE-UE d'au moins 25 kt CO₂e.
- **Étape 2:** au sein des secteurs résultant du filtrage de l'étape 1, des biens présentant des niveaux significatifs de commerce, d'importations et de production associés aux installations SEQE-UE ont été identifiés pour établir une liste restreinte préliminaire.
- **Étape 3:** les biens présélectionnés ont ensuite été analysés pour déterminer s'ils pouvaient être regroupés à un niveau CN plus agrégé et/ou s'il était plus logique d'inclure ou d'exclure certains produits en fonction de facteurs non directement reflétés dans les données statistiques.

Cette approche de filtrage a abouti à des groupes de produits prioritaires pour un examen plus approfondi dans le cadre d'une extension potentielle du champ d'application du MACF,

dans les secteurs de la pâte à papier et du papier, des produits de raffinerie, du verre, de la céramique, des ferro-alliages, de l'aluminium et des métaux non ferreux.

En outre, un nombre important de produits chimiques inorganiques et organiques ainsi que de polymères ont également été identifiés pour une priorisation. Cela n'était cependant pas basé sur l'approche de filtrage statistique. L'approche de filtrage a conduit à des biens répartis dans différentes parties de la chaîne de valeur chimique, qui n'étaient pas congruents avec les processus générant des émissions significatives dans l'industrie chimique. Au lieu de cela, une approche de chaîne de valeur « centrée sur les substances » a été utilisée, c'est-à-dire en se concentrant sur les chaînes de valeur avec des produits chimiques qui sont des substances pures et non des mélanges ou des formulations de plusieurs substances. Les substances produites en grands volumes dans l'UE et/ou entraînant des émissions significatives ont ensuite été sélectionnées pour la priorisation. La liste des produits chimiques et polymères prioritaires comprenait des substances en aval avec des émissions incorporées significatives provenant de biens actuellement inclus dans le MACF, des précurseurs et co-produits clés, et/ou des substances jugées importantes par la Commission pour une analyse plus approfondie.

Principales conclusions de la tâche 2

Cette tâche visait à évaluer la faisabilité technique de l'inclusion de nouveaux biens dans le champ d'application du MACF (Mécanisme d'Ajustement Carbone aux Frontières). L'accent a été mis sur les biens priorisés dans la Tâche 1, en particulier les produits chimiques et les polymères. L'analyse a inclus un examen approfondi du cadre législatif existant du MACF (l'acte d'exécution pour la phase transitoire) et des législations européennes similaires traitant de la déclaration environnementale et des chaînes de valeur. Les critères d'évaluation de la faisabilité technique comprenaient: (1) les biens doivent être clairement identifiables (par code CN ou autre moyen, par exemple un numéro CAS pour les produits chimiques), (2) des règles de surveillance doivent être disponibles, en particulier (a) les processus et voies de production doivent être suffisamment connus pour définir des limites dans la législation, (b) des règles pour attribuer les émissions aux co-produits dans les processus multi-fonctionnels doivent être disponibles, et (c) les problèmes de MRV (Surveillance, Vérification et Déclaration) trop complexes doivent être évités. De plus, (3) la détermination de valeurs par défaut doit être réalisable, et (4) le système MRV doit pouvoir gérer des chaînes de valeur plus longues. Pour les produits chimiques et les polymères, le critère (2) a été complété par une méthodologie permettant de calculer une « mesure de complexité » pour les chaînes de valeur et les processus de production pertinents.

Les résultats de cette tâche suggèrent qu'il est techniquement faisable d'inclure dans le MACF: (1) la plupart des codes CN du secteur de la pâte à papier et du papier, (2) la plupart des codes CN du secteur du verre, (3) plusieurs codes CN du secteur de la céramique, y compris les matériaux réfractaires, (4) dans le secteur des ferro-alliages : le ferrosilicium. Dans ces secteurs, des groupes de biens ont été jugés inadaptés à une inclusion dans le MACF en raison de leur nature trop diverse ou de la complexité excessive de leurs processus de production. Cette dernière raison conduit à considérer le secteur des métaux non ferreux comme trop complexe pour être inclus dans le MACF. Par ailleurs, des groupes de biens ont été identifiés qui ne sont généralement pas produits dans des installations éligibles à l'inclusion dans le SEQUE-UE (Système d'Échange de Quotas d'Émission de l'UE), c'est-à-dire qu'ils seraient considérés comme des « biens en aval » et pourraient être inclus dans le MACF si l'inclusion de leurs précurseurs entraînait un risque de fuite de carbone pour ces biens en aval.

Les produits du secteur de la raffinerie ont été évalués, d'une part, en tant que secteur individuel, et d'autre part, en tant que fournisseurs de précurseurs pour les secteurs des produits chimiques organiques et des polymères. Les conclusions indiquent qu'il serait plutôt

difficile d'inclure les raffineries uniquement pour certains produits, comme le naphta, le gaz de pyrolyse et le reformat en tant que précurseurs pour l'industrie chimique. L'ensemble du secteur a été identifié comme difficile à surveiller, déclarer et vérifier (MRV), mais pas dans une mesure qui rendrait impossible son inclusion dans le MACF, car une MRV similaire est déjà mise en œuvre dans le cadre du SEQUE-UE.

En ce qui concerne les produits chimiques et les polymères, un total de 118 substances ont été évaluées. Cependant, cette évaluation n'a pas abouti à un résultat simple, mais à une situation plus complexe : pour la plupart des substances, s'ils n'ont pas leur propre code CN, il est possible de les identifier, par exemple, par un numéro CAS ou une dénomination UIPAC. Dans certains cas, cela n'a pas été possible avec la méthodologie et les ressources de ce projet. De plus, un simple classement des substances en fonction de la complexité de la chaîne de valeur ne permet pas de trancher clairement en faveur de leur inclusion dans le MACF. Bien que la priorité ait été donnée aux substances déjà couvertes par des références de produits dans le SEQUE-UE et à d'autres produits chimiques à haut volume, un volume élevé ne correspond pas toujours à une bonne faisabilité technique. D'autre part, certains biens plus simples n'ont été inclus dans l'évaluation que parce qu'ils pourraient être des précurseurs de produits chimiques à plus haut volume. Mais si le bien plus complexe était retiré de la liste d'inclusion dans le MACF, ces précurseurs plus simples pourraient également être retirés. De même, certains biens n'ont été inclus que parce qu'ils apparaissent comme co-produits d'autres biens évalués et ne justifieraient pas une inclusion par eux-mêmes. Dans l'ensemble, il a été constaté que l'inclusion de certains produits chimiques et polymères serait techniquement faisable, tout en laissant ouverte la question de la limite de complexité à fixer.

Principales conclusions de la tâche 3

L'objectif de la Tâche 3 était d'identifier les précurseurs des biens relevant du champ d'application actuel du MACF, ainsi que des biens identifiés dans la Tâche 2, et de déterminer s'ils devraient être inclus dans le MACF. Là encore, l'évaluation s'est concentrée sur la faisabilité technique (en utilisant un ensemble de critères similaire à celui de la Tâche 2) et les chaînes de valeur pertinentes. Les biens suivants ont été identifiés comme candidats possibles à l'inclusion dans le MACF en tant que précurseurs du champ d'application existant : chaux, dolomie, magnésie, alumine. Une attention particulière doit être accordée au coke : son inclusion est techniquement réalisable, mais une solution doit être trouvée pour le traitement de ses co-produits et les règles d'attribution des émissions associées. Selon les produits chimiques qui seraient inclus dans le MACF (voir Tâche 2), les co-produits du coke (goudron, huiles légères, aromatiques bruts) deviendraient des précurseurs pour les produits chimiques et devraient alors être explicitement considérés pour leur propre inclusion dans le MACF.

Les produits de raffinerie n'ont pas été spécifiquement évalués dans le cadre de cette tâche, bien qu'ils soient pertinents en tant que précurseurs pour les produits chimiques et les polymères. Cependant, cette évaluation a déjà été réalisée dans la Tâche 2. Pour les produits chimiques et les polymères, aucune évaluation spécifique n'a été menée dans la Tâche 3, car leurs précurseurs avaient déjà été inclus dans la Tâche 2. Enfin, la ferraille peut être un précurseur important pour plusieurs secteurs, tels que le fer et l'acier, l'aluminium, mais aussi pour le verre (calcin) et le papier (pâte recyclée). Cependant, une discussion détaillée sur l'inclusion dans le MACF a été exclue de ce projet, car il a été jugé nécessaire de développer d'abord une classification appropriée des ferrailles pré-/post-consommation et une méthode adaptée pour attribuer les émissions incorporées à la ferraille.

Principales conclusions de la tâche 5

Cette tâche était consacrée au développement d'une méthodologie pour évaluer l'ampleur des émissions de transport non-UE associées aux biens relevant du MACF. Cela inclut une

estimation des émissions de transport non-UE des biens des listes T2 et T3 vers l'UE. Les principales conclusions de cette évaluation sont les suivantes : Il ne semble pas y avoir suffisamment de preuves d'un risque accru de fuite de carbone lié aux obligations de tarification du carbone pour le secteur des transports dans le cadre du SEQUE-UE. De plus, la part des émissions du transport dans les émissions totales reste relativement faible (moins de 5 %). Cette part varie toutefois considérablement selon le pays d'origine et le mode de transport, ce qui souligne l'importance de la distance et des schémas logistiques pour comprendre l'intensité carbone des biens importés dans le cadre du MACF. Cette évaluation pourrait cependant évoluer en fonction de la liste finale des codes NC retenus pour l'extension du champ d'application du MACF.

Principales conclusions de la tâche 6

L'objectif de la Tâche 6 était d'évaluer la faisabilité technique de l'inclusion des émissions des services de transport associés au mouvement des biens MACF en dehors de l'UE, en analysant les exigences en matière de données, en évaluant la disponibilité des données pour les importateurs, et en déterminant l'étendue et le coût de la capacité des autorités de l'UE à surveiller l'exactitude des informations déclarées. Les principales conclusions de cette évaluation sont les suivantes : L'évaluation indique que l'extension du MACF pour inclure les émissions de transport des biens concernés pourrait potentiellement être envisageable du point de vue de la mesure, du stockage et du partage des données d'émissions de transport entre les opérateurs logistiques, bien que certaines limitations subsistent. En outre, cette extension pourrait poser des difficultés aux petits transporteurs des pays tiers, qui sont souvent inconnus même des grands opérateurs logistiques internationaux. Si les institutions européennes disposent des technologies nécessaires pour la déclaration des émissions, le processus de vérification des émissions de transport au niveau du code NC demeure un défi, car il ne s'agit pas d'une pratique courante pour les opérateurs de transport. Les vérificateurs du MACF devraient, pour chaque flux commercial, vérifier les émissions provenant des différents maillons de la chaîne de transport (c'est-à-dire tous les moyens utilisés pour le déplacement d'un lot correspondant à un code NC depuis son origine jusqu'à sa destination), ce qui rendrait le processus de vérification complexe à coordonner et probablement coûteux. Cela nécessiterait en outre de définir des compétences spécifiques pour les vérificateurs ou certificateurs du MACF.

1 Introduction

Article 30 of the CBAM Regulation¹ requires a review and specifies that “the Commission, in consultation with relevant stakeholders, shall collect the information necessary with a view to extending the scope of this Regulation” and Article 30(2) further specifies that “the Commission shall present a report to the European Parliament and to the Council on the application of this Regulation”.

In line with the above review of the CBAM Regulation and expected report of the Commission, the objective of this study was to support the Commission in identifying products currently outside of the scope of the Carbon Border Adjustment Mechanism (CBAM) but in the scope of the EU Emissions Trading System, which could be potentially included in that scope, and to develop the methods to assess embedded emissions in such goods, including those of the transportation of CBAM goods. In other words, to assess the feasibility of extending the scope of the CBAM to:

1. Goods at risk of carbon leakage other than those listed in Annex I of the CBAM Regulation, and specifically organic chemicals and polymers, as required by Article 30 of the CBAM Regulation
2. Input materials (precursor goods) for the goods listed in Annex I of the CBAM Regulation
3. Embedded emissions in the transport of the goods listed in Annex I of the CBAM Regulation, and of new goods with potential for CBAM scope extension.

The overall objective was achieved through the following specific objectives, presented per task:

Task 1 aimed at developing a list of potential goods that are produced by EU ETS installations but not currently in the CBAM scope to assess the feasibility to be added to the current CBAM list, thereby pursuing the objective of mirroring the EU ETS scope.

Task 2 aimed at assessing the technical feasibility of extending the scope to additional CN codes identified in task 1, in particular, organic chemicals and polymers

Task 3 aimed at identifying and specifying relevant precursors given the CBAM's initial scope and its potential horizontal expansion

Task 4 aimed at conducting a consultation with relevant stakeholders to gain an understanding of their perspectives and perception of challenges, concerns, and suggestions related to potential horizontal and vertical (upstream) extension of the scope of the CBAM to the identified sectors, sub-sectors and products

Task 5 aimed at assessing the scale of non-EU transport emissions linked to the import of CBAM goods throughout their supply chain

Task 6 aims to assess the technical feasibility of extending the scope of CBAM to emissions resulting from the transport of CBAM goods outside the EU, based on the CBAM Implementing Regulation for the transitional phase, and evaluate to what extent and at what cost EU authorities can monitor the veracity of reported information

¹ Regulation (EU) 2023/956 of the European Parliament and of the Council of 10 May 2023 establishing a carbon border adjustment mechanism (Text with EEA relevance). <http://data.europa.eu/eli/reg/2023/956/oj>

Task 7 aims to conduct a consultation with relevant stakeholders to understand their perspectives on extending the scope of the CBAM to emissions from non-EU transport emissions.

The study's objectives were achieved through different methods, including desk research, quantitative analysis, input-output modelling, scenario analysis, surveys, and interviews. These methods were refined during the inception phase, with the support of external vital stakeholders and in alignment with the EC's Inter-Secretariat Group (ISG).

2 Task 1. Baseline study of sectors/sub-sectors at risk of carbon leakage

2.1 Objective

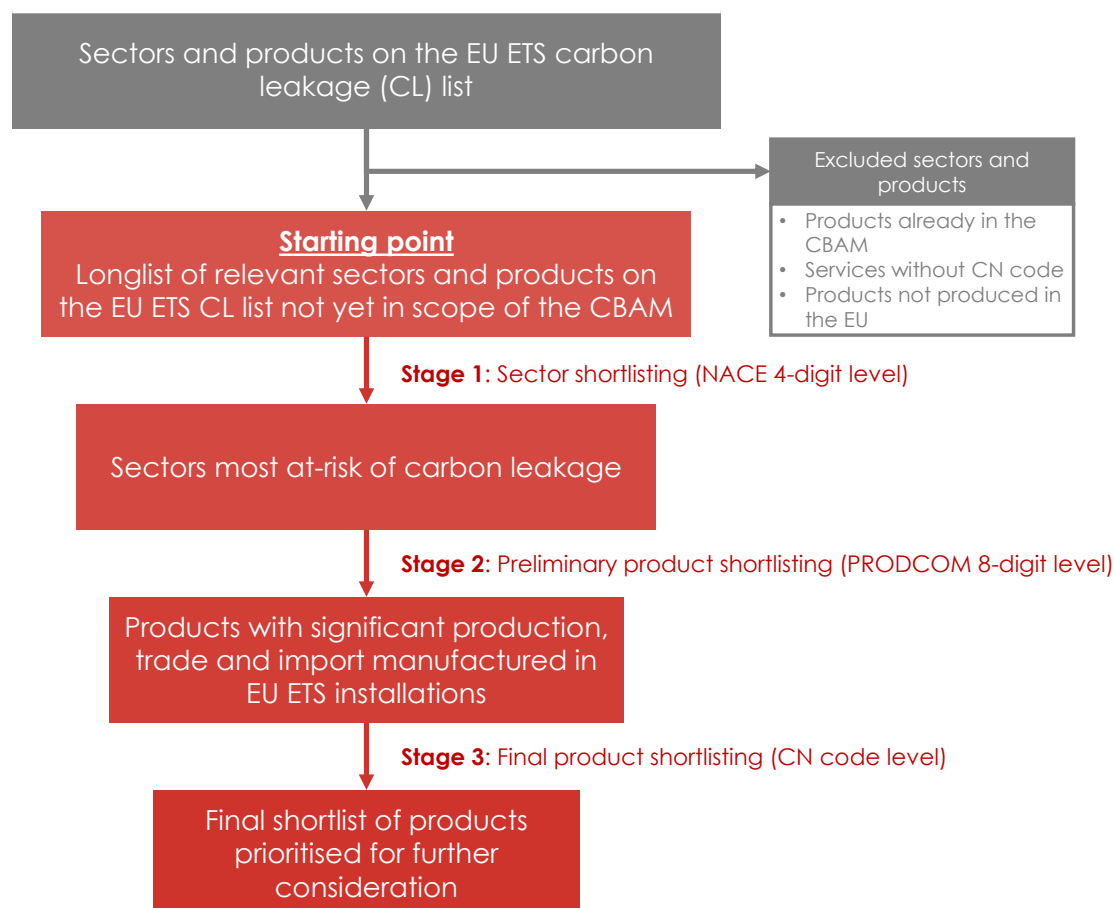
The **objective** of Task 1 of this study was to identify potential segments of CN codes at risk of carbon leakage under the EU ETS that could be treated as a priority for inclusion under the CBAM beyond the ones already included under Annex I of the CBAM Regulation. The segments of CN codes on the final shortlist resulting from the analysis in this chapter were subsequently assessed against their technical feasibility for potential inclusion under the CBAM in Chapter 4 (Task 2).

The rest of this chapter is structured as follows. Section 2.2 presents the methodology for determining the shortlist of CN codes, including the data sources used. The results of each stage are subsequently presented in Section 2.3 with preliminary conclusions in Section 2.6.

2.2 Methodology

We use a three-stage approach for shortlisting products for consideration in the extension of the CBAM to other goods, as shown in Figure 1.

Figure 1 Schematic overview of the three-stage approach for the final shortlist of products



The starting point for the products to extend the CBAM scope to is the **current list of sectors and products deemed at risk of carbon leakage under the EU ETS**, as stipulated in Article 30 (2b) of the CBAM Regulation. This list also includes all sectors and products that are on the

state aid guidelines that can receive compensation for their indirect ETS costs for mitigating carbon leakage.²

The ETS carbon leakage list includes, however, products that are not relevant for the analysis. These were first excluded before filtering the longlist to focus the analysis on the NACE 4-digit sectors at most risk of carbon leakage in Stage 1. In Stage 2, the products with significant production, trade and import manufactured in EU ETS installations among those sectors were identified. This analysis was done on PRODCOM 8-digit level because the necessary data is not available at the CN code level. In the third stage, the PRODCOM 8-digit codes were translated into their corresponding CN code and analysed to determine if certain products can be grouped or whether it was more logical to exclude them due to factors not directly captured in statistics.

The indicators and criteria used in each stage are presented in the subsections below. These indicators have been selected based on the most disaggregated data available at the NACE 4-digit, PRODCOM 8-digit and/or CN 8-digit level (see Annex 10.1.2 for the indicators explored and data sources used).

2.2.1 Starting point – longlist of relevant sectors and products at risk of carbon leakage

The current EU ETS carbon leakage list covers 1086 8-digit PRODCOM codes³, which correspond to 2465 unique CN codes. A portion of these codes are, however, not relevant to the analysis in this chapter due to one of the following reasons:

- **CN codes already included in the CBAM:** a total of 426 CN codes⁴ are already covered under the CBAM, which correspond to 187 unique PRODCOM 8-digit codes. These products are spread over different NACE and PRODCOM codes as shown Table 1.
- **PRODCOM codes relate to a service and not a good:** these PRODCOM codes do not have a corresponding CN code. This is the case for 61 PRODCOM codes in NACE sector 1330 (Finishing of textiles), 2446 (Processing of nuclear fuels) and 2451 (Casting of iron), and PRODCOM 25501134 (Open die forged ferrous parts for transmission shafts, camshafts, crankshafts and cranks, etc.).
- **PRODCOM codes identify a good that is not produced in the EU:** the production value and volume associated with these PRODCOM codes is 0. This is the case for 11 PRODCOM codes corresponding to 11 CN codes⁵.

This results in 852 PRODCOM 8-digit codes (2028 CN codes) remaining on the longlist. See Annex 10.1.1 for the full list of PRODCOM and CN codes.

² Commission Communication 2020/C 317/04

³ Based on the 2023 version of the PRODCOM codes. Does not include special PRODCOM aggregate codes containing "Z" or "T" in the code.

⁴ The total number of CN codes covered under the CBAM is higher, because it also includes products from sectors downstream of the steel and aluminium sectors that are not on the carbon leakage list.

⁵ CN 26161000 - Silver ores and concentrates; CN 26139000 - Molybdenum ores and concentrates (excl. roasted); CN 26171000 Antimony ores and concentrates; CN 28259020 Beryllium oxide and hydroxide; CN 28121400 Phosphorus pentachloride; CN 28121910 Chlorides and chloride oxides of phosphorus (excl. phosphorus oxy-, tri- and pentachloride); CN 28469030 Scandium compounds, inorganic or organic; CN 81109000 Articles of antimony, n.e.s.; CN 81125100 Unwrought thallium; thallium powders; CN 81125900 Articles of thallium, n.e.s.; CN 81124900 Articles of rhenium, n.e.s.).

Table 1 Sectors and products on the EU ETS carbon leakage list in the current CBAM scope

07101020	Iron ores and concentrates. Agglomerated (excluding roasted iron pyrites)	1	1
08122160	Kaolinic clays (ball and plastic clays)	1	1
20111150	Hydrogen	1	1
2015	Manufacture of fertilisers and nitrogen compounds	21	25
2351	Manufacture of cement	3	5
2410	Manufacture of basic iron and steel and of ferro-alloys	101	250
2420	Manufacture of tubes, pipes, hollow profiles and related fittings, of steel	31	73
2431	Cold drawing of bars	12	22
2442	Aluminium production	13	43
2451	Casting of iron	3	5

2.2.2 Stage 1 – Sector shortlisting (NACE 4-digit level)

The sector shortlisting is based on several criteria agreed with DG TAXUD for determining the significance of sectors for this study. This was supplemented with a trade flow analysis at an aggregated product level to shed light on the products and countries that may be affected by the inclusion of the products of the shortlisted sectors.

2.2.2.1 Significance of sectors

Firstly, the shortlist of sectors at the NACE level was based on the following criteria to focus on the most emissions- and trade-intensive sectors in the EU ETS:

- **EU ETS Phase 4 carbon leakage indicator for the sector's NACE code being at least 0.8⁶:** this criterion, agreed with DG TAXUD for this study, enables the analysis to prioritise the sectors at most risk of carbon leakage.
- **At least 1 installation covered by the EU ETS is reporting its main activity under the sector's NACE code⁷:** this is to ensure that any sector to which the CBAM is potentially extended falls in a sector with ETS installations to ensure WTO compatibility at the sector level.
- **Total annual EU ETS emissions of a sector's NACE code being at least 25 ktCO₂e⁸:** This corresponds to the threshold used under the EU ETS to consider whether an installation is a small emitter. Small emitters can be exempted from the EU ETS if they are subject to equivalent measures. If the total emissions of a sector are below the small emitters threshold, this means that the sector only consists of small emitters. These small emitters can mitigate carbon leakage risks by applying to be exempted from the EU ETS and are therefore considered a lower priority in this study for potential extension of the CBAM.

⁶ Based on the carbon leakage indicator established in 2019 based on 2013-2015 data available from https://climate.ec.europa.eu/document/download/de7b6d7e-a152-4796-901d-133e5bd9aefc_en?filename=6_cli-ei-ti_results_en.pdf.

⁷ Based on the anonymised confidential NIMs data for free allocation under the EU ETS for 2021-2025 received from the Commission.

⁸ Based on the average of the latest three years of ETS emissions from the anonymised confidential NIMs data for free allocation under the EU ETS for 2021-2025 received from the Commission, i.e., the years 2016-2018.

Sectors on the EU ETS carbon leakage list that do not meet the abovementioned criteria were not further analysed in Stage 2. The only exceptions were sectors that do not have a carbon leakage indicator of at least 0.8 but share similar or integrated production processes with the sectors meeting the abovementioned criteria or are directly downstream from those sectors.

2.2.2.2 Trade flow and embedded emissions

The trade flow analysis at the aggregated product level entailed conducting two types of analysis, the trade intensity analysis, using Comext data, and the input-output analysis with Exiobase. The trade intensity analysis shed light on the countries and products that would be directly most affected by a CBAM extension. The input-output analysis shed light on the countries and products that in terms of embedded value added and emissions, would be more affected (and broken down by EU vs non-EU origin), considering their supply chain, mainly focusing on upstream echelons.

The trade intensity analysis was assessed and concluded that it is the same indicator used for the CL indicator. The trade intensity in our analysis, however, used updated Comext to 2024, while the original CL indicator published in 2019 uses 2018 data⁹.

For the input-output analysis, we used Exiobase 3.8.2¹⁰, an environmental multiregional input-output database which covers 200 products and 44 countries (EU27 + 17 non-EU countries) with data from 1995 to 2022. Exiobase includes environmental impact factors, which, combined with the input-output data on trade flows, enabled us to shed light on embedded carbon emissions and value added. The advantage of Exiobase compared to other input-output databases is in it is a relatively good product disaggregation, good enough country coverage (so that the EU and its main trading partners are included), and its integration with emissions data.

As a first step, we calculated indicators related to embedded emissions and value added:

- **Embedded emissions** (including EU, non-EU, total, and ratio of non-EU to EU). This is the amount of carbon directly or indirectly emitted to satisfy the production of a given product (considering its entire supply chain). We additionally distinguished between the carbon embedded originating from the EU vs outside the EU. These indicators give a better picture of the environmental impact of a given product, not only looking at its direct emissions, but also capturing the emissions of the products (some of which imported from outside the EU) that were used as intermediate inputs.
- **Origin of value added in international trade**, including EU, non-EU, total, and ratio of non-EU to EU. This indicator is the equivalent to the embedded emissions, but to analyse value added embedded in international trade. This indicator gave an indication of the origin of the value added in trade.

As second step, concentration ratio of the top 5 contributing products or countries was produced, CR5. The concentration ratio was used to proxy the administrative burden by sector. It measures the degree to which a sector's activity is concentrated among a few countries or distributed across many. This ratio ranges from 0% to 100%:

⁹ See EC (2019) Carbon Leakage at https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/free-allocation/carbon-leakage_en#carbon-leakage-list

¹⁰ <https://www.exiobase.eu/>

- A low concentration ratio suggests that a sector's activity is spread across many countries, leading to higher administrative burden due to increased regulatory complexity and coordination efforts.
- A high concentration ratio indicates that a sector is dominated by a few countries, implying lower administrative burden as fewer jurisdictions are involved in regulatory and coordination efforts.

The analysis had **limitations**. While Exiobase is relatively highly disaggregated for an input-output database, it is not disaggregated enough to match all CN codes. Therefore, most CN codes under consideration were mapped to similar Exiobase products, and some dynamics that are specific to individual CN-code level products may not have been properly captured. Additionally, Exiobase has not been updated in recent years, with the input-output data nowcast up to the year 2022. This may also mean that some recent structural changes due to Covid-19 or the war in Ukraine may not be well captured in the data. Instead, the indicators produced will most likely capture longer-term patterns in global value chains. The analysis also indicated that the embedded emissions and origin of value added could not be disaggregated to the level necessary to arrive to definitive conclusions on including or not CN codes in the final short list.

Finally, it is worth noting that Exiobase captures precursors' contribution to embedded emissions, which is something the CL methodology excludes. An assessment of how Exiobase or other Multi-Regional Input-Output analysis of embedded emissions can be used to update the CL methodology, with implications for the sectors to include in CBAM may be needed, but this is outside the scope of this study.

2.2.3 Stage 2 – Preliminary product shortlisting

Within the shortlisted sectors, a preliminary shortlist of products was established based on quantitative criteria agreed with DG TAXUD to focus on products with significant production, trade and import that are also manufactured by EU ETS installations. The following criteria have been applied for this analysis at PRODCOM code level:

- **Goods under the PRODCOM code are produced in at least 1 EU ETS installation:** in Stage 1 the filtering was done based on the NACE code of an EU ETS installation, but the EU ETS installation only manufacture some products that fall under the NACE code. In addition, some EU ETS installations can also manufacture products that fall under NACE codes other than the one the EU ETS installation reports under. An additional filtering is therefore done at product level based on its PRODCOM 8-digit code. The rationale was that for a good to be included in the CBAM, the embedded emissions of the good produced in the EU must be first priced under the EU ETS (and thus produced by an EU ETS installation) to ensure equivalent treatment between good produced in the EU and outside the EU and hence consistency with the Union's non-discrimination obligations under WTO rules. For determining if a product is produced by an EU ETS installation, whether a PRODCOM code is mentioned at least once in the NIMs data for determining free allocation for the EU ETS 2021-2025 is used as an indicator¹¹. It is important to note that operators in the

¹¹ The National Implementation Measures (NIMs) data refers to confidential data used for determining free allocation under the EU ETS for 2021-2025 as received from the Commission in an anonymised manner. The information reported by operators in the NIMs data are for the years 2014-2018. For determining the mention of PRODCOM codes, the 2023 version of the PRODCOM codes have been used. Where PRODCOM codes could not be matched to PRODCOM codes from the 2024 version, a manual matching these PRODCOM codes to the 2010 version of PRODCOM codes (the version specified in the NIMs that EU ETS installations should provide) where possible. For

EU ETS do not have to report PRODCOM under all circumstances. Therefore, the absence of a PRODCOM code in the NIMs only means that there is insufficient data to confirm that a product faces EU ETS costs:

- EU ETS installations may not have reported all relevant PRODCOM codes in the NIMs or reported the wrong PRODCOM codes. PRODCOM codes of products not reported by any ETS installation could not be considered in the analysis.
- Products made in non-ETS installations may face indirect ETS costs from electricity consumption. However, publicly available data is insufficient to determine its significance at the product level, and this was therefore not further investigated.
- Products made in non-ETS installations may face embedded ETS costs from precursors. This would not be visible in the NIMs data.
- EU production value of goods under the PRODCOM code is at least 100 million Euro per year¹²: this is to focus the analysis on the products with a significant production in the EU.
- Extra-EU import value of goods under the PRODCOM code is at least 100 million Euro per year¹³: this is to focus on the most significantly imported products from outside of the EU into the EU.
- Extra-EU trade intensity of goods under the PRODCOM code is at least 20%¹³: the intensity of a given product gives an indication of how easily a good can be traded, with the threshold set at a level to focus on the most significantly traded products. For the trade intensity, the definition used is the same as for establishing the EU ETS carbon leakage list, i.e., the ratio of the sum of extra-EU imports and exports to the sum of the total production and extra-EU imports¹⁴.
- Extra-EU import intensity of goods under the PRODCOM code is at least 10% or Extra-EU import value of goods is at least 250 million Euro per year¹³: these are indicators of direct dependency on imports of a given product from outside of the EU:
 - The trade intensity enables a focus on the products for which a relevant portion or amount is already being imported from outside the EU. The rationale to focus on products with a relevant portion or amount of extra-EU import is that for these products, there are already established import channels that may be scaled up in the short to medium term that could lead to carbon leakage. This would also filter out products that have a high trade intensity largely due to exports. The import intensity is defined as the ratio of imports to the sum of extra-EU imports and total production. The 10%-threshold mirrors the lowest threshold used for the

PRODCOM codes that represent multiple products not specified in any other PRODCOM codes, the mention of the PRODCOM code is only allocated to an equivalent PRODCOM code that covers multiple products not specified elsewhere. Some EU ETS installations only reported the first 4 or 6 digits of the PRODCOM code. If more than one 8-digit PRODCOM code falls under the reported 4- or 6-digit code, the PRODCOM code is not considered as it cannot be confirmed that a specific PRODCOM code product is produced by an EU ETS installation. For more information on free allocation in the EU ETS and the NIMs, see https://climate.ec.europa.eu/eu-action/carbon-markets/eu-emissions-trading-system-eu-ets/free-allocation/allocation-industrial-installations_en.

¹² Based on data from the latest three years available at the time of the analysis, i.e., 2021-2023 from Eurostat database ds-056121 "Total production" available at <https://ec.europa.eu/eurostat/databrowser/view/ds-056121/default/table?lang=en&category=prom>, or if no PRODCOM code exists, based on the Eurostat Structural Business Statistics available at https://ec.europa.eu/eurostat/databrowser/view/sbs_ovw_act/default/table?lang=en.

¹³ Based on data from the latest three years available at the time of the analysis, i.e., 2021-2023 from Eurostat COMEXT database DS-056120: Sold production, exports and imports available at <https://ec.europa.eu/eurostat/comext/newxtweb/>.

¹⁴ Defined as (extra-EU import value + extra-EU export value) / (extra-EU import value + EU production value)

trade intensity in determining the carbon leakage list in Phase III of the EU ETS (2013-2020).

- Some products have substantial imports from outside of the EU but the production in the EU is significantly higher, resulting in a low import intensity. For these products, there would also be well-established import channels that could potentially be scaled up in the short to medium term. To take products with substantial import flows into account, products with more than 250 million Euro per year of extra-EU imports are also included in the preliminary product shortlist. This threshold reflects approximately the 3rd quartile import value of the relevant products on the EU ETS carbon leakage list at PRODCOM 8-digit level from the starting point (Section 2.2.1).

Products meeting all the abovementioned criteria at the PRODCOM 8-digit level were shortlisted for further analysis in Stage 3.

2.2.4 Stage 3 – Final product shortlisting

In the final stage, the PRODCOM products on the Stage 2 preliminary shortlist converted to their CN equivalent¹⁵. The CN products were subsequently analysed to determine if their products could be grouped at a more aggregated CN level or whether it may be more logical to include or exclude them due to factors that are not directly captured in statistics. This was based on a high-level analysis of the description of the CN codes, the import data at CN level, and supplementary desk-based research:

- Grouping products from different CN codes at a higher aggregation level (HS Chapter 2-digit, HS Heading 4-digit or HS Sub-heading 6-digit level)¹⁶ and/or shortlisting products for potential inclusion in the CBAM at the 2-, 4- or 6-digit code level: in some cases, the latter resulted in adding additional CN codes similar to the shortlisted ones that fall under the same HS 4 or 6-digit code but did not get shortlisted in Stage 2. This was done for CN codes on the Stage 2 shortlist cover a representative share of at least 80% in terms of the import volume or value at the higher aggregated HS code level¹⁷. Shortlisting products at the more aggregated level (i.e., shortlisting multiple CN codes based on the HS code level they fall under) could reduce the risk of importers circumventing the CBAM by reporting their products in a similar CN code that is covered by the CBAM.
- Note that there may still be uncovered CN codes through which importers may circumvent the CBAM beyond the ones identified for the final shortlist for further consideration. This was not further investigated in this study as it would have required assessing each individual CN code considered for inclusion under the CBAM and identifying potential codes importers might be able to use to circumvent the CBAM.

¹⁵ Using the PRODCOM 2023 – CN 2023 correspondence table available at <https://circabc.europa.eu/ui/group/c1b49c83-24a7-4ff2-951c-621ac0a89fd8/library/d3056f31-a684-430a-a77d-12d1d0fdfffa>.

¹⁶ The first 6-digits of a CN code correspond to the codes under the Harmonised System (HS) that are uniformly used by customs around the world. For more details, see https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff/combined-nomenclature_en.

¹⁷ Based on import data from the latest three years available at the time of the analysis, i.e., 2021-2023 from Eurostat COMEXT database DS-059332: Extra-EU imports since 2010 by partner and other partner, by HS2-4-6 and CN8 available at <https://ec.europa.eu/eurostat/comext/newxtweb/>.

- Adding products that are very clearly direct downstream products of the shortlisted product for potential extension of the CBAM: this would be relevant for shortlisted raw metal products not yet covered by the CBAM were based on the description of the PRODCOM code it can be clearly determined that a product is (almost) fully made of the product in Stage 2. This would follow the logic applied to steel and aluminium products currently included in the CBAM.
- **Excluding PRODCOM and/or CN codes used for unspecified products not categorised in any other 8-digit PRODCOM and/or CN code if related CN codes for specific products are not shortlisted:** the description of these codes are generally indicated with “n.e.c.” or “other”. Inclusion of such CN codes may make sense if related CN codes with similar products are shortlisted to prevent importers circumventing the CBAM by reporting their goods in such codes for “other” products. However, in sectors where (almost) all products shortlisted in Stage 2 belong to such PRODCOM and/or CN codes such risk are not present. Instead, the Stage 2 shortlisting is mostly the result of the production and import value of these codes being inflated as it represents a combination of multiple products. Also, it cannot be accurately determined if the good specified by the CN code is being produced by an ETS installation as the corresponding PRODCOM code represents widely differing products.
- **Excluding CN codes with a relative low import volume and/or low emission intensities:** for the products with low embedded emissions and/or low emissions intensity, the associated administrative costs of including them in the CBAM may not be proportional to the carbon leakage risk mitigated. However, data for emissions are not available at the product level and emissions intensity only available for a limited number of products (see Annex 10.1.2). To obtain an indication of whether the products involve large, embedded emissions, a high-level assessment is made on whether products involve energy-intensive processes or require a lot of manual work as an indication of a low emissions intensity per product. In addition, where products have a high import value but a limited import volume, this could also be an indication of a low emissions intensity and total amount of total embedded emissions. These products are thus excluded.

The analysis described above resulted in a final shortlist of products for analysis in Chapter 3 in terms of technical feasibility of including them under the CBAM.

2.3 Stage 1 results – Shortlisting of sectors

Stage 1 was the shortlisting of sectors on the longlist of relevant sectors and products on the EU ETS CL list not yet in scope of the CBAM. Section 2.3.1 presents the results of the sector shortlisting and Section 2.3.2 the findings of the supplementary trade flow analysis. Section 2.3.3 concludes with the list of sectors and products for further analysis in Stage 2.

2.3.1 Sector filtering

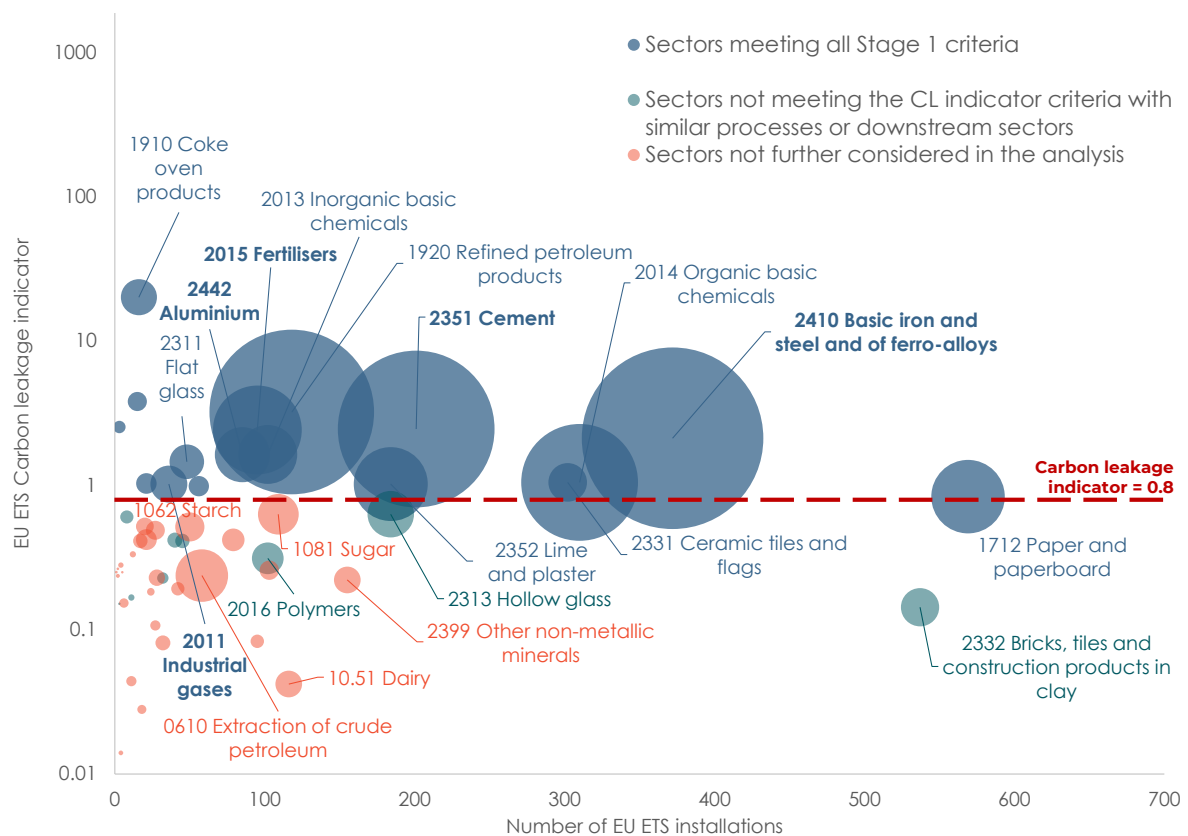
Figure 2 illustrates the results of the Stage 1 sector filtering according to the criteria described in Section 2.2.2. The blue bubbles show the sectors that have a carbon leakage indicator of at least 0.8, at least one EU ETS installation belonging to the NACE code, and the sector having annual EU ETS emissions of at least 25 ktCO₂e. The green bubbles show the sectors added for consideration that have similar or integrated production processes with the sectors meeting the Stage 1 criteria or are directly downstream from those sectors. The orange bubbles depict the sectors not considered for further analysis.

Figure 2 also shows that the sectors with the most significant EU ETS emissions, represented by the relative bubble size, meet all Stage 1 criteria and are therefore considered for further

analysis. Furthermore, by adding the sectors that do not meet all Stage 1 criteria but have similar processes or are directly downstream of the shortlisted sectors, also all sectors with a relative high number of ETS installations are covered. The latter are specifically relevant for sectors in organic chemicals, ceramics and glass. Specifically, these sectors are NACE 2016 (plastics in primary form), 2017 (synthetic rubbers in primary form), 2313 (hollow glass), 2314 (glass fibres), 2320 (refractory products), 2332 (bricks, tiles and construction products in baked clay), 2341 (ceramic household and ornamental articles) and NACE 2342 (ceramics sanitary fixtures).

Finally, the figure shows that some sectors with relative high amount of EU ETS emissions are not considered for further analysis in this study. These sectors have a carbon leakage indicator of less than 0.8, and thus do not meet all Stage 1 criteria, and are also not directly related to the sectors that do meet all Stage 1 criteria. These are mainly food-related sectors such as 1051 (dairy), 1062 (starch) and 1081 (sugar). Other excluded sectors include 0610 (extraction of crude petroleum) and 2399 (other metallic minerals).

Figure 2 Number of EU ETS installations, carbon leakage indicator and annual ETS emissions of the sectors on the EU ETS carbon leakage list at NACE 4-digit level



Note: Includes sectors that are not on the EU ETS carbon leakage list but PRODCOM 6- or 8-digit codes of those sectors that are (e.g., NACE 10.51 Dairy). The bubble sizes represent the relative size of the annual average ETS emissions of the NACE 4-digit sector for 2016-2018. Only the major bubbles in the graph have been identified. The products of the sectors in bold are already (largely) covered under the CBAM.

2.3.2 Supplementary trade flow analysis

A supplementary analysis with EXIOBASE confirmed the relevance of considering the extension of the CBAM to NACE 2016 plastics in primary form (i.e., polymers). When examining the ratio of non-EU embedded emissions to EU embedded emissions across EXIOBASE products,

polymers are among the top five products where a greater share of emissions originates from outside the EU. This means that, compared to the emissions generated within the EU for producing polymers, a significantly higher amount is embedded in the imported inputs used in their production. This highlights a strong dependency on carbon-intensive supply chains located in non-EU countries emissions. The top contributors for the embedded emissions in polymers are products from the chemicals sector, followed by fossil fuel related activities (crude petroleum extraction, electricity by coal and natural gas extraction). This highlights the importance of including polymers despite not having a carbon leakage indicator of at least 0.8, since the carbon leakage indicator does not include upstream embedded emissions of precursors¹⁸. Table 2 presents the concentration of value added (VA) and carbon content (CC) among upstream countries and products for selected NACE sectors. These indicators help identify whether each sector imports from a rather limited or large number of countries and inputs, and how emissions are concentrated in those upstream elements.

Table 2 Exiobase concentration ratio for the sectors NACE2016, NACE2017, NACE2313, NACE2314, NACE 2320, NACE 2341.

2017. Manufacture of synthetic rubber in primary forms	50.5%	68.5%	12.6%	25.4%
2016. Manufacture of plastics in primary forms	3.7%	5.3%	10.4%	28.4%
2313. Manufacture of hollow glass	20.3%	16.1%	4.6%	10.7%
2314. Manufacture of glass fibres	20.3%	16.1%	4.6%	10.7%
2320. Manufacture of refractory products	29.2%	40.2%	8.7%	19.25%
2332. Manufacture of bricks, tiles and construction products, in baked clay	n.a.	n.a.	n.a.	n.a.
2341. Manufacture of ceramic household and ornamental articles	8%	11.9%	5%	13.1%

Source: Based on Exiobase 3.8.2

2.3.3 Shortlist of sectors for further analysis

The analysis in Stage 1 resulted in the shortlist of sectors shown in Table 3 also shows the number of PRODCOM and CN codes for each NACE 4-digit sector for further analysis in Stage 2. In total 530 PRODCOM 8-digit codes, corresponding to 1 175 unique CN codes, remain on the shortlist for analysis under further filtering in Stage 2.

Table 3 Shortlist of sectors and their number of PRODCOM and CN codes for further analysis in Stage 2

NACE code	Description	For analysis in Stage 2	
		No. of PRODCOM 8-digit codes	No. of CN codes
0710	Mining of iron ores	1	1
0899	Other mining and quarrying n.e.c.	11	26
1711	Manufacture of pulp	4	18
1712	Manufacture of paper and paperboard	53	98

¹⁸ No conclusions could be drawn from the EXIOBASE analysis for the other products as they are aggregated as a single EXIOBASE product with their upstream precursors.

NACE code	Description	For analysis in Stage 2	
		No. of PRODCOM 8-digit codes	No. of CN codes
1910	Manufacture of coke oven products	1	2
1920	Manufacture of refined petroleum products	1	44
2011	Manufacture of industrial gases	10	10
2013	Manufacture of other inorganic basic chemicals	114	192
2014	Manufacture of other organic basic chemicals	168	461
2015	Manufacture of fertilisers and nitrogen compounds	9	12
2016	<i>Manufacture of plastics in primary forms</i>	48	96
2017	<i>Manufacture of synthetic rubber in primary forms</i>	2	18
2311	Manufacture of flat glass	8	23
2313	<i>Manufacture of hollow glass</i>	18	43
2314	<i>Manufacture of glass fibres</i>	10	14
2319	<i>Manufacture and processing of other glass, including technical glassware</i>	13	38
2320	<i>Manufacture of refractory products</i>	12	13
2331	Manufacture of ceramic tiles and flags	1	5
2332	<i>Manufacture of bricks, tiles and construction products, in baked clay</i>	5	5
2341	<i>Manufacture of ceramic household and ornamental articles</i>	8	14
2342	<i>Manufacture of ceramic sanitary fixtures</i>	2	2
2352	Manufacture of lime and plaster	6	6
2410	Manufacture of basic iron and steel and of ferro-alloys	13	14
2442	Aluminium production	1	1
2443	Lead, zinc and tin production	11	19
Total		530	1175

Note: sector for which some of the products are already covered under the CBAM are highlighted in red. The sector cement also meets the sector criteria but is not shown on the list because all products under the NACE code of Cement (2351) are already covered under the CBAM. Sectors sharing similar or integrated production processes with the sectors meeting the Stage 1 sector criteria or sectors directly downstream from those sectors are shown in italic. The downstream sectors Cold drawing of bars (2431), Tubes, pipes, hollow profiles and related fittings of steel (2420) and Casting of iron (2451) are not in the table as all 8-digit PRODCOM codes for which corresponding CN codes are available are already covered under the CBAM.

2.4 Stage 2 results – Preliminary shortlisting of products

In Stage 2, the PRODCOM 8-digit products are assessed against their presence in EU ETS installations, their significance in EU production and extra-EU import, and their intensity in trade with and import from extra-EU countries (see Section 2.2.3). This resulted in the number of PRODCOM and CN codes per sector remaining on the list as shown in Table 4 which also shows the sector group they have been categorised under for the analysis in Stage 3.

For some sectors, their products completely disappeared from the shortlist, which are highlighted in grey in Table 4. For NACE 0710 (iron ores) and to a lesser extent 2011 (industrial gases) as well, this is because the PRODCOM code of products falling in those sectors have not been mentioned NIMs data for free allocation (see Section 2.2.3). For NACE 1920 (coke oven products), 2332 (bricks, tiles and construction products in clay), 2352 (lime and plaster) and largely 2011 (industrial gases), products were largely excluded from further consideration due to their low trade and/or import intensity.

This still left 172 PRODCOM 8-digit codes, corresponding to 588 unique CN codes, on the shortlist for further consideration.

Table 4 Stage 2 filtering of products and the sector group they belong to

Sector group	NACE code	Description	No. of PRODCOM 8-digit codes	No. of CN codes	Explanation on sectors without products included in the Stage 2 shortlist
Mining	0710	Mining of iron ores	0	0	No product mentioned in EU ETS installations
Mining	0899	Other mining and quarrying n.e.c.	1	14	
Pulp & paper	1711	Manufacture of pulp	3	14	
Pulp & paper	1712	Manufacture of paper and paperboard	8	16	
Coke oven	1910	Manufacture of coke oven products	0	0	Low import intensity
Refinery	1920	Manufacture of refined petroleum products	1	44	
Industrial gases	2011	Manufacture of industrial gases	0	0	Low trade intensity or not mentioned in EU ETS installations
Inorganic chemicals	2013	Manufacture of other inorganic basic chemicals	15	29	
Organic chemicals & polymers	2014	Manufacture of other organic basic chemicals	81	308	
Fertilisers	2015	Manufacture of fertilisers and nitrogen compounds	2	5	
Organic chemicals & polymers	2016	Manufacture of plastics in primary forms	33	77	
Organic chemicals & polymers	2017	Manufacture of synthetic rubber in primary forms	1	14	
Glass	2311	Manufacture of flat glass	1	4	
Glass	2313	Manufacture of hollow glass	5	18	
Glass	2314	Manufacture of glass fibres	4	5	
Glass	2319	Manufacture and processing of other glass, including technical glassware	4	18	
Ceramics	2320	Manufacture of refractory products	3	3	
Ceramics	2331	Manufacture of ceramic tiles and flags	1	5	
Ceramics	2332	Manufacture of bricks, tiles and construction products, in baked clay	0	0	Low trade and/or import intensity
Ceramics	2341	Manufacture of ceramic household and ornamental articles	1	2	
Ceramics	2342	Manufacture of ceramic sanitary fixtures	2	2	
Lime and plaster	2352	Manufacture of lime and plaster	0	0	Low trade and/or import intensity
Ferro-alloys	2410	Manufacture of basic iron and steel and of ferro-alloys	2	2	
Aluminium	2442	Aluminium production	1	1	
Other non-ferrous metals	2443	Lead, zinc and tin production	3	7	
Total			172	588	

2.5 Stage 3 results – Final shortlist of products prioritised for further consideration

In this last stage, the preliminary shortlist of CN codes was analysed to determine if CN codes should remain on the shortlist and/or if other CN codes should be added (see Section 2.2.4). First, for each sector group with products shortlisted in Stage 2, the shortlisted CN codes are presented below. This is followed by an analysis of the shortlisted CN codes and suggestions on the CN codes as a priority for further consideration. For the sectors where this is applicable, the Stage 2 shortlisted CN codes not prioritised for further consideration are also provided.

2.5.1 Mining

Stage 2 shortlist results

The only shortlisted product in the mining sector group is PRODCOM 08992920 (Other minerals). This includes products from 14 different CN codes not belonging to any other PRODCOM code that are widely different from one another such as quartz, magnesium oxide, asbestos, mica, steatite, feldspar, leucite, nepheline, perlite, chlorites, vermiculite, slag and ash (See Table 56 in Annex 10.1.3).

Prioritisation for further consideration

None of the shortlisted CN level products in this sector are suggested for prioritisation for further consideration (see below).

Not prioritised for further consideration

A further disaggregated analysis of the PRODCOM codes at the CN level of this product group for "other products" reveals that all products have an extra-EU import value below the 100 million Euro criteria used for the Stage 2 analysis. The only exception is magnesium oxide. However, from the available data in the NIMs it cannot be confirmed that magnesium oxide is the product mentioned by EU ETS installations.

In the NIMs, EU ETS installations only reported PRODCOM 08992900, which is the 2010 version of the PRODCOM code for "Other minerals". Based on the same description of the code, PRODCOM 08992900 has been mapped to 2023 version of the PRODCOM code as used in this analysis, which is PRODCOM 08992920¹⁹. As noted above, PRODCOM 08992920 covers many different CN codes beyond magnesium oxide. Thus, it cannot be confirmed with the NIMS data that the embedded emissions of magnesium oxide are priced under the EU ETS such that consistency with the Union's non-discrimination obligations under the WTO rules is ensured. However, the relevant Annex I activity in the EU ETS Directive is "*production of lime or calcination of dolomite or magnesite in rotary kilns or in other furnaces with a production capacity exceeding 50 tonnes per day*" which clearly indicates the relevance of producing calcined

¹⁹ Further inspection of the 2010 version of the PRODCOM 08992900 shows that this code includes various PRODCOM codes in the 2023 version, namely PRODCOM codes 08992910 (Natural graphite), 08992913 (Natural magnesium carbonate (magnesite)), 08992914 (Dead-burned (sintered) magnesia), 08992915 (Fused magnesia and other magnesium oxide) and 08992920 (Other minerals). Of these products, only PRODCOM 08992914 (Dead-burned (sintered) magnesia) and CN 25199010 magnesium oxide within PRODCOM 08992920 (Other minerals) meet the Stage 2 criteria in terms of production, import and trade. However, since EU ETS installations only reported PRODCOM 08992900, which also cover also other products, it cannot be confirmed that magnesium oxide and/or dead-burned (sintered) magnesia is the actual product that is being manufactured in these installations.

magnesium oxide. As one of its most important uses is the production of refractory materials, it is possible that it does not appear in NIMs data due to its integration in the ceramics sector instead of the mining sector. It is therefore discussed in Task 3 as a potential precursor for ceramics.

Based on the analysis discussed above, we suggest to not include any CN codes from this sector as a priority for further consideration.

2.5.2 Pulp & Paper

Stage 2 shortlist results

The shortlisted products from Stage 2 below can be divided into three broad categories based on their production process (see Table 57 in Annex 10.1.3 for the full list of products on the Stage 2 shortlist):

- **Chemical wood pulp:** all CN codes under HS 4702 and 4703 are shortlisted
- **Mechanical wood pulp:** all CN codes under HS 4701 and 4705, 4706 are shortlisted
- **Paper and paperboard:**
 - At the HS 4-digit level, all CN codes under HS 4801 are shortlisted
 - At the HS 6-digit level, all CN codes under HS 480256, 480257, 480261, 480262, 480269, 480300, 480411, 481159 and 481190 are shortlisted.

Prioritisation for further consideration

For wood pulp (HS Chapter 47) at a group level, only CN codes under HS 4704 (Chemical wood pulp, sulphite, other than dissolving grades - unbleached: coniferous) are not shortlisted. This HS 4-digit code only represents 0.5% of the import value and weight of HS Chapter 47. Given that the only differentiation between CN codes in HS 4702 and the other HS codes within HS 47 is dissolving grades, **an extension of the CBAM scope to CN codes under HS 47 could therefore consider an extension to all CN codes under HS 47.**

For paper and paperboard (HS Chapter 48) at a group level, the list of shortlisted products represents 38% in value and 40% in weight. Based on a high-level analysis of the HS and CN code descriptions, there is a clear differentiation in terms of types of products at the HS 4-digit level. This would suggest inclusion of all CN codes under a HS 4-digit code would be ideal to prevent the circumvention risk of reporting in similar uncovered CN codes (and enforcing this in practice). However, the shortlisted CN codes do not cover a representative share (>80%) of the import volume or value of HS Chapter 48 to justify the full inclusion.

The shortlisted CN codes suggested for potential further consideration of extending the CBAM scope under HS Chapter 48 at the HS 4-digit level are as follows:

- **HS 4801 (Newsprint):** all CN codes under the HS 4-digit code are shortlisted;
- **HS 4802 except CN 48021000 (Uncoated paper and paperboard except handmade paper and paperboard):** the shortlisted CN codes represent 82% of the 4-digit HS code's import value and 86% of the imported weight.

Handmade paper and paperboard is proposed to be excluded given the low expected emissions intensity²⁰.

- **HS 4803 (Toilet or facial tissue stock)**: the shortlisted CN codes represent about 90% of the HS 4-digit code's import value and weight that they fall under.
 - In addition, CN codes under **HS 4818 (Toilet paper and similar paper)** should be additionally considered for CBAM extension as they are a direct downstream product under HS 4803.

The shortlisted CN codes suggested to be further considered in this study for the extension of the CBAM scope under the HS 6-digit level are as follows:

- **HS 480411 (Uncoated, unbleached kraftliner)**: the shortlisted PRODCOM 8-digit code covers all CN codes under this HS 6-digit code. The potential extension of the shortlist to all CN codes under the HS 4-digit code 4804 has also been explored, but the shortlisted CN code only represents 43% of the import value of the HS 4-digit code and 51% of the weight; too low to be considered representative.
 - It should be noted that the various other CN codes under the HS 4-digit codes are not shortlisted mostly due to a low import intensity (<10%). Based on the CN code description, it is unclear if including CN codes under HS 480411 under the CBAM could introduce circumvention risk through reporting under similar CN codes such as the ones for uncoated kraftliner that is bleached, semi-bleached or with coloured outer ply (HS 480419) or other kraft paper and paperboard codes (various other CN codes under HS 4804). This requires a product-specific investigation beyond this study.
- **HS 481159 and 481190 (coated, impregnated, covered, surface-coloured, surface-decorated or printed paper, paperboard and other paper articles)**: the shortlisted PRODCOM 8-digit code covers all CN codes under the HS 6-digit code. The potential extension of the shortlist to all CN codes under the HS 4-digit code 4811 has also been explored, but the shortlisted CN codes represent 54% of the import value of the HS 4-digit code and 62% of the weight; too low to be considered representative.
 - Important to note that the products covered under these CN codes are downstream products of other paper and paperboard products before such processing steps. However, not all of the precursors of these CN codes have been shortlisted. Including these CN codes but not their precursors could therefore introduce new carbon leakage risks for these products. If these products were to be included, it is recommended to further investigate these risks for the paper and paperboard sector beyond this study.

²⁰ An indication of handmade paper and paperboard not being emissions intensive is that it is not covered by any of the relevant emissions accounting frameworks for pulp and paper production, which outline the key processing steps and manufacturing facilities to be considered in emissions assessments. This includes the [EMEP/EEA Air Pollutant Emission Inventory Guidebook 2023](#) for the Pulp and Paper Industry, as well as the Confederation of European Paper Industries (CEPI) [Framework for Carbon Footprints of paper and board products](#).

Sectors requiring further analysis

Finally, **a broader consideration** is whether the inclusion of HS Chapter 47 as a whole may lead to carbon leakage risks downstream as wood pulp is the primary precursor of all products under HS Chapter 48. This would require a detailed value chain analysis of the pulp and paper sector that is beyond the scope of this study.

2.5.3 Refinery products

Stage 2 shortlist results

No specific PRODCOM code existing for refinery products. Refinery products have therefore been shortlisted in Stage 2 based on their data for the NACE 4-digit code 1920. The corresponding CN codes for products in this NACE sector are spread over 73 CN codes under HS Chapter 27 Mineral fuels²¹ (see Table 58 in Annex 10.1.2 for the full list of CN codes that corresponds to NACE 1920), which can be grouped as follows:

- **Petroleum oils:** these are all the CN codes that fall under HS 4-digit code 2710, except for HS 6-digit codes 271091 and 271099 that represent waste oils. The latter two codes are not classified under NACE 1920, but under NACE 38.12 for hazardous waste.
- **Petroleum gases:** these are all the CN codes that fall under HS 4-digit code 2711 except 271111 and 271121 that are for liquified and gaseous natural gas. Natural gas falls under NACE 06.20 Extraction of natural gas.
- **Petroleum jelly and wax:** NACE 1920 covers all CN codes that fall under HS 4-digit code 2712 for petroleum jelly; paraffin wax, microcrystalline petroleum wax, slack wax, ozokerite, lignite wax, peat wax, other mineral waxes, and similar products obtained by synthesis or by other processes, whether or not coloured.
- **Petroleum coke:** NACE 1920 covers all CN codes that fall under HS 4-digit code 2713 Petroleum coke, petroleum bitumen and other residues of petroleum oils or of oils obtained from bituminous minerals.
- **Other products with CN codes** that also correspond to NACE 1920, although not being refinery products (but “processed fuels”):
 - CN codes under HS 6-digit code 270212 Briquettes, ovoids and similar solid fuels manufactured from coal;
 - CN codes under HS 6-digit code 270220 Agglomerated lignite (excl. jet).

Prioritisation for further consideration

The shortlisted CN codes suggested to be further considered in this study for the extension of the CBAM scope at the HS 4-digit level are as follows:

- **HS 2710 (petroleum oils):** this covers all CN codes related to petroleum oils resulting from mineral oil refining, including ones for waste oils (under HS 271091 and 271099). While the CN codes for waste oils have not been shortlisted, they are suggested to be included given that it may be very difficult to determine

²¹ Based on the Eurostat correspondence table for CN 2023 - CPA 2.1 available at <https://circabc.europa.eu/ui/group/c1b49c83-24a7-4ff2-951c-621ac0a89fd8/library/b6afc37e-263f-4d8f-ae9c-e510b134087f/details>.

if oils are to be considered as waste. Not including waste oils if non-waste petroleum oils are included under the CBAM could therefore risk importers circumventing the CBAM by reporting imported oils under waste oils.

- HS 2711 except 271111 and 271121 (petroleum gases except liquified and gaseous natural gas): natural gas is not a product from mineral oil refining but extracted from the ground.
 - **A broader consideration** is that if CN code 27111400 (Ethylene, propylene, butylene and butadiene, liquefied, excl. ethylene of a purity of $\geq 95\%$ and propylene, butylene and butadiene of a purity of $\geq 90\%$) under HS 2711 is included, the CN codes for steam cracking products such as ethylene, propylene, butylene and butadiene that fall under HS 4-digit code 2901 may also need to be included to mitigate circumvention risks.
- **HS 2712 (petroleum jelly and wax)**, which are all products that can be obtained from mineral oil refining.
- **HS 2713 (petroleum coke)**, which are all products that can be obtained from mineral oil refining.
 - **A broader consideration** could whether to add all CN codes under HS 4-digit code 2715 related to bituminous mixtures to the CBAM if the CN codes under HS 2713, under which petroleum bitumen falls, is added. The corresponding PRODCOM code for the CN codes under HS 2715 is PRODCOM 23.99.13.10, which covers various bituminous mixtures including those of petroleum bitumen. Trade statistics show that the trade intensity of PRODCOM 23.99.13.10 is 1% and the import intensity less than 1%. The import value is also well below the 100 million Euro threshold used for the Stage 2 analysis. Given these factors, the risk of carbon leakage by not including CN codes under HS 2715 that could include bituminous mixtures with petroleum bitumen appears minimal.

All the CN codes under the abovementioned HS codes cover in total 71 CN codes. An analysis at the CN code level shows that 34 of these CN codes have an extra-EU import value of less than 100 million Euro per year. Following the Stage 2 criteria to focus on the most import-intensive products, these products could be excluded from the shortlist. However, the production of refinery products is a highly integrated and emissions-intensive process²². In 2023, the sector emitted almost 104 Mt CO₂e, equivalent almost 10% of all stationary installations in the EU ETS²³. Developing rules for determining the emissions for each refinery product would require all refinery products to be considered for a potential CBAM extension (which is further explored in Chapter 3).

Not prioritised for further consideration

The shortlisted CN codes that correspond to NACE 1920 suggested not to be further considered in this study for the extension of the CBAM scope are as follows:

²² See for example Ecofys et al. (2009). [Methodology for the free allocation of emission allowances in the EU ETS post 2012 – Sector report for the refinery industry](#).

²³ EEA EU ETS data viewer: <https://www.eea.europa.eu/en/analysis/maps-and-charts/emissions-trading-viewer-1-dashboards>

- CN codes under HS 270212, which do not cover products from mineral oil refining but from coal as per the description of the CN codes;
- CN codes under HS 270220, which is not a product from mineral oil refining but from lignite as per the description of the CN codes.

2.5.4 Inorganic chemicals

Stage 2 shortlist results

In the Stage 2 analysis, a significant number of products have been filtered out compared primarily due to products not being produced in EU ETS installations. This results in the following shortlist, which from a high-level analysis can be broadly categorised as follows (see Table 59 in Annex 10.1.3 for the full list of products on the Stage 2 shortlist):

- **Carbon black:** all CN codes under HS 4-digit code 2803 for carbon black and other forms of carbon are shortlisted.
- **Silicon:** this covers the shortlisted CN codes under HS 6-digit codes 281122 silicon dioxide (silica), 280469 Silicon containing < 99,99% by weight of silicon and 284920 carbides of silicon.
- **Phosphates:** this relates to the shortlisted CN codes under HS 6-digit codes 280920 Phosphoric acid and 283524, 283526, 283529 and 283539 for different types of phosphates.
- **Sodium compounds:** this covers the shortlisted CN codes under HS 6-digit codes 281512 caustic soda, 283620 disodium carbonate (soda ash) and 283630 Sodium hydrogencarbonate (baking soda).
- **Aluminium hydroxide:** this relates to the shortlisted CN codes under HS 6-digit code 281830.
- **Double or complex silicates:** this relates to the shortlisted CN codes under HS 6-digit code 284210.
- **“Other” inorganic chemical groups including ones that are not specified elsewhere,** for which the following “other” inorganic chemical groups have been shortlisted:
 - **Chlorides:** this relates to the shortlisted CN codes under HS 6-digit codes 282720, 282732, and 282739 for the different types of chlorides, including ones that are not specified elsewhere.
 - **Sulphates:** this relates to the shortlisted CN codes under HS 6-digit codes 283311, 283319, 283325 and 283329 for the different types of sulphates, including ones that are not specified elsewhere.
 - **Carbonates:** this relates to the shortlisted CN codes under HS 6-digit codes 283640, 283660, 283692 and 283699 for the different types of carbonates, including ones that are not specified elsewhere.
 - **Other inorganic compounds n.e.c.:** this covers the shortlisted CN codes under HS 6-digit code 285310 Cyanogen chloride and CN code 28539090 for all inorganic compounds not else specified.

Prioritisation for further consideration

Based on a high-level analysis the shortlisted CN codes mentioned above, we suggest **CN code 28183000 (aluminium hydroxide) for potential further consideration in extending**

the CBAM scope. Aluminium hydroxide is an intermediate product in producing alumina, and thus part of the aluminium value chain. Should alumina (CN codes under HS 281820) be included, then CN codes under HS 281830 should also be included.

Further analysis required for consideration

The following CN codes can also be considered for prioritisation in extending the CBAM scope at the HS 4- or 6-digit level, although a broader consideration apply:

- **HS code 2803 (Carbon black):** carbon black already has a product benchmark under the EU ETS and data shows that the production of carbon black is an emissions-intensive process.
- **HS codes 281122, 280461, 280469 and 284920 (silicon):** the shortlisted CN codes under these HS codes are all part of the same value chain, with silica being the main precursor to silicon, and silicon being the main precursor for silicon carbide. Silicon production is also an electro-intensive process. In addition to the shortlisted products in Stage 2, we would suggest adding CN codes under HS 280461 (Silicon containing $\geq 99,99\%$ by weight of silicon) to mitigate any risk of silicon being reported in an uncovered CN code.
 - The inclusion of silica may also support the case for including its downstream product ferro-silicon (CN codes under HS 720221), which is discussed under Section 2.5.9 on ferro-alloys.
- **HS codes 281512, 283620 and 283630 (sodium compounds):** the shortlisted CN codes under these HS codes are all part of the same value chain, with baking soda being a downstream product of soda ash, and soda ash and caustic soda both using sodium chloride (salt) as a precursor. Soda ash already has a product benchmark under the EU ETS and data shows that the production of soda ash is an emissions-intensive process.

The potential inclusion of caustic soda could also justify considering chlorine, a byproduct of the chlor-alkali production process alongside caustic soda, due to its role as a precursor in PVC production. The chlor-alkali production, an energy-intensive process uses brine to produce chlorine, hydrogen (already included in the CBAM) and caustic soda. Should caustic soda be included under the CBAM, a consideration could be to also include CN codes under HS 280110 chlorine to cover all products from the process, even though chlorine is hardly imported (import intensity of 0.1%). However, chlorine is a key precursor to PVC (a key polymer)²⁴. This is one clear example of where additional inorganic chemicals serving as precursors might be considered for the extension of the CBAM beyond the ones shortlisted in this task and more may be identified in the subsequent tasks in this study. A **broader consideration** is thus whether the statistical analysis used in this chapter leads to the most relevant inorganic chemicals used as precursors of organic chemicals and polymers for potential further consideration in the CBAM scope extension to mitigate carbon leakage risks (see also discussion in section 2.5.5 on organic chemicals and polymers). A different approach to conclude which

²⁴ In 2023, 32 % of chlorine in Europe was utilised for producing PVC - a thermoplastic polymer with numerous applications-, while 9% was used for other organic compounds in the European industry. See [Statista Chlorine consumption distribution in Europe 2023, by application](#).

inorganic chemicals could be candidates for potential CBAM inclusion is therefore presented in Chapter 3 as part of the additional analysis on chemicals.

Not prioritised for further consideration

The shortlisted CN codes of the following products suggested not to be further considered in this study for the extension of the CBAM scope are as follows:

- **Phosphates:** phosphoric acid is a precursor to phosphates, with phosphoric acid being produced from an exothermic reaction between sulphuric acid and phosphate rock. For producing phosphates, heat for evaporation of water is needed. However, since sulphuric acid is generally produced onsite and excess heat from those processes can be used, there are limited additional direct emissions involved in the production of phosphates^{25,26}.
- **Double or complex silicates:** complex silicates are a group of products that at least two different metal ions together with silicon and oxygen. These can be different types of metal ions. Double and complex silicates are generally obtained through mining activities and are subsequently processed through a variety of different routes depending on the type of complex silicate and their application. Some of these can be energy-intensive. However, given the complexities in defining clear production routes for the wide possible variety of double or complex silicates²⁷, the CN codes under HS 284210 is not suggested for further consideration.
- **“Other” inorganic chemical groups including ones that are not specified elsewhere:** since the CN codes in this group are a collection of different products, the import value of each CN code was considered. Of the individual CN codes, none of the products had an import value above the Stage 2 filtering criteria of 100 million Euro per year except CN 28332980 (other sulphates not specified elsewhere) and 28539090 (inorganic compounds not specified elsewhere). As these are also CN codes representing a collection of different products, these are also not suggested for further consideration.

2.5.5 Organic chemicals & polymers

Stage 2 shortlist results

Since the value chain, and thus embedded emissions, of NACE 2014 (other organic chemicals), 2016 (plastics in primary form) and 2017 (synthetic rubbers in primary form)

²⁵ [Fertilizers Europe – Booklet No 4 Production of Phosphoric Acid](#) details that when a phosphoric acid plant is connected to a sulfuric acid plant, high-pressure steam from the waste heat recovery boiler is used for power generation, while low-pressure steam is used for phosphoric acid concentration. Additionally, waste heat from other parts of the sulfuric acid plant may reduce steam consumption by being recovered as heated water to help concentrate weak acid. These processes in the production of phosphates configurations translate into limited additional direct emissions.

²⁶ Major EU producers of phosphoric acid also produce sulphuric acid on-site. Examples include [AB Lifosa-Eurochem](#) (one of the largest fertiliser manufacturers in Europe) and [Phosagro](#)

²⁷ Complex silicates can cover a wide range of products and formulations; each produced under specific process conditions and technologies. The [SILMACO](#) (Belgium-based producer of silicates) product catalogue illustrates this complexity with a range of silicates differing in molar ratio and solid content, allowing for tailored applications across various industries.

are strongly interlinked, the shortlisted products from these NACE codes have been considered together as one sector group.

The far majority of the shortlisted products falls within this sector group with 115 PRODCOM 8-digit codes and 399 CN codes (see Table 60 in Annex 10.1.3 for the full list of Stage 2 shortlisted CN codes).

Prioritisation for further consideration

Even when the shortlisted CN codes from Stage 2 are aggregated into HS 6-digit codes, this still leaves 294 HS 6-digit codes shortlisted. An analysis per CN or HS code for potential further consideration is therefore not feasible for this study. Instead, an alternative is to focus on the imported products with the most embedded emissions within this sector group has been explored. Since the embedded emissions are not available from the publicly available data on a CN or HS level, the import volume is used as a proxy instead. Table 5 below shows the top 25 Stage 2 shortlisted imported products by weight at the HS 6-digit level²⁸.

Table 5 HS 6-digit codes of the top 25 imported organic chemicals and polymers from outside of the EU by weight

Top	HS 6-digit code	Description	Extra-EU import weight in ktonnes (average 2021-2023)
1	270799	Oils and other products of the distillation of high temperature coal tar - others	9 626
2	290511	Methanol (methyl alcohol)	6 152
3	382319	Industrial monocarboxylic fatty acids; acid oils from refining; industrial fatty alcohols - others	2 021
4	290110	Acyclic hydrocarbons, Saturated	1 700
5	270750	Other aromatic hydrocarbon mixtures of which 65 % or more by volume (including losses) distils at 250 °C by the ISO 3405 method (equivalent to the ASTM D 86 method)	1 664
6	390120	Polyethylene having a specific gravity of 0,94 or more	1 647
7	390110	Polyethylene having a specific gravity of less than 0,94	1 531
8	390210	Polypropylene	1 525
9	220710	Undenatured ethyl alcohol of an alcoholic strength by volume of 80 % vol or higher	1 075
10	390140	Ethylene-alpha-olefin copolymers, having a specific gravity of less than 0,94	1 073
11	390761	Poly(ethylene terephthalate) - Having a viscosity number of 78 ml/g or higher	1 063
12	291521	Acetic acid	742
13	290531	Ethylene glycol (ethanediol)	720
14	290250	Styrene	642
15	291736	Terephthalic acid and its salts	635
16	290919	Tert-butyl ethyl ether (ethyl-tertio-butyl-ether, ETBE) and other acyclic ethers (excl. diethyl ether)	634

²⁸ Based on an analysis of the description of the shortlisted CN codes, the HS 6-digit level appears to be the appropriate level for the shortlisted CN codes in this sector group where most products that are similar in nature are grouped while allowing for a sufficient differentiation between different types of products.

Top	HS 6-digit code	Description	Extra-EU import weight in ktonnes (average 2021-2023)
17	291532	Vinyl acetate	627
18	440290	Wood charcoal (including shell or nut charcoal), whether or not agglomerated - other	584
19	390410	Poly(vinyl chloride), not mixed with any other substances	563
20	290243	p-Xylene	544
21	390690	Acrylic polymers in primary forms (excl. poly(methyl methacrylate))	469
22	390230	Propylene copolymers	461
23	220720	Ethyl alcohol and other spirits, denatured, of any strength	421
24	290129	Acyclic hydrocarbons, Unsaturated - other	416
25	290211	Cyclohexane	412

The issue with the list Table 5 is that it is not congruent with the processes that lead to highest emissions in the chemical industry. Furthermore, the products on the shortlist are spread out over different parts of the chemical value chain. Some organic chemicals known to have significant embedded emissions are included (such as methanol, polyethylene, polypropylene, PET, styrene, and PVC). However, for many of these products most of their embedded emissions come from their precursors such as ethylene, propylene, and benzene (steam cracking products)²⁹. These precursor products are not on the Stage 2 shortlist due to the relative low trade intensity. At the same time, some precursor products such as ethylene, propylene, butylene, and butadiene are also produced as co-products from mineral oil refining³⁰. The CN codes related to these refinery products have been suggested for prioritisation for CBAM inclusion. Not including the CN codes for the same products from steam cracking could introduce circumvention risks.

To illustrate the relative importance of emissions from precursors in the chemical sector, the annual EU ETS emissions of chemical sector products with a product benchmark shown in Figure 3 can give some insight. The green bars are the ones already covered under the CBAM.

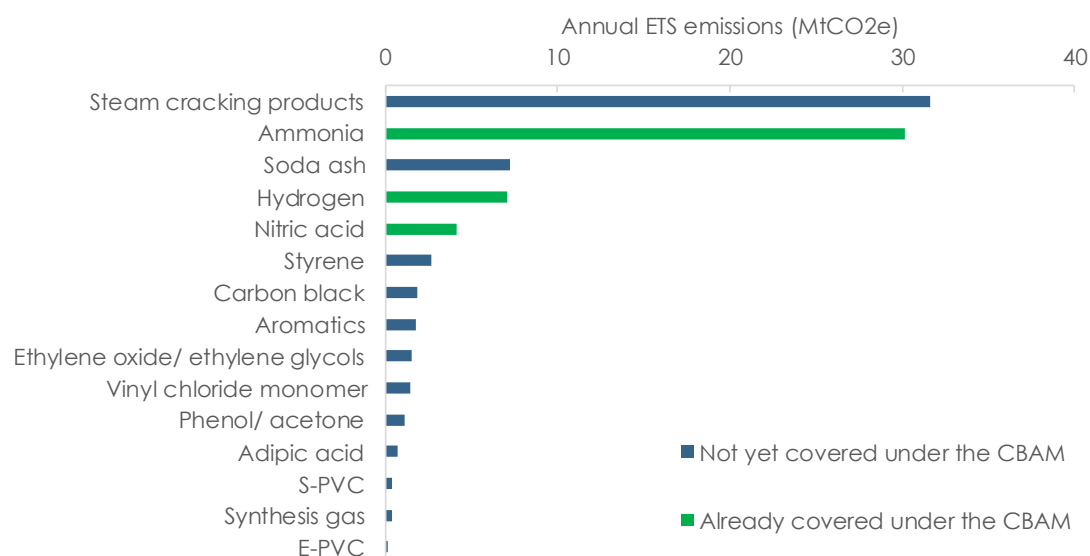
Figure 3 shows that the ETS emissions for steam cracking products are significantly higher than that of their downstream products such as styrene, ethylene oxide / ethylene glycols (precursor of PET) and vinyl chloride monomer (precursor of PVC). Polyethylene and polypropylene do not even have product benchmark under the EU ETS, which is an indicator for their direct manufacturing processes being relatively less emissions-intensive. Only extending the CBAM to the direct manufacturing emissions of downstream chemical products of under the CBAM would therefore insufficiently mitigate the risk of carbon leakage in relation to the embedded emissions of such products.

²⁹ See e.g. Trinomics (2023). [Study on the inclusion of the chemical sector in CBAM](#).

³⁰ JRC (2017). [Best Available Techniques \(BAT\) Reference Document for the Production of Large Volume Organic Chemicals](#).

Instead, given the relative importance of emissions of steam cracking in the overall ETS emissions, an appropriate effort/benefit ratio³¹ could potentially be achieved by only focussing on products from the steam cracking process and their most important downstream products (e.g., polyethylene and polypropylene) in the CBAM.

Figure 3 Annual EU ETS emissions of chemical sector products that have an ETS product benchmark



Based on the average of the latest three years of attributable emissions from the anonymised confidential NIMs data for free allocation under the EU ETS for 2021-2025 received from the Commission, i.e., the years 2016-2018. Steam cracking products are a mix of acetylene, ethylene, propylene, butadiene, benzene and hydrogen exported out of the cracker perimeter, which are covered under one single ETS product benchmark.

Other potential chemical products for inclusion appearing in Table 5 could include methanol to cover a highly imported chemical, which import could further increase in the future an expected prominent hydrogen carrier. For products that are further downstream in the chemical value chain, the incremental environmental benefit of including them under the CBAM decreases while the complexity of determining their embedded emissions increases fast. This is due to the complexity of chemical value chains that can involve multiple precursors for each product. Ultimately, a balance will have to be found between the administrative effort and feasibility of including certain chemical products compared to the environmental benefits gained. This would require a different approach than the statistical analysis used in this chapter. As this also ties into the considerations of technical feasibility, such different approach is explored in Chapter 3.

Further analysis required for consideration

Based on the analysis results of this chapter, it is not possible to determine a list of organic chemical & polymer products for further consideration in this study. Instead, a different

³¹ This refers to the administrative effort for importers and authorities related to inclusion of the product under the CBAM compared to the environmental benefits in terms of carbon leakage mitigation.

approach to conclude on which organic chemicals and polymers was developed for selecting candidate substances for inclusion in the CBAM. This is presented in Chapter 3.

2.5.6 Fertilisers

Stage 2 shortlist results

The shortlisted products from Stage 2 are superphosphates (a type of P-fertiliser) and potassium chloride (a type of K-fertilisers). See Table 53 in Annex 10.1. for the full list of products on the Stage 2 shortlist.

Prioritisation for further consideration

None of the shortlisted CN codes in this sector are suggested for prioritisation for further consideration (see below).

Not prioritised for further consideration

In the Impact Assessment accompanying the proposal for the current CBAM, fertilisers that did not contain nitrogen (i.e., P- and K-fertilisers) were initially considered for inclusion but ultimately excluded. This was on the basis that the processes involved for producing these fertilisers can be considered as not emissions-intensive (including the embedded upstream emissions):

- P-fertilisers are generally produced from an exothermic reaction of concentrated sulphuric acid and phosphate rock. For producing concentrated phosphoric acid for making P-fertilisers, heat for evaporation of water is needed. However, since sulphuric acid is generally produced onsite and excess heat from those processes can be used, there are limited additional direct emissions involved in the production of P-fertiliser (see also Section 2.5.4 on Phosphates)²⁵.
- K-fertilisers are generally produced from potassium salts, which mostly come from mining operations. Such mining operations are generally not included in the EU ETS, and not very energy-intensive³².

We therefore suggest to not include any CN codes from this sector for further consideration. However, should P- and K-fertilisers be considered for inclusion in the CBAM, also P- and K-fertilisers that are not shortlisted should be considered for inclusion to mitigate the risk of circumvention as any P-/K-fertilisers fulfils the same function.

2.5.7 Glass

Stage 2 shortlist results

The shortlisted products in this sector span a wide range of different types of glass products, which can be broadly divided in the following categories (See Table 62 in Annex 10.1.2 for the full list of products on the Stage 2 shortlist):

- **Glass sheets:** CN codes under HS 700529 (other non-wired glass) and HS 700530 (wired glass) are shortlisted.

³² A [CarbonChain's analysis](#) estimates that potassium-based fertilizers emit on average 0.6 CO₂eq per tonne—over four times less than phosphate-based fertilizers (1.7 tonnes) and more than five times less than nitrogen-based fertilizers (2.6 tonnes)

- **Glass bottles and containers for:**
 - **Beverages and foodstuff:** only CN codes representing colourless glass bottles and containers are shortlisted (CN 70109041, 70109043, 70109045, 70109047).
 - **Pharmaceuticals:** all glass bottles and container CN codes for pharmaceuticals are shortlisted (CN 70109071 and 70109079).
 - **Packaging of goods (not specified elsewhere):** the CN code for glass bottles and containers not belonging in any other CN code are shortlisted (CN 70109091 and 70109099).
- **Table/kitchen glassware and drinking glasses:**
 - All CN codes under the HS 6-digit codes 701328 (other stemware drinking glasses), 701349 (other kitchen/table glassware), and 701399 (other glassware) are shortlisted.
 - All CN codes under the HS 6-digit codes 701337 (other drinking glasses) are shortlisted except for CN 70133710 (other drinking glasses of toughened glass).
- **Glass fibres:** the CN codes under HS 6-digit 701911 (chopped strands of ≤ 50 mm), 701912 (rovings), 701913 (other yarns and slivers), 701919 (other slivers, rovings, yarn and chopped strands and mats thereof), 701990 (other glass fibres) are shortlisted.
- **Various technical glassware:**
 - All CN codes under the HS 4-digit codes 7014 (optical glassware), 7017 (Laboratory, hygienic or pharmaceutical glassware), 7018 (glass beads).
 - Almost all CN codes under HS 7020 (other glassware) are also shortlisted (except glass inners).
 - The CN codes under HS 701610 (Glass cubes and other glass smallware) is also shortlisted.

Prioritisation for further consideration

The CN codes suggested for potential further consideration in extending the CBAM scope at the HS 4-digit level are as follows:

- **HS 7013 (Table/kitchen glassware and drinking glasses):** the shortlisted CN codes cover a wide variety of table/kitchen glassware and drinking glasses, which all fall under the HS 4-digit code 7013 (Glassware of a kind used for table, kitchen, toilet, office, indoor decoration or similar purposes). These shortlisted CN codes represent 88% of the import value and 89% of the import volume of the HS 4-digit code 7013, which is why all CN codes under the HS 4-digit code are being suggested to be included for further consideration.
- **HS 7019 except 701980 (Glass fibres except glass wool):** the shortlisted CN codes for glass fibres fall under the HS 4-digit code 7019 (glass fibres), represent 67% of the import value and 73% of the import volume of the 6-digit HS code. However, the CN codes under HS 6-digit code 701980 (glass wool, not shortlisted) are mainly used for insulation and thus has different functionalities than the shortlisted glass fibre CN codes. Excluding glass wool would bring the representation of the shortlisted CN codes under HS 7919 to 74% in import value and 81% in weight. The other uncovered CN codes are primarily downstream

products of the shortlisted CN codes. Therefore, all CN codes under the HS 4-digit code except for glass wool are suggested for further consideration.

- Should glass fibres be included under the CBAM, **downstream products that have a significant share of glass fibres** in their composition may need to be included under the CBAM to mitigate carbon leakage risks downstream. Examples of such products include glass fibre reinforced pipes and glass fibre reinforced profiles. The CN codes for these products could not be identified based on the CN code description alone and would ideally require detailed mapping of the value chain of glass fibres, which is not foreseen in this study. Such a mapping could be provided or validated by the glass fibre industry.

The CN codes suggested for potential further consideration in extending the CBAM scope at the CN code level are as follows:

- **CN 70109041, 70109043, 70109045 and 70109047 (colourless glass bottles for beverages and foodstuff):** this covers all the different CN codes related to colourless glass bottles for beverages and foodstuff, with the CN codes only differentiating the glass bottles based on their nominal capacity.
- **CN 70109071 and 70109079 (pharmaceuticals glass bottles and container):** this covers all CN codes related to glass bottles and container CN codes for pharmaceuticals, with the differentiation in CN codes only being the nominal capacity.
- **CN 70109091 and 70109099 (glass containers for packaging of goods not specified elsewhere):** these CN codes cover all for glass bottles and containers not belonging in any other CN codes, covering both colourless and coloured glass containers.

In addition, there is a **broader consideration** whether to include **all CN codes under the HS 6-digit code 701090 for glass bottles and containers** for prioritisation: the shortlisted CN codes represent 62% of the import value and 46% of the import volume of the HS 6-digit code; too low to be considered representative (>80%) for the whole HS 6-digit code. However, from a high-level analysis of the descriptions of each CN code, the CN codes are mainly differentiated by their intended purpose, followed by capacity and colour. As the intended purpose can be ambiguous, importers may be able to report imported products under the CN codes that have not been included in the Stage 2 shortlist. Furthermore, only including some glass bottles and containers under the CBAM that can fulfil the same purpose as other glass bottles and containers not shortlisted could lead to a market distortion between products. For example, the CN codes for colourless glass bottles for beverages and foodstuff are shortlisted, but the ones for coloured glass bottles are not. This raises the question whether new carbon leakage risks may be introduced if the shortlisted products are included but the other ones are not. Further investigation into these carbon leakage risks for glass bottles and containers beyond this study is therefore recommended if the CN codes above where to be included under the CBAM, including whether covering all CN codes under HS 701090 can resolve these risks.

Not prioritised for further consideration

The shortlisted CN codes of the following products suggested not to be further considered in this study for the extension of the CBAM scope are as follows:

- **Glass sheets (flat glass):** the shortlisted PRODCOM is the code for “other” flat glass products not specified in any other categories. When considering the

products at CN level, the import value of all CN codes are below the 100 million Euro import value criteria used for the Stage 2 filtering. Other flat glass products did not make it on the Stage 2 shortlist due to their low import value and/or import intensity. While most flat glass products do meet the trade intensity, the trade primary relates to exports. This means that none of the flat glass products are suggested for further consideration.

- **Technical glassware:** the shortlisted CN codes cover a wide range of CN codes for technical glassware. However, based on a high-level analysis of the CN code description, these all cover specific technical or special glasses. Such products generally have a low emissions intensity compared to their added value. This is because such products offer higher added value due to their specialised nature (e.g., with a focus on high-quality standards specifically engineered to withstand extreme conditions). The import volumes of almost all of these products are therefore low in both absolute terms and relative to their import value.

2.5.8 Ceramics

Stage 2 shortlist results

The shortlisted products are fragmented across the different CN codes within HS Chapter 69 (ceramic products) as well as one CN code in HS Chapter 68 (articles of stone, plaster, cement, asbestos, mica or similar materials), which can be divided in the following categories (See Table 63 in Annex 10.1.2 for the full list of products on the Stage 2 shortlist):

- **Ceramic flags and tiles:** all CN codes under HS 4-digit code 6907 for ceramic flags and paving, hearth or wall tiles, ceramic mosaic cubes and finishing ceramics are shortlisted.
- **Refractory construction goods:** this relates to the shortlisted CN code 69022099 for refractory construction goods containing > 50% alumina, silica or a mixture or compound of these products.
- **Articles of stone or of other mineral substances:** this relates to the shortlisted CN code 68159100 for various articles of stone or of other mineral substances not specified elsewhere.
- **Other refractory ceramic goods:** this relates to the shortlisted CN code 69039090 for retorts, crucibles, mufflers, nozzles, plugs, supports, cupels, tubes, pipes, sheaths, rods, slide gates and other refractory ceramic goods not specified elsewhere.
- **Tableware and household articles of stoneware:** this relates to the shortlisted CN codes 69120023 for tableware and kitchenware made of stoneware and 69120083 for household and toilet articles made of stoneware.
- **Ceramic sanitary fixtures:** all CN codes under HS 4-digit code 6910 for ceramic sinks, washbasins, washbasin pedestals, baths, bidets, water closet pans, flushing cisterns, urinals and similar sanitary fixtures are shortlisted.

Prioritisation for further consideration

The CN code suggested for potential further consideration in extending the CBAM scope at the HS 4-digit level are as follows:

- **HS 6907 (Ceramic flags and paving):** since all five CN codes under HS 6907 are shortlisted.
- **HS 6910 (Ceramic sanitary fixtures):** since both CN codes under HS 6910 are shortlisted.

The CN code suggested for potential further consideration in extending the CBAM scope at the CN code level are as follows:

- CN 69022099 (Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods, > 50% alumina, silica or a mixture or compound of these products): this is one of the many different types of refractory construction goods being imported into the EU under HS 6902 for refractory ceramics construction goods.
- The potential inclusion of the product at the HS 6-digit or 4-digit level has also been investigated. At the 6-digit level HS 690220, the shortlisted codes represents 77% in value and 64% in weight. At the 4-digit level HS 6902, it represents 39% in value and 28% of HS 6902. While covering a substantial proportion of the import into the EU, it is too low (<80%) to be considered representative.

Not prioritised for further consideration

The shortlisted CN codes suggested not to be further considered in this study for the extension of the CBAM scope are as follows:

- CN 68159100 represents articles of stone or other mineral substances not specified elsewhere containing magnesite, dolomite or chromite, including refractory ceramic products made from those specified minerals but do not fall in any other CN code. This code also covers products other than ceramic products such as products from stone, plaster, cement, asbestos, mica or similar materials that do not fall under any other CN codes in Chapter 68 (Articles of stone, plaster, cement, asbestos, mica or similar materials). The Stage 2 shortlisting of this code will mostly be the result of the production and import value of these codes being inflated, since almost none of the refractory ceramic products are not shortlisted.
- CN 69039090 covers all the refractory ceramic goods in the “other goods”, so it is a collection of various goods that may not have significant imports when considered individually.
- CN 69120023 and 69120083 represent tableware and household articles of stoneware that are likely to have a high value added with relatively large amount of manual work involved, which would indicate a low emissions intensity.³³ In addition, other types of ceramic tableware and household articles have not been shortlisted in Stage 2, which could give rise to circumvention risks.

³³ Tableware ceramic manufacturing, like for household articles made of stoneware, often involves more handcrafting and artisanal techniques. Top European producers of these articles like [Villeroy & Boch AG](#) and [KAHLA/Thüringen Porzellan GmbH](#) emphasise traditional craftsmanship and high added value.

2.5.9 Ferro-alloys

Stage 2 shortlist results

Under the NACE code for iron and steel, only ferro-alloys are not covered under the CBAM. From these ferro-alloys, the shortlisted CN codes fall under HS 6-digit codes for two types of ferro-alloys that both contain silicon (See also Table 64 in Annex 10.1.2):

- **HS 720221** Ferro-silicon, containing by weight > 55% of silicon
- **HS 720230** Ferro-silico-manganese

Prioritisation for further consideration

Given that ferro-silicon and ferro-silico-manganese are being produced by EU ETS installations via electro-intensive processes, CN codes under **HS 720221 and 720230 are suggested for further consideration in extending the CBAM scope.**

If the CBAM were to be extended to the CN codes under these two HS codes, some **broader considerations** need to be considered:

- Other ferro-alloy CN codes that may contain silicon have not been shortlisted. These would include CN 72022910, 72022990, 72028000, 72029100, 72029930, 72025000. These CN codes only represent 6% of the import value and 5% of the import weight in HS 7202. A detailed analysis on each specific product would be required to determine whether carbon leakage risks would be introduced by including only the CN codes under HS 720221 and 720230, which goes beyond the scope of this study.
- Other CN codes in HS 7202 represent a further 26% in import value and 4% in weight. Extending the CBAM to all ferro-alloys may prevent any risk of ambiguity in ferro-alloys covered under the CBAM, since the CBAM currently already covers ferro-alloys that represent 40% in import value and 45% in weight. This means that if the CBAM would be extended to the CN codes under HS 720221 and 720230, only 31% of the import value and 9% of the import weight of HS 7202 would not be covered by the CBAM. In that case, the high proportion of weight covered could be considered as a reason to include all CN codes under HS 7202 in the CBAM. At the same time, the significant difference in uncovered import value compared to the weight might indicate that the remaining CN codes under HS 7202 would have a relatively high added value and thus are less emissions intensive. This would give a reason not to extend the CBAM to those products.

2.5.10 Aluminium

Stage 2 shortlist results

The shortlisted product is aluminium oxide (alumina) as shown in Table 65 in Annex 10.1.3 which is the only PRODCOM under NACE 2442 that is not included under the CBAM. This is a precursor to aluminium.

Prioritisation for further consideration

Given that alumina is being produced by EU ETS installations, and it is a precursor to aluminium that is already covered under the CBAM, codes under **HS 6-digit code 281820 is suggested to be included for further consideration in extending the CBAM scope.**

If aluminium oxide is included in the CBAM scope extension, it may be logical to also include **CN 28183000** aluminium hydroxide (shortlisted under inorganic chemicals), which is an intermediate product in producing alumina. See also Task 3 (section 5.2.4).

2.5.11 Other non-ferrous metals

Stage 2 shortlist results

The shortlisted non-ferrous metal products can be divided in two broad categories (See Table 66 in Annex 10.1.3 for the full list of products on the Stage 2 shortlist):

- **Unwrought lead:** with the CN codes under HS 780110 and 780199 fully shortlisted
- **Unwrought zinc:** with the CN codes under HS 790111 and 790112 fully shortlisted

Prioritisation for further consideration

If the CBAM were to be extended to these shortlisted CN codes, **the CBAM scope extension should be considered at a more aggregated level to include all CN codes under the following HS codes as well as CN codes for downstream products** to mitigate the risk of circumvention. This would follow the same rationale used for steel and aluminium products:

- **HS 7801 (Unwrought lead):** as the shortlisted CN codes cover 90% of the import value and 89% of the weight of the HS 4-digit code and can therefore be considered representative (>80%) of the HS 4-digit code.
 - In addition, the CN codes under **HS 7804 (lead plates, sheets, strip and foil; lead powders and flakes)** and **7806 (other articles of lead)** should be additionally considered for CBAM extension as it is a direct downstream product of products under HS 7801.
- **HS 7901 (Unwrought zinc):** as the shortlisted CN codes cover 88% of the import value and weight of the HS 4-digit code and can therefore be considered representative (>80%) of the whole 4-digit code.
 - In addition, CN codes under HS 7903 (lead dust, powders and flakes), 7904 (zinc bars, rods, profiles and wires) and 7905 (zinc plates, sheets, strip and foil) and 7907 (other articles of zinc) should be additionally considered for CBAM extension as it is a direct downstream product of products under HS 7901.

Additionally, there are **broader considerations** to consider if the CBAM were to be extended to lead and zinc:

- The CN codes suggested for further consideration do not include zinc and lead **waste** and scrap (HS 7802 and 7902). However, this should be considered in the **broader** context whether the CBAM should be extended to scrap/by-products in general, based on objective criteria.
- Zinc and lead ores often appear together with **copper** and they can appear as by-products in each other's production processes. Copper (NACE 2444) was, however, not shortlisted in Stage 1 based on the specified criteria from DG TAXUD of the carbon leakage indicator being >0.8. A quick analysis shows that copper products in CN Chapter 74 meeting the Stage 2 criteria would cover 77% of the import value of CN 74 and 80% of the weight. **A broader**

consideration is therefore whether copper should be further considered for the CBAM scope extension if the scope would be extended to lead and zinc.

2.6 Preliminary conclusions

For the sectors other than chemicals and polymers, the CN codes in Table 6 are suggested for further consideration in the prioritisation of the CBAM extension.

For organic chemicals and polymers, the analysis based on statistics results in 115 PRODCOM 8-digit codes and 399 CN codes. Even when the CN codes are grouped into 6-digit codes, this still leaves 294 HS 6-digit codes shortlisted. Furthermore, 8 PRODCOM 8-digit codes (corresponding to 8 CN codes) were identified for inorganic chemicals³⁴. As this large number cannot be overseen in Task 2, we proposed to analyse the selected chemicals (or production processes) using a different methodology in order to decide if they should or can be included in the CBAM. This is worked out in Chapter 3. For results of that analysis, please see section 3.5.

Table 6 Final shortlist of products prioritised for further consideration

Sector group	CN codes or groups of CN codes (HS codes)	No. of CN codes	Annual extra-EU import value (MEUR)
Pulp & paper	47 (wood pulp) 4801 (newsprint) 4802 except 48021000 (uncoated paper and paperboard except handmade ones) 4803, 4818 (toilet paper and tissue) 480411 (uncoated, unbleached kraftliner) 481159 and 481190 (coated, impregnated, covered, surface-coloured, surface-decorated or printed paper, paperboard and other paper articles)	55	8 153
Refinery products	2710 (petroleum oils) 2711 except 271111 and 271121 (petroleum gases except liquified and gaseous natural gas) 2712 (petroleum jelly and wax) 2713 (petroleum coke)	75	89 389
Glass	7013 (table/kitchen glassware) 7019 except 701980 (glass fibres except glass wool) 70109041, 70109043, 70109045 and 70109047 (colourless glass bottles for beverages and foodstuff) 70109071 and 70109079 (pharmaceuticals glass bottles and container) 70109091 and 70109099 (glass containers for packaging of goods not specified elsewhere)	30	2 929
Ceramics	6907 (Ceramic flags and paving) 69022099 (Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods, > 50% alumina, silica or a mixture or compound of these products) 6910 (Ceramic sanitary fixtures)	8	1 644
Ferro-alloys	720221 (ferro-silicon) 720230 (ferro-silico-manganese)	2	1 631
Aluminium	281820 (alumina) 281830 (aluminium hydroxide)*	2	576

³⁴ Section 2.5.4 identified 9 CN codes shortlisted for prioritisation, which includes CN 28183000 aluminium hydroxide. Since aluminium hydroxide is an intermediate product in producing CN 28182000 alumina, the latter which has been shortlisted under the sector Aluminium, CN 28183000 aluminium hydroxide is therefore included under section Aluminium in the final shortlist.

Non-ferrous metals	7801, 7804, 7806 (lead) 7901, 7903, 7904, 7905, 7907 (zinc)	16	1 807
		Total	188
			106 128

* Note that 281830 aluminium hydroxide was suggested for prioritisation listed as part of Inorganic chemicals analysis but included here under Aluminium as this is also analysed as an aluminium precursor in Task 3.

The annual import value shown is the average for the years 2021-2023 as the summation of the import value at the CN code level based on Eurostat COMEXT database [DS-059332: EU trade since 2002 by HS2-4-6 and CN8](#). Production and emissions data is not available at the CN code level.

In addition to the suggested products for further consideration in the prioritisation of the CBAM extension, there is also a consideration in the broader context on **whether the CBAM should be extended to (pre-consumer) scrap**. The considerations of including scrap in general relates to the methodological discussion whether to attribute embedded emissions to scrap, which is currently not the case under the implementing act for the transitional period of the CBAM. This is further discussed in Chapter 5 (Task 3) in the particular context of aluminium, iron, and steel where scrap can be considered a precursor. Should aluminium and iron and steel (pre-consumer) scraps be included in the CBAM with attributed embedded emissions, then scrap of other relevant sectors (glass, paper, and polymers) may also have to be considered for CBAM extension to ensure a consistent approach across all products.

Furthermore, **additional products have been identified that require a broader consideration** whether to also include for further consideration for the CBAM scope extension:

- Extension to all CN codes under HS Chapter 48 (paper and paperboard), should the main precursor HS Chapter 47 (wood pulp) be included in the CBAM;
- Extension to CN codes under HS 2715 (bituminous mixtures) if the CBAM is extended to CN codes under HS 2713 (which includes petroleum bitumen), although the extra-EU trade and import for products under HS 2715 is minimal;
- Extension to CN codes under HS 2901 (specifically high purity ethylene, propylene, butylene, and butadiene) if the CBAM is extended to CN 27111400 (which covers lower purity ethylene, propylene, butylene, and butadiene);
- Extension to CN codes of all glass bottles and containers under HS 701090, should the CBAM be extended to glass bottles and containers that are on the final shortlist for further consideration;
- Extension to downstream products of glass fibres should the CBAM be extended to glass fibres, for which the downstream products would have to be identified on a product-by-product basis;
- Extension to all ferro-alloys that may contain silicon or even to all CN codes under the HS 4-digit code for ferro-alloys (HS 7202), should the CBAM be extended to ferro-silicon and ferro-silico-manganese;
- Inclusion of copper (HS Chapter 74) if the CBAM is extended to lead and zinc.

For providing a wider basis for a discussion on these points, we have included the broader list of goods in the analysis under Task 2.

3 Additional work under Task 1 regarding chemicals and polymers

For assessing technical feasibility, a focus on a manageable number of substances seems unavoidable, given the high complexity and diversity of chemical products (see section 3.1). However, as mentioned under Task 1 (sections 2.5.4 (inorganic chemicals) and 2.5.5 (organic chemicals and polymers)), the initial methodology for this task did not give a useful result for assessment of technical feasibility: Task 1 results yielded 115 PRODCOM 8-digit codes corresponding to 399 CN codes, covering an unknown much higher number of chemicals. Furthermore, the products on that shortlist are spread out over various parts of the chemical value chain. Some organic chemicals known to have significant embedded emissions are included (such as methanol, polyethylene, polypropylene, PET, styrene, and PVC), but not all of them. On the other hand, the list also contains larger and diverse molecules (e.g. CN 29309040 – 2,2'-Thiodiethyl bis[3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate]) and substances with non-systematic trade names. Those usually require a multi-step chemical synthesis and therefore fall into the category of “fine chemicals” which are produced at rather low volumes globally. This suggests that the assessment of the chemical sector requires a more systematic assessment of value chains for identifying the “most important” chemicals and polymers.

Therefore, significant work beyond the original Task 1 was conducted for identifying a list of chemicals and polymers as candidates for inclusion in the CBAM, as described in this section. This included the following steps:

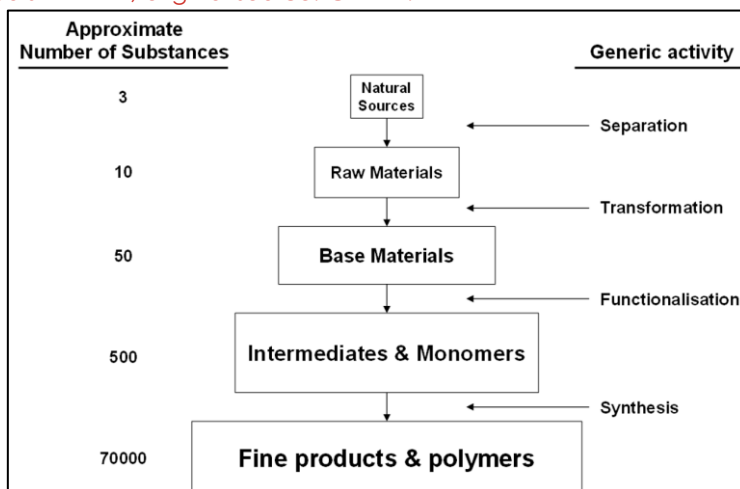
- Starting point is a brief overview of the chemical industry (section 3.1).
- Thereafter it is discussed to what extent the EU ETS can be used to identify important chemicals to be included in the CBAM (section 3.2).
- As another starting point, the main value chains of the large-scale chemicals starting from common raw materials are examined in section 3.3.
- As a result of the above points, criteria for choosing chemicals are discussed in section 3.4.

The resulting shortlist of chemicals and polymers, serving as result of Task 1, is presented in section 3.5. The inorganic and organic chemicals and polymers listed there are the basis for the assessment regarding the technical feasibility of including them in the CBAM under Task 2.3 (see section 4.4).

3.1 Short overview of the chemical industry

The main challenge of assessing the chemicals sector is its sheer size and complexity. Figure 4 provides an insight of its size for the organic chemical industry. Furthermore, the inorganic chemical industry is dealing with a possibly lower number of substances, but a much higher number of raw materials.

Figure 4 Schematic overview of the organic chemicals sector, as presented in the “Large Volume Organic Chemicals” BREF³⁵, original source: CITEPA.



Another issue is that not even the term “chemical industry” is clearly defined. Chemical processes build the backbone of almost all energy-intensive industries, including the production of steel, non-ferrous metals, cement, glass, pulp and paper, etc. Nevertheless, chemical industry is commonly understood as those installations that produce “chemicals”, i.e. substances that are then used as input to diverse applications, where the chemical properties of the product are the reason for using the produced substance, be it the colour, smell, biological activity, reactivity, ability to form structures, etc.

Table 7 lists the subsectors of the chemical industry according to NACE codes (note that a stricter definition would only cover NACE categories starting with “20”, while the wider definition used here also includes 21 and 22). It confirms the wide range of the chemical industry: Some subsectors (e.g. NACE 2011 to 2014) focus on the production of *substances* usually not aiming at giving the product a specific physical appearance, as most often the product is not a consumer good. However, there are exemptions to this rule, e.g. NACE 2015 (fertilisers) produce substances (granules or liquids) which only require packaging for becoming a consumer good. Other subsectors are rather “downstream” sectors which use less energy-intensive production steps, such as preparing consisting of several substances (like glues, paints, pharmaceuticals, etc.). Finally, there are also sectors which use chemical syntheses, but at the same time produce immediately goods where physical appearance matters, such as sheets of rubber (under NACE 2017) or man-made fibres (NACE 2060). The NACE sectors starting with “22” or only loosely considered chemical industry, and rather downstream sectors. Furthermore, there are some sectors that deal with the extraction (and modification) of substances from natural (biological) sources, which lead to goods that are not always fully chemically defined (e.g. essential oils, modified starches).

³⁵ https://eippcb.jrc.ec.europa.eu/sites/default/files/2019-11/JRC109279_LVOC_Bref.pdf, published in 2017

Table 7 Overview of the chemical industry according to NACE codes – a narrower definition of chemical industry would only include the NACE codes starting with “20”. Assessed sectors are marked by bold font.

NACE code	Description
2011	Manufacture of industrial gases
2012	Manufacture of dyes and pigments
2013	Manufacture of other inorganic basic chemicals
2014	Manufacture of other organic basic chemicals
2015	Manufacture of fertilisers and nitrogen compounds
2016	Manufacture of plastics in primary forms
2017	Manufacture of synthetic rubber in primary forms
2020	Manufacture of pesticides and other agrochemical products
2030	Manufacture of paints, varnishes and similar coatings, printing ink and mastics
2041	Manufacture of soap and detergents, cleaning and polishing preparations
2042	Manufacture of perfumes and toilet preparations
2051	Manufacture of explosives
2052	Manufacture of glues
2053	Manufacture of essential oils
2059	Manufacture of other chemical products n.e.c.
2060	Manufacture of man-made fibres
2110	Manufacture of basic pharmaceutical products
2120	Manufacture of pharmaceutical preparations
2211	Manufacture of rubber tyres and tubes; retreading and rebuilding of rubber tyres
2219	Manufacture of other rubber products
2221	Manufacture of plastic plates, sheets, tubes and profiles
2222	Manufacture of plastic packing goods
2223	Manufacture of builders' ware of plastic
2229	Manufacture of other plastic products

For our assessment we use the “substance-centred” approach, i.e. we focus on those goods that are “substances” only, not mixtures / formulations of several substances, as this best fits to the aim of extending the CBAM horizontally, not to downstream products.

As we need to focus on goods rather than the industry sector, our starting point is the system of CN codes (or at higher level, HS headings). This is shown in Table 8. Of the total of 1802 CN codes, our focus is on HS chapter 28 (inorganic chemicals, 264 CN codes), chapter 29 (organic chemicals, 514 CN codes), and plastics (i.e. polymers, HS 39, 214 CN codes³⁶). Fertilisers (HS 31, 31 CN codes) are not further considered, as they have already been discussed in section 2.5.6. These codes are highlighted by bold font in Table 8.

A common element of many CN codes in the chemical sector is that there are whole groups of products covered, with few exceptions where a CN code corresponds to a single chemical substance. Examples for single-substance CN codes include hydrogen, ammonia, ethylene, propylene, etc., which also gives an indication that these are substances of outstanding economic importance.

³⁶ These CN codes do not include only “plastics in primary forms”, i.e. also downstream products which we exclude from our analysis.

Table 8 Overview of product categories under the HS / CN classification system that can be considered "chemical industry". Headings in bold were main target of the assessment.

HS chapter	Description	Number of CN codes
28	Inorganic chemicals; organic or inorganic compounds of precious metals, of rare-earth metals, of radioactive elements or of isotopes	264
29	Organic chemicals	516 ⁽³⁷⁾
30	Pharmaceutical products	55
31	Fertilisers	31
32	Tanning or dyeing extracts; tannins and their derivatives; dyes, pigments and other colouring matter; paints and varnishes; putty and other mastics; inks	59
33	Essential oils and resinoids; perfumery, cosmetic or toilet preparations	49
34	Soap, organic surface-active agents, washing preparations, lubricating preparations, artificial waxes, prepared waxes, polishing or scouring preparations, candles and similar articles, modelling pastes, 'dental waxes' and dental preparations with a basis of plaster	31
35	Albuminoidal substances; modified starches; glues; enzymes	33
36	Explosives; pyrotechnic products; matches; pyrophoric alloys; certain combustible preparations	14
37	Photographic or cinematographic goods	41
38	Miscellaneous chemical products	175 ⁽³⁸⁾
39	Plastics and articles thereof	214
40	Rubber and articles thereof	91
54, 55	Man-made fibres	229
	Total	1802

The complexity of organic chemistry

Figure 4 indicates approximately 70 000 organic chemicals on the market³⁹. For understanding how this is possible, we have a short look at organic chemistry: It is the science of molecules based on hydrocarbons. Hydrocarbons are molecules consisting of various chains, rings or combinations thereof of carbon atoms, containing possible "hetero atoms" (oxygen, nitrogen, sulphur, phosphorous etc.), halogens (chlorine, bromine, fluorine), and hydrogen. The functions of a molecule in the use of the substance are furthermore defined by the existence of double or triple bonds ("unsaturated" hydrocarbons) or aromatic ring systems, and so-called "functional groups" (hydroxy, amine, carboxyl, amide, etc. groups) making them more or less inert or reactive, acidic, or alkaline, etc.

The different size (molecular weight) and the reactivity of the functional groups lead to differences in physical properties, such as melting and boiling point, crystallization or

³⁷ We included here two CN codes (22071000, 22072000) for ethanol, which could be considered "organic chemical), although in the HS/CN system they are listed together with other alcoholic beverages.

³⁸ Here also products are included which should not be considered typical "chemical industry", in particular municipal and other wastes, wasted solvents and sewage sludges.

³⁹ The number of known chemicals is much higher than these estimated 70 000 substances on the market: The Chemicals Abstracts Service (CAS, <https://www.cas.org/cas-data/cas-registry>) claims that their database contains "219 million organic substances, alloys, coordination compounds, minerals, mixtures, polymers, and salts disclosed in publications since the early 1800s."

glass-like properties, colour, etc., and in case of polymers to hardness, elasticity, and the general possibility to give the substance a form in manufacturing by different processes. Furthermore, the molecular structure influences the chemical properties, i.e. the possibility to carry out reactions (from very slow reactions to explosions), which offers the possibility to synthesise other chemicals from each molecule. The molecular structure furthermore makes molecules e.g. robust against oxidation or radiation, leads to physiological reactions in organisms (pharmaceutical use, pesticides, etc.).

This very short overview illustrates why such large variety of chemical substances can be derived from only a few elements. And although in a textbook usually similar molecules are listed together, often in the sequence of increasing molecule's size (e.g. Methanol, $\text{CH}_3\text{-OH}$ is listed together with Ethanol, $\text{C}_2\text{H}_5\text{-OH}$ and 1-Propanol, $\text{C}_3\text{H}_7\text{-OH}$), the industrial production routes of these can be very different, as at an industrial scale, the choice of production method is primarily guided by economic viability. The latter depends not only on the chemical properties of the desired product, but also on the availability of potential precursors. In the above example, methanol is primarily produced from natural gas via synthesis gas obtained by steam reforming (alternatively e.g. from gasification of coal), while ethanol is rather produced by hydration of ethylene (steam cracker product) or via fermentation of sugarcane or various grains. 1-propanol, however, is not produced in the same way as ethanol, i.e. via hydration of propylene, as this gives 2-propanol (iso-propanol). Instead, it is produced via hydrogenation of propanal (propionaldehyde), which itself is produced from ethylene, carbon monoxide, and hydrogen, i.e. ethylene and syngas. Comparison of the production routes of these 3 simple molecules should sufficiently illustrate that production routes of chemicals must be analysed using a strict case-by-case approach, and that no simple analogy of processes should be assumed just because of similarities in the molecular structure.

Since in many cases chemical reactions lead to several co-products from a single process, the economic viability of a production process also depends on the potential use of the co-products. This in turn has led to highly integrated chemical sites producing many different products from a considerable number of interlinked processes. But what is more notable for the CBAM context, it has led to the development of a multitude of production routes for the same substance. Consequently, each substance needs individual assessment regarding the production route(s), which may be a considerable barrier to the technical feasibility to include chemicals in the CBAM. The development of "aggregated goods categories" is hardly possible, which makes the system of MRV rather complex in general for this sector.

3.2 Emissions and substances covered by the EU ETS

As the CBAM should reflect the EU ETS as far as possible, it is logical to see what goods are produced in sectors covered by the EU ETS. In particular, if a substance is covered by a product benchmark, it can be assumed to be a substance both produced at high volumes and leading to significant emissions. Table 9 provides an overview of the chemicals identified using this approach, together with the emissions covered in the EU

ETS (average emissions 2016/17 as used as basis for benchmark development⁴⁰). These emissions are only the emissions of the process directly producing the chemicals in question, i.e. fall short of precursor's contributions to embedded emissions, and are therefore only a first indication for significant emissions for the CBAM. The table also provides production volumes (determined from PRODCOM statistics, average 2021-23) and aggregated trade intensities⁴¹.

The focus on EU ETS product benchmarks would be a very efficient approach: There are only 14 product benchmarks under the EU ETS, but they cover about two-thirds of all emissions of the chemical industry in the EU⁴². They cover 27 chemicals with a total production of 61 million tonnes per year in the EU-27, compared to a total volume of all chemicals being 371 million tonnes. Compared to this, the current CBAM coverage (hydrogen and fertilisers⁴³) reaches only 46,7 million t production and 41,9 million t CO₂e of emissions. In other words, the focus on EU ETS benchmarks would already more than double the emissions coverage, while increasing the covered EU production by only half. This confirms that the highest emissions of the chemical sector can be covered by only focusing on a relatively small number of substances, particularly those having a product benchmark in the EU ETS.

Table 9 Overview of chemicals regarding their EU ETS emissions (if covered by a product benchmark (BM)) and EU-27 production data.

Industry sector	EU ETS BM (if applicable)	Chemicals covered by the EU ETS BM	Emissions [t CO ₂ e]	t CO ₂ e / t product	Production in EU-27 [t]	Trade intensity
Technical gases	Hydrogen BM	Hydrogen	7 051 720	9,50	742 026	0,36%
	Other technical gases				n.a. ⁴⁴	11,68%
Fertilizers	Ammonia BM	Ammonia	30 588 525	10,18	3 003 333	49,30%
	Nitric acid BM	Nitric Acid	4 297 166	4,59	936 381	14,27%
	Other fertilizers				41 950 108	43,60%
Inorganic chemicals	Soda ash BM	Soda ash, Sodium bicarbonate	7 218 816	1,07	6 757 445	44,85%
	Carbon black BM	Carbon black	1 824 820	1,41	1 296 135	61,49%
	Other inorganic chemicals				64 633 166	59,54%
Organic chemicals	HVC BM	Ethylene, propylene, butadiene,	31 393 609	1,55	20 263 222	13,60%

⁴⁰ https://ec.europa.eu/clima/document/download/fd041819-e22e-4e77-a267-6ab4405328aa_en

The emissions for this assessment are required at sub-installation level, which are not published. Data from the EU ETS Registry are available only at installation level and can therefore not be used. More recent data is therefore currently not available.

⁴¹ The aggregated average trade intensity was calculated as follows: First, the average over the years 2021 to 2023 of each PRODCOM value for each parameter (imported value, exported value, production value) was determined. Thereafter, where several PRODCOM codes fall into the same category (e.g. fertilizers), the values of the respective PRODCOM codes were summed up. Finally, Trade intensity is calculated as **Trade intensity=(IMPVAL+EXPVAL)/(IMPVAL+PRODVAL)**

⁴² Eurostat data set "Air emissions accounts by NACE Rev. 2 activity" [env_ac_ainah_r2], years 2021-2023, N₂O and CO₂.

⁴³ Including fertilisers not identifiable by an EU ETS benchmark.

⁴⁴ The table excludes "technical gases" from the sum, as these are not given in tonnes in PRODCOM statistics.

Industry sector	EU ETS BM (if applicable)	Chemicals covered by the EU ETS BM	Emissions [t CO ₂ e]	↑ CO ₂ e / t product	Production in EU-27 [t]	Trade intensity
		acetylene, benzene, H ₂				
	Aromatics BM	Benzene, toluene, xylenes, cumene, cyclohexane	1 896 871	0,18	10 749 696	30,29%
	Styrene BM	Styrene, ethylbenzene	2 676 322	0,76	3 500 000	23,75%
	Phenol / Acetone BM	Phenol, Acetone	1 200 943	0,46	2 597 917	37,27%
	Syngas BM	Methanol, formaldehyde, formic acid ⁴⁵	357 368	0,15	2 443 333	77,20%
	VCM BM	Vinyl Chloride	1 639 059	2,24	733 333	25,93%
	Ethylene glycols BM	Mono, di-, tri-ethylene glycole, ethylene oxide	1 484 310	0,80	1 856 165	34,37%
	Adipic acid BM	Adipic Acid	857 227	1,45	590 314	21,39%
	Other organic chemicals				51 695 459	81,07%
Polymers	PVC BM ⁽⁴⁶⁾	E-PVC, S-PVC	496 485	0,09	5 654 141	34,90%
	Other polymers including rubbers				58 893 892	49,23%
Man-made fibres					1 745 851	56,64%
Downstream chemicals and formulations ⁴⁷					90 847 232	63,60%
Sum covered by EU ETS product benchmarks			92 983 241		61 123 442	
Sum without benchmarks			50 043 401		309 765 707	
Refinery products ⁴⁸			114 809 016	0,19	599 300 000	

The table presents average emissions per tonne of the covered products (note that some benchmarks cover more than one product). For all identified products, a weighted average trade intensity⁴¹ was also calculated. Finally, at the bottom, refinery products are mentioned for completeness. Goods already included in the EU ETS are marked by using *italics*.

⁴⁵ Not according to the FAR, but according to the project team's assessment of main uses of CO/H₂ mixtures, see discussion note on chemicals in annex to this report.

⁴⁶ Production volume includes PVC mixed with other substances and plasticised PVC

⁴⁷ Covering all HS headings from Table 8 which have not been listed separately in Table 9, i.e. all headings under which not the production of single substances in chemical processes, but the mixing (formulation) of substances establishes the main production process.

⁴⁸ As there are no PRODCOM codes for refinery products, production estimate from CONCAWE was used: Report no.15/22: "Estimating the CO₂ intensities of EU refinery products: Statistical Regression Methodology". It can be found here: https://www.concawe.eu/wp-content/uploads/Rpt_22-15.pdf. Therefore, it was also not possible to estimate a trade intensity.

3.3 Main value chains in the chemical industry

Another option to manage the vast number of substances that could be included in the CBAM is to assess the actual value chains of industrial⁴⁹ organic chemistry to identify “most important” substances. The reasoning behind this approach is found in the fact that there are very few raw materials and first production steps (as shown also in Figure 4). This means that these few first products must be produced at the highest volumes, and therefore automatically should qualify as most important. For example, if one substance A is the precursor for 10 substances B₁ to B₁₀, the production volume of A must be higher than the sum of the production volumes of B₁ to B₁₀, since chemical reactions always have yields below 100%.

All organic chemistry starts with the three main sources of fossil carbon which are produced by mining or extraction, which are not systematically covered by the EU ETS and therefore excluded from the CBAM by definition, meaning they do not have any embedded emissions:

- Coal⁵⁰ (various qualities: Hard coal, lignite, etc.);
- Crude oil (various qualities);
- Natural gas (various qualities).

In principle, all three sources are interchangeable: Coal and oil can be “gasified” (by water gas reaction, partial oxidation, pyrolysis), lighter fractions can be converted into higher ones (i.e. heavier molecules can be generated), all three sources can be converted into synthesis gas, which thereafter can be used to produce liquids, in particular fuels via the “Fischer-Tropsch Synthesis”, etc. The main difference is the economic factor: Simple molecules are usually produced from lighter fractions and bigger molecules from the heavier fractions, as the conversion leads to energy losses and additional emissions. Because the biggest source of organic chemicals is oil (petroleum), the term “petrochemicals” is often used for summarizing the first few steps of industrial organic chemicals production. Furthermore, biomass can also be used as raw material for chemical syntheses. The argument would be the same: In principle, all organic sources can be converted into the others. We assume that emissions stemming directly from biomass would be accounted for as zero embedded emissions (monitoring rules reflect those of the EU ETS), and do not further study the related production processes⁵¹. For

⁴⁹ At laboratory scale, an even larger number of production routes is available for each substance, e.g. where special purity of products is required or where technical feasibility to use special precursors is tested.

⁵⁰ Some coal types can be used for coking, producing coke (main use is as precursor for the iron and steel production, and for other metallurgical processes including production of ferro-alloys). Important byproducts of coking are oils and tars that are rich in aromatics, and can therefore be another source for their production, competing with pyrolysis gas from refineries.

⁵¹ In fact, if used for the production of CBAM goods, the production process could have non-zero embedded emissions for various reasons (use of other fuels or chemical precursors), but this would only be relevant in the actual emissions monitoring and would not lead to specific complications. However, the fact needs to be kept in mind that there are always different production routes just because of the possibility of using biomass. Therefore, there will be a residual risk of circumvention and fraud as producers for all organic chemicals and polymers could theoretically claim that they have started their production routes with biomass. However, as there are currently strong incentives (by EU climate policy) to use biomass predominantly for the production of biofuels, while there is little incentive to replace other fossil products like polymers, this risk is low, in particular because for biofuels the RED II requires an elaborated certification system. It is therefore justified to focus the discussion on fossil production routes. More biomass-based production routes will become more important only on the longer term (perhaps in 10 years).

completeness, it needs to be mentioned that some specific chemicals are predominantly produced from biomass via fermentation processes, e.g. bioethanol, citric acid and some pharmaceuticals, and other complex chemicals may be produced via extraction from specific plants, such as e.g. fragrances and flavours. However, these special cases are used at a smaller scale than the petrochemical products discussed in this note, with the one exception of bioethanol, which is produced predominantly for use as biofuel and therefore needs discussion for the case that motor fuels in general could be included in the CBAM.

The first production steps needed in the petrochemical industry are (focussing here on the predominant case of starting from crude oil):

- Removing impurities (such as sulphur and salts) from crude oil and separating the highly diverse mixture of hydrocarbons into narrower fractions – this is a main function of mineral oil refineries;
- Isolating specific simple molecules with defined reactivity from those fractions. The two most important processes in this regard, which play a key role for this chapter, are borderline cases for being considered either still part of the refineries or already chemical industry, as they are found in both types of installations:
 - Steam crackers leading to unsaturated simple molecules, in particular:
 - Olefins (ethylene, propylene, butylene, butadiene), i.e. offering molecules with 2, 3 and 4 carbon atoms as building blocks for further molecules;
 - Benzene (one of the aromatics);
 - Pyrolysis gas ("pygas") as important source of other aromatics, in particular toluene and xylenes;
 - Aromatics production based on different extraction steps starting from pygas or reformat (another refinery fraction).
- The most important chemicals with a single carbon atom are produced predominantly from natural gas in the steam reforming process, which first generates synthesis gas ("syngas", a mixture of carbon monoxide and hydrogen, which is usually not traded, but directly used on site). From there, either separated hydrogen or ammonia from hydrogen can be produced, which are already included⁵² in the CBAM. For the horizontal extension of the CBAM, the most important product of synthesis gas is methanol, which has a vast potential for downstream chemical processes. Also, carbon monoxide can be isolated from syngas, which plays a key role for extending hydrocarbon chains by one carbon atom.
- In some economies (e.g. China), coal plays a more significant role than natural gas. Partial oxidation of coal can be used to produce synthesis gas for the exact same applications as mentioned under the previous bullet point. However, where coal is significantly cheaper than mineral oil, methanol

⁵² Value chains starting from hydrogen and ammonia have been considered in the "downstream extension" project.

produced from coal is also used to produce olefins ("coal to olefins", in particular for ethylene and propylene).

- Furthermore, some organic chemicals (in particular aromatics) can be produced from distillates which are co-products of coke production. Coke is produced from special grades of coal and is discussed as a potential precursor for the existing CBAM scope. The possibility of producing chemicals from coal tar and coal light distillates as co-products of coke adds to the benefits of including coke in the CBAM (see Task 3, section 5.2.2).

These processes provide a comprehensive coverage of the first steps of industrial organic chemistry. The importance of these processes is underlined by the fact that they also correspond to dedicated product benchmarks in the EU ETS with significant emissions (the many diverse processes, which are summarised under "fall-back" allocation approaches in the EU ETS, do not have such high emissions and cannot be singled out in a similar manner).

In principle, the study could end here with identifying **olefins, aromatics and methanol** as the main targets of horizontal CBAM extension since all other organic chemicals can be produced (theoretically) from them. Consequently, all other organic chemicals could be considered downstream goods. However, for a more comprehensive understanding of various production routes and the chemical sector as a whole, the project team looked into a broader range of organic substances with high production volumes or known high emissions (based on product benchmarks, see section 3.2). From a different perspective, also goods already in the existing CBAM scope (hydrogen, ammonia, nitric acid, urea) were considered as "at the beginning of the value chain", and goods produced from them were also considered for assessment⁵³.

Aside from organic chemicals, the mandate for considering the extension of the CBAM in Article 30(2)(a) of the CBAM Regulation explicitly requires a discussion of polymers (which are, in principle, always organic substances and downstream products of their monomers). Therefore, the project team investigated the whole value chain of the most important polymers. Initially, the five **polymers (PE, PP, PS, PVC, PET)** were looked at, which together represent slightly more than half of the total polymer production in the EU in terms of mass, to limit the scope of this study to a manageable number of substances. During the project, the Commission requested the study of the value chains of further polymers, since literature suggested high relevance in terms of embedded emissions and/or production volume (see full list in section 3.5).

All relevant⁵⁴ precursors for the selected substances have also been included in this study. According to the recent "omnibus" amendment of the CBAM Regulation, only

⁵³ These substances were identified in the "CBAM downstream extension" study for DG TAXUD as being worth assessment for CBAM inclusion but considered to better fit into the analytical framework of this "horizontal extension" study. Including them here allowed considering the full range of embedded emissions, including from precursors that are not yet in the CBAM, which would not be the case if those substances were considered only from perspective of being downstream of the existing CBAM scope.

⁵⁴ As precursors can have other precursors themselves, this is a recursive exercise, tracing value chains from the final product back to at least one of the raw materials which are considered to not have embedded emissions themselves.

precursors can be relevant that are CBAM goods themselves⁵⁵. Without these precursors, the embedded emissions of organic chemicals would be incomplete when comparing the coverage of CBAM and EU ETS. Especially embedded emissions of the polymers would be very low, since the polymerisation processes themselves are not very energy- or emissions-intensive. Precursors must be CBAM goods themselves⁵⁶ and would therefore have to be explicitly included in an CBAM scope extension. However, for keeping the study (and a possible future CBAM) reasonably simple, we did not include chemicals in the assessment of production routes which are not used in stoichiometric⁵⁷ amounts, such as polymerisation starters and catalysts, as well as e.g. solvents (which can usually be recovered and reused after the reaction). For efficiency reasons, these are therefore not considered precursors at this stage (which does not prevent including them in the CBAM if further (future) assessments show a need to extend the CBAM scope further).

3.4 Criteria for identifying "most important" chemicals and polymers

From sections 3.1 to 3.3 it has become clear that on the one hand, there is a need to limit any further discussion of a CBAM extension regarding chemicals and polymers to a manageable number of substances. This is in particular important as the assessment of technical feasibility of inclusion has to be done on a case-by-case basis for each substance separately. On the other hand, criteria for selecting these substances need an objective basis. So far, the only objective selection criterion found is whether a substance

⁵⁵ Section 3 of Annex IV of the CBAM Regulation is amended to contain the sentence "Only input materials (precursors) listed in Annex I and originating in third countries and territories that are not exempted pursuant to Annex III, Section 1 are to be considered."

⁵⁶ When developing the current scope of the CBAM (Annex I of the Regulation), the principle was confirmed that every precursor must be a CBAM good itself. The reason is that the CBAM must be non-discriminatory, i.e. the emissions in the EU ETS must lead to similar carbon costs as incurred by goods produced outside the EU and falling under the CBAM.

Example: Good A is a precursor of good B. The following situation can occur:

- A and B produced in the EU ETS: Carbon costs for both production processes A and B are incurred.
- A and B are produced outside the EU ETS: Both production processes contribute to the embedded emissions under the CBAM, both lead therefore to carbon costs.
- A is produced in the EU, then exported and used for producing B, which is then imported under the CBAM: Again, carbon costs incurred for both production processes (although double counting of process A needs to be corrected by deducting the carbon price already paid for product A in the EU when calculating the CBAM obligation. After the "Omnibus" proposal for the CBAM Regulation, product A would simply not contribute as precursor anymore, but the result regarding carbon costs in the CBAM would be the same).
- A is produced outside the EU, and B is produced in the EU: When this precursor A is imported, two situations need to be considered:
 - The precursor A is considered itself a CBAM good: In this case, again all processes are covered by a carbon price, equal treatment to the above cases is given.
 - Precursor A is not considered a CBAM good: In this case, only the process B is covered by a carbon price. The producer of B is situated in the EU and thereby gets a competitive advantage over a non-EU producer of product B. This is an unequal treatment that is *not* WTO-compatible.

⁵⁷ Stoichiometry defines the quantities of inputs and products of a chemical reaction. For example, the equation of the reaction for splitting water by electrolysis is: $2 \text{H}_2\text{O} \rightarrow 2 \text{H}_2 + \text{O}_2$ meaning that 2 moles (approx. 36g) water give 2 moles (approx. 4 g) hydrogen and 1 mole (approx. 32g) oxygen. In many industrial processes, some inputs, like catalysts, are not consumed in the reaction (but they make the reaction faster) and therefore need to be used in small quantities only. Also several co-products occur at non-stoichiometric quantities, which is an important point for attributing emissions to co-products.

is included in the EU ETS with its own product benchmark for free allocation. This indicates that the emissions from this production process across the EU are outstandingly high, so that the definition of a benchmark was found relevant. However, whether the production of a chemical substance is included in the EU ETS at all, cannot serve as criterion, since in principle all chemicals may be produced in EU ETS installations, whether or not a specific activity is defined in Annex I of the EU ETS Directive⁵⁸, or whether a product benchmark is defined in Annex I of the free allocation rules⁵⁹.

Whether a good is “important” for the CBAM depends, in general, on the following factors:

- The sector producing the good must be covered by the EU ETS and exposed to a significant risk of carbon leakage, since the CBAM is designed as instrument for mitigating this carbon leakage risk. However, this criterion is fulfilled for the chemical sector as a whole. Hence, this criterion does not need further consideration.
- For being exposed to carbon leakage, one criterion is that the carbon costs induced by the EU ETS are high enough to put the good produced in the EU at a disadvantage compared to imported goods. Crudely simplified, this can be translated to “the embedded emissions of the produced must be relatively high”. However, determining the embedded emissions of a chemical substance can be very burdensome, as the project team experienced while developing the methodology for this task. In particular, there is a “chicken-and-egg” issue: Determining the embedded emissions of a good requires knowledge of the production routes and precursors of the good and requires detailed information on the emissions of the production processes. This means that more research needs to be done than needed for deciding on the technical feasibility of inclusion in the CBAM, only to decide whether such assessment should be conducted at all. Furthermore, it was found that the embedded emissions of the first few precursors (methanol, olefins, aromatics) are always significant, and therefore also all embedded emissions of substances further down the value chain are significant. Therefore, whether a substance has significant embedded emissions was dropped as criterion after some first tests.
- Whether a good is economically important in the EU can be estimated by the quantities produced or consumed in the EU (at the level of PRODCOM codes). It can furthermore be assumed that this reflects the global importance of the good. If importance of goods for the CBAM is based on whether they are highly traded (imported and exported), trade data based on CN codes can be used (which in the case of chemicals are more disaggregated than PRODCOM codes). However, only few substances have their own CN code. Therefore, such data is available only to a limited extent. Nevertheless, it was agreed with the Commission that production levels in the EU should serve for selecting organic chemicals for assessment (i.e. for putting them on the shortlist

⁵⁸ Directive 2003/87/EC, <http://data.europa.eu/eli/dir/2003/87/2024-03-01>

⁵⁹ Commission Delegated Regulation (EU) 2019/331, http://data.europa.eu/eli/reg_del/2019/331/2024-01-01

being result of Task 1). Chemicals were selected if more than 1 million tonnes per year are produced of that substance in the EU.

- An alternative measure for the importance of production would be the monetary value (revenues from sales) of the produced substances. However, mass was chosen over monetary value, because there is some inverse relationship between these two factors: The cheaper a material is, the more it is used and produced. Emissions are usually linked to mass produced. Therefore, mass produced is the more appropriate parameter.

Summing up the previous arguments, **the following approach was used for selecting chemicals:**

- First, substances covered by EU ETS product benchmarks were selected, as they are clearly associated with significant emissions.
- Secondly, organic chemicals produced in the EU at levels above 1 million tonnes per year were selected⁶⁰.
- Thirdly, the substances identified as candidates under the “downstream extension” project were added, although they are usually produced at lower levels. However, their association to goods already covered by the CBAM implies that their embedded emissions would be significant, which justifies further assessment.
- With regard to polymers, a list of most important ones was agreed with the Commission.
- Finally, during the assessment of technical feasibility (Tasks 2.3 and 2.4, see sections 4.4.10 to 4.4.12) precursors for the already selected substances were identified. Each identified precursor was added to the list, even if produced at much lower levels than 1 million tonnes per year. Where relevant, also the co-products of production processes were added, in particular where they occur in stoichiometric amounts.

3.5 Results – Shortlist of chemicals and polymers for further assessment

Application of the approach described in section 3.4 led to the following list of substances:

- **Inorganic chemicals:**
 - Initially only those chemicals with a product benchmark in the EU ETS and not already included in the CBAM were selected. These are:
 - Soda ash;

⁶⁰ For this selection, COMEXT data (average) 2021-2023 was used, based on PRODCOM 2024 classification. For making this assessment simpler, production data of PRODCOMs covering several CN codes were divided by the number of covered CN codes as approximation for the data at CN level. Furthermore, only CN codes relating to single substances were used, while codes referring to whole substance classes (e.g. “derivatives of xyz”) were excluded, as this may cover an unknown number of substances. An exemption was made in cases like “adipic acid and its salts”, as we assume that in such cases the percentage of adipic acid contained in the imported goods could be specified via a “qualifying parameter” in the CBAM declaration. Finally, CN codes were excluded from the analysis where the covered substance was identified by expert judgment as requiring very complex syntheses, based on the chemical structure of the substance.

- sodium bicarbonate;
- Carbon black.
- Further substances included during the assessment (as precursors or co-products):
 - Chlorine;
 - Hydrogen chloride;
 - Hydrogen cyanide;
 - Carbon monoxide;
 - Nitric oxide;
 - Nitrogen dioxide;
 - Phosgene;
 - Nitrosyl chloride;
 - Nitrosylsulphuric acid;
 - Sodium hydroxide (caustic soda);
 - Hydroxylamine;
 - Hydrogen peroxide;
 - Calcium carbide.
- **Organic chemicals:**
 - Initial selection (EU ETS product benchmarks):
 - Ethylene;
 - Propylene;
 - Butadiene;
 - Acetylene;
 - Benzene;
 - Toluene;
 - Xylenes (mixture, or separately p-xylene, m-xylene, o-xylene);
 - Ethylbenzene;
 - Styrene;
 - Cumene;
 - Ethylene oxide;
 - (Mono-, di-, tri-)ethylene glycol;
 - Phenol;
 - Acetone;
 - Adipic acid;
 - Methanol (Synthesis gas benchmark);
 - Vinylchloride monomer;
 - Substances with production volume > 1 M t per year:
 - Ethanol/ Bioethanol;
 - Propylene oxide;

- Phthalic anhydride;
- Substances proposed in the “downstream extension” project:
 - Melamine;
 - Formic acid;
 - Cyclohexane;
 - Urea resins (polymer);
 - Melamine resins (polymer);
 - Nylon salt;
 - Acrylamide;
 - Dinitrotoluene (DNT);
 - Aniline;
 - Hexamethylenediamine (HMDA)
 - Adiponitrile;
 - Acrylonitrile;
 - Nitrobenzene.
- Necessary precursors if not already listed above:
 - Synthesis gas (“syngas”);
 - Ethane;
 - Propane;
 - Butane / iso-Butane;
 - Naphtha (refinery product);
 - Pyrolysis gas (“pygas,” refinery product);
 - Reformate (refinery product);
 - Coal tar;
 - Light oils from coking;
 - Terephthalic acid (PTA);
 - Dimethyl-terephthalate;
 - Methyl p-Toluate;
 - Ethylene carbonate;
 - Dimethyl ether (DME);
 - Ethylene dichloride (EDC);
 - Ethylbenzene hydroperoxide;
 - Cumene hydroperoxide;
 - Acetophenone;
 - α-Methyl-Benzyl alcohol;
 - α-Methyl styrene;
 - Naphthalene;
 - Diphenyl carbonate (DPC);
 - Dimethyl carbonate (DMC);

- Bisphenol-A;
- Cyclohexene;
- Cyclohexanol;
- Cyclohexanone;
- Cyclohexylbenzene;
- KA-Oil;
- Formaldehyde;
- Methyl formate;
- Sodium formate;
- Formamide;
- Propylene oxide;
- Propylene glycol;
- Tert-butyl hydroperoxide;
- Methylene dianiline (MDA);
- Methylene diphenyl diisocyanate (MDI);
- Toluene Diamine (TDA)
- Toluene diisocyanate (TDI);
- ϵ -Caprolactam;
- MMA (Methyl methacrylate).
- **Polymers:** List agreed with the Commission:
 - PE (Polyethylene), i.e. LDPE (low density PE), LLDPE (Linear low density PE) and HDPE (high density PE);
 - PP (Polypropylene);
 - PVC (Polyvinylchloride), produced via E-PVC (emulsion PVC) and S-PVC (Suspension PVC) production processes;
 - PS (Polystyrene), i.e. EPS (Expansible PS), HIPS (High Impact PS) and GPPS (general purpose PS);
 - PET (Polyethylene terephthalate);
 - PA-6 (Polyamide 6);
 - PA-6,6 (Polyamide 6,6);
 - SAN (Styrene-Acrylonitrile copolymer);
 - ABS (Acrylonitrile-Butadiene-Styrene copolymer);
 - PMMA (Poly Methyl methacrylate);
 - Polyurethanes (PU): The following example substances were assessed:
 - PU made of MDI and PEG;
 - PU made of TDI and PEG;
 - PU made of MDI and PPG;
 - PU made of TDI and PPG;
 - Polycarbonates (PC): As example, only PC made from Bisphenol-A and phosgene was assessed.

- **Polymers** included as precursors:
 - Polybutadiene;
 - Polyethylene glycol;
 - Polypropylene glycol.

4 Task 2. Technical feasibility assessment for extending the CBAM scope

4.1 Objectives and overview of the task

This task aims to identify the technical feasibility of including new goods (horizontal expansion) in the CBAM scope. The focus is on goods identified as priority for inclusion in the CBAM under Task 1, i.e. the shortlist presented in section 2.6, as far as chemicals and polymers are concerned, section 3.5. The criteria for the technical feasibility assessment can be found in section 4.4.1, with specific elements for the chemicals sector added in section 4.4.2. To meet the objectives, the following sub-tasks were carried out:

- Sub-task 2.1 – Comprehensive study of the Implementing Regulation (EU) 2023/1773; results are presented in section 4.2.
- Sub-task 2.2 – Review of EU/international legislation for calculating, measuring, reporting emissions of sectors to be covered; results are presented in section 4.3.
- Sub-task 2.3 – Analyse data and analytical requirements;
- Sub-task 2.4 – Feasibility assessment of extending CBAM to new goods; The methodology, criteria and results of Sub-tasks 2.3 and 2.4 are jointly presented in section 4.4.
- Sub-task 2.5 – Propose a ranking of priority sectors, sub-sectors/products; The result of this Sub-task are presented as conclusions of the whole Task 2 in section 4.5.

4.2 Results of Sub-Task 2.1: Comprehensive study of the Implementing Regulation (EU) 2023/1773

This sub-task was already included in the inception report. The findings there showed that overall, the rules of Implementing Regulation (EU) 2023/1773 are fit for purpose also for inclusion of additional CBAM goods. Throughout this report, we refer to that Regulation also as the “transitional CBAM rules” or “transitional MRV rules”, or simply “the implementing act [on the CBAM monitoring methodology] [for the transitional phase]”, well aware that it is not the only implementing act under the CBAM.

From the analysis of the implementing act, the following points were identified as requiring further attention:

- Unless the general approach of drafting such rules is changed, it is necessary to develop for each good to be included:
 - descriptions of all relevant production routes;
 - boundaries of production processes regarding the emission sources and relevant greenhouse gases;
 - a list of relevant precursors for each production process / route;
 - if possible, aggregated goods categories to which the CN code (or single substance) could be grouped;
- Identification, whether processes can be joint under a so-called “bubble approach”. These points feed into the assessment of data requirements under Sub-Task 2.3 (section 4.4), criterion 2a.

- If relevant, “qualifying parameters” like those regulated in Annex IV section 2 of the transitional implementing act have to be developed. They would contribute to the exact identification of goods, in particular where the CN code is not sufficient for identification. The feasibility to define such parameters and require their reporting contributes to feasibility criterion 1 and is therefore discussed per sector under Sub-Task 2.3 (section 4.4).
- Rules are required for attribution of emissions to co-products, if such production processes occur. A proposal for solving this issue is provided in a separate discussion note which is attached to this report as Annex T2. Some findings of the note are explained in context of the assessment criteria in section 4.4.1.
- Potentially some sector-specific rules from Annex IV of the EU ETS Monitoring and Reporting Regulation (MRR)⁶¹ need to be included, depending on the sectors and goods selected for inclusion (ceramics, paper, etc.).

Furthermore, it was considered whether an “alternative approach”, could be introduced for shortening the CBAM rules and for keeping them more flexible in case of a scope extension. This approach would be to state in the rules “every CBAM good used in a production process is automatically a precursor”. This approach would reduce the implementing act by several pages but might be less clear for the operators when small quantities of precursors or very unusual ones are used. In this approach, for limiting the administrative burden on operators, it should be considered to introduce a threshold for “de-minimis” amounts of embedded emissions resulting from precursor use. This would mean that precursors would have to be monitored only if their used quantity or resulting emissions would exceed that threshold. However, for determining useful thresholds, again some detailed knowledge of the involved production processes would be needed. At this point and for the purpose of this study, it is still assumed that the number of potential precursors and production routes will be manageable by the current approach, and therefore the question of threshold setting is not further pursued.

4.3 Results of Sub-Task 2.2 – Review of EU/international legislation for calculating/measuring /reporting of emissions in the sectors identified in T1

Aiming to prevent carbon leakage resulting from the EU ETS, the CBAM requires authorised declarants to report accurate emissions data, relying on information about the embedded emissions of imported goods from their international suppliers. EU environmental policies operate within a highly interconnected framework, where various regulations and instruments complement and reinforce one another. In addition to the CBAM various other EU legislations require supply chain cooperation with regard to greenhouse gas (GHG) emissions on product level, some of them mandating emissions reporting and data verification. In this chapter the current EU legislations and some related international standards are reviewed, to provide a comprehensive picture on the similarities and differences between them. Hereafter the **Ecodesign for Sustainable Products Regulation**, the **Digital Product Passport**, the **EU Taxonomy Regulation**, the **Green Claims Directive proposal**, and the **Environmental Footprint methods**, are assessed regarding a potential role and synergies to support in the gathering of data of which some are also required by the CBAM and to manage carbon accountability across

⁶¹ Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC, http://data.europa.eu/eli/reg_impl/2018/2066/2025-01-01

borders creating a fair, data-driven approach to carbon pricing. These policies collectively promote sustainability across different sectors and are supported by methodologies such as Life Cycle Assessment (LCA) in general, and Carbon Footprint of Products (CFP) or the Product Environmental Footprint (PEF) in particular, which provide structured approaches to assessing and enhancing product sustainability.

4.3.1 Ecodesign for Sustainable Products Regulation (ESPR)

Regulation (EU) 2024/1781⁶² entered into force in July 2024, establishing a framework for setting of environmental performance and/or information requirements for products sold in the EU. However, as it is a framework regulation, specific ESPR requirements, including the decision to use of and potential information to be conveyed by the Digital Product Passport (DPP), will be determined through product-specific delegated acts⁶³. These delegated acts will define ecodesign requirements for each product group or horizontally for certain product aspects.

Concerning the DPP, while initially introduced by ESPR, the DPP has grown beyond ESPR, and will likely also be introduced in the context of other legislation. For instance, as announced in the Single Market Strategy p.24 *"The Digital Product Passport (DPP) will become the main tool for disclosing and sharing product information across all new and revised product legislation."* and later *"The upcoming NLF reform will fully introduce the DPP to demonstrate compliance with EU product legislation. The potential use of the DPP IT infrastructure to enable the automated re-use of data by companies will also be assessed, to streamline reporting and verification by authorities. It will also explore ways to digitalise processes for product certification, ensuring that conformity assessments are managed in a structured and transparent way, which will save companies and authorities time and money."*

More details on the use of the DPP under the ESPR are found in Articles 9 to 15 of the ESPR, and the potential content of the DPP according to the ESPR is defined in Annex III of that Regulation. An interesting fact for the CBAM is that – provided a DPP is required for the product according to the relevant delegated act – upon import to the EU, the importer must make available a “unique registration identifier of that product”, and under Article 15(3) it is mentioned that there is a data exchange with the “EU Customs Single Window Certificates Exchange System (EU CSW-CERTEX⁶⁴)”. Therefore, it is clear that customs data for the CBAM and for the DPP could be linked electronically, making the tracing through value chains easier than without it. However, that link shall be active “within 4 years after entry into force”, which would be in July 2028 at the latest. It is therefore not possible to immediately build CBAM rules or requirements based on ESPR DPP contents. **However, as is discussed in section 4.3.2, a DPP requirement could be established by the CBAM Regulation itself**, which might work slightly faster, and be more focussed on CBAM

⁶² Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC. <https://eur-lex.europa.eu/eli/reg/2024/1781/oj/eng>

⁶³ See ESPR work plan: https://environment.ec.europa.eu/document/5f7ff5e2-ebe9-4bd4-a139-db881bd6398f_en

⁶⁴ https://taxation-customs.ec.europa.eu/eu-single-window-environment-customs_en

requirements. In this section we discuss only aspects of the ESPR and DPP under the assumption that no specific DPP requirements are established by the CBAM legislation.

Regarding the ESPR's own provisions, Annex III of the ESPR lists potential requirements related to the DPP. The delegated acts adopted pursuant to Article 4 of that regulation will specify what data must or may be included in the DPP from among the listed elements. The following elements are particularly interesting for CBAM purposes:

- TARIC Codes (which are linked to CN codes) can help identify the fact that it is a CBAM good;
- A unique operator identifier and unique facility identifiers: These may correspond to the installation identifier required for the CBAM.

Annex III further states that the delegated acts adopted pursuant to Article 4 shall identify information relevant to ecodesign requirements that manufacturers may include in the digital product passport.

According to Article 18(3) the Commission shall adopt a working plan, which shall set out a list of product groups which are to be prioritised for the setting of ecodesign requirements and the estimated timelines for their setting. The first working plan was published on 16 April 2025.⁶³ Of the included products iron & steel and aluminium products are relevant for the CBAM as such products are already included under the current CBAM scope⁶⁵. The indicative timeline for adoption of the iron & steel products is 2026 and for the aluminium products it is 2027. Usually a transitional phase is foreseen to allow companies to adapt to new requirements. As the definitive phase of the CBAM already starts in 2026, the CBAM methodology cannot yet rely on the DPP.

Article 38 of the ESPR may be used as inspiration for the CBAM, as it states that "when specified in the delegated act adopted pursuant Article 4, supply chain actors shall:

- a) provide, upon request and free of charge, manufacturers, notified bodies and competent national authorities with available relevant information related to the products they supply or the services they provide;
- b) allow, in the absence of information referred to in point (a), manufacturers to assess the products they supply or the services they provide and give access to relevant documents or facilities to those manufacturers; and
- c) enable notified bodies and competent national authorities to verify the accuracy of relevant information related to their activities."

4.3.2 The Digital Product Passport (DPP)

The Digital Product Passport (DPP), established by the ESPR and other legislation (e.g. the Batteries Regulation⁶⁶), will serve as a comprehensive digital record containing information on for example a product's emissions, materials, origin, and recyclability, accessible via data carriers like QR codes. It will be the harmonised common 'vehicle'

⁶³ Furthermore, according to the working plan, the product groups next in line are textiles/apparel (2027), tyres (2027), furniture (2028) and mattresses (2029). These product groups are currently not under the CBAM. However, with future extensions of the CBAM there could be some overlap, or at least some precursors of these product groups might be in the CBAM.

⁶⁶ Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC; <http://data.europa.eu/eli/reg/2023/1542/oj>

(infrastructure) to use in any regulation that wants to make use of the DPP. The DPP can be inter alia a tool to make emissions and life-cycle data available across industries and to regulators, businesses, and consumers alike. For the purpose of the CBAM, the DPP could provide a reliable means to report embedded emissions data in imports, the origin of the good (installation and time of production) and a link to the relevant verification report and possibly even the verification reports of precursors further up the value chain. An option is to provide data on a need-to-know basis for different stakeholders. The DPP might even allow authorities to efficiently assess emissions information and thereby simplify the enforcement of the CBAM's pricing mechanism.

Digital product passports are already part of EU legislation. For example Article 77(1) of the Batteries Regulation requires "[...]each LMT⁶⁷ battery, each industrial battery with a capacity greater than 2 kWh and each electric vehicle battery placed on the market or put into service[...]" to have an electronic record ('battery passport'). Article 78(a) of that regulation states that "the battery passport shall be fully interoperable with other digital product passports required by Union law concerning eco-design, in relation to the technical, semantic and organisational aspects of end-to-end communication and data transfer".

A very prominent appearance of the DPP is in the Ecodesign for Sustainable Products Regulation (ESPR, see chapter 4.3.1). Recital (41) of the ESPR specifies *"To ensure that the digital product passport is flexible, agile and market-driven and evolves in line with business models, markets and innovation, it should be based on a decentralised data system and be set up and managed by economic operators. However, for enforcement and monitoring purposes, it is necessary that competent national authorities and the Commission have direct access to a record of all unique identifiers linked to products placed on the market or put into service. To that end, the Commission should set up and manage a digital product passport registry to store such data (the 'registry'). Where needed to further facilitate enforcement, the Commission should, where appropriate, specify other data included in the digital product passport that need to be stored in the registry."* This means that the DPP is not a single document attached to a good produced or imported, but it is a concept requiring a complex information technology (IT) infrastructure, with technical specifications for the data to be collected by the economic operators (such as producers of goods), the data formats and communication methods of decentralised systems (such as individual databases of economic operators, which might also be set up at country level or by industry sector associations, etc.). It is important that the Commission will have a leading role in the development and setting up of a part of that infrastructure which allows the competent authorities to exert their regulating powers. The last underlined part could be understood as a basis to define e.g. that for CBAM goods and precursors, embedded emissions compliant with CBAM rules could be included as data set in the DPP under the ESPR. However, after discussion with relevant Commission services (DG GROW and DG TAXUD), the more appropriate approach may be to add DPP requirements directly in the CBAM legislation (in the Regulation or in relevant implementing acts). This would overcome mismatching timelines of the ESPR and the CBAM (see chapter 4.3.1) and would also allow a better focus on specific data requirements by the CBAM (e.g. a link to the applicable verification report [in the CBAM Register]).

⁶⁷ light means of transport batteries

According to DG GROW the easiest way for consumers to access the digital product passport will be by scanning the data carrier (for example a QR code) that will be placed on the product or its packaging. The development of a web portal that will allow consumers to search for and compare DPPs is also foreseen in the ESPR. Online “information marketplaces” will be provided with digital means to make the DPP accessible to potential customers where they cannot physically access the product. In the future also private platforms like web portals or mobile applications might allow users to access DPP data for various products. The DPP Registry to be developed by the Commission will serve as central starting point for searches. It will probably only provide the hyperlinks to data stored on other platforms.

How the manipulation of physical markings will be prevented is not yet clear, as the standardisation process and legislative framework for the DPP are still under development. Many operational details will originate from the standardisation work and forthcoming secondary legislation, with adoption expected by mid-2026. These will encompass various technical and procedural aspects crucial for ensuring an integrated implementation of the DPP across Member States.

4.3.3 *EU Taxonomy Regulation*

Regulation (EU) 2020/852⁶⁸ establishes a classification system for environmentally sustainable economic activities, directing investment towards projects that align with EU climate and sustainability goals. By qualifying sustainable activities, the Taxonomy Regulation incentivises companies to adopt low-carbon practices, by aligning financial and production data with EU emissions standards. The Regulation’s rigorous sustainability criteria ensure that investment data on emissions and carbon reduction efforts is consistent and reliable, facilitating data propagation across sectors. Although the standards for which activities can be considered “green” build to some extent on the emissions per produced good (to some extent referring even to EU ETS product benchmarks), there is no direct link to the CBAM.

4.3.4 *Green Claims Directive (Commission Proposal)*

The proposal for the Green Claims Directive⁶⁹, currently in the process of co-decision, aims to enhance the reliability of environmental claims made by companies, requiring them to substantiate claims with rigorous, life-cycle-based evidence and transparent data. This proposed Directive seeks to combat greenwashing by ensuring that environmental claims are accurate and verifiable, to avoid misleading consumers. It mandates companies to back up their claims with science-based methodologies, assess the significant environmental impacts, including a life-cycle assessment if relevant and identify any environmental trade-offs. For climate related claims, companies are to clearly communicate on their GHG emissions, including any offsets used. The proposal also establishes strict requirements for comparative claims and environmental labelling schemes, fostering reliable and transparent schemes across the EU market.

⁶⁸ Regulation (EU) 2020/852 of the European Parliament and of the Council of 18 June 2020 on the establishment of a framework to facilitate sustainable investment and amending Regulation (EU) 2019/2088. <https://eur-lex.europa.eu/eli/reg/2020/852/oj/eng>

⁶⁹ Proposal for a Directive of the European Parliament and of the Council on substantiation and communication of explicit environmental claims (Green Claims Directive). <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52023PC0166>

There are potential synergies between the Green Claims Directive and the CBAM, particularly concerning emissions data accuracy and emissions reporting. The Green Claims Directive's requirement for science-based evidence includes imported goods. The CBAM's clear rules for calculating embedded emissions could, among other methods like the product environmental footprint (PEF, see chapter 4.3.5), serve as a sound basis for emission claims under the Green Claims Directive. The Green Claims Directive's requirement of verification of the data underlying a green claim by a verifier accredited under Regulation (EC) 765/2008 would be another useful synergy.

The Green Claims Directive also mandates that environmental claims rely on widely recognised scientific evidence and relevant international standards (Article 3(1)(b)). Additionally, the Directive requires that any GHG offsets used must be transparently separated from direct emissions, specifying whether they pertain to emission reductions or removals, and ensuring that they are of high integrity and accurately accounted for (Article 3(1)).

We have not identified further points which would have an impact or synergies with the CBAM Regulation.

4.3.5 Product Environmental Footprint (PEF)

The PEF is a method developed by the European Commission to assess the environmental impact of products following an LCA approach. The related guidelines are laid down in Recommendation (EU) 2021/2279 ⁽⁷⁰⁾ for determining the life cycle environmental performance of specific products and for developing Product Environmental Footprint Category Rules (PEFCRs). Even though the EF methodology is currently under revision, adding new impact categories and revising existing ones for the "EF 4.0" reference package, the quantification of GHG emissions will be one of the main impact categories in almost all product categories, providing a potential source of information for CBAM. But at the current stage of implementation, the data availability and comparability of LCA/PEF studies may be not-necessarily given, as explained below.

The determination of embedded emissions of goods imported to the EU under the CBAM strongly reflects the rules for monitoring and reporting of emissions of the EU ETS, as the intention of the CBAM is to provide for comparable carbon costs as would have been incurred if the same good were produced in the EU. Embedded emissions are furthermore inspired by the concept of a **partial carbon footprint** to bridge the gap between the determination of emissions on installation level under the EU ETS and the attribution of emissions to products under the CBAM.

Carbon footprint standards such as the PEF may be used for different purposes and perspectives on products and services, which may need flexibility for the user of the standard. On the other hand, the EU ETS and the CBAM have to attach a calculated financial value to emissions which must be equally robust as other rules in the economy. Methods must exclude flexibilities which could be used for undue gambling by the participants of the ETS or the CBAM. In other words, the same emissions must lead to the same carbon costs for every operator, without any gaps or double counting. The

⁷⁰ PEF refers to the Product Environmental Footprint in accordance with Commission Recommendation of 16.12.2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations, C(2021) 9332 final.
https://environment.ec.europa.eu/publications/recommendation-use-environmental-footprint-methods_en.

methodology must therefore ensure that every tonne of CO₂e must be reported by exactly one operator.

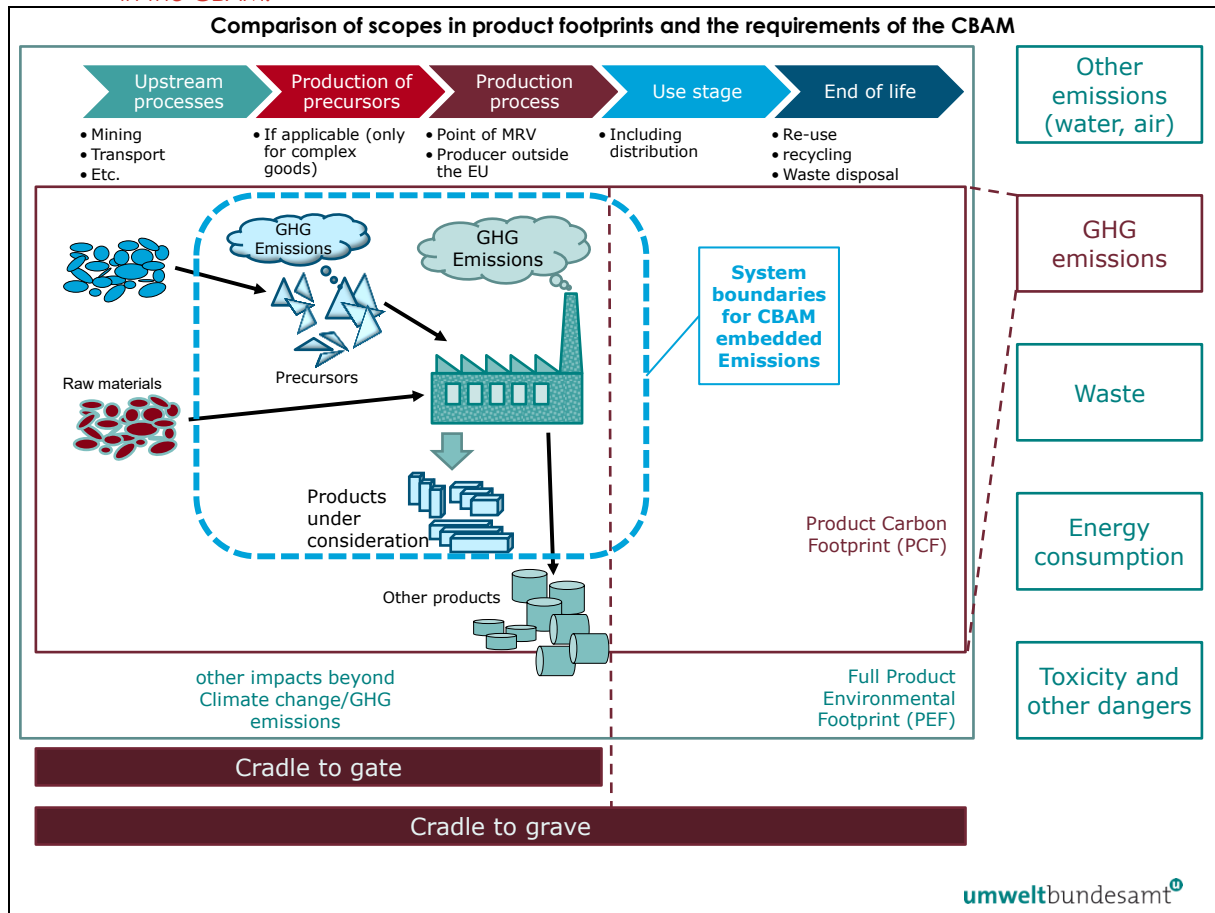
The comparison between GHG emission scopes classification and the CBAM / EU ETS is summarised in Table 10 and Figure 5 (taken from the Commission's CBAM Guidance⁷¹).

Table 10 Comparison of the GHG emission scope of the CBAM, the EU ETS, and the definitions contained in widely used standards (ISO 14064-1 and the 'GHG Protocol')

Parameter	ISO 14064-1 (Annex B)	GHG protocol	EU ETS	CBAM
"Direct emissions" (stationary)	Category 1	Scope 1	Subject to system boundaries of each EU ETS installation	Direct emissions are defined as "Emissions from the production processes of goods including emissions from the production of heating and cooling consumed during the production processes, regardless of the location of the production of the heating and cooling"
"Direct emissions" (mobile, e.g. forklift, cars)			Outside the scope	Outside the scope
"Indirect emissions" (upstream)				
of heating/cooling imported	Category 2	Scope 2	Covered if produced in an EU ETS installation	Included under "direct emissions"
of electricity imported			Covered if produced in an EU ETS installation	Indirect emissions are defined as "Emissions from the production of electricity, which is consumed during the production processes of goods, regardless of the location of the production of the consumed electricity"
of fuels imported	Category 3	Scope 3	Outside the scope	Outside the scope
Transport			Outside the scope	Outside the scope
of (precursor) materials imported	Category 4		Covered if produced in an EU ETS installation	To the extent precursors are defined as relevant in the implementing act
"Indirect emissions" (downstream and other, e.g. use of product, end-of-life emissions)	Category 5		Outside the scope	Outside the scope

⁷¹ For a more detailed comparison of product carbon footprint scopes and the CBAM see Table 6-1 in the Guidance Document for operators producing CBAM goods, <https://taxation-customs.ec.europa.eu/system/files/2023-12/Guidance%20document%20on%20CBAM%20implementation%20for%20installation%20operators%20outside%20the%20EU.pdf>.

Figure 5 Comparison of product environmental footprint, product carbon footprint, and the specific partial carbon footprint that is to be used for determining embedded emissions in the CBAM.



From the above, it should be clear that the CBAM per definition uses narrower system boundaries than the PEF or other LCA standards. Because of this “per definition” difference, it is not a matter of more precise rules, but there is a fundamental difference in the data between CBAM and PEF.

4.3.6 Review of selected international standards

International standards play a crucial role in supporting the accurate calculation, measurement, and reporting of GHG emissions, which is fundamentally in the interest of the CBAM.

The International Organization for Standardization (ISO) has developed a suite of standards that provide a robust framework for GHG quantification, reporting, and verification in many fields. As is under assessment in more detail in another project (MRVA-FA for the definitive phase), these ISO standards may play a role in making the CBAM's MRV system more flexible by providing a robust and transparent approach to emissions accounting, and by using methods that are already well established also outside the EU, thereby perhaps facilitating international acceptance of CBAM by reducing administrative burden.

Firstly, within the ISO 14064 series, ISO 14064-1:2018 establishes rigorous principles and methodologies for the quantification and reporting of GHG emissions and removals at

the organisational level. It mandates the categorisation of emissions into direct and indirect sources and emphasises the importance of consistency, transparency, and accuracy. Furthermore, ISO 14064-1:2018 gives guidance on inventory quality management and verification. The standard's emphasis on third-party verification and robust data management systems certainly helps at international level to support similar rules in the CBAM⁷².

Secondly, ISO 14064-2:2019⁷³ is project-specific and focuses on quantifying, monitoring, and reporting GHG emissions and removals from individual projects or activities. It requires the establishment of baselines and the monitoring of specific emissions reduction initiatives, ensuring that reported mitigation outcomes are real, measurable, and verifiable⁷⁴. This project-level focus makes ISO 14064-2 particularly useful for calculating emissions reductions achieved through specific interventions, such as energy efficiency projects or renewable energy installations. Furthermore, ISO 14064-2 mandates third-party verification of GHG mitigation claims, which enhances the credibility and integrity of emissions reporting. The specific benefit of ISO 14064-2 regarding the CBAM is two-fold: Firstly, installations conducting GHG reduction projects in third countries will be familiar with the standard and its concepts. Such projects may be, for instance, Clean Development Mechanism (CDM) projects or voluntary carbon market projects. It can be expected that some goods falling under the CBAM could be produced by such installations, and the use of the standard could prepare them well for further MRV tasks under the CBAM. Secondly, this standard uses the concept of "validation" in addition to verification. Validation means the forward-looking process for evaluating the reasonableness of the assumptions and methods, which could become useful in the CBAM regarding confirmation of the suitability of the "MMD" (Monitoring methodology Documentation).

Thirdly, ISO 14064-3⁷⁵ outlines requirements for the independent validation and verification of GHG assertions, ensuring the accuracy, completeness, and transparency of emissions data⁷⁶. The standard mandates detailed documentation of data sources, quantification methodologies, and verification procedures, which can serve as input to the development of delegated and implementing acts relevant for the CBAM's accreditation and verification framework (see MRVA-FA project).

Moreover, ISO 14065:2020⁷⁷ establishes requirements for legal entities conducting GHG verification, ensuring impartiality, competence, confidentiality, and accountability throughout the verification process. It mandates that verifiers operate as independent legal entities with adequate resources, facilities, and competent personnel to perform

⁷² ISO 14064-1:2018. Part 1: Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals. <https://www.iso.org/standard/66453.html>

⁷³ ISO 14064-2:2019 Greenhouse gases Part 2: Specification with guidance at the project level for quantification, monitoring and reporting of greenhouse gas emission reductions or removal enhancements. <https://www.iso.org/standard/66454.html>

⁷⁴ ICR. ISO 14064-2. <https://www.carbonregistry.com/iso-14064-2>

⁷⁵ ISO 14064-3:2019 Greenhouse gases Part 3: Specification with guidance for the verification and validation of greenhouse gas statements. <https://www.iso.org/standard/66455.html>

⁷⁶ ICR. ISO 14064-3. <https://www.carbonregistry.com/iso-14064-3>

⁷⁷ ISO 14065:2020 General principles and requirements for bodies validating and verifying environmental information. <https://www.iso.org/standard/74257.html>

GHG verifications⁷⁸. The standard also enforces strict rules against conflicts of interest and prohibits outsourcing critical activities such as independent review and issuance of the verification report. By adopting ISO 14065:2020 as the basis for the CBAM's accreditation framework, an internationally accepted basis for the CBAM and its credibility will be ensured.

Furthermore, ISO 14067:2018⁷⁹ provides a framework for quantifying and reporting the carbon footprint of products throughout their entire life-cycle, from raw material extraction to end-of-life disposal⁸⁰. This comprehensive approach ensures that all relevant emissions are accounted for, which makes this standard an ideal basis for diverse reporting purposes as well as the EU's PEF (Product Environmental Footprint)⁸¹ method.

4.3.7 Conclusions from the legislative review

Several areas of EU legislation have been identified which require the collection and reporting of GHG emissions data, to some extent at the level of the company, but in some cases also at the level of the product. However, there is no legislation applicable to CBAM goods which has similarly exact rules for determining product-level goods, nor rules which have the same system boundaries. Instead, the typical finding is that reference is made to life-cycle emissions as determined using the PEF, ISO 14067 or the GHG Protocol. However, as the system boundaries and monitoring approaches used in LCA/carbon footprint studies are different from the CBAM by definition (see section 4.3.5), only partial overlaps in data and in methodology exist.

The planned DPP (digital product passport) might support the efficient communication between producers and consumers of CBAM goods (if it is a precursor) or producers and importers. However, it is too early to judge the synergies between CBAM and the DPP, as it will strongly depend on whether and how DPP requirements will be included in the CBAM legislation, or whether only requirements will be specified in delegated acts under the EUPR.

4.4 Results of Sub-Tasks 2.3 & 2.4 – Analyse data and analytical requirements, and feasibility of extending CBAM to new goods

The data requirements for monitoring of data for the CBAM include the knowledge of production processes, routes and relevant precursors at the level of the development of the MRV rules. This is a key criterion in this study.

On the side of operators and authorised declarants, data required include either default or actual data for the embedded emissions of goods, for the correlating free allocation in the EU ETS, which can be deducted from the CBAM obligation, and for an effective

⁷⁸ European Commission, 2022. The Accreditation and Verification Regulation - Relation between the AVR and EN ISO 14065. https://climate.ec.europa.eu/system/files/2022-05/quick_guide_verifiers_en.pdf

⁷⁹ ISO 14067:2018 Greenhouse gases — Carbon footprint of products — Requirements and guidelines for quantification. <https://www.iso.org/standard/71206.html>

⁸⁰ Safdie, S. 2024. ISO 14067: Meaning, Standard and Requirements. <https://greenly.earth/en-us/blog/company-guide/iso-14067-meaning-standard-and-requirements>

⁸¹ Commission Recommendation of 16.12.2021 on the use of the Environmental Footprint methods to measure and communicate the life-cycle environmental performance of products and organisations, C(2021) 9332 final. https://environment.ec.europa.eu/publications/recommendation-use-environmental-footprint-methods_en

carbon price paid in a third country, if applicable. However, this project focusses only on the embedded emissions to be determined by operators for demonstrating the monitoring's feasibility, assuming that this is an indication for the feasibility for the other data sets, too.

As the availability of data is at the core of the technical feasibility for including goods in the CBAM, we have implemented sub-tasks 2.3 (analyse data and analytical requirements) and 2.4 (feasibility assessment of extending CBAM to new goods) in one joint assessment, as presented below. The criteria used are described in section 4.4.14.4.1. A specific addition to this methodology used for the sectors of inorganic chemicals, organic chemicals and polymers is described in section 4.4.2. The assessments per sector are then presented in sections 4.4.3 to 4.4.12.

Finally, Sub-task 2.5 (propose a ranking of priority sectors, sub-sectors/products) leads to the conclusions presented in section 4.5.

4.4.1 General considerations on feasibility criteria

Overall, whether or not the CBAM's scope is extended, the situation remains as it has been: Either an operator provides actual (verified) data to the authorised declarant, or the declarant will have to use a default value for the specific embedded emissions⁸² of a good. Hence, technical feasibility of determining embedded emissions boils down to:

- a) The possibility to lay down the MRV rules in a sufficiently detailed and comprehensive way if a sector or individual good requires special rules beyond the existing ones.
- b) The necessary condition to determine default values.

In the proposal (and confirmed in the inception report), the following criteria should serve for identifying the technical feasibility to include goods in the CBAM:

1. **Criterion 1:** Goods must be identifiable (CN code, CAS number for chemicals);
2. **Criterion 2:** Monitoring rules must be available, in particular:
 - **Criterion 2a:** Production processes and routes need to be known (and not overly complex);
 - **Criterion 2b:** Rules for attributing emissions in processes with co-products are required;
 - **Criterion 2c:** Even where MRV rules are sufficient in existing legislation, it may be useful to consider the typical installation in which the good under consideration is produced, and whether these installations are typically complex, so that e.g. more detailed guidance documents might be required. This criterion is specifically applied to refinery products, chemicals and polymers.
3. **Criterion 3:** The determination of default values for embedded emissions must be possible;
4. **Criterion 4:** The MRV system must be able to cover longer value chains. I.e., effective means are needed to pass on data, including embedded emissions,

⁸² Specific embedded emissions are the embedded emissions per tonne of good.

already paid carbon price, and free allocation levels, while ensuring sufficient confidentiality.

In the subsections 4.4.3 to 4.4.12 the criteria will be discussed in detail for each relevant industry sector identified in Task 1. But first, we present here how we interpret the criteria for technical feasibility:

Criterion 1: Goods must be identifiable:

The first thing for reporting an imported good in the CBAM Declaration is to identify it using its CN code. Under the current legislation, there is no other criterion in Annex I of the CBAM Regulation. This creates a couple of issues:

- The analysis leading to the shortlist of goods in Task 1 was carried out based on the “Carbon Leakage List” which identifies sectors at risk of carbon leakage using the NACE/PRODCOM system. Although there are correspondence table available to identify the related CN codes, there are many cases where one PRODCOM correlates with several CN codes, and one CN code can correlate with several PRODCOM codes. Consequently, the analysis produced some heterogenous results, e.g. identifying only very specific goods which sometimes do not cover the whole CN category. This means that sometimes a CBAM good cannot be identified with the whole CN code. In such cases other classification systems may need to be used in addition to the CN code, such as the TARIC code (if available)⁸³, or for chemicals the CAS⁸⁴ Number. Should such more precise classification not be available or not legally possible to use, these goods cannot be included in the CBAM.
- In case of chemicals, some CN codes cover whole substance classes, where individual substances are often produced in very different processes. Often only very few substances are commercially relevant. Many such substance classes are too complicated for inclusion in the CBAM. However, in some cases, individual substances were identified as precursors in specific production routes which may belong to such substance class CN codes. In order to make their inclusions in the CBAM feasible, additional identifiers will be required in Annex I of the CBAM Regulation⁸⁵. Nevertheless, including sub-parts of CN codes is recommended only where a very high relevance of such goods is expected.

⁸³ The TARIC code (Integrated Tariff of the European Union) is the CN code plus two additional digits, and it is used for good to be imported into the EU. See https://taxation-customs.ec.europa.eu/customs-4/calculation-customs-duties/customs-tariff/eu-customs-tariff-taric_en

⁸⁴ CAS: Chemical Abstracts Service, <https://www.cas.org/cas-data/cas-registry>. The CAS Numbers were developed as unique numbers representing complex chemical structures, for example for literature search and various databases. According to that website, there are currently about 219 million (mostly organic) substances registered with CAS numbers. However, CAS numbers have the disadvantage of being only unique but not having any direct correspondence to e.g. the structure formula of the substance. An identification of the correct CAS number is therefore not trivial. The project team used only freely accessible sources for identifying CAS numbers, such as the search function of CAS's web-site, as well as ECHA's database (<https://echa.europa.eu/information-on-chemicals>) and ECICS (https://ec.europa.eu/taxation_customs/dds2/ecics/chemicalsubstance_consultation.jsp?Lang=en). However, due to the complexity of the chemical structures (in particular some of the polymers), the search did not always give robust results.

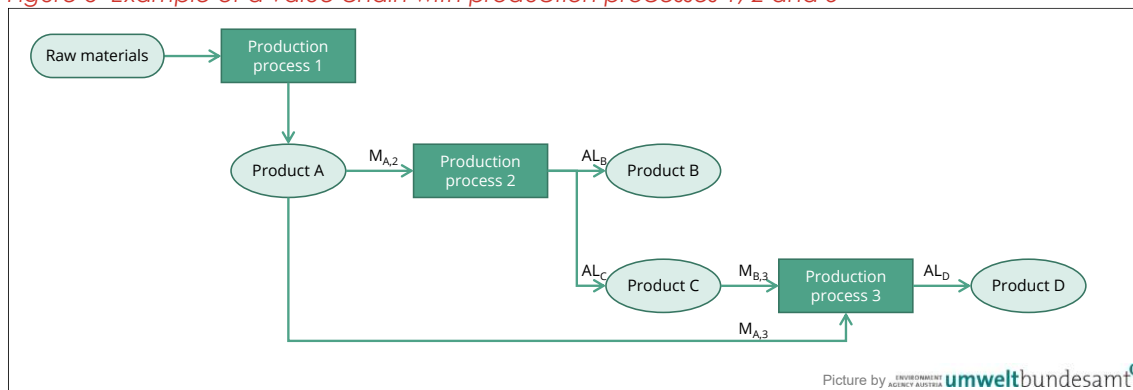
⁸⁵ A first example of such additional identification in Annex I, the omnibus amendments of the CBAM Regulation added in Annex I to CN code “2507 00 80 – Other kaolinic clays” the specification “except non-calcined kaolinic clays”

- On the other end of the complexity range, some CN codes cover at a higher level (HS 4 or 6 digits) goods categories which are relatively homogeneous in terms of production processes, and often many different products are produced in the same installation or at least in very similar installations. Nevertheless, the analysis of Task 1 sometimes identifies only few 8-digit CN codes out of a broader group of similar goods as relevant for inclusion in the CBAM. Our method does not focus specifically on circumvention possibilities. However, it seems possible that in installations producing many different, but sufficiently similar products, just slightly different CN codes could be applied for circumventing the CBAM. Therefore, in some cases it might be advisable to include the whole sector in the CBAM instead of few goods (which are sometimes non-representative for the sector, like if only specific bottle sizes are identified rather than the full container glass sector), or to keep the whole sector outside the CBAM for keeping it simple and focussed on the very high-emission goods. Such relatively homogeneous sectors which are important emissions- and trade-wise are paper, glass and ceramics.

Criterion 2: Monitoring rules must be available:

Criterion 2a. Monitoring rules must be available, in particular **production routes must be known and not overly complex**. This criterion is relevant for both, actual and default values of embedded emissions. Note that by “production route”, the implementing act for the transitional period on the monitoring methodology means the “*specific technology used in a production process to produce goods under an aggregated goods category*”, i.e. it defines only a single production step in a value chain. However, in some cases, especially for chemicals, in this study **by “production route” a whole value chain is meant**. For reasoning, consider the following example value chain in Figure 6:

Figure 6 Example of a value chain with production processes 1, 2 and 3



Production process 1 in the example is a simple one, there are no precursors, but only raw materials involved, i.e. materials (or fuels) which do not have embedded emissions themselves. Hence, Product A is a “simple good” in CBAM terminology. For simplicity, the picture does not show further raw materials, although they may also occur in the other production processes. Production process 2 is an example of a process with two co-products (product B and C), i.e. in this case, an “attribution” of the process’ total emissions to the embedded emissions of those products is required. Production process 3 is now a process that has more than one precursor. Both products A and C are precursors to product D. Depending on how “complexity” of a value chain would be defined, the

complexity for producing product D would be “3” if the number of relevant processes were counted, or only “2”, as only two precursors are relevant, or again “3”, if taking into account that overall there are 3 other products that need to be monitored for determining embedded emissions of product D.

Now, if product D were under discussion for inclusion in the CBAM, the immediate question would be: Are goods A, B and C also included in the CBAM? If the other goods are not included, only the emissions of production process 3 are relevant for the embedded emissions of product D. But if that process leads to little emissions, including it in the CBAM would not be effective, since low embedded emissions may not impact the behaviour of a user of that product whether to purchase that good from an EU installation or to import it. Yet, the inclusion of the good in the CBAM would create some administrative burden, and therefore the cost/benefit ratio for inclusion would be unfavourable. However, if production processes A and B have large emissions, or medium emissions and at least some of the factors $M_{A,2}$, $M_{A,3}$ and $M_{B,3}$ (the masses used in the consecutive processes) are reasonably high, then also the embedded emissions of product D become relatively high if the products A, B and C are included in the CBAM.

An example for such a situation is the production of polyethylene (PE). The first product would be naphtha (a refinery fraction), the second process would then be the steam cracker which produces several goods, amongst which is ethylene, i.e. the monomer which is then used in the polymerisation which would be the third process. This last step⁸⁶ has relatively low emissions, but the embedded emissions of PE would be in a similar order of magnitude as current CBAM goods if also the monomer ethylene and perhaps even naphtha were included in the CBAM.

Therefore, in this chapter we assume that “production route” needs to look at the value chains, and **ideally all products between raw materials and the final good identified in Task 1 should be included in the CBAM** as precursors or normal CBAM goods. However, if each such intermediate product can be produced itself by several production routes, it can mean that the value chains become too complex for complete coverage in the legislation. Furthermore, situations should be avoided where e.g. product A, B and D but not C are included in the CBAM. In particular in such a situation where co-products are involved, omitting one good from the value chain might mean that emissions would be attributed over-proportionally to product B, and the resulting product D would be under-rated regarding embedded emissions. The methodology for the transitional period already avoided such unbalanced attribution by attributing emissions to the non-CBAM co-products of electrolysis processes producing hydrogen. However, for goods to be included in the CBAM in the future, a case-by-case assessment may still be required to identify the relevant products which should be attributed emissions.

More generally, whenever there is a value chain where emissions happen mostly upstream, the efficient way for inclusion in the CBAM is to focus on these upstream goods and include value chains downstream only to an extent that those embedded emissions are not too much “diluted”. Furthermore, a reasonable approach would also be to include those goods where the *imported embedded emissions* (the quantity of good imported multiplied with the specific embedded emissions) are highest. For example, if the precursor is hardly traded, it is better to include the products of the next production

⁸⁶ Note that this cannot be compared to the fictitious example, as the polymerisation has only one precursor.

process in the CBAM, too. Again, this is how the inclusion of polymers might be useful, even if the real emissions happen upstream. Furthermore, where several precursors feed into the same production process, the precursors contributing most to the embedded emissions should be preferably included, while those contributing little should be omitted.

A precursor can contribute significant amounts of embedded emissions either because the specific embedded emissions are high, or because the quantity of precursor consumed in the production process is relatively high.

As has been mentioned earlier, the complexity of the production route has impacts on both, actual emissions and default values.

Regarding actual emissions, the difficulty lies in developing MRV rules in the implementing legislation in a way that all relevant production routes are covered. If the approach in the implementing act for the transitional period is followed, the following elements need to be further defined in the act:

- Which goods (CN codes) together form an “**aggregated goods category**”? For chemicals, unfortunately, there are almost no possibilities to form bigger groups, and many substances will need to be treated as separate “aggregated goods categories”. As the example below shows, for chemicals it can even be necessary to define smaller goods categories, i.e. several categories under a single CN code.
- **What production routes are relevant?** The issue of value chains has been discussed above already. Regarding the “production route” within the meaning of the CBAM implementing act, i.e. the specific technology of a production process, the question is to what extent routes can be described jointly or if they need separate treatment. In concrete case of chemicals, the number of routes can be large. This is because – although many substances are produced overwhelmingly by a few main production routes – they may also be side products of other processes. They may also be produced by some “exotic” (rarely used) processes, e.g. because in single installations there may be a need to make use of by-products from other processes, and which do not fit the main production route.
- What are the **boundaries** of the production process in case of each production route?
- Which **precursors** are relevant to each production route?
- Which **co-products** occur, and what **attribution method**⁸⁷ and factors should be used for the attributed emissions? (Note that this topic only appeared for hydrogen in the implementing act for the transitional period but will be a standard issue for many chemicals).

Regarding defining default values, the complexity of goods’ value chains has also a critical influence. It is discussed further below in this section under Criterion 3.

⁸⁷ This is the topic of a separate discussion note, as mentioned later.

An example of a medium to high complexity, but economically highly relevant⁸⁸ value chain is the one for producing CN 3908 10 00 “Polyamides-6, -11, -12, -6,6, -6,9, -6,10 or -6,12, in primary forms”. Only the two most important Nylon types are described here, and the production routes are simplified by omitting e.g. coal-based value chains:

- Polyamide-6 (PA-6, Nylon 6) production:
 - Mineral oil refining → Naphtha hydrocracking or catalytic reforming → Aromatics extraction in refinery → Benzene
 - Benzene + Hydrogen → Cyclohexane or Naphtha distillation → Cyclohexane
 - Cyclohexane + oxygen (air) → Cyclohexanone
 - Cyclohexanone + hydroxylamine⁸⁹ → Caprolactam⁹⁰ or Cyclohexane + nitrosylchloride → Caprolactam
 - Caprolactam → Nylon 6
- Polyamide-6,6 (PA-6,6, Nylon 6,6) production:
 - Mineral oil refining → Naphtha hydrocracking → Propylene and co-products
 - Ammonia + propylene + Oxygen → Acrylonitrile
 - 2 x Acrylonitrile + Hydrogen → Adiponitrile
 - Adiponitrile + Hydrogen → Hexamethylenediamine (HMDA) – There are also other routes for production of HMDA
 - Ammonia → Nitric acid
 - Nitric acid + Cyclohexanone (s. above) → Adipic acid
 - Adipic Acid + HMDA → Nylon 6,6

In this example, both types of “production routes” appear: There are different value chains involved which arrive at the same final product, and for the production of intermediate products/precursors, there sometimes exist different production routes.

Looking at the two value chains above (which have some overlaps), it becomes clear that the CN code for those polyamides is a case which is already very challenging in terms of determining the embedded emissions, **if all the substances explicitly mentioned are considered as precursors**, i.e. are themselves CBAM goods with their own monitoring of embedded emissions. Since these goods are produced at a very large scale worldwide, the most important production routes are relatively well known and available from chemical textbooks, BREF documents etc., which is not the case for all other chemicals assessed.

However, in the example above, each substance has its own production process, and for many of the involved substances there are several production routes. There is **no**

⁸⁸ According to Task 1 data collection, imports were in the range of 270 000 tonnes/year, relating to a value of 911 million €. Production inside the EU in 25 EU ETS installations, total more than 2 mln tonnes.

⁸⁹ Production routes for hydroxylamine, nitrosylchloride (several routes possible) not included here. Precursor is in general Ammonia.

⁹⁰ In this process, large amounts of ammonium sulphate result as by-product, i.e. formally a good falling into the CBAM fertilizer category.

possibility to define wider “aggregated goods categories” consisting of several similar products. The “bubble approach” might be possible to apply, but as by-products appear, at least some further attribution of embedded emissions may be required. Following the current approach of the implementing act for the monitoring of emissions, only the rules for the two value chains above might take a couple of pages, but then not even the full CN code has been covered yet (but a lot of additional substances as precursors had to be introduced already here). Therefore, despite the principle technical possibility to determine all necessary rules, our team considers the inclusion of CN 3908 10 00 (Polyamides) as too complex for inclusion in the CBAM.

This suggestion takes furthermore into account that the benefit of including that CN code in the CBAM might be limited due to its position in the value chain. Large volumes of polyamides are not directly imported into the EU “in primary forms” as the CN code suggests, but in the form of their downstream products, including polyamide filament and staple fibres, woven and non-woven products, and of course in form of textiles, ropes, rugs etc. According to the data collected for Task 1, only the polyamide fibres imported were worth 0.9 billion Euros, compared to the polyamides in primary form worth about the same amount. Considering the even more downstream goods, the picture would be even clearer, but as the carbon leakage list does not cover more downstream sectors, we have not researched those figures. However, as this example shows, the cost/benefit ratio of including chemical substances of very high complexity is rather low.

As the example shows, it is a very demanding exercise to research all possible production routes of all chemicals which came out of Task 1. Therefore, the project team developed a specific approach to assign a “complexity measure” to value chains of chemicals and polymers. This specific methodology is presented in section 4.4.2.

Criterion 2b. As a second aspect of the criterion “monitoring rules must be available”. Task 2.1 identified that **rules need to be developed for attribution of emissions to co-products**: For this purpose, the project team developed a **discussion note** (attached as Annex 2) covering the following questions:

- If several co-products result jointly from the same production process, which methods can be used to attribute⁹¹ the process' emissions to the different co-products?
- If there are more important “main products”, further by-products of lower (economic) importance, and waste⁹² or scrap⁹³, to which of those categories

⁹¹ The literature and standards dealing with product carbon footprints and life-cycle assessments refer to this splitting of emissions as “allocation”. Due to the different meaning of “allocation” in the context of EU ETS and CBAM, the existing implementing act uses the term “attributed emissions” which is also adequate in this context of co-products.

⁹² For distinguishing “waste” and “scrap”, the definition of waste in accordance with Directive 2008/98/EC (Waste framework directive) may be useful. “‘Waste’ means any substance or object which the holder discards or intends or is required to discard”. In context of a CBAM production process, this means that waste is an (undesired) by-product which has no value to the producer or even causes costs for managing its disposal. On the other side, “scrap” as a by-product may have an economic value, as it can be either recycled in the operator’s own process or sold for recycling in another installation. However, at this stage no decision has been made whether scrap should be attributed embedded emissions or not. Therefore, it needs to be discussed here separately from other co-products.

⁹³ For the specific issue of scrap, on which discussions are ongoing in a different project (the one on defining the MRV methodology for the post-transitional phase), this note will also have to be discussed in that other project.

should the emissions be attributed? Note that for avoiding distraction from the main point, this question was considered of lower priority, and the general term “co-products” was used to cover all products (i.e. “main” products and “by-products”) in a neutral way.

The result of that discussion note is that there are several possibilities to attribute emissions of co-products, however, there is not a single one-fits-all approach. Therefore, it could be envisaged to define a decision tree or a hierarchy of methods to be used, *for example* as follows:

- First, as already defined in the CBAM implementing act, attribution by stoichiometry should be used in all cases where possible.
- In all other cases the attribution by revenue, or other economic attribution rule, should be used, with fixed default price ratios set in the legislation.
- However, if the attribution by revenue is deemed unfitting for the issue at stake, the legislation could require the use of the attribution by mass balance. These cases must be clearly defined in order to avoid uncertainty. Alternatively, for some cases where energy content is deemed suitable (e.g. for refinery products and biofuels), the energy content method could be required.
- “Hybrid approaches” could also be considered, in particular if it can be proven for a production process, that neither of the “pure” approaches above is useable. For example, the co-products could be grouped by ranges of relative market value (e.g. groups of high, medium and lower value products) with fixed “importance” ratios (e.g. 10:7:3), and a second stage of differentiation would then happen within the group based on e.g. mass balance. The caveat is that such approach may give the impression of being arbitrary, even if intended to be simple and pragmatic.

It is crucial that the rules for selection of the approach are very stringent, so that the operator cannot arbitrarily choose what gives the best outcome (lowest embedded emissions). Even better (if the number of goods included in the CBAM remains limited), the applicable approach including the applicable default attribution factors should be given in the legislation for each production process where attribution to co-products is relevant.

As some interviews with stakeholders have shown that they would prefer simplicity in the methodology instead of the most sophisticated approach, it would also be a possibility to just use the mass-balance approach for all cases where stoichiometry is not suitable, or even for really all cases of co-products from a single production process.

Independent of the approach to be chosen in the end, the result of the work in this study is that **in principle it is possible to provide relatively simple and robust rules for attributing emissions to co-products. Therefore, the need for attribution is not to be seen as technical obstacle** to including additional goods in the CBAM. However, since the selection of a specific approach still may require case-by-case discussion (including stakeholders), we still add the need for attribution as a factor contributing to the complexity of adding goods to the CBAM, in particular for the sectors of refineries, chemicals and polymers.

Criterion 2c. Additional criterion for refinery products, chemicals, and polymers. The assessment of the existing legal framework for monitoring, calculating, and reporting of embedded emissions (Implementing Act pursuant to Article 35 of the CBAM Regulation) in Sub-task 2.1 focussed on whether the monitoring rules are in principle suitable for monitoring of new sectors. For chemicals, this “in principle” applies as well. However, it is

typical for the industry sector that the different processes are often, but not always, highly integrated in large complex installations. Therefore, although a “bubble approach” could make monitoring simpler, it cannot be assumed as a given that this will be possible in general. Therefore, even precursors which are seldom traded cannot be simply included in one overarching production process but need to be identified separately as precursors with their own production routes. Secondly, those production processes are very often multi-stage processes, with many fuel- and waste streams and heat exchange between processes. Process gases of undefined composition are often used as fuels. Such special challenges are used under criterion 2c to add to the “complexity measure” in order to identify which substances are easier or more difficult to include in the CBAM from technical perspective.

Criterion 3: The determination of default values must be possible for embedded emissions

Discussions have taken place with experts from the JRC who have developed the existing default values and are currently working on the refinement for the post-transitional phase. According to those experts, complexity of value chains is no obstacle in principle, but a matter of sheer quantity. Especially in the chemical sector the amount of data to be researched would be large due to the number of substances and production routes, requiring for each the determination of the energy consumption and typical fuel mix, quantities of precursors input, and determining the said attribution factors. As said above, the same exercise would be needed for each good to be included in order to cover the whole relevant value chain. Without the ability to form aggregated goods categories, this will become a particularly challenging exercise. This is another reason to focus on the most important goods (meaning those that contribute to the most embedded emissions possibly imported under the CBAM). However, as the difficulty is no absolute reason for exclusion of specific goods or substances from the CBAM, criterion 3 is not further used in our assessment. It could be instead understood as informal limit on the side of the policy makers to focus on most important goods and substances instead of aiming for comprehensiveness.

Criterion 4: The MRV system must be able to cover longer value chains

Effective means are needed to pass on data, including embedded emissions, already paid carbon price, and free allocation levels, while ensuring sufficient confidentiality. Regarding this criterion, the current situation is as follows:

- Authorised declarants can be in situations where:
 - They receive verification reports⁹⁴ (including the relevant parts of the “communication template”, which is equivalent to an emission report) from the producer of the good. This can e.g. be done via email.
 - The operator producing the imported good has disclosed relevant emissions data to the declarant via the CBAM Registry using the mechanism pursuant to Article 10(7) of the CBAM Regulation⁹⁵. Since the declarant is obliged to include the relevant

⁹⁴ Members of the project team are working in a different project (MRVAFA for DG TAXUD) on the development of the required data to be contained in the verification reports.

⁹⁵ “An operator may disclose the information on the verification of embedded emissions referred to in paragraph 5 of this Article to an authorised CBAM declarant. The authorised CBAM declarant shall be entitled to use that disclosed information in order to fulfil the obligation referred to in Article 8.”

verification reports in the CBAM declaration, we assume that again the whole verification report can be obtained via the Registry.

- The declarant does not receive the verification report in one of the two above ways, or there is an issue with the verification, or the good cannot be identified as originating from the said installation. In all those situations, the authorised declarant will have to use default values for the embedded emissions.
- In case of operators purchasing precursors for complex goods from another installation:
 - A verification report from the producer of the precursor can be used.
 - Whether the producer of the precursor can also use the route of disclosing information via the Registry is not clear. The study team assumes that this route is available, particularly for producers whose products are also imported into the EU, i.e. where the disclosing of already available information would only have to be extended from declarants to other selected parties, such as other CBAM producers⁹⁶.
 - If neither of the two routes above work, again there is a need to use default values.

In all the above situations, there are purely 1:1 relationship between providers and users of information. In case of longer value chains, the necessary information trickles down from one verification report to the next, or from one operator's information using default values to the next operator or declarant. Therefore, in principle, the transitional reporting rules are sufficient and do not pose an obstacle to the inclusion of more complex value chains or goods with many precursors in the CBAM. However, further improvements of the CBAM methodology in the future could involve using tools such as the "digital product passport" (DPP, see section 4.3.2) for tracing data, including GHG emissions data, through longer value chains, so that a final user such as the authorised declarants can directly get information without relying on all operators in the value chain to cooperate. Since the DPP is currently only at the planning stage and as the CBAM legislation currently contains no reference to the DPP, there is currently no physical link between the imported good and the operator's data, and no firm link to data across the value chain yet. However, as verification by accredited verifiers will be conducted on data from 2026 onwards, with clear rules under which condition a verifier may accept actual data from other operators, there should not be specific issues regarding data in value chains.

Therefore, comprehensive use of the CBAM Registry, or later the use of the DPP would suffice and would not completely change the range of technical feasibility but would only reduce the administrative burden of the CBAM, making it more acceptable to include additional goods or sectors. The criterion of long value chains is therefore not included in the following assessment regarding technical feasibility of monitoring and reporting. It is, however, considered within the meaning discussed above under criterion 2a, i.e. that a too high number of production steps and production routes makes the development of monitoring rules and default values impractical.

Conclusions on assessment criteria

- **Criterion 1:** Goods must be identifiable: This criterion is now interpreted such that it has to be discussed if the CN code as such is useful, if more differentiation inside a CN code

⁹⁶ Confidentiality concerns still need to be tackled.

is needed, or whether it might be beneficial to include wider groups than just one narrow CN code.

- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): With the clarification that by “production route” two different concepts can be meant: Firstly, the technology used for one production process, as in the transitional CBAM rules, and secondly, longer value chains and relevant precursors to come from raw materials without embedded emissions to the specific good under assessment. In particular for chemicals this will be the most striking criterion. Therefore, a dedicated methodology for calculating a “complexity measure” has been developed (see section 4.4.2).
- **Criterion 2b:** Rules for attributing emissions in processes with co-products are required: We consider this issue as solved and universally applicable. Therefore, no further assessment needed.
- **Criterion 2c:** Identification of special MRV difficulties. This will be added in the calculation of the “complexity measure” of chemicals and polymers. It is also relevant for refinery products.
- **Criterion 3:** The determination of default values for embedded emissions must be possible: We consider this fulfilled for all products (except those that may be excluded based on criterion 2a already). Therefore, no separate assessment is required for individual CN codes.
- **Criterion 4:** The MRV system must be able to cover longer value chains. Again, this is considered fulfilled for all CN codes to be assessed.

In summary, only criteria 1 and 2a, and, where relevant, criterion 2c will need to be applied in the assessment of the technical feasibility for including the CN codes in the CBAM which were identified under Task 1 (section 2.6 and 3.5). This discussion will be presented in the following subsections per industry sector.

Note: Further data sets have to be determined for each good which influence the technical feasibility of monitoring, reporting and verification:

- Data to determine the reduction of the CBAM obligation under Article 31 of the CBAM Regulation by considering the free allocation levels in the EU ETS;
- The carbon price already paid in third countries for the emissions of producing the imported CBAM good (Article 9 of the CBAM Regulation).

However, both issues are outside the scope of this project. Furthermore, we assume that both issues may potentially increase the administrative burden by increasing the number of data sets to be monitored, reported and verified (in case of carbon price: certified), but that the feasibility of determining these data would be not materially different from the determination of embedded emissions. Therefore, we do not consider the influence of these topics relevant for the further assessment.

Also, whether indirect emissions should be included in the CBAM or not does not influence the technical feasibility of inclusions. Just as for the two other points above, it is only the number of data points to be collected that would be influenced by the decision on including indirect emissions.

A note on terminology. In the assessment in the following sub-sections, sometimes it is mentioned that a CN category should be **treated like a “downstream good”**. This means that the production of these goods is often not linked to significant emissions except in the precursors, and/or that the goods do not consist of a single material, which would mean that there is a “dilution” of the material. Therefore, such goods should only be included if a generic approach to the monitoring of emissions for such downstream

goods is provided, in line with the results of the TAXUD project on downstream extension of the CBAM. In that project, it was outlined that the MRV methodology might be simplified for such goods in the way that the emissions of the production process may be omitted, and only the embedded emissions of the precursors need to be considered.

4.4.2 Complexity measure of value chains in the chemicals sectors

For chemicals and polymers, the study team has developed a calculation formula which allows to compare and rank the relative complexity of value chains and their monitoring issues in the following formula.

Note: For the use of this formula, “**production route**” means a single production process for producing a good, not a full value chain, as in other parts of the chemicals' discussion.

$$CM_g = (R_g - 1) + \sum_{r=1}^{R_g} (Prc_r + A_r + CMVR_r) + \sum_{pr=1}^p (CM_{pr}) \quad \text{Equation (1)}$$

$$p = \sum_{r=1}^{R_g} (Prc_r) \quad \text{Equation (2)}$$

Where

CM_g	Complexity Measure for good g
R_g	Number of production Routes for good g – the term (R _g – 1) means that it is zero if only a single route is defined for good g, and every additional route adds to the complexity measure. Alternatively, the weight of having different routes could be increased, e.g. by changing this term to n · (R _g – 1) where n would be a weighting factor, e.g. n = 3.
Prc_r	Number of Precursors for route r. Precursors already included in the CBAM are omitted. If a precursor appears in more than one production route for the same good g, it is counted only once (at first occurrence).
A_r	Attribution needs for route r (A = 1 if any attribution is needed, i.e. if more than one co-product is typically resulting from the production process of this route, otherwise A = 0). Note: Where co-products are treated like waste (which means in organic chemistry that they are usually used as fuel for heat generation), no attribution is assumed necessary. An alternative for this parameter would be to assign the number of co-products instead of 1. However, as not all installation configurations will consider the same by-products as “high value chemicals” (i.e. worth of economical utilization), the method would require some arbitrary selection, which we try to avoid in this determination of complexity. However, it can be considered to increase the parameter in uniform manner for all cases of attribution (e.g. A = 3) in order to emphasise the complexity hidden behind the attribution needs.
CMVR_r	Parameter considering “ Challenging MRV ”, i.e. the degree to which monitoring of the production route r is considered more challenging than for other sectors. 0 means not challenging at all, 3 is considered very challenging. The range of this parameter may be adjusted during the assessment if found necessary.
CM_{pr}	Complexity Measure of the precursor pr . This element in the formula serves for a recursive calculation, i.e. the complexity measure calculated for each precursor using the same equation has to be added until there are no further

precursors. This approach serves for handling branched value chains. The index pr runs from 1 to p , i.e. the total number of direct precursors of good g , which ensures that each direct precursor used in several production processes leading to the same good g will be added only once.

Where the project team found further exploration of precursors exceeding the possibilities of the project (e.g. because a whole substance class is mentioned in literature for the precursor, a default value for CM_{pr} is used in the calculation. This default value should be at least 5...10. In our assessment we used a rather careful value of 5, i.e. aiming at not exaggerating technical complexity.

p Number of all precursors listed directly in production routes for good g (calculated using Equation 2)

For demonstrating how the formula works in practice for determining a complexity measure (Equation (1)), we compare a relatively simple value chain (production of PE (Polyethylene)) and a medium/high complex one (adipic acid). The pictures focus on the direct production path of one final product only, i.e. co-products are only mentioned for the purpose of showing the need to attribute emissions, but alternative routes for these co-products are excluded from the picture in order to not divert attention from the final product.

Figure 7 presents a simplified⁹⁷ value chain for polyethylene (no distinction in principle for LLDPE, LDPE and HDPE). In this example, there is only one linear value chain. Although at two points a need for attributing emissions to co-products occurs, the overall complexity rating calculated using Equation (1) (shown by red circles for selected substances) is low. The potentially largest obstacle is the monitoring of the naphtha production, as this needs the inclusion of refinery products in the CBAM. But the contribution of the refinery process to the total embedded emissions is limited. Hence, this one complexity could be avoided by keeping refinery products outside the CBAM scope.

Figure 7 Example 1: Production route of Polyethylene. The numbers in red circles give the "complexity rating" of each substance.

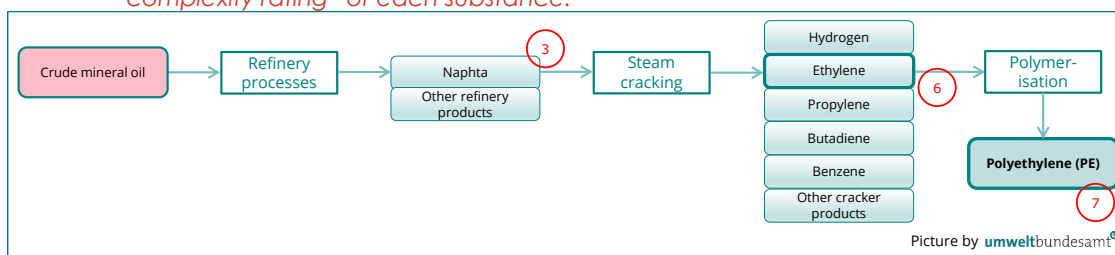


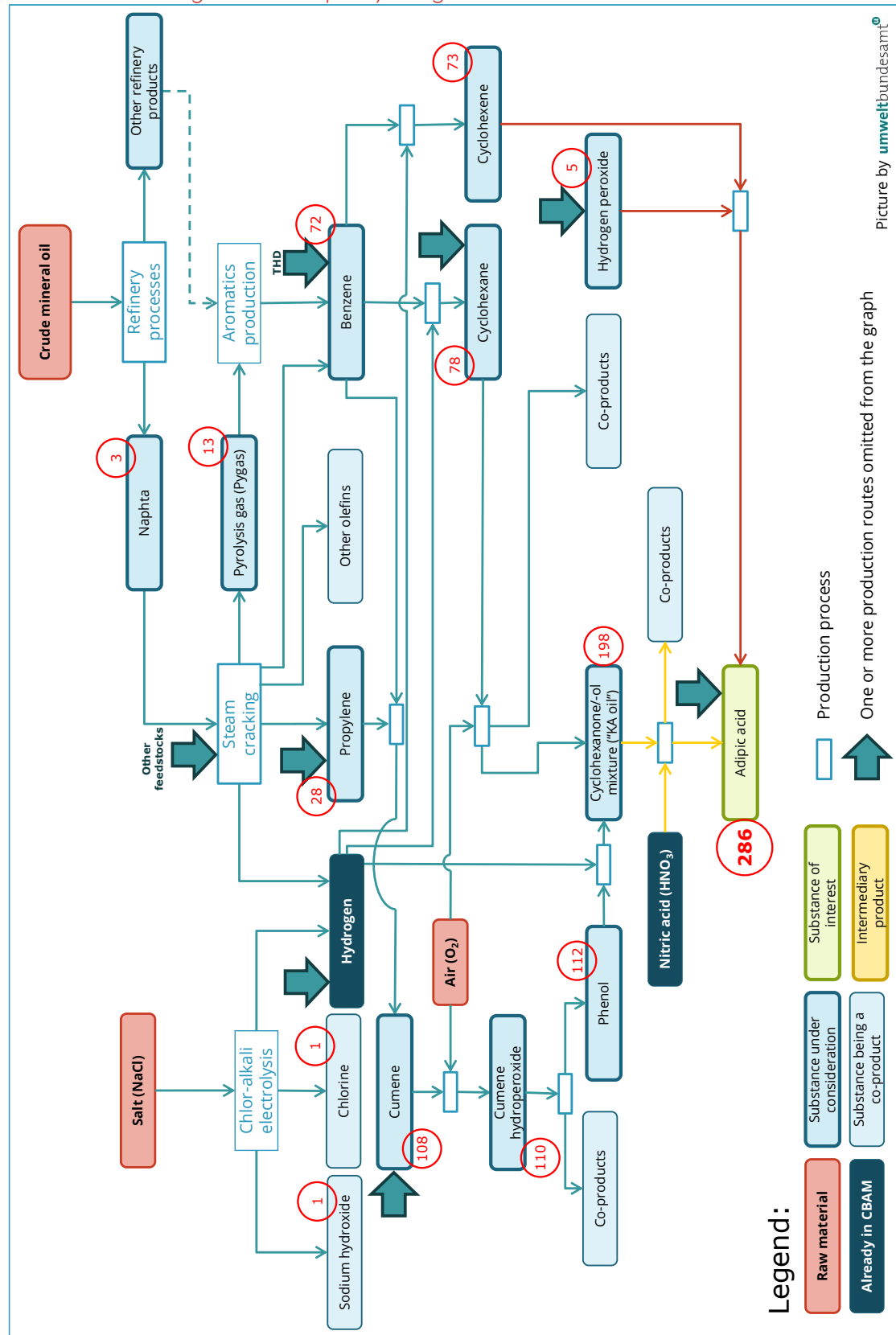
Figure 8 shows the production routes for adipic acid. While the most frequent routes found in literature and probably in practice are the routes using "KA-oil". However, for the full complexity estimate, we included two less common production routes, namely

⁹⁷ This value chain is presented here on purpose in a simpler way than in the actual assessment (section 4.4.1.1 and in the annexed discussion note on chemicals). The aim here is to show how the calculation works in a simple case. In reality, additional production routes are required: The steam cracking can be run using different feedstocks, in particular ethane distilled from natural gas. Secondly, the "coal to olefins" route is omitted. In the complete assessment of value chains, the complexity measure of polyethene is calculated to be 25.

hydrocarboxylation of butadiene and oxidative cleavage of cyclohexene using hydrogen peroxide. For keeping the picture at a manageable size, the route using butadiene is only indicated with an arrow symbol, but it is included in the calculation. Although it is not clear if the latter two routes are actually used at industrial scale, it seems prudent to include them in the assessment (it would be a “conservative” approach to include rather too many options than too few).

KA-oils seems to be not a traded substance (we have not identified a CN code for it). Therefore, it is counted as precursor only to allow for a systematic counting of the production routes: Figure 8 shows three different production processes directly leading to the final product adipic acid, and two processes leading to KA-Oil. Further branching of the full value chain is considered in accordance with Equation (1). Complexity measures for selected substances are shown in red circles. Although the precursors already included in the CBAM have quite complex value chains themselves, they are not considered in order to draw attention purely to the complexity contributed by the substances not yet covered by the CBAM scope.

Figure 8 Example 2: Adipic acid – The different colours of arrows and borders on substances are intended to help the reader to identify the different production routes. The numbers in red circles give the “complexity rating” of each substance.



4.4.3 Pulp & paper

Production processes and value chains

For the assessment, we distinguish pulp (HS 47) and paper & paperboard (HS 48). All paper products have some kind of pulp as precursor. Regarding production processes, it seems possible to create aggregated goods categories (the exact grouping would require information and/or confirmation from stakeholders):

- Pulp:
 - Mechanical pulp;
 - Sulphate (kraft) pulp;
 - Sulphite and other dissolving pulps.
- Paper:
 - In terms of production technology, all paper and cardboard grades could be covered by the same rules, i.e. one single aggregated goods category might be sufficient.
 - Significant differences exist in energy consumption and emissions between integrated pulp and paper installations, and stand-alone paper producers, where the latter uses either recycled or purchased virgin pulp. The difference lies in the fact that in integrated plants most of the emissions will be from biomass and therefore zero-rated under EU ETS rules. This distinction of plants might have an impact on the default values to be used, and on the question whether a bubble approach can be allowed. There is no impact on inclusion of the sector in the CBAM.

Further downstream for the pulp and paper goods there are goods such as HS 4814 to 4817 and 4819 to 4823. The production of decorated articles, bound notebooks and boxes etc. as mentioned in these codes are usually carried out in non-EU ETS installations. It is proposed to not include such goods in the CBAM. Those CN codes could be included under similar conditions as discussed in the “downstream” project, i.e. by assessing if the goods can become at risk of carbon leakage due to the embedded emissions of the upstream paper products. The monitoring could be performed by mere accounting of the precursors’ embedded emissions. At the time of this study, the relevant MRV rules for “downstream” goods are not yet available. The respective goods are therefore considered as “not to be included in the CBAM”. However, if at the time of the horizontal extension of the CBAM, rules for the inclusion of downstream goods are already confirmed (because the extension ‘downstream of existing CBAM’ has already happened), then these goods should be included together with the “horizontal” extension” discussed here.

Since paper in general has relatively low specific embedded emissions due to its biomass origin (i.e. because biomass residues from pulp production are the predominant fuels in integrated pulp and paper production), the carbon leakage risk is expected to be relatively low and it wouldn't be a priority to include pulp and paper in the CBAM in general. However, as the technical feasibility is given, we have rated paper products as “can be included”.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No issues regarding identification found.
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, transposing the

rules in CBAM will be no major obstacle. The production routes and processes and their system boundaries are sufficiently simple to describe. No obstacles found.

Assessment of "other criteria": Table 11 below compares the sector-wide results from Task 1 against the results of the project team under Tasks 2.3 and 2.4 (this chapter).

Table 11 Comparison on the sector-wide results from Task 1 against the results of the project team under Task 2.3 and 2.4 for pulp & paper

HS heading	Description (simplified)	Result of Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Task 2.3 and 2.4
47	Pulp			
4701	Mechanical wood pulp, not chemically treated			yes
4702	Chemical wood pulp, dissolving grades			yes
4703	Chemical wood pulp, soda or sulphate (excl. dissolving grades)			yes
4704	Chemical wood pulp, sulphite (excl. dissolving grades)			yes
4705	Wood pulp obtained by a combination of mechanical and chemical pulping processes			yes
4706	Pulps of fibres derived from recovered "waste and scrap" paper or paperboard or of other fibrous cellulosic material (excl. wood)	yes		yes
4707	Recovered "waste and scrap" paper or paperboard (excl. paper wool)		In analogy to aluminium and iron and steel sectors, this category should be included only if scrap is assigned embedded emissions. However, as "pre-consumer scrap" is of little relevance in this sector, a decision may be appropriate to exclude this category independently of the approach for aluminium and iron and steel.	no
48	Paper and paperboard			
4801	Newsprint	yes		yes
4802	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes	yes, except 480210 handmade papers	Exception of handmade papers justified (not produced in EU ETS installations)	yes, except 480210
4803	Toilet or facial tissue stock, towel or napkin stock and similar	yes		yes
4804	Uncoated kraft paper and paperboard	yes, only 480411 (uncoated, unbleached kraftliner)	Unclear why the other CN-Codes under 4804 should not be included. The whole sector of kraft papers is a very important product group (basis of all kinds of packaging), and many installations will produce a whole range of papers in this group.	Yes, whole 4804
4805	Other paper and paperboard, uncoated	no	For same reasons as 4804 the whole group should be included	yes
4806	Vegetable parchment, greaseproof papers, tracing papers etc.	no	Goods might be composites containing significant additional materials compared to common uncoated or coated paper grades. The goods should remain excluded;	no
4807	Composite paper and paperboard "made by sticking flat layers of paper or paperboard together with an adhesive", not surface-coated	no		no
4808	Corrugated paper and paperboard "with or without glued flat surface sheets", creped, crinkled, embossed or perforated	no	See 4804, whole group should be included	yes

HS heading	Description (simplified)	Result of Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Task 2.3 and 2.4
4809	Carbon paper, self-copy paper and other copying or transfer papers	no	Goods might be composites containing significant additional materials compared to common uncoated or coated paper grades. The goods should remain excluded	no
4810	Paper and paperboard, coated on one or both sides with kaolin "China clay" or other inorganic substances	no	There is no reason for excluding this important paper grade produced and traded at high volumes The content of pulp will be lower than in uncoated papers (i.e. there may be different embedded emissions), but no special monitoring requirements or precursors	yes
4811	Paper, paperboard, cellulose wadding and webs of cellulose fibres, coated, impregnated, covered, surface-coloured, surface-decorated or printed	only 481159 and 481190	Products might be composites containing significant additional materials compared to common uncoated or coated paper grades. The production processes are therefore less clear, and the whole group of goods should remain excluded	no
4812	Filter blocks, slabs and plates, of paper pulp	no	These should be homogeneous products consisting mostly of pulp, i.e. no difficulty in inclusion	yes
4813	Cigarette paper, whether or not cut to size or in the form of booklets or tubes	no	Homogeneous paper grades, no reason for exclusion	yes
4814	Wallpaper and similar wallcoverings of paper; window transparencies of paper	no	These seem to be mostly goods produced downstream from the paper produced in previous CN codes. The production of decorated articles, bound	no
4816	Carbon paper, self-copy paper and other copying or transfer papers	no	notebooks and boxes etc. as mentioned in these codes are usually carried out in non-EU ETS installations.	no
4817	Envelopes, letter cards, plain postcards and correspondence cards, of paper or paperboard; boxes, pouches, wallets etc.	no	Inclusion OK for toilet paper and the like.	no
4818	Toilet paper and similar paper, cellulose wadding or webs of cellulose fibres, of a kind used for household or sanitary purposes; handkerchiefs, cleansing tissues, towels, etc.	yes	However, 481850 (Articles of apparel and clothing accessories, etc.) seem to be downstream products typically not produced in EU ETS installations.	yes, except 481850
4819	Cartons, boxes, cases, bags and other packing containers, of paper, paperboard, cellulose wadding or webs of cellulose fibres, n.e.s.;	no		no
4820	Registers, account books, notebooks, order books, receipt books, letter pads, memorandum pads, diaries and similar articles	no	These seem to be mostly goods produced downstream from the paper produced in previous CN codes. The production of decorated articles, bound	no
4821	Paper or paperboard labels of all kinds, whether or not printed	no	notebooks and boxes etc. as mentioned in these codes are usually carried out in non-EU ETS installations.	no
4822	Bobbins, spools, cops and similar supports of paper pulp, paper or paperboard, whether or not perforated or hardened	no		no
4823	Paper, paperboard, cellulose wadding and webs of cellulose fibres, ... n.e.s.	no		no

4.4.4 Glass

Production processes and value chains

Production of glass is a relatively homogeneous sector regarding the technical processes. In principle, mineral raw materials and/or cullet (recycled glass) are heated until a melt is the result. The liquid glass is then formed to give already the (almost) final product, before some slow and controlled cooling in special ovens finalises the energy-consuming part of the production process and the occurrence of significant emissions. Steps thereafter, such as coating, printing, or painting, do not contribute to significant emissions.

According to the main forming processes, usually the following main product groups are technologically differentiated:

- Container glass (bottles, jars, etc.): The glass is formed by “blowing” – EU ETS benchmarks differentiated coloured and colour-less glass; Installations usually produce a wide range of goods of this category. Consequently, it would be simpler to include the whole CN category.
- Float glass: The glass is poured on liquid tin and cut to sheets – EU ETS benchmark available; Installations usually produce a wide range of goods of this category. Consequently, it would be simpler to include the whole CN category.
- Glass fibres: EU ETS benchmark available; Fibres are often coated with organic material; therefore a separate aggregated goods category should be considered.
- Other glass products: There is a multitude of other glass products, such as table ware, technical glasses, optical and laboratory glasses, etc. Overall, differences include the different chemical composition of the glass, the fact that more manual work may be included (including mouth-blowing for ornamental and small-series products), and grinding and polishing steps may lead to significantly more valuable goods. To some extent our analysis proposes to include those product groups nevertheless, as the overall melting production step can be included in one single aggregated goods category. Only those groups should be excluded that require downstream production processes usually outside the EU ETS.

The detailed assessment is found in Table 12 below.

Regarding the technical feasibility criteria, there is no obstacle for including the glass sector in the CBAM.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No issues regarding identification found.
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, transposing the rules to CBAM will be no major obstacle. The production routes and processes and their system boundaries are sufficiently simple to describe. No obstacles found.

Assessment of “other criteria”: Table 12 below compares the sector-wide results from Task 1 against the results of the project team under Tasks 2.3 and 2.4 (this chapter).

Table 12 Comparison of the sector-wide results from Task 1 against the results of the project team under Task 2.3 for glass

HS heading or CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
70	GLASS AND GLASSWARE			
70010010	Cullet and other waste and scrap of glass	no	In analogy to aluminium and iron and steel sectors, this category should be included only if scrap is assigned embedded emissions. However, as "pre-consumer scrap" is of little relevance in this sector, a decision may be appropriate to exclude this category independently of the approach for aluminium, iron, and steel.	no
700100 91+99	Glass in the mass	no	No reason for non-inclusion.	yes
7002	Glass in balls, rods or tubes, unworked	no	Diverse coatings should not significantly contribute to mass or embedded emissions, hence a single aggregated goods category could suffice"	yes
7003	Cast glass and rolled glass, in sheets or profiles	no		yes
7004	Sheets of glass, drawn or blown	no		yes
7005	Float glass and surface ground or polished glass, in sheets	no		yes
7006	Sheets or profiles of glass, ... bent, edge-worked, engraved, enamelled or otherwise worked, but not framed	no		yes
7007	Safety glass, toughened "tempered", laminated safety glass	no		yes
7008	Multiple-walled insulating units of glass	no	In principle no obstacle, but not 100% of the mass is glass (some minor contribution from seals and insulating gas	yes
7009	Glass mirrors, whether or not framed, incl. rear-view mirrors	no	If framed, it would be already a downstream product. Propose to treat as downstream product (not to be included yet)	no
7010	Carboys, bottles, flasks, jars, pots, phials, ampoules, and other containers, of glass, of a kind used for the conveyance or packing of goods, preserving jars, stoppers, lids and other closures, of glass	no	No reason for non-inclusion for the whole group 7010	yes
70109041	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of ≥ 1 l but $< 2,5$ l	yes	The goods identified under Task 1 are just specific sizes of glass containers, but the total container glass sector is a very important one, many installations producing a large range of different sizes. It would be easier to handle if the whole sector were included.	yes
70109043	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $> 0,33$ l but < 1 l	yes		yes

70109045	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $\geq 0,15$ l to 0,33 l	yes		yes
70109047	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $< 0,15$ l	yes		yes
70109071	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of $> 0,055$ l but $< 2,5$ l	yes		yes
70109079	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of $\leq 0,055$ l	yes		yes
70109091	Carboys, bottles, flasks, jars, pots, phials and other containers, of colourless glass, of a kind used for the commercial conveyance or packing of goods, of a nominal capacity of $< 2,5$ l	yes		yes
70109099	Carboys, bottles, flasks, jars, pots, phials and other containers, of coloured glass, of a kind used for the commercial conveyance or packing of goods, of a nominal capacity of $< 2,5$ l	yes		yes
7011	Glass envelopes, incl. bulbs and tubes, open, and glass parts thereof, without fittings, for electric lamps and light sources, cathode ray tubes or the like	no	These may be different grades of glass, but no reason for exclusion. There may be non-industrial, hand-made goods in this sector, but in principle no reason for exclusion.	yes
7013	Glassware of a kind used for table, kitchen, toilet, office, indoor decoration or similar purposes	yes		yes
7014	Signalling glassware and optical elements of glass, not optically worked	no	These may be different grades of glass, but no reason for exclusion	yes
7015	Clock or watch glasses and similar glasses, glasses for non-corrective or corrective spectacles, etc.	no		yes
7016	Paving blocks, slabs, bricks, squares, tiles and other articles of pressed or moulded glass, whether or not wired, for building or construction purposes	no	No obstacle in principle. However, 70169010 "Leaded lights and the like" are different types of glass, likely requiring special production processes, and should be excluded	yes, except 70169010
7017	Laboratory, hygienic or pharmaceutical glassware, whether or not graduated or	no	These articles may contain a lot of downstream	No

	calibrated (excl. containers for the conveyance or packing of goods, measuring, checking or medical instruments and apparatus of chapter 90)		manufacturing usually not carried out in ETS installations. Better to exclude.	
7018	Glass beads, imitation pearls, imitation precious or semi-precious stones and similar glass smallware, and articles thereof (excl. imitation jewellery); glass eyes (excl. prosthetic articles); statuettes and other ornaments of lamp-worked glass (excl. imitation jewellery); glass microspheres with a diameter of ≤ 1 mm	no	Very diverse products, some of them having downstream character, but some may also include actual glass making (but special glass types). Some installations may be large enough for inclusion in EU EUTS, some not. The heterogenous character of this group makes inclusion in the CBAM difficult.	no
7019	Glass fibres, incl. glass wool, and articles thereof (excl. mineral wools and articles thereof, optical fibres, fibre bundles or cable, electrical insulators or parts thereof, brushes of glass fibres, dolls' wigs)	7019 except 701980 (glass fibres except glass wool)	The whole group 7019 seems very homogeneous and should be included. It should be a separate aggregated goods category, because fibres usually have an organic coating (hence glass content is below 100%)	yes (whole 7019)
7020	Articles of glass, n.e.s.	no	all n.e.s. groups are heterogenous and should be excluded	no

4.4.5 Ceramics

Production processes and value chains

Production of ceramics is a relatively homogeneous sector regarding the technical processes. In principle, mineral raw materials are mixed (often with water), sometimes separately dried before the main heating ("burning") to high temperatures until the chemical structure of the ceramic is reached. Usually, the form cannot be altered after burning due to the high hardness of the product. The process is possible in continuous kilns or batch-wise. Further production steps can be glazing (with a second burning). More complex forms (including artwork, and ornamental tableware) are sometimes produced in small installations usually not covered by the EU ETS. CN categories which may contain significant such goods are proposed for exclusion from the CBAM.

In terms of production volumes, the most important ceramic goods are:

- Building bricks and roof tiles: These goods are relatively heavy compared to their value and are not transported over higher distances. Including them in the CBAM may have minor impact on trade patterns. There is no clear preference based on technical criteria to include the complete CN category in the CBAM or not. However, it is not recommended to include only selected goods out of this group.
- Refractory goods, used e.g. in industry for lining of kilns and furnaces such as cement kilns, blast furnaces, glass furnaces, etc. They play little roles for final consumers but in important industries covered themselves by the CBAM. They usually have a higher value per weight than building bricks and are therefore traded over longer distances. Installations usually produce a wide range of refractory products. Therefore, it would be preferable to include the whole

category in the CBAM, unlike the result of Task 1 suggests. However, the technical criteria give no clear recommendation for it.

Should this category be included in the CBAM, it would have to be assessed if refractories would form a precursor for goods such as cement and iron and steel. However, as the refractory linings usually have to be replaced only every few years, their consumption level is probably too small for justifying the administrative burden of including them in embedded emissions of the main CBAM goods.

- Tiles, flags and paving: this is a product group produced in many, sometimes smaller companies in Europe. The fact that their value depends not only on form and weight but also on design may explain that the trade was sufficiently high for being identified for inclusion in the CBAM under Task 1. From technical feasibility point of view there is no reason for not including this group in the CBAM.
- Other ceramic products: There is a multitude of other products, such as table ware, sanitary products, clay pipes, but also truly diverse technical products such as abrasives (which are found in other CN categories than CN 69), etc. Overall, differences include the different chemical composition of the good, the fact that more manual work may be included (e.g. for ornamental and small-series products). Due to the high diversity and because production may be partly outside the EU ETS, it is recommended to not include these goods.

The detailed assessment is found in the table below.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No issues regarding identification found. The material used might be a useful "qualifying parameter" for refractories to make plausibility checks possible, and to use possible differentiated default values (emissions values may be quite different e.g. for products made from MgO or from Al₂O₃).
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, transposing the rules in CBAM will be no major obstacle. The production routes and processes and their system boundaries are sufficiently simple to describe, too. No obstacles found.

Assessment of "other criteria": Table 13 below compares the sector-wide results from Task 1 against the results of the project team under Tasks 2.3 and 2.4 (this chapter).

Table 13 Comparison of the sector-wide results from Task 1 against the results of the project team under Task 2.3 for Ceramics

HS heading or CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
6901	Bricks, blocks, tiles and other ceramic goods of siliceous fossil meals, e.g. kieselguhr, tripolite or diatomite, or of similar siliceous earths	no		no
6902	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods	no		yes
69022099	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods containing, by weight, > 50% alumina, silica or a mixture or compound of these products (excl. containing, by weight, >= 93% silica, > 7% but < 45% alumina and those of siliceous fossil meals or similar siliceous earths)	yes	Unclear why only a very specific type of refractory blocks is identified. The whole category of refractory products could be included	yes

HS heading or CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
6903	Retorts, crucibles, mufflers, nozzles, plugs, supports, cupels, tubes, pipes, sheaths, rods, slide gates and other refractory ceramic goods	no		yes
6904	Ceramic building bricks, flooring blocks, support or filler tiles and the like	no	Economically important goods, relative homogeneous. Relatively heavy, and therefore little trade?	yes
6905	Roofing tiles, chimney pots, cowls, chimney liners, architectural ornaments and other ceramic constructional goods	no		yes
6906	Ceramic pipes, conduits, guttering and pipe fittings	no		yes
6907	Ceramic flags and paving, hearth or wall tiles; ceramic mosaic cubes etc	yes		yes
6909	Ceramic wares for laboratory, chemical or other technical uses; ceramic troughs, tubs and similar receptacles used in agriculture; ceramic pots, jars and similar articles used for the conveyance or packing of goods	no	Relative heterogeneous product groups, including potentially small installations not covered by the EU ETS. Therefore OK to be kept outside the CBAM	no
6910	Ceramic sinks, washbasins, washbasin pedestals, baths, bidets, water closet pans, flushing cisterns, urinals and similar sanitary fixtures	yes		no
6911	Tableware, kitchenware, other household articles and toilet articles, of porcelain or china	no		no
6912	Tableware, kitchenware, other household articles and toilet articles, of ceramics other than porcelain or china	no		no
6913	Statuettes and other ornamental ceramic articles, n.e.s.	no		no
6914	Ceramic articles, n.e.s.	no		no

4.4.6 Ferro-alloys

Production processes and value chains

Ferro-alloys in principle are already covered by the existing CBAM methodology (implementing act). The production processes of all ferro-alloys are similar. If additional alloys are included in the CBAM, they should be added as potential precursors for the relevant goods of the iron and steel sector where FeNi, FeCr and FeMn are already mentioned as precursors.

Already the consultant's discussion note of January 2023⁹⁸ discussed in the CBAM expert group confirmed the relevance of the two goods identified in Task 1, while other alloys from the CN group 7202 were found as not significantly imported into the EU.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No issues regarding identification found. In line with existing CBAM rules for other alloys, the mass-% of the alloying element (in this case Si and Mn) should be required to be reported as "qualifying parameter".
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, to a large extent already covered in the CBAM implementing act. No obstacles found.

⁹⁸ <https://ec.europa.eu/transparency/expert-groups-register/core/api/front/document/101351/download>

Assessment of “other criteria”: Table 14 below compares the sector-wide results from Task 1 against the results of the project team under Tasks 2.3 and 2.4 (this chapter).

Table 14 Comparison of the sector-wide results from Task 1 against the results of the project team under Task 2.3 and 2.4 for ferro-alloys

HS heading	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
720221	Ferro-silicon, containing by weight > 55% of silicon	yes	no distinction should be made, both	yes
720229	Ferro-silicon, containing by weight <= 55% silicon	no	goods are quite similar in production and use	yes
720230	Ferro-silico-manganese	yes		yes

4.4.7 Aluminium

CN codes proposed after Task 1:

- 281820 Alumina, i.e. aluminium oxide
281830 aluminium hydroxide (identified in Task 1 in the sector “inorganic chemicals”, probably due to different PRODCOM classification)

Production processes and value chains

Both products under discussion are produced in the same production process from the mineral bauxite. The only difference is that alumina is produced by calcination from the hydroxide. Alumina production is covered by the EU ETS Directive and the MRR. The basic emissions involved are only from combustion of fuels, therefore the existing CBAM rules are fully sufficient.

Alumina is a potential precursor for primary aluminium and for aluminous cement. Therefore, if it is included in the CBAM, it should be nominated precursor also in the implementing act. Aluminium hydroxide is more often used in chemical industry (e.g. for the production of zeolites), but as it can fulfil the same role as Alumina, it should be nominated precursor of aluminium, too (this relevance was confirmed by stakeholders based on the discussion note of January 2023⁹⁸). Should ceramics (refractories) be included in the CBAM, alumina would also be a precursor in that sector.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No issues regarding identification found.
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, to a considerable extent already covered in the CBAM implementing act. No obstacles found.

4.4.8 Non-ferrous metals

Note: Task 1 identified only lead and zinc as potential candidates for inclusion in the CBAM. However, in section 2.5.11 it was also mentioned that copper could also be considered for further assessment. It is economically even more important and sometimes occurs in ores together with the other two metals. There are also important alloys of copper and zinc, and production processes are similar to some extent. Therefore, this section includes copper in the assessment.

Production processes and value chains

For all three considered metals, zinc, lead and copper, recycling is at least equally important as primary production from ores. Therefore, any discussion about the treatment of scrap and distinguishing of primary and secondary production, as it is ongoing for aluminium, iron, and steel, will inevitably have to be reflected in the sector of non-ferrous metals. However, unlike for aluminium and steel, materials relevant for recycling do not cover only the scrap and waste of the more or less pure metals. In particular, HS heading 8549 (electrical and electronic waste and scrap) plays a significant role, where in used particular batteries have to be mentioned, as lead and zinc can be used for producing batteries. Also electronic devices are a source for recycling, which is also relevant for copper, as are electrical wires and cables. As concluded for all types of scrap, the inclusion of scrap in the CBAM is technically feasible, albeit difficult. The most consistent dealing with scrap with regard to its embedded emissions is to maintain the current approach (zero embedded emissions), which would mean that it is safe to exclude the scrap-related CN codes from the CBAM. However, at this point in time, we do not propose a specific approach towards scrap.

- Zinc
 - Primary production: Sulfidic ores are roasted; the resulting oxide is reduced using coke. If cadmium is contained in the raw material, it is separated by distillation. Alternatively, zinc can be produced by electrolysis after the oxide has been dissolved in sulphuric acid.
 - Secondary production: Since one of the main applications of zinc is the galvanizing of steel products, and because it has a relatively low boiling point, it can be recovered from the dust of electric arc furnaces when secondary steel is produced using zinc-coated steel scrap. The recovery is done by the “Waelz process” in rotary kilns, which again results in Zinc Oxide, which is usually reduced to Zinc using the electrolysis route.

For producing unwrought zinc, at least two production routes will have to be defined, which – like blast furnace vs. EAF – differ in the input materials and the balance of either direct or indirect emissions being predominant.

- Downstream uses: It is proposed to not include downstream goods of zinc. One main use of zinc is the coating (galvanising) of iron and steel products for corrosion protection. In this case the quantity of zinc in the goods was hitherto considered negligible. The second important use is as an alloying element for copper to produce brass. Thirdly, elemental zinc is used e.g. in the form of sacrificial electrodes in corrosion protection. The latter good may be included in the CBAM just like unwrought zinc, or, if needed, a second aggregated goods category could be defined consisting of “zinc products” (the distinction would be similar to “unwrought aluminium” and “aluminium products”). Other important products of zinc are batteries (e.g. classical zinc-carbon batteries and alkaline batteries).
- Lead
 - Primary production: Smelting is done with coke or other carbon sources in a blast furnace or other furnace. Co-products can be zinc and cadmium (i.e. if both lead and zinc are included in the CBAM, a method for attribution to co-products has to be applied). There are also processes starting from lead/copper concentrates. Again, co-product attribution would be required.

- Secondary production: The main sources are used electrodes of old lead batteries or complete (automotive) batteries. In this case, lead-antimony alloys (“bullion”) are the product. Lead, tin and zinc can also be recovered from the dust of copper production and residues of zinc smelters. Recycling takes place in blast furnaces or rotary kilns; reduction agents are needed like in primary production. Hence, further considerations are needed if in this case the distinction between primary and secondary route is relevant.
- Purification of crude lead bullion can be done either electrolytic (which is expensive) or by pyrometallurgy using various process steps of selective oxidation, distillation etc. Several other non-ferrous metals can be by-products.
- Downstream uses: The main use of lead is in (automotive and industrial) batteries. The lead is significantly “diluted” in this product, so that downstream product rules would apply.
- Copper
 - Primary production: Sulfidic ores do not need additional reduction agents. There are various techniques of smelting. In all cases the resulting crude copper needs electrolytic refining. The unrefined copper is also found in the CN classification and must therefore be considered a CBAM goods category of its own, which means that the production of pure copper for further use needs two separate production processes in the monitoring rules.

There is also a hydrometallurgical route starting from copper ores.

- Secondary production: depending on the raw material, several technologies are used. It would need further consideration if they can be summarised in a single production route.
- Downstream uses of copper:
 - Bars, rods, plates, foils, piping, etc.: Like for aluminium, these would be “copper products”, being one step downstream from refined (unwrought) copper.
 - Electrical wiring: These goods would be “downstream goods” consisting of several materials, therefore diluted. The groups heading 8544 would be relevant, although it is not necessarily copper wires that are covered there (e.g. aluminium and silver could also be covered).
 - Plated goods (staples, nails...): In these cases only a smaller part of the good is copper. They are nevertheless found in the “copper” part of the CN classification. It is suggested not to include these goods. As a counter-example, steel goods galvanised with zinc are included as steel goods, ignoring (at least in the current legislation) the zinc completely.
 - Kitchenware etc.: To be treated like “downstream goods”.

Overall, production routes are not simple and partly intertwined. The fact that secondary materials are very important gives rise to strong concerns about resource shuffling and declaration of the wrong production route. It can be reasonably assumed that installations exist which apply both primary and secondary production processes. Therefore, the inclusion of these goods should be considered of lower priority based on the technical complexity of the sector.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: No major issues regarding identification found.
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): As described above, the production routes are relatively complex and intertwined, and many technologies are involved. For the frequent use of electrolytic refining processes, the inclusion of indirect emissions would make sense. Although it is likely that the EU ETS rules can cover all non-electrolytic processes, and although rules for attribution to co-products may be available, the description of all relevant production routes will be a challenge. **An inclusion of the sector at this stage is considered too difficult.**

The assessment of "other criteria" is included in Table 15 below. These findings may be useful for differentiating the priority for inclusion in the CBAM at a later stage. However, as the overall finding after criterion 2a is already "not feasible", the conclusion in the right column is always "no".

Table 15 Comparison of the sector-wide results from Task 1 against the results of the project team under Task 2.3 for non-ferrous metals.

HS heading	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
7801	Unwrought lead	yes		no
7802	Lead waste and scrap	no	In analogy to aluminium and iron and steel sectors, this category should be included only if scrap is assigned embedded emissions. In such case, for avoiding resource shuffling, HS 854911 (Waste and scrap of lead-acid accumulators; spent lead-acid accumulators) should be considered for inclusion, too.	no
7804	Lead plates, sheets, strip and foil; lead powders and flakes	yes		no
7806	Articles of lead, n.e.s.	yes	This includes the group 78060010 "Containers with an anti-radiation lead covering, for the transport or storage of radioactive materials [Euratom]" In general "Articles of ... n.e.s." groups should be excluded, in particular because they include downstream products not produced in EU ETS installations.	no
7901	Unwrought zinc	yes		no
7902	Zinc waste and scrap	no	In analogy to aluminium and iron and steel sectors, this category should be included only if scrap is assigned embedded emissions. In such case, for avoiding resource shuffling, HS 854912 to 854919 (Primary cells, primary batteries and electric accumulators; spent goods and waste and scrap) should be considered for inclusion, too.	no
7903	Zinc dust, powders and flakes	yes		no

HS heading	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
7904	Zinc bars, rods, profiles and wire, n.e.s.	yes		no
7905	Zinc plates, sheets, strip and foil	yes		no
7907	Articles of zinc, n.e.s.	yes	In general "Articles of ...n.e.s." groups should be excluded, in particular because they include downstream products not produced in EU ETS installations.	no
7401	Copper mattes; cement copper "precipitated copper"	no		no
7402	Copper, unrefined; copper anodes for electrolytic refining	no		no
7403	Copper, refined, and copper alloys, unwrought	no		no
7404	Waste and scrap, of copper	no	In analogy to aluminium and iron and steel sectors, this category should be included only if scrap is assigned embedded emissions.	no
7405	Master alloys of copper	no	This category could contain significant amounts of other elements, unclear precursors. Could be treated as "downstream product"	no
7406	Powders and flakes, of copper	no		no
7407	Bars, rods, and profiles, of copper, n.e.s.	no		no
7408	Copper wire	no		no
7409	Plates, sheets, and strip, of copper, of a thickness of > 0,15 mm	no		no
7410	Copper foil "whether or not printed or backed with paper, paperboard, plastics or similar backing materials", of a thickness "excl. any backing" of <= 0,15 mm	no	Not only copper, perhaps rather a downstream good	no
7411	Copper tubes and pipes	no		no
7412	Copper tube or pipe fittings "e.g., couplings, elbows, sleeves"	no		no
7413	Stranded wire, cables, plaited bands, and the like, of copper (excl. electrically insulated products)	no		no
7415	Nails, tacks, drawing pins, staples, and similar articles, of copper or with shafts of iron or steel and heads of copper, screws, bolts, nuts, screw hooks, rivets, cotters, cotter pins, washers "incl. spring washers" and similar articles, of copper	no	Downstream goods, where potentially copper is not the main constituent, should not be included	no
7418	Table, kitchen or other household articles, sanitary ware, and parts thereof, of copper; pot scourers and scouring or polishing pads,	no	Downstream good, probably not produced in EU ETS installations	no

HS heading	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
	gloves, and the like, of copper			
7419	Other articles of copper, n.e.s.	no	In general "Articles of ...n.e.s." groups should be excluded, in particular because they include downstream products not produced in EU ETS installations.	no

4.4.9 (Mineral oil) Refinery products

In this section we discuss all goods mentioned in Task 1 (section 2.5.3), i.e. all refinery products including liquid fuels, petroleum coke (petcoke), inputs to the chemical industry (naphtha, pyrolysis gas (pygas), aromatics etc.) as well as speciality products like lubricants, waxes, bitumens. However, theoretically, the scope of inclusion into the CBAM could be limited to e.g. only fractions being used in the chemical industry⁹⁹, or only fuels, or any other combination of selection criteria. However, in terms of monitoring and attributing embedded emissions to the goods concerned, the simplest approach would be to cover all goods produced by the refinery sector, since also the full refineries are included in the EU ETS and on the carbon leakage list. **Hence, a split into some parts of refineries continuing to receive free allocation under the EU ETS, and other parts being covered by the CBAM, does not seem useful.**

If nevertheless the question arises whether all refinery products should be included in the CBAM, or e.g. only those needed as input to the chemical industry, it may be useful to consider a classification of refinery products as follows:

- Goods for final consumers (gasoline, diesel, fuel oils, lubricants);
- Intermediates traded¹⁰⁰, but being input for refinery processes, for example different distillation fractions before desulphurisation, or from processes like isomerisation or reforming. Such goods are used e.g. for blending to obtain final fuels, and to balance the needs of a refinery if the desired final products cannot be produced using only the crude oil input alone.
- Overlapping with the previous point are refinery fractions that are used as input to the chemical industry, such as naphtha and pygas.
- Also some goods that can be produced in stand-alone installations of the chemical industry can be directly products of refineries.

The fact that refineries trade intermediates is not a problem for applying a "black box" MRV approach. However, the overlap of these fractions with the precursors for the chemical industry also shows that it would be difficult to argue an inclusion in the CBAM of only those refinery fractions that are used as precursors for the chemical industry. A differentiation between different uses of the traded intermediate would be needed

⁹⁹ For the relative importance of petrochemicals compared to fuels as refinery products, only 14% (13 million barrels per day) of crude oil and 8% (300 billion cubic metres) of natural gas are used for chemical purposes (including fertilisers), as given in "The future of Petrochemicals", OECD/IEA 2018, https://www.oecd.org/en/publications/the-future-of-petrochemicals_9789264307414-en.html

¹⁰⁰ E.g. HS 271019 "Medium oils and preparations, of petroleum or bituminous minerals, not containing biodiesel, n.e.s."

(whether they serve for the production of fuels or chemicals), which is against the current architecture of the CBAM (e.g. ammonia, although listed in Annex I of the CBAM Regulation under “fertilizers” always falls under the CBAM, no matter if used as fertilizer or for other chemical reactions).

Another point, in particular in context of chemicals, is the fact that some raw materials for the organic chemicals sector are not only provided by refineries but may also be the result of coal distillation (i.e. coke production). For these, CN codes exist (under HS heading 2707) that refer to mixtures and less pure substances than usually provided by refineries. For providing a level playing field and avoiding circumvention, inclusion of these coal-based goods would be beneficial. This is discussed together with coke as precursor in section 5.2.2.

Finally, refinery products may be more and more replaced by biomass-derived fuels and other renewable fuels (e-fuels). Although they must have life-cycle emissions significantly lower than those of fossil fuels, in order to be eligible under the RED II¹⁰¹, this does not automatically mean that the embedded emissions are significantly lower than those of refinery-based fuels. For providing a level playing field for fuels, it could be considered to also include these alternative fuels and their blends with fossil fuels in the CBAM¹⁰².

Production processes and value chains

The products most significant by volume from refining are fuels (liquids and petroleum coke (petcoke)). The generic approach to the CBAM methodology is that raw materials coming directly from non-EU ETS sectors (in particular mining and oil extraction) are not considered as precursors with embedded emissions if they are used in installations. This would establish a level playing field for all fuels, i.e. coal, liquid fuels, and natural gas¹⁰³. However, the mineral oil refining sector is not a mining sector and has significant emissions in the EU ETS: In 2023 the sector emitted almost 104 Mt CO₂ or almost 10% of all stationary installations in the EU ETS¹⁰⁴. Furthermore, the sector has been identified as being at risk of carbon leakage, which means its goods could be included in the CBAM. If, however, refinery goods are included in the CBAM, they should also be considered precursors in all processes where they are used. This would have the following consequences:

- All (liquid¹⁰⁵) fuels used in CBAM installations would become “precursors,” i.e. would have to include their embedded emissions. This would add a level of

¹⁰¹ Renewable Energy Directive (recast), Directive (EU) 2018/2001, <http://data.europa.eu/eli/dir/2018/2001/2024-07-16>

¹⁰² This might concern CN codes 38260010 (Fatty-acid mono-alkyl esters, containing by weight => 96,5 % of esters “FAMAE”) and 38260090 (Biodiesel and mixtures thereof, not containing or containing < 70 % by weight of petroleum oils or oils obtained from bituminous minerals). There seem to be no other CN codes suitable to identify other fuels than biodiesel falling under the RED II. Therefore, certification (“Proof of Sustainability”) under the RED II would have to be provided for fully identifying those fuels.

¹⁰³ Coal and gas are not common inputs for refining. But by means of the Fischer-Tropsch synthesis, all carbon-containing raw materials (including biomass) are convertible to each other (this was originally used for “coal liquification”). Although there are significant losses involved, the equivalence of the raw materials in terms of “what can be produced from what” is given. Hence, if refining products were explicitly included in the CBAM, the same goods when being produced from other raw materials would have to be included, too.

¹⁰⁴ Source: EEA EU ETS data viewer: <https://www.eea.europa.eu/en/analysis/maps-and-charts/emissions-trading-viewer-1-dashboards>

¹⁰⁵ Also petcoke and liquified natural gas would be affected in a similar way.

complexity in the MRV methodology across the whole CBAM. Virtually all imported CBAM goods would have higher embedded emissions than without refineries included in the CBAM.

- Should transport emissions be included in the CBAM, these embedded emissions would have to be included in the calculation (i.e. a part of the well-to-tank emissions would have to be added to tank-to-wheel emissions).
- There might be a noticeable effect on consumer prices of transport fuels in the EU (see estimate below), on top of the already incoming ETS2.
- Embedded emissions of almost all organic chemicals and polymers would increase, because most value chains start at refinery products (see section 3.3). One advantage would be that a differentiation between chemical raw materials coming from refineries and from stand-alone chemical plants would be guaranteed to be equally treated, while without refinery products, e.g. some crackers or hydrogen production might be more difficult to monitor if the rest of the refinery would remain excluded.

The installations in the refining sector are exceptionally complex in terms of monitoring of GHG emissions. The mere monitoring of emissions at installation level is challenging as there are usually dozens of emission sources, and the fuels used in the refinery are non-standard but highly fluctuating in composition. Most refineries use process wastes and process gases as fuels. There can be several dozens of processes integrated in a single refinery, with many of them producing mixes of co-products, and many final products are blends of the interim products of the different processes. Furthermore, many refineries are integrated with processes which could also be found as stand-alone processes of the chemical industry, such as performing steam cracking, production of hydrogen, synthesis gas, aromatics, etc. Not all refineries produce the same range of goods. Some are focussing solely on fuels, while others produce inputs for the chemical industry. Some refineries produce specialties such as waxes and lubricant base oils, some produce petroleum coke (petcoke). In other words, no refinery is like the other.

For making the next step in monitoring, i.e. attributing the installation's emissions to the produced goods, to cope with the installations' complexity, it is proposed to consider the complete refinery as one single black box (one single "bubble" of a production process¹⁰⁶), and all potential products need to get the embedded emissions attributed using a single method for co-products. In this context, in an interview performed under Task 4, FuelsEurope agreed in principle with such black box approach including all refinery products, stating that a simple approach to attribution (mass balance) might be preferable over more sophisticated approaches, e.g. if extending the CWT¹⁰⁷ approach from the free allocation rules of the EU ETS.

To estimate the order of magnitude of embedded emissions of refinery products, the project team performed a short literature research and applied the methodology for attributing emissions to co-products developed in a discussion note for this project (see

¹⁰⁶ Should organic chemicals and refineries be included in the CBAM, the organic chemicals would have to fall under separate production processes.

¹⁰⁷ Complexity-Weighted Tonnes (of CO₂)

Annex 2). Data from CONCAWE¹⁰⁸ was used for the production level and product mix of European refineries, as well as the EU ETS emissions of the refining sector¹⁰⁹, and an estimated market value¹¹⁰ of the goods in the respective CN categories. Some uncertainty results from the mapping of CN codes to the product groups used by CONCAWE.

By applying a purely mass-based attribution of emissions to products, the order of magnitude of embedded emissions of *all* refinery products would be ca. 0,17...0,18 t CO₂ per tonne of product. This means that the price (before other taxes) for fossil fuels like gasoline or diesel¹¹¹ would increase by ca. 0,011€ per litre, assuming a CO₂ price of 80€/t. This is lower by orders of magnitude than the actual emissions of combusting such fuels which are in the order of 3 t CO₂ per tonne of fuel. A revenue-based attribution method would lead to more differentiated values (i.e. the CO₂ attributed can be higher or lower than this first estimate depending on the relative market price ratios of the refinery products). However, it can already be said here that the inclusion of refinery products in the CBAM would not have a significant effect on import or energy consumption patterns in the EU. It would neither be a strong driver of decarbonisation outside the EU, in particular when comparing the effect to the administrative burden of monitoring refinery emissions outside the EU and adding embedded emissions of fuels to basically every CBAM good.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: Although it may be difficult for a layman to identify if a liquid fraction from refineries e.g. has a sulphur content of > 0,001% but <= 0,002% by weight (CN 27101946), it can be assumed that refineries and importers of such goods have sufficient experience to classify these goods. The CN definitions are in principle clear for identifying the good, at least if chemical analyses or mass balance information (for mixing with biofuels and other renewable fuels) is available. Hence, the identification of the good is in principle verifiable. We conclude that no major issues regarding identification of the goods prevail.
- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): As has been described above, refinery processes in general are very complex. However, as we consider it feasible to treat a whole refinery as single black box, there would also be only one single production route to be considered overall. The exception would be the production of petrochemicals (organic chemicals) if conducted directly in the refinery, where alternative production routes may have to

¹⁰⁸ Estimating the marginal CO₂ intensities of EU refinery products

<https://www.concawe.eu/publication/estimating-the-marginal-co2-intensities-of-eu-refinery-products-report-117/>

Estimating the CO₂ intensities of EU refinery products: Statistical Regression Methodology,

<https://www.concawe.eu/publication/estimating-the-co2-intensities-of-eu-refinery-products-statistical-regression-methodology/>

¹⁰⁹ Practically all installations are under the EU ETS.

¹¹⁰ As a rough estimate, we assumed that the imported value divided by imported volume gives the average market price of a tonne of good per CN category. For this purpose, we used the three year average values collected from Eurostat for Task 1.

¹¹¹ Because of the simple mass-based attribution, the same value would apply to maritime and aviation fuels as well as to inputs to the chemical industry, such as naphtha.

be defined in addition to the refinery process. We consider the criterion therefore fulfilled.

- **Criterion 2c:** Identification of special MRV difficulties. The MRR for the EU ETS contains sufficient rules for monitoring refineries at installation level. Most of the rules are already contained in the implementing act for the transitional period of the CBAM. Some additional rules (approach to crackers, flaring) would have to be added. The monitoring in general is demanding to operators, and they may require special guidance documents, and verifiers with special qualification would be important for checking the monitoring methodology's completeness. However, this is a case of being difficult, but still possible in principle. The attribution of emissions to goods will also not be necessarily straightforward. However, if e.g. a mass-based approach is selected, it would be sufficiently simple. Hence, we consider criterion 2c not a principal obstacle to including the sector in the CBAM.

Conclusion: While refinery products would increase the overall complexity of the CBAM and its rules¹¹², there are no obstacles of technical nature that would make the inclusion of those goods unfeasible. The result (including a comparison with the outcome of Task 1) is presented in Table 16. The distinction of "intermediate" and "final product" was provided by CONCAWE. However, as discussed above, in the view of the project team, this does not establish a convincing reason for excluding some refinery products from inclusion in the CBAM.

Table 16 Comparison of the sector-wide results from Task 1 against the results of the project team under Task 2.3 and 2.4 for refinery products

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
2701 20 00	Briquettes, ovoids, and similar solid fuels manufactured from coal	(yes)	Not refinery product	no
2702 20 00	Agglomerated lignite (excl. jet)	(yes)	Not refinery product	no
2710 12 11	Light oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)	yes	intermediate	yes
2710 12 15	Light oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)	yes	intermediate	yes
2710 12 21	White spirit	yes	final product	yes
2710 12 25	Special spirits (excl. white spirit) of petroleum or bituminous minerals	yes	final product	yes
2710 12 31	Aviation spirit	yes	final product	yes
2710 12 41	Motor spirit, with a lead content <= 0,013 g/l, with a research octane number "RON" of < 95 (excl. containing biodiesel)	yes	final product	yes
2710 12 45	Motor spirit, with a lead content <= 0,013 g/l, with a research octane number "RON"	yes	final product	yes

¹¹² Their inclusion would also increase the volume of covered goods significantly (average imports 2021 to 2013 amount to ca. 267 million tonnes, corresponding to estimated embedded emissions of 50,7 Mt CO₂e).

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
	of ≥ 95 but < 98 (excl. containing biodiesel)			
2710 12 49	Motor spirit, with a lead content $\leq 0,013$ g/l, with a research octane number "RON" of ≥ 98 (excl. containing biodiesel)	yes	final product	yes
2710 12 50	Motor spirit, with a lead content $> 0,013$ g/l (excl. containing biodiesel, and aviation spirit)	yes	final product	yes
2710 12 70	Jet fuel, spirit type (excl. aviation spirit)	yes	final product	yes
2710 12 90	Light oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. containing biodiesel, for undergoing chemical transformation, and special spirits, motor spirit, and spirit type jet fuel)	yes	intermediate	yes
2710 19 11	Medium oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27	yes	intermediate	yes
2710 19 15	Medium oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27)	yes	intermediate	yes
2710 19 21	Jet fuel, kerosene type	yes	final product	yes
2710 19 25	Kerosene (excl. jet fuel)	yes	final product	yes
2710 19 29	Medium oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. for undergoing chemical transformation, and kerosene)	yes	intermediate	yes
2710 19 31	Gas oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27	yes	intermediate	yes
2710 19 35	Gas oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)	yes	intermediate	yes
2710 19 43	Gas oils of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)	yes	intermediate	yes
2710 19 46	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,002\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)	yes	intermediate	yes
2710 19 47	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,002\%$ but $\leq 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)	yes	intermediate	yes
2710 19 48	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)	yes	intermediate	yes

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
2710 19 51	Fuel oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)	yes	intermediate	yes
2710 19 55	Fuel oils obtained from bituminous materials, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)	yes	intermediate	yes
2710 19 62	Fuel oils obtained from bituminous materials, with a sulphur content of $\leq 0,1\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)	yes	intermediate	yes
2710 19 66	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,1\%$ but $\leq 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)	yes	intermediate	yes
2710 19 67	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)	yes	intermediate	yes
2710 19 71	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing a specific process as defined in Additional Note 5 to chapter 27	yes	intermediate	yes
2710 19 75	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)	yes	intermediate	yes
2710 19 81	Motor oils, compressor lube oils, and turbine lube oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 19 83	Liquids for hydraulic purposes containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 19 85	White oils, liquid paraffin containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 19 87	Gear oils and reductor oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these	yes	intermediate	yes

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
	oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)			
2710 19 91	Metalworking compounds, mould-release oils, anti-corrosion oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 19 93	Electrical insulating oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 19 99	Lubricating oils and other heavy oils and preparations n.e.s., containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)	yes	intermediate	yes
2710 20 11	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight, containing biodiesel	yes	final product	yes
2710 20 16	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,1\%$ by weight, containing biodiesel	yes	final product	yes
2710 20 19	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight, containing biodiesel	yes	final product	yes
2710 20 32	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,5\%$ by weight, containing biodiesel	yes	final product	yes
2710 20 38	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,5\%$ by weight, containing biodiesel	yes	final product	yes
2710 20 90	Oils of $\geq 70\%$ of petroleum or bituminous minerals, containing biodiesel (excl. gas oils and fuel oils)	yes	final product	yes
2711 12 11	Propane of a purity of $\geq 99\%$, for use as a power or heating fuel, liquefied	yes	final product	yes
2711 12 19	Propane of a purity of $\geq 99\%$, liquefied (excl. for use as a power or heating fuel)	yes	final product	yes
2711 12 91	Propane of a purity of $< 99\%$, liquefied, for undergoing a specific process as defined in Additional Note 5 to chapter 27	yes	final product	yes
2711 12 93	Propane of a purity of $< 99\%$, liquefied, for undergoing chemical transformation (excl. processes specified in 2711.12.91)	yes	final product	yes
2711 12 94	Liquid propane of a purity of $> 90\%$, but $< 99\%$ (excl. for undergoing chemical transformation)	yes	final product	yes

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
2711 12 97	Liquefied propane of a purity of <= 90% (excl. for undergoing chemical transformation)	yes	final product	yes
2711 13 10	Butanes for undergoing a specific process as defined in Additional Note 5 to chapter 27, liquefied (excl. of a purity of >= 95% of N-butane or isobutane)	yes	final product	yes
2711 13 30	Butanes for undergoing chemical transformation, liquefied (excl. for specific processes specified in Additional Note 5 to chapter 27 and butanes of a purity of >= 95% of N-butane or isobutane)	yes	final product	yes
2711 13 91	Liquefied butane of a purity of > 90% but < 95% (excl. for undergoing chemical transformation)	yes	final product	yes
2711 13 97	Liquefied butane of a purity of <= 90% (excl. for undergoing chemical transformation)	yes	final product	yes
2711 14 00	Ethylene, propylene, butylene and butadiene, liquefied (excl. ethylene of a purity of >= 95% and propylene, butylene, and butadiene of a purity of >= 90%)	yes	intermediate	yes
2711 19 00	Gaseous hydrocarbons, liquefied, n.e.s. (excl. natural gas, propane, butane, ethylene, propylene, butylene, and butadiene)	yes	intermediate	yes
2711 29 00	Hydrocarbons in gaseous state, n.e.s. (excl. natural gas)	yes	final product	yes
2712 10 10	Crude petroleum jelly	yes	intermediate	yes
2712 10 90	Petroleum jelly (excl. crude)	yes	intermediate	yes
2712 20 10	Synthetic paraffin wax containing < 0,75% by weight of oil and of a molecular weight of >= 460 but <= 1.560	yes	intermediate	yes
2712 20 90	Paraffin wax containing < 0,75% by weight of oil (excl. synthetic paraffin wax of a molecular weight of >= 460 but <= 1.560)	yes	intermediate	yes
2712 90 11	Crude ozokerite, lignite wax or peat wax "natural products"	yes	intermediate	yes
2712 90 19	Ozokerite, lignite wax or peat wax "natural products", whether or not coloured (excl. crude)	yes	intermediate	yes
2712 90 31	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. petroleum jelly, paraffin wax containing < 0,75% by weight of oil, ozokerite, lignite wax and peat wax)	yes	intermediate	yes
2712 90 33	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, petroleum jelly, paraffin wax containing < 0,75% by	yes	intermediate	yes

HS heading / CN code	Description (simplified)	Task 1	Reasons why Task 2 may lead to different result than Task 1, if any	Conclusion Tasks 2.3 and 2.4
2712 90 39	weight of oil, ozokerite, lignite wax and peat wax) Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes (excl. for undergoing chemical transformation, petroleum jelly, paraffin wax containing < 0,75% by weight of oil, ozokerite, lignite wax and peat wax)	yes	intermediate	yes
2712 90 91	Blend of 1-alkenes containing by weight >= 80% of 1-alkenes of a chain-length of >= 24 but <= 28 carbon atoms	yes	intermediate	yes
2712 90 99	Paraffin wax, microcrystalline petroleum wax, slack wax, ozokerite, lignite wax, peat wax, other mineral waxes, and similar products obtained by synthesis or by other processes, whether or not coloured (excl. petroleum jelly, paraffin wax containing < 0,75% by weight of oil and a blend of 1-alkenes containing by weight >= 80% of 1-alkenes of a chain-length of >= 24 but <= 28 carbon atoms)	yes	intermediate	yes
2713 11 00	Petroleum coke, non-calcined	yes	final product	yes
2713 12 00	Petroleum coke, calcined	yes	final product	yes
2713 20 00	Petroleum bitumen	yes	final product	yes
2713 90 10	Residues of petroleum oil or of oil obtained from bituminous minerals for the manufacture of carbon of heading 2803	yes	intermediate	yes
2713 90 90	Residues of petroleum oil or of oil obtained from bituminous minerals (excl. for the manufacture of carbon of heading 2803, petroleum coke and petroleum bitumen)	yes	intermediate	yes

4.4.10 Inorganic Chemicals

A discussion note on chemicals, which covers the relevant production processes and value chains, has been included as Annex 4 to this report.

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: For this purpose we have compiled in Table 17 for each substance to be assessed: The CN code and whether it is clearly related with the substance (if there are more substances covered, we have added the description of that HS heading / CN code). The CAS number is given, where possible. Where the substance is systematically mixed with other substances, a different CAS number may be necessary. We have usually given only the CAS number for the pure substance of the most technically relevant isomer. Not mentioned for individual substances in the table, but to be kept in mind in general is that some additional identification may be needed for the CBAM purpose, e.g. for substances that are traded in aqueous solutions. In such cases, the concentration of the substance will have to be reported by using "qualifying parameters" in the CBAM declaration, except where the customs rules already now require the reporting of the mass of the pure substance instead of the total mass.

As the substance names are often “trivial names” (names commonly used in industry as a kind of jargon, excluding trade names), they often do not give the full information on the structure of the substance. For such cases, we have given also systematic names according to IUPAC¹¹³. Unfortunately, even though there is usually one name that would be the “preferred” name according to IUPAC, these names are not always unique either. We have not tried to give the full list of possible names but given only one or two names that are IUPAC-preferred or at least IUPAC compatible.

- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): The relevant summary parameters of this assessment are given in Table 18. The value chains are presented in more detail in the separate discussion note on chemicals (Annex 4 of the September version). In the table, the following information is given:
 - The complexity measure calculated in accordance with section 4.4.2, first only for the last production step, and then including all precursors from the value chain.
 - The list of all precursors in the value chain, i.e. the list of substances that would have to be included in the CBAM, if the substance indicated would be included in the CBAM.
 - A list of all substances which are produced with this substance as precursor. This gives an indication for how important the substance is for other value chains assessed.
- **Criterion 2c:** Identification of special MRV difficulties: Where relevant, a comment on this criterion is included in Table 18.

Note on the tables: Sorting was done by CN code. The sorting is the same in both tables, but for efficiency, the CN codes and CAS numbers were not repeated.

¹¹³ The International Union of Pure and Applied Chemistry (IUPAC) publishes comprehensive rules for systematic nomenclature of organic substances. The current rules are:

- “Red book” (nomenclature of inorganic chemicals): <https://iupac.org/what-we-do/books/redbook/>
- “Blue book” (nomenclature of organic chemicals): <https://iupac.org/what-we-do/books/bluebook/>
- “Purple book” (for polymers): <https://iupac.org/what-we-do/books/purplebook/>

These rule books have undergone several significant revisions (e.g. blue book: 1993, 2013, 2023). Unfortunately, these rules often allow several approaches to more complex molecules, even though the rules identify a preferred approach. Furthermore, names construed according to older rule books are sometimes more “catchy” (easier to memorise) and therefore persisting in literature. Therefore, IUPAC names given in this report, may sometimes be based on older versions of IUPAC rules.

Table 17 Identification of inorganic chemicals. Where no description of the CN code is given, it is referring to the one relevant substance only. The description is given for clarity only where substance groups are covered.

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Chlorine	280110		7782-50-5	Cl ₂	Dichlorine	
Carbon black	280300		1333-86-4	-		CN code can include other forms of carbon
Hydrogen chloride	280610		7647-01-0	HCl	Hydrogen chloride	CN code: does it include gas or aqueous solution?
Hydrogen cyanide	281112		74-90-8	HCN	Formonitrile; isocyanic acid	CN code: does it include gas or aqueous solution?
Nitric oxide	28112930	Nitrogen oxides	10102-43-9	NO	Nitrogen monoxide	
Nitrogen dioxide (NO ₂)	28112930	Nitrogen oxides	10102-44-0	NO ₂	Nitrogen dioxide	
Carbon monoxide	28112990	Inorganic oxygen compounds of non-metals (excl. diphosphorus pentaoxide, oxides of boron, carbon dioxide, silicon dioxide, sulphur dioxide, sulphur trioxide "sulphuric anhydride", diarsenic trioxide and nitrogen oxides)	630-08-0	CO	Carbon monoxide	
Nitrosylsulphuric acid	28111980	Inorganic acids (excl. hydrogen chloride "hydrochloric acid", chlorosulphuric acid, sulphuric acid, oleum, nitric acid, sulphonitric acids, phosphoric acid, polyphosphoric acids, boric acids, hydrogen fluoride "hydrofluoric acid", hydrogen bromide "hydrobromic acid" and hydrogen cyanide "hydrocyanic acid")	7782-78-7	HNO ₅ S	Nitrosylsulfuric acid	
Phosgene	281211		75-44-5	CCl ₂ O	Carbonyl dichloride	
Nitrosyl chloride	28121990	Chlorides and chloride oxides (excl. of phosphorus, carbonyl dichloride "phosgene", sulphur monochloride, sulphur dichloride and thionyl chloride)	2696-92-6	NOCl	Nitrosyl chloride	
Sodium hydroxide (caustic soda)	281511 and 281512		1310-73-2	NaOH	Sodium hydroxide	CN code: solid and aqueous solution with separate CN code
Hydroxylamine	282510	Hydrazine and hydroxylamine and their inorganic salts	7803-49-8	H ₃ NO	Hydroxylamine; Azinous acid	
Soda ash	283620		497-19-8	CNa ₂ O ₃	Disodium carbonate; Sodium carbonate	
Sodium bicarbonate	283630		144-55-8	CHNaO ₃	Sodium hydrogen carbonate; sodium hydrogencarbonate	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Hydrogen peroxide	284700	Hydrogen peroxide, whether or not solidified with urea	7722-84-1	H2O2	Peroxol; hydrogen peroxide	
Calcium Carbide	28491000	Carbides of calcium, whether or not chemically defined	75-20-7	CaC2	Calcium Acetylide	

Table 18 Information on the complexity of value chains for the inorganic chemicals assessed

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Chlorine	1	1	0		20	Chlorine, Hydrogen chloride, Phosgene, Ethylene dichloride, Vinylchloride, Propylene oxide, Propylene glycol, Dimethyl carbonate, Diphenyl carbonate, Toluene diisocyanate, Methylene diphenyl diisocyanate, Nitrosyl chloride, ε-Caprolactam, PVC, PA-6, PPG, PU (MDI and PEG), PU (TDI and PEG), PU (MDI and PPG), PU (TDI and PPG), PC from BPA
Carbon black	0	0	0		0	Carbon black
Hydrogen chloride	7	8	1	Chlorine,	6	Hydrogen chloride, Ethylene dichloride, Vinylchloride, Nitrosyl chloride, ε-Caprolactam, PVC, PA-6
Hydrogen cyanide	11	38	6	, Syngas, Naphtha, Propane, Ethane, Propylene, Methanol	6	Hydrogen cyanide, Hexamethylenediamine, Nylon salt, Adiponitrile, Methyl methacrylate, PA-6,6, PMMA
Nitric oxide	7	7	0		4	Nitric oxide, Hydroxylamine, Nitrosyl chloride, ε-Caprolactam, PA-6
Nitrogen dioxide	7	7	0		3	Nitrogen dioxide, Nitrosyl chloride, ε-Caprolactam, PA-6
Carbon monoxide	14	21	1	Syngas,	16	Carbon monoxide, Ethylene glycol, Phosgene, Dimethyl carbonate, Diphenyl carbonate, Sodium formate, Formic acid, Toluene diisocyanate, Methylene diphenyl diisocyanate, Methyl methacrylate, PET, PMMA, PU (MDI and PEG), PU (TDI and PEG), PU (MDI and PPG), PU (TDI and PPG), PC from BPA
Nitrosylsulphuric acid	1	1	0		3	Nitrosylsulphuric acid, Nitrosyl chloride, ε-Caprolactam, PA-6
Phosgene	7	29	3	Chlorine, Syngas, Carbon monoxide,	9	Phosgene, Dimethyl carbonate, Diphenyl carbonate, Toluene diisocyanate, Methylene diphenyl diisocyanate, PU (MDI and PEG), PU (TDI and PEG), PU (MDI and PPG), PU (TDI and PPG), PC from BPA
Nitrosyl chloride	14	38	5	Chlorine, Hydrogen chloride, Nitric oxide, Nitrogen dioxide, Nitrosylsulphuric acid,	2	Nitrosyl chloride, ε-Caprolactam, PA-6

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Sodium hydroxide (caustic soda)	1	1	0		9	Sodium hydroxide, Soda ash, Sodium bicarbonate, Propylene oxide, Propylene glycol, Sodium formate, Formic acid, PPG, PU (MDI and PPG), PU (TDI and PPG)
Hydroxylamine	8	15	1	Nitric oxide,	2	Hydroxylamine, ε-Caprolactam, PA-6
Soda ash	10	11	1	Sodium hydroxide,	1	Soda ash, Sodium bicarbonate
Sodium bicarbonate	1	12	2	Sodium hydroxide, Soda ash,	0	Sodium bicarbonate
Hydrogen peroxide	5	5	0		8	Hydrogen peroxide, Propylene oxide, Propylene glycol, Adipic acid, Nylon salt, PA-6,6, PPG, PU (MDI and PPG), PU (TDI and PPG)
Calcium Carbide	0	0	0		3	Calcium Carbide, Acetylene, Vinylchloride, PVC

4.4.11 Organic chemicals

A discussion note on chemicals, which covers the relevant production processes and value chains, has been included as Annex 4 to this report.

The results of the assessment of organic chemicals (including the precursors for the polymers discussed in section 4.4.12) are presented in the same way as for inorganic chemicals: are presented in the same way as for inorganic chemicals:

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: For this purpose we have compiled in Table 19 for each substance to be assessed: The CN code and whether it is clearly related with the substance (if there are more substances covered, we have added the description of that HS heading / CN code). The CAS number is given, where possible. Note that sometimes CAS numbers are different for existing isomers¹¹⁴ and enantiomers¹¹⁵. Also where the substance is systematically mixed with other substances, a different CAS number may be necessary. We have usually given only the CAS number for the pure substance of the most technically relevant isomer.

Not mentioned for individual substances in the table, but to be kept in mind in general is that some additional identification may be needed for the CBAM purpose, e.g. for substances that are traded in aqueous solutions. In such cases, the concentration of the substance will have to be reported by using "qualifying parameters" in the CBAM declaration, except where the customs rules already now require the reporting of the mass of the pure substance instead of the total mass.

As the substance names are often "trivial names" (names commonly used in industry as a kind of jargon, excluding trade names), they often do not give the full information on the structure of the substance. For such cases, we have given also systematic names according to IUPAC. Unfortunately, even though there is usually one name that would be the "preferred" name according to IUPAC, these names are not always unique either. We have not tried to give the full list of possible names but given only one or two names that are IUPAC-preferred or at least IUPAC compatible.

- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): The relevant summary parameters of this assessment are given in Table 20. The value chains are presented in more detail in the separate discussion note on chemicals (Annex 4). In the table, the following information is given:
 - The complexity measure calculated in accordance with section 4.4.2, first only for the last production step, and then including all precursors from the value chain.
 - The list of all precursors in the value chain, i.e. the list of substances that would have to be included in the CBAM, if the substance indicated would be included in the CBAM.
 - A list of all substances which are produced with this substance as precursor. This gives an indication for how important the substance is for other value chains assessed.
- **Criterion 2c:** Identification of special MRV difficulties: Where relevant, a comment on this criterion is included in Table 20.

¹¹⁴ Substances with the same formula, but different structure of the molecule.

¹¹⁵ Asymmetrical substances with the same structural formula, but different mirror-type geometry.

Note on the tables: Sorting was done by CN code. The sorting is the same in both tables, but for efficiency, the CN codes and CAS numbers were not repeated.

Some of the precursor which need separate treatment (e.g. because they are probably not traded, or because they belong to other sectors (refinery, coking), are given in a separate table (Table 21).

Table 19 Identification of organic chemicals. Where no description of the CN code is given, it is referring to the one relevant substance only. The description is given for clarity only where substance groups are covered.

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Ethanol (including Bioethanol)	220710 and 220720		64-17-5	C ₂ H ₆ O	Ethanol	
Ethane	290110	Saturated acyclic hydrocarbons	74-84-0	C ₂ H ₆	Ethane	
Propane	271112		74-98-6	C ₃ H ₈	Propane	HS code for liquefied Propane, CN codes with different purity
Butane	290110	Saturated acyclic hydrocarbons	106-97-8	C ₄ H ₁₀	Butane	"Crude" substance would fall under CN 271113: Butanes, liquefied (excl. of a purity of >= 95% of N-butane or isobutane)
Isobutane	290110	Saturated acyclic hydrocarbons	75-28-5	C ₄ H ₁₀	2-Methylpropane	
Ethylene	290121		74-85-1	C ₂ H ₄	Ethene	
Propylene	290122		115-07-1	C ₃ H ₆	Prop-1-ene; propene	
Butadiene	290124	Buta-1,3-diene, and isoprene	106-99-0	C ₄ H ₆	Buta-1,3-diene	
Acetylene	290129	Hydrocarbons, acyclic, unsaturated (excl. ethylene, propene "propylene", butene "butylene" and isomers thereof and Buta-1,3-diene, and isoprene)	74-86-2	C ₂ H ₂	Acetylene	
Cyclohexane	290211		110-82-7	C ₆ H ₁₂	Cyclohexane	
Cyclohexene	290219	Cyclanes, cyclenes and cycloterpenes (excl. cyclohexane)	110-83-8	C ₆ H ₁₀	Cyclohexene	
Benzene	290220		71-43-2	C ₆ H ₆	Benzene	
Toluene	290230		108-88-3	C ₇ H ₈	Toluene	
Ortho-xylene (o-Xylene)	290241		95-47-6	C ₈ H ₁₀	1,2-xylene; 1,2-dimethylbenzene	
Meta-xylene (m-Xylene)	290242		108-38-3	C ₈ H ₁₀	1,3-xylene; 1,3-dimethylbenzene	
Para-xylene (p-Xylene)	290243		106-42-3	C ₈ H ₁₀	1,4-xylene; 1,4-dimethylbenzene	
Xylenes (mixture)	290244		1330-20-7	C ₈ H ₁₀	Dimethylbenzene	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Styrene	290250		100-42-5	C ₈ H ₈	Ethenylbenzene	
Ethylbenzene	290260		100-41-3	C ₈ H ₁₀	Ethylbenzene	
Cumene	290270		98-82-8	C ₉ H ₁₂	(Propan-2-yl)benzene	
α-methyl styrene	290290	Cyclic hydrocarbons (excl. cyclanes, cyclenes, benzene, toluene, xylenes, styrene, ethylbenzene and cumene)	98-83-9	C ₉ H ₁₀	(Prop-1-en-2-yl)benzene; 2-phenylpropene	
Naphthalene	290290	Cyclic hydrocarbons (excl. cyclanes, cyclenes, benzene, toluene, xylenes, styrene, ethylbenzene and cumene)	91-20-3	C ₁₀ H ₈	Naphthalene	CN refers to pure substance; crude from coking: 270740
Cyclohexylbenzene	29029000	Cyclic hydrocarbons (excl. cyclanes, cyclenes, benzene, toluene, xylenes, styrene, ethylbenzene and cumene)	827-52-1	C ₁₂ H ₁₆	Cyclohexylbenzene, Phenylcyclohexane	
Ethylene dichloride (EDC)	290315		107-06-2	C ₂ H ₄ Cl ₂	1,2-Dichloroethane	
Vinylchloride monomer	290321		75-01-4	C ₂ H ₃ Cl	Chloroethene	
Dinitrotoluene (DNT)	29042000	Derivatives of hydrocarbons containing only nitro or nitroso groups	121-14-2	C ₇ H ₆ N ₂ O ₄	1-Methyl-2,4-dinitrobenzene; '2,4 dinitrotoluene	
Nitrobenzene	29042000	Derivatives of hydrocarbons containing only nitro or nitroso groups	98-95-3	C ₆ H ₅ NO ₂	Nitrobenzene	
Methanol	290511		67-56-1	CH ₃ OH	Methanol	
Monoethylene glycol (MEG)	290531		107-21-1	C ₂ H ₆ O ₂	Ethane-1,2-diol; ethylene glycol	
Propylene glycol	29053200		57-55-6	C ₃ H ₈ O ₂	Propane-1,2-diol	CAS No refers to the racemic mixture
Propylene chlorohydrin	29055998	Halogenated, sulphonated, nitrated or nitrosated derivatives of acyclic alcohols (excl. 2,2-bis(bromomethyl)propanediol and ethchlorvynol "INN")	78-89-7	C ₃ H ₇ ClO	2-Chloropropan-1-ol	
Cyclohexanol	290612	Cyclohexanol, methylcyclohexanols and dimethylcyclohexanols	108-93-0	C ₆ H ₁₂ O	Cyclohexanol	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
α-Methyl-Benzyl alcohol	290629	Aromatic cyclic alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. benzyl alcohol)	98-85-1	C8H10O	1-Phenylethan-1-ol; 1-phenylethanol	
Phenol	290711	Phenol "hydroxybenzene" and its salts	108-95-2	C6H6O	Phenol; Benzenol	
Bisphenol-A	290723	4,4'-Isopropylidenediphenol "bisphenol A, diphenylolpropane" and its salts	80-05-7	C15H16O2	4-[2-(4-Hydroxyphenyl)propan-2-yl]phenol; 4,4'-(Propane-2,2-diyl)diphenol	
Dimethyl ether (DME)	29091990	Acyclic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. diethyl ether and tert-butyl ethyl ether [ethyl-tertio-butyl-ether, ETBE])	115-10-6	C2H6O	Methoxymethane	
Diethylene glycol (DEG)	290941		111-46-6	C4H10O3	2-(2-hydroxyethoxy)ethan-1-ol; 1,1'-oxydiethanol	
Triethylene glycol (TEG)	29094980	Ether-alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. 2,2'-Oxydiethanol "diethylene glycol, digol", 2-(2-Chloroethoxy)ethanol and monoalkylethers of ethylene glycol or of diethylene glycol)	112-27-6	C6H14O4	2-[2-(2-hydroxyethoxy)ethoxy]ethan-1-ol; 2,2'-[Ethane-1,2-diylbis(oxy)]di(ethan-1-ol)	
Cumene hydroperoxide	29096090	Alcohol peroxides, ether peroxides, ketone peroxides and their halogenated, sulphonated, nitrated or nitrosated derivatives	80-15-9	C9H12O2	2-Phenylpropane-2-peroxol; α,α-dimethylbenzyl hydroperoxide	
Ethylbenzene hydroperoxide	29096090	Alcohol peroxides, ether peroxides, ketone peroxides and their halogenated, sulphonated, nitrated or nitrosated derivatives	3071-32-7	C8H10O2	1-Phenylethane-1-peroxol; 1-hydroperoxyethylbenzene	
Tert-butyl hydroperoxide	29096090	Alcohol peroxides, ether peroxides, ketone peroxides and their halogenated, sulphonated, nitrated or nitrosated derivatives	75-91-2	C4H10O2	2-Methylpropane-2-peroxol	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Ethylene oxide	291010		75-21-8	C ₂ H ₄ O	Oxirane; epoxyethane; oxacyclopropane	
Propylene oxide	291020		75-56-9	C ₃ H ₆ O	2-Methyloxirane	
Formaldehyde	291211		50-00-0	CH ₂ O	Formaldehyde	
Acetone	291411		67-64-1	C ₃ H ₆ O	Propan-2-one; Acetone	
Cyclohexanone	291422	Cyclohexanone and methylcyclohexanones	108-94-1	C ₆ H ₁₀ O	Cyclohexanone	
KA-Oil	not traded(?)		-	-	Mixture of Cyclohexanone and cyclohexanol	Needed for MRV and complexity calculation only
Acetophenone	291439	Ketones, aromatic, without other oxygen function (excl. phenylacetone [phenylpropan-2-one])	98-86-2	C ₈ H ₈ O	1-Phenylethan-1-one; 1-phenylethanone	
Formic acid	291511		64-18-6	CH ₂ O ₂	Formic acid	
Sodium formate	291512	Salts of formic acid	141-53-7	CHO ₂ Na	Sodium methanoate	
Methyl formate	291513	Esters of formic acid	107-31-3	C ₂ H ₄ O ₂	Methyl formate	
MMA (Methyl methacrylate)	291614	Esters of methacrylic acid	80-62-6	C ₅ H ₈ O ₂	Methyl 2-methylprop-2-enoate	
Methyl p-Toluate	29163990	Aromatic monocarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. benzoic acid, its salts and esters, benzoyl peroxide, benzoyl chloride, binapacryl [ISO], phenylacetic acid, its salts and esters, and inorganic or organic compounds of mercury whether or not chemically defined)	99-75-2	C ₉ H ₁₀ O ₂	Methyl 4-methylbenzoate	
Adipic acid	291712	Adipic acid, its salts and esters	124-04-9	C ₆ H ₁₀ O ₄	Hexanedioic acid	ECICS lists 41 substances under this CN code
Phthalic anhydride	291735		85-44-9	C ₈ H ₄ O ₃	2-Benzofuran-1,3-dione	
Terephthalic acid (PTA)	291736	Terephthalic acid and its salts	100-21-0	C ₈ H ₆ O ₄	Benzene-1,4-dicarboxylic acid	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Dimethyl-terephthalate	291737	Dimethyl terephthalate	120-61-6	C10H10O4	1,4-Dimethyl benzene-1,4-dicarboxylate	
Dimethyl carbonate (DMC)	29209010	Sulphuric esters and carbonic esters and their salts, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. inorganic or organic compounds of mercury)	616-38-6	C3H6O3	Dimethyl carbonate	
Diphenyl carbonate (DPC)	29209010	Sulphuric esters and carbonic esters and their salts, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. inorganic or organic compounds of mercury)	102-09-0	C13H10O3	Diphenyl carbonate	
Ethylene carbonate	29209010	Sulphuric esters and carbonic esters and their salts, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. inorganic or organic compounds of mercury)	96-49-1	C3H4O3	1,3-dioxolan-2-one	
Hexamethylenediamine (HMDA)	292122	Hexamethylenediamine and its salts	124-09-4	C6H16N2	Hexane-1,6-diamine	
Nylon salt	292122	Hexamethylenediamine and its salts	3323-53-2	-	Adipic acid—hexane-1,6-diamine (1:1)	Nylon salt could also fit under 291712 (Adipic acid and its salts and esters), but it is found in ECICS under this CN code
Aniline	292141	Aniline and its salts (excl. inorganic or organic compounds of mercury)	62-53-3	C6H7N	Aniline	
Toluene Diamine (TDA)	29215119	o-Phenylenediamine, m-phenylenediamine, p-phenylenediamine, diaminotoluenes, and their halogenated, sulphonated, nitrated and nitrosated derivatives; salts thereof (excl. m-phenylenediamine of a purity by weight of >= 99% and containing	95-80-7	C7H10N2	4-Methylbenzene-1,3-diamine	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Methylenedianiline (MDA)	29215990	<= 1% by weight of water, <= 200 mg/kg of o-phenylenediamine and <= 450 mg/kg of p-phenylenediamine) Aromatic polyamines and their derivatives; salts thereof (excl. o-phenylenediamine, m-phenylenediamine, p-phenylenediamine or diaminotoluenes and their derivatives, and salts thereof, m-phenylenebis"methylamine", 2,2'-dichloro-4,4'-methylenedianiline, 4,4'-bi-o-toluidine and 1,8-naphthylenediamine)	101-77-9	C13H14N2	4,4'-Methylenedianiline	
Acrylamide	292419	Acyclic amides, incl. acyclic carbamates, and their derivatives, and salts thereof (excl. meprobamate [INN], fluoroacetamide [ISO], monocrotophos [ISO] and phosphamidon [ISO])	79-06-1	C3H5NO	Prop-2-enamide	
Formamide	292419	Acyclic amides, incl. acyclic carbamates, and their derivatives, and salts thereof (excl. meprobamate [INN], fluoroacetamide [ISO], monocrotophos [ISO] and phosphamidon [ISO])	75-12-7	CH3NO	Formamide; methanamide	
Acrylonitrile	292610		107-13-1	C3H3N	Prop-2-enenitrile	
Adiponitrile	29269070	Nitrile-function compounds (excl. acrylonitrile, 1-cyanoguanidine "dicyandiamide", fenproporex "INN" and its salts, methadone "INN"-intermediate "4-cyano-2-dimethylamino-4,4-diphenylbutane", alpha-Phenylacetoacetonitrile and isophthalonitrile)	111-69-3	C6H8N2	Hexanedinitrile	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
Methylene diphenyl diisocyanate (MDI)	292910	Isocyanates	101-68-8	C15H10N2O2	1,1'-Methylenebis(4-isocyanatobenzene)	
Toluene diisocyanate (TDI)	292910	Isocyanates	584-84-9	C9H6N2O2	2,4-Diisocyanate-1-methylbenzene	
Melamine	293361		108-78-1	C3H6N6	1,3,5-Triazine-2,4,6-triamine	
ε-Caprolactam	293371		105-60-2	C6H11NO	Azepan-2-one	

Table 20 Information on the complexity of value chains for the organic chemicals assessed

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Ethanol (incl. Bioethanol)	6	30	6	Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	0	
Propane	1	1	0		38	Hydrogen cyanide; Propylene; Cyclohexanone; Cyclohexanol; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Diphenyl carbonate; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; ε-Caprolactam; Methyl methacrylate; PP; GPPS and HIPS; EPS; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Ethane	1	1	0		60	Hydrogen cyanide; Ethylene; Propylene; Butadiene; Acetylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Ethylene

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
						dichloride; Vinylchloride; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Diphenyl carbonate; Ethanol; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; ε-Caprolactam; Methyl methacrylate; LLDPE; LDPE; HDPE; PP; Polybutadien; GPPS and HIPS; EPS; PVC; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Butane	1	1	0		16	Isobutane; Butadiene; Tert-butyl hydroperoxide; Propylene oxide; Propylene glycol; Hexamethylenediamine; Nylon salt; Adiponitrile; Methyl methacrylate; Polybutadien; PA-6,6; ABS; PMMA; PPG; PU (MDI and PPG); PU (TDI and PPG)
Isobutane	4	5	1	Butane	15	Butadiene; Tert-butyl hydroperoxide; Propylene oxide; Propylene glycol; Hexamethylenediamine; Nylon salt; Adiponitrile; Methyl methacrylate; Polybutadien; PA-6,6; ABS; PMMA; PPG; PU (MDI and PPG); PU (TDI and PPG)
Ethylene	12	24	5	Syngas; Naphtha; Ethane; Methanol; Dimethyl ether	32	Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Ethylene dichloride; Vinylchloride; Propylene oxide; Propylene glycol; Ethanol; Methyl methacrylate; LLDPE; LDPE; HDPE; GPPS and HIPS; EPS; PVC; PET; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG)
Propylene	15	28	6	Syngas; Naphtha; Propane; Ethane; Methanol; Dimethyl ether	37	Hydrogen cyanide; Cyclohexanone; Cyclohexanol; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene chlorohydrin;

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
						Propylene oxide; Propylene glycol; Diphenyl carbonate; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; ε-Caprolactam; Methyl methacrylate; PP; GPPS and HIPS; EPS; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Butadiene	13	22	4	Naphtha; Ethane; Butane; Isobutane	6	Hexamethylenediamine; Nylon salt; Adiponitrile; Polybutadien; PA-6,6; ABS
Acetylene	11	14	3	Calcium Carbide; Naphtha; Ethane	2	Vinylchloride; PVC
Cyclohexane	7	63	6	Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene	8	Cyclohexanone; Cyclohexanol; Adipic acid; KA-Oil; Nylon salt; ε-Caprolactam; PA-6; PA-6,6
Cyclohexene	7	63	6	Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene	13	Cyclohexanone; Cyclohexanol; Phenol; Bisphenol-A; Diphenyl carbonate; Cyclohexylbenzene; Adipic acid; KA-Oil; Nylon salt; ε-Caprolactam; PA-6; PA-6,6; PC from BPA
Benzene	22	56	5	Reformate; Naphtha; Pygas; Light oil; Toluene	40	Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene oxide; Propylene glycol; Diphenyl carbonate; Cyclohexylbenzene; Adipic acid; KA-Oil; Nylon salt; Nitrobenzene; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Toluene	10	21	4	Reformate; Naphtha; Pygas; Light oil	45	Benzene; Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
						oxide; Propylene glycol; Diphenyl carbonate; Cyclohexylbenzene; Adipic acid; KA-Oil; Nylon salt; Nitrobenzene; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Ortho-xylene (o-Xylene)	1	22	5	Reformate; Naphtha; Pygas; Light oil; Xylenes	1	Phthalic anhydride
Meta-xylene (m-Xylene)	1	22	5	Reformate; Naphtha; Pygas; Light oil; Xylenes	0	
Para-xylene (p-Xylene)	1	22	5	Reformate; Naphtha; Pygas; Light oil; Xylenes	3	Terephthalic acid; Dimethyl-terephthalate; PET
Xylenes (mixture)	10	21	4	Reformate; Naphtha; Pygas; Light oil	8	Ortho-xylene; Meta-xylene; Para-xylene; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Phthalic anhydride; PET
Styrene	5	486	18	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Cumene; Methanol; Dimethyl ether	4	GPPS and HIPS; EPS; SAN; ABS
Ethylbenzene	2	82	11	Syngas; Reformate; Naphtha; Pygas; Ethane; Light oil; Ethylene; Benzene; Toluene; Methanol; Dimethyl ether	13	Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Propylene oxide; Propylene glycol; GPPS and HIPS; EPS; SAN; ABS; PPG; PU (MDI and PPG); PU (TDI and PPG)
Cumene	8	92	12	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Methanol; Dimethyl ether	29	Cyclohexanone; Cyclohexanol; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene oxide; Propylene glycol; Diphenyl carbonate; Adipic acid; KA-Oil; Nylon salt; ε-

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
						Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PPG); PU (TDI and PPG); PC from BPA
a-methyl styrene	2	94	13	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cumene; Methanol; Dimethyl ether	0	
Naphthalene	7	8	1	Coal tar	1	Phthalic anhydride
Cyclohexylbenzene	2	121	7	Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene; Cyclohexene	12	Cyclohexanone; Cyclohexanol; Phenol; Bisphenol-A; Diphenyl carbonate; Adipic acid; KA-Oil; Nylon salt; ε-Caprolactam; PA-6; PA-6,6; PC from BPA
Ethylene dichloride (EDC)	6	39	8	Chlorine; Hydrogen chloride; Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	2	Vinylchloride; PVC
Vinylchloride (VCM)	6	67	11	Chlorine; Hydrogen chloride; Calcium Carbide; Syngas; Naphtha; Ethane; Ethylene; Acetylene; Ethylene dichloride; Methanol; Dimethyl ether	1	PVC
Dinitrotoluene (DNT)	2	23	5	Reformate; Naphtha; Pygas; Light oil; Toluene	4	Toluene Diamine; Toluene diisocyanate; PU (TDI and PEG); PU (TDI and PPG)
Nitrobenzene	2	58	6	Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene	5	Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; PU (MDI and PEG); PU (MDI and PPG)
Methanol	1	8	1	Syngas	69	Hydrogen cyanide; Ethylene; Propylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Ethylene dichloride; Vinylchloride; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Methyl p-toluate; Dimethyl-terephthalate; Dimethyl carbonate; Diphenyl

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
						carbonate; Dimethyl ether; Formaldehyde; Methyl formate; Formamide; Formic acid; Ethanol; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; LLDPE; LDPE; HDPE; PP; GPPS and HIPS; EPS; PVC; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA; UF resin; MF resin
Monoethylene glycol (MEG)	8	104	9	Syngas; Carbon monoxide; Naphtha; Ethane; Ethylene; Ethylene oxide; Ethylene carbonate; Methanol; Dimethyl ether	1	PET
Propylene glycol (PG)	7	502	26	Chlorine; Sodium hydroxide; Hydrogen peroxide; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Cumene; Propylene chlorohydrin; Tert-butyl hydroperoxide; Propylene oxide; Methanol; Dimethyl ether	0	
Propylene chlorohydrin	2	31	8	Chlorine; Syngas; Naphtha; Propane; Ethane; Propylene; Methanol; Dimethyl ether	5	Propylene oxide; Propylene glycol; PPG; PU (MDI and PPG); PU (TDI and PPG)
Cyclohexanol	5	526	19	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; KA-Oil	0	
α-Methyl-benzyl alcohol	16	399	17	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene	10	Styrene; Propylene oxide; Propylene glycol; GPPS and HIPS; EPS; SAN; ABS; PPG; PU (MDI and PPG); PU (TDI and PPG)

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Phenol	10	225	16	hydroperoxide; Acetophenone; Cumene; Methanol; Dimethyl ether Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexene; Cumene; Cumene hydroperoxide; Cyclohexylbenzene; Methanol; Dimethyl ether	11	Cyclohexanone; Cyclohexanol; Bisphenol-A; Diphenyl carbonate; Adipic acid; KA-Oil; Nylon salt; ε-Caprolactam; PA-6; PA-6,6; PC from BPA
Bisphenol-A (BPA)	3	329	18	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Acetone; Cyclohexylbenzene; Methanol; Dimethyl ether	1	PC from BPA
Dimethyl ether (DME)	1	9	2	Syngas; Methanol	57	Hydrogen cyanide; Ethylene; Propylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Ethylene dichloride; Vinylchloride; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Diphenyl carbonate; Ethanol; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; ε-Caprolactam; Methyl methacrylate; LLDPE; LDPE; HDPE; PP; GPPS and HIPS; EPS; PVC; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Diethylene glycol (DEG)	2	33	7	Syngas; Naphtha; Ethane; Ethylene; Ethylene oxide; Methanol; Dimethyl ether	0	
Triethylene glycol (TEG)	2	33	7	Syngas; Naphtha; Ethane; Ethylene; Ethylene oxide; Methanol; Dimethyl ether	0	
Cumene hydroperoxide	2	94	13	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene;	15	Cyclohexanone; Cyclohexanol; Phenol; Acetone; Bisphenol-A; Diphenyl carbonate;

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
				Benzene; Toluene; Cumene; Methanol; Dimethyl ether		Adipic acid; KA-Oil; Nylon salt; ε-Caprolactam; Methyl methacrylate; PA-6; PA-6,6; PMMA; PC from BPA
Ethylbenzene hydroperoxide	2	84	12	Syngas; Reformate; Naphtha; Pygas; Ethane; Light oil; Ethylene; Benzene; Toluene; Ethylbenzene; Methanol; Dimethyl ether	12	Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Propylene oxide; Propylene glycol; GPPS and HIPS; EPS; SAN; ABS; PPG; PU (MDI and PPG); PU (TDI and PPG)
Tert-butyl hydroperoxide	6	11	2	Butane; Isobutane	5	Propylene oxide; Propylene glycol; PPG; PU (MDI and PPG); PU (TDI and PPG)
Ethylene oxide (EO)	7	31	6	Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	8	Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; PET; PEG; PU (MDI and PEG); PU (TDI and PEG)
Propylene oxide (PO)	20	495	25	Chlorine; Sodium hydroxide; Hydrogen peroxide; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Cumene; Propylene chlorohydrin; Tert-butyl hydroperoxide; Methanol; Dimethyl ether	4	Propylene glycol; PPG; PU (MDI and PPG); PU (TDI and PPG)
Formaldehyde	2	10	2	Syngas; Methanol	8	Methylenedianiline; Methylene diphenyl diisocyanate; Methyl methacrylate; PMMA; PU (MDI and PEG); PU (MDI and PPG); UF resin; MF resin
Acetone	7	101	14	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cumene; Cumene hydroperoxide; Methanol; Dimethyl ether	4	Bisphenol-A; Methyl methacrylate; PMMA; PC from BPA
Cyclohexanone	13	655	19	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; KA-Oil	2	ε-Caprolactam; PA-6

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
KA-Oil	8	296	18	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether	7	Cyclohexanone; Cyclohexanol; Adipic acid; Nylon salt; ε-Caprolactam; PA-6; PA-6,6
Acetophenone	13	271	16	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Cumene; Methanol; Dimethyl ether	11	α-Methyl-Benzyl alcohol; Styrene; Propylene oxide; Propylene glycol; GPPS and HIPS; EPS; SAN; ABS; PPG; PU (MDI and PPG); PU (TDI and PPG)
Formic acid	13	139	7	Sodium hydroxide; Syngas; Carbon monoxide; Methanol; Methyl formate; Sodium formate; Formamide	0	
Sodium formate	7	29	3	Sodium hydroxide; Syngas; Carbon monoxide	1	Formic acid
Methyl formate	7	36	3	Syngas; Carbon monoxide; Methanol	2	Formamide; Formic acid
Methyl methacrylate (MMA)	26	262	21	Hydrogen cyanide; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Cumene; Cumene hydroperoxide; Acetone; Methanol; Dimethyl ether; Formaldehyde	1	PMMA
Methyl p-toluate	7	36	7	Syngas; Reformate; Naphtha; Pygas; Light oil; Xylenes; Methanol	2	Dimethyl-terephthalate; PET
Adipic acid	11	375	20	Hydrogen peroxide; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; KA-Oil	2	Nylon salt; PA-6,6

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Phthalic anhydride	4	34	8	Reformate; Naphtha; Pygas; Light oil; Xylenes; Ortho-xylene; Coal tar; Naphthalene	0	
Terephthalic acid (PTA)	2	24	6	Reformate; Naphtha; Pygas; Light oil; Xylenes; Para-xylene	2	Dimethyl-terephthalate; PET
Dimethyl-terephthalate (DMT)	4	72	10	Syngas; Reformate; Naphtha; Pygas; Light oil; Xylenes; Para-xylene; Terephthalic acid; Methyl p-toluate; Methanol	1	PET
Dimethyl carbonate (DMC)	9	67	5	Chlorine; Syngas; Carbon monoxide; Phosgene; Methanol	2	Diphenyl carbonate; PC from BPA
Diphenyl carbonate (DPC)	11	353	21	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Phosgene; Dimethyl carbonate; Cyclohexylbenzene; Methanol; Dimethyl ether	1	PC from BPA
Ethylene carbonate	6	37	7	Syngas; Naphtha; Ethane; Ethylene; Ethylene oxide; Methanol; Dimethyl ether	2	Ethylene glycol; PET
Hexamethylenediamine (HMDA)	2	99	13	Hydrogen cyanide; Syngas; Naphtha; Propane; Ethane; Butane; Isobutane; Propylene; Butadiene; Methanol; Dimethyl ether; Acrylonitrile; Adiponitrile	2	Nylon salt; PA-6,6
Nylon salt	2	476	28	Hydrogen cyanide; Hydrogen peroxide; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Propylene; Butadiene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; Adipic acid; KA-Oil; Hexamethylenediamine; Acrylonitrile; Adiponitrile	1	PA-6,6
Aniline	6	64	7	Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene; Nitrobenzene	4	Methylenedianiline; Methylene diphenyl diisocyanate; PU (MDI and PEG); PU (MDI and PPG)

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Toluene Diamine (TDA)	1	24	6	Reformate; Naphtha; Pygas; Light oil; Toluene; Dinitrotoluene	3	Toluene diisocyanate; PU (TDI and PEG); PU (TDI and PPG)
Methylenedianiline (MDA)	2	76	11	Syngas; Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene; Methanol; Formaldehyde; Nitrobenzene; Aniline	3	Methylene diphenyl diisocyanate; PU (MDI and PEG); PU (MDI and PPG)
Acrylamide	1	31	8	Syngas; Naphtha; Propane; Ethane; Propylene; Methanol; Dimethyl ether; Acrylonitrile	0	
Formamide	4	61	4	Syngas; Carbon monoxide; Methanol; Methyl formate	1	Formic acid
Acrylonitrile	2	30	7	Syngas; Naphtha; Propane; Ethane; Propylene; Methanol; Dimethyl ether	7	Hexamethylenediamine; Nylon salt; Acrylamide; Adiponitrile; PA-6,6; SAN; ABS
Adiponitrile	6	97	12	Hydrogen cyanide; Syngas; Naphtha; Propane; Ethane; Butane; Isobutane; Propylene; Butadiene; Methanol; Dimethyl ether; Acrylonitrile	3	Hexamethylenediamine; Nylon salt; PA-6,6
Methylene diphenyl diisocyanate (MDI)	8	113	15	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Pygas; Light oil; Benzene; Toluene; Phosgene; Methanol; Formaldehyde; Nitrobenzene; Aniline; Methylenedianiline	2	PU (MDI and PEG); PU (MDI and PPG)
Toluene diisocyanate (TDI)	8	61	11	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Pygas; Light oil; Toluene; Phosgene; Dinitrotoluene; Toluene Diamine	2	PU (TDI and PEG); PU (TDI and PPG)
Melamine	1	1	0		1	MF resin
ε-Caprolactam	9	788	27	Chlorine; Hydrogen chloride; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cyclohexanone; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; KA-Oil; Nitric oxide; Hydroxylamine; Nitrogen dioxide; Nitrosylsulphuric acid; Nitrosyl chloride	1	PA-6

Table 21 Information on the complexity for some precursors needed in the assessment of organic chemicals

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Coal tar	1	1	0		2	Naphthalene; Phthalic anhydride
Light oil from coking	1	1	0		55	Benzene; Toluene; Xylenes; Ortho-xylene; Meta-xylene; Para-xylene; Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene oxide; Propylene glycol; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Diphenyl carbonate; Cyclohexylbenzene; Phthalic anhydride; Adipic acid; KA-Oil; Nylon salt; Nitrobenzene; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Naphtha	2	2	0		81	Hydrogen cyanide; Pygas; Ethylene; Propylene; Butadiene; Acetylene; Benzene; Toluene; Xylenes; Ortho-xylene; Meta-xylene; Para-xylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Ethylene dichloride; Vinylchloride; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Diphenyl carbonate; Cyclohexylbenzene; Phthalic anhydride;

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Pyrolysis gas ("pygas")	6	8	1	Naphtha	55	Ethanol; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; Nitrobenzene; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; LLDPE; LDPE; HDPE; PP; Polybutadien; GPPS and HIPS; EPS; PVC; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA Benzene; Toluene; Xylenes; Ortho-xylene; Meta-xylene; Para-xylene; Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A; Propylene oxide; Propylene glycol; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Diphenyl carbonate; Cyclohexylbenzene; Phthalic anhydride; Adipic acid; KA-Oil; Nylon salt; Nitrobenzene; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA
Reformate	2	2	0		55	Benzene; Toluene; Xylenes; Ortho-xylene; Meta-xylene; Para-xylene; Cyclohexane; Cyclohexene; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Bisphenol-A;

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Synthesis gas ("syngas")	7	7	0		74	Propylene oxide; Propylene glycol; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Diphenyl carbonate; Cyclohexylbenzene; Phthalic anhydride; Adipic acid; KA-Oil; Nylon salt; Nitrobenzene; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; GPPS and HIPS; EPS; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA Hydrogen cyanide; Carbon monoxide; Ethylene; Propylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Diethylene glycol; Triethylene glycol; Cyclohexanone; Cyclohexanol; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; α-Methyl-Benzyl alcohol; Styrene; Cumene; α-methyl styrene; Cumene hydroperoxide; Phenol; Acetone; Phosgene; Bisphenol-A; Ethylene dichloride; Vinylchloride; Propylene chlorohydrin; Propylene oxide; Propylene glycol; Methyl p-toluate; Dimethyl-terephthalate; Dimethyl carbonate; Diphenyl carbonate; Methanol; Dimethyl ether; Formaldehyde; Methyl formate; Sodium formate; Formamide; Formic acid; Ethanol; Adipic acid; KA-Oil; Hexamethylenediamine; Nylon salt; Acrylamide; Acrylonitrile; Adiponitrile; Toluene diisocyanate; Methylenedianiline; Methylene diphenyl diisocyanate; ε-Caprolactam; Methyl methacrylate; LLDPE; LDPE; HDPE; PP; GPPS and HIPS; EPS; PVC; PET; PA-6; PA-6,6; SAN; ABS; PMMA; PEG; PPG; PU (MDI and PEG); PU (TDI and PEG); PU (MDI and PPG); PU (TDI and PPG); PC from BPA; UF resin; MF resin

4.4.12 Polymers

For the relevant production processes and value chains please see the separate discussion note on chemicals (Annex 4). The results of the assessment of polymers are presented in the same way as for inorganic and organic chemicals:

Assessment of technical feasibility of MRV

- **Criterion 1:** Goods must be identifiable: For this purpose we have compiled in Table 22 for each substance to be assessed: The CN code and whether it is clearly related with the substance (if there are more substances covered, we have added the description of that HS heading / CN code). The CAS number is given, where possible. Note that sometimes CAS numbers are different for existing isomers and enantiomers. Also where the substance is systematically mixed with other substances, a different CAS number may be necessary. We have usually given only the CAS number for the pure substance of the most technically relevant isomer.

Not mentioned for individual substances in the table, but to be kept in mind in general is that some additional identification may be needed for the CBAM purpose, e.g. for substances that are traded in solutions or other mixtures. In such cases, the concentration of each substance (if it is a CBAM good) in the mixture will have to be reported by using "qualifying parameters" in the CBAM declaration, except where the customs rules already now require the reporting of the mass of the pure substance instead of the total mass. Copolymers represent a special challenge. If the different substances used as monomers (or pre-polymerised components) have very different embedded emissions (which can be assumed to be the case by default), the percentage of each component in the polymer should be provided, together with the fraction of additives (softeners, stabilisers, etc.). Where they are not themselves CBAM goods, it will be sufficient to report the total quantity of such "non-CBAM" fractions. All these additional information can be reported by way of "qualifying parameters" in the CBAM declaration and in the operator's emissions report for verification. Therefore, unclear composition data is not per se a reason for technical infeasibility for inclusion in the CBAM. However, this fact should be considered as additional difficulty in monitoring, reporting and verification, as such parameters may often be proprietary or confidential, or at least unknown to the importer.

As the substance names are often "trivial names" (names commonly used in industry as a kind of jargon, excluding trade names), they often do not give the full information on the structure of the substance. For such cases, we have given also systematic names according to IUPAC. Unfortunately, even though there is usually one name that would be the "preferred" name according to IUPAC, these names are not always unique either. The study team did not try to give the full list of possible names but given only one or two names that are IUPAC-preferred or at least IUPAC compatible.

- **Criterion 2a:** Production processes and routes need to be known (and not overly complex): The relevant summary parameters of this assessment are given in Table 23. The value chains are presented in more detail in the separate discussion note on chemicals (Annex 4). In the table, the following information is given:
 - The complexity measure calculated in accordance with section 4.4.2, first only for the last production step, and then including all precursors from the value chain.
 - The list of all precursors in the value chain, i.e. the list of substances that would have to be included in the CBAM, if the substance indicated would be included in the CBAM.

- A list of all substances which are produced with this substance as precursor. This gives an indication for how important the substance is for other value chains assessed.
- **Criterion 2c:** Identification of special MRV difficulties: Where relevant, a comment on this criterion is included in Table 23.

Note on the tables: Sorting was done by CN code. The sorting is the same in both tables, but for efficiency, the CN codes and CAS numbers were not repeated.

Table 22 Identification of polymers. Where no description of the CN code is given, it is referring to the one relevant substance only. The description is given for clarity only where substance groups are covered.

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
LLDPE (Linear low density PE)	39011010				polyethylene or poly(methylene)	
LDPE (low density PE)	39011090				polyethylene or poly(methylene)	
HDPE (high density PE);	390120				polyethylene or poly(methylene)	
PP (Polypropylene);	390210		9003-07-0		Poly(1-methylethylene)	
Polybutadien	39029090	Polymers of propylene or of other olefins, in primary forms (excl. polypropylene, polyisobutylene, propylene copolymers, and a A-B-A block copolymer of polystyrene, ethylene-butylene copolymer and polystyrene, containing by weight <= 35% of styrene and polybut-1-ene, a copolymer of but-1-ene with ethylene containing by weight <= 10% of ethylene, or a blend of polybut-1-ene with polyethylene and/or polypropylene containing by weight <= 10% of polyethylene and/or <= 25% of polypropylene, in blocks of irregular shape, lumps, powders, granules,	9003-17-2		Poly-1,3-butadiene	Substance included because needed as precursor of ABS. There is also CN 40022000 'Butadiene rubber ""BR"", in primary forms or in plates, sheets or strip', but this should be rather a ""downstream"" product. As we consider Butadiene a precursor for ABS, this CN should not be necessary.

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
EPS (Expansible PS)	390311	flakes and similar bulk forms)	9003-53-6		Poly(1-phenylethylene)	
PS – GPPS (general purpose PS) and HIPS (High Impact PS)	390319		9003-53-6		Poly(1-phenylethylene)	
SAN (Styrene-Acrylonitrile copolymer)	390320	Styrene-acrylonitrile copolymers "SAN", in primary forms	9003-54-7			
ABS (Acrylonitrile-Butadiene-Styrene copolymer)	390330	Acrylonitrile-butadiene-styrene copolymers "ABS", in primary forms	9003-56-9			
PVC (Polyvinylchloride)	390410		9002-86-2		Poly(1-chloroethylene)	CN refers to pure substance; Mixed with plasticisers etc: 390421 and 390422
PMMA (Poly Methyl methacrylate)	390610		9011-14-7		Poly(methyl 2-methylpropenoate)	
Polyethylene glycol (PEG)	39072911	Polyethylene glycols, in primary forms	25322-68-3		α -hydro- ω -hydroxypoly(oxy-1,2-ethanediyl)	
Polypropylene glycol (PPG)	39072920	Polyether alcohols, in primary forms (excl. bis(polyoxyethylene) methylphosphonate and polyethylene glycols)	25322-69-4		α -hydro- ω -hydroxypoly(oxypropylene)	
Polycarbonate from BPA	390740	Polycarbonates, in primary forms	24936-68-3		poly[2,2-propanebis(4-phenyl)carbonate]	Example substance
PET (Polyethylene terephthalate)	390761 and 390769	25038-59-9		Poly(ethylene terephthalate); poly(oxyethyleneoxyterephthaloyl)		
PA-6 (Polyamide 6)	390810	Polyamides-6, -11, -12, 6,6, -6,9, -6,10 or -6,12, in primary forms	25038-54-5		Poly[imino(1-oxo-1,6-hexanediyl)]; Poly(azepan-2-one); poly(hexano-6-lactam); Poly[azanediyl(1-oxohexane-1,6-diyl)]	

Substance name	HS / CN code	Description of CN code	CAS Number	Molecular formula	IUPAC Name(s)	Other comments
PA-6,6 (Polyamide 6,6)	390810	Polyamides-6, -11, -12, -6,6, -6,9, -6,10 or -6,12, in primary forms	32131-17-2		Poly[imino(1,6-dioxo-1,6-hexanediyl)imino-1,6-hexanediyl] ; Poly[imino(1,6-dioxohexamethylene)iminohexamethylene]; Poly(azanediyladipoylazanediylhexane-1,6-diyl)	
Urea formaldehyde resin (UF)	390910	Urea resins and thiourea resins, in primary forms	-			Selected in downstream project
Melamine formaldehyde resin (MF)	390920	Melamine resins, in primary forms	9003-08-1			Selected in downstream project
PU made of MDI and PEG	39095090	Polyurethanes, in primary forms				Example substance
PU made of TDI and PEG	39095090	Polyurethanes, in primary forms				Example substance
PU made of MDI and PPG	39095090	Polyurethanes, in primary forms				Example substance
PU made of TDI and PPG	39095090	Polyurethanes, in primary forms				Example substance

Table 23 Information on the complexity of value chains for the polymers assessed

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Linear low density polyethylene (LLDPE)	1	25	6	Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	0	
Low density polyethylene (LDPE)	1	25	6	Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	0	
High density polyethylene (HDPE)	1	25	6	Syngas; Naphtha; Ethane; Ethylene; Methanol; Dimethyl ether	0	
Polypropylene (PP)	1	29	7	Syngas; Naphtha; Propane; Ethane; Propylene; Methanol; Dimethyl ether	0	

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Polybutadien	1	23	5	Naphtha; Ethane; Butane; Isobutane; Butadiene	1	ABS
Expansible polystyrene (EPS)	1	487	19	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Styrene; Cumene; Methanol; Dimethyl ether	0	
Polystyrene (GPPS and HIPS)	1	487	19	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Styrene; Cumene; Methanol; Dimethyl ether	0	
Styrene-Acrylonitrile copolymer (SAN)	2	518	20	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Styrene; Cumene; Methanol; Dimethyl ether; Acrylonitrile	0	
Acrylonitrile-Butadiene-Styrene copolymer (ABS)	3	542	24	Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Butadiene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Styrene; Cumene; Methanol; Dimethyl ether; Acrylonitrile; Polybutadien	0	
Polyvinylchloride (PVC)	1	68	12	Chlorine; Hydrogen chloride; Calcium Carbide; Syngas; Naphtha; Ethane; Ethylene; Acetylene; Ethylene dichloride; Vinylchloride; Methanol; Dimethyl ether	0	

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Poly methylmethacrylate (PMMA)	1	263	22	Hydrogen cyanide; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Cumene; Cumene hydroperoxide; Acetone; Methanol; Dimethyl ether; Formaldehyde; Methyl methacrylate	0	
Polyethylene glycol (PEG)	1	32	7	Syngas; Naphtha; Ethane; Ethylene; Ethylene oxide; Methanol; Dimethyl ether	2	PU (MDI and PEG); PU (TDI and PEG)
Polypropylene glycol (PPG)	1	496	26	Chlorine; Sodium hydroxide; Hydrogen peroxide; Syngas; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Cumene; Propylene chlorohydrin; Tert-butyl hydroperoxide; Propylene oxide; Methanol; Dimethyl ether	2	PU (MDI and PPG); PU (TDI and PPG)
Polycarbonate from BPA	4	715	24	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexene; Cumene; Cumene hydroperoxide; Phenol; Acetone; Phosgene; Bisphenol-A; Dimethyl carbonate; Diphenyl carbonate; Cyclohexylbenzene; Methanol; Dimethyl ether	0	
Polyethylene terephthalate (PET)	5	205	18	Syngas; Carbon monoxide; Reformate; Naphtha; Pygas; Ethane; Light oil; Ethylene; Xylenes; Para-xylene; Ethylene oxide; Ethylene carbonate; Ethylene glycol; Terephthalic acid; Methyl p-toluate; Dimethyl-terephthalate; Methanol; Dimethyl ether	0	
Polyamide 6 (PA-6)	1	789	28	Chlorine; Hydrogen chloride; Syngas; Reformate; Naphtha; Propane; Pygas;	0	

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
Polyamide 6,6 (PA-6,6)	1	477	29	Ethane; Light oil; Propylene; Benzene; Toluene; Cyclohexane; Cyclohexene; Cyclohexanone; Cumene; Cumene hydroperoxide; Phenol; Cyclohexylbenzene; Methanol; Dimethyl ether; KA-Oil; Nitric oxide; Hydroxylamine; Nitrogen dioxide; Nitrosylsulphuric acid; Nitrosyl chloride; ε-Caprolactam	0	
Urea formaldehyde resin (UF)	2	12	3	Syngas; Methanol; Formaldehyde	0	
Melamine formaldehyde resin (MF)	2	13	4	Syngas; Methanol; Formaldehyde; Melamine	0	
PU made of MDI and PEG	2	147	21	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Pygas; Ethane; Light oil; Ethylene; Benzene; Toluene; Ethylene oxide; Phosgene; Methanol; Dimethyl ether; Formaldehyde; Nitrobenzene; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; PEG	0	
PU made of TDI and PEG	2	95	18	Chlorine; Syngas; Carbon monoxide; Reformate; Naphtha; Pygas; Ethane; Light oil; Ethylene; Toluene; Ethylene oxide; Phosgene; Methanol; Dimethyl ether; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; PEG	0	

Substance name as in report	Complexity without precursors	Complexity WITH precursors	Number of precursors	Full precursor list	number of uses as precursor	Is itself precursor for:
PU made of MDI and PPG	2	611	34	Chlorine; Sodium hydroxide; Hydrogen peroxide; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Cumene; Phosgene; Propylene chlorohydrin; Tert-butyl hydroperoxide; Propylene oxide; Methanol; Dimethyl ether; Formaldehyde; Nitrobenzene; Aniline; Methylenedianiline; Methylene diphenyl diisocyanate; PPG	0	
PU made of TDI and PPG	2	559	32	Chlorine; Sodium hydroxide; Hydrogen peroxide; Syngas; Carbon monoxide; Reformate; Naphtha; Propane; Pygas; Ethane; Light oil; Butane; Isobutane; Ethylene; Propylene; Benzene; Toluene; Ethylbenzene; Ethylbenzene hydroperoxide; Acetophenone; a-Methyl-Benzyl alcohol; Cumene; Phosgene; Propylene chlorohydrin; Tert-butyl hydroperoxide; Propylene oxide; Methanol; Dimethyl ether; Dinitrotoluene; Toluene Diamine; Toluene diisocyanate; PPG	0	

4.5 Conclusions (Sub-task 2.5): Propose a ranking of priority sectors, sub-sectors/products)

4.5.1 Sectors except chemicals and polymers

Considering all assessments of Task 1 (sectors at risk of carbon leakage) and Task 2 (technical feasibility), we conclude on the sectors pulp & paper, glass, ceramics, ferro-alloys, aluminium, non-ferrous metals as follows:

- We assign highest priority for inclusion in the CBAM to the CN codes shortlisted in Task 1 and confirmed by Task 2. This is shown by marking the CN code in dark green in Table 24.
- Second highest priority (light green) is assigned to goods identified in Task 2 to be equally useful for inclusion as those identified in Task 1. They should be considered for inclusion due to strong links to those shortlisted in Task 1 (for avoiding circumvention or unequal treatment of very similar goods).
- Goods of the type like scrap are marked in yellow. This is to indicate that they could be classified like the next (amber) category, but there is currently a decision pending on their treatment in the methodology. Hence, the project team considers them as "difficult to include", but with a different aspect than the purely amber ones.
- Goods which have been identified in Task 1, but which seem more difficult to include due to technical difficulty are marked in amber (low priority).
- Goods which currently appear too difficult for inclusion are marked in red.
- Finally, goods indicated as being potential "downstream" products of other goods in that sector, are indicated in blue. This means that these goods could be included in the CBAM with simplified monitoring requirements (based only on precursors, as discussed in the "CBAM downstream extension project"), provided the other goods in this sector which could be precursors mentioned are already included in the CBAM, and an assessment shows that they become exposed to carbon leakage due to inclusion of the precursors in the CBAM. At this time, these goods marked blue should not be included in the CBAM. However, as soon as rules for the inclusion of downstream goods are become available (because the extension 'downstream of existing CBAM' has already happened), then these goods can be included together with the "clearly horizontal" extension".

Table 24 Result of Task 2: Prioritisation of sectors and goods for inclusion in the CBAM. Colours used: dark green: identified in Task 1, inclusion feasible; light green: feasible; amber: difficult; red: should not be included; yellow: subject to discussion on scrap; blue: downstream products, do not include.

HS heading / CN Code	Description (simplified)
4701	Mechanical wood pulp, not chemically treated
4702	Chemical wood pulp, dissolving grades
4703	Chemical wood pulp, soda or sulphate (excl. dissolving grades)
4704	Chemical wood pulp, sulphite (excl. dissolving grades)
4705	Wood pulp obtained by a combination of mechanical and chemical pulping processes

HS heading / CN Code	Description (simplified)
4706	Pulps of fibres derived from recovered "waste and scrap" paper or paperboard or of other fibrous cellulosic material (excl. wood)
4707	Recovered "waste and scrap" paper or paperboard (excl. paper wool)
4801	Newsprint
4802 except 480210	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes
480210	handmade papers
4803	Toilet or facial tissue stock, towel or napkin stock and similar
480411	Unbleached kraftliner, uncoated, in rolls of a width > 36 cm
4804	Uncoated kraft paper and paperboard
4805	Other paper and paperboard, uncoated
4806	Vegetable parchment, greaseproof papers, tracing papers etc.
4807	Composite paper and paperboard "made by sticking flat layers of paper or paperboard together with an adhesive", not surface-coated
4808	Corrugated paper and paperboard "with or without glued flat surface sheets", creped, crinkled, embossed or perforated
4809	Carbon paper, self-copy paper and other copying or transfer papers
4810	Paper and paperboard, coated on one or both sides with kaolin "China clay" or other inorganic substances
4811	Paper, paperboard, cellulose wadding and webs of cellulose fibres, coated, impregnated, covered, surface-coloured, surface-decorated or printed
4812	Filter blocks, slabs and plates, of paper pulp
4813	Cigarette paper, whether or not cut to size or in the form of booklets or tubes
4814	Wallpaper and similar wallcoverings of paper; window transparencies of paper
4816	Carbon paper, self-copy paper and other copying or transfer papers
4817	Envelopes, letter cards, plain postcards and correspondence cards, of paper or paperboard; boxes, pouches, wallets etc.
4818 except 481850	Toilet paper and similar paper, cellulose wadding or webs of cellulose fibres, of a kind used for household or sanitary purposes; handkerchiefs, cleansing tissues, towels, etc.
481850	Articles of apparel and clothing accessories, etc.
4819	Cartons, boxes, cases, bags and other packing containers, of paper, paperboard, cellulose wadding or webs of cellulose fibres, n.e.s.;
4820	Registers, account books, notebooks, order books, receipt books, letter pads, memorandum pads, diaries and similar articles
4821	Paper or paperboard labels of all kinds, whether or not printed
4822	Bobbins, spools, cops and similar supports of paper pulp, paper or paperboard, whether or not perforated or hardened
4823	Paper, paperboard, cellulose wadding and webs of cellulose fibres, ... n.e.s.
70010010	Cullet and other waste and scrap of glass
700100 91+99	Glass in the mass
7002	Glass in balls, rods or tubes, unworked
7003	Cast glass and rolled glass, in sheets or profiles
7004	Sheets of glass, drawn or blown
7005	Float glass and surface ground or polished glass, in sheets

HS heading / CN Code	Description (simplified)
7006	Sheets or profiles of glass, ... bent, edge-worked, engraved, enamelled or otherwise worked, but not framed
7007	Safety glass, toughened "tempered", laminated safety glass
7008	Multiple-walled insulating units of glass
7009	Glass mirrors, whether or not framed, incl. rear-view mirrors
7010	Carboys, bottles, flasks, jars, pots, phials, ampoules and other containers, of glass, of a kind used for the conveyance or packing of goods, preserving jars, stoppers, lids and other closures, of glass
70109041	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of ≥ 1 l but $< 2,5$ l
70109043	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $> 0,33$ l but < 1 l
70109045	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $\geq 0,15$ l to $0,33$ l
70109047	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of $< 0,15$ l
70109071	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of $> 0,055$ l but $< 2,5$ l
70109079	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of $\leq 0,055$ l
70109091	Carboys, bottles, flasks, jars, pots, phials and other containers, of colourless glass, of a kind used for the commercial conveyance or packing of goods, of a nominal capacity of $< 2,5$ l
70109099	Carboys, bottles, flasks, jars, pots, phials and other containers, of coloured glass, of a kind used for the commercial conveyance or packing of goods, of a nominal capacity of $< 2,5$ l
7011	Glass envelopes, incl. bulbs and tubes, open, and glass parts thereof, without fittings, for electric lamps and light sources, cathode ray tubes or the like
7013	Glassware of a kind used for table, kitchen, toilet, office, indoor decoration or similar purposes
7014	Signalling glassware and optical elements of glass, not optically worked
7015	Clock or watch glasses and similar glasses, glasses for non-corrective or corrective spectacles, etc.
7016 except 70169010	Paving blocks, slabs, bricks, squares, tiles and other articles of pressed or moulded glass, whether or not wired, for building or construction purposes
70169010	Leaded lights and the like
7017	Laboratory, hygienic or pharmaceutical glassware, whether or not graduated or calibrated (excl. containers for the conveyance or packing of goods, measuring, checking or medical instruments and apparatus of chapter 90)
7018	Glass beads, imitation pearls, imitation precious or semi-precious stones and similar glass smallware, and articles thereof (excl. imitation jewellery); glass eyes (excl. prosthetic articles); statuettes and other ornaments of lamp-worked glass (excl. imitation jewellery); glass microspheres with a diameter of ≤ 1 mm
7019	Glass fibres, incl. glass wool, and articles thereof (excl. mineral wools and articles thereof, optical fibres, fibre bundles or cable, electrical insulators or parts thereof, brushes of glass fibres, dolls' wigs)
7020	Articles of glass, n.e.s.

HS heading / CN Code	Description (simplified)
6901	Bricks, blocks, tiles and other ceramic goods of siliceous fossil meals, e.g. kieselguhr, tripolite or diatomite, or of similar siliceous earths
6902	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods
69022099	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods containing, by weight, > 50% alumina, silica or a mixture or compound of these products (excl. containing, by weight, >= 93% silica, > 7% but < 45% alumina and those of siliceous fossil meals or similar siliceous earths)
6903	Retorts, crucibles, mufflers, nozzles, plugs, supports, cupels, tubes, pipes, sheaths, rods, slide gates and other refractory ceramic goods
6904	Ceramic building bricks, flooring blocks, support or filler tiles and the like
6905	Roofing tiles, chimney pots, cowls, chimney liners, architectural ornaments and other ceramic constructional goods
6906	Ceramic pipes, conduits, guttering and pipe fittings
6907	Ceramic flags and paving, hearth or wall tiles; ceramic mosaic cubes etc
6909	Ceramic wares for laboratory, chemical or other technical uses; ceramic troughs, tubs and similar receptacles used in agriculture; ceramic pots, jars and similar articles used for the conveyance or packing of goods
6910	Ceramic sinks, washbasins, washbasin pedestals, baths, bidets, water closet pans, flushing cisterns, urinals and similar sanitary fixtures
6911	Tableware, kitchenware, other household articles and toilet articles, of porcelain or china
6912	Tableware, kitchenware, other household articles and toilet articles, of ceramics other than porcelain or china
6913	Statuettes and other ornamental ceramic articles, n.e.s.
6914	Ceramic articles, n.e.s.
720221	Ferro-silicon, containing by weight > 55% of silicon
720229	Ferro-silicon, containing by weight <= 55% silicon
720230	Ferro-silico-manganese
7801	Unwrought lead
7802	Lead waste and scrap
7804	Lead plates, sheets, strip and foil; lead powders and flakes
7806	Articles of lead, n.e.s.
7901	Unwrought zinc
7902	Zinc waste and scrap
7903	Zinc dust, powders and flakes
7904	Zinc bars, rods, profiles and wire, n.e.s.
7905	Zinc plates, sheets, strip and foil
7907	Articles of zinc, n.e.s.
7401	Copper mattes; cement copper "precipitated copper"
7402	Copper, unrefined; copper anodes for electrolytic refining
7403	Copper, refined, and copper alloys, unwrought
7404	Waste and scrap, of copper
7405	Master alloys of copper
7406	Powders and flakes, of copper
7407	Bars, rods and profiles, of copper, n.e.s.

HS heading / CN Code	Description (simplified)
7408	Copper wire
7409	Plates, sheets and strip, of copper, of a thickness of > 0,15 mm
7410	Copper foil "whether or not printed or backed with paper, paperboard, plastics or similar backing materials", of a thickness "excl. any backing" of <= 0,15 mm
7411	Copper tubes and pipes
7412	Copper tube or pipe fittings "e.g., couplings, elbows, sleeves"
7413	Stranded wire, cables, plaited bands and the like, of copper (excl. electrically insulated products)
7415	Nails, tacks, drawing pins, staples and similar articles, of copper or with shafts of iron or steel and heads of copper, screws, bolts, nuts, screw hooks, rivets, cotters, cotter pins, washers "incl. spring washers" and similar articles, of copper
7418	Table, kitchen or other household articles, sanitary ware, and parts thereof, of copper; pot scourers and scouring or polishing pads, gloves and the like, of copper
7419	Other articles of copper, n.e.s.
27101211	Light oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)
27101215	Light oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)
27101221	White spirit
27101225	Special spirits (excl. white spirit) of petroleum or bituminous minerals
27101231	Aviation spirit
27101241	Motor spirit, with a lead content <= 0,013 g/l, with a research octane number "RON" of < 95 (excl. containing biodiesel)
27101245	Motor spirit, with a lead content <= 0,013 g/l, with a research octane number "RON" of >= 95 but < 98 (excl. containing biodiesel)
27101249	Motor spirit, with a lead content <= 0,013 g/l, with a research octane number "RON" of >= 98 (excl. containing biodiesel)
27101250	Motor spirit, with a lead content > 0,013 g/l (excl. containing biodiesel, and aviation spirit)
27101270	Jet fuel, spirit type (excl. aviation spirit)
27101290	Light oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. containing biodiesel, for undergoing chemical transformation, and special spirits, motor spirit and spirit type jet fuel)
27101911	Medium oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27
27101915	Medium oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27)
27101921	Jet fuel, kerosene type
27101925	Kerosene (excl. jet fuel)
27101929	Medium oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. for undergoing chemical transformation, and kerosene)
27101931	Gas oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27
27101935	Gas oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)

HS heading / CN Code	Description (simplified)
27101943	Gas oils of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101946	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,002\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101947	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,002\%$ but $\leq 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101948	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101951	Fuel oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)
27101955	Fuel oils obtained from bituminous materials, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)
27101962	Fuel oils obtained from bituminous materials, with a sulphur content of $\leq 0,1\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101966	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,1\%$ but $\leq 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101967	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101971	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing a specific process as defined in Additional Note 5 to chapter 27
27101975	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)
27101981	Motor oils, compressor lube oils and turbine lube oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)
27101983	Liquids for hydraulic purposes containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)
27101985	White oils, liquid paraffin containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101987	Gear oils and reductor oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101991	Metalworking compounds, mould-release oils, anti-corrosion oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101993	Electrical insulating oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101999	Lubricating oils and other heavy oils and preparations n.e.s., containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)

HS heading / CN Code	Description (simplified)
27102011	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight, containing biodiesel
27102016	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,1\%$ by weight, containing biodiesel
27102019	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight, containing biodiesel
27102032	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,5\%$ by weight, containing biodiesel
27102038	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,5\%$ by weight, containing biodiesel
27102090	Oils of $\geq 70\%$ of petroleum or bituminous minerals, containing biodiesel (excl. gas oils and fuel oils)
27111211	Propane of a purity of $\geq 99\%$, for use as a power or heating fuel, liquefied
27111219	Propane of a purity of $\geq 99\%$, liquefied (excl. for use as a power or heating fuel)
27111291	Propane of a purity of $< 99\%$, liquefied, for undergoing a specific process as defined in Additional Note 5 to chapter 27
27111293	Propane of a purity of $< 99\%$, liquefied, for undergoing chemical transformation (excl. processes specified in 2711.12.91)
27111294	Liquid propane of a purity of $> 90\%$, but $< 99\%$ (excl. for undergoing chemical transformation)
27111297	Liquefied propane of a purity of $\leq 90\%$ (excl. for undergoing chemical transformation)
27111310	Butanes for undergoing a specific process as defined in Additional Note 5 to chapter 27, liquefied (excl. of a purity of $\geq 95\%$ of N-butane or isobutane)
27111330	Butanes for undergoing chemical transformation, liquefied (excl. for specific processes specified in Additional Note 5 to chapter 27 and butanes of a purity of $\geq 95\%$ of N-butane or isobutane)
27111391	Liquefied butane of a purity of $> 90\%$ but $< 95\%$ (excl. for undergoing chemical transformation)
27111397	Liquefied butane of a purity of $\leq 90\%$ (excl. for undergoing chemical transformation)
27111400	Ethylene, propylene, butylene and butadiene, liquefied (excl. ethylene of a purity of $\geq 95\%$ and propylene, butylene and butadiene of a purity of $\geq 90\%$)
27111900	Gaseous hydrocarbons, liquefied, n.e.s. (excl. natural gas, propane, butane, ethylene, propylene, butylene and butadiene)
27112900	Hydrocarbons in gaseous state, n.e.s. (excl. natural gas)
27121010	Crude petroleum jelly
27121090	Petroleum jelly (excl. crude)
27122010	Synthetic paraffin wax containing $< 0,75\%$ by weight of oil and of a molecular weight of ≥ 460 but ≤ 1.560
27122090	Paraffin wax containing $< 0,75\%$ by weight of oil (excl. synthetic paraffin wax of a molecular weight of ≥ 460 but ≤ 1.560)
27129011	Crude ozokerite, lignite wax or peat wax "natural products"
27129019	Ozokerite, lignite wax or peat wax "natural products", whether or not coloured (excl. crude)
27129031	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. petroleum jelly,

HS heading / CN Code	Description (simplified)
	paraffin wax containing < 0,75% by weight of oil, ozokerite, lignite wax and peat wax)
27129033	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, petroleum jelly, paraffin wax containing < 0,75% by weight of oil, ozokerite, lignite wax and peat wax)
27129039	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes (excl. for undergoing chemical transformation, petroleum jelly, paraffin wax containing < 0,75% by weight of oil, ozokerite, lignite wax and peat wax)
27129091	Blend of 1-alkenes containing by weight $\geq 80\%$ of 1-alkenes of a chain-length of ≥ 24 but ≤ 28 carbon atoms
27129099	Paraffin wax, microcrystalline petroleum wax, slack wax, ozokerite, lignite wax, peat wax, other mineral waxes, and similar products obtained by synthesis or by other processes, whether or not coloured (excl. petroleum jelly, paraffin wax containing < 0,75% by weight of oil and a blend of 1-alkenes containing by weight $\geq 80\%$ of 1-alkenes of a chain-length of ≥ 24 but ≤ 28 carbon atoms)
27131100	Petroleum coke, non-calcined
27131200	Petroleum coke, calcined
27132000	Petroleum bitumen
27139010	Residues of petroleum oil or of oil obtained from bituminous minerals for the manufacture of carbon of heading 2803
27139090	Residues of petroleum oil or of oil obtained from bituminous minerals (excl. for the manufacture of carbon of heading 2803, petroleum coke and petroleum bitumen)

4.5.2 Inorganic and organic chemicals, polymers

Since for chemicals and polymers a different method was applied both for Tasks 1 and 2, also the results need to be presented in a different way. The results of our assessment using the information given in sections 4.4.10 to 4.4.12 is compiled in Table 25. The assessment was performed as follows:

- The parameters used for the assessment were chosen in line with the criteria mentioned already for Task 1 in section 3.4. This means that the existence of an EU ETS product benchmark indicates high priority for inclusion in the CBAM, as does high production volume in the EU-27. The technical complexity of the relevant value chains as calculated using the method described in section 4.4.2 as well as difficulties in clearly identifying the substance upon import were considered.
- As these parameters lead seldom to a black-and-white picture, but to a more differentiated situation, it was not possible to just add up the pros and cons of each case. Instead, a decision-tree like stepwise approach was used as follows:
 - First parameter is the complexity measure calculated for each substance based on the value chains identified. We assigned ranges for the RAG (Red-Amber-Green) rating using Figure 9:
 - Green (fair complexity): Complexity measure 0 to 70;
 - Amber (medium complexity): 71 to 200;
 - Light red (high complexity): 201 to 400;
 - Dark red (too complex): higher than 400.

- If the substance is covered by an EU ETS benchmark, the RAG rating was shifted towards more feasibility for inclusion (leading in some cases to a “dark green”, i.e. best rating).
- The production volume of the substance in the EU-27 was used to shift the RAG rate to greener if the production volume was very high (i.e. above 1 million tonnes per year), redder ore, if very low or unknown. As we used “sold quantity produced” as parameter, production volumes sometimes appeared unusually low. To avoid bias of this kind, we sometimes deviated from this number using expert judgment, i.e. considering that the substance is produced actually in higher volumes but immediately consumed by using it as a precursor.
- Difficulties in identifying the good using CN code and/or CAS number, or where the “substance” referred to a too wide or undefined substance class, the assessment result we changed to dark red, as this is a hard knock-out criterion for inclusion in the CBAM.

It has to be mentioned that the assessment of value chains has some limitations when it comes to substances appearing as intermediate in production processes. Theoretically, some intermediates can be tradable substances on their own if isolated from the reaction mixture and purified, e.g. cyclohexanone oxime in the production of caprolactam. However, this production process is usually highly integrated and optimised to produce only the desired end product. Therefore, we have omitted such intermediates from the list of precursors. However, in a few cases, it may be possible to omit further intermediates shown¹¹⁶ in the value chain graphics presented in the chemicals discussion note in annex. This also may apply to goods that are indicated to be only co-products in the value chains assessed. For the CBAM, the consequence will be that the production processes for monitoring (implementing act pursuant to Article 7 of the CBAM Regulation) will have to be defined sufficiently encompassing such that the emissions of the process also include the production and use of the intermediate substance. This is particularly important for synthesis gas, which is covered by an EU ETS product benchmark and used for methanol and a plethora of other syntheses but usually produced on-site and not internationally traded. Also for KA-oil (an intermediate in adipic acid production), we could not identify a CN code or CAS number. As it may not be appropriate to assume that the pure components of KA-oil (cyclohexanol and cyclohexanone) be used instead, the KA-oil should be included directly in the system boundaries of the production process of adipic acid (if included in the CBAM).

Finally, it needs to be kept in mind that the assessment results compiled in Table 25 do not directly represent a list of substances that can be immediately used for a CBAM extension, e.g. by selecting the colour codes which should be included. Instead, first a decision has to be made which chemicals and polymers should be included in the CBAM, and then their full value chains have to be considered. This means that e.g. if the decision is made that PET (polyethylene terephthalate) should *not* be included, the list should be amended such that all substances that appear only in the value chain of PET can be removed, except if there are other criteria applicable (e.g. coverage by the EU ETS). We have included comments in the table regarding a few example cases where such exclusion of rarely used precursors would be applicable.

¹¹⁶ Such substances are indicated by a specific colour in those graphs, e.g. Ethylbenzene hydroperoxide, α-Methyl-Benzyl alcohol, α-methyl styrene, cumene hydroperoxide.

Figure 9 Illustration of the intervals for the complexity measure for chemicals and polymers used. Substances were sorted by increasing complexity as shown by the green line referring to the left axis. The yellow bars (referring to the right axis) indicate the annual production volumes of the substance.

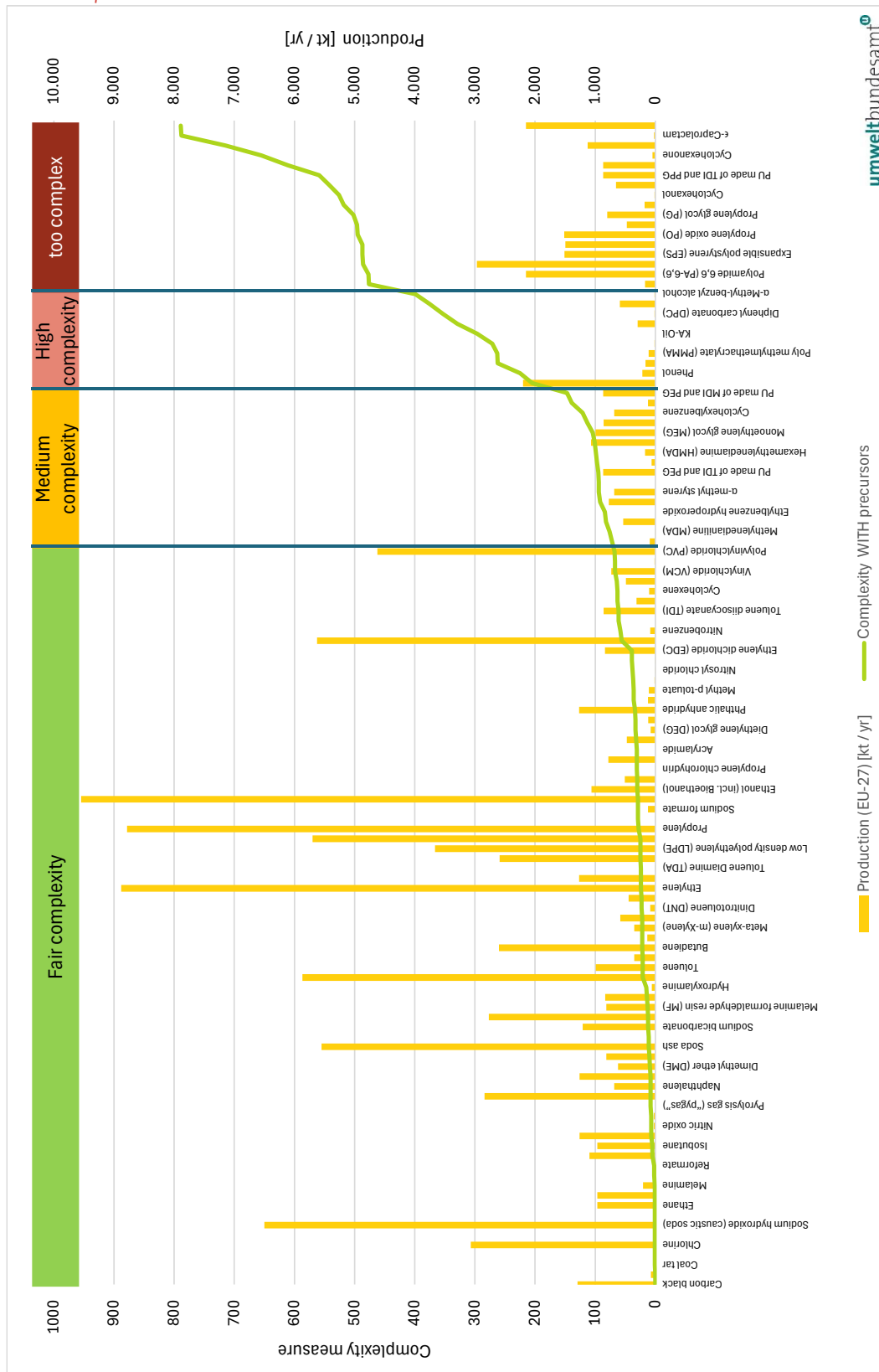


Figure 10 Zoomed-in version of Figure 9.

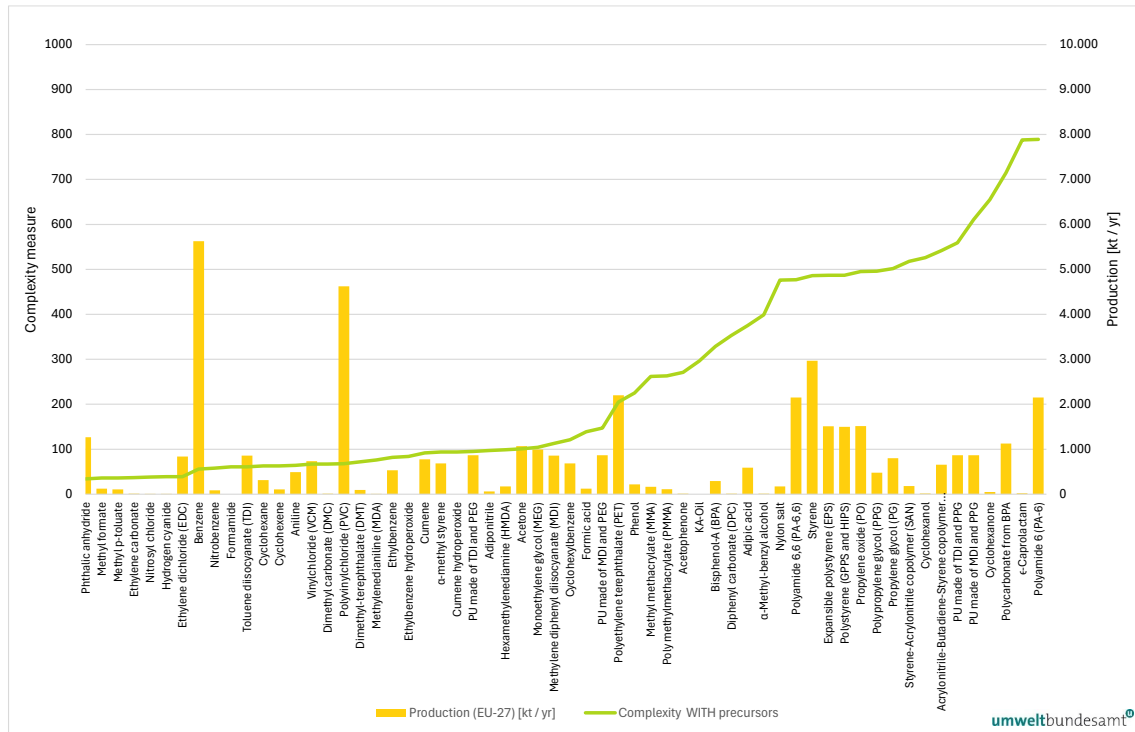
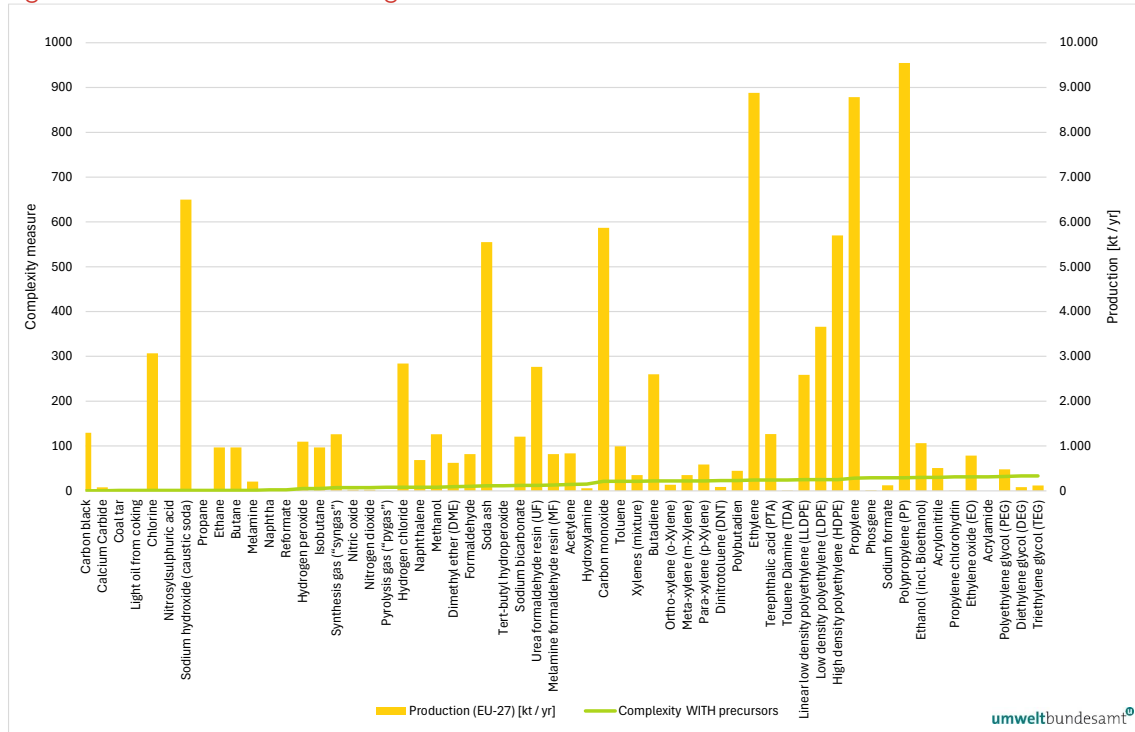


Table 25 Result of Task 2: Prioritisation of sectors and goods for inclusion in the CBAM. For colour codes please see text above. The sorting was made to show increasing complexity. Overall result: 1= high priority for inclusion (green), 5=lowest priority (red). Result 0 means that the substance belongs to a different sector and the result is already presented in Table 24.

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Carbon black	0	0	0	yes	1.296	> 1Mt		1	
Calcium Carbide	0	0	3		75	low volume, low priority		2	Only needed for older routes EDC/VCM / PVC
Coal tar	1	0	2		no data	other sector		0	Would be included with coke
Light oil from coking	1	0	55		no data	other sector		0	Would be included with coke
Chlorine	1	0	21		3.067	> 1Mt		2	Inclusion only relevant with indirect emissions
Nitrosylsulphuric acid	1	0	3		7	low volume, low priority		4	Required for PA-6
Sodium hydroxide (caustic soda)	1	0	9		6.500	> 1Mt		2	Inclusion only relevant with indirect emissions
Propane	1	0	38		no data	other sector		0	Would be included with refinery products
Ethane	1	0	60		967	< 0,5 Mt		2	
Butane	1	0	16		967	< 0,5 Mt		2	
Melamine	1	0	1		207	< 0,5 Mt		2	
Naphtha	2	0	81		no data	other sector	tbd	0	Would be included with refinery products
Reformate	2	0	55		no data	other sector	tbd	0	Would be included with refinery products
Hydrogen peroxide	5	0	8		1.095	> 1Mt		2	Production complexity likely under-estimated; indirect emissions would be important
Isobutane	5	1	15		967	< 0,5 Mt		2	
Synthesis gas ("syngas")	7	0	74	yes	1.260	> 1Mt	difficult	n.a.	Important for emissions but not traded. Emissions could be included directly in production

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
									processes, instead of listing separately as precursor (like in Hydrogen and Ammonia production)
Nitric oxide	7	0	4		24	low volume, low priority		4	Required for PA-6
Nitrogen dioxide	7	0	3		24	low volume, low priority		4	Required for PA-6
Pyrolysis gas ("pygas")	8	1	55		no data	other sector	tbd	0	Would be included with refinery products
Hydrogen chloride	8	1	6		2.841	> 1Mt		2	Indirect emissions can be important for some production routes
Naphthalene	8	1	1		684	> 0,5 Mt		2	
Methanol	8	1	69	yes	1.260	> 1Mt		1	
Dimethyl ether (DME)	9	2	57		622	> 0,5 Mt		2	Appears only as intermediate in integrated process, low priority
Formaldehyde	10	2	8		817	> 0,5 Mt		2	
Soda ash	11	1	1	yes	5.551	> 1Mt		1	
Tert-butyl hydroperoxide	11	2	5		no data	< 0,5 Mt		3	Appears only as intermediate in integrated process, low priority
Sodium bicarbonate	12	2	0	yes	1.207	> 1Mt		1	
Urea formaldehyde resin (UF)	12	3	0		2.767	> 1Mt	difficult	4	Hard to identify
Melamine formaldehyde resin (MF)	13	4	0		817	> 0,5 Mt	difficult	4	Hard to identify, in practice not final resin, but pre-polymerised substance used
Acetylene	14	3	2	yes	833	> 0,5 Mt		1	

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Hydroxylamine	15	1	2		58	low volume, low priority		3	Only precursor in one route (PA-6)
Carbon monoxide	21	1	18		5.867	> 1Mt		2	
Toluene	21	4	45	yes	992	> 0,5 Mt		1	
Xylenes (mixture)	21	4	8	yes	350	< 0,5 Mt		1	
Butadiene	22	4	6	yes	2.599	> 1Mt		1	
Ortho-xylene (o-Xylene)	22	5	1	yes	134	< 0,5 Mt		1	
Meta-xylene (m-Xylene)	22	5	0	yes	350	< 0,5 Mt		1	
Para-xylene (p-Xylene)	22	5	3	yes	585	> 0,5 Mt		1	
Dinitrotoluene (DNT)	23	5	4		86	low volume, low priority		3	Only relevant if polyurethanes included
Polybutadiene	23	5	1		445	< 0,5 Mt		2	Only relevant if ABS included
Ethylene	24	5	32	yes	8.880	> 1Mt		1	
Terephthalic acid (PTA)	24	6	2		1.267	> 1Mt		2	
Toluene Diamine (TDA)	24	6	3		8	low volume, low priority		3	Only relevant if polyurethanes included
Linear low density polyethylene (LLDPE)	25	6	0		2.589	> 1Mt		1	High importance polymer
Low density polyethylene (LDPE)	25	6	0		3.662	> 1Mt		1	High importance polymer
High density polyethylene (HDPE)	25	6	0		5.700	> 1Mt		1	High importance polymer
Propylene	28	6	37	yes	8.784	> 1Mt		1	
Phosgene	29	3	9		9	low volume, low priority		2	
Sodium formate	29	3	1		122	low volume, low priority		3	Only precursor in one route
Polypropylene (PP)	29	7	0		9.547	> 1Mt		1	High importance polymer
Ethanol (incl. Bioethanol)	30	6	0		1.063	> 1Mt		2	

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Acrylonitrile	30	7	7		508	> 0,5 Mt		2	
Propylene chlorohydrin	31	8	5		3	low volume, low priority		3	Appears only as intermediate in integrated process, low priority
Ethylene oxide (EO)	31	6	8	yes	783	> 0,5 Mt		1	
Acrylamide	31	8	0		no data	< 0,5 Mt		3	Not used as precursor, can be omitted
Polyethylene glycol (PEG)	32	7	2		477	< 0,5 Mt		3	Only relevant if polyurethanes included
Diethylene glycol (DEG)	33	7	0	yes	79	< 0,5 Mt		1	Appears only as co-product in value chain, low priority
Triethylene glycol (TEG)	33	7	0	yes	118	< 0,5 Mt		1	Appears only as co-product in value chain, low priority
Phthalic anhydride	34	8	0		1.267	> 1Mt		2	
Methyl formate	36	3	2		122	low volume, low priority		2	Appears only as intermediate in integrated process, low priority
Methyl p-toluate	36	7	2		107	low volume, low priority		2	Appears only as intermediate in integrated process, low priority
Ethylene carbonate	37	7	2		13	low volume, low priority		2	Important for PET and PU
Nitrosyl chloride	38	5	2		9	low volume, low priority		3	Only relevant if PA-6 is included
Hydrogen cyanide	39	7	6		7	low volume, low priority		2	
Ethylene dichloride (EDC)	39	8	2		837	> 0,5 Mt		2	
Benzene	56	5	40	yes	5.626	> 1Mt		1	
Nitrobenzene	58	6	5		86	low volume, low priority		3	

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Formamide	61	4	1		no data	< 0,5 Mt		3	Appears only as intermediate in integrated process, low priority
Toluene diisocyanate (TDI)	61	11	2		860	< 0,5 Mt		2	Only needed for PU
Cyclohexane	63	6	8		313	< 0,5 Mt		3	
Cyclohexene	63	6	13		105	low volume, low priority		3	
Aniline	64	7	4		491	< 0,5 Mt		3	
Vinylchloride (VCM)	67	11	1	yes	733	> 0,5 Mt		1	
Dimethyl carbonate (DMC)	67	5	2		13	low volume, low priority		2	needed for polycarbonates
Polyvinylchloride (PVC)	68	12	0	yes	4.623	> 1Mt		1	
Dimethyl-terephthalate (DMT)	72	10	1		94	low volume, low priority		3	
Methylenedianiline (MDA)	76	11	3		8	low volume, low priority		3	
Ethylbenzene	82	11	13		533	> 0,5 Mt		3	
Ethylbenzene hydroperoxide	84	12	12		no data	< 0,5 Mt		3	Appears only as intermediate in integrated process, low priority
Cumene	92	12	29	yes	775	> 0,5 Mt		1	
α-methyl styrene	94	13	0		684	> 0,5 Mt		3	Appears only as co-product in value chain, low priority
Cumene hydroperoxide	94	13	15		no data	< 0,5 Mt		3	Appears only as intermediate in integrated process, low priority
PU made of TDI and PEG	95	18	0		867	> 0,5 Mt	difficult	4	substance only for illustration purposes. PU are hard to identify
Adiponitrile	97	12	3		62	low volume, low priority		3	for PA-6,6

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Hexamethylenediamine (HMDA)	99	13	2		171	low volume, low priority		3	for PA-6,6
Acetone	101	14	4	yes	1.069	> 1Mt		1	
Monoethylene glycol (MEG)	104	9	1	yes	993	> 0,5 Mt		1	
Methylene diphenyl diisocyanate (MDI)	113	15	2		860	< 0,5 Mt		3	
Cyclohexylbenzene	121	7	12		684	> 0,5 Mt		3	Appears only as intermediate in integrated process, low priority
Formic acid	139	7	0		122	low volume, low priority		3	
PU made of MDI and PEG	147	21	0		867	> 0,5 Mt	difficult	4	substance only for illustration purposes. PU are hard to identify
Polyethylene terephthalate (PET)	205	18	0		2.200	> 1Mt		3	
Phenol	225	16	11	yes	218	> 0,5 Mt		2	
Methyl methacrylate (MMA)	262	21	1		164	low volume, low priority		4	
Poly methylmethacrylate (PMMA)	263	22	0		110	low volume, low priority		4	
Acetophenone	271	16	11		14	low volume, low priority		3	Appears only as co-product in value chain, low priority
KA-Oil	296	18	7		no data	< 0,5 Mt		n.a.	not traded substance, could be included directly in production routes (see syngas)
Bisphenol-A (BPA)	329	18	1		293	< 0,5 Mt		4	
Diphenyl carbonate (DPC)	353	21	1		13	low volume, low priority		4	

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
Adipic acid	375	20	2	yes	590	> 0,5 Mt		4	Inclusion would be priority 1 (EU ETS), but this would apply only to the direct process without precursors
α-Methyl-benzyl alcohol	399	17	10		13	low volume, low priority		3	Appears only as intermediate in integrated process, low priority
Nylon salt	476	28	1		171	low volume, low priority		5	
Polyamide 6,6 (PA-6,6)	477	29	0		2.150	> 1Mt		5	
Styrene	486	18	4	yes	2.967	> 1Mt		3	Inclusion would be priority 1 (EU ETS), but this would apply only to the direct process without precursors
Expansible polystyrene (EPS)	487	19	0		1.511	> 1Mt		3	As monomer is in EU ETS, despite high complexity, high volume could still justify inclusion
Polystyrene (GPPS and HIPS)	487	19	0		1.497	> 1Mt		3	As monomer is in EU ETS, despite high complexity, high volume could still justify inclusion
Propylene oxide (PO)	495	25	4		1.514	> 1Mt		4	
Polypropylene glycol (PPG)	496	26	2		477	< 0,5 Mt	difficult	4	Only relevant if polyurethanes included
Propylene glycol (PG)	502	26	0		800	< 0,5 Mt		4	Possibly needed for PU
Styrene-Acrylonitrile copolymer (SAN)	518	20	0		180	low volume, low priority		4	
Cyclohexanol	526	19	0		15	low volume, low priority		5	
Acrylonitrile-Butadiene-Styrene copolymer (ABS)	542	24	0		655	> 0,5 Mt		4	

Substance name as in report	Complexity WITH precursors	Number of precursors	number of uses as precursor	Is important (EU ETS benchmark)?	Production (EU-27) [kt / yr]	Is important (>1Mt/a)? Expert judgment	Can be identified?	Overall result	Comments
PU made of TDI and PPG	559	32	0		867	> 0,5 Mt	difficult	4	Substance only for illustration purposes. Pu are hard to identify
PU made of MDI and PPG	611	34	0		867	> 0,5 Mt	difficult	4	Substance only for illustration purposes. Pu are hard to identify
Cyclohexanone	655	19	2		47	low volume, low priority		5	
Polycarbonate from BPA	715	24	0		1.127	> 1Mt	difficult	5	
ε-Caprolactam	788	27	1		21	low volume, low priority		5	Production volume argument overridden by use in high-volume PA-6
Polyamide 6 (PA-6)	789	28	0		2.150	> 1Mt		5	

5 Task 3. Identification of relevant precursors to be included in CBAM

5.1 Objectives and summary of the task

The objective of Task 3 is to identify precursors of goods under the existing CBAM scope, and of the goods identified under Task 2, and whether they should be included in the CBAM.

For this purpose, the value chains of CBAM goods and goods newly identified under Task 2 were mapped as a first step. Next, an estimate of production emissions of the precursors were used to identify precursors contributing significantly to the emissions throughout the value chain, making it environmentally desirable to include the precursor in the CBAM. A similar trade analysis as under Task 1.2 was not conducted, as all results were already included in Task 1. However, the findings of Task 1 were used to differentiate the priority for inclusion. In the final step of Task 3, the assessment of technical feasibility for inclusion (using the methodology developed under Task 2) was carried out to result in a prioritised list of precursors to be potentially included in the CBAM.

Regarding how far upstream the value chain should be followed, the Commission confirmed that the boundaries of the EU ETS should serve as the main guiding principle. This means that, for example, mining and fuel extraction should be omitted. The CBAM should be kept as simple as possible.

5.2 Results: Precursors of current CBAM scope

The assessment of precursors' contribution to embedded emissions to the current scope of the CBAM builds on work performed for developing a discussion note (shared with the CBAM expert group in January 2023)¹¹⁷ under a previous project for DG TAXUD. In this Task we add the assessment of criteria regarding the technical feasibility of MRV, as in Task 2 (only criteria 1 and 2a, and sometimes 2c are relevant, as has been explained in section 4.4.1).

In the graphs on production processes, blue fields indicate the precursors under discussion in this chapter. Green fields mean materials which are considered to be formal precursors according to the current scope of the CBAM, i.e. that their embedded emissions need to be considered, while materials in red fields are normal process inputs which are not considered precursors, so that only their direct emissions occurring during the production process have to be accounted for, not any embedded emissions. Only the most relevant inputs are shown.

¹¹⁷ <https://ec.europa.eu/transparency/expert-groups-register/core/api/front/document/101351/download>

5.2.1 Cement

Calcined lime, dolime and magnesia¹¹⁸ could theoretically be precursors, although usually the non-calcined minerals are used in the process. Using such inputs could reduce embedded emissions of the produced cement only if those precursors are *not* included. Therefore, their declaration as formal precursors would ensure equal treatment of installations performing the calcination as part of the clinker burning process and installation which use already burned minerals. However, as separate burning would be less energy efficient, it is not highly likely that separate calcination would become a common practice. The inclusion of these goods as precursors is therefore low priority for the cement sector. However, they would also be relevant in the iron and steel sector (see section 5.2.2).

Furthermore, it is worth mentioning that calcined lime and dolime as individual CBAM goods would have an impact in the building industry and chemical industry, and magnesia in the production of refractories (see ceramics, section 4.4.5).

Technical feasibility assessment

Criterion 1: Goods must be identifiable: No issues regarding identification found. As “qualifying parameters”, the degree of calcination (e.g. expressed as residual CO₂ in the material) could be added.

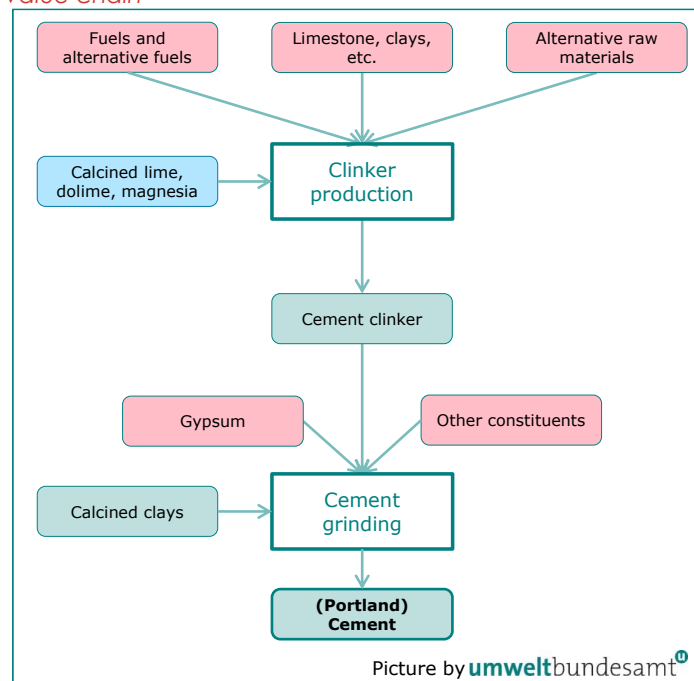
Criterion 2a: Production processes and routes need to be known (and not overly complex): All production processes are covered by EU ETS MRV rules, and processes' system boundaries are simple. No obstacles found.

¹¹⁸ CN 2518 20 00 Dolime: Calcined or sintered dolomite (excl. broken or crushed dolomite for concrete aggregates, road metalling or railway or other ballast)

2519 90 Magnesia: Fused magnesia; dead-burned "sintered" magnesia, whether or not containing small quantities of other oxides added before sintering; other magnesium oxide (excl. natural magnesium carbonate "magnesite")

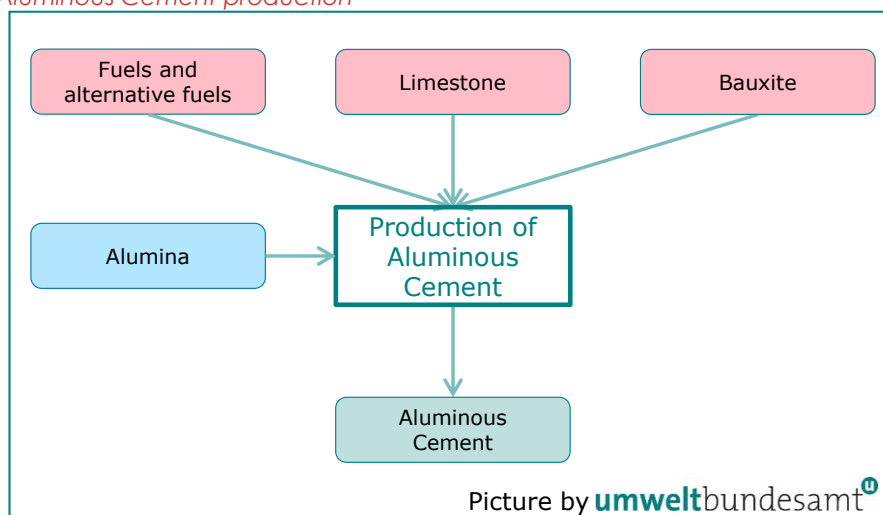
2522 Lime: Quicklime, slaked lime and hydraulic lime (excl. pure calcium oxide and calcium hydroxide)

Figure 11 Cement value chain



For aluminous cement: **Alumina** (or even aluminium hydroxide) would be relevant as precursor: The technical feasibility of inclusion has already been discussed as new good to be added to the aluminium sector (see section 4.4.7). No obstacles were identified there. In terms of relevance to produce aluminous cement the argumentation would be the same as above for calcined minerals for clinker production: If the precursor were included in the CBAM, the embedded emissions would be always the same independent of calcination directly in the process or as a separate production step.

Figure 12 Aluminous Cement production



5.2.2 Iron & Steel

Possible precursors are:

Coke¹¹⁹ (HS heading 2704): Relevance: Embedded emissions of coke would increase the embedded emissions of steel products by 5 to 6%. Coke can also be a precursor to other products, in particular ferro-alloys and non-ferrous metals (if included in the CBAM).

It is important to note that the production of coke may lead to marketable co-products: Coke production consists of heating coal¹²⁰ under exclusion from air. While the non-fugitive carbon remains in the coke, fugitive substances are collected as coke oven gas (usually used as fuel in the process and in the adjacent steel plant), as well as liquid fractions (coal oils and tar). While some coke producers use these fractions as fuels, these fractions can be used as raw materials for certain organic chemicals (in particular aromatics such as benzene, toluene, xylenes, naphthalene and higher aromatics). Some coke producers sell only the oils or tars, while others extract or distil some more valuable fractions therefrom. The HS/CN classification lists such specific fractions, which have to be distinguished from e.g. pure benzene. If a coke producer produces also pure benzene, then the respective CN code from the chemicals sector would have to be applied (see section 4.4.11 on organic chemicals). Therefore, the inclusion of coke in the CBAM should be accompanied with the inclusion of the mentioned co-products, which would correspond to the goods listed in Table 26 (without CN codes falling under the chemicals sector).

Table 26 Relevant co-products from coke production

HS / CN code	Description
2706	Tar distilled from coal, from lignite or from peat, and other mineral tars, whether or not dehydrated or partially distilled, incl. reconstituted tars
2707	Oils and other products of the distillation of high temperature coal tar; similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents
270710	Benzol "benzene" containing > 50% of benzene (excl. chemically defined)
270720	Toluol "toluene" containing > 50% of toluene (excl. chemically defined)
270730	Xylol "xylenes" containing > 50% of xylenes (excl. chemically defined)
270740	Naphthalene containing > 50% of naphthalene (excl. chemically defined)
270750	Aromatic hydrocarbon mixtures of which >= 65% by volume, incl. losses, distils at 250°C by the ASTM D 86 method (excl. chemically defined compounds)
270791	Creosote oils (excl. chemically defined)
270799	Oils and other products of the distillation of high temperature coal tars; similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents (excl. chemically-defined compounds, benzol "benzene", toluol "toluene", xylol "xylenes", naphthalene, aromatic hydrocarbon mixtures of subheading 2707.50, and creosote oils)
27079911	Crude light oils from the distillation of high temperature coal tars, of which >= 90% by volume distils at temperatures of up to 200°C (excl. chemically defined)

¹¹⁹ HS 2704 00 "Coke and semi-coke of coal, of lignite or of peat, whether or not agglomerated; retort carbon" – Petroleum coke (petcoke) under HS codes 2713 11 and 2713 12 would not be applicable, as it is not used in the steel industry for technical reasons.

¹²⁰ Only certain qualities of coal are useable.

HS / CN code	Description
27079919	Crude light oils from the distillation of high temperature coal tars (excl. those of which $\geq 90\%$ by volume distils at temperatures of up to 200°C and chemically defined compounds)
27079920	Anthracene (excl. chemically defined); sulphuretted toppings from the first distillation of high temperature coal tars
27079950	Pyridine, quinoline, acridine, aniline bases and other basic products of the first distillation of high temperature coal tars, n.e.s.
27079980	Phenols containing $> 50\%$ of phenols (excl. chemically defined)
27079991	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, for the manufacture of carbon in heading 2803
27079999	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, n.e.s.
2708	Pitch and pitch coke, obtained from coal tar or from other mineral tars
270810	Pitch obtained from coal tar or from other mineral tars
270820	Pitch coke obtained from coal tar or from other mineral tars

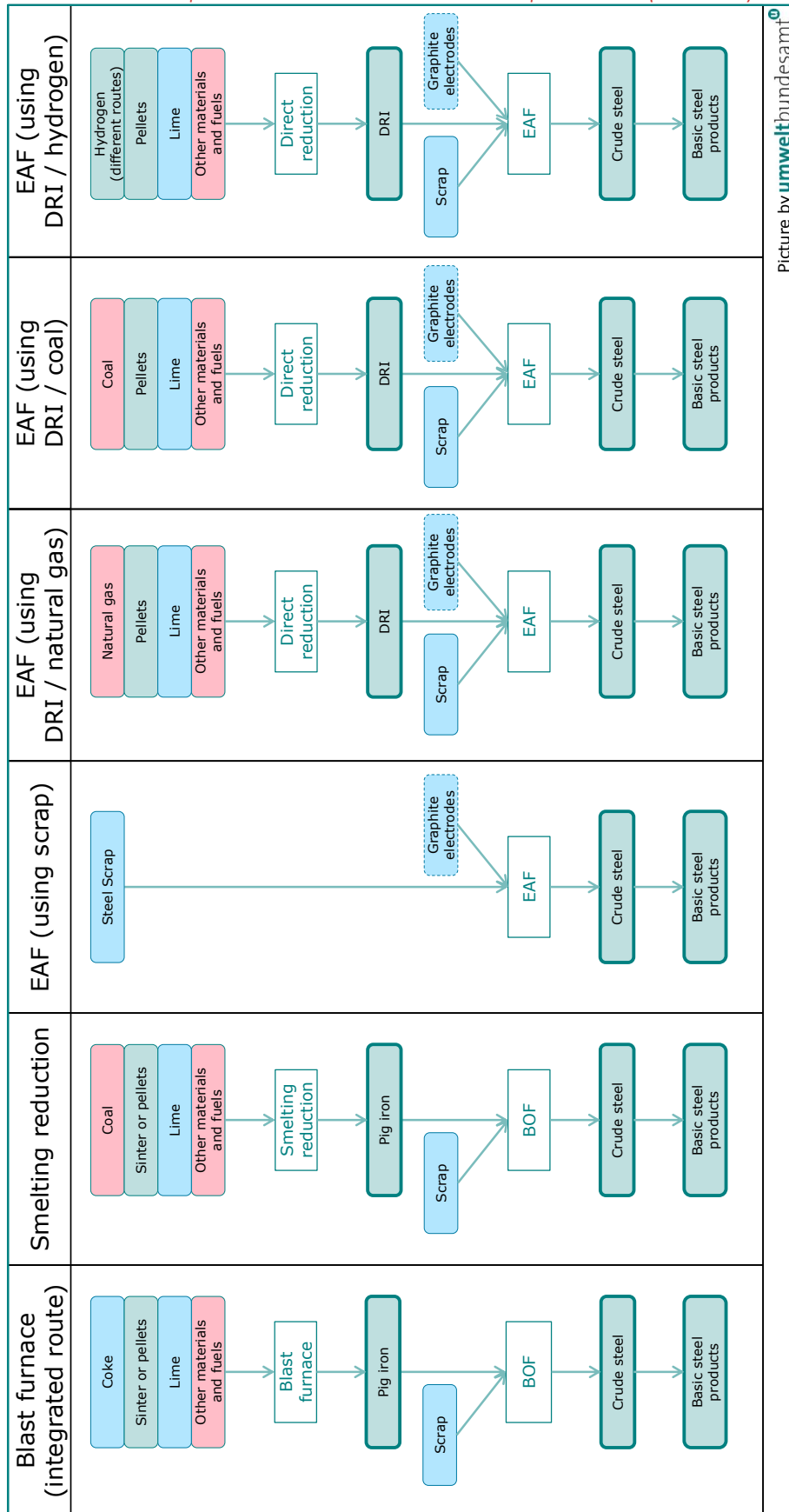
Lime (and dolime, magnesita): Relevance: The inclusion as precursor could increase the embedded emissions of steel products by 3 to 4%. These precursors are relevant also for other sectors/goods. The technical feasibility was already assessed in section 5.2.1 (cement).

Alloys: For technical feasibility see section 4.4.6 regarding CN 720221 (ferro-silicon) and 720230 (ferro-silico-manganese). The relevance for inclusion would be given for high-alloy steels at a similar order of magnitude as for the already included alloys FeNi, FeCr and FeMn (no data on embedded emissions available).

For the production of some ferro-alloys already included in the CBAM, a theoretical possibility is the production via silicothermic reduction, for which silicon would be a precursor. However, as this is an older route to low-carbon ferro-manganese only, it is not further considered.

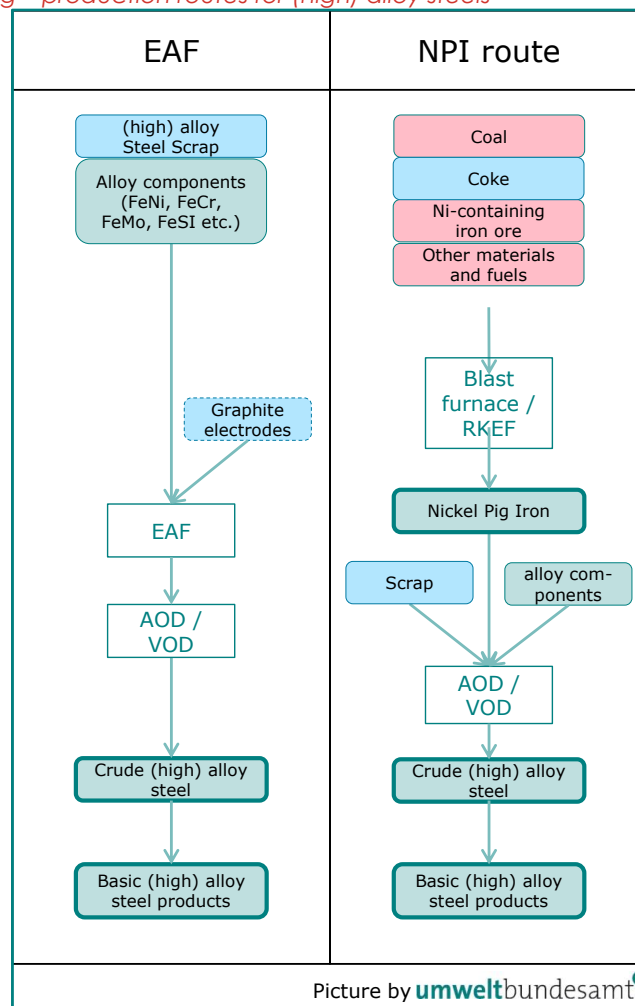
Graphite electrodes for electric arc furnaces: Relevance rather low since the consumption levels of 2 – 6 kg / t steel do not lead to a significant increase of embedded emissions of steel. Therefore, they are not further considered for inclusion.

Figure 13 Value chains and production routes in carbon steel production (overview)



Picture by [umweltbundesamt](#)⁹

Figure 14 Steelmaking – production routes for (high) alloy steels



Scrap could also be a precursor. Depending on the approach for attributing emissions to co-products, the inclusion of scrap in the CBAM is technically feasible, albeit difficult. The most consistent dealing with scrap with regard to its embedded emissions is to maintain the current approach (zero embedded emissions), which would mean that it is safe to exclude the scrap-related CN codes from the CBAM. Care should be taken that a uniform approach to scrap (and other waste for recycling, such as cullet for glass) is adopted. However, in the steel sector, the differentiation between recycled and virgin material is less pronounced than in the aluminium sector, as even in primary steel production routes, scrap is always added at the stage of the basic oxygen converter (BOF).

At this point in time, we do not propose a specific approach towards scrap.

Technical feasibility assessment for coke

Criterion 1: Goods must be identifiable: No issues regarding identification found.

Criterion 2a: Production processes and routes need to be known (and not overly complex): The production process is covered by EU ETS MRV rules, and processes' system

boundaries are simple. There is a product benchmark with definition of system boundaries. No obstacles found.

Criterion 2c: Identification of special MRV difficulties. Under this criterion, it has to be mentioned that a methodology for attributing embedded emissions to co-products is required, considering that some of the co-products are not necessarily sold, but used immediately in the process. Not all coke producers have the same range of co-products. A mass balance approach seems feasible. Although also an energy-based approach would be feasible, it seems less useful, as there may be significant differences in the value of co-products beyond their energy content but being the basis for chemicals production. The varying product range may be a reason for making the selection of an attribution method more difficult than for other cases mentioned in this report.

Conclusion for coke: On the side of technical feasibility, there is no absolute obstacle for inclusion in the CBAM. However, the link of its co-products to the chemicals sector may require equal treatment in particular with the refinery sector, as both coke co-products and refining products serve as raw materials or precursors for organic chemicals (and therefore to polymers).

5.2.3 Fertilizers

N.A. (no relevant precursors identified compared to the current situation).

5.2.4 Aluminium

Alumina¹²¹ is a logical precursor for aluminium. As already discussed under Task 2 (section 4.4.7), also aluminium hydroxide¹²² should be treated alike. The technical feasibility assessment has been carried out in section 4.4.7. In terms of relevance, embedded emissions of alumina would contribute 30–40% to the embedded emissions of (primary) unwrought aluminium if only direct emissions are considered, but only about 10% if indirect emissions are considered, too.

Prebaked anodes¹²³: Relevance: Contribution to embedded emissions was estimated to be 2,5 % (if indirect emissions included), or 10% if only direct emissions were relevant for the aluminium sector. The emissions of this product in the EU are less than a million tonnes of CO₂. Inclusion of the electrodes in the CBAM as good (as condition for treating it as precursor) is therefore not a priority. However, if included, the electrodes' own potential precursors would be petroleum coke (petcoke¹²⁴), which would be a CBAM good if refinery products were included in the CBAM, and coal tar and pitch, which are co-products of coke (see section 5.2.2) on iron and steel). Furthermore, if those precursors are included in the CBAM, also "electrode paste"¹²⁵ (for Söderberg cells) should be included for equal treatment of all aluminium production technologies.

¹²¹ CN 28182000

¹²² CN 28183000

¹²³ CN 8545 19 00 "Electrodes of graphite or other carbon, for electrical purposes (excl. those used for furnaces)"

¹²⁴ CN 2713 11 00 and 2713 12 00

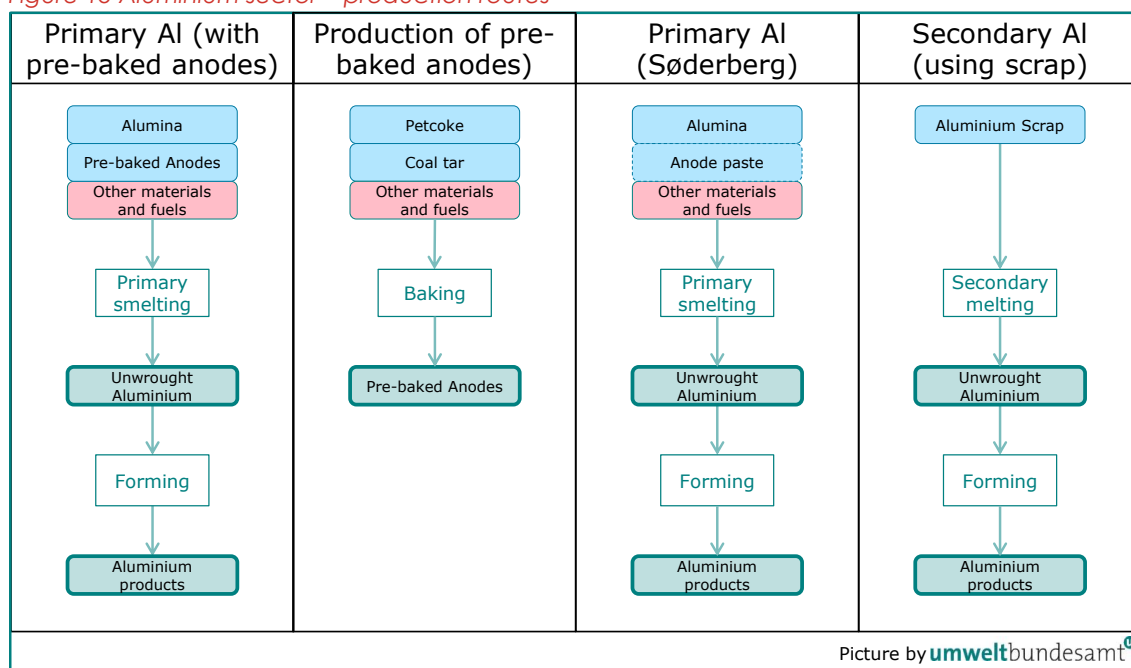
¹²⁵ CN 3801 30 00 – Carbonaceous pastes for electrodes and similar pastes for furnace linings

In terms of technical feasibility there would be no obstacles, as the rules of the EU ETS (including a product benchmark and system boundaries) can be used for pre-baked electrodes.

Aluminium scrap: The inclusion of scrap in the CBAM is technically feasible, albeit difficult. The most consistent dealing with scrap with regard to its embedded emissions is to maintain the current approach (zero embedded emissions), which would mean that it is safe to exclude the scrap-related CN codes from the CBAM. Care should be taken that a uniform approach to scrap (and other waste for recycling, such as cullet for glass) is adopted for all relevant sectors. At this point in time, we do not propose a specific approach towards scrap.

Alloying elements: These are added after the production of primary aluminium. Although theoretically it would be possible to produce alloys at the stage of producing unwrought aluminium, it is more common that the alloy is produced only specifically for each type of aluminium product (e.g. before producing cast products). Most important alloying elements for aluminium are magnesium, silicon, copper, zinc, but many more elements are used in smaller quantities. However, unlike steel, typical overall alloy contents are low¹²⁶. As the analysis of the non-ferrous metals sector (to which these elements would belong) has shown in general that it is currently too complex for inclusion, it does not appear feasible to include alloying elements for aluminium at this stage.

Figure 15 Aluminium sector – production routes



¹²⁶ Usually lower than 5% in total alloying elements content, with some few exceptions for e.g. aerospace applications. A list of alloys is found e.g. in <https://www.aluminum.org/sites/default/files/2021-11/TealSheet.pdf>

5.2.5 Hydrogen

N.A. (no relevant precursors identified compared to the current situation).

5.3 Results: Precursors for sectors to be potentially introduced following Task 2

5.3.1 Pulp and paper

The production of **pulp** (for CN codes see section 4.4.3) is based either on biomass or on recycled pulp (i.e. the paper equivalent to scrap). Therefore, depending on a decision on whether scrap should be treated as precursor with embedded emissions or not, recycled pulp would be the only precursor of relevance. As the monitoring of the recycling process is covered by EU ETS rules and having a specific product benchmark, there are no technical difficulties to be expected.

Paper is produced from diverse grades of pulp (including recycled pulp). There are no additional precursors of relevance.

5.3.2 Glass

Glass is produced from mixtures of inorganic substances and minerals, and possibly cullet (the glass equivalent of scrap). The exact recipe depends on the intended use of the glass. The high-volume glass products identified under Task 2 (see section 4.4.4) usually consist of silica sand (mining product), soda ash (an inorganic chemical, assessed in section 4.4.10), limestone or burnt lime or dolime, varying amounts of cullet (from process or post-consumer), and smaller amounts of additives for colouring or decolouring.

For soda ash, no technical difficulties are expected (covered by the EU ETS. Note that soda ash could be included in the CBAM as stand-alone good, even if no other chemicals were included in the CBAM, due to its role as precursor for the glass sector, or because the sector has significant emissions on its own. For burnt lime and dolime there are no technical obstacles either (see section 5.2.2 on iron and steel precursors). With regard to cullet, the general decision on whether scrap should be treated as precursor with embedded emissions or not, cullet would be the only other precursor of relevance.

5.3.3 Ceramics

Building bricks and rooftiles are produced predominantly from clay, which as a mining product would not be considered a precursor. **Fine ceramics** (tiles, tableware, sanitary ware) are usually produced from higher quality clays including kaolinic clays. It is possible that clays falling into the category "calcined clays" are used too. This product is already included in the CBAM as precursor in the cement sector. For being sure to cover all possible precursors, these calcined clays should be considered as precursor in the ceramics sector, too. Under the EU ETS, also "spray dried powder" is defined as ceramic product with its own benchmark. However, this is an intermediate and seems not to be traded (except within the same company) – we could not identify a specific CN code for it. It might be in the same group as calcined clays, or by CN 25087000 (Chamotte or dinas earths).

Regarding recycling, the majority of post-consumer wastes of bricks and the like are used in the building sector finely crushed as additives to cement and concrete), but not at large scale in the production of new bricks. We therefore do not consider the topic of "scrap" as relevant in this sector.

Refractories: Depending on the composition of the product, alumina (see section 4.4.7) can be a precursor. Also calcined or dead-burnt magnesia and dolime (assessed under cement, section 5.2.1) can be precursors for refractories.

5.3.4 Ferro-Alloys

For Ferro-silicon and Ferro-silicon-manganese, identified in Task2, only the respective ores (mining products) are used as raw materials. Coke used as reducing agent and graphite electrodes used in electric arc furnaces could be relevant precursors. They are discussed for the steel sector in section 5.2.2.

5.3.5 Non-Ferrous Metals

N.A. (identified goods were considered too complex for inclusion in the CBAM at this stage).

5.3.6 Refineries

N.A. (no relevant precursors identified, as raw materials come directly from mining processes, i.e. extraction of mineral oil / oil shales).

5.3.7 Chemicals & Polymers

There are no precursors that have not yet been discussed in other parts of this report. The main starting points of the value chains of all organic chemicals have been identified in section 3.3 to be:

- Natural gas: Mining product, not considered precursor.
- Mineral oil: In this case, certain refinery products (in particular naphtha, pygas and reformat, assessed in section 4.4.9 and 4.5.2) may be precursors.
- Coal: Mining product not considered precursor.
- Coke-oven co-products: Some of them (coal tar, light oils, but also distillates like raw benzene) can be precursors for chemicals and polymers, see sections 4.5.2 and 5.2.2.
- Biomass: As a raw material, the production is not an industrial process covered by the EU ETS, like mining products.

Beyond these raw materials, all chemicals appearing as precursors of the chemicals and polymers originally selected for assessment have been included themselves in the assessment of Task 2.3 and 2.4. Therefore, no further precursors remain for discussion here in Task 3.

5.4 Conclusions on precursors

Summing up sections 5.2 and 5.3, the project team's conclusions on the possible inclusion of precursors in the CBAM is that there are no top-priority substances, but a couple of goods that may be useful to be included to provide a better coverage of embedded emissions of the existing CBAM scope. If goods like coke were included, arguments beyond the technical feasibility should be considered, like the equal treatment of precursors for the chemical industry, i.e. co-products of the coking process (tars, oils, aromatics) and refinery products. As several goods identified as precursors of the existing CBAM scope have an application in several sectors, we present the conclusions in a sector-independent list.

Table 27 presents the result for the precursors of the current CBAM scope, and Table 28 for the scope considered for inclusion under Task 2, i.e. the horizontal CBAM extension. We use a similar colour code as for the conclusions on Task 2 in section 4.5.1:

- Light green indicates that there are no technical obstacles for inclusion in the CBAM (although, as mentioned in the beginning of this section, there is no absolute priority need for including them).
- Goods of the type like scrap are marked in yellow. This is to indicate that they could be classified like the next (amber) category, but there is currently a decision pending on their treatment in the methodology. Hence, the project team considers them as "difficult to include", but with a different aspect than the purely amber ones.
- Goods which seem more difficult to include are marked in amber. However, this low prioritisation is mostly not due to technical difficulty, but rather to the fact that a wider decision is needed on including those goods. This applies to coke and its co-products and refinery products. Here the decision is needed to include either all of them or none of them for equal treatment as precursors in the chemical sector. The alternative would be to include coke as stand-alone precursor (mostly for the iron and steel sector and ferro-alloys) and treat chemicals (if included in the CBAM at all) without precursors from the refinery and coke production.
- Goods which currently appear too difficult for inclusion would be marked in red, but none have been identified.

*Table 27 Precursors for the existing CBAM scope and their prioritisation for inclusion in the CBAM.
For colour code see text above*

Precursor	HS / CN code	Description	Sector of the precursor's possible use
Lime	252210	Quicklime	Cement; steel; Chemical industry ¹²⁷
	252220	Slaked lime	Chemicals ¹²⁷
	252230	Hydraulic lime (excl. pure calcium oxide and calcium hydroxide)	Cement, steel
Dolime	251820	Calcined or sintered dolomite (excl. broken or crushed dolomite for concrete aggregates, road metalling or railway or other ballast)	Cement, steel, refractories (ceramics)
Magnesia	251990	Fused magnesia; dead-burned "sintered" magnesia, whether or not containing small quantities of other oxides added before sintering; other magnesium oxide (excl. natural magnesium carbonate "magnesite")	Cement, steel, refractories (ceramics)

¹²⁷ Soda ash, calcium carbide: In both cases, limestone is more likely to be used. However, calcined lime (calcium oxide), both as solid and as slaked lime, can be used for many processes where alkalines are required, although this was not specifically mentioned in the note annexed to this report, as lime was considered currently out of the scope of the CBAM.

Precursor	HS / CN code	Description	Sector of the precursor's possible use
	25199010	Magnesium oxide, whether or not pure (excl. calcined natural magnesium carbonate)	Steel, refractories (ceramics), chemicals ¹²⁸
	25199030	Dead-burned "sintered" magnesia, whether or not containing small quantities of other oxides added before sintering	Refractories (ceramics)
	25199090	Fused magnesia	Refractories (ceramics),
Alumina			
	281820	Aluminium oxide (excl. artificial corundum)	Aluminium, cement ¹²⁹
Coke			
	2704	Coke and semi-coke of coal, of lignite or of peat, whether or not agglomerated; retort carbon	Steel, ferro-alloys, chemicals ¹³⁰
Coke co-products			
	2706	Tar distilled from coal, from lignite or from peat, and other mineral tars, whether or not dehydrated or partially distilled, incl. reconstituted tars	Chemicals ¹³¹
	270710	Benzol "benzene" containing > 50% of benzene (excl. chemically defined)	Chemicals ¹³¹
	270720	Toluol "toluene" containing > 50% of toluene (excl. chemically defined)	Chemicals ¹³¹
	270730	Xylol "xylenes" containing > 50% of xylenes (excl. chemically defined)	Chemicals ¹³¹
	270740	Naphthalene containing > 50% of naphthalene (excl. chemically defined)	Chemicals ¹³²
	270750	Aromatic hydrocarbon mixtures of which >= 65% by volume, incl. losses, distils at 250°C by the ASTM D 86 method (excl. chemically defined compounds)	Chemicals ¹³¹
	270791	Creosote oils (excl. chemically defined)	Chemicals ¹³³
	270799	Oils and other products of the distillation of high temperature coal tars; similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents (excl. chemically-defined compounds, benzol "benzene", toluol "toluene", xylol "xylenes", naphthalene, aromatic hydrocarbon mixtures of subheading 2707.50, and creosote oils)	Chemicals ¹³¹
	27079911	Crude light oils from the distillation of high temperature coal tars, of which >= 90% by volume distils at temperatures of up to 200°C (excl. chemically defined)	Chemicals ¹³¹
	27079919	Crude light oils from the distillation of high temperature coal tars (excl. those of which >= 90% by volume distils at temperatures of up to 200°C and chemically defined compounds)	Chemicals ¹³¹
	27079920	Anthracene (excl. chemically defined); sulphuretted toppings from the first distillation of high temperature coal tars	Chemicals ¹³³

¹²⁸ In the current assessment there were no processes with magnesium oxide involved.

¹²⁹ As precursor to aluminous cement.

¹³⁰ For example soda ash, calcium carbide.

¹³¹ In particular BTX: Benzene, Toluene, Xylenes, but also other chemicals not assessed.

¹³² In particular Naphthalene

¹³³ No specific substance in our assessment

Precursor	HS / CN code	Description	Sector of the precursor's possible use
	27079950	Pyridine, quinoline, acridine, aniline bases and other basic products of the first distillation of high temperature coal tars, n.e.s.	Chemicals ¹³³
	27079980	Phenols containing > 50% of phenols (excl. chemically defined)	Chemicals ¹³⁴
	27079991	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, for the manufacture of carbon in heading 2803	Chemicals ¹³⁵
	27079999	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, n.e.s.	Chemicals
	270810	Pitch obtained from coal tar or from other mineral tars	(¹³⁶)
	270820	Pitch coke obtained from coal tar or from other mineral tars	(¹³⁶)
Graphite electrodes (for Iron & steel and ferro-alloys)			
	854511	Electrodes of graphite or other carbon, for electric furnaces	Steel, ferro-alloys, chemicals ¹³⁷
Prebaked Anodes (for aluminium production)			
	854519	Electrodes of graphite or other carbon, for electrical purposes (excl. those used for furnaces)	Primary aluminium
Electrode paste (for Aluminium production)			
	380130	Carbonaceous pastes for electrodes and similar pastes for furnace linings	Primary aluminium, ferro-alloys
Precursors for anodes			
		Petcoke - see refinery products	Primary aluminium, ferro-alloys
		Coal tar and pitch - see coke byproducts	Primary aluminium, ferro-alloys
Scrap - Iron & Steel			
	720410	Waste and scrap, of cast iron (excl. radioactive)	Steel
	720421	Waste and scrap of stainless steel (excl. radioactive, and waste and scrap of batteries and electric accumulators)	Steel
	720429	Waste and scrap of alloy steel (excl. stainless steel, and waste and scrap, radioactive, or waste and scrap from batteries and electric accumulators)	Steel
	720430	Waste and scrap of tinned iron or steel (excl. radioactive, and waste and scrap of batteries and electric accumulators)	Steel
	720441	Turnings, shavings, chips, milling waste, sawdust, filings, trimmings and stampings of iron or steel, whether or not in bundles (excl. those of cast iron, alloy steel or tinned iron or steel)	Steel
	720449	Waste and scrap of iron or steel (excl. slag, scale and other waste of the production of iron and steel; radioactive waste and scrap; fragments of pigs, blocks or other primary forms of pig iron or	Steel

¹³⁴ Phenol, but more likely other phenols not included in our assessment

¹³⁵ Carbon black

¹³⁶ This might be a precursor to the production of anodes in the Aluminium sector or for graphite electrodes (steel, ferro-alloys, calcium carbide), i.e. for products themselves only under consideration and not yet in the CBAM

¹³⁷ Calcium carbide

Precursor	HS / CN code	Description	Sector of the precursor's possible use
		spiegeleisen; waste and scrap of cast iron, alloy steel or tinned iron or steel; turnings, shavings, chips, milling waste, sawdust, filings, trimmings and stampings; waste and scrap of primary cells, primary batteries and electric accumulators)	
	720450	Remelting scrap ingots of iron or steel (excl. products whose chemical composition conforms to the definitions of pig iron, spiegeleisen, or ferro-alloys)	
Scrap - Aluminium			
	76020011	Turnings, shavings, chips, milling waste, sawdust and filings, of aluminium; waste of coloured, coated or bonded sheets and foil, of a thickness "excl. any backing" of <= 0,2 mm of aluminium	
	76020019	Waste of aluminium, incl. faulty workpieces and workpieces which have become unusable in the course of production or processing (excl. slag, scale and other waste from the production of iron or steel, containing recyclable aluminium in the form of silicates, ingots and other primary forms, of smelted waste or scrap, of aluminium, ash or the residues of the production of aluminium, and waste in heading 7602.00.11)	
	76020090	Scrap of aluminium (excl. slags, scale and the like from iron and steel production, containing recoverable aluminium in the form of silicates, ingots or other similar unwrought shapes, of remelted waste and scrap, of aluminium, and ashes and residues from aluminium production)	

Table 28 Precursors for goods considered for horizontal CBAM extension, and their prioritisation for inclusion in the CBAM. For colour code see text above. Precursors for goods considered for horizontal CBAM extension, and their prioritisation for inclusion in the CBAM. For colour code see text above

Precursor	HS / CN code	Description	Sector of the precursor's possible use
Paper waste and scrap			
	470710	Recovered "waste and scrap" paper or paperboard of unbleached Paper kraft paper, corrugated paper or corrugated paperboard	
	470720	Recovered "waste and scrap" paper or paperboard made mainly of bleached chemical pulp, not coloured in the mass	Paper
	47073010	Old and unsold newspapers and magazines, telephone directories, Paper brochures and printed advertising material	
	47073090	"Waste and scrap" of paper or paperboard made mainly of mechanical pulp (excl. old and unsold newspapers and magazines, telephone directories, brochures and printed advertising material)	Paper
	47079010	Unsorted, recovered "waste and scrap" paper or paperboard (excl. Paper paper wool)	
	47079090	Sorted, recovered "waste and scrap" paper or paperboard (excl. waste and scrap of unbleached kraft paper or kraft paperboard, or of corrugated paper or corrugated paperboard, that of paper or paperboard made mainly of bleached chemical pulp not coloured in the mass, that of paper or paperboard made mainly of mechanical pulp, and paper wool)	Paper
Cullet (Glass)			
	70010010	Cullet and other waste and scrap of glass (excl. glass in the form of powder, granules or flakes, and activated glass of heading 8549)	Glass
Soda ash for glass			
	283620	Disodium carbonate	Glass
Refinery products for chemicals			

Precursor	HS / CN code	Description	Sector of the precursor's possible use
		Naphtha	Chemicals ¹³⁸
		Pyrolysis gas	Chemicals ¹³⁹
		Reformate	Chemicals ¹³⁹
Coking co-products for chemicals			
	2706	Coal tar	Chemicals ¹³⁹
	2707	Light oils from coking	Chemicals ¹³⁹

¹³⁸ Large number of organic chemicals and polymers (olefins and their downstream substances)

¹³⁹ Large number of organic chemicals and polymers (aromatics and their downstream substances)

6 Task 4. Stakeholder consultation related to horizontal and vertical scope extensions

6.1 Objectives of the task

The **objective** of Task 4 is to gather insights on the horizontal and vertical expansion of the CBAM scope beyond sectors already covered in Annex I of the CBAM Regulation through stakeholder engagement activities. These activities are organised in three stages, with only the first stage concluded so far:

1. **Scoping interviews** with a select stakeholder group have been conducted during the inception phase. The outcomes of these interviews were summarised in the inception report and provided again in the T4 Annex section.
2. **A stakeholder survey** to ensure feedback from a broad range of stakeholders while still obtaining manageable responses, facilitating the incorporation of valuable insights. The survey is set to gather input on challenges, concerns, and suggestions related to the expansion of the CBAM to other goods not in Annex I of the CBAM Regulation. This will build on the insights already obtained through the scoping interviews. Stakeholder input will also be sought regarding specific CN codes/sectors suggested to be potentially included in the CBAM, including argumentation and supporting evidence for these suggestions.
3. **Targeted in-depth interviews** with stakeholders following the stakeholder survey to delve deeper into qualitative information obtained from the survey as well as technical questions not suitable for a survey. This aims to obtain in-depth insights and information gaps that the desk-based research and analysis could not fulfil or to validate certain findings. These in-depth interviews will primarily be with industry associations. Still, they may also include respondents in the survey who may have relevant insights worth further exploring relevant for the outcomes of the priority sectors and products for potential CBAM inclusion and have indicated a willingness to participate in in-depth interviews.

6.2 Results

All activities under Task 4 were conducted in coordination with Task 7 to ensure a coherent and efficient approach in engaging stakeholders relevant for both Task 4 and 7. For the results of this task, we refer to the *T4 Annexes* enclosed at the end of this report.

7 Task 5: Assessment of the scale of emissions from transport services linked to the transport of CBAM goods outside the EU

7.1 Summary of the objectives and results of task 5

This task was devoted to developing a methodology for assessing the scale of non-EU transport emissions associated with CBAM goods. This includes an estimate of the non-EU transport emissions of goods in the T2 and T3 list into the EU.

The subtasks are the following:

T5.1. The analysis of common transport modes for CBAM goods throughout their supply chains from the step they become CBAM onwards, shows larger shares of road and rail transport modes used for CBAM goods than for Precursors. Precursors to the EU tend to be moved more by sea and rail. Implication: the use of sea transport for the movement of precursors may contribute to strengthen climate objectives, which relates to low risk of carbon leakage.

T5.2 – The analysis of transport modes for CBAM goods outside EU ETS scope: characterising emission profiles and considering key factors. There are no internationally accepted emission intensity databases where transport operators can rely on. Yet the available information clearly indicates that transport emissions outside the EU tend to be slightly higher, and Air transport is the most intensive mode outside the EU, followed by Road, Rail, and Sea. Implication: There is need to set up an internationally accepted emission intensity database for the transport sector, in order to more reliably assess the likelihood of carbon leakage based on CBAM transport emissions.

T5.3 – The analysis of typical emission intensities for freight transport modes (g CO₂/km-tonne) across various geographical regions shows that emission intensities differ across regions globally, especially for rail and air transport. For road and sea transport the differences across regions are marginal, and in fact, international sea shipping is the same. The EU has lower emission intensity in rail and air transport, whilst for road and sea, not significant differences can be seen. Implication: should CBAM extended goods prioritise road transport more than sea transport, the likelihood of increased emissions on transport due to horizontal scope extension increases. If production is transferred to geographical locations such as Asia, where to ship goods to the EU is substantially based on sea transport and rail transport, then there would be a low probability of carbon leakage.

T5.4 – The emission intensities and transportation profiles of the newly proposed CN codes for extension may have larger share of transport emissions in total emissions compared to the current CBAM goods. There is a significant increase of road transport of the new CBAM extension codes than the previous CBAM goods in scope. Implications: increased transport emissions related to the extension of CBAM goods horizontally, requires further assessment at the level of the selected CN codes to be extended and their specific transport emission profiles, in order to assess the overall effect on carbon leakage.

T5.5 – The risk assessment for carbon leakage based on the impact of EU ETS expansion to maritime and road transport on CBAM goods, with scenario analysis for carbon price evolution shows a low probability of carbon leakage, and a difficulty to decouple the effect of EU Member States' Carbon Price Instruments (CPIs) from the EU ETS. The increase transport emission carbon prices in the EU may lead to shift of CBAM production to third

countries without CPIs for transport emissions in place, however, whether the shift could be to third countries near or far the EU, it is not possible to determine. A shift in production to distant geographies implies the use of sea and rail transport to ship CBAM goods to the EU; and the shift of production to non-EU countries means that most of them already have CPIs in place as shown in Task 6.6. or countries in the process of adopting comprehensive CPIs such as Turkey, Albania, Ukraine, Morocco). Implication: The risk of carbon leakage as a result of the transport sector being included in the EU ETS remains low.

Task 5 concluding remarks. The assessment does not indicate increased risk of carbon leakage via shift of production of CBAM goods outside the EU. In addition, the share of transport emissions in total emissions is rather low (less than 5%). This share varies considerably depending on the country of origin and mode of transport, highlighting the importance of distance and logistics patterns in understanding the carbon intensity of goods imported under the CBAM. This assessment, however, could change depending on the final list of CN codes for extension of the CBAM scope.

7.2 T5.1 – Analysis of common transport modes for CBAM goods throughout their supply chains from the step they become CBAM onwards

This sub-task aimed at analysing and defining the profiles of the transport modes used for each CBAM good, as well as for selected goods, considering all logistical stages from the time the goods become CBAM classified until their arrival at the EU border¹⁴⁰.

Visual maps of the supply chains and the transport modes were developed following a structured approach that is described below.

1. First, the relevant CBAM goods were identified from the CBAM regulation, establishing that in-scope products include **cement, hydrogen, iron and steel, fertilisers and aluminium**, with their corresponding CN codes.
2. Then, the raw materials necessary to produce each specific good were determined, following the guidance document on CBAM implementation for installation operators outside the EU¹⁴¹.
3. After this identification process, the transport stages were broken down, analysing the origin of CBAM goods outside the EU, intermediate export countries and final destinations within the EU.

¹⁴⁰ The final assessment in this section includes transport work carried from origin to capital cities of the importing countries. The reason for this is the unavailability of data on international trade on where border crossings were base in case. For instance data from the UN/LOCODE dataset includes land border crossings and ports, but no way of calculating a weighted average distance based on tons or value of goods imported. Based on this data, any adjustment for this capital versus EU border discrepancy would be inaccurate and is therefore omitted.

¹⁴¹ European Commission. (2023) *Guidance document on CBAM implementation for installation operators outside the EU*. Directorate-General for Taxation and Customs Union.

4. The operational characteristics of each route were also analysed, which allowed for the construction of the corresponding transport profiles and the identification of the transport mode used in every step of the supply chain.

Transport profiles were developed leveraging the database created throughout this task and following the four steps described above. In table 29, the characteristics of the transport profile of CBAM transport into the EU are given.

Table 29 Transport profile of CBAM transport into the EU

The CN code of the CBAM good
The origin and destination, specifying the region and the corresponding country
The mode of transport used, and the total weight transported in that mode, expressed in both kilograms and tons
The average weight in tons (calculated as the average between the years 2019 and 2021)
The total distance travelled by each mode of transport from origin to destination
The share of total transport weight, in other words, the proportion of weight transported by each mode of transport to the total transport used

To create the database, which can be found in Annexes – Task 5, we used publicly available data from **UNCTADstat**, more specifically we used the trade and transport dataset ¹⁴². This dataset provides information on the quantity of goods traded (in kilograms) and the estimated transport work involved (measured in tonne-kilometres), broken down by country of origin, destination, and mode of transport (road, rail, maritime, air). We retrieved this data for the years 2019 to 2021¹⁴³ and focused on a selected list of product categories, identified using their CN codes, and which corresponds to all CBAM goods in-scope. To broaden the analyses, we also mapped all products identified under task 2 and 3, and only those with a corresponding CN code were added to the database.

Note on transport profiles

The visuals developed under this task are the CBAM goods already in scope. For the goods identified under tasks 2 and 3, we refer to the database in Annex 10.4.1, which provides the transport profiles for all CBAM goods in scope as well as those potentially included in the CBAM scope extension.

In Table 30, an example of this database is presented, with the CBAM good **Kaolin and other kaolinic clays, whether or not calcined**, with CN code 2507, transported between India (Asia) and Belgium (Europe). In this specific example, it can be observed that 93.3% of the weight of this CBAM good is primarily transported by road from India to Belgium, followed by railway and sea in a much smaller share

¹⁴² <https://unctadstat.unctad.org/datacentre/dataviewer/US.TransportCosts>

¹⁴³ The 2019 to 2021 period was selected due to availability of transport work data per transport mode per CN code (at four digit level). More recent years (2023 and 2024) did not offer complete transport work disaggregation for all CN codes.

Table 30 Database development under task 5.1

CN Code	Origin (Region)	Destination (Region)	Origin (country)	Destination (country)	Transport Mode	Total weight (tons 2019 – 2021)	Yearly Average weight (tons 2019-2021)	Share of tons per transport mode
2507	Asia	Europe	India	Belgium	Railway	177,427	59,142	4.4%
2507	Asia	Europe	India	Belgium	Road	3737,78	1245,92	93.3%
2507	Asia	Europe	India	Belgium	Sea	89,76	29,92	2.2%

Source: the study team

The shares above account for the full transport chain, and it should not be considered as an origin and a destination only served by a single transport mode, although this possibility cannot be excluded. The most likely configuration in the example above is that multiple Indian producers/exporters move CBAM goods by road and or rail to consolidation/transshipment point (e.g. Port), CBAM goods are transported by sea to an EU port, and CBAM goods are moved by road and or rail to destination in the EU. The large shares of road transport in this case may indicate large number of suppliers at origin moving goods to a consolidation point and or large distances from origin to consolidation point, as well as large number of destinations in the EU and or distance from port of arrival.

Figure 16 shows the supply chains of cement, hydrogen, iron and steel, fertilisers and aluminium, respectively, as described in the guidance document on CBAM implementation for installation operators outside the EU¹⁴⁴. This document includes multiple examples for each CBAM good, from which we have selected a few, as illustrative examples, as agreed upon with the EC during progress calls during the interim phase. It is important to highlight that not all raw materials used in the production of a CBAM good are, in turn, CBAM. Therefore, we have indicated next to each good considered CBAM its corresponding CN code, which allowed us to identify its transport profile and to analyse the different modes used at each stage of the supply chain. Additionally, we have also included the specific stages of the supply chain where transport emissions will be considered when calculating the corresponding emissions.

The reason for this decision is that in CBAM declarations, only CBAM and precursors (with embedded emissions) are reported. Therefore, despite a potential extension of CBAM goods' transport emissions, it is impractical to include materials' transport emissions when said materials are not included in the CBAM declarations. Thus, of the entire CBAM supply chain, only CBAM goods and its precursors declared in CBAM may include transport emissions.

As an example, Figure 16(a) shows the Portland cement supply chain (CN 2523 2100). Portland cement requires clinker (CN 2523 1000) and calcined clays (CN 2507 0080). These precursors must be transported to the place where cement grinding is carried out, a key point in the process where CO₂ emissions derived from transportation are generated. Gypsum and other constituents are also incorporated at this stage, and they

¹⁴⁴ European Commission. (2023) *Guidance document on CBAM implementation for installation operators outside the EU*. Directorate-General for Taxation and Customs Union.

also can generate transport emissions, depending on the production process. From the production site of the Portland cement to the destination, there are also transport emissions. Thus, while transport emissions are in all material and precursors moved within the supply chain and its production process, it is not possible to capture all the transport works' emissions associated to the Portland cement. But it is feasible to link transport emissions from precursors and CBAM goods included in the CBAM declarations. The implication for Portland cement, is that **at least 2 in 6 transport works' emissions are not able to be accounted in CBAM declarations.**

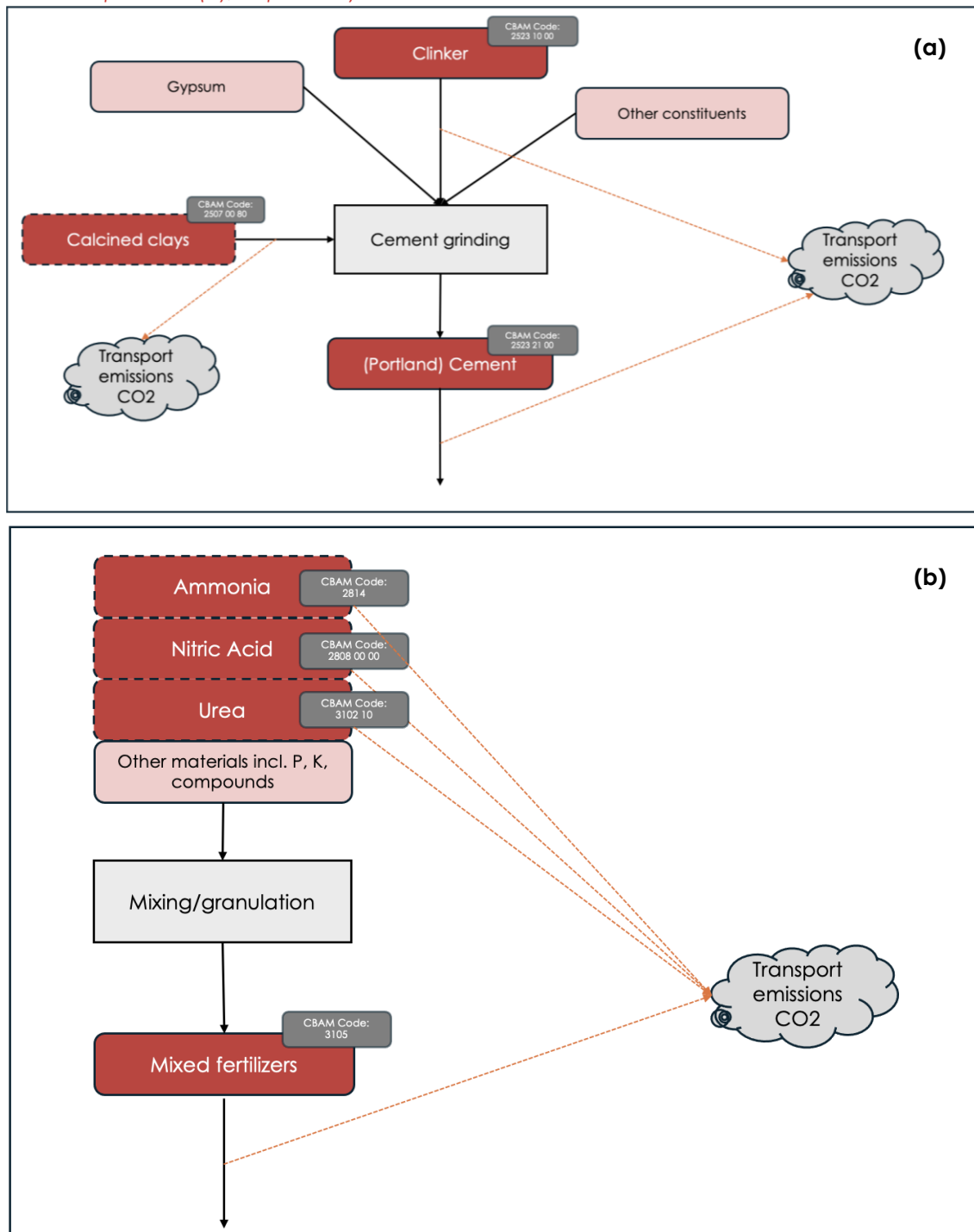
Figure 16(b) illustrates the supply chain of mixed fertilisers (CN 3105). For the production of this CBAM good, at least three CBAM goods are required (precursors): ammonia, nitric acid and urea. These components, together with other materials, are produced and transported, generating CO₂ emissions derived from transportation, until they reach the mixing/granulation process, where the final product, mixed fertilisers, is obtained. Finally, the mixed fertilisers are transported to their destination, which again involves the generation of CO₂ emissions associated with transportation. In this supply chain, at least **1 in 5 transport works' emissions cannot be accounted for in CBAM declarations.**

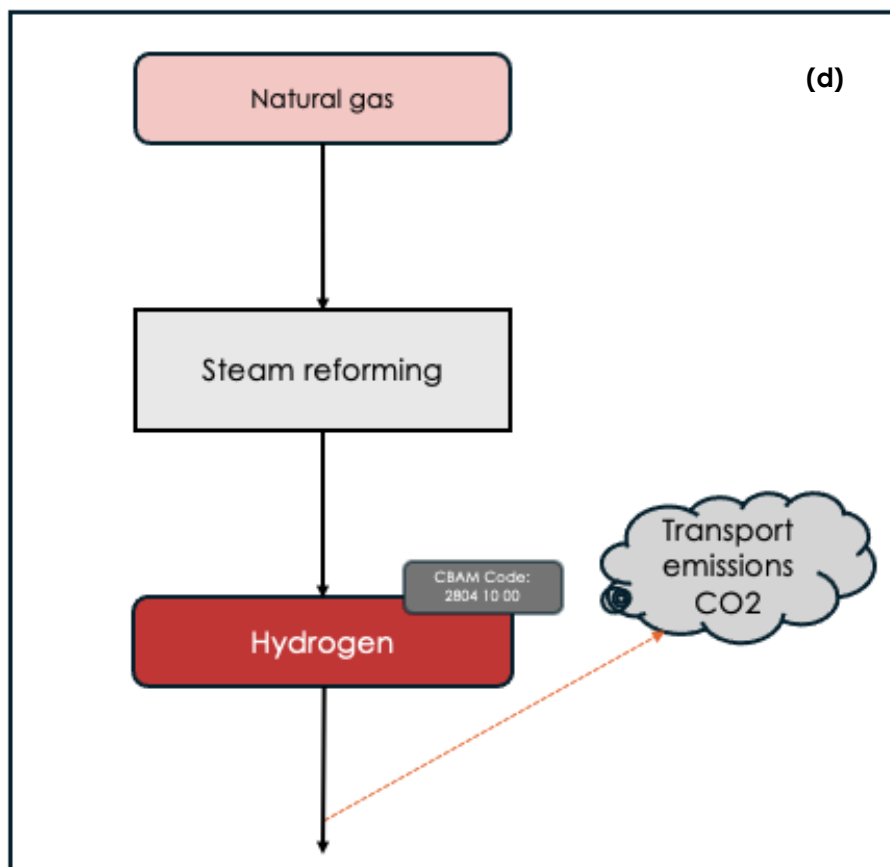
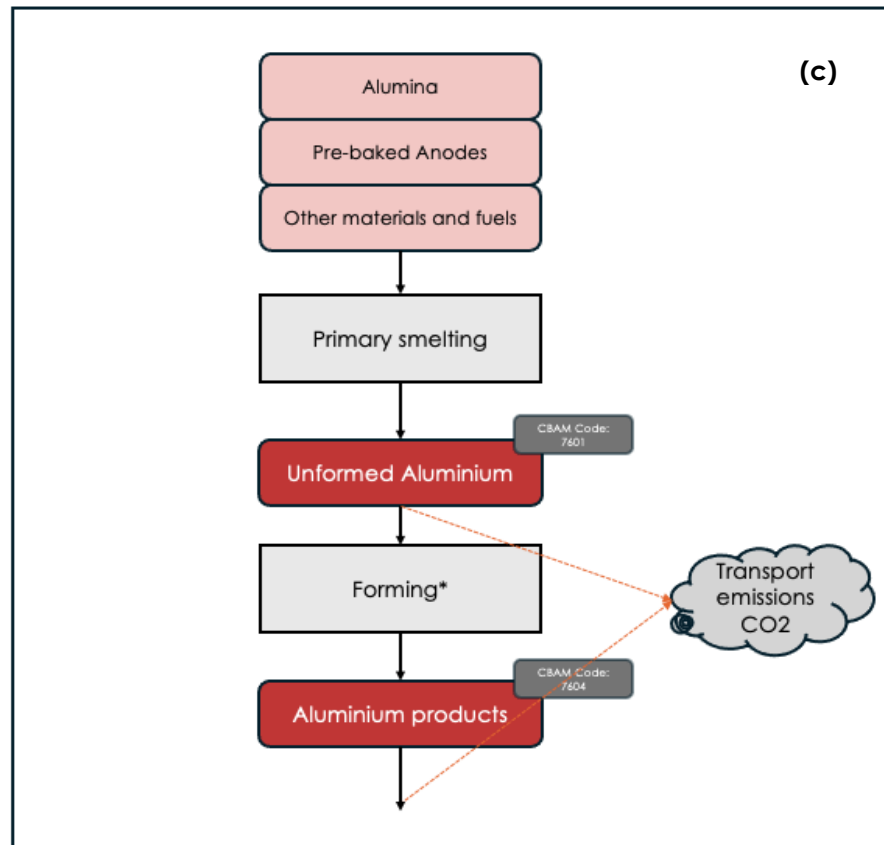
Figure 16(c) describes the supply chain of aluminium products (CN 7604...). In this case, it is observed that the only CBAM good (precursor) is needed for producing unformed aluminium (CN 7601...), which in turn is transported to the forming process site, to produce aluminium products. Once the final product is obtained, it is transported to its destination again. These precursors and final CBAM good can have their transport emissions accounted. On the other hand, to produce unformed aluminium, materials such as Alumina, pre-baked Anodes, Fuel, and other materials need to be brought from elsewhere into the production process. However, their transport emissions cannot be accounted. This shows that **at least 3 in 5 transport works' emissions are not possible to be accounted for.**

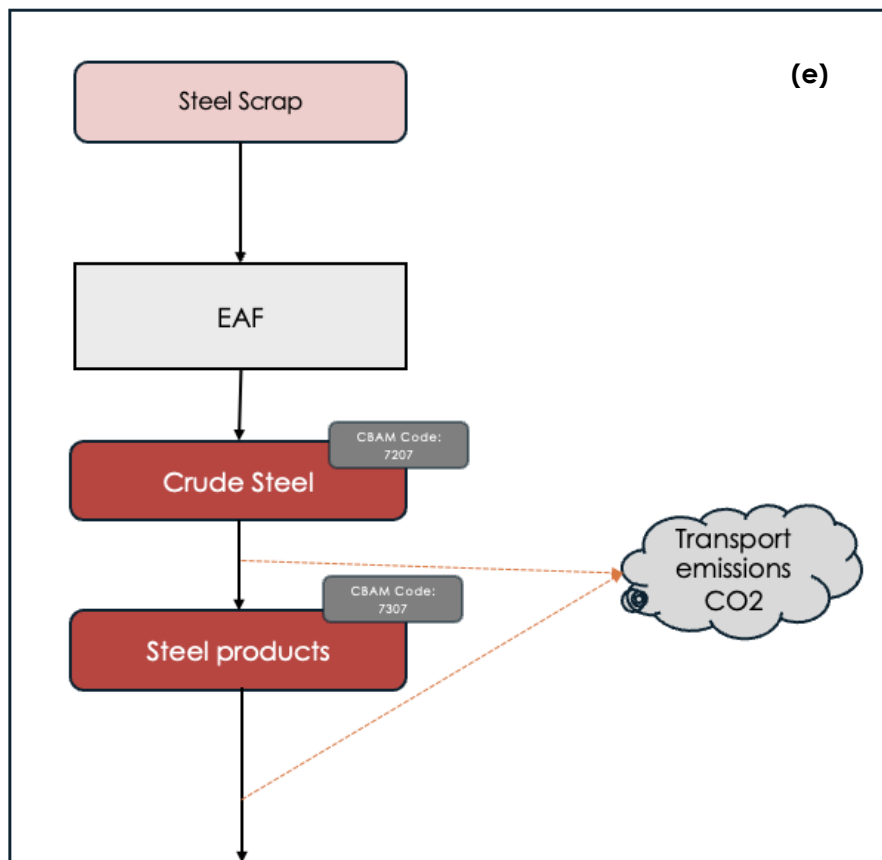
Figure 16(d) shows the hydrogen supply chain (CN 2804 1000). In this case, it can be observed that the process begins with natural gas, which undergoes a steam reforming process to produce the hydrogen CBAM good. Due to the configuration of this process, the CO₂ emissions derived from the transportation of the whole process of this CBAM good are the ones that happen once the hydrogen is produced, and it is ready for its distribution. Since natural gas is not considered a precursor, its transport work's emissions from upstream production to the hydrogen production site cannot be accounted. Thus, only the transport work's emissions of hydrogen itself can be accounted, leaving **1 in 2 transport works unaccounted in CBAM declarations.** On the other hand, if ammonia (CN 2814...) was used as hydrogen carrier into a hydrogen cracker, then both the ammonia and the hydrogen transport work's emissions would be accounted, then **100% of the transport work's emissions would be accounted in the CBAM declaration.**

Finally, Figure 16(e) illustrates the supply chain of steel products (CN 7307...). The process begins transporting steel scrap to the Electric Arc Furnace (EAF), a method used for melting scrap steel to produce crude steel. Once crude steel is produced, it is transported to the next stage, where it is processed into steel products. Both crude steel and steel products are classified as CBAM goods, meaning that CO₂ emissions from their transport works may be accounted in CBAM declarations, which cannot be said the same for scrap steel which is considered material and not a precursor with embedded emissions, leaving **1 in 3 transport works' emissions unaccounted for.**

Figure 16 Supply chains of cement (a), mixed fertilizers (b), aluminium (c), hydrogen (d) and steel products (e), respectively







As it was illustrated above, it is only practical to assess transport work's emissions ¹⁴⁵ from CBAM goods and precursors that can be included in the CBAM declaration. This leads to the following steps to filter the database developed beforehand, to include only the supply chains described in Figure 16. For each CBAM good, we estimated the share of transport mode for the movement of CBAM goods from non-EU countries into the EU. This analysis is presented in Figures 17-21, represents the transport profile for Portland Cement. As explained earlier in Figure 16(a), the production of Portland Cement requires both clinker and calcined clay. On the left side of Figure 5, the transport profile of these two CBAM goods outside Europe is shown. Clinker is mainly transported by maritime transport (54.02%), followed by road transport (27.57%). Similarly, calcined clay is primarily transported by sea (74.64%), with road transport (24.21%) as the second mode. These percentages were calculated by filtering the database by its CN code and excluding any region except Europe, allowing for an analysis of their transport behaviour outside the continent. On the right side of the figure, the transport profile of these materials *into* Europe is displayed, showing significant differences. In the case of clinker, the most common transport mode to Europe is railway (42.83%), followed by maritime transport

¹⁴⁵ Transport work is in ton-km. It corresponds to the product of the distance over which traded goods are transported and their total weight in kilograms. Distance refers to the average minimum distance needed to transport goods from main city centres of the origin to main city centres of the destination country. Source: UNCTAD (2024) Trade-and-Transport Dataset (experimental), annual, 2016 onward. Accessible at <https://unctadstat.unctad.org/datacentre/reportInfo/US.TransportCosts>

(35.05%) and road transport (22.11%). On the other hand, calcined clay is transported to Europe almost exclusively by maritime transport (95.65%).

Once these materials are processed, the final product, (Portland) cement, is primarily transported to Europe by railway (42.83%), followed by maritime transport (35.05%) and road transport (22.11%). It is worth noting that both Portland (Cement) and clinker cement have the same transport profiles when shipped to Europe. This occurs because, despite their specific classifications (Portland Cement: 2523 21 00, Clinker: 2523 10 00), the developed database considers CBAM goods based on the **first four digits** of their CN codes (2523), leading to the same transport profiles¹⁴⁶.

Figure 17 Transport profile of CBAM good (Portland) Cement

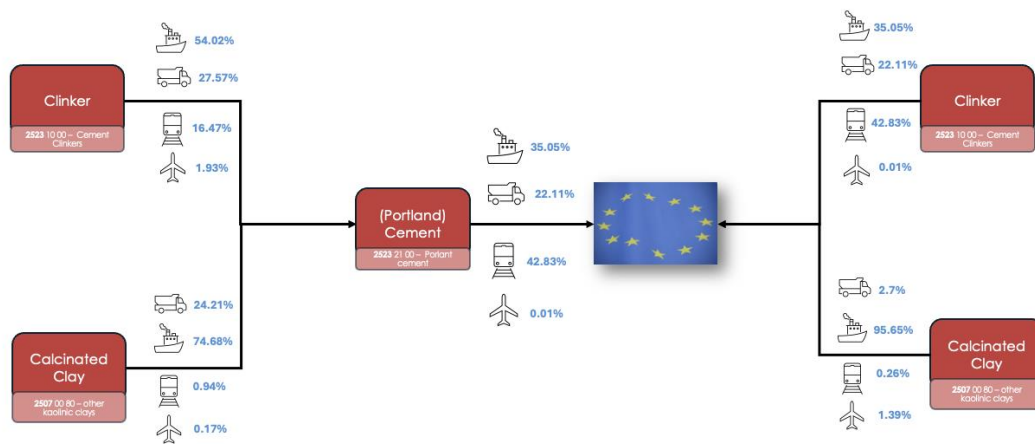
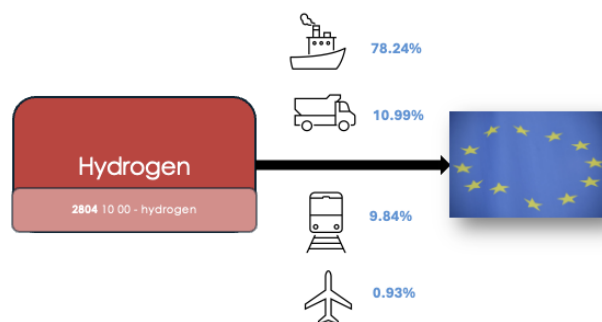


Figure 18, the transport profile of the CBAM good hydrogen is shown. It can be observed that hydrogen is predominantly transported to Europe via maritime transport (78.24%), followed by road transport (10.99%) and railway transport (9.84%).

Figure 18 Transport profile of CBAM good Hydrogen



In the case of CBAM goods fertilisers, the production process follows a similar structure to what was explained in Figure 14(c), involving urea, ammonia, and nitric acid as CBAM

¹⁴⁶ While the share of CBAM goods transported by air to the EU is rather small, it still a possibility. Note that while the volumes are very small, the tonne-km is large due to transported distance.

goods used to produce the final product, mixed fertilisers. The same approach was taken to analyse how these materials are transported outside Europe and how they are transported into Europe, revealing significant differences. In Figure 18, the transport profiles of the CBAM good of Mixed Fertilisers can be found. In the case of urea, outside Europe, it is primarily transported by maritime transport (85.4%), followed by railway (9.78%). However, when urea arrives in Europe, it is still predominantly transported by maritime transport and rail, but with different proportions: almost 70% by sea and 26.28% by rail. Something similar can be observed for ammonia.

An interesting case can be observed with nitric acid: outside Europe, nitric acid is mostly transported by maritime transport (45.62%), followed very closely by railway (40.55%), and even air transport (10.26%) plays a role. However, when nitric acid is transported into Europe, the transport profile changes drastically, with road transport becoming the dominant mode (89.31%), while maritime, rail, and air transport are barely used. For the final product, mixed fertilisers, the transport profile into Europe shows that they are mainly shipped via maritime transport (58.04%), followed by railway (32.11%).

Figure 19 Transport profile of CBAM good mixed fertilizers

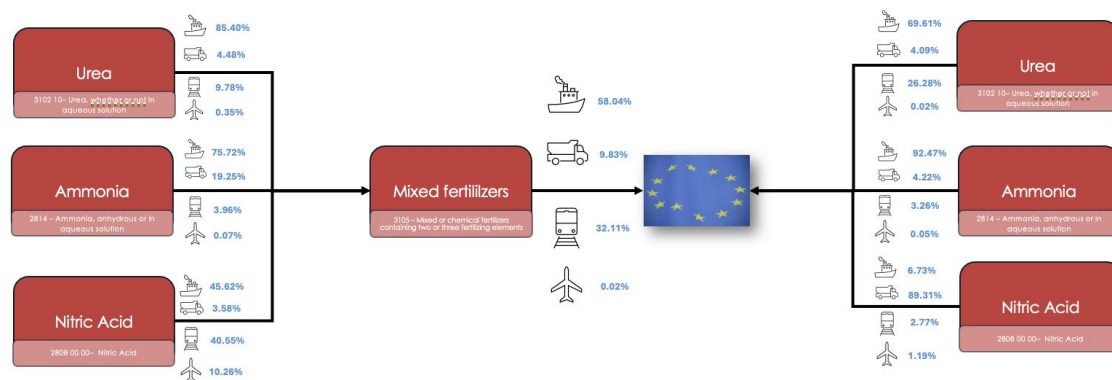
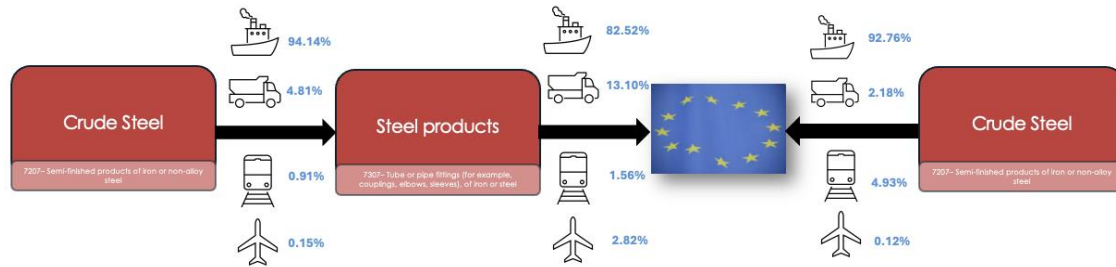


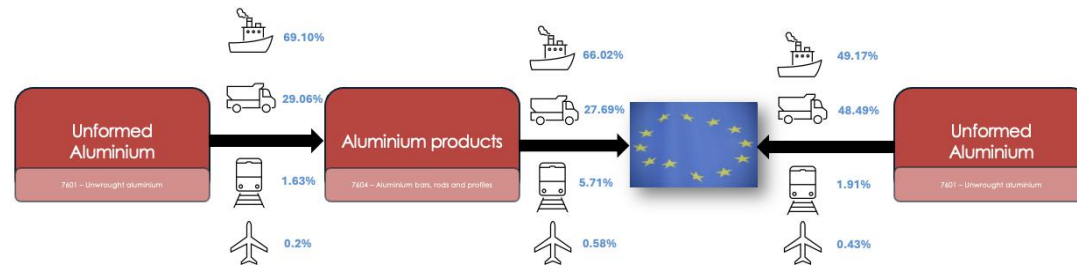
Figure 19 shows the transport profile of the CBAM good of steel products. It can be observed that crude steel has a very similar transport profile both outside Europe and when arriving at Europe. In both cases, maritime transport is the predominant mode. However, the second transport mode differs outside Europe, road transport follows maritime transport, whereas when crude steel reaches Europe, rail transport becomes the second most used mode. For steel products being transported to Europe, maritime transport is again the most used mode (82.52%), followed by road transport (13.10%).

Figure 20 Transport profile of CBAM good Steel Products



Finally, Figure 21 shows the transport profile of the CBAM good aluminium products. It can be observed that unformed aluminium outside Europe is transported by maritime transport (69.10%), followed by road transport (29.06%). However, when this CBAM good reaches Europe, its transport profile changes significantly, being transported almost equally by maritime transport (49.17%) and road transport (48.49%). For the final product, aluminium products, the main transport mode into Europe is maritime transport (66.02%), followed by road transport (27.69%) and rail transport (5.71%).

Figure 21 Transport profile of CBAM good Aluminium Products



7.3 T5.2 – Analysis of transport modes for CBAM goods outside EU ETS scope: characterising emission profiles and considering key factors

This subtask describes the emissions intensity profiles of each transportation mode for the current CBAM goods. The emission profiles for selected potential goods and precursors were assessed, and delivered in separate Excel file, for technical inclusion under horizontal and vertical extension in tasks 2 and 3, identifying transport modes outside the EU ETS scope.

Given that different modes of transport are used in each country with varying levels of emissions, influenced not only by the type of vehicle, but also by the availability and type of fuel in each region, we have developed a specific emissions profile for each combination of vehicle and region. This analysis has been conducted using the GLEC

v3.1 methodological framework¹⁴⁷ and the Ecotransit emission calculation tool¹⁴⁸. This makes it possible to standardise and assess the environmental impact of transport of the CBAM goods' supply chains.

- **Road**

Multiple factors determine the emission profile of transport modes. This analysis considers two key factors, the fuel intensity of the vehicle, and the emission intensity of the fuel.

The fuel intensity determines the fuel consumption, and the emission intensity the amount of CO₂ emissions released by the fuel combustion by the transport. The fuel intensity varies according to the characteristics of the transport, and the emission intensity depends on the carbon content of the fuel supplied in the regions where the vehicles are operated. The typical fuel intensity measure used by transport of goods is kg per ton – kilometre, or litres per ton-kilometre. The emission intensity measures used by transport of goods are well to tank emissions, tank to wheel emissions, or the sum of the two, i.e. well to wheel emissions.

There are multiple sources of information with fuel and emissions intensity of road transport at regional level. For instance, the Environmental Protection Agency of the United States (EPA), produces a yearly emissions performance per transport mode and transport characteristics¹⁴⁹. The table below shows the modes and characteristics included in their performance, based on self-reporting.

Table 31 Transport modes and characteristics included

Truck Carriers	Logistics Business Units	Other Carriers
Auto	All Logistics	Rail
Drayage	Primarily Air	Barge
Expedited	Primarily Barge	Air
Flatbed	Primarily Rail	
Heavy/Bulk	Primarily Truck	
LTL/Dry Van	Mixed	
Mixed		
Moving & Utility		
Package Delivery		
Refrigerated		
Specialized		
Tanker		
TL/Dry Van		

Source: Year 2022 is the most recent available year. Accessible at: <https://www.epa.gov/smartway/smartway-carrier-performance-ranking>

In sections T5.2 and T5.3, the average of the transport mode by all categories of transport is used. Ecotransit and GLEC v3.1 ensure the most recent and comprehensive use of data sources. The process followed to ensure consistency in the emission intensities of road transport was as follows: using the Ecotransit emission calculation tool mentioned before,

¹⁴⁷ Smart Freight Centre. (2023). *Calculate & report – GLEC framework*. Retrieved from <https://www.smartfreightcentre.org/en/our-programs/emissions-accounting/global-logistics-emissions-council/calculate-report-glec-framework/>

¹⁴⁸ <https://emissioncalculator.ecotransit.world/calculator>

¹⁴⁹ Year 2022 is the most recent available year. Accessible at: <https://www.epa.gov/smartway/smartway-carrier-performance-ranking>

we identified the countries belonging to each macro-region, and simulated road transport. From this, we extracted the WTW values (gCo2e/ton-km) per country and calculated the average across all countries within each macro-region, which was then used as the representative WTW value.

Since Ecotransit does not provide information on WTT and TTW values, we relied on the GLEC framework, which offers emission data for a wide range of vehicle types. In order to make this applicable, we researched the most typical vehicles by macro-region, which is shown in the table below, and used the GLEC database to extract their corresponding WTT and TTW values to calculate proportions relative to WTW. Finally, these proportions were applied to the WTW values derived from Ecotransit in order to ensure consistency and accuracy.

Table 32 Road Transport modes depending on the region

Transport Mode	North America	Europe	Asia	Africa	South America
Road	Heavy Bulk	Articulated truck up to 72 GVW (Container)	Articulated Truck HDV above 49.0 t GVW	Articulated truck up to 72 GVW (Container)	Articulated truck up to 72 GVW (Container)

Source: GLEC v3.1

• Rail

As a first step, we have reviewed several relevant documents. The International Energy Agency (IEA) report on The Future of Rail, analyses the degree of electrification of trains by region. This document shows that in North America, the share of electric trains in the total tonne-km transported is 0%, which indicates that all freight rail transport in the region uses diesel. Based on this information, we have consulted data from the United States Environmental Protection Agency's SmartWay Carrier Data¹⁵⁰, from which we have extracted a value of 81.8 grams of CO₂ per ton-km for all reported rail vehicles in the US, that corresponds to the total emissions, in this case, being the Well-to-Wheel emissions. Based on similarities in rail infrastructure, this value could also be applied to Latin America and the Caribbean.

However, in order to ensure consistency, we ultimately decided to rely exclusively on Ecotransit as the data source for WTW values, and the GLEC framework for the proportions of WTT and TTW, following the same approach as for road transport.

• Sea

The emission intensity of the maritime transport also varies according to the transport profile. There are key factors affecting emission intensity, namely ship type, year of construction, size category, and fuel type.

¹⁵⁰ The SmartWay carrier performance Data is compiled by EPA for ranking of best performers. An implication is that the list consists of the best performers. The implication is that taking the emissions from this data to represent the US carriers' environmental performance will normally be underestimating the industry average, and it is not possible to determine the extent of the underestimation. See EPA's SmartWay Carrier Performance Ranking: <https://www.epa.gov/smartway/smartway-carrier-performance-ranking>

An analysis of existing 'reliable' sources such as EU MRV EU ETS for shipping, International Maritime Organisation GreenVoyage 2050 calculator, and the GLEC v3.1¹⁵¹, show several limitations:

- None of the frameworks distinguish vessels emissions by trade lane;
- The EU MRV EU ETS and the IMO GreenVoyage2050 calculator do not distinguish vessels' emissions per vessel's fuel type used;
- The EU MRV EU ETS and the IMO GreenVoyage2050 calculator do not distinguish vessels' emissions per source of emissions, WtT, TtW, WtW;

This leaves our analysis with the GLEC v3.1 as the only source of emissions per vessel type, size, and emission sources. However, it is not possible to reliably determine the vessels emission of CBAM goods by trade lanes (origin - destination), due to the need to use typical **bulk carriers as the vessel type**. We averaged the emission intensities that are shown in the GLEC framework by the three types of fuel (HFO, VLSFO and MDO) of the biggest bulk carrier, and we used that as the emission intensities. Table 33 shows the vessel type used across the different geographies.

Table 33 Maritime transport modes depending on the region

Transport mode	North America	South America	Asia	Oceania	Europe	Africa
Sea	Bulk carrier of size (200000 -+ dwt)	Bulk carrier of size (200000 -+ dwt)	Bulk carrier of size (200000 -+ dwt)	Bulk carrier of size (200000 -+ dwt)	Bulk carrier of size (200000 -+ dwt)	Bulk carrier of size (200000 -+ dwt)

Source: GLEC v3.1

• Air

For the air transportation emission intensity, also regional differentiations were found. This is because emissions from air freight depend on multiple factors, such as aircraft type, route specifics, and load factors, and this also seem to be reflected in differences geographical location. For our calculations, we have selected the freighter (long-haul >1500 km) category, as it provides the most representative emission factor for global freight transportation.

The tables presented in task 5.3 provide the emissions in grams of CO₂ equivalent for each specific mode of transport, being road, air, sea, and rail, respectively, in each region, broken down into WTT (well-to-tank), TTW (tank-to-wheel), and WTW (well-to-wheel), when available. The emissions have been mapped according to the selected vehicle type and the specific region that is being considered.

It is important to note that, before calculating the global average excluding Europe for each transport mode, it is necessary to determine the emissions of each transport type regionally. This step was conducted in the next task 5.3, where the emissions associated with each mode of transport per region can be found. The table shows the global average emission profile excluding European Emissions intensity from the calculations to isolate the impact of transport emissions outside the European context.

¹⁵¹ While GLEC does not differentiate emissions non-containerized cargo, which is assumed to be the typical vessel type for CBAM goods.

Table 34 Global emission intensity of transport modes excluding Europe (gCO₂e/t-km)

Transport Mode	Average WTT emissions (g CO ₂ e/t-km)	Average TTW emissions (g CO ₂ e/t-km)	Average WTW emissions (g CO ₂ e/t-km)
Road	13.67	50.93	64.6
Rail	5.24	18.88	24.12
Air	168.23	805.94	974.17
Sea¹⁵²	0.26	1.33	1.59

Source: Elaborated based on data from *International Energy Agency (2019)*, *GLEC v3.1 Framework*, *EcoTransit* and *US Department of Transportation*.

7.4 T5.3: Identify and review the typical emissions intensities of different transport modes in (g CO₂/km-tonne) for the different modes of transport, if appropriate, differentiating by geographic regions.

This subtask reports the regional-level standard emissions intensity for each mode of transport, quantified in gCO₂e/t-km for different regions. As indicated above, due to the need to distinguish emissions by source in the transport work, we used the GLEC v3.1 framework. We proceeded to map the most used transport modes in each region as shown in subtask 5.2. After identifying these transport modes along with their fuel types¹⁵³, the corresponding emissions were mapped and the results of each mode of transport and region can be found in the following tables.

It is important to reiterate the assumptions stated in task 5.2 as these provide the necessary context to understand the difference in emissions intensities by region.

Table 35 Road transport emission intensity by region (gCO₂/t-km)

Region	WTT (g CO ₂ e/t-km)	TTW (g CO ₂ e/t-km)	WTW (g CO ₂ e/t-km)
North America	13.06	48.64	61.7
Europe	13.49	50.23	63.78
Asia	13.76	51.25	65.01
Africa	14.43	53.76	68.19
South America	13.44	50.06	63.50
Oceania	13.39	49.86	63.25

Source: EcoTransIT, GLEC 3.1

¹⁵² Since this task focuses on identifying transport modes outside the EU ETS scope, only 50% of maritime transport emissions will be considered. Therefore, the maritime transport emissions factor mapped in the GLEC Framework will be divided by 2.

¹⁵³ For sea shipping, rather than assuming a fuel type to assign emissions, we used the average of GLEC emissions of bulk carriers across fuel types.

Table 36 Air transport emission intensity by region (gCO₂/t-km)

Region	WTT (g CO ₂ e/t-km)	TTW (g CO ₂ e/t-km)	WTW (g CO ₂ e/t-km)
North America	165.97	795.1	961.07
Europe	157.41	754.06	911.47
Asia	129.75	621.55	751.3
Africa	139.12	666.43	805.55
South America	233.11	1116.71	1349.83
Oceania	184.07	881.82	1065.9

Source: EconTransIT, GLEC 3.1

For maritime transport, we have taken the average of the three fuel types listed in the GLEC Framework and calculated a single representative value. As shown in the table below, the data does not distinguish by region, following the approach explained in Task 5.2.

Table 37 Maritime transport emission intensity by region (gCO₂/t-km)

Region	WTT (g CO ₂ e/t-km)	TTW (g CO ₂ e/t-km)	WTW (g CO ₂ e/t-km)
North America	0.266	1.33	1.58
Europe	0.266	1.33	1.58
Asia	0.266	1.33	1.58
Africa	0.266	1.33	1.58
South America	0.266	1.33	1.58
Oceania	0.266	1.33	1.58

Source: EcoTransIT, GLEC v3.1

In the table below, the emissions disaggregated by WTT (Well-To-Tank), TTW (Tank-To-Wheel), and WTW (Well-To-Wheel) for railways in different regions are presented. These values have been calculated following the reasoning and sources explained in Task 5.2.

Table 38 Railway transport emission intensity by region (gCO₂/t-km)

Region	WTT (g CO ₂ e/t-km)	TTW (g CO ₂ e/t-km)	WTW (g CO ₂ e/t-km)
North America	5.14	18.52	23.66
Europe	2.50	9.00	11.50
Asia	5.20	18.74	23.94
Africa	5.46	19.64	25.10
South America	5.17	18.61	23.78
Oceania	4.67	16.83	21.5

Source: EcoTransIT, GLEC 3.1

7.5 T5.4: Provide a quantitative estimate of emissions from the transport of each CBAM good taking place outside the EU and not covered by the EU ETS

For Task 5.4, we leverage the database developed under this task, as well as the work done in **Tasks 5.1, 5.2, and 5.3**, where both the transport profiles for each origin–destination pair and CN code were established, and the typical emission profiles for each region were derived.

The first step was to extract the distances and tons transported by each mode of transport for every CBAM good, as well as for those goods identified under Tasks 2 and 3, in case they fall under the CBAM scope extension, from our database. Then, since we already have the emission profiles for each transport mode in each region, these were mapped to every observation in our database. Given that these emission intensities are expressed in gCO₂eq/ton-km, the second step was to multiply each emission intensity by the ton-km. However, it has been considered necessary to calculate an **adjustment factor to discount the portion of the distance travelled by road¹⁵⁴ and rail¹⁵⁵ within the European Union (EU) to then calculate and adjusted transport-work**. This is because, when a product arrives from a non-EU country to a Member State by road and railway, it is necessary to identify the nearest EU border and, from that point onwards, the kilometres travelled within the EU must be discounted, as the distance and associated emissions are already accounted for in the EU ETS. In contrast, maritime and air transport do not require this adjustment since goods are shipped or flown directly to the port¹⁵⁶ or airport in the EU border.

To produce the above-mentioned adjustment factor, it is necessary to correctly account for the distance travelled by the specific good. In order to do so, an average distance between the border crossing point and the capital of the Member State of destination has been estimated, considering origins in Africa, Asia, the Middle East, and the United Kingdom. These points of entry are Algeciras (Spain), Brest (Belarus, border with Poland), Svilengrad (Bulgaria), and Calais (France) respectively. This distance as a fraction of the total average distance travelled by road or rail is thus the adjustment factor. This factor is then applied to calculate the adjusted distance and adjusted transport work, using (1-adjustment factor) multiplied by the original distance and transport work. Once the adjusted ton-km was obtained, each record could then be multiplied by its corresponding WTT, TTW, and WTW emission values (those mapped under task 5.2 and 5.3), from which the total transport emissions (WTW) were calculated.

7.5.1 Additional task under task 5.4

As it was considered necessary to estimate the **share of transport-related emissions in related to the total emissions of CBAM goods**, it was required to incorporate in the database the production emissions of the CBAM goods. To do so, information related to the default values of the production process of the CBAM goods that are already in scope (tonne CO₂e/tonne goods) was retrieved from the document 'default values for

¹⁵⁴ While road transport is not currently included in the EU ETS scope, the analysis is done considering ETS II in place

¹⁵⁵ While diesel rail is not in the EU ETS, the EU is largely electrified and would indirectly be covered by the EU ETS.

¹⁵⁶ Half of the emissions of the maritime sector have been also adjusted for, to account for the fact that EU ETS covers half of the incoming international sea shipping.

the transitional period of the CBAM between 1 October 2023 and 31 December 2025¹⁵⁷. As in our records, the information available for each CBAM good is based on a 4-digit CN code, to estimate the default production values, we calculated an average for each 4-digit CN code by aggregating all products that share the same first four digits. In other words, when several products (with more than four digits) belong to the same 4-digit category, their values were averaged to obtain a representative default value for that 4-digit CN code. Results can be found in table 40. For the CBAM goods under potential scope extension, we leveraged the work developed in Task 1, where direct, indirect, and total emissions were estimated for the CN codes potentially to be included. We followed the same averaging approach to derive the default values for these goods. Results can be found in table 39.

Table 39 Estimation of default values for production of CBAM goods scope extension

Table 39 Estimation of default values for production of CBAM goods scope extension			
CBAM scope extension	Production default values (tonco2e/ton)		
cn code	Direct emissions (tonCO2/ton)	Indirect emissions (tonCO2/ton)	Total emissions (tonCO2/ton)
2518	0,78	0,02	0,8
2519	0,1	0,01	0,11
2522	0,78	0,02	0,8
2708	5,23	-0,02	5,21
2801	0,18	0,14	0,32
2803	0,18	0,14	0,32
2806	0,18	0,14	0,32
2811	0,128	0,396	0,528
2812	0,18	0,14	0,32
2815	0,18	0,14	0,32
2818	0,14	0,42	0,53
2825	0,35	0,30	0,65
2836	0,18	0,14	0,32
2847	0,18	0,14	0,32
2849	0,18	0,14	0,32
3801	0,02	0,02	0,05
3901	0,07	0,28	0,35
3902	0,07	0,28	0,35
3903	0,07	0,28	0,35
3904	0,07	0,28	0,35
3906	0,07	0,28	0,35
3907	0,07	0,28	0,35
3908	0,07	0,28	0,35
3909	0,07	0,28	0,35

¹⁵⁷ <https://taxation-customs.ec.europa.eu/system/files/2023-12/Default%20values%20transitional%20period.pdf>

CBAM scope extension	Production default values (tonco2e/ton)		
4701	0,05	0,2	0,26
4702	0,05	0,2	0,26
4703	0,05	0,2	0,26
4704	0,05	0,2	0,26
4705	0,05	0,2	0,26
4706	0,05	0,2	0,26
4801	0,31	0,31	0,62
4802	0,31	0,31	0,62
4803	0,31	0,31	0,62
4804	0,31	0,31	0,62
4805	0,31	0,31	0,62
4806	0,31	0,31	0,62
4807	0,31	0,31	0,62
4808	0,31	0,31	0,62
4809	0,31	0,31	0,62
4810	0,31	0,31	0,62
4811	0,31	0,31	0,62
4813	0,31	0,31	0,62
6901	0,16	0,07	0,23
6902	0,16	0,07	0,23
6903	0,16	0,07	0,23
6904	0,01	0	0,01
6905	0,01	0	0,01
6906	0,01	0	0,01
6907	0,09	0,03	0,12
6911	0,08	0,29	0,37
6912	0,08	0,29	0,37
6913	0,08	0,29	0,37
7001	0,58	0,43	1,01
7002	0,58	0,43	1,01
7003	0,65	0,14	0,79
7004	0,65	0,14	0,79
7005	0,65	0,14	0,79
7010	0,13	0,06	0,19
7011	0,58	0,43	1,01
7013	0,11	0,04	0,15
7014	0,58	0,43	1,01
7015	0,58	0,43	1,01
7016	0,58	0,43	1,01
7017	0,58	0,43	1,01

CBAM scope extension	Production default values (tonco2e/ton)		
7018	0,58	0,43	1,01
7019	0,45	0,89	1,34
7020	0,42	0,3	0,72
7202	0,92	0,33	1,24
7204	0,92	0,33	1,24
7401	0,17	0,09	0,26
7402	0,17	0,09	0,26
7403	0,17	0,09	0,26
7405	0,17	0,09	0,26
7406	0,17	0,09	0,26
7407	0,17	0,09	0,26
7408	0,17	0,09	0,26
7409	0,17	0,09	0,26
7410	0,17	0,09	0,26
7411	0,17	0,09	0,26
7412	0,17	0,09	0,26
7801	0,36	0,69	1,05
7804	0,36	0,69	1,05
7901	0,36	0,69	1,05
7903	0,36	0,69	1,05
7904	0,36	0,69	1,05
7905	0,36	0,69	1,05

Source: the study team based on Task 1

Table 40 Default values for production of CBAM good in scope

CBAM in scope	Production default values (tonco2e/ton)		
cn code	direct emissions	indirect emissions	total emissions
2507	0,23	0,08	0,32
2523	1,285	0,0975	1,3825
2601	0,31	0,05	0,36
2804	10,4	0	10,4
2808	2,56	0,05	2,6
2814	2,68	0,14	2,82
2834	1,83	0,06	1,88
3102	1,896	0,083	1,977
3105	0,97	0,09	1,06
7201	1,9	0,17	2,07
7202	2,33	2,76	5,07
7203	4,81	0	4,81
7205	1,9	0,17	2,07
7206	2,245	0,23	2,475
7207	2,14	0,42	2,56

CBAM in scope	Production default values (tonco2e/ton)		
7208	2,01	0,27	2,28
7209	2,03	0,36	2,29
7210	1,97	0,39	2,25
7211	2,02	0,315	2,335
7212	1,97	0,39	2,35
7213	1,89	0,32	2,21
7214	2,042	0,38	2,422
7215	1,89	0,32	2,21
7216	1,89	0,32	2,21
7217	1,9325	0,505	2,4375
7218	2,345	2	4,345
7219	2,19	1,94	4,13
7220	2,195	1,945	4,135
7221	2,14	2,17	4,3
7222	2,214	2,156	4,362
7223	2,13	2,36	4,49
7224	2,088	0,517	2,605
7225	1,947	0,457	2,401
7226	1,95	0,44	2,39
7227	1,86	0,57	2,43
7228	2,04	0,64	2,68
7229	1,84	0,75	0,59
7301	2,03	0,36	2,39
7302	1,93	0,29	2,21
7303	2,21	0,35	2,56
7304	1,89	0,32	2,2
7305	2,03	0,36	2,39
7306	1,982	0,396	2,373
7307	1,9	0,46	2,36
7308	2,46	2,55	5,01
7309	1,97	0,39	2,35
7310	1,97	0,39	2,35
7311	1,89	0,32	2,21
7318	1,92	0,54	2,46
7326	2,248	0,48	2,727
7601	2,36	8,14	10,49
7603	2,48	8,4	10,88
7604	2,562	8,576	11,144
7605	2,31	7,49	9,8
7606	2,86	9,25	12,11

CBAM in scope	Production default values (tonco2e/ton)		
7607	2,86	9,25	12,11
7608	2,73	9,3	12,04
7609	2,73	9,3	12,04
7610	2,73	9,3	12,04
7611	2,86	9,25	12,11
7612	2,86	9,25	12,11
7613	2,86	9,25	12,11
7614	2,31	7,49	9,8
7616	2,765	9,0375	11,8025

Source: own elaboration based on European Commission, "Default values for the transitional period of the CBAM between 1 October 2023 and 31 December 2025", December 2023.

In order to give a clear view of the differentiation of transport emissions for those CBAM good in scope and those in potential for extension, tables 41 and 42 have been developed separately and will be analysed separately.

Hence, table 41 shows the average tons traded, the total transport emissions and the total production emission emissions, total emissions per origin or CBAM good, ultimately giving the share of transport and production emissions over the total emissions. It can be observed that the main exporters of CBAM goods (without differentiating by individual product) are Russia, Ukraine, Turkey, Brazil, Canada, China, and the United Kingdom. However, despite being the largest exporters, these countries are **not necessarily those with the highest transport emissions**. As explained throughout task 5, transport emissions depend on the ton-km factor, that is, not only the quantity of goods transported from origin to destination impacts the emissions, but also the distance to the destination itself.

When looking closely to the total transport emissions (tonCO₂), regardless of transport mode, the countries with the highest transport emissions are the **UK, China, Taiwan, Iceland, Liberia, the Russian Federation, Turkey, and Brazil, respectively**. This distribution highlights the combined influence of volume of goods, geographical distance and transport mode in determining transport related emissions. Countries closer to the EU, such as Russia, Ukraine, or Turkey, show lower transport emission shares relative to the total emissions compared with China, Taiwan, Iceland or Liberia, since distances are shorter and the adjusted ton-kilometres are reduced after entering the EU border. In addition to distance, the choice of transport mode plays a crucial role in transport emissions. Taking a closer look to the data by filtering by transport mode, data shows the following:

- Ukraine, Turkey, Russia and UK respectively are the countries that move the most quantity of goods by road
- Ukraine, Russia, Turkey, Belarus, Liberia and UK, are the countries that rely the most on railway to move CBAM goods.
- By contrast, top exporters, like Brazil or Canada rely mainly on maritime transport to import CBAM goods to the EU.

Taking into consideration the emissions intensities of road and railway transport (we refer to task 5.3), the choice of these modes of transport to move heavy goods explains why some countries that are close to EU still appear among the top transport emitters.

Overall, when all origins and modes of transport are added together, transport related emissions account for approximately 4.88% of total emissions, compared to 95.12% for production. This indicates that, although transport contributes relatively little to total emissions, its impact varies considerably depending on the country of origin and mode of transport, highlighting the importance of distance and logistics patterns in understanding the carbon intensity of goods imported under the CBAM.

Table 41 Average imports of CBAM good in scope, and emission shares by non-EU countries ,

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Russian Federation	21500769,29	1115157,042	58452091,73	59567248,78	1,87%	98,13%
Ukraine	16937414,72	364885,5056	23137693,59	23502579,09	1,55%	98,45%
Türkiye	11700757,93	441465,4546	28624076,2	29065541,65	1,52%	98,48%
Brazil	10595140,32	417031,0577	6485662,64	6902693,698	6,04%	93,96%
Canada	10058242,68	236231,9554	6475316,986	6711548,941	3,52%	96,48%
China	5712828,831	2040918,511	23411075,11	25451993,62	8,02%	91,98%
United Kingdom	5439503,654	2427270,385	17783219,59	20210489,98	12,01%	87,99%
Liberia	3981066,413	1569011,919	1433307,567	3002319,486	52,26%	47,74%
South Africa	3584563,181	302844,3316	4040927,898	4343772,23	6,97%	93,03%
Norway	3395532,186	145614,1829	20473839,91	20619454,09	0,71%	99,29%
India	3306310,31	354398,52	10411726,56	10766125,08	3,29%	96,71%
Egypt	3032496,723	101929,2749	9454960,168	9556889,443	1,07%	98,93%
Belarus	2760056,92	53821,62661	5475275,722	5529097,349	0,97%	99,03%
Korea, Republic of	2681178,964	327637,0165	7764508,497	8092145,513	4,05%	95,95%
Algeria	2433698,495	134061,175	5800422,718	5934483,893	2,26%	97,74%
Switzerland, Liechtenstein	2158205,187	58062,1955	10033161,43	10091223,62	0,58%	99,42%
Morocco	1845086,91	139190,0497	2165194,683	2304384,733	6,04%	93,96%
China, Taiwan	1659364,263	1890583,594	4335555,668	6226139,262	30,37%	69,63%
Province of Serbia	1423771,749	37409,16799	4651615,527	4689024,695	0,80%	99,20%
United States of America	1382129,33	200908,7297	4181487,175	4382395,905	4,58%	95,42%
Trinidad and Tobago	1007565,2	12750,01791	2998402,631	3011152,649	0,42%	99,58%
Viet Nam	940025,155	162413,4866	2628963,191	2791376,677	5,82%	94,18%
Mauritania	836971,041	57881,41321	301429,7711	359311,1843	16,11%	83,89%
Iceland	831396,355	1881204,989	8690357,185	10571562,17	17,79%	82,21%
Bosnia and Herzegovina	772309,269	20027,09467	2205133,792	2225160,887	0,90%	99,10%
United Arab Emirates	733007,191	117767,9609	6224077,007	6341844,968	1,86%	98,14%
Tunisia	650292,581	46042,25791	1097708,373	1143750,631	4,03%	95,97%
North Macedonia	627017,247	16381,46198	1768149,385	1784530,847	0,92%	99,08%
Japan	556678,808	108469,4232	1485755,749	1594225,173	6,80%	93,20%
Mozambique	501252,967	176973,0714	5237046,164	5414019,235	3,27%	96,73%
Saudi Arabia	402441,25	39324,15157	1177851,764	1217175,915	3,23%	96,77%
Indonesia	377758,858	265831,8785	1297393,303	1563225,182	17,01%	82,99%

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Moldova, Republic of	343770,223	6304,327393	761051,4449	767355,7723	0,82%	99,18%
Venezuela (Bolivarian Rep. of)	332814,458	6905,904682	1292481,367	1299387,272	0,53%	99,47%
Mexico	283836,396	15079,07129	339730,0292	354809,1005	4,25%	95,75%
Malaysia	262457,54	104437,2964	1113904,484	1218341,781	8,57%	91,43%
Chile	256429,86	14350,08007	486508,8491	500858,9292	2,87%	97,13%
Bahrain	243786	43278,68644	2551743,358	2595022,044	1,67%	98,33%
Georgia	237592,444	8039,301295	656966,9187	665006,2199	1,21%	98,79%
Australia	227466,376	51899,27458	450299,8981	502199,1726	10,33%	89,67%
Oman	220662,599	17629,15623	893278,2823	910907,4385	1,94%	98,06%
Israel	215352,453	10873,82215	543997,4677	554871,2899	1,96%	98,04%
Albania	209036,483	10258,93201	431977,799	442236,731	2,32%	97,68%
Kazakhstan	187968,772	18408,37654	1891213,926	1909622,303	0,96%	99,04%
Libya	181984,734	6156,709368	779289,7996	785446,509	0,78%	99,22%
Colombia	177230,644	4358,089385	218682,9425	223041,0318	1,95%	98,05%
Turkmenistan	165618,443	9309,795423	444446,523	453756,3184	2,05%	97,95%
Iran (Islamic Republic of)	141796,838	6725,364553	433286,2043	440011,5688	1,53%	98,47%
Thailand	131380,234	70172,00738	512140,1617	582312,1691	12,05%	87,95%
Jordan	125163,838	4223,71577	225855,474	230079,1897	1,84%	98,16%
Qatar	68957,218	3874,845842	429906,2306	433781,0765	0,89%	99,11%
Montenegro	68666,35	2738,661731	439302,1011	442040,7629	0,62%	99,38%
Bahamas	65130,465	1072,176159	672870,1939	673942,37	0,16%	99,84%
Argentina	64394,213	7598,317929	85637,07126	93235,38919	8,15%	91,85%
Cameroon	45278,432	15727,9198	473217,1946	488945,1144	3,22%	96,78%
Ghana	39573,617	19634,50274	413961,5195	433596,0222	4,53%	95,47%
Azerbaijan	37714,025	1470,688551	165118,3084	166588,9969	0,88%	99,12%
Uzbekistan	32516,919	2112,992618	113479,8769	115592,8695	1,83%	98,17%
Pakistan	29804,417	4061,661558	50323,44773	54385,10928	7,47%	92,53%
China, Hong Kong SAR	22714,697	7386,669954	119006,2643	126392,9342	5,84%	94,16%
Armenia	19966,846	1417,34542	241279,7889	242697,1344	0,58%	99,42%
New Zealand	18496,375	10866,84587	163517,6334	174384,4792	6,23%	93,77%
Tajikistan	16339,916	6035,656918	170662,584	176698,241	3,42%	96,58%
Singapore	12011,874	11632,65359	79426,94431	91059,5979	12,77%	87,23%
Philippines	6254,963	7255,543292	21951,05088	29206,59417	24,84%	75,16%
Dominican Republic	6110,083	307,3920515	7866,914353	8174,306405	3,76%	96,24%
Lebanon	4719,587	144,5345595	26996,06766	27140,60222	0,53%	99,47%
Cambodia	3349,031	2095,430685	7929,867123	10025,29781	20,90%	79,10%
Mauritius	3046,995	66,3691908	21347,32793	21413,69712	0,31%	99,69%
Uruguay	2462,299	1480,232877	4759,104493	6239,33737	23,72%	76,28%
Panama	2286,786	229,6499136	4443,646757	4673,296671	4,91%	95,09%
Niger	1946,592	163,5891396	5117,835279	5281,424419	3,10%	96,90%

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Peru	1858,63	1980,296654	8196,615712	10176,91237	19,46%	80,54%
San Marino	1319,633	55,87436572	12060,9998	12116,87416	0,46%	99,54%
China, Macao SAR	1267,401	545,1877528	9713,954681	10259,14243	5,31%	94,69%
Honduras	1040,74	83,08199999	10774,28657	10857,36856	0,77%	99,23%
Kuwait	788,499	52,16784895	5596,977983	5649,145832	0,92%	99,08%
Marshall Islands	723,303	25,32096754	1613,017287	1638,338255	1,55%	98,45%
Turks and Caicos Islands	703,172	15,3272545	3473,402382	3488,729636	0,44%	99,56%
Tokelau	682,456	23,02454748	1570,752299	1593,776846	1,44%	98,56%
Costa Rica	641,28	14,80562266	1927,024985	1941,830608	0,76%	99,24%
Senegal	641,129	86,0439887	1753,879396	1839,923385	4,68%	95,32%
Gibraltar	567,953	47,19035279	2407,643184	2454,833537	1,92%	98,08%
Angola	554,874	32,56636515	1668,832107	1701,398472	1,91%	98,09%
Barbados	485,148	6,512694242	586,572235	593,0849292	1,10%	98,90%
Gabon	463,616	124,5665337	1635,205231	1759,771765	7,08%	92,92%
Cuba	426,742	12,64545762	2011,563019	2024,208477	0,62%	99,38%
Sri Lanka	377,422	739,0094282	1997,472565	2736,481993	27,01%	72,99%
Andorra	373,621	28,07869499	1327,374289	1355,452984	2,07%	97,93%
Bangladesh	336,57	101,9862636	1551,540645	1653,526909	6,17%	93,83%
British Virgin Islands	298,099	3,476799124	626,503477	629,9802761	0,55%	99,45%
Cote d'Ivoire	245,166	49,26810793	870,697031	919,9651389	5,36%	94,64%
Suriname	241,505	3,752077125	572,666943	576,4190201	0,65%	99,35%
Guinea	240,59	55,82747549	604,413999	660,2414745	8,46%	91,54%
Sierra Leone	233,276	51,33440948	862,55224	913,8866495	5,62%	94,38%
Benin	232,925	20,74073317	559,287635	580,0283682	3,58%	96,42%
Ecuador	192,663	41,08883451	716,147186	757,2360205	5,43%	94,57%
Curacao	179,643	2,457445869	824,667542	827,1249879	0,30%	99,70%
Botswana	169,774	30,85700986	1518,511787	1549,368797	1,99%	98,01%
Namibia	168,205	65,82672403	453,250788	519,077512	12,68%	87,32%
Saint Barthélemy	159,574	2,410068245	483,638802	486,0488702	0,50%	99,50%
Falkland Islands (Malvinas)	150,206	36,3986345	478,335563	514,7341975	7,07%	92,93%
Solomon Islands	148,964	5,266596654	353,90985	359,1764467	1,47%	98,53%
Faroe Islands	144,381	2,053001063	617,308948	619,3619491	0,33%	99,67%
Lao People's Dem. Rep.	141,369	43,20530761	433,418724	476,6240316	9,06%	90,94%
Grenada	140,75	1,677046773	363,29054	364,9675868	0,46%	99,54%
Haiti	135,608	6,323944015	169,478122	175,802066	3,60%	96,40%
Myanmar	130,262	28,73448589	1158,845711	1187,580197	2,42%	97,58%
Cabo Verde	127,597	2,802934283	644,206427	647,0093613	0,43%	99,57%
Sint Maarten (Dutch part)	126,356	1,736864696	754,105785	755,8426497	0,23%	99,77%
Yemen	115,447	9,066732651	190,662797	199,7295297	4,54%	95,46%

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Kenya	108,712	39,4559396	427,17862	466,634614	8,46%	91,54%
El Salvador	105,813	47,22174024	292,333035	339,5547752	13,91%	86,09%
Nigeria	104,505	34,15013028	441,251868	475,4019983	7,18%	92,82%
Afghanistan	94,391	13,31159393	289,902314	303,2139079	4,39%	95,61%
Ethiopia	90,158	39,25316201	855,962021	895,215183	4,38%	95,62%
Aruba	88,859	1,316676115	737,034653	738,3513291	0,18%	99,82%
Madagascar	83,355	18,79373199	609,550529	628,344261	2,99%	97,01%
Fiji	81,031	2,804503733	435,302049	438,1065527	0,64%	99,36%
Paraguay	75,773	78,19078315	412,557536	490,7483191	15,93%	84,07%
Eritrea	66,974	1,738338994	152,690952	154,429291	1,13%	98,87%
Korea, Dem. People's Rep. of	61,652	11,63830674	151,153865	162,7921717	7,15%	92,85%
Iraq	59,714	7,996464298	98,360614	106,3570783	7,52%	92,48%
Congo, Dem. Rep. of the	55,482	74,09201776	320,510545	394,6025628	18,78%	81,22%
Nicaragua	47,265	39,39790627	192,229357	231,6272633	17,01%	82,99%
Djibouti	46,209	12,21930235	110,501854	122,7211564	9,96%	90,04%
Belize	44,094	2,96951151	442,897019	445,8665305	0,67%	99,33%
Mali	41,955	13,27027813	81,821469	95,09174713	13,96%	86,04%
Eswatini	39,681	12,59806306	351,483096	364,0811591	3,46%	96,54%
American Samoa	34,292	1,118048543	111,926896	113,0449445	0,99%	99,01%
Guatemala	32,74	34,27500187	110,329774	144,6047759	23,70%	76,30%
Saint Lucia	32,612	0,553614219	97,065147	97,61876122	0,57%	99,43%
Nepal	32,499	27,07681225	88,667802	115,7446143	23,39%	76,61%
Guyana	28,288	52,75150987	165,683701	218,4352109	24,15%	75,85%
Zimbabwe	28,081	26,02986314	51,188038	77,21790114	33,71%	66,29%
Bolivia (Plurinational State of)	26,713	22,66949455	122,952036	145,6215306	15,57%	84,43%
Malawi	26,352	7,66782895	58,195989	65,86381795	11,64%	88,36%
Burkina Faso	25,66	6,572124132	95,898837	102,4709611	6,41%	93,59%
Uganda	24,503	23,92826756	95,967726	119,8959936	19,96%	80,04%
Nauru	22,765	1,774448519	67,638915	69,41336352	2,56%	97,44%
Bermuda	22,123	0,28526016	48,040217	48,32547716	0,59%	99,41%
Antigua and Barbuda	21,941	0,287609682	78,783348	79,07095768	0,36%	99,64%
New Caledonia	20,24	0,625840665	60,783909	61,40974966	1,02%	98,98%
Equatorial Guinea	19,09	0,762804098	42,642891	43,4056951	1,76%	98,24%
Syrian Arab Republic	18,96	2,744371982	75,430355	78,17472698	3,51%	96,49%
Cayman Islands	17,462	0,293228231	71,502687	71,79591523	0,41%	99,59%
Guinea-Bissau	16,163	0,0996294	38,624648	38,7242774	0,26%	99,74%
Gambia	15,871	21,52878175	101,062616	122,5913977	17,56%	82,44%

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Tanzania, United Republic of	15,01	11,7329372	86,619434	98,3523712	11,93%	88,07%
Greenland	14,685	0,194629082	53,74836	53,94298908	0,36%	99,64%
Seychelles	14,357	0,312840533	57,847834	58,16067453	0,54%	99,46%
Kyrgyzstan	14,129	2,443839631	63,843296	66,28713563	3,69%	96,31%
Chad	13,6	2,845671003	67,672165	70,517836	4,04%	95,96%
Jamaica	12,655	0,708918065	35,676279	36,38519706	1,95%	98,05%
French Polynesia	12,313	0,32331747	50,490945	50,81426247	0,64%	99,36%
Mongolia	11,595	7,215285078	33,009658	40,22494308	17,94%	82,06%
Sao Tome and Principe	8,933	0,147550947	76,402749	76,55029995	0,19%	99,81%
Bhutan	7,844	0,963056288	20,885198	21,84825429	4,41%	95,59%
Cook Islands	7,279	0,22983439	20,318745	20,54857939	1,12%	98,88%
Central African Republic	6,414	1,21202061	47,953263	49,16528361	2,47%	97,53%
Rwanda	6,398	1,773681551	29,284069	31,05775055	5,71%	94,29%
Togo	6,376	1,420936483	14,55735	15,97828648	8,89%	91,11%
Saint Pierre and Miquelon	6,234	0,141151771	14,55354	14,69469177	0,96%	99,04%
Brunei	6,083	28,00693998	28,292871	56,29981098	49,75%	50,25%
Darussalam	6,008	0,135098844	28,15725	28,29234884	0,48%	99,52%
Saint Kitts and Nevis	5,68	1,538697142	16,046813	17,58551014	8,75%	91,25%
Northern Mariana Islands	5,536	6,36556516	12,23944	18,60500516	34,21%	65,79%
Western Sahara	5,424	0,419536719	17,498067	17,91760372	2,34%	97,66%
Sudan	5,313	65,79558108	11,768361	77,56394208	84,83%	15,17%
Tuvalu	5,059	0,937762839	11,184294	12,12205684	7,74%	92,26%
Timor-Leste	4,897	5,970678169	11,613159	17,58383717	33,96%	66,04%
Zambia	3,803	0,087810071	31,466804	31,55461407	0,28%	99,72%
Dominica	2,723	0,275687547	6,20688	6,482567547	4,25%	95,75%
Burundi	2,229	0,313781403	6,421089	6,734870403	4,66%	95,34%
Somalia	2,1	0,031026731	4,998	5,029026731	0,62%	99,38%
Anguilla	1,951	0,077036949	19,211369	19,28840595	0,40%	99,60%
State of Palestine	1,784	1,763350546	6,20027	7,963620546	22,14%	77,86%
Papua New Guinea	1,102	0,031894753	8,748775	8,780669753	0,36%	99,64%
Christmas Island	1,005	0,019165715	4,7105	4,729665715	0,41%	99,59%
Lesotho	0,932	2,834379023	2,05972	4,894099023	57,91%	42,09%
Micronesia (Federated States of)	0,688	0,250841121	1,677309	1,928150121	13,01%	86,99%
Saint Vincent and the Grenadines	0,553	0,007041051	2,922631	2,929672051	0,24%	99,76%
Bonaire, Sint Eustatius and Saba						

Origin	Yearly Average Weight (Tons 2019 - 2021)	Total transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Niue	0,326	1,180770058	0,723528	1,904298058	62,01%	37,99%
Guam	0,214	0,006841054	0,546218	0,553059054	1,24%	98,76%
Tonga	0,153	0,004838636	0,644201	0,649039636	0,75%	99,25%
Comoros	0,1	0,00208605	0,132592	0,13467805	1,55%	98,45%
Samoa	0,096	1,423835309	0,262295	1,686130309	84,44%	15,56%
Vanuatu	0,067	0,019446154	0,187814	0,207260154	9,38%	90,62%
Palau	0,012	0,023096674	0,033972	0,057068674	40,47%	59,53%
South Sudan	0,009	0,008856398	0,024858	0,033714398	26,27%	73,73%
Montserrat	0,005	0,032452631	0,01105	0,043502631	74,60%	25,40%
Saint Helena	0,003	6,62505E-05	0,008493	0,008559251	0,77%	99,23%
Grand Total	128359882,2	15784551,18	307540378,6	323324929,7	4,88%	95,12%

* Ranked by Average yearly tonnes of good.

Table 42 shows the average imports of CBAM scope extension goods (those of tasks 2 and 3) and the emissions shares of transport and production relative to total emissions, by non-EU countries. It can be observed that the total share of transport emissions is higher than for the CBAM goods that are in scope, increasing from 4.88% to 16.11%. This change in emissions share can be explained by the difference between the default values for the production of goods already covered by the CBAM and those that could be covered by the extension of its scope. Looking at the averages, CBAM goods that are in-scope have a total average emissions intensity related to production of 4.65 tons of CO₂ per ton of good, while goods covered by the extension of the CBAM's scope have a much lower average of 0.57 tonnes of CO₂ per tonne of goods.

Table 42 Average imports of CBAM scope extension goods, and emission shared by non-EU countries, ranked in descendent order average imports of CBAM goods to the EU (from non-EU countries)

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
United Kingdom	5610609,029	2706922,004	4188326,999	6895249,003	39,26%	60,74%
Türkiye	4994140,504	193943,064	1879906,976	2073850,04	9,35%	90,65%
Norway	4498167,256	144102,9001	6362888,404	6506991,304	2,21%	97,79%
United States of America	4259610,646	259703,5297	2138346,244	2398049,774	10,83%	89,17%
Russian Federation	4009704,353	320691,3793	2765936,374	3086627,753	10,39%	89,61%
Switzerland, Liechtenstein	3507105,53	86581,27424	2666776,442	2753357,716	3,14%	96,86%
China	3156027,912	802949,8367	1528725,52	2331675,357	34,44%	65,56%
Brazil	2216619,463	625128,0622	1505036,224	2130164,286	29,35%	70,65%
Saudi Arabia	2195620,652	143022,1308	777210,978	920233,1087	15,54%	84,46%
Ukraine	2181095,337	72175,30854	3870762,762	3942938,071	1,83%	98,17%
Korea, Republic of	2111133,005	355982,4734	1021234,091	1377216,565	25,85%	74,15%
India	1719718,298	251054,0615	1631795,291	1882849,353	13,33%	86,67%
Serbia	1460548,805	32464,12362	593445,2333	625909,3569	5,19%	94,81%

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Bosnia and Herzegovina	1197857,647	25259,13018	633892,5595	659151,6897	3,83%	96,17%
South Africa	1024158,795	141240,5869	3040133,509	3181374,095	4,44%	95,56%
Egypt	922426,7	47628,88748	663794,3653	711423,2528	6,69%	93,31%
Chile	559368,311	432623,7146	168459,4196	601083,1342	71,97%	28,03%
Uruguay	506402,931	245283,7981	133010,3419	378294,1401	64,84%	35,16%
Canada	452707,495	33176,31007	293415,7942	326592,1042	10,16%	89,84%
Qatar	443856,018	25115,99745	153176,6521	178292,6496	14,09%	85,91%
Malaysia	382288,879	91743,05135	967989,4933	1059732,545	8,66%	91,34%
Israel	376223,964	17497,3106	238295,5144	255792,825	6,84%	93,16%
Belarus	361180,527	11731,82456	216218,3336	227950,1582	5,15%	94,85%
Japan	357605,183	67489,75845	195420,5064	262910,2649	25,67%	74,33%
Indonesia	348046,225	218364,099	189576,8715	407940,9705	53,53%	46,47%
Mexico	340712,845	29910,66889	226265,7437	256176,4126	11,68%	88,32%
Kazakhstan	334922,714	24575,60634	1061367,694	1085943,301	2,26%	97,74%
Jamaica	325671,608	8761,165844	187409,4445	196170,6104	4,47%	95,53%
United Arab Emirates	306439,548	39550,81062	93819,64833	133370,4589	29,65%	70,35%
China, Taiwan	267511,697	330227,8455	151845,5643	482073,4099	68,50%	31,50%
Province of North Macedonia	259778,762	4407,225518	265122,6495	269529,8751	1,64%	98,36%
Lebanon	252619,883	1040,831175	306494,3562	307535,1874	0,34%	99,66%
Iceland	248524,069	209722,9087	746349,0139	956071,9226	21,94%	78,06%
Thailand	227017,919	42627,32034	105948,6475	148575,9679	28,69%	71,31%
Iran (Islamic Republic of)	195956,068	16202,05593	167243,5119	183445,5678	8,83%	91,17%
Viet Nam	157171,87	26361,77583	59401,62795	85763,40378	30,74%	69,26%
Singapore	145233,77	31960,60422	62441,44171	94402,04593	33,86%	66,14%
Moldova, Republic of	144062,77	4140,638856	35251,21151	39391,85037	10,51%	89,49%
Congo, Dem. Rep. of the	140960,739	13961,95826	36663,1398	50625,09807	27,58%	72,42%
Zambia	99018,777	26132,52828	243359,1832	269491,7114	9,70%	90,30%
Kuwait	95684,163	14506,29077	63238,10318	77744,39394	18,66%	81,34%
Albania	95300,319	2395,007098	280410,7754	282805,7825	0,85%	99,15%
Venezuela (Bolivarian Rep. of)	90612,935	2971,224764	108374,0625	111345,2873	2,67%	97,33%
Tunisia	81527,773	5868,426085	43967,20427	49835,63035	11,78%	88,22%
Namibia	80366,951	45630,40002	27357,09875	72987,49876	62,52%	37,48%
Peru	74981,861	19593,95871	44679,97042	64273,92913	30,49%	69,51%
Australia	72679,941	13881,22037	58686,39575	72567,61612	19,13%	80,87%
Pakistan	57853,273	6178,107244	18801,53384	24979,64109	24,73%	75,27%
New Caledonia	51023,245	1762,928995	185196,456	186959,385	0,94%	99,06%
Morocco	44928,596	3858,62157	39285,87886	43144,50043	8,94%	91,06%
Colombia	43578,705	2950,835588	105992,1343	108942,9699	2,71%	97,29%

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Algeria	41944,087	188,1271091	34401,31311	34589,44022	0,54%	99,46%
Dominican Republic	37231,564	628,0863456	127740,0502	128368,1365	0,49%	99,51%
Philippines	35896,586	1578,983862	11111,44216	12690,42602	12,44%	87,56%
Georgia	34113,413	2071,645045	103258,3385	105329,9836	1,97%	98,03%
Oman	32311,41	2367,395648	24643,28508	27010,68072	8,76%	91,24%
Guatemala	31002,869	674,2002525	111209,8493	111884,0496	0,60%	99,40%
Uzbekistan	28334,282	3908,136702	12885,05382	16793,19052	23,27%	76,73%
Gabon	27272,051	11451,33609	97705,78903	109157,1251	10,49%	89,51%
Jordan	24939,269	872,9339167	11083,59689	11956,53081	7,30%	92,70%
Bahrain	24866,253	2922,050513	30794,19291	33716,24342	8,67%	91,33%
Zimbabwe	23335,324	2907,747744	81298,70699	84206,45474	3,45%	96,55%
Argentina	19201,307	19918,45453	29265,04963	49183,50416	40,50%	59,50%
Azerbaijan	14948,97	863,018681	24741,55042	25604,5691	3,37%	96,63%
Nigeria	13921,306	2739,2178	12787,40327	15526,62107	17,64%	82,36%
China, Hong Kong SAR	13244,175	3716,019731	6024,034112	9740,053843	38,15%	61,85%
Bangladesh	13206,585	3120,86019	4227,290705	7348,150895	42,47%	57,53%
Libya	13051,626	37,35785106	16147,71442	16185,07227	0,23%	99,77%
Armenia	12966,118	877,0195653	45878,68223	46755,7018	1,88%	98,12%
Bhutan	12069,029	2325,300413	43797,28591	46122,58633	5,04%	94,96%
Tanzania, United Republic of	8751,944	671,4829402	2196,68039	2868,16333	23,41%	76,59%
Gibraltar	8608,86	237,7945458	10452,37998	10690,17453	2,22%	97,78%
Greenland	6843,132	92,82721517	8478,35707	8571,184285	1,08%	98,92%
Mozambique	6393,664	968,3517781	21736,62104	22704,97282	4,26%	95,74%
Montenegro	5742,197	120,1973152	5367,559506	5487,756821	2,19%	97,81%
Andorra	5527,244	354,9118689	5128,282006	5483,193875	6,47%	93,53%
Suriname	5418,676	73,04246079	6529,16298	6602,205441	1,11%	98,89%
Bahamas	5353,129	83,81910322	6759,128518	6842,947621	1,22%	98,78%
Paraguay	5286,046	225,7593442	11014,48619	11240,24554	2,01%	97,99%
Cuba	5015,324	99,07872278	6068,807871	6167,886594	1,61%	98,39%
Faroe Islands	4047,413	13,26939148	3726,801395	3740,070786	0,35%	99,65%
Ecuador	3622,751	174,9642738	2775,719111	2950,683385	5,93%	94,07%
Mali	3364,097	599,0203385	12124,93014	12723,95048	4,71%	95,29%
Ghana	3239,867	503,4919313	3241,083095	3744,575026	13,45%	86,55%
Sudan	2752,741	779,748961	2874,809	3654,557961	21,34%	78,66%
Trinidad and Tobago	2715,328	34,8996457	1719,562713	1754,462359	1,99%	98,01%
New Zealand	2453,832	503,5970221	975,767054	1479,364076	34,04%	65,96%
Curacao	2388,056	36,17212844	1014,52356	1050,695688	3,44%	96,56%
Ethiopia	2221,161	184,9668504	765,389097	950,3559474	19,46%	80,54%
Kyrgyzstan	1938,572	233,0482307	2100,435738	2333,483969	9,99%	90,01%
Cameroon	1906,647	352,5240874	1961,206799	2313,730886	15,24%	84,76%

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Panama	1883,473	60,49039515	2052,230614	2112,721009	2,86%	97,14%
Sri Lanka	1611,76	3064,855395	567,752829	3632,608224	84,37%	15,63%
Cabo Verde	1605,261	11,80487484	1499,782293	1511,587168	0,78%	99,22%
Turkmenistan	1451,964	78,86606456	506,90537	585,7714346	13,46%	86,54%
Bolivia (Plurinational State of)	1365,93	70,76191158	1393,055355	1463,817267	4,83%	95,17%
Mauritius	1150,595	25,19966942	539,637849	564,8375184	4,46%	95,54%
Syrian Arab Republic	962,308	14,19249028	618,833191	633,0256813	2,24%	97,76%
Angola	856,33	126,0103206	262,46625	388,4765706	32,44%	67,56%
Korea, Dem. People's Rep. of	796,755	189,2993956	278,741072	468,0404676	40,45%	59,55%
Costa Rica	763,393	14,06304546	821,935089	835,9981345	1,68%	98,32%
Central African Republic	753,919	82,46185491	580,174521	662,6363759	12,44%	87,56%
Sierra Leone	726,928	98,62628958	410,74018	509,3664696	19,36%	80,64%
Somalia	708,006	97,37053157	431,254925	528,6254566	18,42%	81,58%
Cote d'Ivoire	691,097	60,6985761	690,749085	751,4476611	8,08%	91,92%
Myanmar	684,475	68,18317526	822,690029	890,8732043	7,65%	92,35%
Senegal	681,271	99,84852521	585,526801	685,3753262	14,57%	85,43%
Bonaire, Sint Eustatius and Saba	649,66	8,760585709	803,329985	812,0905707	1,08%	98,92%
Kenya	632,513	97,65504316	200,59591	298,2509532	32,74%	67,26%
Mongolia	565,811	130,0274744	110,436475	240,4639494	54,07%	45,93%
Sao Tome and Principe	477,712	7,329449541	518,720046	526,0494955	1,39%	98,61%
Honduras	447,507	23,17989386	258,12605	281,3059439	8,24%	91,76%
Guyana	445,585	855,0144632	446,62038	1301,634843	65,69%	34,31%
Aruba	374,99	5,930006882	462,339484	468,2694909	1,27%	98,73%
El Salvador	372,107	20,41335569	388,194355	408,6077107	5,00%	95,00%
China, Macao SAR	337,953	94,80985503	93,684825	188,49468	50,30%	49,70%
Niger	313,037	47,76062654	148,465365	196,2259915	24,34%	75,66%
American Samoa	280,735	9,744785056	89,666282	99,41106706	9,80%	90,20%
Liberia	275,688	37,86146315	332,9285	370,7899632	10,21%	89,79%
Marshall Islands	265,145	8,481854485	261,856208	270,3380625	3,14%	96,86%
Saint Barthelemy	258,633	2,857465637	273,860784	276,7182496	1,03%	98,97%
Equatorial Guinea	247,519	3,087035177	304,3099	307,3969352	1,00%	99,00%
Nicaragua	239,992	10,51549755	114,953547	125,4690446	8,38%	91,62%
Benin	212,809	24,82092077	177,69604	202,5169608	12,26%	87,74%
Haiti	211,432	56,49427452	95,869715	152,3639895	37,08%	62,92%
Uganda	208,496	31,97314568	73,61217	105,5853157	30,28%	69,72%
Botswana	205,427	38,17512035	78,56414	116,7392603	32,70%	67,30%
Tokelau	175,526	5,023811984	45,21603	50,23984198	10,00%	90,00%

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Iraq	168,964	15,89353247	61,561305	77,45483747	20,52%	79,48%
Cayman Islands	167,421	3,875863639	143,281955	147,1578186	2,63%	97,37%
Sint Maarten (Dutch part)	139,692	1,65465567	164,873815	166,5284707	0,99%	99,01%
Burkina Faso	134,148	15,4134848	47,144095	62,5575798	24,64%	75,36%
Belize	129,83	4,643137485	130,79353	135,4366675	3,43%	96,57%
Chad	111,961	18,39233784	59,887026	78,27936384	23,50%	76,50%
Djibouti	106,876	31,47768059	44,938125	76,41580559	41,19%	58,81%
Grenada	90,956	1,082341419	72,448355	73,53069642	1,47%	98,53%
Eritrea	90,493	16,08196731	19,17601	35,25797731	45,61%	54,39%
Yemen	84,816	3,889971064	69,85788	73,74785106	5,27%	94,73%
Barbados	83,711	1,432462709	31,727055	33,15951771	4,32%	95,68%
San Marino	77,331	4,734070661	28,039345	32,77341566	14,44%	85,56%
Nepal	75,555	144,1599635	37,303324	181,4632875	79,44%	20,56%
Lao People's Dem. Rep.	58,729	13,60622243	23,39855	37,00477243	36,77%	63,23%
Fiji	56,481	1,879414049	22,74219	24,62160405	7,63%	92,37%
Tajikistan	53,884	11,4647139	109,826851	121,2915649	9,45%	90,55%
Dominica	46,182	0,553804863	158,700734	159,2545389	0,35%	99,65%
Antigua and Barbuda	45,896	0,465585306	55,26728	55,73286531	0,84%	99,16%
Nauru	43,861	1,54028804	60,17679	61,71707804	2,50%	97,50%
Mauritania	40,36	6,011836108	29,248684	35,26052011	17,05%	82,95%
Togo	39,104	5,339589214	12,10614	17,44572921	30,61%	69,39%
French Polynesia	37,218	0,925057267	13,052742	13,97779927	6,62%	93,38%
Tuvalu	37,039	164,3914906	12,24866	176,6401506	93,07%	6,93%
Gambia	36,068	2,223458259	44,59965	46,82310826	4,75%	95,25%
Guinea	33,901	3,109955895	20,441247	23,5512029	13,21%	86,79%
Madagascar	33,265	3,204590598	19,337959	22,5425496	14,22%	85,78%
Solomon Islands	30,85	1,092636778	10,72775	11,82038678	9,24%	90,76%
Turks and Caicos Islands	29,659	1,878238609	12,82287	14,70110861	12,78%	87,22%
Cambodia	22,284	7,063720417	11,762541	18,82626142	37,52%	62,48%
Seychelles	17,736	0,39771651	8,52685	8,92456651	4,46%	95,54%
Falkland Islands (Malvinas)	16,764	0,808447986	14,20781	15,01625799	5,38%	94,62%
State of Palestine	16,05	1,845899635	5,88978	7,735679635	23,86%	76,14%
Afghanistan	13,874	3,785448757	6,550574	10,33602276	36,62%	63,38%
Western Sahara	13,713	1,479580922	4,86516	6,344740922	23,32%	76,68%
Eswatini	13,599	14,2706944	8,05879	22,3294844	63,91%	36,09%
Bermuda	12,988	0,150608639	13,857275	14,00788364	1,08%	98,92%
South Sudan	11,641	1,602970364	5,30204	6,905010364	23,21%	76,79%
Burundi	9,519	4,288902827	3,13881	7,427712827	57,74%	42,26%

Origin	Average Weight (Tons 2019 - 2021)	Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Sum of Total Emissions (Production +Transport)	Share (%) transport emissions	Share (%) production emissions
Saint Helena	5,836	0,124320358	3,56004	3,684360358	3,37%	96,63%
Maldives	3,88	0,981066352	4,664634	5,645700352	17,38%	82,62%
British Virgin Islands	3,303	0,042607366	4,03917	4,081777366	1,04%	98,96%
Guinea-Bissau	2,913	0,056312828	2,06664	2,122952828	2,65%	97,35%
Saint Lucia	2,615	0,028084609	3,212549	3,240633609	0,87%	99,13%
Saint Pierre and Miquelon	1,154	0,026127211	0,30004	0,326167211	8,01%	91,99%
Saint Vincent and the Grenadines	1,062	0,025835112	0,20709	0,232925112	11,09%	88,91%
Cook Islands	0,966	0,032351266	0,37436	0,406711266	7,95%	92,05%
Brunei Darussalam	0,947	0,30203647	0,46213	0,76416647	39,52%	60,48%
Christmas Island	0,768	0,021044335	0,24048	0,261524335	8,05%	91,95%
Papua New Guinea	0,487	0,043338673	0,22616	0,269498673	16,08%	83,92%
Anguilla	0,41	0,009078294	0,2542	0,263278294	3,45%	96,55%
Saint Kitts and Nevis	0,251	0,005385266	0,08296	0,088345266	6,10%	93,90%
Malawi	0,188	0,059746718	0,06954	0,129286718	46,21%	53,79%
Palau	0,138	0,003842403	0,0207	0,024542403	15,66%	84,34%
Kiribati	0,114	0,003523925	0,01368	0,017203925	20,48%	79,52%
Samoa	0,068	0,48027988	0,02018	0,50045988	95,97%	4,03%
Northern Mariana Islands	0,063	0,089898054	0,0093	0,099198054	90,62%	9,38%
Timor-Leste	0,049	0,615815817	0,01813	0,633945817	97,14%	2,86%
Guam	0,032	0,001128663	0,01984	0,020968663	5,38%	94,62%
Tonga	0,012	0,000385185	0,00444	0,004825185	7,98%	92,02%
Rwanda	0,007	0,001458203	0,00237	0,003828203	38,09%	61,91%
Niue	0,006	0,000191767	0,00156	0,001751767	10,95%	89,05%
Montserrat	0,006	0,026601642	0,0017	0,028301642	93,99%	6,01%
Grand Total	53659960,61	8356768,184	43513719	51870487,18	16,11%	83,89%

* Ranked by Average yearly tonnes of good.

Source: own elaboration based on UNData, EcoTransit, Glec 3.1

Overall, the share of transport emissions of both CBAM extension and in scope products equals 6.43%, as it can be observed in the next table.

Table 43 Weighted average of both CBAM in scope goods and CBAM extension, and share of transport and production emissions

Scope / Extension	Average Weight (Tons 2019 - 2021)	Total Transport Emissions (tonCO2e)	Total Production Emissions (tonCO2e)	Total Emissions (Production +Transport)	share (%) transport emissions	share (%) production emissions
CBAM Extension	53659960,61	8356768,184	43513719	51870487,18	16,11%	83,89%
CBAM Scope	128359882,2	15784551,18	307540378,6	323324929,7	4,88%	95,12%
Grand Total	182019842,8	24141319,36	351054097,6	375195416,9	6,43%	93,57%

Source: own elaboration based on UNData, EcoTransit, Glec 3.1

Further analysis is required by CN code and their mode of transport from third countries into the EU, to better understand the overall effect on total emissions and the percentage of total emissions from the CBAM sector. This analysis is recommended with the final selection of CN codes for CBAM extension, as road transport may be the preferred mode for goods with origin near the EU, while sea and rail transport may be the predominant mode of transport for goods originating in countries farther away from the EU.

7.6 T5.5: Risk Assessment for Carbon Leakage: Impact of EU ETS Expansion to Maritime, and Road Transport of CBAM Goods, with Scenario Analysis for Carbon Price Evolution

The objective of this section is to analyse the impact of EU ETS expansion to Maritime and Road transport of CBAM goods with simulation of likely carbon price evolutions. The analysis is done in two steps, first, the effect of carbon price evolutions on the CBAM carbon leakage is assessed, without considering the transport sector, and in the second step the transport sector is added, under the four scenarios listed below.

- a) **Baseline.** No EU ETS carbon price on transport, EU CPIs on transport, Global CPIs on transport:
 - a. Global CPIs Price - Low
 - b. Global CPIs Price - Medium
 - c. Global CPIs Price - High
- b) **Pre EU ETS Transport.** No EU ETS carbon price on transport, No EU CPIs on transport, No Global CPIs on transport:
 - a. Carbon price zero
- c) **Post EU ETS Transport.** EU ETS carbon price on transport, EU CPIs on transport, Global CPIs on transport:
 - a. Global CPIs Price - Low
 - b. CBAM transport emissions scope extended=Global CPIs prices = EU ETS
 - c. Global CPIs Price - High
- d) **Post EU ETS changing price.** EU ETS carbon price on transport varies, Global CPIs on transport remain at current levels:
 - a. EU ETS Price - Low
 - b. EU ETS Price - Current
 - c. EU ETS Price - High, Global CPIs Price - 25

The analysis has been done based on data on transport, emissions, and prices in Task 5 and Task 6. It is important to consider that carbon leakage studies tend to use an indicator that builds on trade intensity, production values, and emission intensity. One of the few scientific publication with a practical indicator, ad-hoc for this study, is that of Misch, F., & Wingender, P. (2024)¹⁵⁸ who follows the definition by Branger, F., & Quirion, P. (2014)¹⁵⁹, and which measure the effect of global emissions in function of changes in carbon price evolution.

The modelling results suggest that the extension of the EU ETS to the transport sector reinforces existing cost differentials when global carbon prices remain low, thus accelerating production and emission leakage. Conversely, harmonisation of carbon pricing between the EU and third countries, either through aligned CPIs or an extended CBAM, effectively reduces leakage risks. Over time, increasing ETS prices and wider carbon pricing coverage are likely to limit leakage, though structural declines in EU CBAM production are expected to persist due to demand factors beyond carbon costs.

A description of the model, and a detailed analysis for each scenario is provided in the sections following the table below which summarises the different scenarios.

¹⁵⁸ Misch, F., & Wingender, P. (2024). Revisiting carbon leakage. *Energy Economics*, 140, 107786. <https://doi.org/10.1016/j.eneco.2024.107786>

¹⁵⁹ Branger, F., & Quirion, P. (2014). Climate policy and the 'carbon haven' effect. *Wiley Interdisciplinary Reviews: Climate Change*, 5(1), 53-71. <https://doi.org/10.1002/wcc.245>

Table 44 Summary of scenario settings and main effects on CBAM carbon leakage and transport emissions

Scenario	Carbon price setting	Main effect on production leakage	Main effect on carbon leakage / transport emissions	Overall interpretation
(a) Baseline No EU ETS on transport; EU and global CPIs apply	EU ETS Transport = €0; EU CPIs = €48; RoW CPIs = €5–100	- Low RoW price: Significant relocation of CBAM production to third countries (within ≈11 years). - Medium RoW price: Very slight, temporary increase in non-EU production. - High RoW price: No notable relocation from EU to third countries; some production leakage from third countries to the EU due to the high global CPI for about 3 years; EU production decline mainly demand-driven.	- Low: Growth in third-country transport emissions. - Medium: Minor temporary increase, followed by stabilisation. - High: Stable global emissions; continuing EU decline.	Under low global prices, EU competitiveness weakens and leakage rises. Higher global prices promote stability in both trade and emissions.
(b) Pre-EU ETS Transport No carbon pricing anywhere	EU ETS Transport = €0; EU CPIs = €0; RoW CPIs = €0	Slight decrease in third-country production during first 4–5 years, followed by stabilisation. EU production continues to fall due to declining CBAM demand.	Transport emissions abroad remain stable; EU emissions fall, reducing global transport emissions overall.	Without carbon pricing, production shifts are demand-driven rather than cost-driven. Leakage remains limited.
(c) Post-EU ETS Transport ETS applies to transport; global CPIs vary	EU ETS Transport = €55; RoW CPIs = €5–100	- Low RoW price (€5): Rapid shift of production abroad (≈10 years), faster than baseline (≈14 years). - Equal price (€55): Small, short-term growth of third-country production, stabilising by year ≈6–7. - High RoW price (€100): Stable non-EU production; faster EU adjustment.	- Low: Global transport emissions decline but with accelerated EU production loss. - Equal: Emission cost harmonisation reduces leakage. - High: Similar emissions trend to baseline; faster EU production correction.	When RoW carbon prices are low, ETS introduction accelerates leakage; global price alignment mitigates this effect.

Scenario	Carbon price setting	Main effect on production leakage	Main effect on carbon leakage / transport emissions	Overall interpretation
(d) Post-EU ETS – Changing price – ETS varies; global CPIs fixed	EU ETS Transport= low–high; RoW CPIs = €25	EU production continues to fall but at a slower rate as ETS price rises; third-country production converges by ≈2030.	Carbon leakage peaks before 2030, then declines after CBAM transport scope extension. Long-term leakage remains limited.	Gradual ETS price increases help moderate leakage but do not reverse structural competitiveness loss.

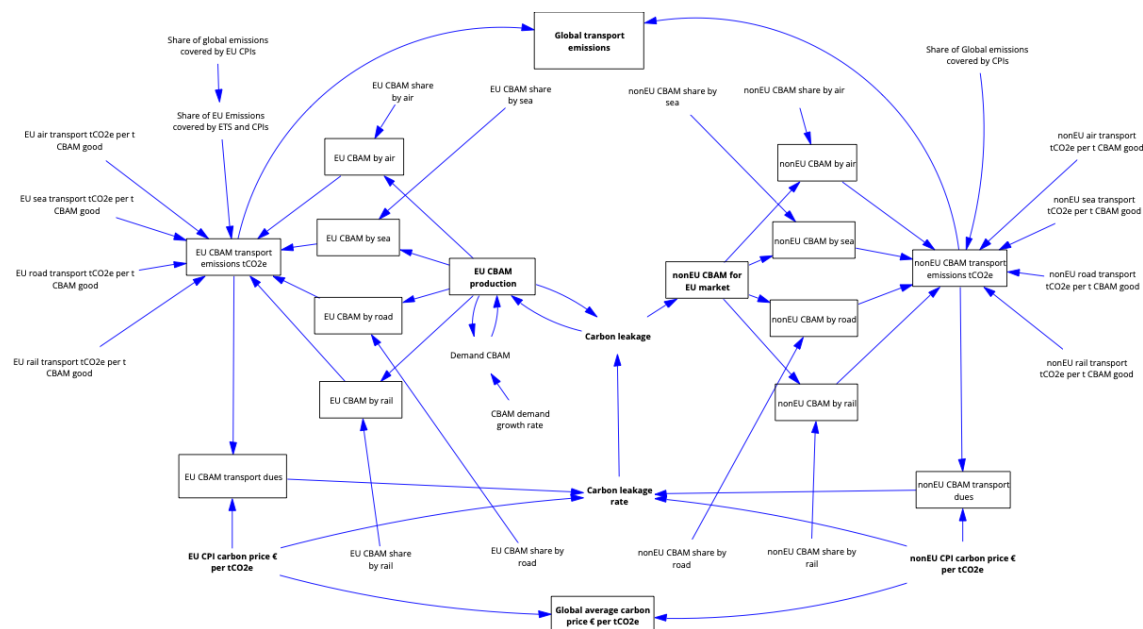
Source: Study team elaboration based on Task 5.5 model results (Vensim PLE).

7.6.1 The EU ETS on transport based CBAM carbon leakage model

To assess the extent to which emissions are exported to third countries as a result of the implementation of the EU ETS on the transport sector (road and maritime), a system dynamics model is used, based on *Vensim*. The model captures the carbon leakage by transfer of production from the EU to the rest of the world with EU as final market, as a result of the obligations by the EU ETS carbon price on EU transport of CBAM goods. The transfer of production to the rest of the world, leads to releasing emissions from the transport of CBAM goods to the EU. Provided there are CPIs implemented targeting directly or throughout upstream carbon price the transport sector, operators have obligation to pay a carbon price per tCO₂e. The increase in cost by the producer may drive the decision to relocate the source of production to locations with lower costs (incl. cost of emissions released by the transport of CBAM goods). Carbon leakage refers to the situation where efforts to reduce emissions within the EU lead to a corresponding increase in emissions elsewhere in the world.

The relocation to farther places, causes transport emissions to increase. Operators can use multiple modes of transport for the movement of CBAM goods to the EU markets. The transport mix determine the amount of emissions and the potential to face carbon price for such emissions. This model described, assumes production costs equal across regions, and assesses only the effect of carbon price paid on transport emissions on the CBAM goods carbon leakage. Thus, the two key causal indicators are the carbon price per tCO₂e of the existing CPIs that directly or indirectly targets the transport sector, and the amount of emissions released by the transport sector for the movement of CBAM goods into the EU. It is important to highlight that this analysis consider that carbon leakage occurs when source of production is shifted from the EU to third countries, only if the final destination of the relocated production is the EU (imports). The figure below, allows the visualise the complex and dynamic causal effects of the carbon leakage. Further down, a table lists the variables and equations for the baseline scenario.

Figure 22 System dynamics model of EU ETS on Transport effect on EU CBAM goods carbon leakage



Source: The study team

Table 45 System dynamics model equations of EU ETS on Transport effect on EU CBAM goods carbon leakage

Variable names	Equations
Carbon leakage	EU CBAM production * Carbon leakage rate
Carbon leakage rate	IF THEN ELSE (EU CBAM transport dues=1, ((EU CPI carbon price € per tCO ₂ eq - nonEU CPI carbon price € per tCO ₂ eq) * 0.0058) / (EU CPI carbon price € per tCO ₂ eq + nonEU CPI carbon price € per tCO ₂ eq), ((EU CBAM transport dues-nonEU CBAM transport dues) * 0.058)/(EU CBAM transport dues + nonEU CBAM transport dues))
CBAM demand growth rate	-0.035
Demand CBAM	CBAM demand growth rate * EU CBAM production
EU CBAM by air	EU CBAM production * EU CBAM share by air
EU CBAM by rail	EU CBAM production * EU CBAM share by rail
EU CBAM by road	EU CBAM production * EU CBAM share by road
EU CBAM by sea	EU CBAM production * EU CBAM share by sea
EU CBAM production ¹⁶⁰	(3.5276e+08) + Demand CBAM-Carbon leakage
EU CBAM share by air	0.0001
EU CBAM share by rail	0.3341
EU CBAM share by road	0.6658
EU CBAM share by sea	0.0001
EU CBAM transport dues	EU CBAM transport emissions tCO ₂ eq * EU CPI carbon price € per tCO ₂ eq
EU CBAM transport emissions tCO ₂ e	((EU CBAM by air * EU air transport tCO ₂ eq per t CBAM good) + (EU CBAM by rail * EU rail transport tCO ₂ eq per t CBAM good) + (EU CBAM by road * EU road transport tCO ₂ eq per t CBAM good) + (EU CBAM by sea * EU sea transport tCO ₂ eq per t CBAM good)) * Share of EU Emissions covered by ETS and CPIs
EU CPI carbon price € per tCO ₂ e	49.775
EU rail transport tCO ₂ e per t CBAM good	0.0889
EU road transport tCO ₂ e per t CBAM good	0.8541
EU sea transport tCO ₂ e per t CBAM good	0.04
EU air transport tCO ₂ e per t CBAM good	15.83
Global average carbon price € per tCO ₂ e	(EU CPI carbon price € per tCO ₂ e+nonEU CPI carbon price € per tCO ₂ e)
Global transport emissions	(EU CBAM transport emissions tCO ₂ e+nonEU CBAM transport emissions tCO ₂ e)
nonEU CBAM by air	nonEU CBAM for EU market * nonEU CBAM share by air
nonEU CBAM by rail	nonEU CBAM for EU market * nonEU CBAM share by rail
nonEU CBAM by road	nonEU CBAM for EU market * nonEU CBAM share by road
nonEU CBAM by sea	nonEU CBAM for EU market * nonEU CBAM share by sea
nonEU CBAM for EU market	(4.6089e+07)+Carbon leakage
nonEU CBAM share by air	0.004
nonEU CBAM share by rail	0.147
nonEU CBAM share by road	0.292
nonEU CBAM share by sea	0.558
nonEU CBAM transport dues	nonEU CBAM transport emissions tCO ₂ eq * nonEU CPI carbon price € per tCO ₂ e

¹⁶⁰ EU production refers to the production of CBAM goods in EU destined to EU demand, and not to cover EU exports.

Variable names	Equations
nonEU CBAM transport emissions tCO₂e	$((\text{nonEU CBAM by air} * \text{nonEU air transport tCO}_2\text{eq per t CBAM good}) + (\text{nonEU CBAM by rail} * \text{nonEU rail transport tCO}_2\text{eq per t CBAM good}) + (\text{nonEU CBAM by road} * \text{nonEU road transport tCO}_2\text{eq per t CBAM good}) + (\text{nonEU CBAM by sea} * \text{nonEU sea transport tCO}_2\text{eq per t CBAM good})) * \text{Share of Global emissions covered by CPIs}$
nonEU CPI carbon price € per tCO₂e	24.69
nonEU rail transport tCO₂e per t CBAM good	0.089
nonEU road transport tCO₂e per t CBAM good	0.844
nonEU sea transport tCO₂e per t CBAM good	0.04
nonEU air transport tCO₂e per t CBAM good	17.137
Share of EU Emissions covered by ETS and CPIs	1 * Share of global emissions covered by EU CPIs
Share of Global emissions covered by CPIs	0.198
Share of global emissions covered by EU CPIs	0.032 for EU ETS / 0.009 for other EU CPIs
Time	1
TIME STEP	1
INITIAL TIME	0
FINAL TIME	15

Source: The study team

7.6.2 Baseline. No EU ETS carbon price on transport, EU CPIs on transport, Global CPIs on transport

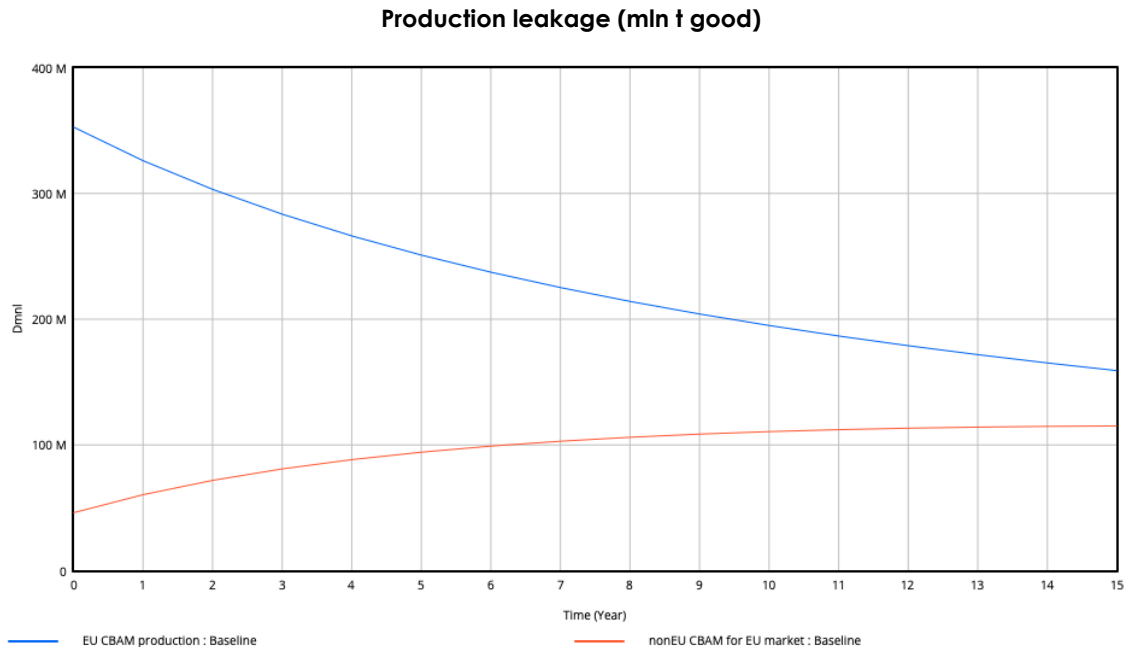
The baseline or current situation scenario has set the year 2021 as the initial time and the year 2035 as the final time. Aggregating data from Cembureau, European Aluminium, Eurofer, and Fertilizers Europe, the EU27 demand for CBAM goods in the year 2021 was near 350 mln tonnes and by the year 2023 it was 283 mln tonnes, representing a change in CBAM goods demand of - 3.5% yearly. The EU production was estimated in the same period to change by - 6.8% yearly during the same period, starting from 352 mln tonnes of CBAM goods in the year 2021. Imports of CBAM goods into the EU was estimated to change by 5.9% yearly, starting from 46 mln tonnes in the year 2023. These rates of change have been applied across scenarios.

The baseline scenario considers the EU ETS on transport sector not enforced and with no carbon price paid by the transport sector based on the EU ETS¹⁶¹. The baseline scenario assumes the current situation to continue for the next 15 years. It assumes that in third countries, CPIs carbon prices for the transport sector are in place, given the status as

¹⁶¹ It is important to consider that the transport sector in the EU is subject to carbon prices due to the presence of other CPIs at member state level. For instance, at the baseline year of this analysis, 2021, the EU ETS price per tCO₂e was € 49.7 while for the rest of the EU countries (Denmark carbon tax, Finland carbon tax, France carbon tax, Germany ETS, Ireland carbon tax, Luxembourg carbon tax, Poland carbon tax, Slovenia carbon tax, Sweden carbon tax), with pricing information in the World Bank's Carbon Pricing Dashboard (<https://carbonpricingdashboard.worldbank.org/>), and directly regulating the transport sector, the average carbon price was € 48.4 in 2021. In the Baseline however, this prices are not included, but they are included in the Pre-EU ETS-Transport (PREST) scenario.

implemented CPI with transport sector in scope in the World Banks' Carbon Price Dashboard¹⁶², as analysed in Task 6.6. The model also considers that the EU ETS covers 3.2% of the global emissions and the rest of the EU CPIs cover 0.9% of the global emissions, and the rest of the world CPIs cover up to 19.8% of the global emissions¹⁶³. These rates are assumed to apply in the same proportion for the global transport sector emissions. In the EU the average carbon price applied by CPIs that cover transport emissions is €48 per tCO₂eq, while the rest of the world's average carbon price by instruments that cover transport emissions is €24. The results of the system of equations gives the following output:

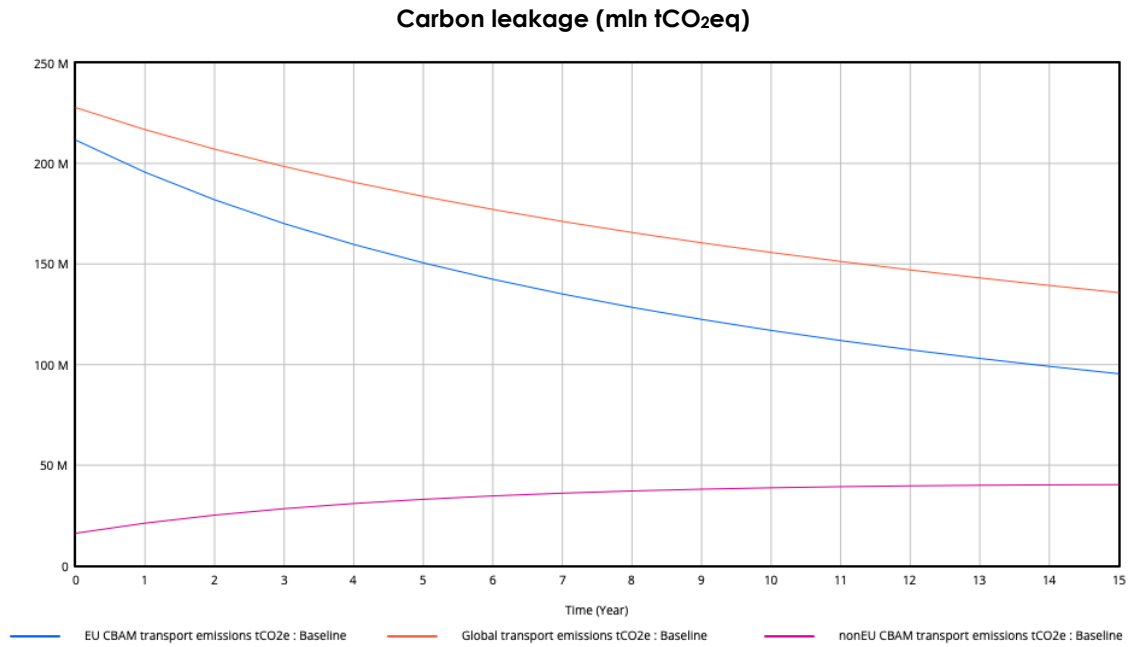
a. Global CPIs Price - Low (EU ETS =€0; EU CPIs=€48, RoW CPIs=€5)



Source: study team

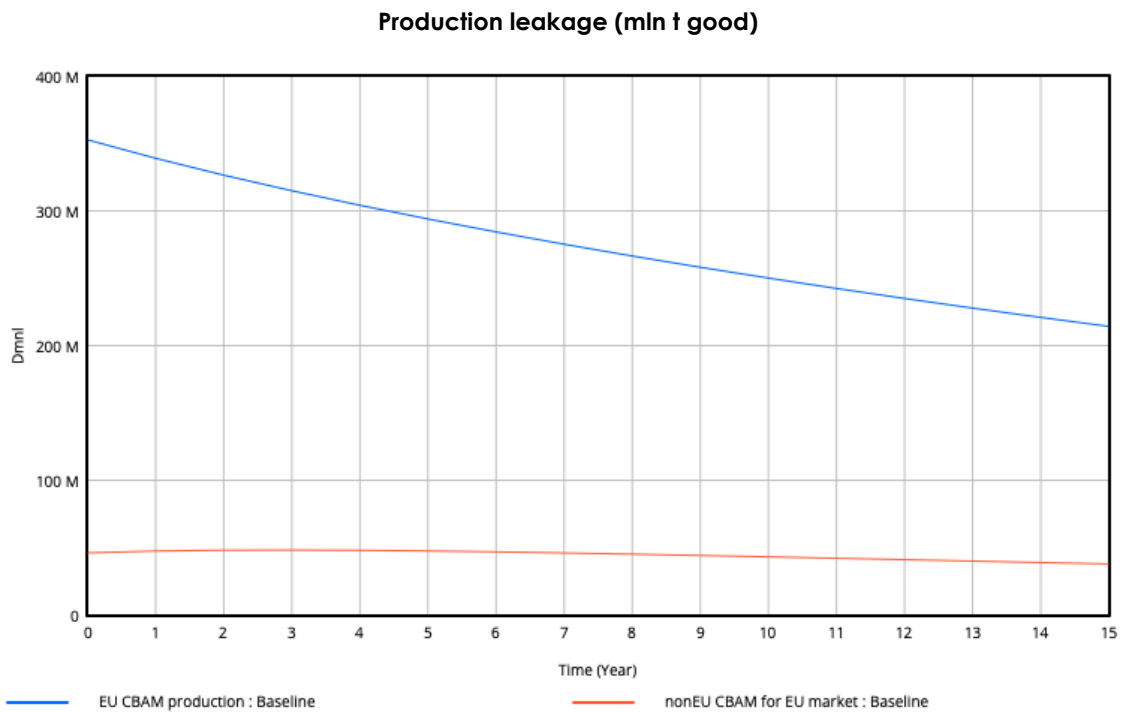
¹⁶² <https://carbonpricingdashboard.worldbank.org/>

¹⁶³ EU ETS share (3.2%) used in scenarios where EU ETS > 0 and EU CPI = 0. EU CPI share (0.9%) used where EU ETS = 0 and EU CPI > 0.

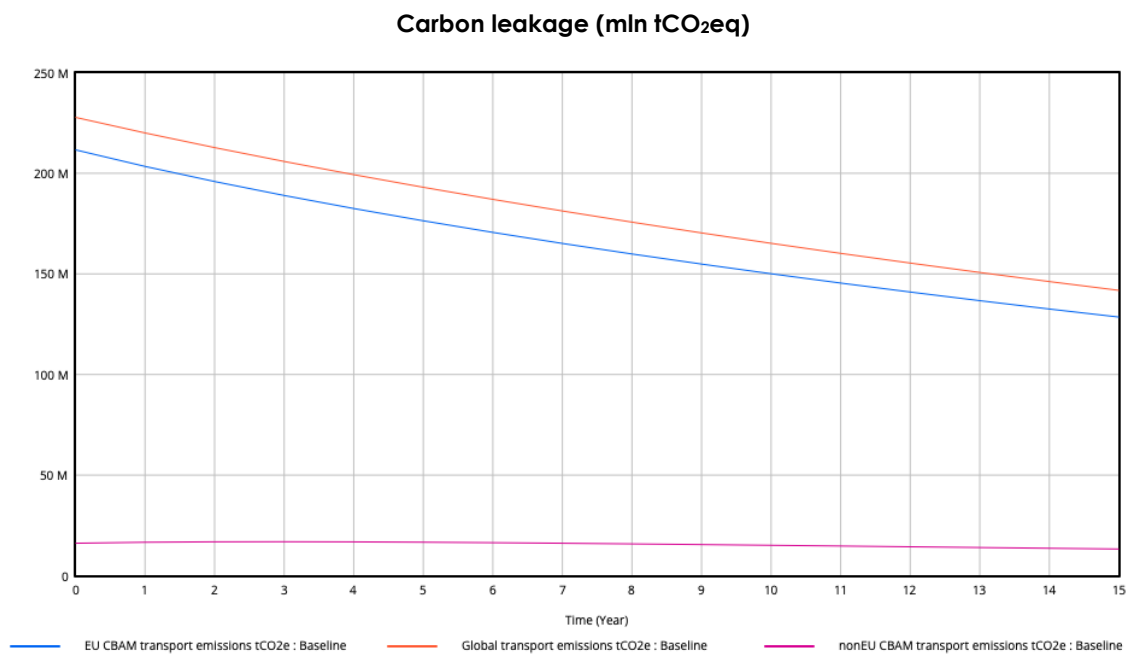


Source: study team

b. Global CPIs Price - Medium (EU ETS =€0; EU CPIs=€48, RoW CPIs=€25)

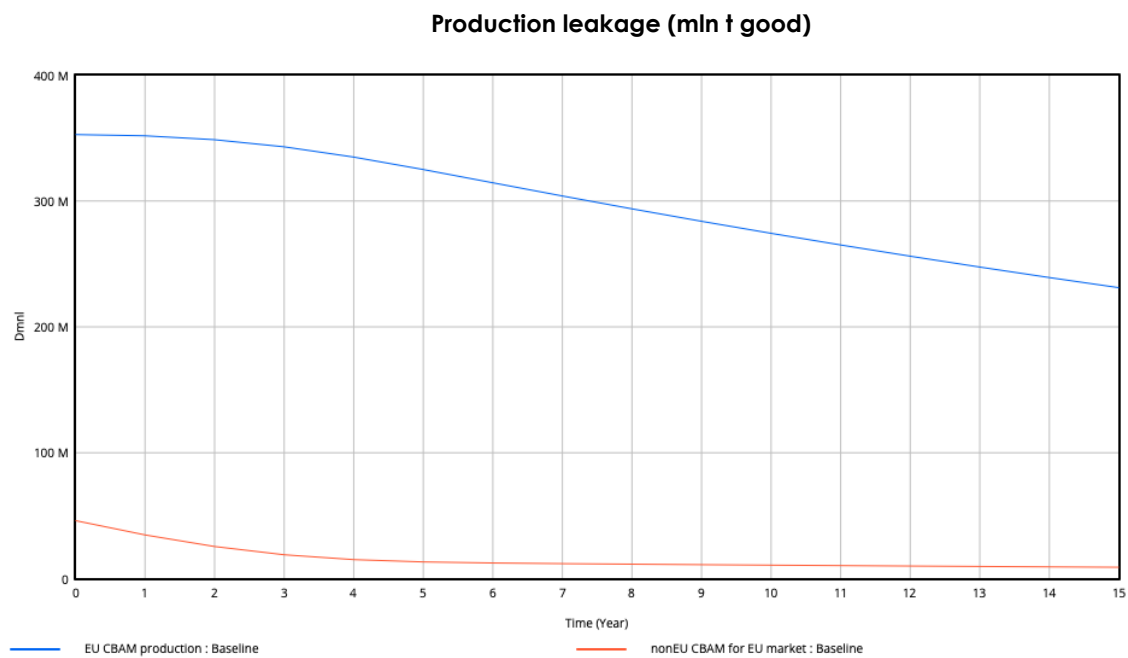


Source: study team

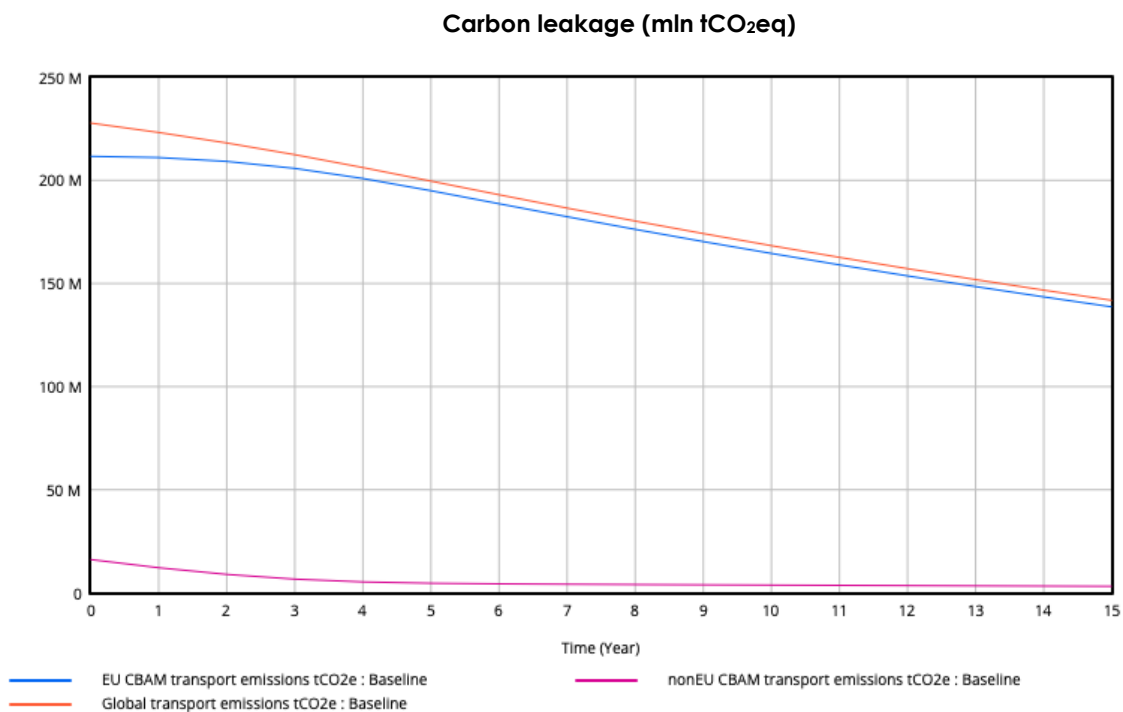


Source: study team

c. Global CPIs Price - High (EU ETS =€0; EU CPIs=€48, RoW CPIs=€100)



Source: study team



Source: study team

Under the **Low price scenario** show that a possible effect would be the shift of production from the EU to third countries, and a considerable increase in the share of global transport emissions by third countries. Under this scenario, it is possible that over the next 11 years, the EU gives away to third countries a significant portion of the production for the EU markets. Global transport emissions continue to drop, as a result of the EU reduction of transport emissions over time due to declining demand.

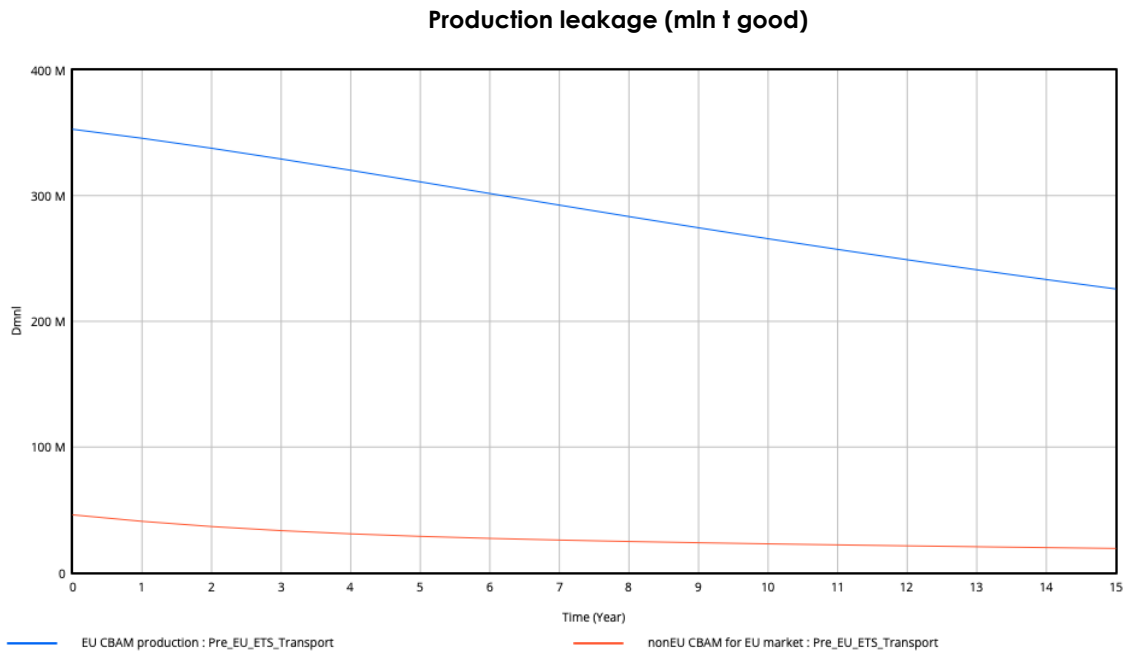
Under the **Medium price scenario**, i.e. current global CPI average of €25 Euro, a very slight temporary increase of third countries production for the EU market can be expected, as well as a slight increase in global transport emissions by third countries. However, under this scenario, the EU shift of production to third countries is not substantial and is eventually overshadowed by the decrease in demand.

Under the **High price scenario**, a global CPI average of €100 Euro was applied, leading to no increase of production in third countries for EU markets, despite the EU production continue to decline, due to mainly reduction of the general demand for CBAM goods in the EU. In fact, there is even some production leakage from third countries to the EU due to the high global CPI. This stabilises after three years, after which further declines are due to the change in demand.

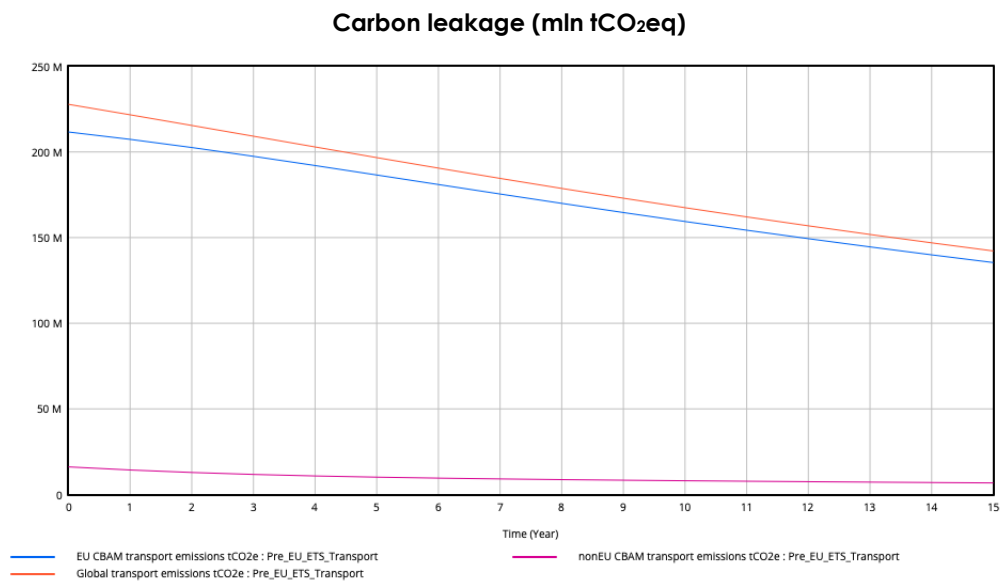
7.6.3 Pre EU ETS Transport. No EU ETS carbon price on transport, No EU CPIs on transport, No Global CPIs on transport

The Pre EU ETS Transport scenario assumes no CPIs on transport emissions anywhere in the world. All the rest of the assumptions remain the same with respect to production and demand trends. Under this scenario the output is as follows:

α. Global CPIs Price - Low (EU ETS =€0; EU CPIs=€0, RoW CPIs=€0)



Source: study team



Source: study team

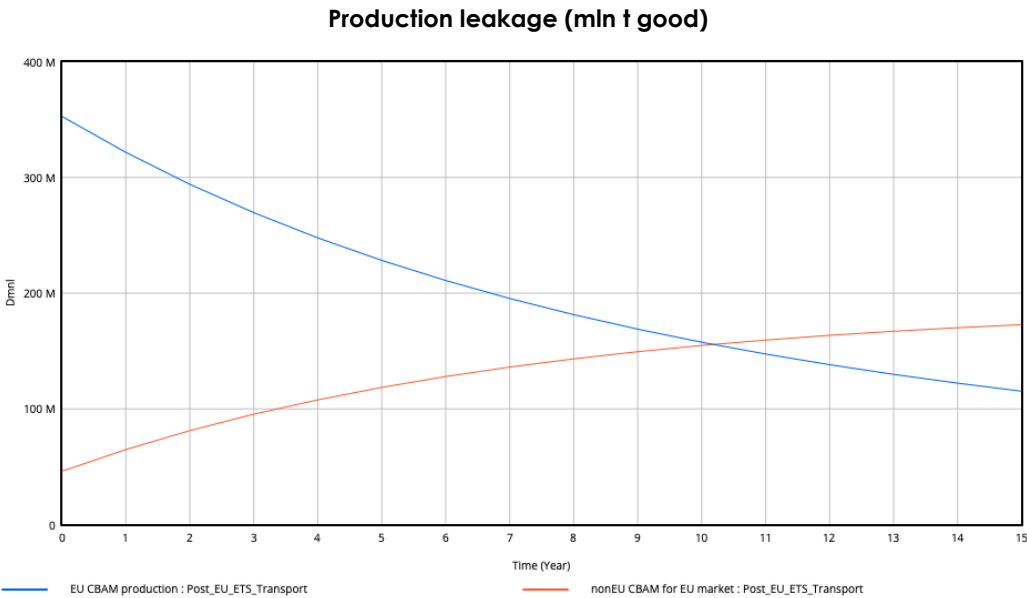
Under the **Pre EU ETS Transport scenario**, the absence of carbon prices for transport emissions, leads to a slight decrease of production in third countries during the first 4 to 5 years, and from the 5th year, production in third countries remains stable. The EU production continues to decline mainly due to the reduction of aggregate CBAM goods demand in the EU. Transport emissions by third countries shipping CBAM goods to the EU

also decreases over time along with EU transport emissions decline, also bringing the global transport emissions of CBAM goods for the EU market downwards.

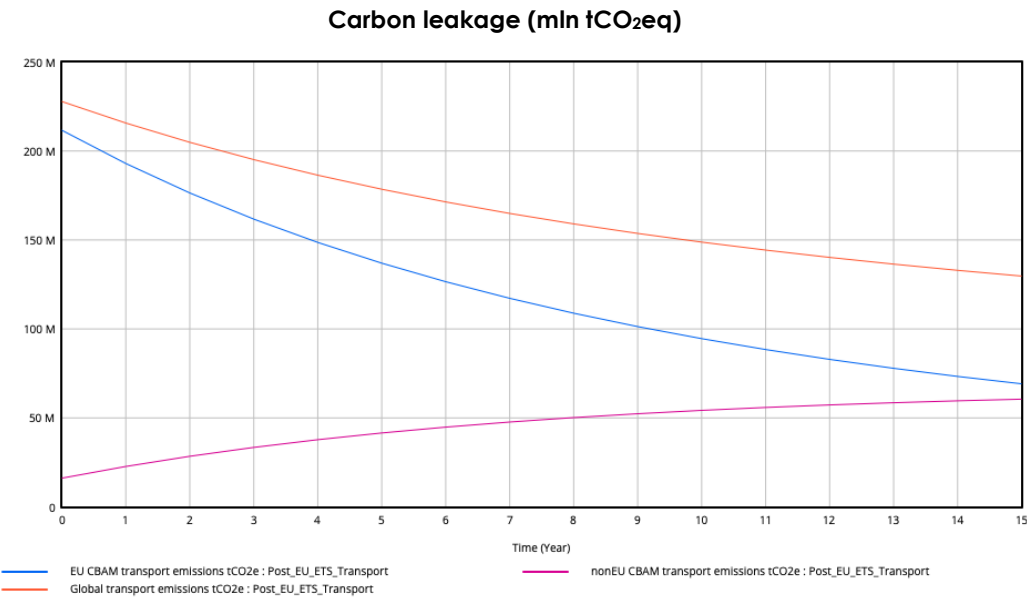
7.6.4 Post EU ETS Transport. EU ETS carbon price on transport, EU CPIs on transport, Global CPIs on transport

The post EU ETS transport scenario, analyses three situations, low, EU ETS level prices, and High prices by third countries CPIs on transport emissions. The EU ETS prices apply in these three scenarios.

a) Global CPIs Price - Low (EU ETS =€55; RoW CPIs=€5)



Source: Study team

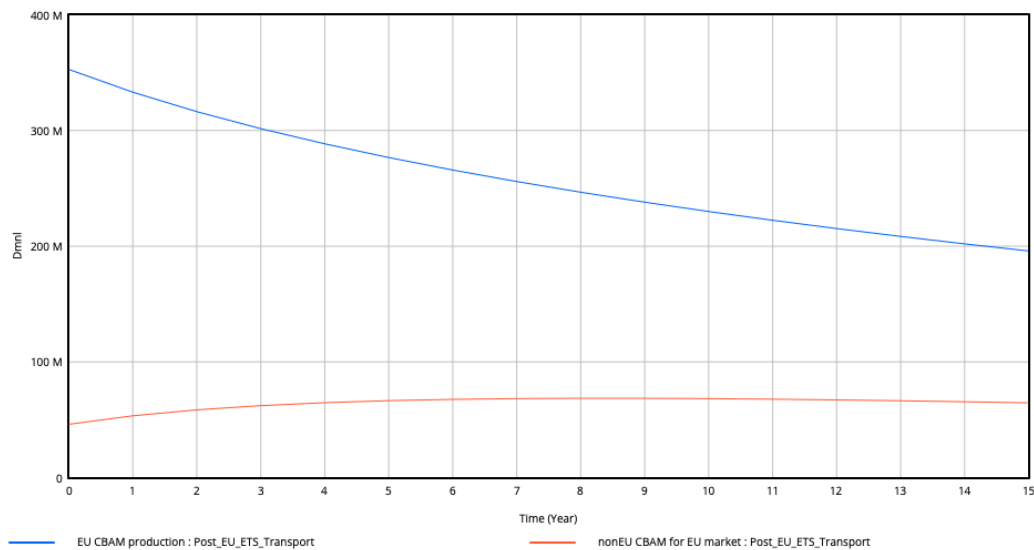


Source: Study team

Post EU ETS–Low price scenario (EU ETS = €55; RoW CPIs = €5) describes the case where the ETS is introduced while CPIs in third countries are maintained very low. The model indicates how this price disparity would result in a shift of **production** from the EU to third countries, where carbon costs are lower. As a result, within approximately **10 years**, the EU could lose most of its production for the EU market to external producers. Global transport emissions continue to drop, due to reduced production and transport emissions in the EU. In comparison with the **baseline scenario**, where EU ETS is not in place but the CPIs in third countries are also maintained very low, the same loss of production happens in approximately **14 years**. Hence, the model is suggesting how the introduction of EU ETS when RoW CPIs are low appears to accelerate the loss of EU production up to two years. Regarding global transport emissions, both baseline and post EU ETS scenarios show very similar trends.

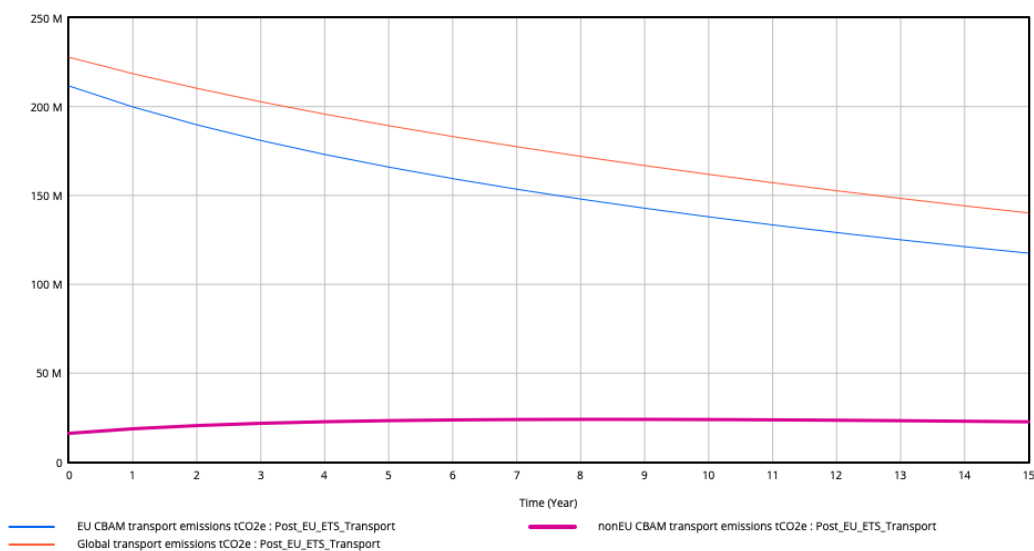
b) CBAM transport emissions scope extended=Global CPIs prices=EU ETS

Production leakage (mln t good)



Source: Study team

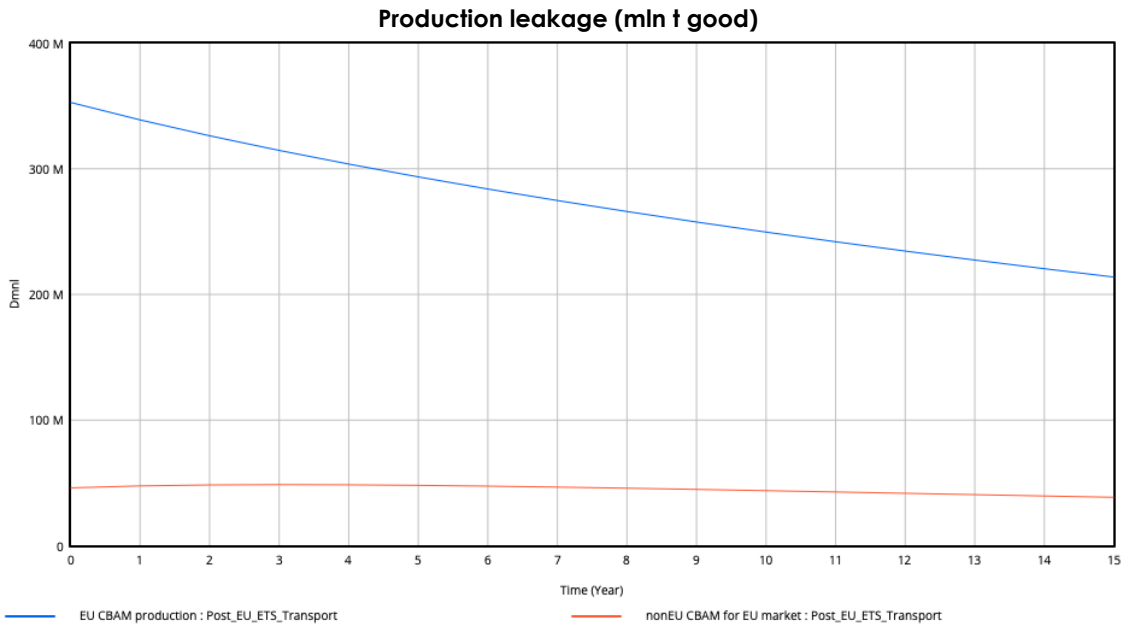
Carbon leakage (mln tCO₂e)



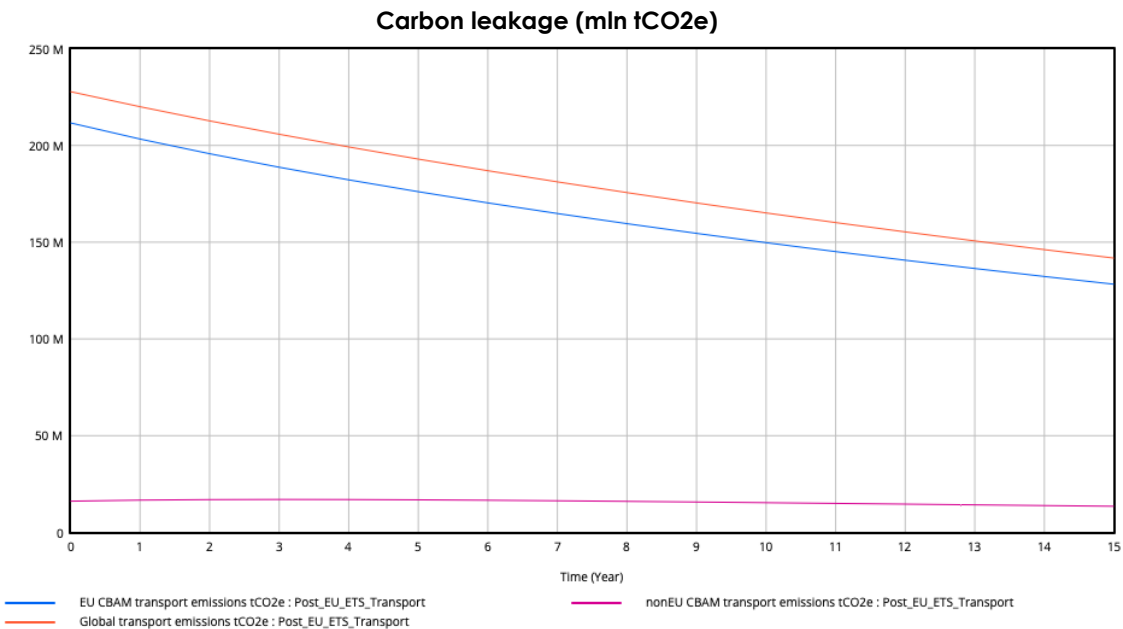
Source: Study team

Under the **CBAM implementation scenario**, i.e. where the current EU ETS price of €54 per tCO₂e applies also to third countries' transport emissions, the production of CBAM goods by third countries for the EU market grows slightly during the next 6 years, as well as their transport emissions. Comparing this scenario with the **baseline scenario**, where the EU ETS is 0, the EU CPIs is €48 per tCO₂e and the RoW CPI is €25 per tCO₂e, in this scenario the production of non-EU CBAM for EU markets grows slightly more due to the larger share of global emissions covered by the EU ETS than the EU CPIs., while still never exceeding 100M.

c) Global CPIs Price – High (EU ETS =€55; RoW CPIs=€100)



Source: Study team



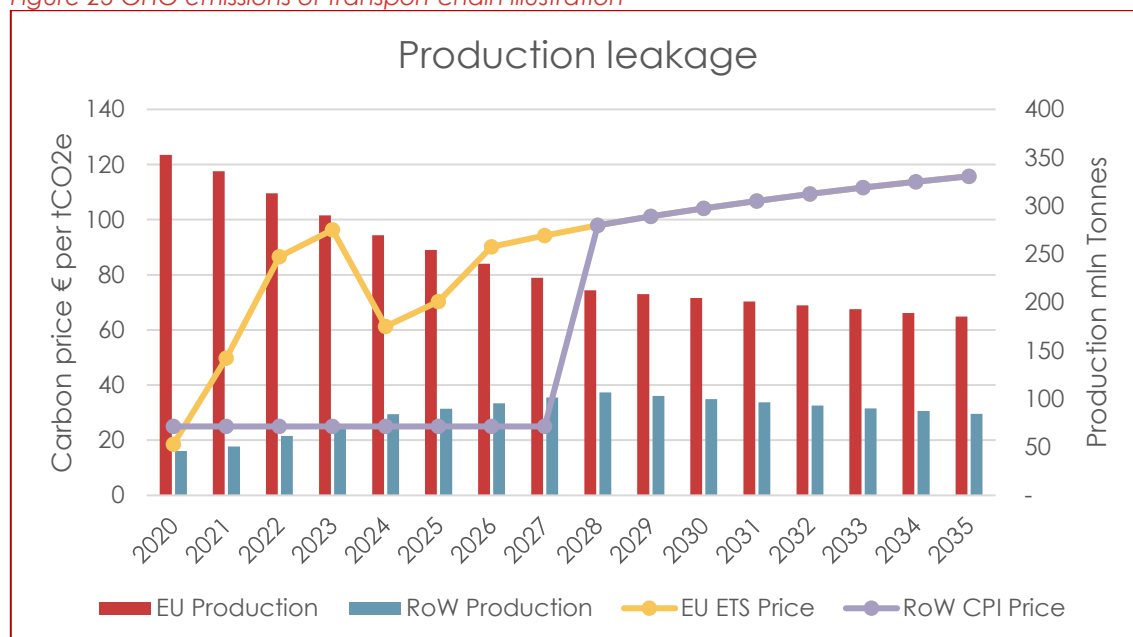
Source: Study team

Under the **high price scenario**, a global CPI average of €100 Euro was applied, the production of CBAM goods by third countries for the EU market remains stable, exhibiting very little production leakage. Compared to the **baseline scenario** (where EU ETS is not in place, but EU CPIs=€48, RoW CPIs=€100), however, the decline of EU production happens earlier and less explosively, mainly driven by the decline in demand. Regarding transport emissions, both scenarios display similar and comparable trends, remaining stable over time. Contrary to the baseline, there is no significant production leakage from third countries to the EU.

7.6.5 Post EU ETS changing price. EU ETS carbon price on transport varies, Global CPIs on transport remain at current levels

The analysis under this scenario focuses on the understanding of the production and carbon leakages according to various EU ETS prices. The rest of the world carbon price is kept constant at €25, until the year 2027, when the CBAM carbon price would be due in a hypothetical scenario of scope extension.

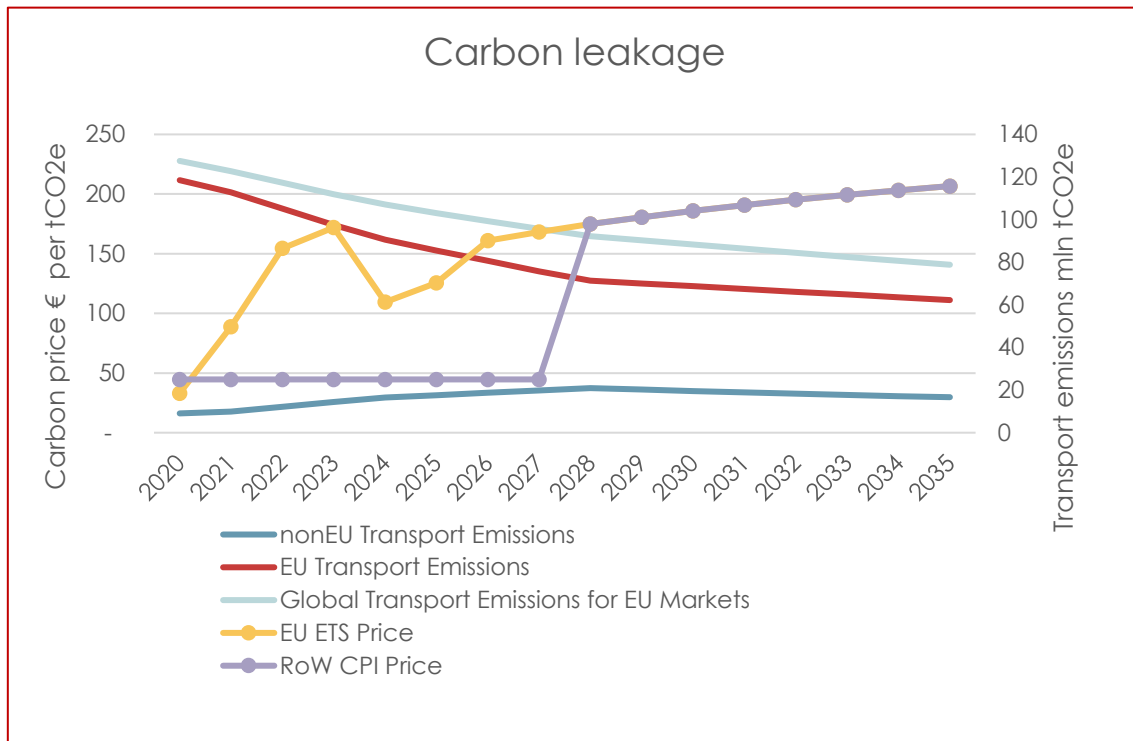
Figure 23 GHG emissions of transport chain illustration



Source: Study team

The system dynamics output indicates that the EU CBAM production continues to decline in the EU, while the production in third countries for the EU market slowly converges towards the year 2030, eventually dropping due to declining demand as EU and RoW prices converge. The carbon price paid on third countries' emissions at the time of CBAM is implemented for transport emissions, does not seem to correct the trajectory. There may not be enough higher carbon prices in the rest of the world to cause a production leakage in favour of the EU. Compared to the medium price baseline scenario, some similarities can be perceived, as both show similar trends on the reduction on EU production. However, in the Post EU ETS changing price scenario, the rise of the EU ETS price over time slightly slows down the decline.

Figure 24 GHG emissions of transport chain illustration



Source: Study team

The output for carbon leakage indicates the possibility of carbon leakage trend based on the increased emissions of the transport sector moving CBAM goods to the EU, while the EU continues to reduce the transport emissions. This trend however interrupts in the year 2028; three years after the adoption of the CBAM scope extension to the transport sector. From that year onwards, the transport emissions generated for the shipment of CBAM goods for the EU markets, continue to decline from both EU transport operators and third countries operators.

However, taking a close look at the long-term trend, the general level of emissions and production in the EU and in third countries start much before the EU ETS implementation on Road and Maritime transport sectors, and there is low probability of these trends to change in the long term. It is important to highlight the fact that emissions in the Rest of the World production (for EU markets) and related transport emissions do not increase substantially by the year 2035.

8 Task 6. Technical feasibility assessment for extending the scope of CBAM to transport emissions from the transport of CBAM goods outside the EU

8.1 Summary of the objectives and results of the task 6

The objective of Task 6 was to assess the technical feasibility of including emissions from transport services associated with the movement of CBAM goods outside the EU, analysing data requirements, evaluating data availability for importers, and determining the extent and cost of EU authorities' ability to monitor the accuracy of reported information. The results of each one of the subtasks that are detailed ahead, as the following:

Task T6.1, development of a methodology for tracing and calculating transport emissions of CBAM goods, was based on CountEmissions EU and ISO14083 standards. The task shows the different approaches for calculating and reporting transport emissions with regards to sources of information and recognised calculation standards. From the calculation viewpoint, it is feasibility the measurement and reporting of transport emissions along the logistics chain. However, the feasibility of measurement and reporting may be less for small transport operators, which is why Task 6.5 developed default values for CBAM goods transport emissions at the CN code level, under different information availability scenarios.

The assessment of (Task 6.2) the data collection challenges for importers regarding transport emissions of CBAM goods, indicate that challenges at three main levels, Data sharing long the logistics chain up to the importer, lack of internationally recognised emission intensity and factors, and inability importers to access information from small operators in the context of complex international logistics operations. Implication: there is a need to set up internationally accepted sources of emissions intensity and agreed by the EC and third countries.

The assessment of (Task 6.3) verification approaches for transport emissions reporting, shows that already existing verification mechanisms can be used for transport emissions verification. Similar procedures of submission of reports for desk assessment, verification of transport emissions data at the logistics operator' site, can be performed. On the other hand, while transport operators are acquainted to reporting standards, there is a need to align multiple emerging regulations requiring emissions reporting from the road, maritime and air transport sectors. This implies that a transitional period for the transport sector would be required to allow operators to adapt measuring and reporting obligations.

The assessment of (Task 6.4) the institutional and infrastructural implications on EU authorities, shows that while the technological and information systems means across EU institutions to receive and check data shared by economic operators is well developed, specific competences on transport operations and emissions are required, to be able to design and implement a verification system applicable to multiple logistics operators located in multiple international locations. Implications: there is a substantial level of new expertise to be acquitted by verifiers and EU authorities receiving CBAM report with transport emissions attributed at the CN code level.

The assessment (Task 6.6) of Carbon Price Instruments (CPIs) coverage for different transport modes per region, indicates that the transport sector both at the EU and in third

countries the transport sector is generally in the CPIs scope, either directly (point source) or indirectly (upstream carbon price or through electricity generation used by the transport sector-i.e. rail). Given that several Member States already have CPIs implemented, the future entry into force of the EU ETS II for the road transport may need to be further analysed, to assess the risk of double carbon pricing in the EU. It is important to notice that CPIs are also present in countries near the EU, such that any potential shift of production to those regions, would in any case have been covered by CPIs in the transport sector. However, the effective carbon prices in those schemes are often substantially lower than the expected prices under the EU ETS II, meaning that carbon leakage risk cannot be ruled out.

Task 6 concluding remarks. The assessment indicates that the extension to include transport emission of CBAM goods could potentially be possible from the view point of the measurement, storing and sharing of transport emissions data along logistics operators, with some limitations. Moreover, it could present challenges for small transport operators in third countries, who are usually not known even by the international logistics operators. While EU institutions have the information technologies available for reporting, the verification process of transport emissions at the level of CN code remains a challenge, as it is not a practice that transport operators are used to. CBAM verifiers would require per single trade flow to verify emissions in different transport chain elements (all those used for the movement of a CN code lote from origin to destination) rendering the verification process complex to coordinate, and probably costly, as it would also require specific competences to be defined for the CBAM verifier/certifier.

8.2 Introductory note to task 6

As it will be shown in the next sections, several stakeholders indicated the need to align any potential extension to transport emissions to existing legislations, especially with the proposed CountEmissions EU Regulation and the ReFuelEU Aviation¹⁶⁴. Considering these findings the following note was developed for further consideration during T6.

The Proposal for a Regulation of the European Parliament and of the Council on the accounting of greenhouse gas emissions of transport services ¹⁶⁵ (CountEmissionsEU).

The proposed Regulation CountEmissions EU establishes a harmonised framework for calculating and disclosing emissions for transport services starting or ending on Union territory. CountEmissions EU applies an "opt-in" approach: while it does not mandate the calculation and reporting of GHG emissions, it obliges those entities sharing their calculation to any third parties to follow a dedicated set of rules. The proposed regulation adopts the Well-to-Wheel/Wake EN ISO 14083:2023 method and prioritises primary data while permitting secondary data (Emission Factors and default intensity values) from EU databases or approved third-party datasets.

¹⁶⁴ Consolidated text: Regulation (EU) 2023/2405 of the European Parliament and of the Council of 18 October 2023 on ensuring a level playing field for sustainable air transport (ReFuelEU Aviation) (Text with EEA relevance)Text with EEA relevance. <http://data.europa.eu/eli/reg/2023/2405/2023-10-31>

¹⁶⁵ European Commission, 2023. Proposal for a Regulation of the European Parliament and of the Council on the accounting of greenhouse gas emissions of transport services. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52023PC0441>

A potential key synergy with the CBAM lies in the harmonised set of rules for GHG emissions calculations and disclosure provided by the proposed Regulation. By integrating transport services-related emissions data, CBAM can more accurately reflect the GHG emissions of imported products, enhancing the environmental integrity of the carbon pricing mechanism¹⁶⁶. This is particularly relevant given the potential scale of emissions from transport services linked to the movement of CBAM goods.

The strategic synergies between CBAM and the proposed Regulation are also significant in the context of a potential CBAM scope extension to other goods and transportation services. The detailed input and output data requirements mandated by the proposed Regulation (Articles 5–9) provide a robust framework for calculating emissions in a consistent and transparent manner. This framework could support the expansion of CBAM to cover GHG emissions from the transportation of goods, thereby enhancing the mechanism's environmental effectiveness and contributing to the EU's decarbonisation objectives.

Both regulations emphasise transparency and verification. The 'Proposal for a Regulation Of The European Parliament and of The Council on the accounting of greenhouse gas emissions of transport services' mandates clear communication of emissions data and verification by accredited bodies (Articles 10–13), which supports CBAM's requirement for credible and verifiable emissions reporting (Article 8). The digital integration of emissions data further facilitates interoperability with CBAM's digital infrastructure, enhancing traceability and customs procedures.

Pending the final agreement of the co-legislators, the update of the EU databases under the proposed Regulation (Articles 5 and 8) should ensure accurate and consistent emission factors, supporting CBAM's emissions calculations. CBAM can maintain high data integrity and transparency standards by aligning with these databases, promoting global decarbonisation¹⁶⁷. Furthermore, to remain coherent with the EU ETS II, CBAM declarants would need from the European Commission central database to include:

- Metrics disaggregating CO₂e into Carbon dioxide (CO₂), Methane (CH₄), and Nitrous oxide (N₂O). This is due to road and aviation transport's GHG emissions in EU ETS scope being CO₂ and maritime transport being CO₂, CH₄ and N₂O. Potential scope extension to CBAM goods' transport emissions needs alignment with EU ETS transport emissions.
- Transport chain distance and volume per CN code¹⁶⁸:
 - For CBAM goods. From origin (non-EU country's CBAM good installation) to destination (EU Member State's port of entry) by transport mode. If multimodal, the average distance per transport chain element¹⁶⁹.

¹⁶⁶ CountEmissions EU adopts a WtW and not an LCA approach. It is advisable to care for the Carbon Footprint, as it can be misleading as it often refers to LCA.

¹⁶⁷ The update is proposed to take place every 2 years to enhance stability of the calculations as well as predictability.

¹⁶⁸ Transport chain is the sequence of elements related to freight that, when taken together, constitutes its movement from an origin to a destination (ISO 14083:2023).

¹⁶⁹ Transport chain element is a section of a transport chain within which the freight is carried by a single vehicle or transits through a single hub (ISO 14083:2023).

- For CBAM precursors. Same as per CBAM goods.¹⁷⁰
- Transport/vehicle type and size (e.g. Bulk carrier, size (200000 +- dwt))

- Carbon price effectively paid

A limitation lies in the fact that CBAM only requires CBAM goods and precursors with embedded emissions to be reported, and therefore, only transport emissions related to those CN codes can be accounted for. It was shown in section T5.1 how transport emissions can only be partially reported, and it is not possible to account for the whole supply chain-related transport emissions. Harmonised and consistent transport emission intensity is missing for several countries, meaning CBAM declarants will often be required to rely on default values. Furthermore, as rail transport with diesel trains is not in the EU ETS; consequently, in a multimodal setting, rail transport activity and its emissions may need to be excluded from CBAM reporting, leaving a potentially important part of the transport chain unaccounted. While the electricity used for rail is included in the EU ETS scope, the regulated industry in this case is the 'electricity and heat generation' industries, not the rail sector specifically. If the rail sector were included in the CBAM scope, it may raise concerns internationally over potential unfair treatment of the third countries' rail transport sector, because rail is not regulated by the EU ETS directly. If the CBAM scope was to include the rail in the future, there would be a need also to include the rail in the EU ETS to avoid unfair treatment of third countries. This would allow to regulate not only the emissions of the electricity consumed by the rail sector in the EU and in third countries, but also the emissions from fuel (diesel) consumption of the rail sector in third countries.

In conclusion, while the harmonisation of GHG emissions accounting methodologies and the integration of transport-related GHG emissions data into CBAM's reporting framework offer strategic opportunities to enhance the accuracy and environmental integrity of both regulations, emissions will not account for the full CBAM supply chain transport works. Strategic coordination and possible regulatory adjustments would be required to fully align the two regulations by updating the CBAM reporting structure, potentially linking to future CountEmissions EU databases.

Given the full compliance status of CBAM, and the opt-in binding approach¹⁷¹ of the CountEmissions EU, all CBAM declarants will automatically be required to comply using the CountEmissions EU reporting standards, to declare transport emissions of their CBAM goods declared. Thus, the following potential scenarios can be foreseen.

The Omnibus simplification proposal for CBAM estimates that 200k EU importers are subject to CBAM declarations¹⁷². The CountEmissions EU, if adopted, will require freight

¹⁷⁰ Rail distance and its emissions could be reported as optional.

¹⁷¹ The 'opt-in binding' means that whether by compliance or voluntary, any transport emissions data communication/report is required to follow the CountEmissions EU measurement and reporting standards.

¹⁷² Table 1 of the COMMISSION STAFF WORKING DOCUMENT Accompanying the document Proposal for a Regulation of the European Parliament and of the Council amending Regulation (EU) 2023/956 as regards simplifying and strengthening the carbon border adjustment mechanism. Accessible at https://commission.europa.eu/publications/omnibus-i_en

transport, wholesalers, retailers, and manufacturing to perform GHG accounting¹⁷³, which may allow CBAM declarants to have readily available information from CountEmissions EU compliant operators¹⁷⁴. Thus, the following scenarios may apply:

- **Scenario 1.** CBAM scope is not extended to transport emissions and CountEmissions EU is not adopted. Under this scenario, only freight transport, wholesalers, retailers, and manufacturing that are already performing GHG accounting will continue to do so, and no change climate targets may be foreseen.
- **Scenario 2.** CBAM scope extension is not extended to transport emissions and CountEmissions EU is adopted. Under this scenario, the impact assessment of the CountEmissions EU applies, and will apply only to companies voluntarily accounting and reporting for GHG emissions. Climate targets will be based on progress based on voluntary uptake by operators.
- **Scenario 3.** CBAM scope is extended to transport emissions and CountEmissions is not adopted. Under this scenario, it is expected that 200k CBAM declarants will need to collect transport information and report transport data. Transport monitoring and reporting specifications can be tailored for CBAM declarations and use the CBAM registry/platform. Climate targets will be based on CBAM declarants and CBAM importers' transport or logistics operators.
- **Scenario 4.** CBAM scope is extended to transport emissions and CountEmissions is adopted. Under this scenario, it is expected that 200k CBAM declarants will need to collect transport information and report transport data based on the CountEmissions EU specifications. If CountEmissions EU sets up its own central registry, CBAM declarants could be allowed to access CountEmissions EU registry to compile the necessary information for CBAM declarations. Climate targets will be based on CBAM declarants and CBAM importers' transport or logistics operators.

Regulation (EU) 2015/757 on the monitoring, reporting and verification of greenhouse gas emissions from maritime transport.

Directive (EU) 2023/959¹⁷⁵ further reinforces this integration by allocating allowances and imposing surrender requirements on maritime transport emissions, covering 50% of emissions from international voyages and 100% of emissions from intra-EU voyages and

¹⁷³ Table 23 and 25 of the SWD(2023) 441 final. COMMISSION STAFF WORKING DOCUMENT IMPACT ASSESSMENT REPORT. Accompanying the document. Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND THE COUNCIL on the accounting of greenhouse gas emissions of transport services. Accessible at https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13217-Count-your-transport-emissions-CountEmissions-EU_en

¹⁷⁴ CBAM required Importers or an authorised CBAM declarant to present CBAM declarations. When actual values are used, the installation in the third country communicates the relevant emissions data through a communication template to the CBAM Declarant. CountEmissions EU on the other hand, may oblige all economic operators (i.e. CBAM supply and logistics chain), not only CBAM Declarants, to follow the specifications of the CountEmissions EU regulation.

¹⁷⁵ Directive (EU) 2023/959 of the European Parliament and of the Council of 10 May 2023 amending Directive 2003/87/EC establishing a system for greenhouse gas emission allowance trading within the Union and Decision (EU) 2015/1814 concerning the establishment and operation of a market stability reserve for the Union greenhouse gas emission trading system. <https://eur-lex.europa.eu/eli/dir/2023/959/oj/eng>

within ports under the jurisdiction of a Member State (Article 3ga). This Directive (EU) 2023/959 also mandates monitoring and reporting of emissions at the company level in accordance with Regulation (EU) 2015/757 (Article 3gd) (MRV Regulation) and aligns verification and accreditation rules for maritime emissions (Article 3ge).

Regulation (EU) 2015/757¹⁷⁶ on the monitoring, reporting, and verification (MRV Regulation) of GHG emissions from maritime transport establishes a framework for accurate monitoring and reporting of emissions from ships arriving at, within, or departing from EU ports. It applies to ships of 5,000 gross tonnage and above, as well as to smaller cargo and offshore ships from January 2025 onwards (Article 2). The monitoring, reporting, and verification (MRV) covers CO₂, methane (CH₄), and nitrous oxide (N₂O) emissions from fuel combustion during voyages and at berth, requiring consistent and transparent data reporting (Articles 4–5).

A strategic potential synergy could be explored by leveraging the MRV Regulation's detailed emissions data for maritime transport to estimate transport-related emissions linked to CBAM goods. The MRV Regulation requires companies to monitor and report emissions for each ship on a per-voyage and annual basis (Articles 8–10), offering granular data that could enhance the accuracy of transport-related emissions estimates for CBAM goods. This would support the assessment of the scale of emissions from transport services linked to the movement of CBAM goods, providing valuable insights into their carbon footprint.

Moreover, while the MRV Regulation mandates verified emissions data (Articles 13–15) and emphasises consistency and transparency in reporting (Article 4), its life-cycle approach differs from CBAM's production-based accounting. This could limit the direct applicability of MRV data for CBAM, as emissions from maritime transport are not included in CBAM's embedded emissions calculations. For meaningful synergies, coordination would be needed to harmonise emissions scopes and reporting methodologies. This for instance include the need to estimate well to tank and tank to wheel emissions.

Nevertheless, the MRV Regulation, could indirectly support CBAM by enhancing the overall transparency and comparability of emissions data within the EU market. Their robust verification framework and continuous improvement mechanisms contribute to a reliable data ecosystem, potentially facilitating future regulatory alignment. Additionally, integrating MRV data with CBAM's digital infrastructure, including the CBAM registry (Article 14), could enhance traceability and transparency in emissions reporting for imported goods.

In conclusion, the MRV Regulation, provides a detailed and verified dataset for maritime emissions, which could be valuable for estimating transport-related emissions of CBAM goods. This would be particularly beneficial for assessing the scale of emissions from transport services linked to the movement of CBAM goods. However, in addition to estimating WtT and TtW emissions, strategic coordination and regulatory adjustments

¹⁷⁶ Regulation (EU) 2015/757 of the European Parliament and of the Council of 29 April 2015 on the monitoring, reporting and verification of greenhouse gas emissions from maritime transport, and amending Directive 2009/16/EC. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02015R0757-20250101>

would be required to harmonise emissions boundaries, ensuring accurate and consistent emissions accounting in support of the EU's decarbonisation objectives.

8.3 T6.1 - Methodology development for tracing and calculating transport emissions of CBAM goods

This chapter focuses on the methodology for tracing and calculating the greenhouse gas (GHG) emissions associated with the transport of CBAM goods, which helps the assessment of the **technical feasibility** of extending the scope of CBAM to transport emissions.

8.3.1 Definition of transport chain elements

The first step in developing a methodology to calculate emissions related to the transport of CBAM goods, is to decompose the transport chain into individual **transport chain elements (TCEs)**, as defined in ISO 14083 framework. A TCE is defined as either the movement of freight by a single vehicle or the processing of freight within a single hub. Each time a shipment is transferred to a new vehicle, switches transport mode, or passes through a different hub, a new TCE must be defined. **This segmentation is essential to ensure that each element of the transport chain is identified and can be assigned a corresponding GHG emissions.**

The transport chain of a CBAM good, including not only the transport of the CBAM good itself but also the upstream transport steps associated with its precursors¹⁷⁷, comprises the full sequence of operations from the consignor's point of departure to the consignee's point of arrival within the EU, and **each stage in the whole process should be identified as distinct transport chain element**. However, this identification is especially complex in the context of international transport chains, where goods often transit through multiple countries, involve several modes of transport, and pass-through various handling or storage stages before reaching their destination. Nonetheless, it is essential that every operation within the transport chain, regardless of mode, geography, or operator, is recorded as a separate TCE, in line with ISO 14083 principles of completeness and transparency, to ensure consistent and accurate quantification of GHG emissions across the whole transport chain.

Note on scope for Precursors

While the methodology allows for calculation of all transport chain elements, from precursors to CBAM good's final EU destination, it is necessary for the EC to clarify whether the declarant should focus on transport emissions that only include CBAM goods imported into the EU (i.e. transport chain elements from CBAM exporter to EU entry import point) or also its precursors. The study team's understanding is that precursors are also CBAM goods and theoretically, could also be in scope. However, the amount of data points and evidence to be produced on the precursors' transport chain elements may increase the administrative complexity substantially.

¹⁷⁷ Pending internal decision from the EC.

Each TCE is classified as follows:

- A transport operation category (TOC) covers the active movement of freight, whether by road, rail, sea, inland waterway, air, or pipeline, and includes emissions generated directly by the vehicle or transport equipment in operation. Each TOC groups transport activities that share similar characteristics and are sufficiently homogeneous to allow comparable emissions accounting.
- A hub operation category (HOC) captures the emissions associated with stationary activities, such as loading, unloading, handling, transshipment, or temporary storage of goods in logistics hubs. Similar as TOCs, each HOC groups operations occurring at fixed locations that share similar emission characteristics (i.e. energy source, handling equipment used).

Note on scope for hub operation category (HOC)

*While the methodology allows for calculation of transport and hub operations' emissions, as agreed with the EC, **hubs' emissions are not included in the emissions reporting**. Thus, while for completeness the methodology describes hubs operations, the hubs' emissions are not accounted for.¹⁷⁸*

8.3.2 Tracing and structuring of the transport chain

Following the identification of all relevant **transport chain elements (TCEs)**, the next step is the **structuring of the transport chain** in a way that enables emissions to be calculated. This involves **mapping out the entire route take by a CBAM good, and of its precursors¹⁷⁹**, from its point of origin to the EU border, and defining the characteristics of each TCE.

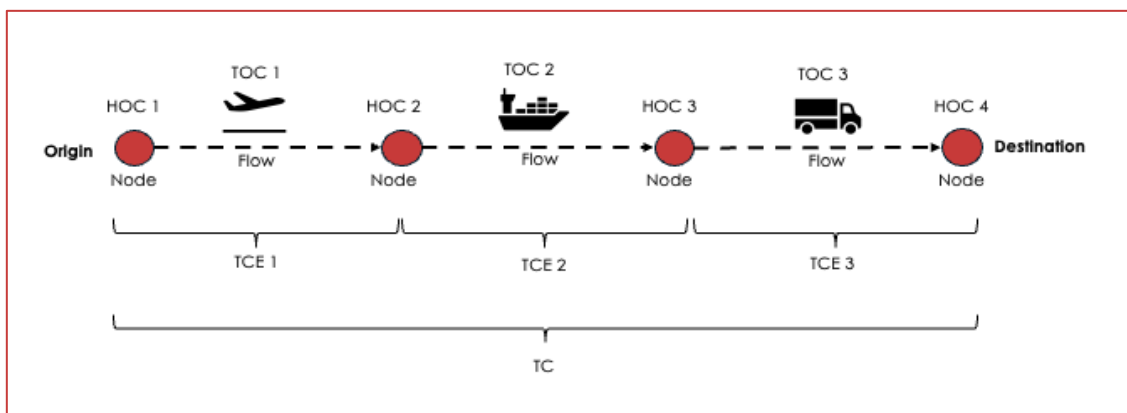
As previously explained, every transport operation and change in mode or vehicle must be recognised as a distinct TCE and included in the overall emissions profile. This applies regardless of whether operations are carried out by the same or different operators, or whether they occur within a single country or across borders.

The below illustration shows a simplified example of a transport chain (TC) moving a CBAM good and its precursors from its origin to its destination (i.e. the EU border). The chain is composed of three distinct Transport Chain Elements (TCEs), each representing a unique segment of the journey. Within each TCE, there is a **Transport Operation Component (TOC)**, or flow, which represents the active movement of goods by air, sea, train or road, and a **Hub Operation Component (HOC)**, or node, marking the handling or transfer points at nodes such as airports, ports or distribution centres.

¹⁷⁸ HOC operations require specific complex data from hub operators that are rarely available to transport operators involved in the movement of CBAM goods. Moreover, assuming that operators do not provide operational data for HOCs emissions, there are no available databases providing default values on HOCs, while for transport emissions there are databases and average values that can be used. As an example of the complexity, a supporting project for the CountEmissions EU is the CLEVER project (<https://www.etp-logistics.eu/alice-projects/clever-project/>), which itself is not developing HOCs emissions factors, because it considers it outside the project scope. Thus, this could substantially limit the capacity of CBAM declarants to rely on robust HOC default values.

¹⁷⁹ Pending internal decision from the EC.

Figure 25 Transport chain illustration



Each of these components must be identified and assessed individually to calculate the full GHG emissions profile of the CBAM transport route, which will be explained in the following sections.

8.3.3 Key concepts and definitions

Before presenting the methodology for calculating GHG emissions, it is essential to introduce a set of key concepts and data requirements on which the reporting of transport emissions is based.

Once the transport chain has been structured into TCEs, specific parameters must be defined for each element to be able to consistently attribute GHG emissions. These include the following:

- The **mode of transport** used (e.g. road, rail, sea, air)
- The **vehicle type**, including specifications (e.g. heavy-duty truck, container ship, freight train)
- The **fuel used** (e.g. diesel, electricity, LNG, biodiesel)
- The **weight of goods transported** in each segment, expressed in tons.
- The **distance travelled** in each segment, ideally based on the actual transport route or an approximation.

8.3.3.1 Transport activity (ton-km)

Transport activity refers to the **amount of goods moved and the distance over which they are transported, expressed in ton-kilometre (t-km)**, and it is calculated with the product of the weight of the shipment (expressed in tons) and the distance travelled (kilometres), and **it is necessary to be calculated at TOC level**.

Transport activity is an essential variable when applying the activity-based method for calculating emissions, specifically when fuel consumption data is not available. These methods will be explained in the following section.

8.3.3.2 GHG Emission Intensity (gCO₂e/ton-km)

GHG Emission intensity is defined as a coefficient that links GHG emissions with a specific transport activity. It quantifies the emissions generated per unit of transport service provided. For freight, it is commonly expressed in grams of CO₂ equivalent per ton-kilometre (gCO₂eq/ton-km).

In the following table, the influencing factors affecting the GHG emission intensities are explained and linked with the parameters that are necessary to define by TCE, as explained at the beginning of the section.

Table 46 Influencing factors affecting the GHG emission intensities

Influencing Factor	Why it is influenced
Mode of transport	Different modes have varying energy and emissions profiles
Energy Carrier	Cleaner fuels emit less CO ₂ e per energy unit
Vehicle type	Older or heavier vehicles may have higher fuel consumption per ton-km
Load factor	Better utilization lowers the emissions per unit transported
Distance and routing efficiency	Longer or inefficient routes can increase the total emissions
Regional energy mix	The GHG emissions associated with the energy used (electricity or fuel vary by region due to differences in how energy is produced)

8.3.3.3 Emission factor (gCO₂e/unit)

An emission factor is a coefficient that quantifies the amount of GHG emissions released per unit of energy or fuel consumed, typically expressed in grams of CO₂ equivalent per litre, kilogram, or kilowatt-hour (gCO₂eq/unit), and they allow for the estimation of GHG emissions based on energy or fuel usage.

In the following table, the influencing factors affecting the emission factors are explained and linked with the parameters that are necessary to define by TCE, as explained at the beginning of the section.

Table 47 Influencing factors affecting the emission factors

Influencing Factor	Why it is influenced
Energy carrier¹⁸⁰	Different energy carriers (e.g., fossil fuels, electricity, or biofuels) have different ghg intensities per unit consumed due to chemical composition. Biofuels, for example, can have significantly lower emission factors depending on their source and production method.
Upstream emissions	The upstream emissions related to energy production and delivery can vary depending on the production method itself and the source location

¹⁸⁰ It is important to note that different vehicles have different levels of efficiency and emissions. A heavy-duty diesel truck, for example, will emit more GHG per litre of fuel burned than a light van.

Influencing Factor	Why it is influenced
Electricity generation mix (region specific) ¹⁸¹	Regional emission factors are needed as GHG intensity depends on how the fuel was produced
Blending and composition	The composition of the energy carrier affects its emission factors

8.3.3.4 Data availability

ISO 14083 emphasizes that the transparency of GHG emission calculation depend on the quality and availability of the data. To ensure flexibility across method, the framework allows the use of different data sources, classified as primary, secondary and default data, based on what is available and feasible for the reporting. Definitions are found below.

- **Option A: Primary data** includes directly measured values from operations, such as actual **fuel** use, measured cargo **weights**, or GPS-derived transport **distances**. This type of data is considered the most accurate and is preferred where available.
- **Option B: Secondary data** is derived from internal estimations based on operational records or business systems and may include planned transport distances or shipment weights from booking data.
- **Option C: Default data** consists of generalised values published in standardised databases. These values are generally conservative and represent average operations for a given transport profile.

When primary data is not available or accessible, emissions may be estimated using secondary data or default values provided by recognized sources.¹⁸²

8.3.4 Calculating GHG Emissions

The calculation of GHG emissions for a transport chain follows a **bottom-up approach** beginning at the Transport Operation Category (TOC) level and aggregating up to the overall Transport Chain (TC) level. This process requires the collection of various variables at each stage, which are then used to determine emission intensities, transport activities, and lastly the total GHG emissions across the entire transport chain.

There are two main calculation approaches defined under ISO 14083, **fuel-based method** and **activity-based method**, as explained in the following sections.

When fuel consumption data is not accessible, the **activity-based method** is used. This approach allows to estimate emissions based on the transport performance of the TOC, measured in tonne-kilometres. The activity of a TOC is calculated by multiplying the weight of the goods transported, including packaging, by the distance they were moved. The resulting tonne-kilometres are then multiplied by an **emission intensity**.

¹⁸¹ Emission factor also depends on the region, due to variations in energy sources or fuel quality. The ISO 14083 explicitly recognizes this regional variability and requires transparency in justifying the choice of factors and/or intensities in the final reports.

¹⁸² See Annex J of ISO 14083

8.3.4.1 Fuel-based method for GHG emissions calculation

The **fuel-based method** uses actual energy consumption data to estimate emissions. In this case, the primary input is the **quantity of fuel or energy consumed, such as litres of fuel or kilowatt-hours of electricity, during the transport operation**¹⁸³. This amount is then multiplied by the corresponding emission factor.

Once the TCEs of the transport chain have been identified and the TOCs have been defined with the necessary characteristics explained in the section 1.3, emissions can be calculated based on the fuel consumption data (classified as an option A data source under ISO 14083), through the following approach.

As previously explained, the total emissions of a transport operation are equal to the sum of vehicle operation emissions and vehicle energy provision emissions, as it can be seen in (1)

$$GHG_{TOC} = GHG_{VO,TOC} + GHG_{VEP,TOC} \quad (1)$$

Where:

$$GHG_{VO,TOC} = \sum_i (Q_{A_i} \times EF_{VO,A_i}) \quad (2)$$

$$GHG_{VEP,TOC} = \sum_i (Q_{A_i} \times EF_{VEP,A_i}) \quad (3)$$

$GHG_{VO,TOC}$ and $GHG_{VEP,TOC}$, as defined in (2) and (3), are composed of the following parameters:

- 1) i : index that corresponds to a transport activity within a vehicle operation.
- 2) A_i : specific transport activity
- 3) Q_{A_i} : Quantity of GHG activity type A_i (if the activity A_i is combustion of diesel fuel, then Q_{A_i} is the amount (in kg, or liters) of diesel consumed)
- 4) EF_{VO,A_i} : Emission factor for vehicle operations related to A_i . This represents the amount of GHG emissions (in kg CO₂e) per unit of fuel consumed during the operation of the vehicle. If the activity A_i involves the combustion of diesel fuel, this factor indicates the kg CO₂e per kg of a specific type of fuel burned by the specific vehicle.
- 5) EF_{VEP,A_i} : Emission factor for vehicle energy provision related to A_i . This represents the GHG emissions (in kg CO₂e) per unit of fuel or energy produced and supplied to the vehicle. If the activity A_i is related to diesel fuel use, this factor indicates the kg CO₂e emitted per kg of diesel fuel due to its production, refining, and distribution (upstream emissions).

Once the GHG_{TOC} have been determined, and **as we are considering only TOCs as TCEs**, then GHG emissions of the transport chain will be calculated as the summation of all the GHG emissions of each TCE, as represented in equation (4)

¹⁸³ To be discussed with EC the division of the fuel or energy consumed by CBAM good and non-CBAM good, as this method does not consider the tons transported but an overall fuel consumed.

$$GHG_{Transport\ Chain} = \sum_{j=1}^n GHG_{TCE_j} \quad (4)$$

Where:

N: total number of TCE within the transport chain

$j \in \{1, \dots, N\}$ index of each TCE.

8.3.4.2 Activity-based method for GHG emissions calculation

When fuel consumption data is not accessible, the activity-based method can be used. This approach estimates emissions based on the transport *performance* of the TOC, measured in ton-kilometres. As previously explained in section 1.3., the transport activity of a TOC is calculated by the product between the weight of the CBAM good transported and the distance they were moved. This can be done for each identified TOC, as shown in equation (5)

$$T_{TOC_j} = Weight_{TOC_j} \times Distance_{TOC_j} \quad (5)$$

The total transport activity of the transport chain ($T_{Transport\ Chain}$) is the summation of the transport activities of all TOCs within the transport chain, as shown in equation (6).

$$T_{Transport\ Chain} = \sum_{j=1}^N T_{TOC,j} \quad (6)$$

Where:

N: total number of TCE within the transport chain

$j \in \{1, \dots, N\}$ index of each TCE.

Therefore, the GHG emissions of a TOC can be calculated by multiplying the transport activity by its emission intensity. When a transport chain consists of multiple TCEs using different modes of transport, each segment requires a specific emission intensity based on its unique fuel type and vehicle characteristics.

$$GHG_{TOC,i} = EI_{TOC,i} \times T_{TOC,i} \quad (7)$$

It is important to mention that, as explained in ISO 14083, emission intensities used in activity-based method can reflect tank-to-wheel emissions (Vehicle operation) or well-to-wheel emissions (vehicle energy provision), as shown in equation (8):

$$GHG_{TOC,i} = (EI_{VO,i} + EI_{VEP,i}) \times T_{TOC,i} \quad (8)$$

Consequently, the total GHG emissions of the entire transport chain are determined by summing the emissions of all TCEs, as indicated in equation (9):

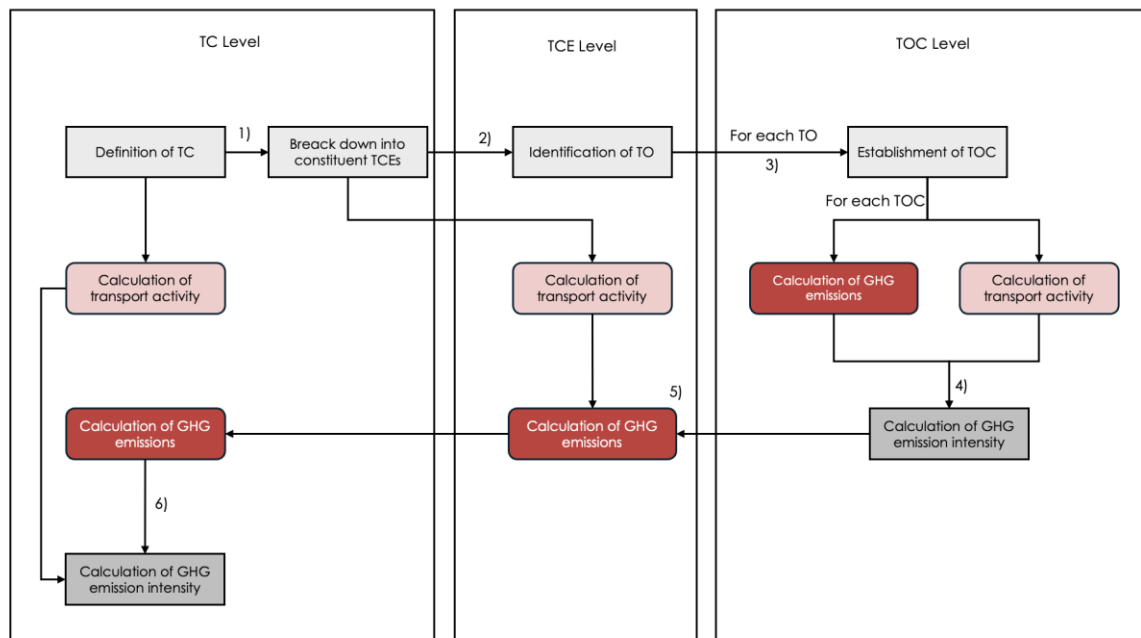
$$GHG_{Transport\ Chain} = \sum_{i=1}^n EI_{TOC,i} \times T_{TOC,i} \quad (9)$$

8.3.5 Reporting of total GHG emissions

This section describes a proposed guideline for reporting GHG total emissions of the transport chain of CBAM goods in line with the ISO 14083 framework. Reporting of GHG emissions must **not only cover GHG emissions from the transport chain of the final CBAM good to the EU border, but also from the transport chain of its precursors**. This means that any upstream transport of materials classified as precursors and used in the manufacture of the CBAM product may also be identified, quantified and included in the overall transport emissions report. However, and as agreed with the EC, it is important to note that the transport of precursors should be analysed from origin to the EU, not from origin to CBAM producer in third countries. This is due to complexities and lack of reliable data for multiple precursors in complex CBAM goods.

To support this, the figure below provides a workflow for the reporting process. This roadmap has been directly derived from ISO and show how GHG emissions are progressively calculated and aggregated from TOC to Transport Chain level.

Figure 26 GHG emissions of transport chain illustration



Source: The study team elaborated based on ISO

As it can be observed in the figure 26, the flowchart organizes the emissions accounting process in different levels of hierarchy:

- TOC Level: at this level, is where the GHG intensity if established per transport operation category
- TCE Level: in this study, and as already explained throughout this chapter, only TCE with TOC category will be considered, hence, TOC and TCE level is equal.
- TC Level: at this level, the full transport chain's emissions are aggregated and the emission intensity is calculated

The following is a step-by-step explanation of the reporting structure and the information to be provided at each step.

Step 0 – Identification of Transport Chain Elements within the Transport Chain.

The process starts with an identification of the transport chain elements (TCE) within the transport chain, that is, defining the order in which the transport occurred as already explained in previous sections. This step is essential to ensure that the emissions are assigned at the correct level.

This step applies both to the transport chain of **the final CBAM good** and to the transport chains of any **precursors** used in its production.

Step 1 – Classification of Transport Chain Elements

As previously mentioned, only transport chain elements with transport operation category (TOC) will be considered in this study (flows). However, hub operations ¹⁸⁴, remain important for reconstructing the structure of the transport chain. Identifying where a hub operation occurs helps to determine the boundaries between TCEs: if a hub is known, it implies that a new TCE classified as TOC must follow or precede it. Knowing **where and how** goods are transferred between transport modes at the nodes allows to visualise the full chain (not only the flow of goods, but also where transitions occur). The transport providers must provide this information.

For example, a provider might send product by ship (flow), which is unloaded at a port (node), from where it may be transferred to a truck (flow) and delivered to the importer's facility in the EU (node) ¹⁸⁵. It is necessary to apply this logic to precursors of CBAM goods: each leg of their transportation must be identified and classified, including possible modal transitions at hubs (nodes).

Step 2 – Information to gather at TOC level

For each TOC's transport chain, there are two main variables that are necessary to provide: **the GHG emissions and the transport activity data (ton-km)**, previously defined in section 1.3. and from which its calculation procedure has been provided. These variables will be then used to calculate the **GHG emission intensity** of the TOC.

The following information is mandatory for each TOC involved in the transport of both the CBAM good and its precursors¹⁸⁶:

- 1) Mode of transport: whether it was road, rail, sea, or air.

¹⁸⁴ Hub part presented for completeness, not to be used in the final methodological calculations. Despite their importance, hub operations are recommended for exclusion from the emissions calculation due to the unavailability of data and the potential introduction inconsistencies and reduction of overall feasibility of the study.

¹⁸⁵ However, it is important to note that the leg from the EU port of entry to the destination within EU territory should be excluded from the emissions calculation under this methodology. This is because, once the goods enter the EU, they fall under the scope of the EU ETS, where carbon costs are already accounted for. As such, emissions reporting for CBAM purposes should be limited to transport operations occurring **prior to entry into the EU customs territory**.

¹⁸⁶ Precursors data pending decision of the EC.

- 2) Distance travelled (in kilometres): which is defined as the distance over which the specific goods were transported
 - a) The origin and destination points: the provider needs to transport data on start and end points in a clear and traceable way
 - b) The method used to calculate the distance: based on ISO, there are three accepted methods for calculating the distance, and this method must be disclosed
 - a. When actual distance (AD) is used, it must be corrected by a distance adjustment factor (DAF)
 - c) In order to calculate the actual distance travelled to arrive to the EU border, we leverage the explanation made in task 5.4, where an **adjustment factor to discount the portion of the distance travelled by road¹⁸⁷ and rail¹⁸⁸ within the European Union (EU) is used to then calculate and adjusted transport-work**. This is because, when a product arrives from a non-EU country to a Member State by road and railway, it is necessary to identify the nearest EU border and, from that point onwards, the kilometres travelled within the EU must be discounted, as the distance and associated emissions are already accounted for in the EU ETS. In contrast, maritime and air transport do not require this adjustment since goods are shipped or flown directly to the port¹⁸⁹ or airport in the EU border.
 - 3) Weight of transported goods (in tons), and excluding packaging, pallets or containers used only for transport
 - 4) GHG emission intensity (gCO₂eq/ton-km) of the transport mode
 - 5) Method used to derive the emission intensity, selected from
 - a. Option A: which corresponds to primary data, direct measurement
 - b. Option B: model-based estimation
 - c. Option C: default values from official databases
- 5.1) If option A has been chosen (primary data), additional data must be gathered:
- i. Quantity and type of fuel/energy consumed
 - ii. Relevant emission factor used (both operation and energy provision)
 - iii. Allocation logic if the vehicle carries mixed loads.

¹⁸⁷ While road transport is not currently included in the EU ETS scope, the analysis is done considering ETS II in place

¹⁸⁸ While diesel rail is not in the EU ETS, the EU is largely electrified and would indirectly be covered by the EU ETS.

¹⁸⁹ Half of the emissions of the maritime sector have been also adjusted for, to account for the fact that EU ETS covers half of the incoming international sea shipping.

5.2) If option B has been chosen,

- i. The sources of the secondary data must be disclosed (such as databases, industry averages, historical data).

5.2) If option C has been chosen,

- i. The sources of the default values must be disclosed (such as databases, industry averages, historical data).

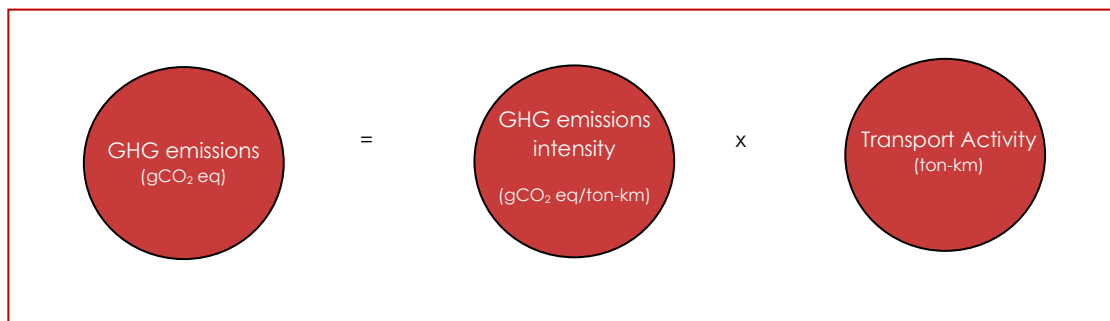
Step 3: Calculation of GHG emissions of the transport chain

With all the required information gathered from transport providers, the total GHG emissions of the entire transport chain can be calculated. This calculation follows a bottom-up approach based on ISO 14083's methodology:

Step 3.1. Calculate GHG emissions per TOC

Based on the detailed information on GHG emission intensity and transport activity, it is possible to calculate the GHG emissions for each transport operation using the following formula, which refers to the developed equation (7) previously explained in section 1.4.

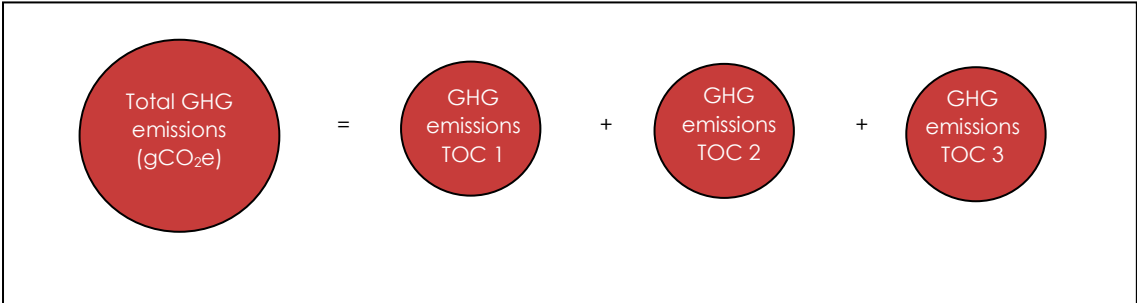
Figure 27 Formula to calculate the GHG emissions at TOC level given the GHG emission intensity and the transport activity data



Step 3.2. Calculate GHG emissions across all TOCs, to determine the total GHG emissions of the transport chain.

Aggregate the results from all TOCs to determine the total transport-related GHG emissions of your chain and report the final result.

Figure 28 Formula to calculate the GHG emissions of the transport chain, based on GHG emissions of the different TCEs

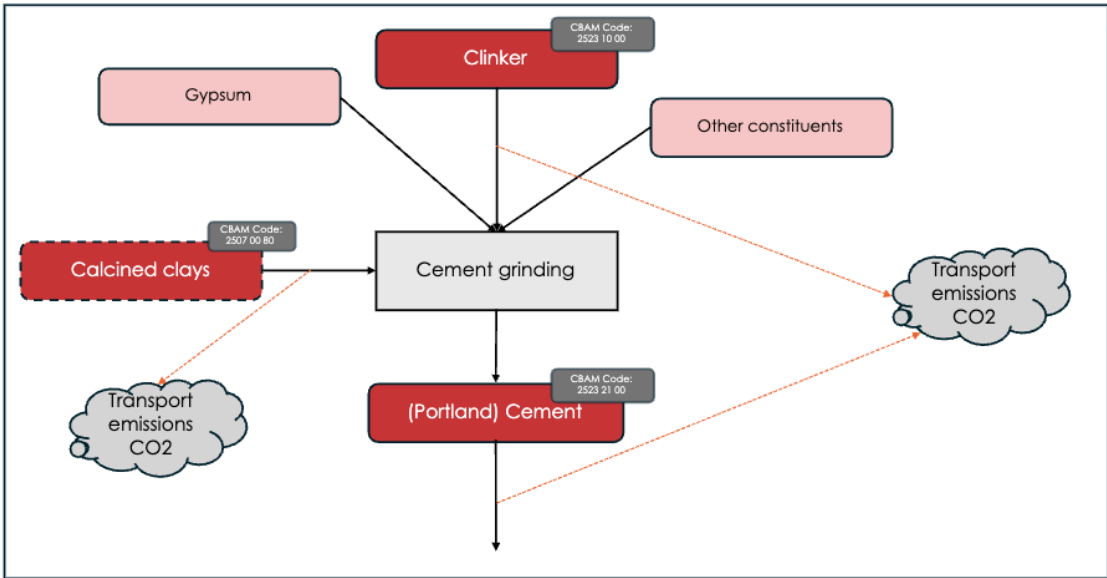


Source: The study team elaborated based on ISO

Practical example:

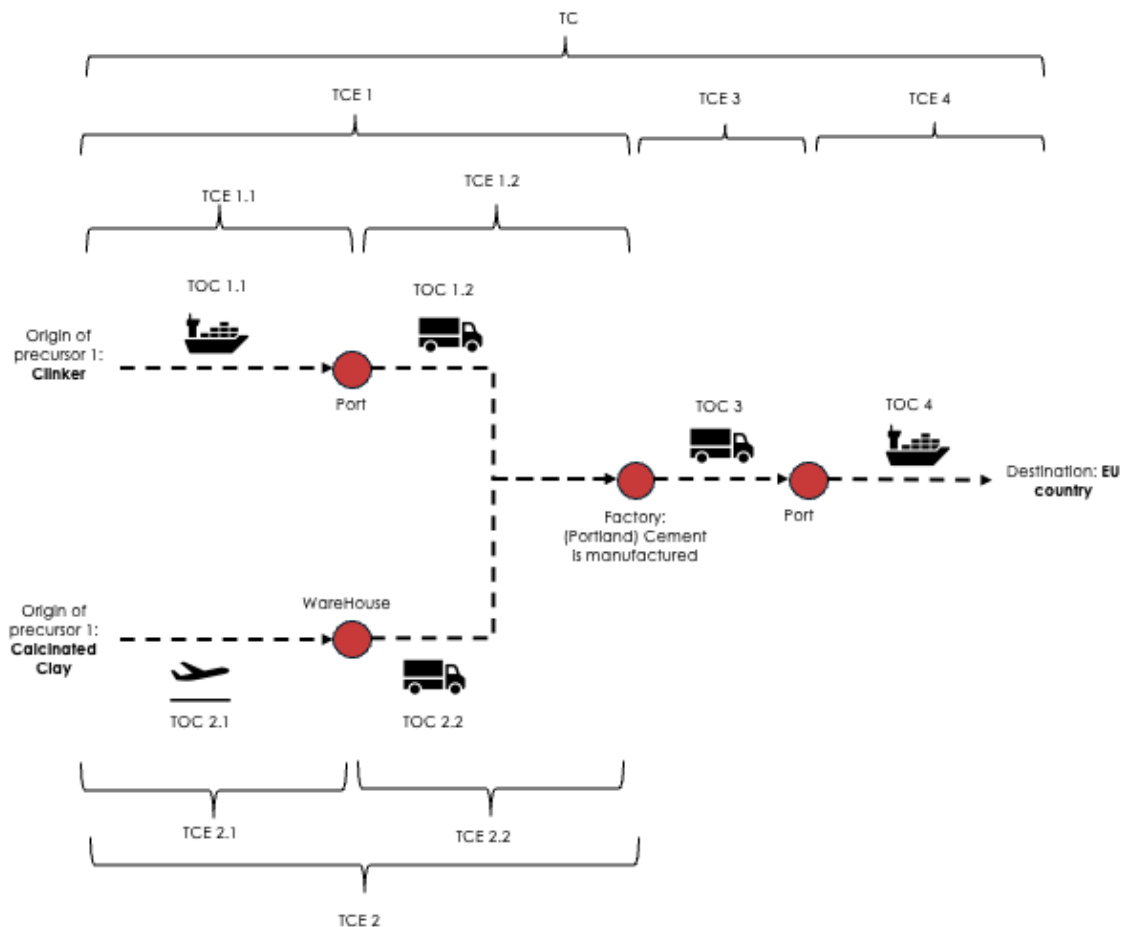
This is an example to illustrate the GHG emissions calculation of a CBAM product: **Portland Cement**. As shown in the image below, this cement product requires two key precursors before entering the cement grinding process: **Clinker and Calcined Clay**.

Figure 29 Production process of (Portland) Cement



This production process has been translated into a hypothetical transport chain, which is illustrated in the figure below (based on Section 5 of this report). Since both precursors are needed simultaneously to produce Portland cement, the picture clearly shows two parallel transport flows: the upper flow corresponds to the transport of Clinker, and the lower flow corresponds to the transport of Calcined Clay.

Figure 30 Hypothetical Transport Chain for the product (Portland) Cement



The above diagram makes it easy to identify: each TOC (Transport Operation Component) (each transport mode), each TCE (Transport Chain Element) (each grouped segment of transport activity), and the complete TC (Transport Chain). In a potential future extension of CBAM, importers of final goods may be required to report GHG emissions from their entire transport chain, including all TOCs and TCEs, from the origin of each precursor to the EU border.

Step-by-step breakdown of the transport chain

Precursor 1: Clinker. This precursor is transported in **TCE 1**, which consists of two TOCs:

- **TOC 1.1:** Clinker is shipped to port (mode: sea)
- **TOC 1.2:** Then transported by truck from the port to the factory (mode: road)

Precursor 2: Calcinated Clay. This precursor is transported in **TCE 2**, which consists of two TOCs:

- **TOC 2.1:** Calcinated Clay is shipped to warehouse (mode: air)
- **TOC 2.2:** Then transported by truck from the warehouse to the factory (mode: road)

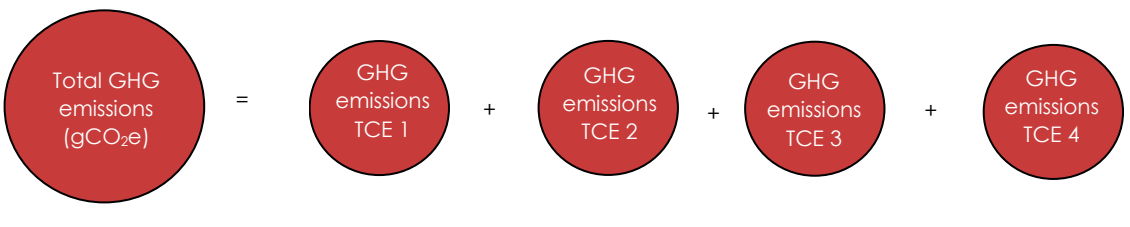
After manufacturing at the factory, the **final product** enters:

- **TOC 3:** Road transport from factory to port
- **TOC 4:** Sea transport from the port to the destination country at the EU border

For the six mentioned TOC, the information displayed in the following box need to be provided by transport/logistics operators to the CBAM declarant.

- Transport operation category (TOC)
- Mode (road, rail, air, sea)
- Distance (km)
- Weight of the specific load (ton)
- GHG emission intensity (gCO₂eq/ton-km)
- Method to derive GHG emission intensity
- Primary data

Once all relevant information for each TOC is collected from the four TCEs, the calculation of the GHG emissions of the transport chain can be done by adding up the emissions of all TOCs along the entire transport chain.



8.4 T6.2 – Data collection challenges for importers regarding transport emissions of CBAM goods

Importers must collect data across three main categories: **emission factors, transport details, and supporting documentation**. Each category presents specific challenges with respect to data availability, consistency, accuracy, and traceability.

Emissions factors

Emission factors are essential tools used to quantify the environmental impact of transport activities. They represent the **amount of greenhouse gas emissions** (typically CO₂, CH₄, and N₂O) **released per unit of activity**, often expressed in kilograms of CO₂ equivalent per tonne-kilometre (kg CO₂eq/t-km). These factors play a pivotal role in calculating the carbon footprint of transported goods and must be appropriately selected to reflect specific transport modes, fuel types, and regional contexts.

To apply the calculation framework correctly, **emission factors must be collected either directly from the transport operator or derived from verified sources**. However, the **variability of emission factors across regions, fuel types, and transport modes, coupled with the lack of a centralised database**, is a major challenge for importers. Reliable and

disaggregated data, especially for operations in third countries or less regulated sectors, is often missing. The fast-evolving nature of transport technologies and fuel blends further complicates factor selection.

When available, primary data (such as verified fuel use or operator-provided emissions coefficients) should be used in order to collect the correct emission factors. However, as it will be explained in the next section, collecting primary data derived from transport details is complex, making it considerably difficult to gather the appropriate information to accurately calculate emissions.

Due to the complexity of gathering primary data, the use of default values can be applied. These are provided in internationally recognised databases, including ISO 14083 (Annexes K and Q) and they are designed to be conservative and are considered acceptable proxies when primary values are unavailable. However, the existence of numerous default values in different databases, with a variable quality and scope, is also a major challenge for importers. The lack of consistency limits comparability and reduces confidence in the calculation of transport emissions.

Transport details

Transport details encompass all **logistical information** necessary for calculating emissions. This includes the **transport mode** (road, rail, sea, air), **fuel type** (e.g. diesel, electricity, LNG), **distances travelled**, **vehicle specifications**, and **load factors**. These details enable the emission intensity of each transport segment to be contextualized and accurately assessed.

However, importers often struggle with access to complete and consistent transport data. This is particularly problematic for **complex supply chains involving multimodal routes or subcontracted providers**. Differences in data format, proprietary systems, and inconsistent record-keeping among logistics partners make it difficult to collect reliable values for distance, vehicle type, or fuel consumption.

Declarants may have access to such details through transport contracts, delivery notes, customs declarations (e.g. the Single Administrative Document), bills of lading, and digital tracking systems. However, while larger companies may gather this data via integrated logistics platforms or enterprise resource planning systems, smaller firms and those relying on third-party logistics may lack visibility, particularly for upstream legs or multimodal journeys.

In many cases, challenges lie not only in the access but also in the quality of the data. Records may be incomplete, fragmented or even unverifiable, leading to a high degree of uncertainty on the calculations. Besides, some logistics partners are often reluctant to share sensitive information such as fuel consumption, actual distances, or load factors, which complicates the task of calculating transport emissions accurately.

In some cases, declarants may lack access to upstream data entirely. To mitigate this, the use of **standardised route profiles and default mode/fuel combinations**, such as those foreseen by ISO 14083, can offer a pragmatic solution. These values allow for the calculation of conservative estimates where full traceability is not possible. Contractual arrangements, supplier questionnaires, or digital data sharing systems may also help in gathering necessary information.

The digitalisation of the transport sector through interoperable platforms is seen as a promising way to improve the traceability of the data, reduce format fragmentation and facilitate the access to reliable data.

Supporting documentation

Supporting documentation serves as **proof of the data used in emissions calculations**. This includes fuel receipts, GPS tracking logs, third-party verification reports, maintenance records, and certificates from transport providers. These documents are critical in reinforcing the credibility of emission declarations and are necessary to comply with CBAM reporting obligations.

The feasibility of accessing this documentation depends heavily on **the level of digitalisation and integration within the supply chain**. Some declarants may be able to automate documentation retrieval through centralised systems, while others may need to rely on manual exchanges of records or request documentation from upstream suppliers, which can be challenging when those suppliers are located outside jurisdictions with mandatory GHG reporting.

Even when data is known, gathering sufficient supporting documentation remains a challenge. In many cases, fuel receipts, route logs, or third-party emission reports are not routinely shared between supply chain actors. Smaller firms may not have procedures in place to store such records or may lack the resources to request and verify them.

To address this, importers can encourage the adoption of paperless systems and standardised data formats within their supplier networks. Where needed, they should provide guidance to third-country suppliers on what data is required and why. The use of ISO 14083-certified providers and verified digital platforms can also strengthen the reliability of documentation. Where full traceability is not possible, a clear justification of assumptions and a quality classification of each data source should be included in the emission report to enable risk assessment.

Available data sources for importers

Importers responsible for reporting embedded transport emissions under CBAM may already have **partial access to several key datasets, even before the introduction of dedicated obligations**. Understanding what data is currently available, and from which sources, is essential for assessing the feasibility of integrating transport emissions into CBAM compliance procedures. Available sources range from **official customs records and contractual logistics documentation to international databases and certification systems**, each offering varying degrees of reliability, granularity, and accessibility.

At the core of currently available data is the **transitional CBAM registry**, which already collects basic but useful information for each CBAM Report. This includes the **country of origin, the EU destination, the volume and quantity of goods**, and, in some cases, supporting documentation for production emissions. Although the registry does not yet require detailed transport data, this core geographic and quantitative information can help structure the transport chain and assign emissions based on default distances, trade lanes, and mode-specific profiles.

Complementing this, **commercial logistics documentation** such as transport contracts, delivery notes, shipping manifests, customs declarations (notably the Single Administrative Document), and bills of lading (including FIATA variants) may contain relevant transport details. These documents often include **transport modes, estimated distances, shipment weights, and routes**. When declared in a standardised way and collected consistently, this information can be used to reconstruct the various Transport Chain Elements (TCEs) and define the calculation boundaries for embedded emissions.

Further support can come from sourcing goods from companies certified under **ISO 14000 environmental management systems**, particularly those also adhering to **ISO 14083** for transport emissions. These companies are typically required to collect and report on energy use, fuel consumption, and emission-related data, which can greatly facilitate the provision of primary or secondary data to importers. ISO certification also provides a layer of third-party verification that enhances the credibility of reported values and supports traceability under CBAM requirements.

Standardised classification tools such as **UN/LOCODE**¹⁹⁰ play an important role in simplifying data collection. These coding systems enable consistent identification of transport locations (e.g. ports, airports, hubs) and commodity types, allowing importers to structure the transport chain and apply emission profiles with a higher degree of automation. Their adoption also improves interoperability with customs databases and international trade reporting tools.

In terms of transport-specific data, several **sectoral tools and industry alliances** have developed resources that may be valuable to importers. For instance:

- The **European Automobile Manufacturers Association (ACEA)**¹⁹¹ publishes average emission factors and **fuel consumption data for different vehicle** types across the EU, which can support mode-specific transport emission calculations.
- The **Eurocontrol Small Emitter Tool (SET)**¹⁹² allows for the estimation of CO₂ emissions from air transport by providing **fuel burn and emissions data for flights** covered under the EU ETS. This tool is especially useful for aviation segments within multimodal supply chains.

Importers may also draw on **international or national databases** for fallback emission values, secondary data, or regional factors. These include:

- The **Greenhouse Gas Protocol**¹⁹³, which offers methodological guidance and sector-specific emission factors.
- The **European Environment Agency (EEA)**¹⁹⁴, which provides air pollutant inventories and greenhouse gas data and emission factors across transport modes.¹⁹⁵
- The **International Energy Agency (IEA)**¹⁹⁶, with global transport CO₂ emissions datasets and fuel use statistics.
- **National databases**, often hosted by ministries or statistical agencies, which provide detailed information on domestic transport infrastructure, emission

¹⁹⁰ Available at: <https://unece.org/trade/cefact/unlocode-code-list-country-and-territory>

¹⁹¹ <https://www.acea.auto/reliable-data-statistics>

¹⁹² <https://www.eurocontrol.int/tool/small-emitters-tool>

¹⁹³ <https://ghgprotocol.org/calculation-tools-and-guidance>

¹⁹⁴ <https://www.eea.europa.eu/en/analysis/indicators/greenhouse-gas-emissions-from-transport>

¹⁹⁵ <https://www.eea.europa.eu/en/analysis/publications/emep-eea-guidebook-2023>

¹⁹⁶ <https://www.iea.org/reports/transport-energy-and-co2>

factors, and fuel quality, important for aligning default values with actual regional practices.

In addition, the forthcoming **CountEmissions EU** initiative is expected to play a central role in supporting importers. As a harmonised EU-wide framework for transport emissions calculation and reporting, CountEmissions EU will provide default emission values, methodologies, and reporting templates aligned with ISO 14083. It is designed to apply across freight and passenger transport sectors and will offer a coherent standard to support both compliance and comparability across supply chains.

Under Task 6.5, a comprehensive database of default values has been developed, where default values have been estimated by grouping emission factors and intensities by trade lane (origin/destination), mode of transport and CBAM goods group. **This database is different from existing sources, as it corrects for the distance that transport work in the EU, emissions in the EU are already covered under EU ETS, and therefore, they should not be accounted for in CBAM declarations.** This will provide a reliable fallback for declarants unable to obtain complete primary data, while encouraging them to progressively shift towards higher-quality reporting as data systems improve.

Transitional period

Transport operators have expressed through interviews and survey the need for a transitional period that reflects the need to not only to adapt to a potential CBAM obligation in case the scope is extended to include their emissions, but also to allow for multiple regulations and international initiatives to be adopted and aligned.

- As it was the case for importers and installations, the transport sector will require guidance documents and EC organized workshops dedicated to the attribution of transport emissions to CBAM goods, as well as to adopt communication systems (authorisations, registrations, templates, transmissions, etc.) to share data with CBAM declarants or upload to a CBAM registry. A transitional period will allow the transport sector to adopt systems and coordinate operators across the multiple transport chain elements, including inclusion of contractual clauses on data sharing for CBAM reporting purposes in their yearly or multi-year contractual agreements.
- The emerging regulations at the EU and international level such as CountEmissions EU, IMO's maritime carbon pricing, CORSIA's aviation carbon pricing, schemes, etc. implies the need of the transport sector to comply, potentially multiplying the number of reporting instruments, verification reports, methodologies for calculating and attributing emissions along transport chain elements, and so on. A transitional period that allow for such initiatives to be clarified may allow for CBAM declarants to leverage from such emerging systems, and re-use data for CBAM Declarations.

8.5 T6.3 – Evaluation of verification approaches for transport emissions reporting

As indicated previously, CBAM Declarants using actual values for reporting CBAM goods emissions and paying the carbon price, shall rely on CBAM verified data shared by installations. This same principle shall apply to transport emissions of CBAM goods. A CBAM verification report of transport emissions need to be issued, for it to be used by CBAM Declarants. It is assumed that in case of default values, such verification report are not necessary.

Mapping the CBAM verification event

The following process illustrates when in time the verification of transport emissions and their verification may take place.

- 1. CBAM goods are transported and imported into the EU from a third country,**
 - a. Transport activity (tonnes and km) and or transport emissions (sensors) data is generated and stored at the level of transport chain element
 - b. If economic operators are in CPI jurisdiction, activity and emission data are verified by CPIs verifiers and CPI obligations met.
- 2. CBAM Declarant request CBAM verification report to third country operators and transport providers.**
 - a. Economic operators (CBAM installation and transport operators) prepare emissions data. Economic operators in CPI jurisdictions may use CPI verification reports and complement with additional data required by CBAM verifiers.
 - i. Installations attribute emissions per CN code
 - ii. Transport operators attribute emissions per CN code
 - b. CBAM Verifiers issue a verification reports to installations and to transport operators
 - c. CBAM installations and transport operators communicate emissions data and CBAM verification report with attributed emissions per CN code to CBAM Declarant.
 - i. Transport emissions are attributed to CBAM goods per CN code shall include transport activity data (origin, destination, transport mode, distance, tonnes, emissions (split by GHG gas), per transport chain, per CN code).
- 3. CBAM Declarant calculates and reports total CBAM goods and transport verified emissions.**

CBAM verification of transport emissions

CBAM declarants require verification reports to contain emissions and activity data per CN code. The type of transport data required is described in section 8.2.5 Reporting of total GHG emissions. It is however important to highlight that emissions at the transport chain level, need to be attributed to CN code level.

Transport operators can use fuel purchase receipts, travel logs, and data extracts from telematic systems. This data needs to be available to CBAM verifiers as well as the method and tools used for the calculation of emissions for each Transport Chain, and the way they attribute emissions to specific CBAM CN codes.

If a CBAM good is imported using multiple transport modes, the transport operator should provide the documentary evidence of the attribution of emissions of each transport chain per CN code, in a way that can be added up across transport chains:

$$GHG_{Transport\ Chain} = \sum_{i=1}^n EI_{TOC,i} \times T_{TOC,i} \quad (9)$$

$$GHG_{Transport\ Chain\ CNi=1} = GHG_{Transport\ Chain} - \sum_{i=2}^n GHG_{Transport\ Chain\ CNi} \quad (10)$$

Notation:

$GHG_{Transport\ Chain\ CNi=1}$: *Transport chain emissions attributed to CBAM good CN code 1*

$GHG_{Transport\ Chain}$: *Total Transport Chain emissions*

$\sum_{i=2}^n GHG_{Transport\ Chain\ CNi}$: *Transport chain emissions attributed of all goods, excl. CBAM good CN code 1*

The verification approaches that CBAM verifiers for transport emissions consist of desk reviews, audits, automated digital tracker systems for data from supply chain operations, and scheduled audits by CBAM verifiers. The criteria for the evaluation of verification approaches are Efficacy, Documentary evidence, and their potential drawbacks.

8.5.1 Desk Reviews

- **Efficacy:** Desk review allows for setting up an initial dossier of the CBAM goods transport emissions, prepared and submitted by the CBAM Declarant. This dossier should contain when available the: the ISO 14083 certification report, CPI verified transport emissions report, importer and import locations data, transport mode per transport chain per CN code, and exporting installation data.

Stakeholders indicated during interviews that much of the information necessary for the desk review of the transport and logistics chain depends on very much on the importer's visibility of the supply chain, as well as on the incoterms used for the international trade. For instance, an importer purchasing from a third country broker may not have knowledge of the country of origin or countries of origins if the broker consolidated CBAM goods before export to the EU. Also, when the international trade is organised by the exporter (Delivered Duty Paid-DDP), the importer and CBAM Declarant, may have limited access to the information to be shared with the CBAM transport emission verifier. When the trade is organised by the importer (Ex Works-EXW), it is more likely for the importer and the CBAM declarant to have access to the necessary information to share with CBAM transport emissions verifiers; and even then, if the logistics and transport operations are outsourced to a logistics service provider, the importer and CBAM Declarant need to rely on the logistics service provider to provide such information to the importer and CBAM Declarant. Incoterms between EXW and DDP, imply partial access to necessary data by the importer and the CBAM Declarant to share with the CBAM transport emissions verifier.

- **Documentary Evidence:**
 - **Emission Intensity:** Stakeholders indicated that there are only a limited number of sources to access emissions intensity data, GLEC, ISO 14083, IPCC, IMO, and CORSIA data. In third countries, there are limited number of sources, and the most reliable information is transport characteristics data, which specifies the estimated emissions per unit of fuel used. While these sources are normally updated for maritime and aviation, they are less for road and rail. In case of rail, the emission intensity depends on the degree of electrification of the rail systems, and when whether the electricity used for electrification is based on

renewable sources or not. This is data not available to the freight train operator, as it is information that rests with the rail infrastructure operator.

- **Transport Details:** Stakeholders indicated that no challenge exists regarding the possibility of accessing and sharing invoices from transport companies, tracking documents, shipping manifests, fuel receipts, etc, provided confidentiality agreements are signed among the parties.
- **Limitations:** Desk reviews require collection of data from multiple sources to verify that in fact the lots of CN codes included in the CBAM Declarations were imported with the declared transport mode, from said origin, and to verify that share of total transport chain emissions were correctly attributed to the specific CN codes. Rather than verifying emissions from a limited number of installations (exporters), CBAM verifiers need to account for multiple transport operators along the full international transport chain. When emissions data is missing from a single transport chain element, it is difficult to validate the total transport emissions of the CN code. Large logistics service providers with full control of the international transport chains, are more able to provide the relevant data for verifications, whilst SMEs transport operators with limited resources and data sharing capabilities, may find the compilation of data and sharing with CBAM Declarants and complex and costly task.

8.5.2 CBAM transport emissions verification/audit:

- **Efficacy:** CBAM transport emissions verifications are vital for deterring misreporting and ensuring data integrity. They enable in-depth examination, including requests for additional data and cross-checks with existing EU systems, e.g. upcoming CountEmissions EU. It is crucial to verify that transport emissions were correctly attributed to the CN code imported into the EU. For instance, a shipment containing multiple CN codes, some of which non CBAM goods, need to be correctly attributed per CBAM CN codes, and CBAM transport emissions verifiers need to verify that data used by transport operators had been correctly extracted and reported. CBAM verifier can also check and request for clarification regarding CPIs carbon price paid for verified transport emissions on the CBAM goods imported into the EU. This should allow the CBAM Declarant to subtract from the number of CBAM certificates to surrender.
- **Documentary Evidence:** Similar to desk reviews, the verification process checks that all data provided for desk review is backed up with the sources of data by the economic operators.
 - Internal data management systems generating data points for the calculation of transport activity and emissions per CN code I with destination EU. This should include:
 - i. Origin of the cargo loading of CBAM good with CN code i
 - ii. Volume of loading CBAM good with CN code i
 - iii. Transport mode and transport characteristics (e.g. vehicle size category, fuel type, containerised vs general/bulk, etc.)
 - iv. Transportation route from Origin to transshipment point
 - v. Location of transshipment point (incl. distance from origin to transshipment point)

- vi. If available: emissions data and fuel consumption data from vehicle sensors.
- vii. Unloading volumes from previous transport mode of CBAM good CN code i in transshipment point
- viii. Volume of loading of CBAM good with CN code i in transshipment point
- ix. Transport mode and transport characteristics of re-loading of CBAM good code i in transshipment point.
- x. Transportation route from Origin to EU Port of Entry of CBAM good CN code i (incl. distance from origin to transshipment point).
- xi. If available: emissions data and fuel consumption data from vehicle sensors.
- o Link to source of specific emission factor databases used by transport operators and or importer for estimating total transport operation emissions, should include:
 - i. Emissions per vehicle, per vehicle type, per region, per trade lane if available
 - ii. Emissions per t-km, per vehicle type, per region, per trade lane if available
 - iii. Emissions per fuel consumption, per fuel type, per vehicle type, per region
- o Data exchange protocols between the importer, transport providers, exporter.

- **Limitations:**

- o **Limited scope:** CBAM transport emissions verifications can happen upon a request by the CBAM Declarant. CBAM transport emissions will require data that is limited to the transport chain system boundaries. The CBAM goods Verifiers for the installation emissions cannot be the same as the CBAM transport emissions verifiers, due to differences in scope of the verification.
- o **Resource-intensive:** Verification process requires data to be provided by multiple economic operators, which means that it is not occurring in a single location (i.e. origin, transshipment point, destination, headquarters of the logistics service provider managing the full logistics operations from origin to destination). When the incoterm is chosen to share the responsibility of the international trade (e.g. Free On Board), at least two logistics service providers are involved, it may require at least two verification events.
- o **Detection challenges:** The amount of data points required for CBAM transport emissions verification is rather complex and may be subject to misreporting or manipulation if not structure reporting templates are not used.
- o **Time lag:** The CBAM verification period should happen after the calendar year has passed. Economic operators may have limited access to data of transport events that occurred during the previous year. The verification is one in a year event, and not a continuous monitoring of the CBAM goods transport emissions.

8.5.3 Automated Digital Tracking Systems:

- **Efficacy:** While Automated Digital Tracking and Tracing Systems offer significant potential for real-time monitoring, enhanced data accuracy, and streamlined verification processes, its implementation requires careful consideration of cost, data privacy, standardisation, and industry collaboration, given that data sharing needs to be part of the logistics service providers contractual agreements, trickling down to all their individual transport operators.
- **Documentary Evidence:**
 - Extracts from telematics systems has greatly been improved over the years in the transport industry, by all transport modes. Thus, it is safe to assume that data can be available.
 - Sensor readings (e.g., fuel consumption data, driving behaviour, distance, and emissions data) are increasingly embedded in vehicle telematics.
 - Digitally signed, electronic transmitted, and verifiable.
- **Limitations:**
 - **Cost and Implementation:** Setting up and maintaining such systems can be expensive, potentially posing challenges for smaller operators. Not all transport operators, SMEs in third countries, have yet adopted new technologies.
 - **Cybersecurity and AI Risks:** Robust safeguards are necessary to protect against cyberattacks and ensure data integrity. The International Road Transport Union (IRU) is increasingly advocating the use of AI, for instance expressed in public statements: "AI solutions are delivering concrete value. While successful implementation requires investment and change management, solutions are increasingly designed to integrate with existing systems, making wider adoption more feasible"¹⁹⁷. While the EC is still in the process of setting up the AI risk assessment frameworks, early adoption of AI for monitoring, and reporting transport activity (incl. emissions) need to be taken with care.

8.5.4 General assessment of verification approaches

An effective CBAM goods transport emissions verification may require the following considerations:

- **Combining Desk Reviews and Verifications.** Desk review should allow the CBAM transport emissions verifier to review the methods used to attribute emissions to CN codes i. Verifications should be used to review that data generation originated from the transport operations, and transport activity was correctly registered by transport operators. While the use of automated systems can be helpful, it is unlikely for them to be adopted by smaller transport operators in third countries, and therefore, there may be a risk of substantial data gaps in the full transport chain. Thus, automated systems can be allowed to be optional, and

¹⁹⁷ IRU News : <https://www.iru.org/news-resources/newsroom/insights-ground-ai-and-leading-trends-trucking>

exclusively to substitute documentary evidence during the CBAM transport emissions verification processes.

- **Leveraging existing infrastructure:** CBAM registry, can leverage from IMO, CORSIA, and ISO14083 certified emission factors databases. There should be an integration with any registry that may emerge from the CountEmissions EU regulation. The Clever project is generating emissions factors for fuels used by the transport sector, at an international level, and it could be a benchmark tool for CBAM transport emissions verifiers when assessing the emission factors databased used by transport operators.
- **Stakeholder collaboration:** The main stakeholders for an effective CBAM transport emissions verification are the ones coordinating the import-export logistics operations according to the incoterms agreed between the importer and exporter. If DDP it is the exporter who should be involved in the CBAM transport emissions verification process, if ExWork, it is the Importer who should be involved in the verification, and anything incoterm in between implies that both the importer and exporter should participate in the transport emissions verification process.

Upon consultation with stakeholders, CBAM Declarants indicated the need to identify correctly where the liability of monitoring and reporting is among the international supply chain. The Incoterms is the framework that allows for the identification and allocation of responsibilities. The transport industry, by all accounts is willing and able collaborate with the rest of the economic operators, to the extent that data is available. Yet, there is a clear position that the liabilities should remain with the CBAM importers, which are the ones subject of the CBAM Regulation.

8.6 T6.4 – Institutional and infrastructural implications on EU authorities.

The scope of this sub-task 6.4 is to assess the institutional and infrastructural implications for EU authorities overseeing the system, ensuring feasibility without imposing undue burdens. The analysis encompasses three key areas: **technological needs**, exploring existing tools and systems while considering potential developments; **human resource requirements**, assessing personnel needs and training; and **infrastructure demands**, evaluating IT upgrades to handle anticipated data volume.

The assessment of the institutional and infrastructural implications on EU authorities, is based on proposed regulations and existing regulations such as 1) CountEmissions EU¹⁹⁸,

¹⁹⁸ Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on the accounting of greenhouse gas emissions of transport services. COM(2023) 441 final. Available: <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=celex:52023PC0441>

eFTI Regulation¹⁹⁹, the EU MRV Maritime GHG emissions regulation²⁰⁰, and based on the knowledge of the CBAM – Declarant Portal.

8.6.1 *Technology and data collection needs*

CBAM Declarants currently use CBAM registry and submits reports to EU authorities. A key assumption would be the need to adapt the registration and CBAM Declarant Portal to include Transport emissions reports of CBAM goods. It is important that any CBAM report of transport emissions can be built on existing systems. For instance, 1) CountEmissions EU will requires statistical offices dealing with transport statistics to collect statistical data on transport operations and emissions, 2) the electronic Freight Transport Information Regulation requires National Authorities to adopt IT systems to receive freight transport data electronically, 3) the EU MRV for Maritime GHG emissions reporting, require the EC and each Member State of the flags of the ship to receive electronic reports of emissions per ship on annual basis, and 4) CBAM Registry allows for reports to be submitted electronically. This suggests that both economic operators and EU Authorities should already have the technological means to receive, store, and transmit information related to CBAM goods' transport emissions.

8.6.2 *Communication and Collaboration*

Existing registries, portals, and communication systems, have been designed for specific regulatory purposes; which implies that interoperability and authorisation of data exchange must exists between systems to allow for the implementation of a key stakeholder request, reuse of data for compliance purposes. Stakeholders highlighted concerns on the increasing number of regulations requiring monitoring and reporting. CBAM goods transport emissions is considered to add significant complexity to their already reporting obligations. For instance, transport operators are expected to share data on emissions at transport level, EU MRV for maritime requires reports at annual level per ship. CBAM on the other hand, requires emissions to be attributed to the CN code level. This is a level of specificity that requires substantial upgrading of the communication system to be used effectively by EU Authorities.

8.6.3 *Human resource requirements*

The expanding scope of CBAM to include transport emissions requires a workforce equipped with the necessary knowledge and skills. There are three main groups of personnel that need to have specific knowledge. Firstly, CBAM verifiers will need to have substantial knowledge on the ISO 14083 and the reporting standards, as well as on the third countries emission intensity and emission factor databases. Secondly, CBAM Registry and CBAM Declarant portal's administrators will require to integrate into the system reporting templates and guidelines that aligned with emerging regulations, such as the CountEmissions EU. Thirdly, the EC will require to update default values for transport emissions more frequently than with CBAM goods embedded emissions, due to the more dynamic evolution of the transport sector.

¹⁹⁹ Regulation (EU) 2020/1056 of the European Parliament and of the Council of 15 July 2020 on electronic freight transport information. Available: <http://data.europa.eu/eli/reg/2020/1056/oj>

²⁰⁰ REGULATION (EU) 2015/757 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 29 April 2015 on the monitoring, reporting and verification of greenhouse gas emissions from maritime transport, and amending Directive 2009/16/EC. Available: <http://data.europa.eu/eli/reg/2015/757/2024-01-01>

8.6.4 Policy and legal expertise

The rapidly evolving regulatory landscape of the transport sector, with regards to the increased focus on their decarbonisation, require close attention to the coherence of policy measures, over the next few years, as already indicated based on CORSIA, IMO, increasing number of countries adopting Carbon pricing instruments at global level. It will be particularly important to provide guidance in the future as to how transport sector will manage the surrendering of CBAM certificates and their claims to deduct carbon price already paid under their jurisdiction.

Transport systems and modes are different for each type of good and by origin of the CBAM good. The CBAM Diplomacy Task Force may need to play an increasingly important role on liaising with third countries on CBAM goods transport emissions topics, as per the number of stakeholders will significantly increase with the inclusion of transport emissions in CBAM scope. One key area is the discussion with third countries the identification of their national institute of transport accepted emission intensities per transport mode, and emission factors of fuel by type of fuel. Expectedly, third countries with CPIs on transport sectors should have identified emissions factors accepted at their in their jurisdictions.

Collaboration and Knowledge Sharing within the Inter-Secretariat Group with DG Environment, DG Move, and DG TAXUD is vital. Emissions data will need to be shared with DG environment, and transport activity data due to CountEmissions EU will be managed by DG Move. This creates the need for a close collaboration with DG TAXUD for the assessment of reports, identification of outliers on transport activity, emissions report, and so on. This should allow the EC to request clarification to declarants on their CBAM goods transport emissions declarations per CN code.

Finally, legal experts will be required for the drafting of an Implementing Act on transport emissions and their CBAM obligations should the scope be extended. Such an Implementing Act will need to include the method for attributing emissions to CN codes, methodological approaches for the definition of the carbon price effectively paid, and eligibility criteria for acceptance of offsets, rebates, and compensations.

8.7 T6.5 – Establishment of default values for transport emissions of CBAM goods.

To set default values for the transport of CBAM goods, a database was created of all the modes of transports used to trade CBAM goods between countries. To create the database, we used publicly available data from UNCTADstat, more specifically we used the trade and transport dataset²⁰¹. This dataset provides information on the quantity of goods traded (in kilograms) and the estimated transport work involved (measured in tonne-kilometres), broken down by country of origin, destination, and mode of transport. We retrieved this data for the years 2019 to 2021²⁰² and focused on a selected list of product categories, identified using their CN codes. We then processed the data to

²⁰¹ <https://unctadstat.unctad.org/datacentre/dataviewer/US.TransportCosts>

²⁰² The 2019 to 2021 period was selected due to availability of transport work data per transport mode per CN code (at four digit level). More recent years (2023 and 2024) did not offer complete transport work disaggregation for all CN codes.

summarise the most relevant transport patterns for each product, including total weights, transport work, and modal shares between trading partners.

The next step was to quantify the emissions associated with different modes of transport at the regional level for the different CBAM goods. This was done with the work previously completed under Task 5, where well-to-tank (WTT), tank-to-wheel (TTW), and well-to-wheel (WTW) emission factors were provided per mode of transport and region and expressed in grams and tons of CO₂e per tonne-kilometre.

Using these emission factors, total emissions were calculated for each record by multiplying the TTW emissions factor and the WTW emissions factor by the corresponding transport activity, defined by the specific combination of CBAM good, mode of transport and region of origin and destination. Hence, the resulting values represent the total emissions (in tons of CO₂e) for each transport flow provided in function of TTW and WTW. Providing results in both dimensions allow for consistency with the EU ETS which relies on TTW values, while at the same time allowing the use of WTW as an alternative. This means that we can offer default values under both approaches, with WTW acting as a “mark-up” that clearly illustrates the additional costs associated with using default values instead of primary data.

In the table below, an example of the above-mentioned calculation can be seen for an example of a specific good (CN code 2601, which corresponds to Iron ores and concentrates, including roasted iron pyrites) for different origin-destination pairs and mode of transport. The table includes the information of the transport activity (weight, transport-work) as well as the corresponding TTW and WTW value expressed in gCO₂e/ton-km and the total emissions in function of WTW.

Table 48 Emission calculation for CN code 2601 by Origin-Destination pair and transport mode extracted from the database

CN code	Origin (country)	Destination (country)	Mode of transport	Average weight (ton) 2019-2021	Adjusted Transport work (ton-km)	TTW (gCO ₂ e/ton-km)	WTW (gCO ₂ e/ton-km)	Total transport emission (ton CO ₂ e) based on WTW
2601	USA	Czechia	Sea	35.32	450945.33	1.33	1.59	0.71
2601	Uruguay	Estonia	Air	0.178	2972.66	1116.71	1349.83	3.32

Source: The study team

8.7.1 Options for selecting default values depending on available information

As highlighted throughout this section, one of the main challenges faced by importers of CBAM goods is the lack of detailed information regarding how their shipments were transported, especially when multimodal transport is involved. This lack of traceability makes it challenging to determine transport emissions by mode and region with accuracy, and therefore to mirror the EU ETS system. It is important to highlight that such designing a mechanisms that mirrors the EU ETS for road transport, for instance, would imply regulating fuel distributors, not directly the road transport service providers, as the point regulation of the EU ETS for road transport is upstream, not point source. This would render enforcement an administratively complex process.

To address this issue, in the following sections, four options are proposed for calculating default values depending on available data:

- **Option 1:** Default values when complete transport activity data is available. (gCO₂e/ton-km);
- **Option 2:** Default values based on transport mode, weight and distance (gCO₂e/ton-km);
- **Option 3:** Default values based on weight only (gCO₂e/ton).
- **Option 4:** Default values by CN code and exporting country (gCO₂e/ton).

It is important to mention that all default values provided below will be given in function of TTW and WTW as previously explained. The study team considers that while “mark-up” could consider the worst performing transport characteristic, an approach that would be most consistent and available throughout the world would be to use WTW emission factors, as it penalises by adding Well to Tank emissions (WTT) to the Tank to Wheel emissions (TTW).

8.7.1.1 Option 1: Default values when complete transport activity data is available. (gCO₂e/ton-km)

Although obtaining accurate and reliable transport activity data remains a significant barrier for importers, this option was developed to be available when they do have the complete transport activity data. In such cases, the importer would have access to accurate and verified information of the specific transport mode used to transport the CBAM good to the UE, as well as the different steps that took place.

Hence, the transport operators or logistic service providers would provide the default value for the specific transport – or transports - mode selected, or the importer could extract the default value from recognised sources (such as the ones already provided throughout this report). In such a case, the total emissions would be calculated using the developed formula provided in section 6.1 of this report, expressed in both TTW and WTW.

8.7.1.2 Option 2: Default values based on transport mode, weight, and distance (gCO₂e/ton-km)

When complete transport activity is not available, but the transport mode used, the weight of the CBAM good produced, the intermediate steps and the destination of such good is known, the default values produced are those developed in this option.

This approach establishes default values for each product category (CN code), calculated as the average emissions per ton-kilometre, disaggregated by origin of the good, transport mode and region. However, it should be noted that in practice these default values correspond directly to the well-to-wheel factors already mapped in Task 5, and the reason is explained below.

The calculation of a default value starts by multiplying the WTW emission factor (expressed gCO₂e/ton-km) by the ton-kilometres of the good to finally obtain the emissions in gCO₂e. This gives the total transport emissions in gCO₂e. To express the current result on an intensity basis (gCO₂e/ton-km), the total transport emissions are then divided by the total ton-kilometres of the good, that is, the same ton-kilometres. As a result, the operation returns the same original WTW emission factor. Therefore, the default values to be used under this option are the already mapped in Task 5, that is, default values that are region-dependent, as previously explained, and which are given in TTW and WTW.

The default values that have been developed under this option are CBAM good-dependent and transport mode-dependent, just to give to the importer a clear step by step guidance.

For instance, an importer introducing in the EU a CBAM good with CN code 2601 (corresponding to iron ores and concentrates, including roasted iron pyrites), would then consult the reference table provided in the database to identify the relevant default values.

The importer under this option would know, for instance, that the good was transported first by road from Egypt to Morocco and then by sea from Morocco to the EU. Then the default values to be used are the ones provided in the table below, both for TTW and WTW.

Table 49 Default emission factors for CN code 2601 by exporting country and mode of transport (TTW and WTW) based on transport activity (ton-km)

CN code	Origin (Exporting) Country	Mode of Transport	Default Emission Factor based on TTW (tCO ₂ eq/ton-km)	Default Emission Factor based on WTW (tCO ₂ eq/ton-km)
2601	Egypt	Road	0,00005376	0,00006819
2601	Morocco	Sea	0,00000133	0,00000159

Source: The Study Team

To calculate the total transport emissions, the importer must multiply the applicable default value by the transported weight of the CBAM good imported and by the relevant transport distance transported. In this specific example, that calculation would require determining the distance by road from Egypt to Morocco, and the maritime distance from Morocco to the EU border entry point. The sum of these segments (which in practice are transport chain elements (TCE)) would provide the total emissions associated to the imported good with CN code 2601.

In the database, a dedicated sheet called 'default values per ton-km' was created with pivot tables, where default values can be easily explored.

8.7.1.3 Option 3: Default values based on weight only (gCO₂e/ton)

This approach produces default values for each product category (CN code) as an average of the emissions per tonne of goods transported, disaggregated by transport mode and region. It relies on historical data and transport patterns extracted from the database named '5_1_CBAM transport work and emissions database' previously explained and shared with TAXUD.

Since the database includes the mass of goods transported (in tonnes), it is possible to calculate default emission values by dividing the total emissions by the total mass imported. Each default value represents the average emission intensity across all records associated with a given origin-destination pair, mode of transport, and CN code, thus reflecting the transport emission performance for that route and product based on historical data. This option applies when the specific transport activity (meaning, the detailed route or intermediate stops) is not available, and only general information is known, such as, for instance, the good was transported by air from Australia to the EU. This results in a product-specific and transport-specific default value expressed in **tCO₂eq/ton of CBAM good** by transport mode, region, and CN code and, to ensure consistency, the default values are given in function of TTW and WTW. As previously explained, in the database, a dedicated sheet called 'default values per ton of goods' has been created with pivot tables, allowing to explore these values.

See an example in the table below when the transport **mode selected is Air and the good with CN code 2601 come from Australia.**

Table 50 Default emission values for CN code 2601 by mode of transport and exporting country (air, Australia) in function of tons of good.

CN code	Origin (Exporting) Country	Mode of Transport	Default Emission Value (tCO ₂ eq/ton of good) based on TTW.	Default Emission Value (tCO ₂ eq/ton of good) based on WTW.
2601	Australia	Air	18.14	21.93

Source: The Study Team

8.7.1.4 Option 4: Default values by CN code and exporting country (tCO₂e/ton)

This simpler approach calculates average emissions per tonne of goods for each CN code across all transport modes, without disaggregating by mode. It also still relies on the same historical dataset but offers a single average emission factor per product and OD pair. **This option may be particularly useful for importers with no access to transport information.** However, it may overestimate and underestimate emissions for goods that did not use high or low emissions transport mode. It is important to highlight that a mark-up is already set, by considering default values based on WTW emissions, rather than TTW. This would reduce the risk of importers using energy-intensive transport modes could rely on default values to lower their declaration of emissions, and better opt for declaring actual emissions, which should be based on TTW actual values.

To give an example when there exist and underestimate, let us say the CBAM good with CN code 7211 is imported from Australia. Under this option 4, the importer will say they do not have available information on how this good did arrived at the EU and hence use the value that appears in the below under option 4. However, if it is the case that the good was transported by air, which is the row that is presented in the table below under option 3, **there would be an underestimation of 7.38 tons of CO₂eq per ton of good with CN code 7211 imported.**

Table 51 Average default emission values for CN code 7211 by exporting country (all transport modes combined)

Option	CN code	Origin (Exporting) Country	Mode of transport	Default Emission Value (tCO ₂ eq/ton of good) based on TTW	Default Emission Value (tCO ₂ eq/ton of good) based on WTW
3	7211	Australia	Air	18.14	21.93
4	7211	Australia	Unknown	8.06	9.7

Source: the study team

8.7.2 Clarification on exporting country vs producing country

It is important to clarify that the country of export is not always the same as the country where the CBAM product was produced. In global supply chains, intermediate stages are common. For example, a product may be produced in one country, stored or processed in another, and then exported to the EU from a third country. This distinction is important because default emission values are assigned using transport data linked to the exporting country, not necessarily the country of origin (production). If the country of production is different from the exporter, the emissions related to transport might be underestimated.

Example 1: CN code 2601 (Iron ore), produced and exported from Brazil

Iron ore is mined and loaded at the port of Rio de Janeiro and exported directly to Belgium by sea. In this case, Brazil is both the producer and the exporter, and the default value used corresponds to the Brazil – EU sea route, which corresponds to 0.095 tCO₂eq/ton. (see database)

Example 2: CN code 2601 (Iron ore), produced in Peru and exported from Chile.

Iron ore is mined in Peru and transported by truck to Chile, from where it is shipped to the EU. In this case, the exporter is Chile, so the default value used would correspond to Chile–EU by sea, which looking at the database is equal to 0.128 tCO₂eq/ton. However, this default would not account for the overland transport from Peru to Chile. If the importer has no data on this leg, emissions may be **underestimated**. One possible solution is to apply the highest default emission factor. In this case, as the default value of the transport Peru-EU by sea is equal to 0.11 tCO₂eq/ton, then it will be useful to use one of the following options:

On the one hand, under option 1 previously explained, the importer has access to transport information and can then calculate emissions for each leg of the journey. In this case, they would sum the emissions from the truck leg (Peru to Chile) and the sea leg (Chile to the EU) to reflect the full transport chain.

On the other hand, under option 2, the importer lacks specific transport data. In this case, they would refer to the database and compare the default emission values for Peru–EU and Chile–EU, then select the highest of the two, this way to ensure that the total transport emissions are not underestimated.

8.8 T6.6 - Assessment of Carbon Price Instruments coverage for different transport modes per region

This section presents an assessment of the global distribution and application of carbon pricing instruments, that is, Emissions Trading Systems (ETS) and carbon taxes, in the transport sector, based on data from both national and supranational initiatives.

According to the World Bank's Carbon Pricing Dashboard²⁰³, there are currently 133 carbon pricing instruments active worldwide (including ETS and carbon taxes). Of the total, **65 CPIs cover the transport sector, either directly (by mode type, including aviation) or indirectly (through the fuels used)**. There are two main type of CPIs, Carbon taxes and Emission Trading Systems (ETS)²⁰⁴.

²⁰³ <https://carbonpricingdashboard.worldbank.org/about#download-data>

²⁰⁴ <https://ourworldindata.org/emissions-by-sector>

Table 52 Countries and sub-national regions where implemented carbon pricing instruments (CPI) apply directly or indirectly (through fuels and electricity) to the transport sector (April 2025)

Region / Country / CPI	CPI (NUMBER AND TYPE)
North America	26
Canada	8
Canada federal OBPS (National)	ETS
Alberta TIER	ETS
BC OBPS	ETS
New Brunswick OBPS	ETS
Newfoundland and Labrador PSS	ETS
Nova Scotia OBPS	ETS
Ontario EPS	ETS
Quebec CaT	ETS
Mexico	12
Mexico carbon tax (National)	Carbon Tax
Colima carbon tax	Carbon Tax
Durango carbon tax	Carbon Tax
Guanajuato carbon tax	Carbon Tax
Mexico City carbon tax	Carbon Tax
Mexico ETS	ETS
Morelos carbon tax	Carbon Tax
San Luis Potosí carbon tax	Carbon Tax
State of Mexico carbon tax	Carbon Tax
Tamaulipas carbon tax	Carbon Tax
Yucatan carbon tax	Carbon Tax
Zacatecas carbon tax	Carbon Tax
United States	6
California CaT	ETS
Colorado GHG crediting trading system	ETS
Massachusetts ETS	ETS
Oregon ETS	ETS
RGGI	ETS
Washington CCA	ETS
EU	16
Austria (Austria ETS)	ETS
Denmark (Denmark carbon tax)	Carbon Tax
Estonia (Estonia carbon tax)	Carbon Tax
European Union – 27 Member States (EU ETS II)	ETS
Finland (Finland carbon tax)	Carbon Tax
France (France carbon tax)	Carbon Tax
Germany (Germany ETS)	ETS
Hungary (Hungary carbon tax)	Carbon Tax
Ireland (Ireland carbon tax)	Carbon Tax
Latvia (Latvia carbon Tax)	Carbon Tax

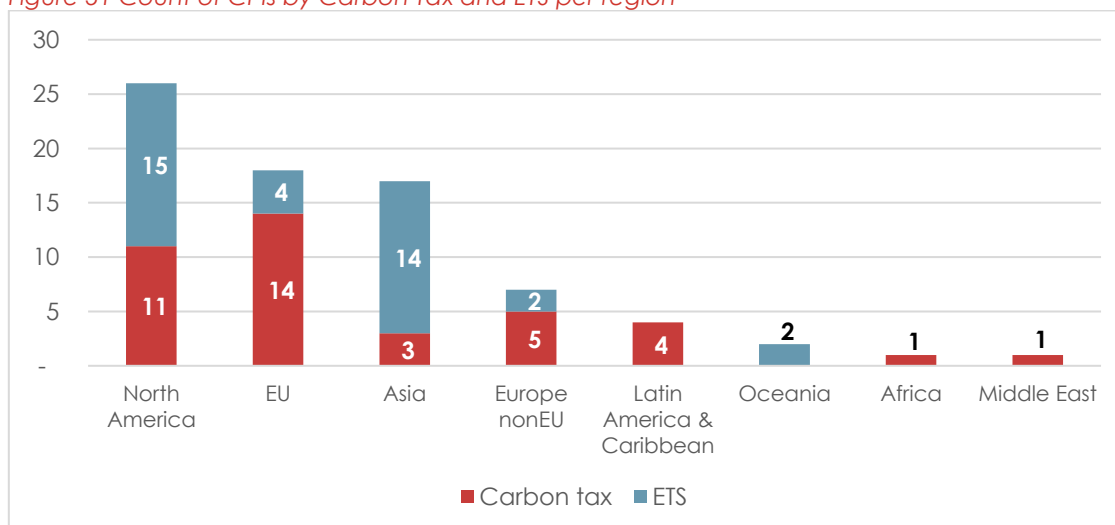
Region / Country / CPI	CPI (NUMBER AND TYPE)
Luxembourg (Luxembourg carbon tax)	Carbon Tax
Netherlands (Netherlands carbon tax)	Carbon Tax
Poland (Poland carbon tax)	Carbon Tax
Portugal (Portugal carbon tax)	Carbon Tax
Slovenia (Slovenia carbon tax)	Carbon Tax
Sweden (Sweden carbon tax)	Carbon Tax
Asia	17
China	10
China national ETS (National)	ETS
Beijing pilot ETS	ETS
Chongqing pilot ETS	ETS
Fujian pilot ETS	ETS
Guangdong pilot ETS	ETS
Hubei pilot ETS	ETS
Shanghai pilot ETS	ETS
Shenzhen pilot ETS	ETS
Taiwan, China carbon fee	Carbon tax
Tianjin pilot ETS	ETS
Japan	3
Japan carbon tax (National)	Carbon Tax
Saitama ETS	ETS
Tokyo CaT	ETS
Indonesia (Indonesia ETS)	ETS
Kazakhstan (Kazakhstan ETS)	ETS
Singapore (Singapore carbon tax)	Carbon Tax
South Korea (Korea ETS)	ETS
Europe nonEU	9
United Kingdom (UK ETS)	ETS
United Kingdom (Carbon price support)	Carbon Tax
Iceland (Iceland carbon tax)	Carbon Tax
Liechtenstein (Liechtenstein carbon tax)	Carbon Tax
Montenegro (Montenegro ETS)	ETS
Norway (Norway carbon tax)	Carbon Tax
Switzerland (Switzerland carbon tax)	Carbon Tax
Switzerland (Switzerland ETS)	ETS
Ukraine (Ukraine carbon tax)	Carbon Tax
Latin America & Caribbean	4
Argentina (Argentina carbon tax)	Carbon Tax
Chile (Chile carbon tax)	Carbon Tax
Colombia (Colombia carbon tax)	Carbon Tax
Uruguay (Uruguay CO2 tax)	Carbon Tax
Oceania	2
Australia (Australia Safeguard Mechanism)	ETS

Region / Country / CPI	CPI (NUMBER AND TYPE)
New Zealand (New Zealand ETS)	ETS
Africa	1
South Africa (South Africa carbon tax)	Carbon Tax
Middle East	1
Israel (Israel Carbon Tax)	Carbon Tax
Grand Total	76

Source: Study team elaboration based on World Bank's carbon pricing dashboard (<https://carbonpricingdashboard.worldbank.org/>)

Regarding the distinction of carbon tax and ETS schemes worldwide, the next figure illustrates the **distribution** of these 76 CPIs applicable to transport across regions, disaggregated by type: carbon taxes (red) and emissions trading systems (blue)²⁰⁵. The figure shows that North American leads by number of CPIs Implemented followed by Asia and the EU. Carbon taxes are the frequent CPI types used to regulate transport emissions directly or indirectly, with 39 CPIs, while there are 37 Emissions Trading Systems worldwide that regulate transport emissions.

Figure 31 Count of CPIs by Carbon Tax and ETS per region

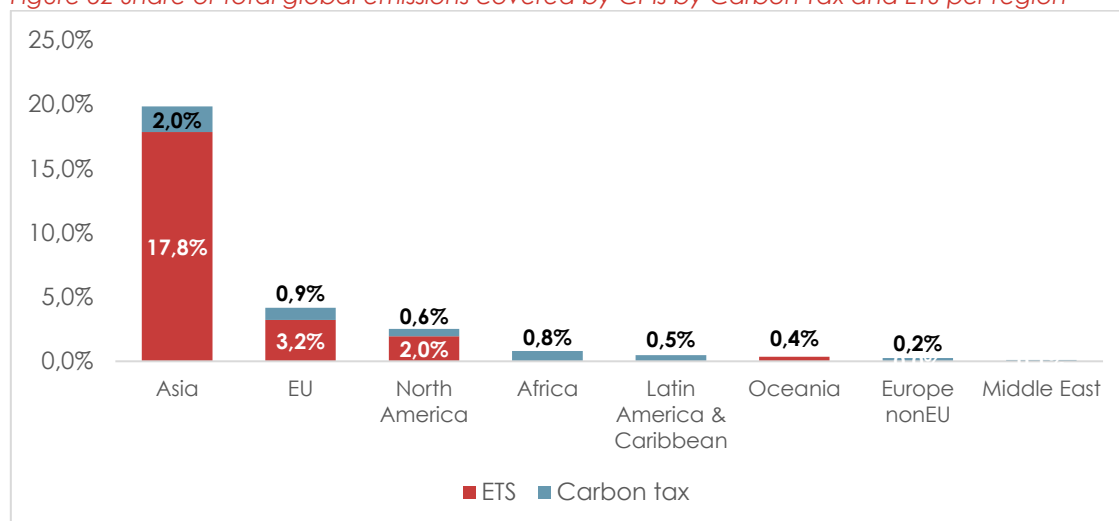


Source: Study team elaboration based on World Bank's carbon pricing dashboard (<https://carbonpricingdashboard.worldbank.org/>)

²⁰⁵ According to the World Bank's CPI dashboard, in an emissions trading system (ETS), the government places a limit on the amount of GHG emissions from covered entities. Entities must surrender emission units (or "allowances") to cover their emissions within a compliance period. Each emission unit represents the right to emit a certain volume of emissions (typically 1 tCO₂e) and can be traded between covered entities or sometimes with other traders. The carbon price in these systems is usually a function of supply and demand for emission units. Through a carbon tax a government levies a fee on covered entities for their GHG emissions, providing a financial incentive to reduce emissions. Under a carbon tax, the government sets the price of emissions (the tax rate). The resulting volume of emissions reductions achieved by the policy is determined by the response of the emitting entities to the set price. See <https://carbonpricingdashboard.worldbank.org/about#download-data>

While the number of CPIs implemented for regulating transport emissions is important, it is also the share of global emissions covered by the CPIs, as it helps understanding the extent to which the CPI may address climate targets. The figure below, indicates the largest share of global emissions concentration per region. It shows that Asia's CPIs covers the largest share of global emissions, followed by the EU's CPIs, and North America's CPIs.

Figure 32 Share of total global emissions covered by CPIs by Carbon Tax and ETS per region²⁰⁶



Source: Study team elaboration based on World Bank's carbon pricing dashboard (<https://carbonpricingdashboard.worldbank.org/>)

It is worth noting from table 33 that CPIs vary from supranational (e.g. EU ETS), National (e.g. China, South Africa), subnational/Federal State/local levels (Alberta, Tokyo, etc.). In the following section, a more detailed analysis of these instruments is provided, focusing on key dimensions such as carbon pricing levels, fuel coverage and the specific transport modes included.

An **implication regarding the countries with CPIs covering transport emissions**: CBAM Declarations of goods originating in countries with CPIs covering transport emissions, may be more likely to have transport activity and emissions data than countries without CPIs, and some form of verification report of the CPI of the third country may be available, potentially able to be shared with CBAM Declarant. This consideration however applies to CPIs directly regulating the transport sector, as for the case of coverage through upstream fuel tax, the regulated entity is hardly even the transport operator, but their suppliers (fuels distributors and electricity generation).

8.8.1 Carbon price of CPIs covering transport sector directly or through upstream CPIs

Carbon pricing systems vary considerably worldwide, showing not only large disparities between regions, but also across countries and jurisdictions. Based on data from the World Bank's Carbon Pricing Dashboard²⁰⁷ from the 1st of April 2025, for the instruments

²⁰⁶ The share of total global emissions correspond to emissions released by all industries and sectors worldwide. The CPIs refer to all sectors, including the transport sector. CPIs not regulating the transport sector directly or through fuels or electricity (for electrified road transport), are not included in this analysis.

²⁰⁷ <https://carbonpricingdashboard.worldbank.org/about#download-data>

with status “implemented”, the following carbon pricing patterns can be seen. While third countries' carbon price effectively paid tend to be lower in the Rest of the World, there are not many insights for all the transport related CPIs. The available information on carbon price effectively paid is exclusively for Road Transport in the OECD database. But there is yet to be produced a transport related carbon price effectively paid rates for all transport modes.

Based on the World Bank's Carbon pricing Dashboard 2024 average ETS carbon prices, the EU ETS stands out with an average price of 60 €/tCO₂eq²⁰⁸. In East Asia and the Pacific, where six ETS include transport sector, prices are considerably lower, with an average of 12.72€/tCO₂eq, and highly variable, with the lowest case of Shenzhen (China) standing out at just 5.55€/tCO₂eq compared to 27.21€/tCO₂eq in New Zealand. In North America, ETS prices are also variable, but generally higher, with an average of 34.23€/tCO₂eq. North America includes one of the highest regional carbon prices under ETS observed worldwide, with Washington at 42.5€/tCO₂eq.

Carbon taxes, in the year 2024 according to the average prices in the World Bank's Carbon Pricing Dashboard, show greater heterogeneity. In Europe, Nordic countries stand out with very high rates: Sweden (123 €/tCO₂eq), Liechtenstein (115 €/tCO₂eq) and Norway (113 €/tCO₂eq), which are considerable above the EU ETS average carbon price. In contrast, Japan applies a symbolic tax (1.64 €/tCO₂eq) and Mexico applies a carbon tax of 3.35€/tCO₂eq. The Latin America & Caribbean's carbon taxes shows extreme disparity. Argentina and Colombia have carbon tax rates varying from 4 to 6 €/tCO₂eq. Uruguay has a remarkable price at 134 €/tCO₂eq, higher than Sweden's carbon tax rate, the highest of Europe²⁰⁹. In Africa, only South Africa imposes a carbon tax of €11/tCO₂eq, which is modest but significant in the regional context. In general, carbon taxes tend to show greater regional variation, with some countries setting carbon prices well above EU ETS levels (such as Sweden and Uruguay), while others remain well below (such as Japan or Mexico). ETS prices tend to be more consistent within regions, especially in structured markets such as the EU or North America.

On average carbon prices at regional level show that non-EU European countries impose higher carbon prices through taxation, while through emission trading systems it is the EU ETS the CPI with the higher carbon price.

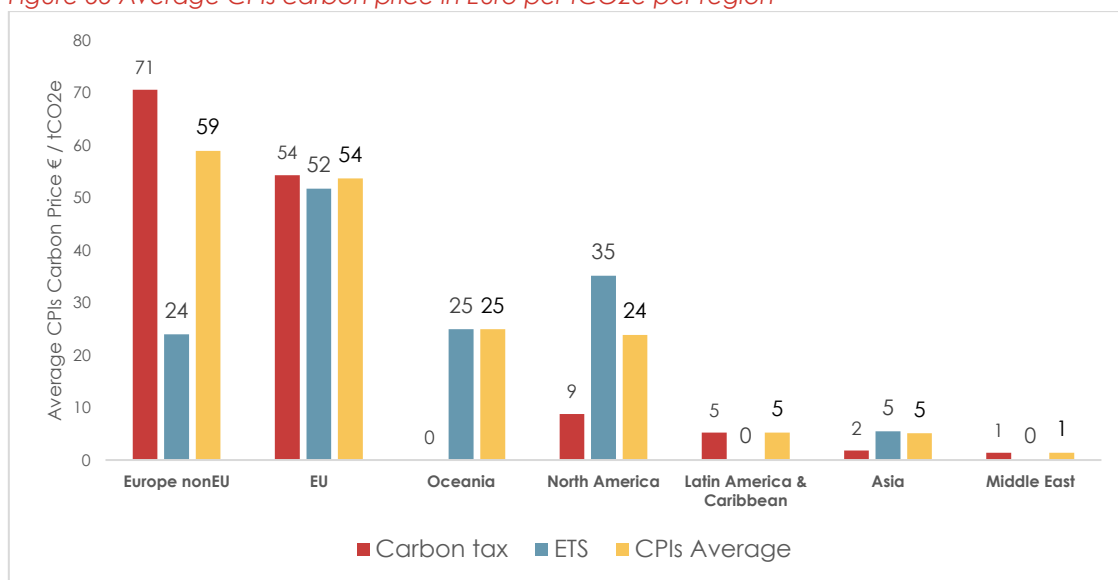
Notably, Asian countries' CPIs, in terms of carbon price imposed through taxation as well as by means of emission trading systems, are the ones with the lowest carbon prices, which contrasts with the share of global emissions covered by their CPIs.

An **implication regarding the carbon price paid**. For CBAM Declarations of transport emissions while transport emissions may have been paid for a carbon price in many Asian, Latin American & Caribbean and Middle East countries according to their CPIs, the price paid for those emissions is substantially lower than the EU ETS, which implies that their CBAM certificates to deduct from surrendering will be very low, despite having paid transport emissions dues in their CPIs jurisdictions, due to the low price paid for emissions.

²⁰⁸ Price for April 1st, 2025. Converted into Euro with the European Central Bank average exchange rate for the same day.

²⁰⁹ According to the world bank's carbon price dashboard, Sweden's carbon tax apply to the following fuels: Diesel, Other oil products, Gasoline, Kerosene, LPG, Natural gas, Jet fuel, Coal.

Figure 33 Average CPLs carbon price in Euro per tCO₂e per region²¹⁰



Source: Study team elaboration based on World Bank's carbon pricing dashboard (<https://carbonpricingdashboard.worldbank.org/>)

Comparing the type of fuels covered by ETS and carbon taxes in the transport sector shows that in the EU, both the ETS and carbon taxes tend to cover a similar range of traditional fossil fuels, such as diesel, petrol, kerosene, LPG, natural gas, coal and other oil products. However, there are some countries, such as the Nordic countries, including Sweden, Norway, Denmark and Finland, or countries like Hungary, extend this coverage through their carbon taxes to also include waste as fuel and jet fuel, reinforcing their environmental approach. In Africa, the only country with a carbon tax, South Africa, adopts an extensive coverage, in line with those fuels covered by the Nordic countries.

Asia and Oceania, the situation is more fragmented. The ETS of countries such as China, Australia and New Zealand cover broadly similar fossil fuels, but only some include jet fuel (such as Shanghai, Australia, and New Zealand), while Beijing, South Korea and Shenzhen exclude it. In contrast, the Japanese carbon tax is limited to conventional fuels (such as diesel, gasoline, natural gas, coal) and completely excludes jet fuel.

In Latin America & Caribbean, the coverage varies significantly between countries as well. Argentina and Colombia apply taxes to a wide range of fuels, including jet fuel and various oil fuels, while Uruguay applies its tax only to gasoline, which considerably limits its potential environmental impact.

Finally, in North America, both ETS and carbon taxes show significant internal variations. Within the US, the ETS of California, Oregon and Washington differ notably: only Washington includes jet fuel, while Oregon excludes several key fuels such as kerosene, other oil products, or Coal. In Canada, Quebec's ETS shows a broad and standard coverage, covering the same fuels as California. As for the Mexican carbon tax, it covers

²¹⁰ The share of total global emissions correspond to emissions released by all industries and sectors worldwide. The CPLs refer to all sectors, including the transport sector. CPLs not regulating the transport sector directly or through fuels or electricity (for electrified road transport), are not included in this analysis.

in detail a variety of oil products (including fuel oil, coke, crude oil), showing a higher level of specificity, in addition to jet fuel.

8.8.2 *IMO and CORSIA Carbon pricing*

As previously stated, IMO and ICAO have agreed on a carbon pricing schemes for maritime and aviation transport, respectively.

The IMO Net-zero Framework was approved by the Marine Environment Protection Committee during its 83rd session from 7 to 11 of April 2025,²¹¹ and it is the first framework to combine mandatory emission limits and GHG pricing globally across the entire shipping industry, aiming for a net-zero emissions by 2050 by the acceleration of near zero GHG fuels, technologies and energy.

Under this draft regulation, that may be adopted by 2026²¹², the measures will be applied to global fuel standards, i.e. and obligation for reduction of ships' annual GHG fuel intensity (GFI); and global economic measures, i.e. acquisition of remedial units to balance ships emitting more than a specific GFI threshold, or acquisition of financial rewards otherwise. This contribution will be collected by the IMO Net-Zero Fund, and the revenues will be distributed among diverse initiatives, namely: support on innovation, research, infrastructure and just transition initiative in developing countries, mitigation of negative impacts on vulnerable states, among others.

On the other hand, CORSIA, which stands for Carbon offsetting and Reduction Scheme for International Aviation, is a global initiative by ICAO to address CO2 emissions from international aviation. It aims to achieve carbon neutral growth by offsetting emissions above 85% of 2019 levels. CORSIA use a monitoring, reporting and verification approach in which airlines are obliged to track and report CO2 emissions on international routes. Anything over those limits must be offset, airlines must buy carbon credits of those extra emissions that will go to projects such as renewable energy or reforestation.

So far, 130 countries have volunteered in CORSIA. The initiative has been divided in different phases and, currently, it is on a voluntary phase until 2026. From 2027 onwards, CORSIA will become mandatory, with some exceptions applying to less develop countries, countries with low aviation activity or small island developing states.

²¹¹ <https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-approves-netzero-regulations.aspx>

²¹² <https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-net-zero-shipping-talks-to-resume-in-2026.aspx>

9 Task 7. Consultation with stakeholders to collect feedback on potential challenges, concerns and suggestions regarding the extension of CBAM scope to embedded emissions in the transport outside the EU of the goods under scope (extended emission scope)

The objective of Task 7 was to map stakeholders and conduct consultations to collect, analyse and report primary data, insights and feedback on the **transport of CBAM goods into the EU**. It was focus on identifying and reporting challenges, potential concerns, and suggestions for extending the embedding scope of the CBAM to non-EU transport emissions. Evidence is collected to provide additional quantitative and qualitative data to cross-reference the information from the desk research and address data gaps identified in previous tasks (T1, T5.1, T5.2, T6.2). The consultations become factual summaries, and a synopsis report annexed to the draft final report (D3).

In common with Task 4, the approach is staged, comprising the following types of activities:

1. **Scoping interviews** were conducted with a selected group of stakeholders from July to September 2024. The scoping interview guidelines are in the Appendix A of this Inception report.
2. A **stakeholder survey** to ensure feedback from a broad range of stakeholders while still obtaining manageable responses, facilitating the incorporation of valuable insights. The survey is set to gather input on challenges, concerns, and suggestions related to the expansion of specific CN codes/sectors in a structured manner.
3. **Targeted in-depth interviews** with stakeholders following the stakeholder survey to delve deeper into qualitative information, providing in-depth insights and information gaps that could not be filled through the study. These in-depth interviews will primarily be with industry associations. Still, they will also include respondents in the survey who may have further interesting insights worth exploring in more detail and have indicated a willingness to participate in in-depth interviews.

9.1 Results

All activities under Task 4 were conducted in coordination with Task 7 to ensure a coherent and efficient approach in engaging stakeholders relevant for both Task 4 and 7.

The completion of this task has resulted in the Annexes - Task 7 of this final report, where the results of both task 4 and 7 have been merged to give an overall view of the stakeholder consultation. Annex 7 encompass the following elements:

- A stakeholder consultation analysis report, providing an in-depth analysis of the whole consultation process, called: 'Annex 7_1_Stakeholder Consultation Analysis'
- Factual summaries of the in-depth interviews, called: 'Annex 7_2_In-depth Interviews - Factual Summary Report'

- Factual summaries of the survey interviews, called: 'Annex 7_3_Survey - Factual Summary Report'
- Instead of a separate synopsis report, the analysis of the main findings from the stakeholder consultation, highlighting recurring concerns, insights, and areas of consensus, is reflected in the factual summaries of the survey and the interviews. For further details, refer to the stakeholder consultation analysis.
- Quantifying survey results to gather opinions and concerns regarding the potential inclusion of new goods within the expanded scope.
- Mapping of industry stakeholders, detailing the survey questions and providing an in-depth analysis of the input received from stakeholders.
- Both were produced together in the same document, entitled *Annex 7_4_Survey responses*:
 - An Excel file containing detailed (anonymised) survey responses and the corresponding analysis, where the corresponding names, emails and name of organisation has been removed, and appears in the first tab 'content'
 - An Excel file presenting the quantitative results derived from the survey and interviews. In the same Excel file, there is a tab per question with the analysis.

10 Annexes

10.1 T1 annexes

10.1.1 Longlist of assessed products

The table below presents the longlist of 8-digit PRODCOM codes and accompanying CN codes used as the starting point for the analysis in Chapter 0.

Table 53 List of 8-digit PRODCOM and CN codes assessed

PRODCOM Code	PRODCOM description	CN codes
05100000	Hard coal	27012000
06100000	Crude petroleum	27090010; 27090090
07101010	Iron ores and concentrates. Non-agglomerated (excluding roasted iron pyrites)	26011100
07291100	Copper ores and concentrates	26030000
07291200	Nickel ores and concentrates	26040000
07291300	Aluminium ores and concentrates	26060000
07291420	Other precious-metal ores and concentrates (excl. silver ores and concentrates)	26169000
07291510	Lead ores and concentrates	26070000
07291520	Zinc ores and concentrates	26080000
07291530	Tin ores and concentrates	26090000
07291905	Cobalt ores and concentrates	26050000
07291910	Chromium ores and concentrates	26100000
07291915	Manganese ores and concentrates, including ferruginous manganese ores and concentrates with a manganese content of 20 % or more, calculated on the dry weight	26020000
07291920	Tungsten ores and concentrates	26110000
07291925	Molybdenum ores and concentrates. Roasted.	26131000
07291930	Titanium ores and concentrates	26140000
07291935	Zirconium ores and concentrates	26151000
07291940	Niobium, tantalum or vanadium ores and concentrates	26159000
07291990	Other ores and concentrates n.e.s.	26179000
08122140	Kaolin	25070020
08911100	Natural calcium phosphates; natural aluminium calcium phosphates and phosphatic chalk	25101000; 25102000
08911220	Unroasted iron pyrites	25020000
08911230	Crude or unrefined sulphur (including recovered sulphur)	25030010
08911910	Natural barium sulphate (barytes) and carbonate (witherite)	25111000; 25112000
08911911	Natural borates and concentrates thereof (not calcined), but not including borates separated from natural brine; natural boric acid containing not more than 85 % of H ₃ BO ₃ calculated on the dry weight	25280000

PRODCOM Code	PRODCOM description	CN codes
08911912	Fluorspar	25292100; 25292200
08911913	Kieserite, epsomite (natural magnesium sulphates)	25302000
08911919	Other mineral substances n.e.s. (excluding vermiculite, perlite and chlorites, unexpanded and kieserite, epsomite (natural magnesium sulphates))	25309030; 25309040; 25309050; 25309070
08931000	Salt (including denatured salt but excluding salt suitable for human consumption) and pure sodium chloride, whether or not in aqueous solution or containing added anti-caking or free-flowing agents	25010031; 25010051; 25010099
08991000	Natural bitumen and natural asphalt; asphaltites and asphaltic rocks	27149000
08992110	Diamonds unworked or simply sawn or roughly shaped	71021000; 71023100
08992120	Precious stones (other than diamonds) and semi-precious stones, unworked or simply sawn or roughly shaped	71031000
08992210	Industrial diamonds. Unworked or simply sawn, cleaved or bruted, but not mounted or set	71022100
08992220	Pumice stone	25131000
08992230	Emery, natural corundum, natural garnet and other natural abrasives, whether or not heat-treated	25132000
08992910	Natural graphite	25041000; 25049000
08992913	Natural magnesium carbonate (magnesite);	25191000
08992914	Dead-burned (sintered) magnesia	25199030
08992915	Fused magnesia and other magnesium oxide	25199090
08992920	Other minerals	25061000; 25120000; 25199010; 25241000; 25249000; 25251000; 25252000; 25253000; 25261000; 25262000; 25291000; 25293000; 25301000; 26219000
10311130	Frozen potatoes, prepared or preserved (including potatoes cooked or partly cooked in oil and then frozen; excluding by vinegar or acetic acid)	20041010; 20041091; 20041099
10311300	Dried potatoes in the form of flour, meal, flakes, granules and pellets	11051000; 11052000
10391725	Concentrated tomato puree and paste	20029031; 20029039; 20029091; 20029099
10411100	Lard stearin, lard oil, oleostearin, oleo-oil and tallow oil (excluding emulsified, mixed or otherwise prepared)	15030011; 15030019; 15030030; 15030090
10411200	Fats and oils and their fractions of fish or marine mammals (excluding chemically modified)	15041010; 15041091; 15041099; 15042010; 15042090; 15043010; 15043090
10411900	Other animal fats and oils and their fractions, whether or not refined (excluding chemically modified)	15060000
10412110	Crude groundnut oil and its fractions (excluding chemically modified)	15081010; 15081090
10412210	Virgin olive oil and its fractions (excluding chemically modified)	15092000; 15093000; 15094000
10412221	Crude olive pomace oil and its fractions not chemically modified (including those blended with virgin olive oil or its fractions) (excluding virgin olive oil)	15101000

PRODCOM Code	PRODCOM description	CN codes
10412300	Crude sunflower-seed and safflower oil and their fractions (excluding chemically modified)	15121110; 15121191; 15121199
10412410	Crude rape, colza or mustard oil and their fractions (excluding chemically modified)	15141110; 15141190; 15149110; 15149190
10412510	Crude palm oil and its fractions (excluding chemically modified)	15111010; 15111090
10412910	Other vegetable oils, crude (excluding chemically modified oils)	15071010; 15071090; 15122110; 15122190; 15131110; 15131191; 15131199; 15132110; 15132130; 15132190; 15151100; 15155011; 15155019; 15156011; 15156059; 15159021; 15159029; 15159040; 15159059
10413000	Cotton linters	14042000
10414130	Oilcake and other solid residues resulting from the extraction of soya-bean oil	23040000
10414150	Oilcake and other solid residues resulting from the extraction of sunflower seed fats or oils	23063000
10414170	Oilcake and other solid residues resulting from the extraction of rape or colza seed fats or oils	23064100; 23064900
10414190	Oilcake and other solid residues from extraction of vegetable fats/oils (including cotton seeds, linseed, coconut, copra, palm nuts or kernels; excluding soya beans, sunflower, rape or colza seeds)	23050000; 23061000; 23062000; 23065000; 23066000; 23069005; 23069011; 23069019; 23069090
10414200	Flours and meals of oil seeds or oleaginous fruits (excluding of mustard)	12081000; 12089000
10415100	Refined soya-bean oil and its fractions (excluding chemically modified)	15079010; 15079090
10415200	Refined groundnut oil and its fractions (excluding chemically modified)	15089010; 15089090
10415310	Refined olive oil and its fractions (excluding chemically modified)	15099000
10415331	Other oils and their fractions obtained solely from olives not chemically modified (including those blended with virgin olive oil or its fractions) (excluding crude olive pomace oil, virgin olive oil)	15109000
10415400	Refined sunflower-seed and safflower oil and their fractions (excluding chemically modified)	15121910; 15121990
10415500	Refined cotton-seed oil and its fractions (excluding chemically modified)	15122910; 15122990
10415600	Refined rape, colza or mustard oil and their fractions (excluding chemically modified)	15141910; 15141990; 15149910; 15149990
10415700	Refined palm oil and its fractions (excluding chemically modified)	15119011; 15119019; 15119091; 15119099
10415800	Refined coconut (copra) oil and its fractions (excluding chemically modified)	15131911; 15131919; 15131930; 15131991; 15131999
10415900	Other oils and their fractions, refined but not chemically modified; fixed vegetable fats and other vegetable oils (except maize oil) and their fractions n.e.c. refined but not chemically modified	15132911; 15132919; 15132930; 15132950; 15132990; 15151910; 15151990; 15153010; 15153090; 15155091; 15155099; 15156051; 15156060; 15156091; 15156099; 15159011; 15159031; 15159039; 15159051; 15159060; 15159091; 15159099
10416030	Animal fats and oils and their fractions partly or wholly hydrogenated, inter-esterified, re-esterified or elaidinised, but not further prepared (including refined)	15161010; 15161090

PRODCOM Code	PRODCOM description	CN codes
10416050	Vegetable fats and oils and their fractions partly or wholly hydrogenated, inter-esterified, re-esterified or elaidinised, but not further prepared (including refined)	15162010; 15162091; 15162095; 15162096; 15162098; 15163091; 15163098
10417100	Vegetable waxes (including refined) (excluding triglycerides)	15211000
10417200	Degras; residues resulting from the treatment of fatty substances or animal or vegetable waxes	15220010; 15220031; 15220039; 15220091; 15220099
10512130	Skimmed milk powder (milk and cream in solid forms, of a fat content by weight of $\leq 1,5\%$), in immediate packings of $\leq 2,5$ kg	04021011; 04021091
10512160	Skimmed milk powder (milk and cream in solid forms, of a fat content by weight of $\leq 1,5\%$), in immediate packings of $> 2,5$ kg	04021019; 04021099
10512230	Whole milk powder or full cream powder (milk and cream in solid forms, of a fat content by weight of $> 1,5\%$), in immediate packings of $\leq 2,5$ kg	04022111; 04022191; 04022911; 04022915; 04022991
10512260	Whole milk powder or full cream powder (milk and cream in solid forms, of a fat content by weight of $> 1,5\%$), in immediate packings of $> 2,5$ kg	04022118; 04022199; 04022919; 04022999
10515300	Casein and caseinates	35011010; 35011050; 35011090
10515400	Lactose and lactose syrup (including chemically pure lactose)	17021100; 17021900
10515530	Whey and modified whey in powder, granules or other solid forms, whether or not concentrated or containing added sweetening matter	04041002; 04041004; 04041006; 04041012; 04041014; 04041016; 04041026; 04041028; 04041032; 04041034; 04041036; 04041038
10621111	Wheat starch	11081100
10621113	Maize (corn) starch	11081200
10621115	Potato starch	11081300
10621119	Starches (including rice, manioc, arrowroot and sago palm pith) (excluding wheat, maize (corn) and potato)	11081400; 11081910; 11081990
10621130	Inulin	11082000
10621150	Wheat gluten (excluding wheat gluten prepared for use as a glue or as a glazing or dressing for the textile industry)	11090000
10621170	Dextrins and other modified starches (including esterified or etherified, soluble starch, pregelatinised or swelling starch, dialdehyde starch, starch treated with formaldehyde or epichlorohydrin)	35051010; 35051050; 35051090
10621200	Tapioca and substitutes therefor prepared from starch, in the form of flakes, grains, pearls, siftings or similar forms	19030000
10621310	Glucose and glucose syrup (excluding with added flavouring or colouring matter)	17023010; 17023050; 17023090; 17024010; 17024090
10621320	Chemically pure fructose in solid form; fructose and fructose syrup, containing in the dry state $> 50\%$ of fructose; isoglucose excluding with added flavouring or colouring matter	17025000; 17026010; 17026080; 17026095; 17029030
10621330	Maltodextrin and maltodextrin syrup (excluding with added flavouring or colouring matter)	17029050
10621390	Other sugars (including invert sugar) n.e.c.	17029010; 17029080; 17029095
10621430	Crude maize (corn) oil and its fractions (excluding chemically modified)	15152110; 15152190

PRODCOM Code	PRODCOM description	CN codes
10621460	Refined maize (corn) oil and its fractions (excluding chemically modified)	15152910; 15152990
10622000	Residues of starch manufacture and similar residues	23031011; 23031019; 23031090
10811100	Raw cane and beet sugar in solid form, not containing added flavouring or colouring matter	17011210; 17011290; 17011310; 17011390; 17011410; 17011490
10811230	Refined white cane or beet sugar in solid form	17019910
10811290	Refined cane or beet sugar in a solid form (excluding white sugar)	17019990
10811300	Refined cane or beet sugar, containing added flavouring or colouring matter; maple sugar and maple syrup	17019100; 17022010; 17022090
10811430	Cane molasses	17031000
10811450	Molasses obtained from the extraction or refining of sugar (excluding cane molasses)	17039000
10812000	Beet-pulp, bagasse and other sugar manufacturing waste (including defecation scum and filter press residues)	23032010; 23032090
10891334	Bakers' yeast	21021031; 21021039
11061030	Malt, not roasted (excluding alcohol duty)	11071011; 11071019; 11071091; 11071099
11061050	Roasted malt (excluding alcohol duty, products which have undergone further processing, roasted malt put up as coffee substitutes)	11072000
13101000	Wool grease and fatty substances derived therefrom, including lanolin	15050010; 15050090
13102100	Raw silk (not thrown)	50020000
13102200	Wool, degreased or carbonised, not carded or combed	51012100; 51012900; 51013000
13102300	Noils of wool or fine animal hair	51031010; 51031090
13102400	Wool or animal hair, carded or combed (including wool tops)	51051000; 51052100; 51052900; 51053100; 51053900; 51054000
13102500	Cotton, carded or combed	52030000
13102600	Jute and other textile fibres (except flax, true hemp and ramie), processed but not spun	53039000
13102900	Other vegetable textile fibres, processed but not spun	53012100; 53012900; 53013000; 53029000; 53050000
13103100	Synthetic staple fibres, carded, combed or otherwise processed for spinning	55061000; 55062000; 55063000; 55064000; 55069000
13103200	Artificial staple fibres, carded, combed or otherwise processed for spinning	55070000
13104010	Silk yarn, n.p.r.s. (excluding spun from silk waste)	50040010; 50040090
13104030	Yarn spun from silk waste, n.p.r.s.	50050010; 50050090
13104050	Silk yarn and silk waste yarn, p.r.s.; silk-worm gut	50060010; 50060090
13105010	Yarn of carded wool or fine animal hair, n.p.r.s.	51061010; 51061090; 51062010; 51062091; 51062099; 51081010; 51081090
13105030	Yarn of combed wool or fine animal hair, n.p.r.s.	51071010; 51071090; 51072010; 51072030; 51072051; 51072059; 51072091; 51072099; 51082010; 51082090
13105050	Yarn of wool or fine animal hair, p.r.s.	51091010; 51091090; 51099000
13106132	Yarn of uncombed cotton, n.p.r.s., for woven fabrics (excluding for carpets and floor coverings)	52051100; 52051200; 52051300; 52051400; 52051510; 52051590;

PRODCOM Code	PRODCOM description	CN codes
		52053100; 52053200; 52053300; 52053400; 52053500; 52061100; 52061200; 52061300; 52061400; 52061500; 52063100; 52063200; 52063300; 52063400; 52063500
13106133	Yarn of uncombed cotton, n.p.r.s., for knitted fabrics and hosiery	52051100; 52051200; 52051300; 52051400; 52051510; 52051590; 52053100; 52053200; 52053300; 52053400; 52053500; 52061100; 52061200; 52061300; 52061400; 52061500; 52063100; 52063200; 52063300; 52063400; 52063500
13106135	Yarn of uncombed cotton, n.p.r.s., for other uses (including carpets and floor coverings)	52051100; 52051200; 52051300; 52051400; 52051510; 52051590; 52053100; 52053200; 52053300; 52053400; 52053500; 52061100; 52061200; 52061300; 52061400; 52061500; 52063100; 52063200; 52063300; 52063400; 52063500
13106152	Yarn of combed cotton, n.p.r.s., for woven fabrics (excluding for carpets and floor coverings)	52052100; 52052200; 52052300; 52052400; 52052600; 52052700; 52052800; 52054100; 52054200; 52054300; 52054400; 52054600; 52054700; 52054800; 52062100; 52062200; 52062300; 52062400; 52062500; 52064100; 52064200; 52064300; 52064400; 52064500
13106153	Yarn of combed cotton, n.p.r.s., for knitted fabrics and hosiery	52052100; 52052200; 52052300; 52052400; 52052600; 52052700; 52052800; 52054100; 52054200; 52054300; 52054400; 52054600; 52054700; 52054800; 52062100; 52062200; 52062300; 52062400; 52062500; 52064100; 52064200; 52064300; 52064400; 52064500
13106155	Yarn of combed cotton, n.p.r.s., for other uses (including carpets and floor coverings)	52052100; 52052200; 52052300; 52052400; 52052600; 52052700; 52052800; 52054100; 52054200; 52054300; 52054400; 52054600; 52054700; 52054800; 52062100; 52062200; 52062300; 52062400; 52062500; 52064100; 52064200; 52064300; 52064400; 52064500
13106160	Cotton yarn, p.r.s. (excluding sewing thread)	52071000; 52079000
13106200	Cotton sewing thread	52041100; 52041900; 52042000
13107110	Flax yarn, n.p.r.s.	53061010; 53061030; 53061050; 53062010
13107120	Flax yarn, p.r.s.	53061090; 53062090
13107200	Yarn of vegetable or bast fibres (excluding flax); paper yarn	53071000; 53072000; 53081000; 53082010; 53082090; 53089012; 53089019; 53089050; 53089090
13108110	Multiple or cabled synthetic filament yarn, n.p.r.s.	54026100; 54026200; 54026300; 54026900
13108130	Multiple or cabled yarn of artificial filaments, n.p.r.s. (excluding sewing thread)	54034100; 54034200; 54034900
13108150	Man-made filament yarn, p.r.s. (excluding sewing thread)	54060000

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13108210	Yarn (other than sewing thread) containing $\geq 85\%$ by weight of synthetic staple fibres, n.p.r.s.	55091100; 55091200; 55092100; 55092200; 55093100; 55093200; 55094100; 55094200
13108250	Yarn (other than sewing thread) containing $\geq 85\%$ by weight of synthetic staple fibres, p.r.s.	55111000
13108320	Yarn containing $< 85\%$ by weight of polyester staple fibres (other than sewing thread), mixed with artificial fibres, n.p.r.s.	55095100
13108333	Yarn containing $< 85\%$ by weight of synthetic staple fibres (other than sewing thread) mixed with carded wool or fine animal hair, n.p.r.s.	55095200; 55096100; 55099100
13108336	Yarn containing $< 85\%$ by weight of synthetic staple fibres, mixed with combed wool or fine animal hair, n.p.r.s.	55095200; 55096100; 55099100
13108340	Yarn containing $< 85\%$ by weight of synthetic staple fibres (other than sewing thread), mixed with cotton, n.p.r.s.	55095300; 55096200; 55099200
13108380	Other yarns, containing $< 85\%$ by weight of synthetic staple fibres (other than sewing thread), n.p.r.s., n.e.c.	55095900; 55096900; 55099900
13108390	Yarn containing $< 85\%$ by weight of synthetic staple fibres (other than sewing thread), p.r.s.	55112000
13108410	Yarn (other than sewing thread) of artificial staple fibres, n.p.r.s.	55101100; 55101200; 55102000; 55103000; 55109000
13108430	Yarn (other than sewing thread) of artificial staple fibres, p.r.s.	55113000
13108510	Sewing thread of man-made filaments	54011012; 54011014; 54011016; 54011018; 54011090; 54012010; 54012090
13108550	Sewing thread of man-made staple fibres	55081010; 55081090; 55082010; 55082090
13951010	Non-wovens of a weight $\leq 25 \text{ g/m}^2$ (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	56031190; 56039190
13951020	Non-wovens of a weight of $> 25 \text{ g/m}^2$ but $\leq 70 \text{ g/m}^2$ (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	56031290; 56039290
13951030	Non-wovens of a weight of $> 70 \text{ g/m}^2$ but $\leq 150 \text{ g/m}^2$ (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	56031390; 56039390
13951050	Non-wovens of a weight of $> 150 \text{ g/m}^2$ (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	56031490; 56039490
13951070	Non-wovens, coated or covered (including articles made from non-wovens) (excluding articles of apparel)	56031110; 56031210; 56031310; 56031410; 56039110; 56039210; 56039310; 56039410
14111000	Articles of apparel of leather or of composition leather (including coats and overcoats) (excluding clothing accessories, headgear, footwear)	42031000
16211100	Plywood, veneered panels and similar laminated wood, of bamboo	44121000
16211200	Particle board, of wood	44101110; 44101130; 44101150; 44101190
16211316	Oriented strand board (OSB), of wood	44101210; 44101290
16211419	Waferboard and similar board, of wood (excluding particle board and oriented strand board [OSB])	44101900
16211450	Particle board and similar board of ligneous materials (excluding wood)	44109000

PRODCOM Code	PRODCOM description	CN codes
16211523	Medium density fibreboard (MDF), of wood or other ligneous materials, whether or not bonded with resins or other organic substances, of a thickness not exceeding 5 mm	44111210; 44111292; 44111294
16211526	Medium density fibreboard (MDF), of wood or other ligneous materials, whether or not bonded with resins or other organic substances, of a thickness exceeding 5 mm but not exceeding 9 mm	44111310; 44111392; 44111394
16211529	Medium density fibreboard (MDF), of wood or other ligneous materials, whether or not bonded with resins or other organic substances, of a thickness exceeding 9 mm	44111410; 44111492; 44111495; 44111497
16211543	Fibreboard (excluding medium density fibreboard [MDF]), of wood or other ligneous materials, whether or not bonded with resins or other organic substances, of a density exceeding 0,8 g/cm ³	44119210; 44119290
16211546	Fibreboard (excluding medium density fibreboard [MDF]), of wood or other ligneous materials, whether or not bonded with resins or other organic substances, of a density exceeding 0,5 g/cm ³ but not exceeding 0,8 g/cm ³	44119300
16211549	Fibreboard of wood or other ligneous materials (excluding medium density fibreboard [MDF]), whether or not bonded with resins or other organic substances, of a density not exceeding 0,5 g/cm ³	44119410; 44119490
16211611	Other plywood, veneered panels and similar laminated wood, of coniferous wood	44123900; 44124900; 44125900; 44129910; 44129990
16211712	Plywood consisting solely of sheets of wood (excluding of bamboo), each ply not exceeding 6 mm thickness, with at least one outer ply of tropical wood	44123110; 44123190; 44124199; 44125190; 44129110; 44129199
16211811	Other plywood, veneered panels and similar laminated wood, of other wood	44123310; 44123320; 44123330; 44123390; 44123400; 44124191; 44124200; 44125110; 44125200; 44129191; 44129210; 44129290
16212100	Densified wood, in blocks, plates, strips or profile shapes	44130000
16212210	Veneer sheets and sheets for plywood and other wood sawn lengthwise, sliced or peeled, of a thickness ≤ 6 mm of coniferous wood	44081015; 44081091; 44081098
16212300	Veneer sheets and sheets for plywood and other wood, sawn lengthwise, sliced or peeled, of a thickness ≤ 6 mm of tropical wood	44083111; 44083121; 44083125; 44083130; 44083915; 44083921; 44083930; 44083955; 44083970; 44083985; 44083995
16212400	Veneer sheets and sheets for plywood and other wood sawn lengthwise, sliced or peeled, of a thickness ≤ 6 mm of other wood than coniferous or tropical wood	44089015; 44089035; 44089085; 44089095
17111100	Chemical wood pulp, dissolving grades	47020000
17111200	Chemical wood pulp, soda or sulphate, other than dissolving grades	47031100; 47031900; 47032100; 47032900
17111300	Chemical wood pulp, sulphite, other than dissolving grades	47041100; 47041900; 47042100; 47042900
17111400	Mechanical wood pulp; semi-chemical wood pulp; pulps of fibrous cellulosic material other than wood	47010010; 47010090; 47050000; 47061000; 47062000; 47063000; 47069100; 47069200; 47069300
17121100	Newsprint in rolls or sheets	48010000

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17121200	Handmade paper and paperboard	48021000
17121300	Paper and paperboard used as a base for photo-sensitive, heat-sensitive or electro-sensitive paper; carbonising base paper; wallpaper base	48022000; 48024010; 48024090
17121410	Graphic paper, paperboard : mechanical fibres <= 10 %, weight < 40 g/m ²	48025400
17121435	Graphic paper, paperboard : mechanical fibres <= 10 %, weight >= 40 g/m ² but <= 150 g/m ² , in rolls	48025515; 48025525; 48025530; 48025590
17121439	Graphic paper, paperboard : mechanical fibres <= 10 %, weight >= 40 g/m ² but <= 150 g/m ² , sheets	48025620; 48025680; 48025700
17121450	Graphic paper, paperboard : mechanical fibres <= 10 %, weight > 150 g/m ²	48025810; 48025890
17121470	Graphic paper, paperboard : mechanical fibres > 10 %	48026115; 48026180; 48026200; 48026900
17122030	Cellulose wadding for household or sanitary purposes, in rolls of a width > 36 cm or in rectangular (including square sheets) with at least one side > 36 cm in an unfolded state	48030010
17122055	Creped paper and webs of cellulose fibres for household/ sanitary purposes, in rolls, width > 36 cm, rectangular sheets min. one side > 36 cm in unfolded state, weight <= 25 g/m ² /ply	48030031
17122057	Creped paper and webs of cellulose fibres for household/sanitary purposes, in rolls, width > 36 cm, rectangular sheets min. one side > 36 cm in unfolded state, weight > 25 g/m ² /ply	48030039
17122090	Paper stock for household : others	48030090
17123100	Uncoated, unbleached kraftliner in rolls or sheets (excluding for writing, printing or other graphic purposes, punch card stock and punch card tape paper)	48041111; 48041115; 48041119; 48041190
17123200	Uncoated kraftliner in rolls or sheets (excluding unbleached, for writing; printing or other graphic purposes, punch card stock and punch card tape paper)	48041912; 48041919; 48041930; 48041990
17123300	Semi-chemical fluting	48051100
17123400	Recycled fluting and other fluting	48051200; 48051910; 48051990
17123520	Uncoated testliner (recycled liner board), weight <= 150 g/m ² , in rolls or sheets	48052400
17123540	Uncoated testliner (recycled liner board), weight > 150 g/m ² , in rolls or sheets	48052500
17124120	Uncoated, unbleached sack kraft paper (excluding for writing, printing or other graphic purposes, punch card stock and punch card tape paper)	48042110; 48042190
17124140	Uncoated sack kraft paper (excluding unbleached, for writing, printing or other graphic purposes, punch card stock and punch card tape paper)	48042910; 48042990
17124160	Uncoated kraft paper and paperboard weighing <= 150 g/m ² (excluding kraftliner, sack kraft paper, for writing, printing and other graphic purposes, etc.)	48043151; 48043158; 48043180; 48043951; 48043958; 48043980
17124180	Creped or crinkled sack kraft paper; in rolls or sheets	48084000
17124220	Sulphite wrapping paper in rolls or sheets	48053000
17124240	Other uncoated paper and paperboard, in rolls or sheets, weight <= 150 g/m ² (excluding products of	48059100

PRODCOM Code	PRODCOM description	CN codes
17124260	HS 4802, fluting paper, testliner, sulphite wrapping paper, filter or felt paper and paperboard) Other uncoated paper and paperboard, in rolls or sheets, weight > 150 g/m ² and < 225 g/m ² (excluding products of HS 4802, fluting paper, testliner, sulphite wrapping paper, filter or felt paper and paperboard)	48059200
17124280	Other uncoated paper and paperboard, in rolls or sheets, weight >= 225 g/m ² (excluding products of HS 4802, fluting paper, testliner, sulphite wrapping paper, filter or felt paper and paperboard)	48059320; 48059380
17124330	Uncoated filter paper and paperboard in rolls or sheets	48054000
17124360	Uncoated felt paper and paperboard in rolls or sheets	48055000
17124400	Cigarette paper (excluding in the form of booklets or tubes), in rolls > 5 cm wide	48139010; 48139090
17125110	Uncoated, inside grey paperboard	48045990
17125910	Other uncoated paperboard	48044191; 48044198; 48044200; 48044900; 48045100; 48045200; 48045910
17126000	Vegetable parchment, greaseproof papers, tracing papers and glassine and other glazed transparent or translucent papers	48061000; 48062000; 48063000; 48064010; 48064090
17127100	Composite paper and paperboard in rolls or sheets (including strawpaper and paperboard) (excluding surface coated or impregnated)	48070030; 48070080
17127200	Paper and paperboard, creped, crinkled, embossed or perforated	48084000; 48089000
17127336	Coated bases for paper and paperboard of a kind used for: photo-, heat- and electro-sensitive paper and having 10 % or less of mechanical and chemical fibres, and paper and paperboard of a kind used for writing, printing or other graphic purposes, which weighs less than or equal to 150 g/m ²	48101300; 48101400; 48101900
17127360	Lightweight coated paper for writing, printing, graphic purposes, m.f. > 10 %	48102200
17127375	Other coated mechanical graphic paper for writing, printing, graphic purposes, m.f. > 10 %, rolls	48102930
17127379	Other coated mechanical graphic paper for writing, printing, graphic purposes, m.f. > 10 %, sheets	48102980
17127400	Kraft paper (other than that of a kind used for writing, printing or other graphic purposes), coated with kaolin or with other inorganic substances	48103100
17127500	Kraft paperboard (other than that of a kind used for writing, printing or other graphic purposes), coated with kaolin or with other inorganic substances	48103210; 48103290
17127600	Carbon paper, self-copy paper and other copying or transfer paper, in rolls or sheets	48092000; 48099000
17127710	Tarred, bituminised or asphalted paper and paperboard in rolls or sheets	48111000
17127733	Self-adhesive paper and paperboard in rolls or sheets	48114120; 48114190
17127735	Gummed paper and paperboard in rolls or sheets (excluding self-adhesives)	48114900

PRODCOM Code	PRODCOM description	CN codes
17127755	Bleached paper and paperboard in rolls or sheets, coated, impregnated or covered with plastics weighing > 150 g/m ² (excluding adhesives)	48115100
17127759	Paper and paperboard in rolls or sheets, coated, impregnated or covered with plastics (excluding adhesives, bleached and weighing > 150 g/m ²)	48115900
17127770	Paper and paperboard in rolls or sheets, coated, impregnated or covered with wax, paraffin wax, stearin, oil or glycerol	48116000
17127780	Other paper, paperboard, coated..., n.e.c.	48119000
17127820	Kraft paper and paperboard, coated on one or both sides with kaolin or other inorganic substances, in rolls or in square or rectangular sheets, of any size (excluding that for writing, printing or other graphic purposes; paper and paperboard bleached uniformly in the mass and containing > 95 % chemically processed wood fibres by weight in relation to the total fibre content)	48103900
17127850	Multi-ply paper and paperboard, coated, others	48109290
17127953	Multi-ply paper and paperboard, coated, of which each layer in bleached	48109210
17127955	Multi-ply paper and paperboard, coated, with one bleached outer layer	48109230
17127970	Paper/paperboard in rolls or sheets, coated on one/both sides with kaolin or other inorganic substances excluding of a kind used for any graphic purposes, multi-ply paper/paperboard	48109910; 48109980
19103000	Pitch and pitch coke, obtained from coal tar or from other mineral tars	27081000; 27082000
19200000	Refined petroleum products	27101211; 27101215; 27101221; 27101225; 27101231; 27101241; 27101245; 27101249; 27101250; 27101270; 27101290; 27101911; 27101915; 27101921; 27101925; 27101929; 27101931; 27101935; 27101943; 27101946; 27101947; 27101948; 27101951; 27101955; 27101962; 27101966; 27101967; 27101971; 27101975; 27101981; 27101983; 27101985; 27101987; 27101991; 27101993; 27101999; 27102011; 27102016; 27102019; 27102032; 27102038; 27102090; 27109100; 27109900
20111120	Argon	28042100
20111140	Helium	28042910
20111160	Nitrogen	28043000
20111170	Oxygen	28044000
20111190	Other rare gases (neon, krypton and xenon)	28042990
20111230	Carbon dioxide	28112100
20111250	Sulphur trioxide (sulphuric anhydride); diarsenic trioxide	28112910
20111270	Nitrogen oxides	28112930

PRODCOM Code	PRODCOM description	CN codes
20111290	Inorganic oxygen compounds of non-metals (excluding sulphur trioxide (sulphuric anhydride); diarsenic trioxide, nitrogen oxides, silicon dioxide, sulphur dioxide, carbon dioxide)	28112990
20111300	Liquid air; compressed air	28539030
20121130	Zinc oxide; zinc peroxide	28170000
20121150	Titanium oxides	28230000
20121210	Manganese oxides. Manganese dioxide	28201000
20121220	Manganese oxides. Manganese oxide containing by weight 77 % or more of manganese	28209010
20121230	Manganese oxides. Manganese oxide containing by weight less than 77 % of manganese	28209090
20121240	Chromium, lead and copper oxides and hydroxides	28191000; 28199010; 28199090; 28241000; 28249000; 28255000
20121910	Iron oxides and hydroxides; earth colours containing 70 % or more by weight of combined iron evaluated as Fe ₂ O ₃	28211000; 28212000
20121930	Cobalt oxides and hydroxides; commercial cobalt oxides	28220000
20121955	Lithium oxide and hydroxide	28252000
20121960	Vanadium oxides and hydroxides	28253000
20121965	Nickel oxides and hydroxides	28254000
20121970	Germanium oxides and zirconium dioxide	28256000
20121973	Molybdenum oxides and hydroxides	28257000
20121975	Antimony oxides	28258000
20121981	Tungsten oxides and hydroxides	28259040
20121996	Tin oxides and hydroxides	28259085
20121997	Other inorganic bases; other metal oxides, hydroxides and peroxides (excluding tin oxides and hydroxides), n.e.c.	28259011; 28259019; 28259060; 28259085
20122110	Disperse dyes and preparations based thereon	32041100
20122120	Acid and mordant dyes and preparations based thereon	32041200
20122130	Basic dyes and preparations based thereon	32041300
20122140	Direct dyes and preparations based thereon	32041400
20122150	Other synthetic organic colouring matters	32041500; 32041600; 32041700; 32041800; 32041900; 32049000
20122160	Synthetic organic products used as fluorescent brightening agents	32042000
20122170	Colour lakes; preparations based on colour lakes	32050000
20122250	Tanning extracts of vegetable origin; tannins and their salts, ethers, esters and other derivatives	32011000; 32012000; 32019020; 32019090
20122270	Colouring matter of vegetable or animal origin and preparations based thereon (including dyeing extracts) (excluding animal black)	32030010; 32030090
20122330	Synthetic organic tanning substances	32021000
20122350	Inorganic tanning substances; tanning preparations; enzymatic preparations for pre-tanning	32029000

PRODCOM Code	PRODCOM description	CN codes
20122415	Pigments and preparations based on titanium dioxide containing ≥ 80 % by weight of titanium dioxide	32061100
20122419	Pigments and preparations based on titanium dioxide (excluding those containing ≥ 80 % by weight of titanium dioxide)	32061900
20122450	Other colouring matter; pigments and preparations based on inorganic or mineral colouring matter; inorganic products of a kind used as luminophores	32062000; 32064100; 32064200; 32064910; 32064970; 32065000
20132111	Chlorine	28011000
20132116	Iodine; fluorine; bromine	28012000; 28013010; 28013090
20132120	Sulphur, sublimed or precipitated; colloidal sulphur	28020000
20132130	Carbon (carbon blacks and other forms of carbon, n.e.c.)	28030000
20132141	Boron	28045010
20132142	Tellurium	28045090
20132160	Silicon. Containing by weight not less than 99,99 % of silicon	28046100
20132170	Silicon. Other than containing by weight not less than 99,99 % of silicon	28046900
20132181	Phosphorus	28047010; 28047090
20132185	Arsenic; selenium	28048000; 28049000
20132210	Phosphorus oxychloride	28121200
20132220	Phosphorus trichloride	28121300
20132237	Halides and halide-oxides of non-metals (excluding chlorides and chloride oxides of phosphorus)	28121100; 28121500; 28121600; 28121700; 28121990; 28129000
20132270	Phosphorus sulphides, commercial phosphorus trisulphide	28139010
20132280	Sulphides of non-metals (excluding phosphorus sulphides; commercial phosphorus trisulphide)	28131000; 28139090
20132310	Cerium, lanthanum, praseodymium, neodymium and samarium	28053021; 28053029
20132311	Europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium and yttrium	28053031; 28053039
20132312	Scandium metal	28053040
20132319	Alkali or alkaline-earth metals; rare-earth metals; mercury (excluding cerium, lanthanum, praseodymium, neodymium and samarium, europium, gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium and yttrium and scandium metal)	28051100; 28051200; 28051910; 28051990; 28053010; 28053080; 28054010; 28054090
20132413	Hydrogen chloride (hydrochloric acid)	28061000
20132415	Chlorosulphuric acid	28062000
20132434	Sulphuric acid; oleum	28070000
20132453	Diphosphorus pentaoxide	28091000
20132455	Phosphoric acid and polyphosphoric acids	28092000
20132462	Diboron trioxide	28100010
20132465	Oxides of boron and boric acids (excl. diboron trioxide)	28100090

PRODCOM Code	PRODCOM description	CN codes
20132469	Other inorganic acids and other inorganic oxygen compounds of non-metals (excluding hydrogen fluoride)	28111200; 28111910; 28111980
20132473	Hydrogen fluoride (hydrofluoric acid)	28111100
20132475	Silicon dioxide	28112200
20132477	Sulphur dioxide	28112905
20132525	Sodium hydroxide (caustic soda), solid	28151100
20132527	Sodium hydroxide in aqueous solution (soda lye or liquid soda)	28151200
20132530	Potassium hydroxide (caustic potash)	28152000
20132550	Peroxides of sodium or potassium	28153000
20132563	Hydroxide and peroxide of magnesium	28161000
20132565	Oxides, hydroxides and peroxides, of strontium or barium	28164000
20132570	Aluminium hydroxide	28183000
20132580	Hydrazine and hydroxylamine and their inorganic salts	28251000
20133115	Fluorides; fluorosilicates; fluoroaluminates and other complex fluorine salts (Lithium hexafluorophosphate (1-), Lithium difluorophosphate, Lithium hexafluoroarsenate monohydrate, Lithium tetrafluoroborate)	28269080
20133119	Fluorides; fluorosilicates; fluoroaluminates and other complex fluorine salts (excl. sodium hexafluoroaluminate "synthetic cryolite"; hexafluorophosphate (1-), difluorophosphate, hexafluoroarsenate monohydrate and tetrafluoroborate of lithium; inorganic or organic compounds of mercury)	28261200; 28261910; 28261990; 28263000; 28269010
20133131	Magnesium chlorides	28273100
20133132	Nickel chlorides	28273500
20133133	Tin chlorides	28273910
20133134	Cobalt chlorides	28273930
20133135	Lithium chloride	28273985
20133138	Other chlorides (excluding lithium, ammonium, magnesium, nickel, tin and cobalt chloride) n.e.s.	28272000; 28273200; 28273920; 28273985
20133150	Chloride oxides and chloride hydroxides of copper and other metals	28274100; 28274910; 28274990
20133172	Lithium bromide	28275900
20133173	Bromides and bromide oxides (excl. bromides of sodium, of potassium and of lithium)	28275900
20133179	Bromides of sodium or of potassium; iodides and iodide oxides	28275100; 28276000
20133230	Hypochlorites; commercial calcium hypochlorite; chlorites; hypobromites	28281000; 28289000
20133251	Lithium perchlorate	28299010
20133259	Chlorates and perchlorates; bromates and perbromates; iodates and periodates (excluding lithium perchlorate)	28291100; 28291900; 28299010; 28299040; 28299080
20134111	Sulphides of calcium, of antimony or of iron	28309011

PRODCOM Code	PRODCOM description	CN codes
20134120	Sulphides; polysulphides, whether or not chemically defined; dithionites and sulfoxylates (excluding of calcium, antimony and iron)	28301000; 28309085; 28311000; 28319000
20134133	Sulphites	28321000; 28322000
20134135	Thiosulphates	28323000
20134150	Sulphates of aluminium	28332200
20134155	Sulphates of barium	28332700
20134160	Sulphates of magnesium	28332100
20134161	Sulphates of nickel	28332400
20134162	Sulphates of cobalt; of titanium	28332930
20134165	Sulphates (excluding those of aluminium, barium, magnesium, nickel, cobalt, titanium)	28331100; 28331900; 28332500; 28332920; 28332960; 28332980
20134173	Alums	28333000
20134175	Peroxosulphates (persulphates)	28334000
20134212	Nitrates of copper and other (excluding those of potassium, of barium, of beryllium, of cadmium, of cobalt, of nickel, of lead)	28342940; 28342980
20134213	Nitrates of barium; of beryllium; of cadmium; of lead	28342920
20134214	Nitrate of cobalt	28342920
20134215	Nitrate of nickel	28342920
20134220	Phosphinates (hypophosphites) and phosphonates (phosphites)	28351000
20134230	Phosphates of mono- or di-sodium	28352200
20134240	Calcium hydrogenorthophosphate (dicalcium phosphate)	28352500
20134270	Sodium triphosphate (sodium tripolyphosphate)	28353100
20134281	Lithium iron phosphate	28352990
20134289	Phosphates (excluding calcium hydrogenorthophosphate and mono- or disodium phosphate, and lithium iron phosphate); polyphosphates (excluding sodium triphosphate)	28352400; 28352600; 28352910; 28352930; 28352990; 28353900
20134310	Disodium carbonate	28362000
20134320	Sodium hydrogencarbonate (sodium bicarbonate)	28363000
20134340	Calcium carbonate	28365000
20134351	Lithium carbonates with a minimum content of 99,5% of Li ₂ CO ₃ or more	28369100
20134352	Lithium carbonates with a content of less than 99,5% of Li ₂ CO ₃	28369100
20134396	Cobalt carbonates	28369917
20134399	Other carbonates (excluding cobalt carbonates)	28364000; 28366000; 28369200; 28369911; 28369917; 28369990
20135115	Tungstates	28418000
20135120	Manganites, manganates and permanganates; molybdates	28416100; 28416900; 28417000
20135125	Chromates and dichromates; peroxochromates	28413000; 28415000
20135178	Salts of oxometallic and peroxometallic acids: zincates and vanadates	28419030

PRODCOM Code	PRODCOM description	CN codes
20135180	Tantalates	28419085
20135181	Other salts of oxometallic and peroxometallic acids (excluding tantalates)	28419085
20135183	Silver nitrate	28432100
20135185	Colloidal precious metals; compounds and amalgams of precious metals (excluding silver nitrate)	28431010; 28431090; 28432900; 28433000; 28439010; 28439090
20135250	Distilled and conductivity water and water of similar purity	28539010
20135270	Compounds, inorganic or organic, of mercury, chemically defined as mercury (excluding amalgams)	28521000
20135275	Compounds, inorganic or organic, of mercury, not chemically defined as mercury (excluding amalgams)	28529000
20135290	Other inorganic compounds n.e.c.; amalgams (excluding distilled and conductivity water and water of similar purity, liquid air and compressed air, those of precious metals)	28531000; 28539090
20136100	Heavy water (deuterium oxide); isotopes and their compounds (excluding radioactive and the fissile or fertile chemical isotopes)	28451000; 28452000; 28453000; 28454000; 28459010; 28459090
20136220	Cyanides; cyanide oxides and complex cyanides	28371100; 28371900; 28372000
20136230	Borates; peroxoborates (perborates)	28401100; 28401910; 28401990; 28402010; 28402090; 28403000
20136240	Silicates; commercial alkali metal silicates	28391100; 28391900; 28399000
20136270	Double or complex silicates	28421000
20136280	Salts of inorganic acids or peroxyacids (excluding azides and double or complex silicates)	28429010; 28429080
20136300	Hydrogen peroxide	28470000
20136410	Carbides of silicon, whether or not chemically defined	28492000
20136420	Carbides of boron, whether or not chemically defined	28499010
20136430	Carbides of tungsten, whether or not chemically defined	28499030
20136451	Carbides of aluminium, of chromium, of molybdenum, of vanadium, of tantalum; of titanium, whether or not chemically defined	28499050
20136480	Phosphides (excluding ferrophosphorus), whether or not chemically defined; hydrides, nitrides, azides, silicides and borides, whether or not chemically defined, other than compounds which are also carbides of heading 20.13.64.50	28500020; 28500060; 28500090; 28539090
20136490	Other carbides n.e.s.	28491000; 28499090
20136510	Compounds, inorganic or organic, of rare-earth metals or of mixtures of these metals. Cerium compounds	28461000
20136520	Compounds, inorganic or organic, of rare-earth metals or of mixtures of these metals. Compounds of lanthanum, praseodymium, neodymium, samarium	28469040; 28469050
20136550	Compounds, inorganic or organic, of rare-earth metals or of mixtures of these metals. Compounds of europium, gadolinium, terbium, dysprosium,	28469060; 28469070

PRODCOM Code	PRODCOM description	CN codes
	holmium, erbium, thulium, ytterbium, lutetium or yttrium	
20136585	Compounds, inorganic or organic, of rare-earth metals or of mixtures of these metals. Compounds of mixtures of metals	28469090
20136600	Sulphur (excluding crude, sublimed, precipitated and colloidal)	25030090
20136700	Roasted iron pyrites	26012000
20136800	Synthetic or reconstructed precious or semi-precious stones, unworked or simply sawn or roughly shaped	71041000; 71042100; 71042900
20141120	Saturated acyclic hydrocarbons	29011000
20141130	Ethylene	29012100
20141140	Propene (propylene)	29012200
20141150	Butene (butylene) and isomers thereof	29012300
20141160	Buta-1,3-diene and isoprene	29012400
20141190	Unsaturated acyclic hydrocarbons (excluding ethylene, propene, butene, buta-1,3-diene and isoprene)	29012900
20141213	Cyclohexane	29021100
20141215	Cyclanes; cyclenes and cycloterpenes (excluding cyclohexane)	29021900
20141223	Benzene	29022000
20141225	Toluene	29023000
20141243	o-Xylene	29024100
20141245	p-Xylene	29024300
20141247	m-Xylene and mixed xylene isomers	29024200; 29024400
20141250	Styrene	29025000
20141260	Ethylbenzene	29026000
20141270	Cumene	29027000
20141290	Other cyclic hydrocarbons	29029000
20141313	Chloromethane (methyl chloride) and chloroethane (ethyl chloride)	29031100
20141315	Dichloromethane (methylene chloride)	29031200
20141323	Chloroform (trichloromethane)	29031300
20141325	Carbon tetrachloride	29031400
20141353	1,2-Dichloroethane (ethylene dichloride)	29031500
20141357	Saturated chlorinated derivatives of acyclic hydrocarbons, n.e.c.	29031900
20141371	Vinyl chloride (chloroethylene)	29032100
20141374	Trichloroethylene; tetrachloroethylene (perchloroethylene)	29032200; 29032300
20141379	Unsaturated chlorinated derivatives of acyclic hydrocarbons (excluding vinyl chloride, trichloroethylene, tetrachloroethylene)	29032900
20141450	Derivatives of hydrocarbons containing only sulpho groups; their salts and ethyl esters	29041000
20141470	Derivatives of hydrocarbons containing only nitro or only nitroso groups	29042000

PRODCOM Code	PRODCOM description	CN codes
20141490	Derivatives of hydrocarbons (excluding those containing only sulpho groups; their salts and ethyl esters, those containing only nitro or only nitroso groups)	29043100; 29043200; 29043300; 29043400; 29043500; 29043600; 29049100; 29049900
20141910	Fluorinated; brominated or iodinated derivatives of acyclic hydrocarbons	29034100; 29034200; 29034300; 29034400; 29034500; 29034600; 29034700; 29034800; 29034910; 29034930; 29034990; 29035100; 29035900; 29036100; 29036200; 29036911; 29036919; 29036980
20141930	Halogenated derivatives of acyclic hydrocarbons containing ≥ 2 different halogens	29037100; 29037200; 29037300; 29037400; 29037500; 29037610; 29037620; 29037690; 29037760; 29037790; 29037800; 29037930; 29037980
20141950	Halogenated derivatives of cyclanic, cyclenic or cycloterpenic hydrocarbons	29038100; 29038200; 29038300; 29038910; 29038980
20141970	Halogenated derivatives of aromatic hydrocarbons	29039100; 29039200; 29039300; 29039400; 29039910; 29039980
20142100	Industrial fatty alcohols	38237000
20142210	Methanol (methyl alcohol)	29051100
20142220	Propan-1-ol (propyl alcohol) and propan-2-ol (isopropyl alcohol)	29051200
20142230	Butan-1-ol (n-butyl alcohol)	29051300
20142240	Butanols (excluding butan-1-ol (n-butyl alcohol))	29051410; 29051490
20142263	Octanol (octyl alcohol) and isomers thereof	29051620; 29051685
20142265	Lauryl alcohol; cetyl alcohol; stearyl alcohol and other saturated monohydric alcohols (excluding methyl, propyl and isopropyl, n-butyl, other butanols, octyl)	29051700; 29051900
20142270	Unsaturated monohydric alcohols	29052200; 29052910; 29052990
20142310	Ethylene glycol (ethanediol)	29053100
20142320	Propylene glycol (propane-1,2-diol)	29053200
20142333	D-glucitol (sorbitol)	29054411; 29054419; 29054491; 29054499
20142337	Diols and polyhydric alcohols (excluding ethylene glycol and propylene glycol, D-glucitol and tetramethylene glycol having a bio-based carbon content of 100 % by mass)	29053920; 29053928; 29053930; 29053995; 29054100; 29054200; 29054300; 29054900
20142338	Butane-1,4-diol or tetramethylene glycol (1,4-butanediol) having a bio-based carbon content of 100 % by mass	29053926
20142350	Halogenated, sulphonated, nitrated or nitrosated derivatives of acyclic alcohols	29055100; 29055991; 29055998
20142360	Glycerol (including synthetic; excluding crude, waters and lyes)	29054500
20142373	Cyclanic, cyclenic or cycloterpenic alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives	29061100; 29061200; 29061310; 29061390; 29061900
20142375	Aromatic alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives	29062100; 29062900
20142410	Monophenols	29071100; 29071200; 29071300; 29071510; 29071590; 29071910; 29071990

PRODCOM Code	PRODCOM description	CN codes
20142433	4,4-Isopropylidenediphenol (bisphenol A; diphenylolpropane) and its salts	29072300
20142439	Polyphenols (including salts; excluding 4,4 isopropylidenediphenol) and phenol-alcohols	29072100; 29072200; 29072900
20142450	Halogenated, sulphonated, nitrated or nitrosated derivatives of phenols or phenol-alcohols	29081100; 29081900; 29089100; 29089200; 29089900
20143120	Industrial stearic acid	38231100
20143130	Industrial oleic acid	38231200
20143150	Industrial tall oil fatty acids	38231300
20143195	Industrial monocarboxylic fatty acids distilled (excluding stearic, oleic tall oil)	38231910
20143197	Industrial monocarboxylic fatty acids (excluding stearic, oleic, tall oil, distilled)	38231930; 38231990
20143215	Ethyl acetate	29153100
20143219	Esters of acetic acid (excluding ethyl acetate)	29153200; 29153300; 29153600; 29153900
20143220	Mono-, di- or tri-chloroacetic acids; propionic, butanoic and pentanoic acids; their salts and esters	29154000; 29155000; 29156011; 29156019; 29156090
20143235	Palmitic acid, stearic acid, their salts and esters	29157040; 29157050
20143250	Formic acid, its salts and esters	29151100; 29151200; 29151300
20143271	Acetic acid	29152100
20143277	Acetic anhydride	29152400
20143278	Salts of acetic acid	29152900
20143280	Lauric acid and others; salts and esters	29159030; 29159070
20143310	Acrylic acid and its salts and other monocarboxylic acid	29161100; 29161600; 29161910; 29161940; 29161995; 29162000
20143320	Esters of acrylic acid	29161200
20143330	Methacrylic acid and its salts	29161300
20143340	Esters of methacrylic acid	29161400
20143350	Oleic, linoleic or linolenic acids; their salts and esters	29161500
20143363	Benzoic acid; its salts and esters	29163100
20143365	Benzoyl peroxide and benzoyl chloride	29163200
20143367	Phenylacetic acid; its salts and esters	29163400; 29163910
20143370	Aromatic monocarboxylic acids, (anhydrides), halides, peroxides, peroxyacids, derivatives excluding benzoic acid, phenylacetic acids their salts/esters, benzoyl peroxide, benzoyl chloride	29163990
20143381	Oxalic, azelaic, malonic, other, cyclanic, cylenic or cycloterpenic polycarboxylic acids, salts (excluding butanedioic acid having a bio-based carbon content of 100 % by mass)	29171100; 29171310; 29171390; 29171910; 29171980; 29172000
20143382	Ethane-1,2-dicarboxylic acid or butanedioic acid (succinic acid) having a bio-based carbon content of 100 % by mass	29171920
20143385	Adipic acid; its salts and esters	29171200
20143387	Maleic anhydride	29171400
20143410	Dibutyl and dioctyl orthophthalates	29173200; 29173400
20143420	Other esters of orthophthalic acid	29173300; 29173400

PRODCOM Code	PRODCOM description	CN codes
20143430	Phthalic anhydride; terephthalic acid and its salts	29173500; 29173600
20143440	Aromatic polycarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excluding esters of orthophthalic acid, phthalic anhydride, terephthalic acid and its salts)	29173700; 29173920; 29173935; 29173985
20143473	Citric acid and its salts and esters	29181400; 29181500
20143475	Carboxylic acid with alcohol, phenol, aldehyde or ketone functions	29181100; 29181200; 29181300; 29181600; 29181700; 29181800; 29181930; 29181940; 29181998; 29182900; 29183000; 29189100; 29189940; 29189990
20144113	Methylamine; di- or trimethylamine and their salts	29211100
20144119	Other acyclic monoamines and their derivatives; salts thereof	29211200; 29211300; 29211400; 29211940; 29211950; 29211999
20144123	Hexamethylenediamine and its salts; ethylenediamine and its salts	29212100; 29212200
20144129	Other acyclic polyamines and their derivatives; salts thereof	29212900
20144130	Cyclanic, cyclenic or cycloterpenic mono- or polyamines, and their derivatives; salts thereof	29213010; 29213091; 29213099
20144151	Aniline and its salts (excluding derivatives)	29214100
20144153	Aniline derivatives and their salts	29214200
20144159	Other aromatic monoamines and their derivatives; salts thereof	29214300; 29214400; 29214500; 29214600; 29214900
20144170	Aromatic polyamines and their derivatives; salts thereof	29215111; 29215119; 29215190; 29215950; 29215990
20144233	Monoethanolamine and its salts	29221100
20144235	Diethanolamine and its salts	29221200
20144238	Amino-alcohols, their ethers and esters with only 1 oxygen function and their salts (excluding monoethanolamine and its salts, diethanolamine and its salts)	29221400; 29221500; 29221600; 29221700; 29221800; 29221900
20144290	Oxygen-function amino-compounds (excluding amino-alcohols, their esters and ethers and salts thereof, lysine and its salts and esters, glutamic acid its salts and esters)	29222100; 29222900; 29223100; 29223900; 29224300; 29224400; 29224920; 29224985; 29225000
20144310	Ureines and their derivatives; salts thereof	29242100
20144320	Saccharin and its salts	29251100
20144330	Imides and their derivatives, and salts thereof (excluding saccharin and its salts)	29251200; 29251920; 29251995
20144340	Imines and their derivatives; and salts thereof	29252100; 29252900
20144350	Acrylonitrile	29261000
20144360	1-Cyanoguanidine (dicyandiamide)	29262000
20144370	Nitrile-function compounds (excluding acrylonitrile, 1-cyanoguanidine (dicyandiamide))	29263000; 29264000; 29269020; 29269070
20144420	Diazo-, azo- or azoxy-compounds	29270000
20144430	Organic derivatives of hydrazine or of hydroxylamine	29280010; 29280090
20144450	Isocyanates	29291000
20144490	Compounds with other nitrogen function (excluding isocyanates)	29299000

PRODCOM Code	PRODCOM description	CN codes
20145133	Thiocarbamates and dithiocarbamates; thiuram mono-, di- or tetrasulphides; methionine	29302000; 29303000; 29304010; 29304090
20145139	Other organo-sulphur compounds	29301000; 29306000; 29307000; 29308000; 29309013; 29309016; 29309030; 29309040; 29309050; 29309098
20145151	Organo-inorganic compounds (excluding organo-sulphur compounds)	29311000; 29312000; 29314100; 29314200; 29314300; 29314400; 29314500; 29314600; 29314700; 29314800; 29314910; 29314920; 29314930; 29314940; 29314990; 29315100; 29315200; 29315300; 29315400; 29315910; 29315990; 29319000
20145215	Tetrahydrofuran, 2-Furaldehyde, Furfuryl alcohol, Tetrahydrofurfuryl alcohol and Piperonal	29321100; 29321200; 29321300; 29329300
20145225	Heterocyclic compounds with oxygen only hetero-atom(s) (excluding other lactones)	29321400; 29321900; 29329100; 29329200; 29329400; 29329500; 29329600; 29329900
20145230	Heterocyclic compounds with nitrogen only hetero-atom(s); containing an unfused imidazole ring (excluding hydantoin and its derivatives)	29332910; 29332990
20145260	Melamine	29336100
20145280	Compounds containing in the structure an unfused pyridine ring or a quinoline or isoquinoline ring-system, not further fused; lactames; other heterocyclic compounds with nitrogen hetero-atom(s) only (excluding compounds containing in the structure an unfused pyrazole ring, an unfused imidazole ring, a pyrimidine ring, a piperazine ring or an unfused triazine ring)	29333100; 29333200; 29333300; 29333400; 29333500; 29333600; 29333700; 29333910; 29333920; 29333925; 29333935; 29333940; 29333945; 29333950; 29333955; 29333999; 29334100; 29334910; 29334930; 29334990; 29337100; 29337200; 29337900; 29339110; 29339190; 29339200; 29339920; 29339950; 29339980
20145290	Nucleic acids and other heterocyclic compounds - thiazole, benzothiazole, other cycles	29341000; 29342020; 29342080; 29349100; 29349200; 29349960; 29349990; 29398000
20145350	Phosphoric esters; and their salts (including lactophosphates; their halogenated; sulphonated; nitrated or nitrosated derivatives)	29191000; 29199000
20145380	Esters of other inorganic acids of non-metals (excluding esters of hydrogen halides) and their salts; their halogenated, sulphonated, nitrated or nitrosated derivatives	29201100; 29201900; 29202100; 29202200; 29202300; 29202400; 29202900; 29203000; 29209010; 29209070
20146111	Methanal (formaldehyde)	29121100
20146113	Ethanal (acetaldehyde)	29121200
20146115	Butanal (butyraldehyde; normal isomer)	29121900
20146119	Acyclic aldehydes, without other oxygen function (excluding methanal (formaldehyde), ethanal (acetaldehyde), butanal (butyraldehyde; normal isomer))	29121900
20146120	Cyclic aldehydes; without other oxygen function	29122100; 29122900
20146135	Aldehyde-alcohols, Aldehyde-ethers, aldehyde-phenols and aldehydes with other oxygen function	29124100; 29124200; 29124900
20146150	Cyclic polymers of aldehydes	29125000
20146160	Paraformaldehyde	29126000

PRODCOM Code	PRODCOM description	CN codes
20146170	Halogenated, sulphonated, nitrated or nitrosated derivatives of cyclic polymers of aldehydes or paraformaldehyde	29130000
20146211	Acetone	29141100
20146213	Butanone (methyl ethyl ketone)	29141200
20146215	4-Methylpentan-2-one (methyl isobutyl ketone)	29141300
20146219	Acyclic ketones; without other oxygen function (excluding acetone, butanone (methyl ethyl ketone), 4-methylpentan-2-one (methyl isobutyl ketone))	29141910; 29141990
20146231	Camphor; aromatic ketones without other oxygen function; ketone-alcohols; ketone-aldehydes; ketone-phenols and ketones with other oxygen function	29142900; 29143100; 29143900; 29144010; 29144090; 29145000
20146233	Cyclohexanone and methylcyclohexanones	29142200
20146235	Ionones and methylionones	29142300
20146239	Cyclanic, cyclenic or cycloterpenic ketones without other oxygen function (excluding camphor, cyclohexanone and methylcyclohexanones, ionones and methylionones)	29142900
20146260	Quinones	29146100; 29146200; 29146910; 29146980
20146270	Halogenated; sulphonated; nitrated or nitrosated derivatives of ketones and quinones	29147100; 29147900
20146310	Acyclic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives	29091100; 29091910; 29091990
20146323	Cyclanic, cyclenic or cycloterpenic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives	29092000
20146325	Aromatic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives	29093010; 29093031; 29093035; 29093038; 29093090
20146333	2,2-Oxydiethanol (diethylene glycol; digol)	29094100
20146339	Ether-alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives (excluding 2,2-Oxydiethanol)	29094300; 29094400; 29094911; 29094980
20146350	Ether-phenols; ether-alcohol-phenols and their halogenated, sulphonated, nitrated or nitrosated derivatives	29095000
20146361	Alcohol peroxides, ether peroxides, acetal and hemiacetal peroxides, ketone peroxides and their halogenated, sulphonated, nitrated or nitrosated derivatives	29096010; 29096090
20146373	Oxirane (ethylene oxide)	29101000
20146375	Methyloxirane (propylene oxide)	29102000
20146379	Epoxides, epoxyalcohols, -phenols, epoxyethers, with a 3-membered ring and their halogenated, sulphonated, nitrated/nitrosated derivatives excluding oxirane, methyloxirane (propylene oxide)	29103000; 29104000; 29105000; 29109000
20146381	Acetals and hemiacetals and their halogenated; sulphonated; nitrated or nitrosated derivatives	29110000
20146430	Other organic compounds, n.e.c.	29420000
20146450	Rennet and concentrates thereof	35071000

PRODCOM Code	PRODCOM description	CN codes
20146470	Enzymes; prepared enzymes (not elsewhere specified or included) (excluding rennet and concentrates)	35079030; 35079090
20147120	Activated natural mineral products; animal black	38029000
20147130	Tall oil; whether or not refined	38030010; 38030090
20147140	Gum, wood or sulphate turpentine oils, pine oil and other alike	38051010; 38051030; 38051090; 38059010; 38059090
20147150	Rosin and resin acids; and derivatives; rosin spirit and oils; run gums	38061000; 38062000; 38063000; 38069000
20147170	Wood tar; wood tar oils; wood creosote; wood naphtha; vegetable pitch; brewers' pitch and similar preparations based on rosin, resin acids or on vegetable pitch	38070010; 38070090
20147200	Wood charcoal whether or not agglomerated (including shell or nut charcoal)	44021000; 44022000; 44029000
20147320	Benzol (benzene), toluol (toluene) and xylol (xylenes)	27071000; 27072000; 27073000
20147340	Naphthalene and other aromatic hydrocarbon mixtures (excluding benzole, toluole, xylol)	27074000; 27075000
20147360	Phenols	27079980
20147390	Other oils and oil products, n.e.c.	27079100; 27079911; 27079919; 27079920; 27079950; 27079991; 27079999
20147400	Undenatured ethyl alcohol of an alcoholic strength by volume $\geq 80\%$ (important: excluding alcohol duty)	22071000
20147500	Denatured ethyl alcohol and other denatured spirits; of any strength	22072000
20148000	Residual lyes from the manufacture of wood pulp, excluding tall oil	38040000
20152030	Ammonium chloride	28271000
20152080	Nitrites	28341000
20154100	Superphosphates (excluding potassic, in tablets or similar forms or in packages of a weight of ≤ 10 kg)	31031100; 31031900
20154900	Mineral or chemical fertilisers, phosphatic n.e.c.	31039000
20155100	Potassium chloride (excluding in tablets or similar forms or in packages of a weight of ≤ 10 kg)	31042010; 31042050; 31042090
20155200	Potassium sulphate (excluding in tablets or similar forms or in packages of a weight of ≤ 10 kg)	31043000
20155900	Mineral or chemical fertilisers, potassic, n.e.c.	31049000
20157500	Mineral or chemical fertilisers containing the two fertilising elements phosphorus and potassium	31056000
20158000	Animal or vegetable fertilisers	31010000
20161035	Linear polyethylene having a specific gravity $< 0,94$, in primary forms	39011010
20161039	Polyethylene having a specific gravity $< 0,94$, in primary forms (excluding linear)	39011090
20161050	Polyethylene having a specific gravity of $\geq 0,94$, in primary forms	39012010; 39012090
20161070	Ethylene-vinyl acetate copolymers, in primary forms	39013000
20161090	Polymers of ethylene, in primary forms (excluding polyethylene, ethylene-vinyl acetate copolymers)	39014000; 39019030; 39019080
20162035	Expansible polystyrene, in primary forms	39031100

PRODCOM Code	PRODCOM description	CN codes
20162039	Polystyrene, in primary forms (excluding expansible polystyrene)	39031900
20162050	Styrene-acrylonitrile (SAN) copolymers, in primary forms	39032000
20162070	Acrylonitrile-butadiene-styrene (ABS) copolymers, in primary forms	39033000
20162090	Polymers of styrene, in primary forms (excluding polystyrene, styrene-acrylonitrile (SAN) copolymers, acrylonitrile-butadiene-styrene (ABS) copolymers)	39039010; 39039020; 39039090
20163010	Polyvinyl chloride, not mixed with any other substances, in primary forms	39041000
20163023	Non-plasticised polyvinyl chloride mixed with any other substance, in primary forms	39042100
20163025	Plasticised polyvinyl chloride mixed with any other substance, in primary forms	39042200
20163040	Vinyl chloride-vinyl acetate copolymers and other vinyl chloride copolymers, in primary forms	39043000; 39044000
20163060	Fluoropolymers	39046100; 39046910; 39046920; 39046980
20163090	Polymers of halogenated olefins, in primary forms, n.e.c.	39045010; 39045090; 39049000
20164013	Polyacetals, in primary forms	39071000
20164015	Polyethylene glycols and other polyether alcohols, in primary forms	39072100; 39072911; 39072920
20164020	Polyethers, in primary forms (excluding polyacetals, polyether alcohols)	39072991; 39072999
20164030	Epoxide resins, in primary forms	39073000
20164040	Polycarbonates, in primary forms	39074000
20164050	Alkyd resins, in primary forms	39075000
20164062	Polyethylene terephthalate in primary forms having a viscosity number of ≥ 78 ml/g	39076100
20164064	Other polyethylene terephthalate in primary forms	39076900
20164070	Unsaturated liquid polyesters, in primary forms (excluding polyacetals, polyethers, epoxide resins, polycarbonates, alkyd resins, polyethylene terephthalate)	39079110
20164080	Unsaturated polyesters, in primary forms (excluding liquid polyesters, polyacetals, polyethers, epoxide resins, polycarbonates, alkyd resins, polyethylene terephthalate)	39079190
20164090	Polyesters, in primary forms (excluding polyacetals, polyethers, epoxide resins, polycarbonates, alkyd resins, polyethylene terephthalate, other unsaturated polyesters)	39077000; 39079905; 39079910; 39079980
20165130	Polypropylene, in primary forms	39021000
20165150	Polymers of propylene or of other olefins, in primary forms (excluding polypropylene)	39022000; 39023000; 39029010; 39029020; 39029090
20165230	Polymers of vinyl acetate, in aqueous dispersion, in primary forms	39051200; 39052100
20165250	Polymers of vinyl acetate, in primary forms (excluding in aqueous dispersion)	39051900; 39052900
20165270	Polymers of vinyl esters or other vinyl polymers, in primary forms (excluding vinyl acetate)	39053000; 39059100; 39059910; 39059990
20165350	Polymethyl methacrylate, in primary forms	39061000

PRODCOM Code	PRODCOM description	CN codes
20165390	Acrylic polymers, in primary forms (excluding polymethyl methacrylate)	39069010; 39069020; 39069030; 39069040; 39069050; 39069060; 39069090
20165450	Polyamide -6, -11, -12, -6,6, -6,9, -6,10 or -6,12, in primary forms	39081000
20165490	Polyamides, in primary forms (excluding polyamide -6, -11, -12, -6,6, -6,9, -6,10 or -6,12)	39089000
20165550	Urea resins and thiourea resins, in primary forms	39091000
20165570	Melamine resins, in primary forms	39092000
20165630	Amino resins, in primary forms (excluding urea and thiourea resins, melamine resins)	39093100; 39093900
20165650	Phenolic resins, in primary forms	39094000
20165670	Polyurethanes, in primary forms	39095010; 39095090
20165700	Silicones, in primary forms	39100000
20165920	Petroleum resins, coumarone-indene resins, polyterpenes, polysulphides, polysulphones, etc., n.e.c., in primary forms	39111000; 39112000; 39119011; 39119013; 39119019; 39119092; 39119099
20165945	Sodium carboxymethyl cellulose	39123100
20165950	Cellulose and its chemical derivatives, in primary forms, excluding sodium carboxymethyl cellulose and n.e.c.	39121100; 39121200; 39122011; 39122019; 39122090; 39123920; 39123985; 39129010; 39129090
20165955	Natural polymers and modified natural polymers, not elsewhere specified or included, in primary forms .	39131000
	Alginic acid, its salts and esters	
20165965	Natural polymers and modified natural polymers, e.g. hardened proteins, chemical derivatives of natural rubber, n.e.s., in primary forms (excl. alginic acid and its salts and esters)	39139000
20165970	Ion-exchangers based on synthetic or natural polymers, in primary forms	39140000
20171050	Synthetic latex rubber	40021100; 40024100; 40025100; 40029100
20171090	Synthetic rubber (excluding latex)	40021910; 40021920; 40021930; 40021990; 40022000; 40023100; 40023900; 40024900; 40025900; 40026000; 40027000; 40028000; 40029910; 40029990
20302150	Vitrifiable enamels and glazes, engobes (slips) and similar preparations for ceramics, enamelling or glass	32072010; 32072090
20302170	Liquid lustres and similar preparations; glass frit and other glass in powder; granules or flakes	32073000; 32074040; 32074085
20601110	Aramids staple, not carded, combed or otherwise processed for spinning	55031100
20601120	Other polyamide tow and staple, not carded, combed or otherwise processed for spinning	55011100; 55011900; 55031900
20601130	Polyester tow and staple, not carded, combed or otherwise processed for spinning	55012000; 55032000
20601140	Acrylic tow and staple, not carded, combed or otherwise processed for spinning	55013000; 55033000
20601150	Polypropylene synthetic tow and staple not carded, combed or otherwise processed for spinning	55014000; 55034000
20601190	Other synthetic tow and staple not carded, combed or otherwise processed for spinning	55019000; 55039000
20601220	High-tenacity filament yarn of aramids (excluding sewing thread and yarn put up for retail sale)	54021100

PRODCOM Code	PRODCOM description	CN codes
20601240	High-tenacity filament yarn of nylon or other polyamides (excluding sewing thread, yarn put up for retail sale and high-tenacity filament yarn of aramids)	54021900
20601260	High-tenacity filament yarn of polyesters (excluding that put up for retail sale)	54022000
20601310	Polyamide textile filament yarn, n.p.r.s. (excluding sewing thread)	54023100; 54024500; 54025100
20601320	Polyamide carpet filament yarn, n.p.r.s. (excluding sewing thread)	54023200
20601330	Polyester textile filament yarn, n.p.r.s. (excluding sewing thread)	54023300; 54024600; 54024700; 54025200
20601340	Polypropylene filament yarn, n.p.r.s. (excluding sewing thread)	54023400; 54024800; 54025300
20601350	Elastomeric filament yarn, n.p.r.s.	54024400
20601390	Other synthetic filament yarn, n.p.r.s. (excluding sewing thread)	54023900; 54024900; 54025900
20601420	Polypropylene monofilament of ≥ 67 decitex and with a cross-sectional dimension of ≤ 1 mm (excluding elastomers)	54041200
20601440	Synthetic monofilament of ≥ 67 decitex and with a cross-sectional dimension of ≤ 1 mm (excluding polypropylene monofilament); strip and the like (for example, artificial straw), of synthetic textile materials, of an apparent width ≤ 5 mm	54041100; 54041900; 54049010; 54049090
20602140	Artificial filament tow, of acetate	55021000
20602150	Artificial filament tow and staple fibres (not carded, combed or otherwise processed for spinning, excluding the ones of acetate)	55029000; 55041000; 55049000
20602200	High tenacity filament yarn of viscose rayon, n.p.r.s. (excluding sewing thread)	54031000
20602320	Yarn of viscose rayon filament, including monofilament of < 67 decitex, single, n.p.r.s. (excluding sewing thread and high-tenacity yarn)	54033100; 54033200
20602340	Filament yarn of cellulose acetate, including monofilament of < 67 decitex, single, n.p.r.s. (excluding sewing thread and high-tenacity yarn)	54033300
20602390	Other artificial filament yarn, including artificial monofilament of < 67 decitex, single, n.p.r.s. (excluding sewing thread)	54033900
20602400	Artificial monofilament of ≥ 67 decitex and of which the cross-sectional dimension ≤ 1 mm; strip and the like of artificial textile materials of an apparent width ≤ 5 mm	54050000
21101030	Salicylic acid and its salts	29182100
21101050	O-acetylsalicylic acid; its salts and esters	29182200
21101070	Esters of salicylic acid and their salts (excluding of O-acetylsalicylic acid)	29182300
21102010	Lysine and its esters, and salts thereof	29224100
21102020	Glutamic acid and its salts	29224200
21102040	Quaternary ammonium salts and hydroxides; lecithins and other phosphoaminolipids, whether or not chemically defined	29231000; 29232000; 29233000; 29234000; 29239000
21102060	Acyclic amides and their derivatives, and salts thereof (including acyclic carbamates)	29241100; 29241200; 29241900

PRODCOM Code	PRODCOM description	CN codes
21102070	Cyclic amides and their derivatives, and salts thereof (including cyclic carbamates) (excluding ureines and their derivatives, and salts thereof)	29242300; 29242400; 29242500; 29242910; 29242970
21103117	Phenolphthalein; 1-Hydroxy-4-[1-(4-hydroxy-3-methoxycarbonyl-1-naphthyl)-3-oxo-1H,3H-benzo[de]isochromen-1-yl]-6-octadecyloxy-2-naphthoic acid; 3'-Chloro-6'-cyclohexylaminospiro[isobenzofuran-1 (3H),9'-xanthen]-3-one; 6'-(N-Ethyl-p-toluidino)-2'-methylspiro[isobenzofuran-1 (3H),9'-xanthen]-3-one; Methyl-6-docosyloxy-1-hydroxy-4-[1-(4-hydroxy-3-methyl-1-phenanthryl)-3-oxo-1H,3H-naphtho[1,8-cd]pyran-1-yl]naphthalene-2-carboxylate, gamma-Butyrolactone	29322010; 29322020
21103119	Lactones (excluding phenolphthalein; 1-Hydroxy-4-[1-(4-hydroxy-3-methoxycarbonyl-1-naphthyl)-3-oxo-1H,3H-benzo[de]isochromen-1-yl]-6-octadecyloxy-2-naphthoic acid; 3'-Chloro-6'-cyclohexylaminospiro[isobenzofuran-1 (3H),9'-xanthen]-3-one; 6'-(N-Ethyl-p-toluidino)-2'-methylspiro[isobenzofuran-1 (3H),9'-xanthen]-3-one; Methyl-6-docosyloxy-1-hydroxy-4-[1-(4-hydroxy-3-methyl-1-phenanthryl)-3-oxo-1H,3H-naphtho[1,8-cd]pyran-1-yl]naphthalene-2-carboxyla, gamma-Butyrolactone, and inorganic or organic compounds of mercury)	29322090
21103130	Compounds containing an unfused pyrazole ring (whether or not hydrogenated) in the structure	29331110; 29331190; 29331910; 29331990
21103140	Hydantoin and its derivatives	29332100
21103155	Malonylurea (barbituric acid) and its derivatives, and salts thereof	29335200; 29335310; 29335390; 29335400
21103159	Compounds containing a pyrimidine ring (whether or not hydrogenated) or piperazine ring in the structure (excluding malonylurea (barbituric acid) and its derivatives)	29335500; 29335910; 29335920; 29335995
21103170	Compounds containing an unfused triazine ring (whether or not hydrogenated) in the structure (excluding melamine)	29336910; 29336940; 29336980
21103180	Compounds containing a phenothiazine ring-system (whether or not hydrogenated); not further fused	29343010; 29343090
21103200	Sulphonamides	29351000; 29352000; 29353000; 29354000; 29355000; 29359030; 29359090
21104000	Sugars, pure (excluding glucose, etc.); sugar ethers and salts, etc.	29400000
21105100	Provitamins and vitamins, natural or reproduced by synthesis (including natural concentrates), derivatives thereof used primarily as vitamins, and intermixtures of the foregoing, whether or not in any solvent	29362100; 29362200; 29362300; 29362400; 29362500; 29362600; 29362700; 29362800; 29362900; 29369000
21105200	Hormones, prostaglandins, thromboxanes and leukotrienes, natural or reproduced by synthesis; derivatives and structural analogues thereof, including chain modified polypeptides, used primarily as hormones	29371100; 29371200; 29371900; 29372100; 29372200; 29372300; 29372900; 29375000; 29379000
21105300	Glycosides and vegetable alkaloids, natural or reproduced by synthesis, and their salts, ethers, esters and other derivatives	29381000; 29389010; 29389030; 29389090; 29391100; 29391900; 29392000; 29393000; 29394100;

PRODCOM Code	PRODCOM description	CN codes
		29394200; 29394300; 29394400; 29394500; 29394900; 29395100; 29395900; 29396100; 29396200; 29396300; 29396900; 29397200; 29397910; 29397990
21105400	Antibiotics	29411000; 29412030; 29412080; 29413000; 29414000; 29415000; 29419000
21106020	Extracts of glands or other organs or of their secretions (for organo-therapeutic uses)	30012010; 30012090
21106040	Glands and other organs or substances for therapeutic or prophylactic use, n.e.c. (excluding blood and extracts of glands or other organs)	30019020; 30019091; 30019098
21106051	Cell therapy products	30025100
21106052	Other cell cultures, whether or not modified	30025900
21106053	Human blood; animal blood prepared for therapeutic, prophylactic or diagnostic uses; other products n.e.c. in the HS heading 3002	30029010; 30029030
21106060	Toxins, cultures of micro-organisms and similar products (excl. yeasts and vaccines)	30024900
23111110	Non-wired sheets, of cast or rolled glass, whether or not with absorbent, reflecting or non-reflecting layer, but not otherwise worked	70031210; 70031291; 70031299; 70031910; 70031990
23111130	Wired sheets or profiles, of cast or rolled glass, whether or not with absorbent, reflecting or non-reflecting layer, but not otherwise worked	70032000; 70033000
23111150	Sheets, of drawn glass or blown glass, whether or not having an absorbent, reflecting or non-reflecting layer, but not otherwise worked	70042010; 70042091; 70042099; 70049010; 70049080
23111212	Non-wired sheets, of float, surface ground or polished glass, having a non-reflecting layer	70051005
23111214	Non-wired sheets, of float, surface ground or polished glass, having an absorbent or reflective layer, of a thickness ≤ 3,5 mm	70051025
23111217	Non-wired sheets, of float, surface ground or polished glass, having an absorbent or reflecting layer, not otherwise worked, of a thickness > 3,5 mm	70051030; 70051080
23111230	Non-wired sheets, of float, surface ground or polished glass, coloured throughout the mass, opacified, flashed or merely surface ground	70052125; 70052130; 70052180
23111290	Other sheets of float/ground/polished glass, n.e.c.	70052925; 70052935; 70052980; 70053000
23131110	Glass preserving jars, stoppers, lids and other closures (including stoppers and closures of any material presented with the containers for which they are intended)	70102000; 70109010
23131120	Containers made from tubing of glass (excluding preserving jars)	70109021
23131130	Glass containers of a nominal capacity ≥ 2,5 litres (excluding preserving jars)	70109031
23131140	Bottles of colourless glass of a nominal capacity < 2,5 litres, for beverages and foodstuffs (excluding bottles covered with leather or composition leather, infant's feeding bottles)	70109041; 70109043; 70109045; 70109047
23131150	Bottles of coloured glass of a nominal capacity < 2,5 litres, for beverages and foodstuffs (excluding bottles)	70109051; 70109053; 70109055; 70109057

PRODCOM Code	PRODCOM description	CN codes
	covered with leather or composition leather, infant's feeding bottles)	
23131160	Glass containers for beverages and foodstuffs of a nominal capacity < 2,5 litres (excluding bottles, flasks covered with leather or composition leather, domestic glassware, vacuum flasks and vessels)	70109061; 70109067
23131170	Glass containers for pharmaceutical products of a nominal capacity < 2,5 litres	70109071; 70109079
23131180	Glass containers of a nominal capacity < 2,5 litres for the conveyance or packing of goods (excluding for beverages and foodstuffs, for pharmaceutical products, containers made from glass tubing)	70109091; 70109099
23131220	Drinking glasses (including stemware drinking glasses), other than of glass ceramics, of lead crystal, gathered by hand	70132210; 70133311; 70133319
23131240	Drinking glasses (including stemware drinking glasses), other than of glass ceramics, of lead crystal, gathered mechanically	70132290; 70133391; 70133399
23131260	Drinking glasses (excluding stemware drinking glasses and products of glass ceramics or lead crystal), of toughened glass	70133710
23131290	Other drinking glasses	70132810; 70132890; 70133751; 70133759; 70133791; 70133799
23131310	Table or kitchen glassware of lead crystal gathered by hand (excluding of glass-ceramics, of toughened glass, drinking glasses)	70134110; 70139110
23131330	Table or kitchen glassware of lead crystal gathered mechanically (excluding of glass ceramics, of toughened glass, drinking glasses)	70134190; 70139190
23131350	Table/kitchen glassware with linear coefficient of expansion $\leq 5 \times 10^{-6}/K$, temperature range of 0 °C to 300 °C excluding of glass-ceramics, lead crystal/toughened glass, drinking glasses	70134200
23131360	Glass-ceramic table, kitchen, toilet, office, indoor decoration or similar purpose glassware	70131000
23131390	Table/kitchen glassware (excluding drinking), toughened glass	70134910; 70134991; 70134999; 70139900
23131400	Glass inners for vacuum flasks or for other vacuum vessels (including unfinished and finished)	70200007; 70200008
23141110	Glass fibre threads cut into lengths of at least 3 mm but ≤ 50 mm (chopped strands)	70191100
23141130	Glass fibre filaments (including rovings)	70191200
23141151	Yarn and slivers of glass fibres (excl. in chopped strands of a length of ≤ 50 mm, and rovings) and glass fibre cakes "textile cakes" and chopped glass fibre strands of a length > 50 mm	70191300; 70191900
23141220	Mats of irregularly laminated glass fibres (excl. glass wool)	70191400; 70191500
23141240	Thin sheets "voiles" of irregularly laminated glass fibres (excl. glass wool)	70197100
23141252	Closed fabrics of rovings of glass fibres, mechanically bonded (excl. woven)	70196200
23141253	Stitched or needled fabrics of glass fibres (excl. closed fabrics of rovings)	70196910
23141254	Chemically bonded glass fibres and articles thereof (for example, yarn, rovings, woven fabrics)	70197100; 70197200; 70197300
23141296	Glass fibres and articles thereof (excl. staple fibres, rovings, yarn, chopped strands, woven fabrics, also	70199000

PRODCOM Code	PRODCOM description	CN codes
	narrow fabrics, thin sheets "voiles", webs, mats, mattresses and boards and similar nonwoven products, stone or slag wool and articles thereof, electrical insulators or parts thereof, optical fibres, fibre bundles or cable, brushes of glass fibres, and dolls' wigs, glass wool and articles of glass wool)	
23141298	Glass wool and articles of glass wool	70198010; 70198090
23191110	Glass in the mass (excluding glass in the form of powder, granules or flakes)	70010091; 70010099
23191130	Unworked glass in balls or rods (excluding glass balls as toys, glass balls which have been ground after shaping, used as stoppers for bottles, glass microspheres <= 1 mm in diameter)	70021000; 70022010; 70022090
23191150	Unworked glass tubes (including tubes which have had fluorescent material added to them in the mass) (excluding tubes coated inside with fluorescent material)	70023100; 70023200; 70023900
23191200	Paving blocks, bricks, tiles and other articles of pressed or moulded glass; leaded lights and the like; multicellular or foam glass in blocks, plates or similar forms	70169010; 70169040; 70169070
23192100	Open glass envelopes for electric lamps, cathode-ray tubes or the like	70111000; 70112000; 70119000
23192200	Clock or watch glasses, glasses for spectacles, not optically worked	70151000; 70159000
23192330	Laboratory, hygienic or pharmaceutical glassware whether or not graduated	70171000; 70172000; 70179000
23192350	Glass ampoules used for the conveyance or packing of goods	70101000
23192400	Glass parts for lamps and lighting fittings, etc.	94059110; 94059190
23192500	Glass electrical insulators (excluding insulating fittings (other than insulators) for electrical machinery, appliances or equipment)	85461000
23192640	Signalling glassware and optical elements of glass, not optically worked; glass cubes and other glass smallware, for mosaic or similar decorative purposes (excluding finished panels and other decorative motifs made from mosaic cubes)	70140000; 70161000
23192670	Glass smallware (including beads, imitation pearls/stones, etc.)	70181011; 70181019; 70181030; 70181051; 70181059; 70181090; 70182000; 70189010; 70189090
23192690	Other articles of glass, n.e.c.	70200005; 70200010; 70200030; 70200080
23201100	Ceramic goods of siliceous fossil meals or earths including bricks, blocks, slabs, panels, tiles, hollow bricks, cylinder shells and pipes excluding filter plates containing kieselguhr and quartz	69010000
23201210	Refractory ceramic constructional goods containing >50 % of MgO, CaO or Cr ₂ O ₃ including bricks, blocks and tiles excluding goods of siliceous fossil meals or earths, tubing and piping	69021000
23201233	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods containing, by weight, >= 93% silica (excl. those of siliceous fossil meals or similar siliceous earths)	69022010
23201235	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods containing, by	69022091

PRODCOM Code	PRODCOM description	CN codes
	weight, > 7 % but < 45 % alumina, but > 50 % by weight combined with silica	
23201237	Refractory bricks, blocks..., weight > 50 % Al ₂ O ₃ and/or SiO ₂ : others	69022099
23201290	Refractory bricks, blocks, tiles, etc., n.e.c.	69029000
23201300	Refractory cements, mortars, concretes and similar compositions (including refractory plastics, ramming mixes, gunning mixes) (excluding carbonaceous pastes)	38160090
23201411	Articles of stone or of other mineral substances, n.e.c. containing magnesite, magnesia in the form of periclase, dolomite including in the form of dolime, or chromite	68159100
23201430	Refractory ceramic goods, n.e.c., by weight > 25 % graphite or other forms of carbon	69031000; 69039010
23201455	Refractory ceramic goods, n.e.c., alumina or silica or mixture > 50 % : alumina < 45 %	69032010
23201459	Refractory ceramic goods, n.e.c., alumina or silica or mixture > 50 % : alumina >= 45 %	69032090
23201490	Refractory ceramic goods, n.e.c.	69039090
23311000	Ceramic tiles and flags	69072100; 69072200; 69072300; 69073000; 69074000
23321110	Non-refractory clay building bricks (excluding of siliceous fossil meals or earths)	69041000
23321130	Non-refractory clay flooring blocks, support or filler tiles and the like (excluding of siliceous fossil meals or earths)	69049000
23321250	Non-refractory clay roofing tiles	69051000
23321270	Non-refractory clay constructional products (including chimneypots, cowls, chimney liners and flue-blocks, architectural ornaments, ventilator grills, clay-lath; excluding pipes, guttering and the like)	69059000
23321300	Ceramic pipes, conduits, guttering and pipe fittings: drain pipes and guttering with fittings	69060000
23411130	Porcelain or china tableware and kitchenware (excluding electro-thermic apparatus, coffee or spice mills with metal working parts)	69111000
23411150	Household and toilet articles, n.e.c., of porcelain or china	69119000
23411210	Ceramic tableware, other household articles : common pottery	69120021; 69120081
23411230	Ceramic tableware, other household articles : stoneware	69120023; 69120083
23411250	Ceramic tableware, other household articles : earthenware or fine pottery	69120025; 69120085
23411290	Ceramic tableware, other household articles : others	69120029; 69120089
23411330	Statuettes and other ornamental articles, of porcelain or china	69131000
23411350	Ceramic statuettes and other ornamental articles	69139010; 69139093; 69139098
23421030	Ceramic sinks, etc. and other sanitary fixtures, of porcelain or china	69101000
23421050	Ceramic sinks, wash basins, baths... and other sanitary fixtures, n.e.c.	69109000
23521033	Quicklime	25221000
23521035	Slaked lime	25222000

PRODCOM Code	PRODCOM description	CN codes
23521050	Hydraulic lime	25223000
23522000	Plasters consisting of calcined gypsum or calcium sulphate (including for use in building, for use in dressing woven fabrics or surfacing paper, for use in dentistry)	25202000
23523030	Calcined and sintered dolomite, crude, roughly trimmed or merely cut into rectangular or square blocks or slabs	25182000
23523051	Dolomite ramming mix	38160010
23991100	Fabricated asbestos fibres; mixtures with a basis of asbestos and magnesium carbonate; articles of such mixtures, or of asbestos; friction material for brakes, clutches and the like, not mounted	68128010; 68128090; 68129100; 68129910; 68129990; 68132000; 68138100; 68138900
23991255	Articles of asphalt or of similar materials, e.g. petroleum bitumen or coal tar pitch, in rolls	68071000
23991290	Products based on bitumen (excluding in rolls)	68079000
23991310	Bituminous mixtures based on natural and artificial aggregate and bitumen or natural asphalt as a binder	27150000
23991320	Pre-coated aggregates	25173000
23991400	Artificial graphite, colloidal, semi-colloidal graphite, and preparations	38011000; 38012010; 38012090; 38013000; 38019000
23991500	Artificial corundum (excluding mechanical mixtures)	28181011; 28181019; 28181091; 28181099
23991910	Slag wool, rock wool and similar mineral wools and mixtures thereof, in bulk, sheets or rolls	68061000
23991920	Exfoliated vermiculite, expanded clays, foamed slag and similar expanded mineral materials and mixtures thereof	68062010; 68062090
23991930	Mixtures and articles of heat/sound-insulating materials n.e.c.	68069000
23991940	Calcined kaolin	25070020
23991950	Worked mica and articles of mica	68141000; 68149000
23991970	Non-electrical articles of graphite or other carbon	68151100; 68151200; 68151300; 68151900
23991980	Articles of peat (including sheets, cylinder shells and plant pots) (excluding textile articles of peat fibre)	68152000
23991991	Articles of stone or other mineral substances, n.e.c.	68159900
24101235	Ferro-silicon, containing by weight > 55% of silicon	72022100
24101236	Ferro-silicon, containing by weight ≤ 55% silicon and ≥ 4% but ≤ 10% of magnesium	72022910
24101239	Other ferro-silicon, containing by weight ≤ 55% silicon (excl. that containing by weight ≥ 4% but ≤ 10% of magnesium)	72022990
24101245	Ferro-silico-manganese	72023000
24101250	Ferro-tungsten and ferro-silico-tungsten	72028000
24101255	Ferro-titanium and ferro-silico-titanium	72029100
24101265	Ferro-vanadium	72029200
24101270	Ferro-niobium	72029300
24101275	Ferro-molybdenum	72027000
24101280	Ferro-phosphorus	72029910

PRODCOM Code	PRODCOM description	CN codes
24101285	Ferro-silico-magnesium	72029930
24101295	Other ferro alloys nowhere else specified or included	72025000; 72029980
24101420	Remelting scrap ingots of iron or steel (excluding products whose chemical composition conforms to the definitions of pig iron, spiegeleisen, or ferro-alloys)	72045000
24421200	Aluminium oxide (excluding artificial corundum)	28182000
24431130	Refined unwrought lead (excluding lead powders or flakes)	78011000
24431150	Unwrought lead containing antimony (excluding lead powders or flakes)	78019100
24431190	Unwrought lead (excluding lead powders or flakes, unwrought lead containing antimony, refined)	78019910; 78019990
24431230	Unwrought non-alloy zinc (excluding zinc dust, powders and flakes)	79011100; 79011210; 79011230; 79011290
24431250	Unwrought zinc alloys (excluding zinc dust, powders and flakes)	79012000
24431330	Unwrought non-alloy tin (excluding tin powders and flakes)	80011000
24431350	Unwrought tin alloys (excluding tin powders and flakes)	80012000
24432100	Lead plates, sheets, strip and foil; lead powders and flakes (excluding lead powders or flakes prepared as colours; paints or the like, insulated electric strip)	78041100; 78041900; 78042000
24432200	Zinc dust, powders and flakes (excluding zinc dust powders or flakes prepared as colours, paints or the like, zinc pellets)	79031000; 79039000
24432300	Zinc bars, rods, profiles, wire, plates, sheets, strip and foil	79040000; 79050000
24432400	Tin bars, rods, profiles and wires	80030000
24441100	Copper mattes; cement copper (precipitated copper) (excluding copper powder)	74010000
24441200	Unrefined copper, copper anodes for electrolytic refining (including blister copper) (excluding electrocopper-plating, electroplating anodes)	74020000
24441330	Unwrought unalloyed refined copper (excluding rolled, extruded or forged sintered products)	74031100; 74031200; 74031300; 74031900
24441370	Unwrought copper alloys (excluding rolled, extruded or forged sintered products); master alloys of copper (including alloys which are not usefully malleable) (excluding copper phosphide (phosphor copper) containing > 15 % by weight of phosphorous)	74032100; 74032200; 74032900; 74050000
24442100	Copper powders and flakes excluding cement copper, powders/flake powders used in the preparation of paints such as bronzes/golds, (chemical compounds), refined copper shot	74061000; 74062000
24442200	Copper and copper alloy bars, rods, profiles and hollow profiles (excluding bars and rods obtained by casting or sintering, copper wire rod in coils)	74071000; 74072110; 74072190; 74072900
24442330	Copper wire refined (transv. section > 6 mm), of copper alloy	74081100; 74082100; 74082200; 74082900
24442350	Copper wire with cross-sectional dimension > 0,5 mm, <= 6 mm (excluding twine or cord reinforced with wire, stranded wire and cables)	74081910

PRODCOM Code	PRODCOM description	CN codes
24442370	Copper wire with cross-sectional dimension ≤ 0,5 mm (excluding twine or cord reinforced with wire, stranded wire and cables)	74081990
24442400	Copper and copper alloy plates, sheets and strip of a thickness > 0,15 mm (excluding expanded copper metal, insulated electric strip)	74091100; 74091900; 74092100; 74092900; 74093100; 74093900; 74094000; 74099000
24442500	Copper foil, of a thickness (excluding any backing) ≤ 0,15 mm	74101100; 74101200; 74102100; 74102200
24442630	Copper tubes and pipes	74111010; 74111090; 74112110; 74112190; 74112200; 74112900
24442650	Copper and copper alloy tube/pipe fittings including couplings, elbows, sleeves, tees and joints excluding bolts and nuts used for assembling/fixing pipes/tubes, fittings with taps, cocks, valves	74121000; 74122000
24451110	Nickel, not alloyed, unwrought	75021000
24451120	Unwrought nickel alloys	75022000
24451210	Nickel mattes	75011000
24451220	Nickel oxide sinters and other intermediate products of nickel metallurgy	75012000
24452100	Nickel powders and flakes (excluding nickel oxide sinters)	75040000
24452200	Nickel and nickel alloy bars, rods, profiles and wires (excluding prepared bars, rods or profiles for use in structures, insulated electric bars and wire, enamelled wire)	75051100; 75051200; 75052100; 75052200
24452300	Nickel and nickel alloy plate, sheet, strip and foil (excluding expanded metal)	75061000; 75062000
24452400	Nickel tubes, pipes and tube or pipe fittings	75071100; 75071200; 75072000
24453014	Tungsten powder and unwrought tungsten, incl. bars and rods obtained simply by sintering	81011000; 81019400
24453015	Tungsten and articles thereof; bars and rods, other than those obtained simply by sintering, profiles, wire, plates, sheets, strip and foil	81019600; 81019910; 81019990
24453017	Molybdenum and articles thereof (excluding waste and scrap), n.e.c.	81021000; 81029400; 81029500; 81029600; 81029900
24453020	Tantalum, unwrought, incl. bars and rods of tantalum obtained simply by sintering; tantalum powder	81032000
24453021	Tantalum and articles thereof. Bars and rods, other than those obtained simply by sintering, profiles, wire, plates, sheets, strip and foil	81039910
24453022	Other articles of tantalum, including crucibles (excluding waste and scrap), n.e.s.	81039100; 81039990
24453024	Unwrought magnesium, containing ≥ 99,8% by weight of magnesium	81041100
24453026	Unwrought magnesium, containing < 99,8% by weight of magnesium	81041900
24453028	Other articles of magnesium (excluding waste and scrap), n.e.s.	81043000; 81049000
24453034	Unwrought cadmium; cadmium powders (excluding waste and scrap)	81126910
24453035	Cobalt mattes and other intermediate products of cobalt metallurgy; unwrought cobalt; powders (excl. cobalt waste and scrap)	81052000
24453036	Articles of cobalt, n.e.s.	81059000

PRODCOM Code	PRODCOM description	CN codes
24453037	Other cadmium and articles thereof (excluding waste and scrap, cadmium powders and unwrought cadmium), n.e.c.	81126990
24453038	Bismuth and articles thereof containing more than 99,99 % of bismuth, by weight (excluding waste and scrap)	81061010; 81061090
24453039	Bismuth and articles thereof containing ≤ 99,99 % of bismuth, by weight (excluding waste and scrap)	81069010; 81069090
24453041	Titanium, unwrought; powders	81082000
24453042	Titanium and articles thereof; bars and rods, profiles, wire, plates, sheets, strip and foil, tubes and pipes, other (excluding waste and scrap)	81089030; 81089050; 81089060; 81089090
24453045	Antimony. Unwrought antimony; powders	81101000
24453049	Unwrought zirconium; zirconium powders (excluding waste and scrap)	81092100; 81092900
24453050	Other zirconium and articles thereof (excluding waste and scrap, zirconium powders and unwrought zirconium), n.e.c.	81099100; 81099900
24453058	Germanium and articles thereof (excluding waste and scrap, germanium powders and unwrought germanium), n.e.c.	81129940
24453059	Unwrought niobium; niobium powders (excluding waste and scrap)	81129240
24453060	Unwrought beryllium; beryllium powders	81121200
24453061	Articles of beryllium, n.e.s. (excluding waste and scrap)	81121900
24453067	Niobium (columbium) and articles thereof (excluding waste and scrap, niobium powders and unwrought niobium), n.e.c.	81129950
24453068	Unwrought hafnium; hafnium powders (excluding waste and scrap)	81123100
24453069	Other hafnium and articles thereof (excluding waste and scrap, hafnium powders and unwrought hafnium), n.e.c.	81123900
24453070	Unwrought indium; indium powders	81129281
24453072	Indium and articles thereof (other than unwrought, waste and scrap and powders) n.e.c.	81129970
24453073	Unwrought gallium; gallium powders	81129289
24453074	Gallium and articles thereof (other than unwrought, waste and scrap and powders) n.e.c.	81129970
24453075	Vanadium and articles thereof (other than unwrought, waste and scrap and powders) n.e.c.	81129970
24453076	Unwrought vanadium; vanadium powders (excl. ash and residues containing vanadium)	81129291
24453079	Unwrought germanium; germanium powders	81129295
24453085	Manganese. Unwrought manganese; powders	81110011
24453086	Manganese waste and scrap (excl. ash and residues containing manganese)	81110019
24453087	Articles of manganese, n.e.s.	81110090
24453091	Cermets unwrought	81130020
24453092	Cermets and articles thereof (excluding unwrought)	81130090
24453093	Chromium unwrought; powders	81122110; 81122190

PRODCOM Code	PRODCOM description	CN codes
24453094	Chromium and articles thereof (excluding unwrought and powders)	81122900
24453097	Unwrought rhenium; rhenium powders (excluding waste and scrap)	81124190

10.1.2 Indicators and data sources

A key criterion for establishing the current CBAM scope is the **risk of carbon leakage**. At the same time, the **relevance in terms of emissions** and a **balance between a broad coverage and limiting complexity and administrative burden** are also equally important.²¹³ This is because if the production and/or related emissions of a sector, subsector or product are very limited in the EU, the administrative burden of including the product in the CBAM may be disproportional to the carbon leakage risk prevented, particularly if data availability to determine emissions of the products is limited. These sectors, subsectors or products may therefore be a lower priority for the horizontal scope extension of the CBAM.

Therefore, various indicators for determining the significance of sectors and products for potential extension of the CBAM have been explored for the analysis in Chapter 2. These are shown in Table 54²¹⁴:

- **EU ETS and emissions data:** the availability of direct emissions data at the NACE 4-digit level is comprehensive for EU ETS installations but poor for emissions not reported by EU ETS installations. At the PRODCOM 8-digit level, the availability of emissions data is poor and only limited to the PRODCOM codes mentioned by EU ETS installations in their application for free allocation under the EU ETS.
- **Economic data:** the data for production volume and value is available down to the PRODCOM 8-digit level, which can be aggregated to determine the same data at NACE 4-digit level. The data for gross value added (GVA) is only available at the NACE 4-digit level. It could be approximated at the PRODCOM 8-digit level using the production value share of the PRODCOM code in the NACE sector.
- **Emissions intensity data:** these data at the NACE 4-digit level are available as a share of GVA from the data used to determine the EU ETS carbon leakage list for 2021-2030, but only for the years 2013-2015. At the PRODCOM 8-digit level, the data are only available for a very limited number of products that are not yet covered under the CBAM.

We can conclude from Table 54 that at the NACE 4-digit sector level, only data of EU ETS installations can be considered sufficiently robust for the shortlisting of sectors. At the PRODCOM 8-digit level, only the mentions of PRODCOM codes by EU ETS installations and economic data related to production volume and value are sufficiently robust for shortlisting of products. The availability of emission intensity data is relatively poor, particularly for products at PRODCOM 8-digit level. However, experience with the implementation of the current CBAM has shown that, as long as production processes

²¹³ Regulation (EU) 2023/956, recital (31).

²¹⁴ No data is publicly available at the CN code level for the indicators presented in the table and thus the CN code level is not mentioned in the table.

can be clearly identified and defined, the absence of emissions intensity data does not prevent inclusion of the product under the CBAM²¹⁵. It simply means that more time is required to develop the necessary rules (including default values) for determining the embedded emissions of imported products covered under the CBAM.

Table 54 Data sources used and level of data disaggregation available for mapping the significance of sectors and products in the EU

Indicators	NACE 4-digit level	PRODCOM 8-digit level
A No. of EU ETS installations	Confidential NIMs data from DG CLIMA of 2014-2018 for determining free allocation of 2021-2025, supplemented with data available from EUETS.info and gapfilling based on desk research ²¹⁶	Only mentions of PRODCOM codes based on confidential NIMs data from DG CLIMA of 2014-2018 for determining free allocation of 2021-2025
Direct EU ETS emissions	Confidential NIMs data from DG CLIMA for 2014-2018 for determining free allocation of 2021-2025, supplemented with data from EU ETS Verified Emissions for 2023	Only for a limited number of PRODCOM codes based on confidential NIMs data from DG CLIMA for 2014-2018 for determining free allocation of 2021-2025
Direct emissions (total)	Estimates based on emission intensity * GVA (see B and C below)	Only estimates for some PRODCOM codes based on available direct emissions intensity factors * tonnes of production (see B and C below)
Indirect emissions	Estimates based on emission intensity * GVA & Eurostat for NACE 4-digit (2 sectors)	Only estimates for some PRODCOM codes based on available indirect emissions intensity factors * tonnes of production (see B and C below)
B Production value and volume	Estimates based on the sum of PRODCOM level data	Based on Eurostat total production , or if no PRODCOM code exists, based on the Eurostat Structural Business Statistics .
Gross Added value (GVA)	Based on Eurostat gross added value	Estimates based on share of production value of NACE 4-digit * GVA of NACE 4-digit
C Emission intensity (direct)	Intensity as share of GVA based on EU ETS Phase 4 Preliminary CL list – CL indicator underlying data (2013-2015 average)	Only available for 71 of 925 PRODCOM codes not yet covered under the CBAM based on ETS benchmark values (2016-2017 average)
Emission intensity (indirect)	Intensity as share of GVA based on EU ETS Phase 4 Preliminary CL list – CL indicator underlying data (2013-2015 average)	Only available for 61 of 925 PRODCOM codes not yet covered under the CBAM based on ETS electricity efficiency benchmarks (2017-2019 estimates)

In addition to the significance of a product in the EU, it is also important to consider its relevance in trade flows of the EU for potential CBAM inclusion, particularly with regards to import. This analysis is based on two primary datasets: Comext Trade Data from

²¹⁵ Leading up to the start of the transitional phase of the CBAM, the Joint Research Centre of the Commission estimated GHG emissions intensities for products currently covered under the CBAM for the EU and its main trading partners. See Vidovic, D., Marmier, A., Zore, L. and Moya, J., Greenhouse gas emission intensities of the steel, fertilisers, aluminium and cement industries in the EU and its main trading partners, Publications Office of the European Union, Luxembourg, 2023, doi:10.2760/359533, JRC134682.

²¹⁶ For the EU ETS installations without a NACE code, these were found using online sources such as [kompass.com](#), to identify the NACE code affiliation of the installation. These are particularly needed for installations which were added after 2020 and not included in the EUETS.info database. This accounts for about 2% of EU ETS emissions for 2023.

Eurostat (DS-059322 or similar) and PRODCOM Production Data from Eurostat (DS-056121). Comext provides international trade data for the EU, including imports, exports, and trade balances categorised by CN and PRODCOM codes. PRODCOM contains annual EU production statistics, measured in volume and value, classified by PRODCOM codes.

Table 55 Data sources used for the mapping of trade flow indicators

Indicators	Level of data disaggregation	Sources
Tradability (Import, export and production value)	PRODCOM LEVEL	Eurostat COMEXT database DS-056120: Sold production, exports and imports
Import value and volume	PRODCOM and CN code level	Eurostat COMEXT database DS-059332: EU trade since 2002 by HS2-4-6 and CN8
Trade flows	EXIOBASE product level, which equivalent to NACE 2-digit level for industrial products (except for some products, which are mainly refinery products)	EXIOBASE 3
Extra-EU Import Value	CN code level	Eurostat COMEXT database DS-059332: EU trade since 2002 by HS2-4-6 and CN8
Extra-EU Import Intensity	PRODCOM and CN code level	Eurostat COMEXT database DS-059332: EU trade since 2002 by HS2-4-6 and CN8 ; Eurostat COMEXT database DS-056121 Total production
Top 5 Non-EU Countries	CN code level	Eurostat COMEXT database DS-059332: EU trade since 2002 by HS2-4-6 and CN8

10.1.3 Stage 2 shortlisted PRODCOM and CN codes

The tables below present the PRODCOM 8-digit and CN 8-digit codes that were preliminary shortlisted in Stage 2 of the Chapter 0 analysis. For the refinery products sector, only the CN codes are presented as no PRODCOM 8-digit codes exist for the refinery sector. For the products in the Organic chemicals & polymers sector (i.e., products falling under NACE codes 2014, 2016 and 2017), also only CN 8-digit codes from the Stage 2 analysis are shown due to the extensiveness of the list.

Table 56 PRODCOM and CN codes of Stage 2 shortlisted products in the mining sector

PRODCOM code	PRODCOM Description	CN code	CN description
08992920	Other minerals	25061000	Quartz (excl. quartz sands)
		25120000	Siliceous fossil meals, e.g. kieselguhr, tripolite and diatomite, and similar siliceous earths, whether or not calcined, of an apparent specific gravity of ≤ 1
		25199010	Magnesium oxide, whether or not pure (excl. calcined natural magnesium carbonate)
		25241000	Crocidolite asbestos (excl. products made from crocidolite)

PRODCOM code	PRODCOM Description	CN code	CN description
		25249000	Asbestos (excl. crocidolite and products made from asbestos)
		25251000	Crude mica and mica rifted into sheets or splittings
		25252000	Mica powder
		25253000	Mica waste
		25261000	Natural steatite, whether or not roughly trimmed or merely cut, by sawing or otherwise, into blocks or slabs of a square or rectangular shape, and talc, uncrushed or unpowdered
		25262000	Natural steatite and talc, crushed or powdered
		25291000	Feldspar
		25293000	Leucite, nepheline and nepheline syenite
		25301000	Vermiculite, perlite and chlorites, unexpanded
		26219000	Slag and ash, incl. seaweed ash "kelp" (excl. slag, incl. granulated, from the manufacture of iron or steel, ashes and residues containing arsenic, metals or metal compounds and those from the incineration of municipal waste)

Table 57 PRODCOM and CN codes of Stage 2 shortlisted products in the pulp and paper sector

PRODCOM code	PRODCOM Description	CN code	CN description
17111100	Chemical wood pulp, dissolving grades	47020000	Chemical wood pulp, dissolving grades
17111200	Chemical wood pulp, soda or sulphate, other than dissolving grades	47031100	Unbleached coniferous chemical wood pulp, soda or sulphate (excl. dissolving grades)
		47031900	Unbleached non-coniferous chemical wood pulp, soda or sulphate (excl. dissolving grades)
		47032100	Semi-bleached or bleached coniferous chemical wood pulp, soda or sulphate (excl. dissolving grades)
		47032900	Semi-bleached or bleached non-coniferous chemical wood pulp, soda or sulphate (excl. dissolving grades)
17111400	Mechanical wood pulp; semi-chemical wood pulp; pulps of fibrous cellulosic material other than wood	47010010	Thermo-mechanical wood pulp, not chemically treated
		47010090	Mechanical wood pulp, not chemically treated (excl. thermo-mechanical wood pulp)
		47050000	Wood pulp obtained by a combination of mechanical and chemical pulping processes
		47061000	Pulp of cotton linters
		47062000	Pulps of fibres derived from recovered "waste and scrap" paper or paperboard
		47063000	Pulps of fibrous cellulosic bamboo material

PRODCOM code	PRODCOM Description	CN code	CN description
		47069100	Mechanical pulp of fibrous cellulosic material (excl. that of bamboo, wood, cotton linters and fibres derived from recovered [waste and scrap] paper or paperboard)
		47069200	Chemical pulp of fibrous cellulosic material (excl. that of bamboo, wood, cotton linters and fibres derived from recovered [waste and scrap] paper or paperboard)
		47069300	Semi-chemical pulp of fibrous cellulosic material (excl. that of bamboo, wood, cotton linters and fibres derived from recovered [waste and scrap] paper or paperboard)
17121100	Newsprint in rolls or sheets	48010000	Newsprint as specified in Note 4 to chapter 48, in rolls of a width > 28 cm or in square or rectangular sheets with one side > 28 cm and the other side > 15 cm in the unfolded state
17121439	Graphic paper, paperboard : mechanical fibres <= 10 %, weight >= 40 g/m ² but <= 150 g/m ² , sheets	48025620	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in rectangular sheets with one side measuring 297 mm and the other side 210 mm "A4-format", not containing fibres obtained by a mechanical or chemi-mechanical process or of which <= 10% by weight of the total fibre content consists of such fibres, and weighing 40 g to 150 g/m ² , n.e.s.
		48025680	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in square or rectangular sheets with one side <= 435 mm and the other side <= 297 mm in the unfolded state, not containing fibres obtained by a mechanical or chemi-mechanical process or of which <= 10% by weight of the total fibre content consists of such fibres, and weighing 40 g to 150 g/m ² , n.e.s. (excl. with one side measuring 297 mm and the other side 210 mm in the unfolded state "A4-format")
		48025700	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in square or rectangular sheets with one side > 435 mm or with one side <= 435 mm and the other side > 297 mm in the unfolded state, not containing fibres obtained by a mechanical or chemi-mechanical process or of which <= 10% by weight of the total fibre content consists of such fibres, and weighing 40 g to 150 g/m ² , n.e.s.
17121470	Graphic paper, paperboard : mechanical fibres > 10 %	48026115	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in rolls of any size, weighing < 72 g/m ² and of which > 50% by weight of the total fibre content consists of fibres obtained by a mechanical process, n.e.s.
		48026180	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in rolls of any size, of which > 10% by weight of the total fibre content consists of fibres obtained by a mechanical or chemi-mechanical process, n.e.s. (excl. products weighing < 72 g/m ² and of which > 50% by weight of the

PRODCOM code	PRODCOM Description	CN code	CN description
17122055	Creped paper and webs of cellulose fibres for household/ sanitary purposes, in rolls, width > 36 cm, rectangular sheets min. one side > 36 cm in unfolded state, weight <= 25 g/m²/ply		total fibre content consists of fibres obtained by a mechanical process)
		48026200	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in square or rectangular sheets with one side <= 435 mm and the other side <= 297 mm in the unfolded state, of which > 10% by weight of the total fibre content consists of fibres obtained by a mechanical or chemi-mechanical process, n.e.s.
		48026900	Uncoated paper and paperboard, of a kind used for writing, printing or other graphic purposes, and non-perforated punchcards and punch-tape paper, in square or rectangular sheets with one side > 435 mm or with one side <= 435 mm and the other side > 297 mm in the unfolded state, of which > 10% by weight of the total fibre content consists of fibres obtained by a mechanical or chemi-mechanical process, n.e.s.
		48030031	Creped paper for household or sanitary purposes and webs of cellulose fibres "tissues", in rolls of a width > 36 cm or in square or rectangular sheets with one side > 36 cm and the other side > 15 cm in the unfolded state, weighing <= 25 g/m² per ply
17122090	Paper stock for household : others	48030090	Toilet or facial tissue stock, towel or napkin stock and similar paper for household or sanitary purposes, whether or not creped, crinkled, embossed, perforated, surface-coloured, surface-decorated or printed, in rolls of a width > 36 cm or in square or rectangular sheets with one side > 36 cm and the other side > 15 cm in the unfolded state (excl. cellulose wadding, creped paper and webs of cellulose fibres "tissues")
17123100	Uncoated, unbleached kraftliner in rolls or sheets (excluding for writing, printing or other graphic purposes, punch card stock and punch card tape paper)	48041111	Unbleached kraftliner, uncoated, in rolls of a width > 36 cm, containing >= 80% coniferous wood sulphate or soda pulp by weight in relation to the total fibre content and weighing < 150 g/m² (excl. goods of heading 4802 and 4803)
		48041115	Unbleached kraftliner, uncoated, in rolls of a width > 36 cm, containing >= 80% coniferous wood sulphate or soda pulp by weight in relation to the total fibre content and weighing >= 150 g to < 175 g/m² (excl. goods of heading 4802 and 4803)
		48041119	Unbleached kraftliner, uncoated, in rolls of a width > 36 cm, containing >= 80% coniferous wood sulphate or soda pulp by weight in relation to the total fibre content and weighing >= 175 g/m² (excl. goods of heading 4802 and 4803)
		48041190	Unbleached kraftliner, uncoated, in rolls of a width > 36 cm (excl. that containing > 80% coniferous wood sulphate or soda pulp by weight in relation to the total fibre content and goods of heading 4802 and 4803)

PRODCOM code	PRODCOM Description	CN code	CN description
17127759	Paper and paperboard in rolls or sheets, coated, impregnated or covered with plastics (excluding adhesives, bleached and weighing > 150 g/m²)	48115900	Paper and paperboard, surface-coloured, surface-decorated or printed, coated, impregnated or covered with artificial resins or plastics, in rolls or in square or rectangular sheets, of any size (excl. bleached and weighing > 150 g/m², and adhesives)
17127780	Other paper, paperboard, coated..., n.e.c.	48119000	Paper, paperboard, cellulose wadding and webs of soft cellulose, coated, impregnated, covered, surface-coloured, surface-decorated or printed, in rolls or in square or rectangular sheets, of any size (excl. goods of heading 4803, 4809, 4810 and 4818, and of subheading 4811.10 to 4811.60)

Table 58 CN codes of Stage 2 shortlisted products in the refinery sector

CN code	CN description
27012000	Briquettes, ovoids and similar solid fuels manufactured from coal
27022000	Agglomerated lignite (excl. jet)
27101211	Light oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)
27101215	Light oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)
27101221	White spirit
27101225	Special spirits (excl. white spirit) of petroleum or bituminous minerals
27101231	Aviation spirit
27101241	Motor spirit, with a lead content ≤ 0,013 g/l, with a research octane number "RON" of < 95 (excl. containing biodiesel)
27101245	Motor spirit, with a lead content ≤ 0,013 g/l, with a research octane number "RON" of ≥ 95 but < 98 (excl. containing biodiesel)
27101249	Motor spirit, with a lead content ≤ 0,013 g/l, with a research octane number "RON" of ≥ 98 (excl. containing biodiesel)
27101250	Motor spirit, with a lead content > 0,013 g/l (excl. containing biodiesel, and aviation spirit)
27101270	Jet fuel, spirit type (excl. aviation spirit)
27101290	Light oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. containing biodiesel, for undergoing chemical transformation, and special spirits, motor spirit and spirit type jet fuel)
27101911	Medium oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27

CN code	CN description
27101915	Medium oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27)
27101921	Jet fuel, kerosene type
27101925	Kerosene (excl. jet fuel)
27101929	Medium oils and preparations, of petroleum or bituminous minerals, n.e.s. (excl. for undergoing chemical transformation, and kerosene)
27101931	Gas oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27
27101935	Gas oils of petroleum or bituminous minerals, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)
27101943	Gas oils of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101946	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,002\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101947	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,002\%$ but $\leq 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101948	Gas oils of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight (excl. containing biodiesel, and for undergoing chemical transformation)
27101951	Fuel oils of petroleum or bituminous minerals for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. containing biodiesel)
27101955	Fuel oils obtained from bituminous materials, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, and containing biodiesel)
27101962	Fuel oils obtained from bituminous materials, with a sulphur content of $\leq 0,1\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101966	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,1\%$ but $\leq 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101967	Fuel oils obtained from bituminous materials, with a sulphur content of $> 0,5\%$ by weight (excl. for undergoing chemical transformation, and containing biodiesel)
27101971	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing a specific process as defined in Additional Note 5 to chapter 27
27101975	Lubricating oils and other preparations containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations, for undergoing chemical transformation (excl. specific processes specified in Additional Note 5 to chapter 27)
27101981	Motor oils, compressor lube oils and turbine lube oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)

CN code	CN description
27101983	Liquids for hydraulic purposes containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)
27101985	White oils, liquid paraffin containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101987	Gear oils and reductor oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101991	Metalworking compounds, mould-release oils, anti-corrosion oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101993	Electrical insulating oils containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituent of the preparations (excl. for undergoing chemical transformation)
27101999	Lubricating oils and other heavy oils and preparations n.e.s., containing by weight $\geq 70\%$ of petroleum oils or of oils obtained from bituminous minerals, these oils being the basic constituents of the preparations (excl. for undergoing chemical transformation)
27102011	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,001\%$ by weight, containing biodiesel
27102016	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,001\%$ but $\leq 0,1\%$ by weight, containing biodiesel
27102019	Gas oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,1\%$ by weight, containing biodiesel
27102032	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $\leq 0,5\%$ by weight, containing biodiesel
27102038	Fuel oils of $\geq 70\%$ of petroleum or bituminous minerals, with a sulphur content of $> 0,5\%$ by weight, containing biodiesel
27102090	Oils of $\geq 70\%$ of petroleum or bituminous minerals, containing biodiesel (excl. gas oils and fuel oils)
27111211	Propane of a purity of $\geq 99\%$, for use as a power or heating fuel, liquefied
27111219	Propane of a purity of $\geq 99\%$, liquefied (excl. for use as a power or heating fuel)
27111291	Propane of a purity of $< 99\%$, liquefied, for undergoing a specific process as defined in Additional Note 5 to chapter 27
27111293	Propane of a purity of $< 99\%$, liquefied, for undergoing chemical transformation (excl. processes specified in 2711.12.91)
27111294	Liquid propane of a purity of $> 90\%$, but $< 99\%$ (excl. for undergoing chemical transformation)
27111297	Liquefied propane of a purity of $\leq 90\%$ (excl. for undergoing chemical transformation)
27111310	Butanes for undergoing a specific process as defined in Additional Note 5 to chapter 27, liquefied (excl. of a purity of $\geq 95\%$ of N-butane or isobutane)

CN code	CN description
27111330	Butanes for undergoing chemical transformation, liquefied (excl. for specific processes specified in Additional Note 5 to chapter 27 and butanes of a purity of $\geq 95\%$ of N-butane or isobutane)
27111391	Liquefied butane of a purity of $> 90\%$ but $< 95\%$ (excl. for undergoing chemical transformation)
27111397	Liquefied butane of a purity of $\leq 90\%$ (excl. for undergoing chemical transformation)
27111400	Ethylene, propylene, butylene and butadiene, liquefied (excl. ethylene of a purity of $\geq 95\%$ and propylene, butylene and butadiene of a purity of $\geq 90\%$)
27111900	Gaseous hydrocarbons, liquefied, n.e.s. (excl. natural gas, propane, butane, ethylene, propylene, butylene and butadiene)
27112900	Hydrocarbons in gaseous state, n.e.s. (excl. natural gas)
27121010	Crude petroleum jelly
27121090	Petroleum jelly (excl. crude)
27122010	Synthetic paraffin wax containing $< 0,75\%$ by weight of oil and of a molecular weight of ≥ 460 but ≤ 1.560
27122090	Paraffin wax containing $< 0,75\%$ by weight of oil (excl. synthetic paraffin wax of a molecular weight of ≥ 460 but ≤ 1.560)
27129011	Crude ozokerite, lignite wax or peat wax "natural products"
27129019	Ozokerite, lignite wax or peat wax "natural products", whether or not coloured (excl. crude)
27129031	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing a specific process as defined in Additional Note 5 to chapter 27 (excl. petroleum jelly, paraffin wax containing $< 0,75\%$ by weight of oil, ozokerite, lignite wax and peat wax)
27129033	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes, for undergoing chemical transformation (excl. for specific processes specified in Additional Note 5 to chapter 27, petroleum jelly, paraffin wax containing $< 0,75\%$ by weight of oil, ozokerite, lignite wax and peat wax)
27129039	Crude paraffin wax, microcrystalline petroleum wax, slack wax, other mineral waxes, and similar products obtained by synthesis or by other processes (excl. for undergoing chemical transformation, petroleum jelly, paraffin wax containing $< 0,75\%$ by weight of oil, ozokerite, lignite wax and peat wax)
27129091	Blend of 1-alkenes containing by weight $\geq 80\%$ of 1-alkenes of a chain-length of ≥ 24 but ≤ 28 carbon atoms
27129099	Paraffin wax, microcrystalline petroleum wax, slack wax, ozokerite, lignite wax, peat wax, other mineral waxes, and similar products obtained by synthesis or by other processes, whether or not coloured (excl. petroleum jelly, paraffin wax containing $< 0,75\%$ by weight of oil and a blend of 1-alkenes containing by weight $\geq 80\%$ of 1-alkenes of a chain-length of ≥ 24 but ≤ 28 carbon atoms)
27131100	Petroleum coke, non-calcined
27131200	Petroleum coke, calcined
27132000	Petroleum bitumen

CN code	CN description
27139010	Residues of petroleum oil or of oil obtained from bituminous minerals for the manufacture of carbon of heading 2803
27139090	Residues of petroleum oil or of oil obtained from bituminous minerals (excl. for the manufacture of carbon of heading 2803, petroleum coke and petroleum bitumen)

Table 59 PRODCOM and CN codes of Stage 2 shortlisted products in the inorganic chemicals sector

PRODCOM code	PRODCOM Description	CN code	CN description
20132130	Carbon (carbon blacks and other forms of carbon, n.e.c.)	28030000	Carbon "carbon blacks and other forms of carbon", n.e.s.
20132170	Silicon. Other than containing by weight not less than 99,99 % of silicon	28046900	Silicon containing < 99,99% by weight of silicon
20132455	Phosphoric acid and polyphosphoric acids	28092000	Phosphoric acid; polyphosphoric acids, whether or not chemically defined
20132475	Silicon dioxide	28112200	Silicon dioxide
20132527	Sodium hydroxide in aqueous solution (soda lye or liquid soda)	28151200	Sodium hydroxide "caustic soda" in aqueous solution "soda lye or liquid soda"
20132570	Aluminium hydroxide	28183000	Aluminium hydroxide
20133138	Other chlorides (excluding lithium, ammonium, magnesium, nickel, tin and cobalt chloride) n.e.s.	28272000	Calcium chloride
		28273200	Aluminium chloride
		28273920	Iron chlorides
		28273985	Chlorides (excl. ammonium, calcium, magnesium, aluminium, iron, cobalt, nickel, tin and mercury chloride)
		28331100	Disodium sulphate
20134165	Sulphates (excluding those of aluminium, barium, magnesium, nickel, cobalt, titanium)	28331900	Sodium sulphates (excl. disodium)
		28332500	Sulphates of copper
		28332920	Sulphates of cadmium, of chromium and of zinc
		28332960	Sulphates of lead
		28332980	Sulphates (excl. of sodium, magnesium, aluminium, nickel, copper, barium, cadmium, chromium, zinc, cobalt, titanium, lead and mercury)
		28352400	Phosphates of potassium
20134289	Phosphates (excluding calcium hydrogenorthophosphate and mono- or disodium phosphate, and lithium iron phosphate); polyphosphates	28352600	Phosphates of calcium (excl. calcium hydrogenorthophosphate "dicalcium phosphate")
		28352910	Phosphate of triammonium

PRODCOM code	PRODCOM Description	CN code	CN description
	(excluding sodium triphosphate)	28352930	Phosphate of trisodium
		28352990	Phosphates (excl. phosphates of triammonium, monosodium, disodium, trisodium, of potassium, of calcium and of mercury)
		28353900	Polyphosphates, whether or not chemically defined (excl. sodium triphosphate "sodium tripolyphosphate", and inorganic or organic compounds of mercury whether or not chemically defined)
20134310	Disodium carbonate	28362000	Disodium carbonate
20134320	Sodium hydrogencarbonate (sodium bicarbonate)	28363000	Sodium hydrogencarbonate "sodium bicarbonate"
		28364000	Potassium carbonates
		28366000	Barium carbonate
		28369200	Strontium carbonate
		28369911	Carbonates of magnesium and of copper
20134399	Other carbonates (excluding cobalt carbonates)	28369917	Carbonates; commercial ammonium carbonate containing ammonium carbamate (excl. disodium carbonate, sodium hydrogencarbonate "sodium bicarbonate", potassium carbonates, calcium carbonate, barium carbonate, lithium carbonates, strontium carbonate, carbonates of magnesium and of copper and inorganic or organic compounds of mercury)
		28369990	Peroxocarbonates "percarbonates"
	Other inorganic compounds n.e.c.; amalgams (excluding distilled and conductivity water and water of similar purity, liquid air and compressed air, those of precious metals)	28531000	Cyanogen chloride "chlorcyan"
20135290		28539090	Inorganic compounds, n.e.s.; amalgams (excl. of precious metals)
20136270	Double or complex silicates	28421000	Double or complex silicates of inorganic acids or peroxyacids, incl. aluminosilicates whether or not chemically defined (excl. inorganic or organic compounds of mercury whether or not chemically defined)
20136410	Carbides of silicon, whether or not chemically defined	28492000	Carbides of silicon, whether or not chemically defined

Table 60 CN codes of Stage 2 shortlisted products in the organic chemicals & polymers sector

CN code	CN description
22071000	Undenatured ethyl alcohol, of actual alcoholic strength of $\geq 80\%$
22072000	Denatured ethyl alcohol and other spirits of any strength
27071000	Benzol "benzene" containing $> 50\%$ of benzene (excl. chemically defined)
27072000	Toluol "toluene" containing $> 50\%$ of toluene (excl. chemically defined)
27073000	Xylol "xylenes" containing $> 50\%$ of xylenes (excl. chemically defined)
27074000	Naphthalene containing $> 50\%$ of naphthalene (excl. chemically defined)
27075000	Aromatic hydrocarbon mixtures of which $\geq 65\%$ by volume, incl. losses, distils at 250°C by the ASTM D 86 method (excl. chemically defined compounds)
27079100	Creosote oils (excl. chemically defined)
27079911	Crude light oils from the distillation of high temperature coal tars, of which $\geq 90\%$ by volume distils at temperatures of up to 200°C (excl. chemically defined)
27079919	Crude light oils from the distillation of high temperature coal tars (excl. those of which $\geq 90\%$ by volume distils at temperatures of up to 200°C and chemically defined compounds)
27079920	Anthracene (excl. chemically defined); sulphuretted toppings from the first distillation of high temperature coal tars
27079950	Pyridine, quinoline, acridine, aniline bases and other basic products of the first distillation of high temperature coal tars, n.e.s.
27079991	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, for the manufacture of carbon in heading 2803
27079999	Oils and other products of the distillation of high temperature coal tars and similar products in which the weight of the aromatic constituents exceeds that of the non-aromatic constituents, n.e.s.
29011000	Saturated acyclic hydrocarbons
29012900	Hydrocarbons, acyclic, unsaturated (excl. ethylene, propene "propylene", butene "butylene" and isomers thereof and Buta-1,3-diene and isoprene)
29021100	Cyclohexane
29021900	Cyclanes, cyclenes and cycloterpenes (excl. cyclohexane)
29024100	o-Xylene
29024300	p-Xylene
29025000	Styrene
29029000	Cyclic hydrocarbons (excl. cyclanes, cyclenes, benzene, toluene, xylenes, styrene, ethylbenzene and cumene)
29031500	Ethylene dichloride "ISO" "1,2 dichloroethane"
29034100	Trifluoromethane "HFC-23"
29034200	Difluoromethane "HFC-32"
29034300	Fluoromethane "HFC-41", 1,2-difluoroethane "HFC-152" and 1,1-difluoroethane "HFC-152a"
29034400	Pentafluoroethane "HFC-125", 1,1,1-trifluoroethane "HFC-143a" and 1,1,2-trifluoroethane "HFC-143"
29034500	1,1,1,2-Tetrafluoroethane "HFC-134a" and 1,1,2,2-tetrafluoroethane "HFC-134"
29034600	1,1,1,2,3,3,3-Heptafluoropropane "HFC-227ea", 1,1,1,2,2,3-hexafluoropropane "HFC-236cb", 1,1,1,2,3,3-hexafluoropropane "HFC-236ea" and 1,1,1,3,3,3-hexafluoropropane "HFC-236fa"
29034700	1,1,1,3,3-Pentafluoropropane "HFC-245fa" and 1,1,2,2,3-pentafluoropropane "HFC-245ca"
29034800	1,1,1,3,3-Pentafluorobutane "HFC-365mfc" and 1,1,1,2,2,3,4,5,5,5-decafluoropentane "HFC-43-10mee"
29034910	Pentafluoropropanes, hexafluoropropanes and heptafluoropropanes (excl. 1,1,1,2,3,3,3-Heptafluoropropane "HFC-227ea", 1,1,1,2,2,3-hexafluoropropane "HFC-236cb", 1,1,1,2,3,3-hexafluoropropane "HFC-236ea", 1,1,1,3,3,3-hexafluoropropane "HFC-236fa", 1,1,1,3,3-Pentafluoropropane "HFC-245fa" and 1,1,2,2,3-pentafluoropropane "HFC-245ca")
29034930	Saturated perfluorinated derivatives of acyclic hydrocarbons

CN code	CN description
29034990	Saturated fluorinated derivatives of acyclic hydrocarbons (excl. products of 2903.41 to 2903.49.30)
29035100	2,3,3,3-Tetrafluoropropene "HFO-1234yf", 1,3,3,3-tetrafluoropropene "HFO-1234ze" and (Z)-1,1,1,4,4,4-hexafluoro-2-butene "HFO-1336mzz"
29035900	Unsaturated fluorinated derivatives of acyclic hydrocarbons (excl. 2,3,3,3-tetrafluoropropene "HFO-1234yf", 1,3,3,3-tetrafluoropropene "HFO-1234ze" and (Z)-1,1,1,4,4,4-hexafluoro-2-butene "HFO-1336mzz")
29036100	Methyl bromide "bromomethane"
29036200	Ethylene dibromide "ISO" "1,2-dibromoethane"
29036911	Dibromomethane
29036919	Brominated derivatives of acyclic hydrocarbons (excl. methyl bromide "bromomethane", ethylene dibromide "ISO" "1,2-dibromoethane" and dibromomethane)
29036980	Iodinated derivatives of acyclic hydrocarbons
29039100	Chlorobenzene, o-dichlorobenzene and p-dichlorobenzene
29039200	Hexachlorobenzene (ISO) and DDT (ISO) (clofenotane (INN), 1,1,1-trichloro-2,2-bis(p-chlorophenyl)ethane)
29039300	Pentachlorobenzene "ISO"
29039400	Hexabromobiphenyls
29039910	2,3,4,5,6-Pentabromoethylbenzene
29039980	Halogenated derivatives of aromatic hydrocarbons (excl. chlorobenzene, o-dichlorobenzene, p-dichlorobenzene, hexachlorobenzene [ISO], DDT [ISO] "clofenotane [INN], 1,1,1-trichloro-2,2-bis[p-chlorophenyl]ethane", pentachlorobenzene "ISO", hexabromobiphenyls and 2,3,4,5,6-Pentabromoethylbenzene)
29051100	Methanol "methyl alcohol"
29051200	Propan-1-ol "propyl alcohol" and propan-2-ol "isopropyl alcohol"
29051300	Butan-1-ol "n-butyl alcohol"
29051700	Dodecan-1-ol "lauryl alcohol", hexadecan-1-ol "cetyl alcohol" and octadecan-1-ol "stearyl alcohol"
29051900	Saturated monohydric acyclic alcohols (excl. methanol "methyl alcohol", propan-1-ol "propyl alcohol", propan-2-ol "isopropyl alcohol", butanols, octanol "octyl alcohol" and isomers thereof, dodecan-1-ol "lauryl alcohol", hexadecan-1-ol "cetyl alcohol" and octadecan-1-ol "stearyl alcohol")
29052200	Acyclic terpene alcohols
29052910	Allyl alcohol
29052990	Unsaturated monohydric acyclic alcohols (excl. allyl alcohol and acyclic terpene alcohols)
29053100	Ethylene glycol "ethanediol"
29053920	Butane-1,3-diol
29053928	Butane-1,4-diol (excl. having a bio-based carbon content of 100%)
29053930	2,4,7,9-Tetramethyldec-5-yne-4,7-diol
29053995	Acyclic diols (excl. ethylene glycol "ethanediol", propylene glycol "propane-1,2-diol", butane-1,3-diol, butane-1,4-diol and 2,4,7,9-tetramethyldec-5-yne-4,7-diol)
29054100	2-Ethyl-2-"hydroxymethyl" propane-1,3-diol "trimethylolpropane"
29054200	Pentaerythritol
29054300	Mannitol
29054900	Tri- and other polyhydric acyclic alcohols (excl. 2-ethyl-2-"hydroxymethyl" propane-1,3-diol "trimethylolpropane", pentaerythritol, mannitol, d-glucitol "sorbitol" and glycerol)
29061100	Menthol
29061200	Cyclohexanol, methylcyclohexanols and dimethylcyclohexanols
29061310	Sterols
29061390	Inositols

CN code	CN description
29061900	Cyclanic, cyclenic or cycloterpenic alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. menthol, cyclohexanol, methylcyclohexanols, dimethylcyclohexanols, sterols and inositols)
29071100	Phenol "hydroxybenzene" and its salts
29071200	Cresols and their salts
29071300	Octylphenol, nonylphenol and their isomers; salts thereof
29071510	1-Naphthol
29071590	Naphthols and their salts (excl. 1-naphthol)
29071910	Xylenols and their salts
29071990	Monophenols (excl. phenol "hydroxybenzene" and its salts, cresols and their salts, octylphenol, nonylphenol and their isomers and salts thereof, xylenols and their salts and naphthols and their salts)
29072100	Resorcinol and its salts
29072200	Hydroquinone "quinol" and its salts
29072300	4,4'-Isopropylidenediphenol "bisphenol A, diphenylolpropane" and its salts
29072900	Polyphenols and phenol-alcohols (excl. resorcinol and hydroquinone "quinol" and their salts, and 4,4'-isopropylidenediphenol "bisphenol A, diphenylolpropane" and its salts)
29091100	Diethyl ether
29091910	Tert-butyl ethyl ether (ethyl-tertio-butyl-ether, ETBE)
29091990	Acyclic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. diethyl ether and tert-butyl ethyl ether [ethyl-tertio-butyl-ether, ETBE])
29093010	Diphenyl ether
29093031	Pentabromodiphenyl ether; 1,2,4,5-tetrabromo-3,6-bis"pentabromophenoxy"benzene
29093035	1,2-Bis"2,4,6-tribromophenoxy"ethane for the manufacture of acrylonitrile-butadiene-styrene [ABS]
29093038	Brominated derivatives of aromatic ethers (excl. pentabromodiphenyl ether, 1,2,4,5-tetrabromo-3,6-bis"pentabromophenoxy"benzene and 1,2-bis"2,4,6-tribromophenoxy"ethane for the manufacture of acrylonitrile-butadiene-styrene [ABS])
29093090	Aromatic ethers and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. diphenyl ether and brominated derivatives)
29094300	Monobutyl ethers of ethylene glycol or of diethylene glycol
29094400	Monoalkylethers of ethylene glycol or of diethylene glycol (excl. monobutyl ethers)
29094911	2-"2-Chloroethoxy"ethanol
29094980	Cyclic ether-alcohols and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. 2-(2-Chloroethoxy)ethanol)
29103000	1-Chloro-2,3-epoxypropane "epichlorohydrin"
29104000	Dieldrin "ISO" "INN"
29105000	Endrin "ISO"
29109000	Epoxides, epoxyalcohols, epoxyphenols and epoxyethers, with a three-membered ring, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. oxirane "ethylene oxide", methyloxirane "propylene oxide", 1-Chloro-2,3-epoxypropane "epichlorohydrin", dieldrin [ISO] [INN] and endrin "ISO")
29121900	Acyclic aldehydes, without other oxygen function (excl. methanal [formaldehyde] and ethanal [acetaldehyde])
29124100	Vanillin "4-hydroxy-3-methoxybenzaldehyde"
29124200	Ethylvanillin "3-ethoxy-4-hydroxybenzaldehyde"
29124900	Aldehyde-alcohols, aldehyde-ethers, aldehyde-phenols and aldehydes with other oxygen function (excl. ethylvanillin "3-ethoxy-4-hydroxybenzaldehyde" and vanillin "4-hydroxy-3-methoxybenzaldehyde")
29147100	Chlordecone "ISO"

CN code	CN description
29147900	Halogenated, sulphonated, nitrated or nitrosated derivatives of ketones or quinones (excl. chlordecone "ISO" and inorganic or organic compounds of mercury)
29152100	Acetic acid
29153200	Vinyl acetate
29153300	n-Butyl acetate
29153600	Dinoseb acetate "ISO"
29153900	Esters of acetic acid (excl. ethyl, vinyl, n-butyl and dinoseb [ISO] acetates)
29154000	Mono- di- or trichloroacetic acids, their salts and esters
29155000	Propionic acid, its salts and esters
29156011	1-Isopropyl-2,2-dimethyltrimethylene diisobutyrate
29156019	Butanoic acids and their salts and esters (excl. 1-isopropyl-2,2-dimethyltrimethylene diisobutyrate)
29156090	Pentanoic acids and their salts and esters
29157040	Palmitic acid and its salts and esters
29157050	Stearic acid and its salts and esters
29159030	Lauric acid and its salts and esters
29159070	Saturated acyclic monocarboxylic acids, their anhydrides, halides, peroxides and peroxyacids; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. formic acid and acetic acid, mono-, di- or trichloroacetic acids, propionic acid, butanoic and pentanoic acids, palmitic, stearic and lauric acids and their salts and esters, and acetic anhydride)
29161100	Acrylic acid and its salts
29161200	Esters of acrylic acid
29161400	Esters of methacrylic acid
29161600	Binapacryl (ISO)
29161910	Undecenoic acids and their salts and esters
29161940	Crotonic acid
29161995	Unsaturated acyclic monocarboxylic acids, their anhydrides, halides, peroxides, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. acrylic acid and its salts and esters, methacrylic acid and its salts and esters, oleic, linoleic or linolenic acids, their salts and esters, undecenoic acids and their salts and esters, crotonic acid and binapacryl [ISO])
29162000	Cyclanic, cyclenic or cycloterpenic monocarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. inorganic or organic compounds of mercury)
29163990	Aromatic monocarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. benzoic acid, its salts and esters, benzoyl peroxide, benzoyl chloride, binapacryl [ISO], phenylacetic acid, its salts and esters, and inorganic or organic compounds of mercury whether or not chemically defined)
29171100	Oxalic acid, its salts and esters (excl. inorganic or organic compounds of mercury)
29171200	Adipic acid, its salts and esters
29171310	Sebacic acid
29171390	Azelaic acid, their salts and esters and salts and esters of sebacic acid
29171910	Malonic acid, its salts and esters
29171980	Acyclic polycarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. oxalic acid, adipic acid, azelaic acid, sebacic acid, malonic acid, their salts and esters, maleic anhydride, inorganic or organic compounds of mercury, and ethane-1,2-dicarboxylic acid or butanedioic acid "succinic acid" having a bio-based carbon content of 100%)
29172000	Cyclanic, cyclenic or cycloterpenic polycarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their derivatives
29173500	Phthalic anhydride

CN code	CN description
29173600	Terephthalic acid and its salts
29173700	Dimethyl terephthalate
29173920	Ester or anhydride of tetrabromophthalic acid; benzene-1,2,4-tricarboxylic acid; isophthaloyl dichloride, containing by weight 0,8 % or less of terephthaloyl dichloride; naphthalene-1,4,5,8-tetracarboxylic acid; tetrachlorophthalic anhydride; sodium 3,5-bis(methoxycarbonyl)benzenesulphonate
29173935	Bis(2-ethylhexyl) benzene-1,4-dicarboxylate "DOTP"
29173985	Aromatic polycarboxylic acids, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. esters of orthophthalic acid, phthalic anhydride, terephthalic acid and its salts, dimethyl terephthalate, ester or anhydride of tetrabromophthalic acid, benzene-1,2,4-tricarboxylic acid, isophthaloyl dichloride containing by weight <= 0,8% of terephthaloyl dichloride, naphthalene-1,4,5,8-tetracarboxylic acid, tetrachlorophthalic anhydride, sodium 3,5-bis(methoxycarbonyl)benzenesulphonate and bis(2-ethylhexyl) benzene-1,4-dicarboxylate "DOTP")
29181100	Lactic acid, its salts and esters (excl. inorganic or organic compounds of mercury)
29181200	Tartaric acid
29181300	Salts and esters of tartaric acid
29181400	Citric acid
29181500	Salts and esters of citric acid (excl. inorganic or organic compounds of mercury)
29181600	Gluconic acid, its salts and esters
29181700	2,2-Diphenyl-2-hydroxyacetic acid "benzilic acid"
29181800	Chlorobenzilate "ISO"
29181930	Cholic acid and 3-alpha, 12-alpha-dihydroxy-5-beta-cholan-24-oic acid "deoxycholic acid", their salts and esters
29181940	2,2-Bis(hydroxymethyl)propionic acid
29181998	Carboxylic acids with alcohol function but without other oxygen function and their anhydrides, halides, peroxides and peroxyacids; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. lactic acid, tartaric acid, citric acid, gluconic acid, cholic acid, 3-alpha, 12-alpha-dihydroxy-5-beta-cholan-24-oic acid "deoxycholic acid" and their salts and esters, and 2,2-bis(hydroxymethyl)propionic acid and chlorobenzilate [ISO] and 2,2-Diphenyl-2-hydroxyacetic acid (benzilic acid))
29182900	Carboxylic acids with phenol function but without other oxygen function, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. salicylic acid and o-Acetylsalicylic acid, and their salts and esters)
29183000	Carboxylic acids with aldehyde or ketone function but without other oxygen function, their anhydrides, halides, peroxides, peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives
29189100	2,4,5-T "ISO" "2,4,5-trichlorophenoxyacetic acid", its salts and esters
29189940	2,6-Dimethoxybenzoic acid; dicamba (ISO); sodium phenoxyacetate
29189990	Carboxylic acids with additional oxygen function, their anhydrides, halides, peroxides and peroxyacids and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. only with alcohol, phenol, aldehyde or ketone function, and 2,6-dimethoxybenzoic acid, dicamba [ISO], sodium phenoxyacetate, and 2,4,5-T [ISO] [2,4,5-trichlorophenoxyacetic acid] and its salts and esters)
29191000	Tris(2,3-dibromopropyl) phosphate
29199000	Phosphoric esters and their salts, incl. lactophosphates; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. tris(2,3-dibromopropyl) phosphate)
29201100	Parathion "ISO" and parathion-methyl "ISO" "methyl-parathion"

CN code	CN description
29201900	Thiophosphoric esters "phosphorothioates" and their salts; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. parathion [ISO] and parathion-methyl [ISO] [methyl-parathion])
29202100	Dimethyl phosphite
29202200	Diethyl phosphite
29202300	Trimethyl phosphite
29202400	Triethyl phosphite
29202900	Phosphite esters and their salts; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. dimethyl, diethyl, trimethyl and triethyl phosphites)
29203000	Endosulfan "ISO"
29209010	Sulphuric esters and carbonic esters and their salts, and their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. inorganic or organic compounds of mercury)
29209070	Esters of inorganic acids of non-metals and their salts; their halogenated, sulphonated, nitrated or nitrosated derivatives (excl. esters of hydrogen halides, phosphoric esters, phosphite esters, sulphuric esters, carbonic esters and thiophosphoric esters "phosphorothioates" and their salts, and their halogenated, sulphonated, nitrated or nitrosated derivatives, and endosulfan "ISO")
29211200	2-"N,N-Dimethylamino"ethylchloride hydrochloride
29211300	2-"N,N-Diethylamino"ethylchloride hydrochloride
29211400	2-(N,N-Diisopropylamino)ethylchloride hydrochloride
29211940	1,1,3,3-Tetramethylbutylamine
29211950	Diethylamine and its salts
29211999	Acyclic monoamines and their derivatives; salts thereof (excl. methylamine, dimethylamine, trimethylamine, diethylamine, and their salts, 1,1,3,3-tetramethylbutylamine, 2-(N,N-Diethylamino)ethyl chloride hydrochloride, 2-(N,N-diisopropylamino)ethyl chloride hydrochloride and 2-(N,N-dimethylamino)ethyl chloride hydrochloride)
29212100	Ethylenediamine and its salts
29212200	Hexamethylenediamine and its salts
29214100	Aniline and its salts (excl. inorganic or organic compounds of mercury)
29214300	Toluidines and their derivatives; salts thereof
29214400	Diphenylamine and its derivatives; salts thereof
29214500	1-Naphthylamine "alpha-naphthylamine", 2-naphthylamine "beta-naphthylamine" and their derivatives; salts thereof
29214600	Amfetamine "INN", benzfetamine "INN", dexamfetamine "INN", etilamfetamine "INN", fencamfetamine "INN", lefetamine "INN", levamfetamine "INN", mefenorex "INN" and phentermine "INN", and salts thereof
29214900	Aromatic monoamines and derivatives; salts thereof (excl. aniline, toluidines, diphenylamine, 1-naphthylamine "alpha-naphthylamine", 2-naphthylamine "beta-naphthylamine" and their derivatives, and salts thereof, and amfetamine "INN", benzfetamine "INN", dexamfetamine "INN", etilamfetamine "INN", fencamfetamine "INN", lefetamine "INN", levamfetamine "INN", mefenorex "INN" and phentermine "INN", and salts thereof)
29215111	m-Phenylenediamine of a purity by weight of >= 99% and containing <= 1% by weight of water, <= 200 mg/kg of o-phenylenediamine and <= 450 mg/kg of p-phenylenediamine
29215119	o-Phenylenediamine, m-phenylenediamine, p-phenylenediamine, diaminotoluenes, and their halogenated, sulphonated, nitrated and nitrosated derivatives; salts thereof (excl. m-phenylenediamine of a purity by weight of >= 99% and containing <= 1% by weight of water, <= 200 mg/kg of o-phenylenediamine and <= 450 mg/kg of p-phenylenediamine)
29215190	Derivatives of o-phenylenediamine, m-phenylenediamine, p-phenylenediamine or diaminotoluenes; salts thereof (excl. halogenated, sulphonated, nitrated and nitrosated derivatives, and salts thereof)
29215950	m-Phenylenebis(methylamine); 2,2'-dichloro-4,4'-methylenedianiline; 4,4'-bi-o-toluidine; 1,8-naphthylenediamine

CN code	CN description
29215990	Aromatic polyamines and their derivatives; salts thereof (excl. o-phenylenediamine, m-phenylenediamine, p-phenylenediamine or diaminotoluenes and their derivatives, and salts thereof, m-phenylenebis"methylamine", 2,2'-dichloro-4,4'-methylenedianiline, 4,4'-bi-o-toluidine and 1,8-naphthylenediamine)
29221400	Dextropropoxyphene "INN" and its salts
29221500	Triethanolamine
29221600	Diethanolammonium perfluorooctane sulphonate
29221700	Methyldiethanolamine and ethyldiethanolamine
29221800	2-"N,N-Diisopropylamino"ethanol
29221900	Amino-alcohols, their ethers and esters; salts thereof (other than those containing > one kind of oxygen function and excl. monoethanolamine, diethanolamine, dextropropoxyphene "INN", their salts, triethanolamine, diethanolammonium perfluorooctane sulphonate, methyldiethanolamine, ethyldiethanolamine and 2-(N,N-Diisopropylamino)ethanol)
29222100	Aminohydroxynaphthalenesulphonic acids and their salts
29222900	Amino-naphthols and other amino-phenols, their ethers and esters; salts thereof (excl. those containing > one kind of oxygen function; aminohydroxynaphthalenesulphonic acids and their salts)
29223100	Amfepramone "INN", methadone "INN" and normethadone "INN", and salts thereof
29223900	Amino-aldehydes, amino-ketones and amino-quinones; salts thereof (excl. those containing > one kind of oxygen function, and amfepramone "INN", methadone "INN" and normethadone "INN", and salts thereof)
29224300	Anthranilic acid and its salts
29224400	Tilidine "INN" and its salts
29224920	beta-Alanine
29224985	Amino-acids and their esters; salts thereof (excl. those containing > one kind of oxygen function, lysine and its esters, and salts thereof, and glutamic acid, anthranilic acid, tilidine "INN" and their salts and beta-alanine)
29225000	Amino-alcohol-phenols, amino-acid-phenols and other amino-compounds with oxygen function (excl. amino-alcohols, amino-naphthols and other amino-phenols, their ethers and esters and salts thereof, amino-aldehydes, amino-ketones and amino-quinones, and salts thereof, amino-acids and their esters and salts thereof)
29261000	Acrylonitrile
29263000	Fenproporex "INN" and its salts; methadone "INN"-intermediate "4-cyano-2-dimethylamino-4,4-diphenylbutane"
29264000	alpha-Phenylacetoacetonitrile
29269020	Isophthalonitrile
29269070	Nitrile-function compounds (excl. acrylonitrile, 1-cyanoguanidine "dicyandiamide", fenproporex "INN" and its salts, methadone "INN"-intermediate "4-cyano-2-dimethylamino-4,4-diphenylbutane", alpha-Phenylacetoacetonitrile and isophthalonitrile)
29280010	N,N-Bis"2-methoxyethyl"hydroxylamine
29280090	Organic derivatives of hydrazine or of hydroxylamine (excl. N,N-bis"2-methoxyethyl"hydroxylamine)
29291000	Isocyanates
29301000	2-(N,N-Dimethylamino) ethanethiol
29302000	Thiocarbamates and dithiocarbamates (excl. inorganic or organic compounds of mercury)
29303000	Thiuram mono-, di- or tetrasulphides
29304010	Methionine "INN"
29304090	Methionine (excl. methionine "INN")
29306000	2-(N,N-Diethylamino)ethanethiol
29307000	Bis(2-hydroxyethyl)sulfide (thiodiglycol (INN))

CN code	CN description
29308000	Aldicarb (ISO), Captafol (ISO) and methamidophos (ISO)
29309013	Cysteine and cystine
29309016	Derivatives of cysteine or of cystine
29309030	DL-2-hydroxy-4-"methylthio"butyric acid
29309040	2,2'-Thiodiethyl bis[3-"3,5-di-tert-butyl-4-hydroxyphenyl"propionate]
29309050	Mixture of isomers consisting of 4-methyl-2,6-bis"methylthio"-m-phenylenediamine and 2-methyl-4,6-bis"methylthio"-m-phenylenediamine
29309098	Organo-sulphur compounds (excl. thiocarbamates and dithiocarbamates, thiuram mono-, di- or tetrasulphides, methionine, aldicarb [ISO], captafol [ISO], methamidophos [ISO], cysteine or cystine and their derivatives, thiodiglycol [INN] [bis(2-hydroxyethyl)sulfide], DL-2-hydroxy-4-"methylthio"butyric acid, 2,2'-thiodiethyl bis[3-"3,5-di-tert-butyl-4-hydroxyphenyl"propionate], a mixture of isomers consisting of 4-methyl-2,6-bis"methylthio"-m-phenylenediamine and 2-methyl-4,6-bis"methylthio"-m-phenylenediamine, 2-(N,N-Diethylamino)ethanethiol and 2-(N,N-Dimethylamino) ethanethiol)
29311000	Tetramethyl lead and tetraethyl lead
29312000	Tributyltin compounds
29314100	Dimethyl methylphosphonate
29314200	Dimethyl propylphosphonate
29314300	Diethyl ethylphosphonate
29314400	Methylphosphonic acid
29314500	Salt of methylphosphonic acid and (aminoiminomethyl)urea (1: 1)
29314600	2,4,6-Tripropyl-1,3,5,2,4,6-trioxatriphosphinane 2,4,6-trioxide
29314700	(5-Ethyl-2-methyl-2-oxido-1,3,2-dioxaphosphinan-5-yl) methyl methyl methylphosphonate
29314800	3,9-Dimethyl-2,4,8,10-tetraoxa-3,9-diphosphaspiro[5.5] undecane 3,9-dioxide
29314910	Sodium 3-(trihydroxysilyl)propyl methylphosphonate
29314920	Bis[(5-ethyl-2-methyl-2-oxido-1,3,2-dioxaphosphinan-5-yl)methyl] methylphosphonate
29314930	Etidronic acid (INN) (1-hydroxyethane-1,1-diphosphonic acid) and its salts
29314940	(Nitrilotrimethanediyl)tris(phosphonic acid), {ethane-1,2-diylbis[nitrilobis(methylene)]}tetrakis(phosphonic acid), [(bis{2-[bis(phosphonomethyl)amino]ethyl}amino)methyl]phosphonic acid, {hexane-1,6-diylbis[nitrilobis(methylene)]}tetrakis(phosphonic acid), {[2-hydroxyethyl]imino}bis(methylene)}bis(phosphonic acid), and [(bis{6-[bis(phosphonomethyl)amino]hexyl}amino)methyl]phosphonic acid; salts thereof
29314990	Separate chemically defined non-halogenated organo-phosphorous derivatives, n.e.s.
29315100	Methylphosphonic dichloride
29315200	Propylphosphonic dichloride
29315300	O-(3-chloropropyl) O-[4-nitro-3-(trifluoromethyl)phenyl] methylphosphonothionate
29315400	Trichlorfon "ISO"
29315910	Methylphosphonoyl difluoride "methylphosphonic difluoride"
29315990	Separate chemically defined halogenated organo-phosphorous derivatives, n.e.s.
29319000	Separate chemically defined organo-inorganic compounds (excl. organo-sulphur, mercury, tetramethyl lead, tetraethyl lead and tributyltin compounds, and organo-phosphorous derivatives)
29321400	Sucralose
29321900	Heterocyclic compounds with oxygen hetero-atom[s] only, containing an unfused furan ring, whether or not hydrogenated, in the structure (excl. tetrahydrofuran, 2-furaldehyde "furfuraldehyde", furfuryl alcohol, tetrahydrofurfuryl alcohol and sucralose)
29329100	Isosafrole
29329200	1-"1,3-Benzodioxol-5-yl"propan-2-one

CN code	CN description
29329400	Safrole
29329500	Tetrahydrocannabinols "all isomers"
29329600	Carbofuran "ISO"
29329900	Heterocyclic compounds with oxygen hetero-atom[s] only (excl. compounds containing unfused furan ring, whether or not hydrogenated, in the structure, and lactones, isosafrole, 1-[1,3-benzodioxol-5-yl]propan-2-one, piperonal, safrole, tetrahydrocannabinols "all isomers", carbofuran, and inorganic or organic compounds of mercury)
29332910	Naphazoline hydrochloride "INNM" and naphazoline nitrate "INNM"; phentolamine "INN"; tolazoline hydrochloride "INNM"
29332990	Heterocyclic compounds with nitrogen hetero-atom[s] only, containing an unfused imidazole ring, whether or not hydrogenated, in the structure (excl. hydantoin and its derivatives, naphazoline hydrochloride "INNM", naphazoline nitrate "INNM", phentolamine "INN" and tolazoline hydrochloride "INNM")
29333100	Pyridine and its salts
29333200	Piperidine and its salts
29333300	Alfentanil "INN", anileridine "INN", bezitramide "INN", bromazepam "INN", carfentanil "INN", difenoxin "INN", diphenoxylate "INN", dipipanone "INN", fentanyl "INN", ketobemidone "INN", methylphenidate "INN", pentazocine "INN", pethidine "INN", pethidine "INN" intermediate A, phencyclidine "INN" "PCP", phenoperidine "INN", pipradol "INN", piritramide "INN", propiram "INN", remifentanyl "INN", and trimeperidine "INN", and salts thereof
29333400	Fentanyls and their derivatives, with nitrogen hetero-atom[s] only, containing an unfused pyridine ring, whether or not hydrogenated, in the structure (excl. products of 2933.33)
29333500	3-Quinuclidinol
29333600	4-Anilino-N-phenethylpiperidine "ANPP"
29333700	N-Phenethyl-4-piperidone "NPP"
29333910	lproniazid "INN"; ketobemidone hydrochloride "INNM"; pyrodoxigmine bromide "INN"
29333920	2,3,5,6-Tetrachloropyridine
29333925	3,6-Dichloropyridine-2-carboxylic acid
29333935	2-Hydroxyethylammonium-3,6-dichloropyridine-2-carboxylate
29333940	2-Butoxyethyl"3,5,6-trichloro-2-pyridyloxy"acetate
29333945	3,5-Dichloro-2,4,6-trifluoropyridine
29333950	Fluroxypyr "ISO" methyl ester
29333955	4-Methylpyridine
29333999	Heterocyclic compounds with nitrogen hetero-atom[s] only, containing an unfused pyridine ring, whether or not hydrogenated, in the structure, n.e.s.
29334100	Levorphanol "INN" and its salts
29334910	Halogen derivatives of quinoline; quinolinecarboxylic acid derivatives
29334930	Dextromethorphan "INN" and its salts
29334990	Heterocyclic compounds with nitrogen hetero-atom[s] only, containing in the structure a quinoline or isoquinoline ring-system, whether or not hydrogenated, but not further fused (excl. levorphanol "INN", dextromethorphan "INN", and their salts, halogen derivatives of quinoline, quinolinecarboxylic acid derivatives, and inorganic or organic compounds of mercury)
29336100	Melamine
29337100	6-Hexanelactam "epsilon-caprolactam"
29337200	Clobazam "INN" and methypylon "INN"
29337900	Lactams (excl. 6-hexanelactam "epsilon-caprolactam", clobazam "INN", methypylon "INN", and inorganic or organic compounds of mercury)
29339110	Chlorodiazepoxide "INN"
29339190	Alprazolam "INN", camazepam "INN", clonazepam "INN", clorazepate, delorazepam "INN", diazepam "INN", estazolam "INN", ethyl loflazepate "INN", fludiazepam "INN", flunitrazepam "INN", flurazepam "INN", halazepam "INN",

CN code	CN description
	lorazepam "INN", lormetazepam "INN", mazindol "INN", medazepam "INN", midazolam "INN", nimetazepam "INN", nitrazepam "INN", nordazepam "INN", oxazepam "INN", pinazepam "INN", prazepam "INN", pyrovalerone "INN", temazepam "INN", tetrazepam "INN" and triazolam "INN", salts thereof, and salts of chlordiazepoxide "INN"
29339200	Azinphos-methyl "ISO"
29339920	Indole, 3-methylindole "skatole", 6-allyl-6,7-dihydro-5H-dibenz"c,e"azepine "azapetine", phenindamine "INN" and their salts; imipramine hydrochloride "INNM"
29339950	2,4-Di-tert-butyl-6-"5-chlorobenzotriazol-2-yl"phenol
29339980	Heterocyclic compounds with nitrogen hetero-atom[s] only (excl. those containing an unfused pyrazole, imidazole, pyridine or triazine ring, whether or not hydrogenated, a quinoline or isoquinoline ring-system, not further fused, whether or not hydrogenated, a pyrimidine ring, whether or not hydrogenated, or piperazine ring in the structure, lactams, alprazolam "INN", camazepam "INN", chlordiazepoxide "INN", clonazepam "INN", clorazepate, delorazepam "INN", diazepam "INN", estazolam "INN", ethyl loflazepate "INN", fludiazepam "INN", flunitrazepam "INN", flurazepam "INN", halazepam "INN", lorazepam "INN", lormetazepam "INN", mazindol "INN", medazepam "INN", midazolam "INN", nimetazepam "INN", nitrazepam "INN", nordazepam "INN", oxazepam "INN", pinazepam "INN", prazepam "INN", pyrovalerone "INN", temazepam "INN", tetrazepam "INN", triazolam "INN", salts thereof, indole, 3-methylindole "skatole", 6-allyl-6,7-dihydro-5H-dibenz"c,e"azepine "azapetine", phenindamine "INN" and their salts, imipramine hydrochloride "INNM", 2,4-di-tert-butyl-6-"5-chlorobenzotriazol-2-yl"phenol and azinphos-methyl "ISO")
29341000	Heterocyclic compounds containing an unfused thiazole ring, whether or not hydrogenated, in the structure
29342020	Di"benzothiazol-2-yl"disulphide; benzothiazol-2-thiol "mercaptobenzothiazole" and its salts
29342080	Heterocyclic compounds containing in the structure a benzothiazole ring-system, whether or not hydrogenated, but not further fused (excl. di"benzothiazol-2-yl"disulphide; benzothiazole-2-thiol "mercaptobenzothiazole" and its salts, and inorganic or organic compounds of mercury)
29349100	Aminorex "INN", brotizolam "INN", clotiazepam "INN", cloxazolam "INN", dextromoramide "INN", haloxazolam "INN", ketazolam "INN", mesocarb "INN", oxazolam "INN", pemoline "INN", phendimetrazine "INN", phenmetrazine "INN" and sufentanil "INN", and salts thereof
29349200	Fentanyl and their derivatives (excl. with nitrogen hetero-atom[s] only, and products of 2934.91)
29349960	Chlorprothixene (INN); thenalidine (INN) and its tartrates and maleates; furazolidone (INN); 7-aminocephalosporanic acid; salts and esters of (6R,7R)-3-acetoxymethyl-7-[(R)-2-formyloxy-2-phenylacetamido]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid; 1-[2-(1,3-Dioxan-2-yl)ethyl]-2-methylpyridinium bromide
29349990	Nucleic acids and their salts, whether or not chemically defined; heterocyclic compounds (excl. those with oxygen or nitrogen hetero-atom's" only, compounds containing in the structure an unfused thiazole ring or a benzothiazole or phenothiazine ring-system or further fused, aminorex "INN", brotizolam "INN", clotiazepam "INN", cloxazolam "INN", dextromoramide "INN", haloxazolam "INN", ketazolam "INN", mesocarb "INN", oxazolam "INN", pemoline "INN", phendimetrazine "INN", phenmetrazine "INN", sufentanil "INN", salts thereof, other fentanyl and their derivatives, chlorprothixene "INN", thenalidine "INN" and its tartrates and maleates, furazolidone "INN", 7-aminocephalosporanic acid, salts and esters of "6R, 7R"-3-acetoxymethyl-7-["R"-2-formyloxy-2-phenylacetamido]-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylic acid, 1-[2-"1,3-dioxan-2-yl"ethyl]-2-methylpyridinium bromide, and inorganic or organic compounds of mercury)
29398000	Non-vegetal alkaloids, natural or reproduced by synthesis, and their salts, ethers, esters and other derivatives
35079030	Lipoprotein lipase and aspergillus alkaline protease

CN code	CN description
35079090	Enzymes and prepared enzymes, n.e.s. (excl. rennet and concentrates thereof, lipoprotein lipase and Aspergillus alkaline protease)
38030010	Crude tall oil
38030090	Tall oil, whether or not refined (excl. crude tall oil)
38061000	Rosin and resin acids
38062000	Salts of rosin, of resin acids or of derivatives of rosin or resin acids (excl. salts of rosin adducts)
38063000	Ester gums
38069000	Derivatives of rosin, incl. salts of rosin adducts, and of resin acids, light and heavy resin oils and modified natural resins obtained by heat treatment "run gums" (excl. salts of rosin, of resin acids or salts of derivatives of rosin or resin acids, and ester gums)
38231100	Stearic acid, industrial
38231910	Fatty acids, distilled
38231930	Fatty acid distillate
38231990	Fatty acids, industrial, monocarboxylic; acid oils from refining (excl. stearic acid, oleic acid and tall oil fatty acids, distilled fatty acids and fatty acid distillate)
38237000	Fatty alcohols, industrial
39011010	Linear polyethylene with a specific gravity of < 0,94, in primary forms
39011090	Polyethylene with a specific gravity of < 0,94, in primary forms (excl. linear polyethylene)
39012010	Polyethylene in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms, of a specific gravity of $\geq 0,958$ at 23°C, containing ≤ 50 mg/kg of aluminium, ≤ 2 mg/kg of calcium, of chromium, of iron, of nickel and of titanium each and ≤ 8 mg/kg of vanadium, for the manufacture of chlorosulphonated polyethylene
39012090	Polyethylene with a specific gravity of $\geq 0,94$, in primary forms (excl. polyethylene in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms, of a specific gravity of $\geq 0,958$ at 23°C, containing ≤ 50 mg/kg of aluminium, ≤ 2 mg/kg of calcium, of chromium, of iron, of nickel and of titanium each and ≤ 8 mg/kg of vanadium, for the manufacture of chlorosulphonated polyethylene)
39013000	Ethylene-vinyl acetate copolymers, in primary forms
39014000	Ethylene-alpha-olefin copolymers, having a specific gravity of < 0,94 , in primary forms
39019030	Ionomer resin consisting of a salt of a terpolymer of ethylene with isobutyl acrylate and methacrylic acid, in primary forms, and A-B-A block copolymer of ethylene of polystyrene, ethylene-butylene copolymer and polystyrene, containing by weight $\leq 35\%$ of styrene, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms
39019080	Polymers of ethylene, in primary forms (excl. polyethylene, ethylene-vinyl acetate copolymers, ethylene-alpha-olefins copolymers having a specific gravity of < 0,94, ionomer resin consisting of a salt of a terpolymer of ethylene with isobutyl acrylate and methacrylic acid and A-B-A block copolymer of ethylene of polystyrene, ethylene-butylene copolymer and polystyrene, containing by weight $\leq 35\%$ of styrene, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms)
39021000	Polypropylene, in primary forms
39022000	Polyisobutylene, in primary forms
39023000	Propylene copolymers, in primary forms
39029010	A-B-A block copolymer of propylene or of other olefins, of polystyrene, ethylene-butylene copolymer and polystyrene, containing by weight $\leq 35\%$ of styrene, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms
39029020	Polybut-1-ene, a copolymer of but-1-ene with ethylene containing by weight $\leq 10\%$ of ethylene, or a blend of polybut-1-ene with polyethylene and/or polypropylene containing by weight $\leq 10\%$ of polyethylene and/or $\leq 25\%$ of

CN code	CN description
	polypropylene, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms
39029090	Polymers of propylene or of other olefins, in primary forms (excl. polypropylene, polyisobutylene, propylene copolymers, and a A-B-A block copolymer of polystyrene, ethylene-butylene copolymer and polystyrene, containing by weight $\leq$ 35% of styrene and polybut-1-ene, a copolymer of but-1-ene with ethylene containing by weight $\leq$ 10% of ethylene, or a blend of polybut-1-ene with polyethylene and/or polypropylene containing by weight $\leq$ 10% of polyethylene and/or $\leq$ 25% of polypropylene, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms)
39031900	Polystyrene, in primary forms (excl. expansible)
39033000	Acrylonitrile-butadiene-styrene copolymers "ABS", in primary forms
39039010	Copolymer solely of styrene with allyl alcohol, of an acetyl value of $\geq$ 175, in primary form
39039020	Brominated polystyrene containing by weight $\geq$ 58% but $\leq$ 71% of bromine, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms
39039090	Polymers of styrene, in primary forms (excl. polystyrene, styrene-acrylonitrile copolymers "SAN", acrylonitrile-butadiene-styrene "ABS", copolymer solely of styrene with allyl alcohol, of an acetyl value of $\geq$ 175 and brominated polystyrene, containing by weight $\geq$ 58% but $\leq$ 71% of bromine, in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms)
39041000	Poly"vinyl chloride", in primary forms, not mixed with any other substances
39046100	Polytetrafluoroethylene, in primary forms
39046910	Poly"vinyl fluoride" in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms
39046920	Fluoroelastomers FKM, in primary forms
39046980	Fluoropolymers of vinyl chloride or of other halogenated olefins, in primary forms (excl. fluoroelastomers FKM, polytetrafluoroethylene, poly(vinyl fluoride) in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms)
39051900	Poly"vinyl acetate", in primary forms (excl. in aqueous dispersion)
39052900	Vinyl acetate copolymers, in primary forms (excl. in aqueous dispersion)
39053000	Poly"vinyl alcohol", in primary forms, whether or not containing unhydrolyzed acetate groups
39059100	Copolymers of vinyl, in primary forms (excl. vinyl chloride-vinyl acetate copolymers and other vinyl chloride copolymers, and vinyl acetate copolymers)
39059910	Poly"vinyl formal" in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms, of a molecular weight of $\geq$ 10.000 but $\leq$ 40.000 and containing by weight $\geq$ 9,5% but $\leq$ 13% of acetyl groups evaluated as vinyl acetate and $\geq$ 5% but $\leq$ 6,5% of hydroxy groups evaluated as vinyl alcohol
39059990	Polymers of vinyl esters and other vinyl polymers, in primary forms (excl. those of vinyl chloride or other halogenated olefins, poly"vinyl acetate", copolymers and poly"vinyl alcohol", whether or not containing unhydrolysed acetate groups, and poly"vinyl formal" in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms, of a molecular weight of $\geq$ 10.000 but $\leq$ 40.000 and containing by weight $\geq$ 9,5% but $\leq$ 13% of acetyl groups evaluated as vinyl acetate and $\geq$ 5% but $\leq$ 6,5% of hydroxy groups evaluated as vinyl alcohol)
39061000	Poly"methyl methacrylate", in primary forms
39069010	Poly[N-"3-hydroxyimino-1,1-dimethylbutyl"acrylamide], in primary forms
39069020	Copolymer of 2-diisopropylaminoethyl methacrylate with decyl methacrylate in the form of a solution in N,N-dimethylacetamide, containing by weight $\geq$ 55% of copolymer
39069030	Copolymer of acrylic acid with 2-ethylhexyl acrylate containing by weight $\geq$ 10% but $\leq$ 11% of 2-ethylhexyl acrylate, in primary forms
39069040	Copolymer of acrylonitrile with methyl acrylate, modified with polybutadiene-acrylonitrile "NBR", in primary forms
39069050	Polymerization product of acrylic acid with alkyl methacrylate and small quantities of other monomers, for use as a thickener in the manufacture of textile printing pastes

CN code	CN description
39069060	Copolymer of methyl acrylate with ethylene and a monomer containing a non-terminal carboxy group as a substituent, containing by weight $\geq 50\%$ of methyl acrylate, whether or not compounded with silica, in primary forms
39069090	Acrylic polymers in primary forms (excl. poly(methyl methacrylate), poly[N-(3-hydroxyimino-1,1-dimethylbutyl)acrylamide], copolymer of 2-diisopropylaminoethyl methacrylate with decyl methacrylate in the form of a solution in N,N-dimethylacetamide, containing by weight $\geq 55\%$ of copolymer, copolymer of acrylic acid with 2-ethylhexyl acrylate containing by weight $\geq 10\%$ but $\leq 11\%$ of 2-ethylhexyl acrylate, copolymer of acrylonitrile with methyl acrylate modified with polybutadiene-acrylonitrile "NBR", polymerization product of acrylic acid with alkyl methacrylate and small quantities of other monomers for use as a thickener in the manufacture of textile printing pastes and copolymer of methyl acrylate with ethylene and a monomer containing a non-terminal carboxy group as a substituent, containing by weight $\geq 50\%$ of methyl acrylate, whether or not compounded with silica)
39071000	Polyacetals, in primary forms
39072100	Bis(polyoxyethylene) methylphosphonate, in primary forms
39072911	Polyethylene glycols, in primary forms
39072920	Polyether alcohols, in primary forms (excl. bis(polyoxyethylene) methylphosphonate and polyethylene glycols)
39072991	Copolymer of 1-chloro-2,3-epoxypropane with ethylene oxide, in primary forms
39072999	Polyethers in primary forms (excl. polyether alcohols, polyacetals and copolymer of 1-chloro-2,3-epoxypropane with ethylene oxide)
39073000	Epoxide resins, in primary forms
39074000	Polycarbonates, in primary forms
39076100	Poly(ethylene terephthalate), in primary forms, having a viscosity number of ≥ 78 ml/g
39076900	Poly(ethylene terephthalate), in primary forms, having a viscosity number of < 78 ml/g
39077000	Poly(lactic acid), in primary forms
39079110	Unsaturated liquid polyesters, in primary forms (excl. polycarbonates, alkyd resins, poly(ethylene terephthalate) and poly(lactic acid))
39079905	Thermoplastic liquid crystal aromatic polyester copolymers, saturated, in primary forms
39079910	Poly(ethylene naphthalene-2,6-dicarboxylate), saturated, in primary forms
39079980	Polyesters, saturated, in primary forms (excl. polycarbonates, alkyd resins, poly(ethylene terephthalate), poly(lactic acid), poly(ethylene naphthalene-2,6-dicarboxylate) and thermoplastic liquid crystal aromatic polyester copolymers)
39081000	Polyamides-6, -11, -12, -6,6, -6,9, -6,10 or -6,12, in primary forms
39089000	Polyamides, in primary forms (excl. polyamides-6, -11, -12, -6,6, -6,9, -6,10 and -6,12)
39093100	Poly(methylene phenyl isocyanate) "crude MDI, polymeric MDI", in primary forms
39093900	Amino-resins, in primary forms (excl. urea, thiourea and melamine resins and MDI)
39095010	Polyurethane of 2,2'-(tert-butylimino)diethanol and 4,4'-methylenedicyclohexyl diisocyanate, in the form of a solution in N,N-dimethylacetamide, containing by weight $\geq 50\%$ of polymer
39095090	Polyurethanes in primary forms (excl. polyurethane of 2,2'-(tert-butylimino)diethanol and 4,4'-methylenedicyclohexyl diisocyanate, in the form of a solution in N,N-dimethylacetamide)
39100000	Silicones in primary forms
39111000	Petroleum resins, coumarone, indene or coumarone-indene resins and polyterpenes, in primary forms
39112000	Poly(1,3-phenylene methylphosphonate), in primary forms
39119011	Poly(oxy-1,4-phenylenesulphonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene) in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms, whether or not chemically modified

CN code	CN description
39119013	Poly"thio-1,4-phenylene", whether or not chemically modified, in primary forms
39119019	Condensation or rearrangement polymerization products, whether or not chemically modified, n.e.s., in primary forms (excl. poly"oxy-1,4-phenylenesulphonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene" in blocks of irregular shape, lumps, powders, granules, flakes and similar bulk forms and poly"thio-1,4-phenylene)
39119092	Hydrogenated copolymers of vinyltoluene and alfa-methylstyrene, and copolymer of p-cresol and divinylbenzene, in the form of a solution in N,N-dimethylacetamide containing by weight 50 % or more of polymer, produced by chemical synthesis, in primary forms
39119099	Polymer and prepolymer plastics produced by chemical synthesis, n.e.s., in primary forms (excl. copolymer of p-cresol and divinylbenzene in the form of a solution in N,N-dimethylacetamide containing by weight >= 50% of polymer, hydrogenated copolymers of vinyltoluene and alfa-methylstyrene, and poly(1,3-phenylene methylphosphonate))
39121100	Non-plasticised cellulose acetates, in primary forms
39121200	Plasticised cellulose acetates, in primary forms
39122011	Non-plasticised collodions and celloidin, in primary forms
39122019	Non-plasticised cellulose nitrates, in primary forms (excl. collodions and celloidin)
39122090	Plasticised cellulose nitrates, incl. collodions, in primary forms
39123920	Hydroxypropylcellulose in primary forms
39123985	Cellulose ethers in primary forms (excl. carboxymethylcellulose and its salts and hydroxypropylcellulose)
39129010	Cellulose esters, in primary forms
39129090	Cellulose and chemical derivatives thereof, n.e.s., in primary forms (excl. cellulose acetates, cellulose nitrates, cellulose ethers and cellulose esters)
39140000	Ion-exchangers based on polymers of heading 3901 to 3913, in primary forms
40021910	Styrene-butadiene rubber produced by emulsion polymerisation "E-SBR", in bales
40021920	Styrene-butadiene-styrene block copolymers produced by solution polymerisation "SBS, thermoplastic elastomers", in granules, crumbs or powders
40021930	Styrene-butadiene rubber produced by solution polymerisation "S-SBR", in bales
40021990	Styrene-butadiene rubber "SBR" and carboxylated styrene-butadiene rubber "XSBR", in primary forms or in plates, sheets or strip (excl. E-SBR and S-SBR in bales, SBS thermoplastic elastomers in granules, crumbs or powder and latex)
40022000	Butadiene rubber "BR", in primary forms or in plates, sheets or strip
40023100	Isobutylene isoprene rubber "IIR", in primary forms or in plates, sheets or strip
40023900	Halo-isobutene-isoprene rubber "CIIR" or "BIIR", in primary forms or in plates, sheets or strip
40024900	Chloroprene "chlorobutadiene rubber, CR", in primary forms or in plates, sheets or strip (excl. latex)
40025900	Acrylonitrile-butadiene rubber "NBR", in primary forms or in plates, sheets or strip (excl. latex)
40026000	Isoprene rubber "IR", in primary forms or in plates, sheets or strip
40027000	Ethylene-propylene diene rubber "EPDM", non-conjugated, in primary forms or in plates, sheets or strip
40028000	Mixtures of natural rubber, balata, gutta-percha, guayule, chicle or similar types of natural rubber with synthetic rubber or factice, in primary forms or in plates, sheets or strip
40029910	Natural rubber products modified by the incorporation of plastics (excl. depolymerised natural rubber)
40029990	Synthetic rubber and factice derived from oils, in primary forms or plates, sheets or strip (excl. latex; styrene-butadiene "SBR", carboxylated styrene-butadiene "XSBR", butadiene "BR" butyl "IIR", halo-isobutene-isoprene "CIIR" or "BIIR", chloroprene -chlorobutadiene- "CR", acrylonitrile-butadiene "NBR", isoprene "IR" or ethylene-propylene-non-conjugated diene "EPDM" rubber; products modified by the incorporation of plastics)

CN code	CN description
44021000	Bamboo charcoal, incl. shell or nut charcoal, whether or not agglomerated (excl. used as a medicament, mixed with incense, activated bamboo charcoal and in the form of crayons)
44022000	Wood charcoal of shell or nut, whether or not agglomerated (excl. of bamboo, used as a medicament, mixed with incense, activated charcoal and in the form of crayons)
44029000	Wood charcoal, whether or not agglomerated (excl. of bamboo or shell or nut, charcoal used as a medicament, charcoal mixed with incense, activated charcoal and charcoal in the form of crayons)

Table 61 PRODCOM and CN codes of Stage 2 shortlisted products in the fertiliser sector

PRODCOM code	PRODCOM Description	CN code	CN description
20154100	Superphosphates (excluding potassic, in tablets or similar forms or in packages of a weight of <= 10 kg)	31031100	Superphosphates containing by weight => 35 % of diphosphorus pentaoxide "P2O5" (excl. such products in tablets or similar forms, or in packages with a gross weight of <= 10 kg)
		31031900	Superphosphates (excl. such products containing by weight => 35 % of diphosphorus pentaoxide, or in tablets or similar forms, or in packages with a gross weight of <= 10 kg)
20155100	Potassium chloride (excluding in tablets or similar forms or in packages of a weight of <= 10 kg)	31042010	Potassium chloride containing <= 40% potassium monoxide in relation to the weight of the dry product (excl. that in tablets or similar forms, or in packages with a gross weight of <= 10 kg)
		31042050	Potassium chloride containing > 40% but <= 62% potassium monoxide in relation to the weight of the dry product (excl. that in tablets or similar forms, or in packages with a gross weight of <= 10 kg)
		31042090	Potassium chloride containing > 62% potassium monoxide in relation to the weight of the dry product (excl. that in tablets or similar forms, or in packages with a gross weight of <= 10 kg)

Table 62 PRODCOM and CN codes of Stage 2 shortlisted products in the glass sector

PRODCOM code	PRODCOM Description	CN code	CN description
23111290	Other sheets of float/ground/polished glass, n.e.c.	70052925	Float glass and surface ground and polished glass, in sheets, but not otherwise worked, of a thickness of <= 3,5 mm (excl. wired glass or glass coloured throughout the mass "body tinted", opacified, flashed or merely surface ground, or glass having an absorbent, reflecting or non-reflecting layer)
		70052935	Float glass and surface ground and polished glass, in sheets, but not otherwise worked, of a thickness of > 3,5 mm but <= 4,5 mm (excl.

PRODCOM code	PRODCOM Description	CN code	CN description
			wired glass or glass coloured throughout the mass "body tinted", opacified, flashed or merely surface ground, or glass having an absorbent, reflecting or non-reflecting layer)
		70052980	Float glass and surface ground and polished glass, in sheets, but not otherwise worked, of a thickness of > 4,5 mm (excl. horticultural glass or wired glass or glass coloured throughout the mass "body tinted", opacified, flashed or merely surface ground, or glass having an absorbent, reflecting or non-reflecting layer)
		70053000	Float glass and surface ground and polished glass, in sheets, whether or not having an absorbent, reflecting or non-reflecting layer, wired, but not otherwise worked
23131140	Bottles of colourless glass of a nominal capacity < 2,5 litres, for beverages and foodstuffs (excluding bottles covered with leather or composition leather, infant's feeding bottles)	70109041	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of >= 1 l but < 2,5 l
		70109043	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of > 0,33 l but < 1 l
		70109045	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of >= 0,15 l to 0,33 l
		70109047	Bottles of colourless glass, of a kind used for the commercial conveyance or packing of foodstuffs and beverages, of a nominal capacity of < 0,15 l
23131170	Glass containers for pharmaceutical products of a nominal capacity < 2,5 litres	70109071	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of > 0,055 l but < 2,5 l (excl. ampoules, containers made from tubing, glass inners for containers, with vacuum insulation)
		70109079	Bottles, flasks, phials and other containers, of glass, of a kind used for the commercial conveyance or packing of pharmaceutical products, of a nominal capacity of <= 0,055 l (excl. ampoules, containers made from tubing, glass inners for containers, with vacuum insulation)
23131180	Glass containers of a nominal capacity < 2,5 litres for the conveyance	70109091	Carboys, bottles, flasks, jars, pots, phials and other containers, of colourless glass, of a kind used for the commercial conveyance or

PRODCOM code	PRODCOM Description	CN code	CN description
	or packing of goods (excluding for beverages and foodstuffs, for pharmaceutical products, containers made from glass tubing)		packing of goods, of a nominal capacity of < 2,5 l (excl. containers for foodstuffs, beverages or pharmaceutical products, ampoules, containers made from tubing, glass inners for containers, with vacuum insulation, perfume atomizers, flasks, bottles etc. for atomizers)
		70109099	Carboys, bottles, flasks, jars, pots, phials and other containers, of coloured glass, of a kind used for the commercial conveyance or packing of goods, of a nominal capacity of < 2,5 l (excl. containers for foodstuffs, beverages or pharmaceutical products, ampoules, containers made from tubing, glass inners for containers, with vacuum insulation, perfume atomizers, flasks, bottles etc. for atomizers)
		70132810	Drinking glasses, stemware, gathered by hand (excl. of glass ceramics or of lead crystal)
		70132890	Drinking glasses, stemware, gathered mechanically (excl. of glass ceramics or of lead crystal)
		70133751	Drinking glasses, gathered by hand, cut or otherwise decorated (excl. glasses of glass ceramics, lead crystal or toughened glass and stemware)
23131290	Other drinking glasses	70133759	Drinking glasses, gathered by hand (excl. glasses cut or otherwise decorated, or of glass ceramics, lead crystal or toughened glass and stemware)
		70133791	Drinking glasses, gathered mechanically, cut or otherwise decorated (excl. glasses of glass ceramics, lead crystal or toughened glass and stemware)
		70133799	Drinking glasses, gathered mechanically (excl. glasses cut or otherwise decorated, or of glass ceramics, lead crystal or toughened glass and stemware)
23131390	Table/kitchen glassware (excluding drinking), toughened glass	70134910	Glassware for table or kitchen purposes, of toughened glass (excl. glass having a linear coefficient of expansion $\leq 5 \times 10^{-6}$ per kelvin within a temperature range of 0 to 300°C, glassware of glass ceramics or lead crystal, articles of heading 7018, drinking glasses, glass preserving jars "sterilising jars", vacuum flasks and other vacuum vessels)
		70134991	Glassware of a kind used for table or kitchen purposes, gathered by hand (excl.

PRODCOM code	PRODCOM Description	CN code	CN description
			toughened glass and glass having a linear coefficient of expansion $\leq 5 \times 10^{-6}$ per kelvin within a temperature range of 0 to 300°C, glassware of glass ceramics or lead crystal, articles of heading 7018, drinking glasses, glass preserving jars "sterilising jars", vacuum flasks and other vacuum vessels)
		70134999	Glassware of a kind used for table or kitchen purposes, gathered mechanically (excl. toughened glass and glass having a linear coefficient of expansion $\leq 5 \times 10^{-6}$ per kelvin within a temperature range of 0 to 300°C, glassware of glass ceramics or lead crystal, articles of heading 7018, drinking glasses, glass preserving jars "sterilising jars", vacuum flasks and other vacuum vessels)
		70139900	Glassware of a kind used for toilet, office, indoor decoration or similar purposes (excl. glassware of lead crystal or of a kind used for table or kitchen purposes, articles of heading 7018, mirrors, leaded lights and the like, lighting fittings and parts thereof, atomizers for perfume and the like)
23141110	Glass fibre threads cut into lengths of at least 3 mm but ≤ 50 mm (chopped strands)	70191100	Glass fibre threads "chopped strands", cut into lengths ≤ 50 mm
23141130	Glass fibre filaments (including rovings)	70191200	Rovings of glass fibres
23141151	Yarn and slivers of glass fibres (excl. in chopped strands of a length of ≤ 50 mm, and rovings) and glass fibre cakes "textile cakes" and chopped glass fibre strands of a length >50 mm	70191300	Yarn and slivers of glass fibres (excl. in chopped strands of a length of ≤ 50 mm, and rovings)
		70191900	Glass fibre cakes "textile cakes" and chopped glass fibre strands of a length >50 mm
23141296	Glass fibres and articles thereof (excl. staple fibres, rovings, yarn, chopped strands, woven fabrics, also narrow fabrics, thin sheets "voiles", webs, mats, mattresses and boards and similar nonwoven products, stone or slag wool and articles thereof, electrical insulators or parts thereof, optical	70199000	Glass fibres and articles thereof, n.e.s.

PRODCOM code	PRODCOM Description	CN code	CN description
	fibres, fibre bundles or cable, brushes of glass fibres, and dolls' wigs, glass wool and articles of glass wool)		
23192330	Laboratory, hygienic or pharmaceutical glassware whether or not graduated	70171000	Laboratory, hygienic or pharmaceutical glassware, whether or not graduated or calibrated, of fused quartz or other fused silica (excl. containers for the conveyance or packing of goods, measuring, checking or medical instruments and apparatus of chapter 90)
		70172000	Laboratory, hygienic or pharmaceutical glassware, whether or not graduated or calibrated, having a linear coefficient of expansion $\leq 5 \times 10^{-6}$ per kelvin within a temperature range of 0°C to 300°C (excl. glass of fused quartz or other fused silica, containers for the conveyance or packing of goods, measuring, checking or medical instruments and apparatus of chapter 90)
		70179000	Laboratory, hygienic or pharmaceutical glassware, whether or not graduated or calibrated (excl. glass having a linear coefficient of expansion $\leq 5 \times 10^{-6}$ per kelvin within a temperature range of 0°C to 300°C or of fused quartz or other fused silica, containers for the conveyance or packing of goods, measuring, checking or medical instruments and apparatus of chapter 90)
23192640	Signalling glassware and optical elements of glass, not optically worked; glass cubes and other glass smallware, for mosaic or similar decorative purposes (excluding finished panels and other decorative motifs made from mosaic cubes)	70140000	Signalling glassware and optical elements of glass, not optically worked (excl. clock or watch glasses, glasses for non-corrective or corrective spectacles, incl. hollow glass spheres and their segments, for the manufacture of such glasses, microspheres, loose, lighting fittings and parts thereof)
		70161000	Glass cubes and other glass smallware, whether or not on a backing, for mosaics or similar decorative purposes (excl. finished panels and other finished decorative motifs, made from glass cubes for mosaics)
23192670	Glass smallware (including beads, imitation pearls/stones, etc.)	70181011	Glass beads, cut and mechanically polished (excl. articles thereof)
		70181019	Glass beads (excl. beads, cut and mechanically polished, and articles thereof)
		70181030	Imitation pearls of glass (excl. articles thereof)

PRODCOM code	PRODCOM Description	CN code	CN description
23192690	Other articles of glass, n.e.c.	70181051	Imitation precious and semi-precious stones of glass, cut and mechanically polished (excl. articles thereof)
		70181059	Imitation precious and semi-precious stones of glass (excl. beads, cut and mechanically polished, and articles thereof)
		70181090	Imitation coral and similar glass smallware (excl. articles thereof and imitation pearls, precious and semi-precious stones)
		70182000	Glass microspheres <= 1 mm in diameter
		70189010	Glass eyes, articles of glass beads, or of imitation pearls, imitation precious or semi-precious stones, or of other glass smallware (excl. prosthetic articles and imitation jewellery)
		70189090	Statuettes and other ornaments of lamp-worked glass (excl. imitation jewellery)
		70200005	Quartz reactor tubes and holders designed for insertion into diffusion and oxidation furnaces for production of semiconductor materials
		70200010	Glassware of fused quartz or other fused silica, n.e.s.
		70200030	Articles of glass having a linear coefficient of expansion <= 5 x 10 ⁻⁶ per kelvin within a temperature range of 0°C to 300°C, n.e.s. (excl. glassware of fused quartz or other fused silica)
		70200080	Articles of glass, n.e.s.

Table 63 PRODCOM and CN codes of Stage 2 shortlisted products in the ceramics sector

PRODCOM code	PRODCOM Description	CN code	CN description
23311000	Ceramic tiles and flags	69072100	Ceramic flags and paving, hearth or wall tiles, of a water absorption coefficient by weight <= 0,5 % (excl. refractory, mosaic cubes and finishing ceramics)
		69072200	Ceramic flags and paving, hearth or wall tiles, of a water absorption coefficient by weight > 0,5 % but <= 10 % (excl. refractory, mosaic cubes and finishing ceramics)
		69072300	Ceramic flags and paving, hearth or wall tiles, of a water absorption coefficient by weight > 10 % (excl. refractory, mosaic cubes and finishing ceramics)

PRODCOM code	PRODCOM Description	CN code	CN description
23201237	Refractory bricks, blocks..., weight > 50 % Al ₂ O ₃ and/or SiO ₂ : others	69073000	Ceramic mosaic cubes and the like, whether or not on a backing (excl. refractory and finishing ceramics)
		69074000	Finishing ceramics (excl. refractory)
		69022099	Refractory bricks, blocks, tiles and similar refractory ceramic constructional goods containing, by weight, > 50% alumina, silica or a mixture or compound of these products (excl. containing, by weight, >= 93% silica, > 7% but < 45% alumina and those of siliceous fossil meals or similar siliceous earths)
23201411	Articles of stone or of other mineral substances, n.e.c. containing magnesite, magnesia in the form of periclase, dolomite including in the form of dolime, or chromite	68159100	Articles of stone or other mineral substances, n.e.s. containing magnesite, magnesia in the form of periclase, dolomite incl. in the form of dolime, or chromite
23201490	Refractory ceramic goods, n.e.c.	69039090	Retorts, crucibles, mufflers, nozzles, plugs, supports, cupels, tubes, pipes, sheaths, rods, slide gates and other refractory ceramic goods (excl. those of siliceous fossil meals or of similar siliceous earths, articles of heading 6902, articles containing carbon, alumina or silica of subheading 6903.10.00 and 6903.90.10)
23411230	Ceramic tableware, other household articles : stoneware	69120023	Tableware and kitchenware, of stoneware (excl. statuettes and other ornamental articles, pots, jars, carboys and similar receptacles for the conveyance or packing of goods, and coffee grinders and spice mills with receptacles made of ceramics and working parts of metal)
		69120083	Household articles and toilet articles, of stoneware (excl. tableware, kitchenware, baths, bidets, sinks and similar sanitary fixtures, statuettes and other ornamental articles, pots, jars, carboys and similar receptacles for the conveyance or packing of goods)
23421030	Ceramic sinks, etc. and other sanitary fixtures, of porcelain or china	69101000	Ceramic sinks, washbasins, washbasin pedestals, baths, bidets, water closet pans, flushing cisterns, urinals and similar sanitary fixtures of porcelain or china (excl. soap dishes, sponge holders, tooth-brush holders, towel hooks, and toilet paper holders)

PRODCOM code	PRODCOM Description	CN code	CN description
23421050	Ceramic sinks, wash basins, baths... and other sanitary fixtures, n.e.c.	69109000	Ceramic sinks, washbasins, washbasin pedestals, baths, bidets, water closet pans, flushing cisterns, urinals and similar sanitary fixtures (excl. of porcelain or china, soap dishes, sponge holders, tooth-brush holders, towel hooks and toilet paper holders)

Table 64 PRODCOM and CN codes of Stage 2 shortlisted products in the ferro-alloys sector

PRODCOM code	PRODCOM Description	CN code	CN description
24101235	Ferro-silicon, containing by weight > 55% of silicon	72022100	Ferro-silicon, containing by weight > 55% of silicon
24101245	Ferro-silico-manganese	72023000	Ferro-silico-manganese

Table 65 PRODCOM and CN codes of Stage 2 shortlisted products in the aluminium sector

PRODCOM code	PRODCOM Description	CN code	CN description
24421200	Aluminium oxide (excluding artificial corundum)	28182000	Aluminium oxide (excl. artificial corundum)

Table 66 PRODCOM and CN codes of Stage 2 shortlisted products in the other non-ferrous metals sector

PRODCOM code	PRODCOM Description	CN code	CN description
24431130	Refined unwrought lead (excluding lead powders or flakes)	78011000	Unwrought lead, refined
24431190	Unwrought lead (excluding lead powders or flakes, unwrought lead containing antimony, refined)	78019910	Unwrought lead, containing by weight >= 0,02% of silver, for refining "bullion lead"
		78019990	Unwrought lead (excl. lead containing by weight antimony as the principal other element, lead for refining containing by weight >= 0,02% of silver (bullion lead) and refined lead)
24431230	Unwrought non-alloy zinc (excluding zinc dust, powders and flakes)	79011100	Unwrought zinc, not alloyed, containing by weight >= 99,99% of zinc
		79011210	Unwrought zinc, not alloyed, containing by weight >= 99,95% but < 99,99% of zinc
		79011230	Unwrought zinc, not alloyed, containing by weight >= 98,5% but < 99,95% of zinc

PRODCOM code	PRODCOM Description	CN code	CN description
		79011290	Unwrought zinc, not alloyed, containing by weight >= 97,5% but < 98,95% of zinc

10.1.4 Metadata for EXIOBASE

EXIOBASE is a Multi-Regional Environmentally Extended Input-Output (MRIO) database that provides detailed economic, trade, and environmental impact data. It integrates monetary and physical supply-use tables with environmental satellite accounts, making it one of the most comprehensive databases for analysing global environmental footprints and trade-related emissions.

Below an overview of indicators created from the EXIOBASE database is given.

Embedded Emissions: The total amount of greenhouse gas (GHG) emissions associated with a product or sector, including both direct emissions (from production) and indirect emissions (from upstream inputs and trade).

- **EU and non-EU Embedded Emissions per Value Added Euro:** The amount of embedded emissions (kg CO₂eq) generated within the EU per €1 of value added by an economic sector.
- **Total Embedded Emissions per EU Value Added:** The sum of EU and non-EU embedded emissions per €1 of value added within the EU.

EU Value Added in EXIOBASE: The total economic value added (€) by EU-based sectors, as recorded in EXIOBASE.

Top Upstream Countries: This represents the top non-EU countries affected by CBAM through the ultimate origin of the selected product

- **Upstream Countries Value Added:** The total economic value added (€) by non-EU countries in the supply chain of a selected product.
- **Upstream Countries VA Concentration Ratio:** The percentage of total upstream value added accounted for by the top 5 non-EU countries in the supply chain.
- **Upstream Countries Carbon Content:** The total embedded carbon emissions (kg CO₂eq) from non-EU countries contributing to the selected product.
- **Upstream Countries CC Concentration Ratio:** The percentage of total upstream carbon content contributed by the top 5 non-EU countries.

Top Upstream Products: This represents the top non-EU products which are affected by CBAM through the ultimate origin of the selected product

- **Upstream Products Value Added:** The total economic value added (€) by non-EU products that serve as inputs for the selected product.
- **Upstream Products VA Concentration Ratio:** The percentage of total upstream value added contributed by the top 5 non-EU products in the supply chain.
- **Upstream Products Carbon Content:** The total embedded carbon emissions (kg CO₂eq) from non-EU products contributing to the selected product.
- **Upstream Products CC Concentration Ratio:** The percentage of total upstream carbon content contributed by the top 5 non-EU products.

Top Downstream Countries: This shows the EU countries whose costs may be impacted by the CBAM extension, both directly and indirectly.

- **Downstream Countries Value Added:** The total economic value added (€) in the EU economy from the selected product.
- **Downstream Countries VA Concentration Ratio:** The percentage of total downstream value added accounted for by the top 5 EU countries.
- **Downstream Countries Carbon Content:** The total embedded carbon emissions (kg CO₂eq) in the selected product, impacting EU industries and consumers.
- **Downstream Countries CC Concentration Ratio:** The percentage of total downstream carbon content accounted for by the top 5 EU countries.

Top Downstream Products: This shows the EU products whose costs may be impacted by the CBAM extension, both directly and indirectly.

- **Downstream Products Value Added:** The total economic value added (€) by EU products that use the selected product as an input.
- **Downstream Products VA Concentration Ratio:** The percentage of total downstream value added generated by the top 5 EU downstream products.
- **Downstream Products Carbon Content:** The total embedded carbon emissions (kg CO₂eq) in EU downstream products.
- **Downstream Products CC Concentration Ratio:** The percentage of total downstream carbon content accounted for by the top 5 EU downstream products.

10.2 T2 annexes

See files attached with this Final Report named '**Annex 2 - Discussion note on Attribution to By-Products**' and "**Annex 4 – Note on chemicals**"

10.3 T4 annexes

The annexes related to this task are provided together with the annexes of task 7, that is, the factual summaries of both the interviews and the survey.

10.4 T5 annexes

10.4.1 T5.1. Annex - CBAM transport work and emissions database

See Excel file attached with this Final Report named **'Task 5- Full database.'** Specifically **columns A to S**

10.4.2 T5.4 Annex - Quantitative estimate of emissions from the transport of each CBAM good taking place outside the EU and not covered by the EU ETS

See Excel file attached with this Final Report named **'Task 5- Full database.'** Specifically **columns AF to AH.**

10.4.3 T5.5 Carbon leakage model additional outputs

10.5 T6 annexes

See Excel file attached with this Final report named: '**6_5_Default_values**'.

10.6 T7 annexes

See file attached with this Final report:

- 'Annex 7_1_Stakeholder Consultation Analysis'
- 'Annex 7_1_In-depth Interviews - Factual Summary Report'
- 'Annex 7_2_Survey - Factual Summary Report'
- 'Annex 7_4_Survey responses, that encompass the following deliverables:
 - An Excel file
 - Containing detailed (anonymised) survey responses and the corresponding analysis, where the corresponding names, emails and name of organisation has been removed, and appears in the first tab 'content'
 - Presenting the quantitative results derived from the survey and interviews. In the same Excel file, there is a tab per question with the analysis.

10.7 External reviewers' comments

10.7.1 Task 1 to 3

External review of the Study on a potential CBAM scope extension to other goods and transportation services. Specific Contract No TAXUD/2024/DE/311 implementing framework contract No GROW/2021/OP/0001 Lot 2

Evaluation form

Name of Reviewer (Institution): Michael Mehling (MIT)		
Date of review: 20 October 2025		
Report title: Study on a potential CBAM scope extension to other goods and transportation services		Score 4.5
1 The report structure (score 0-5):		
1.1	The report is well structured, logical, clear and complete (e.g., background and objectives are presented before findings, findings are presented before conclusions and recommendations).	4.5
2. Study objectives and scope (score 0-5):		
2.1	The report presents a clear and full description of the 'objective' of the study.	5
2.2	Relevance of stakeholders involved in the study/ project implementation, including the consortium partners, external stakeholders, and their roles.	4.5
2.3	The report identifies the implementation status of the project, including phase of the task, and any significant changes (e.g. plans, strategies, methodologies) that have occurred and explains the implications of those changes for the project.	4
3 Methodology (score 0-5):		
3.1	The report presents a transparent description of the methodology applied, explains how the study was specifically designed to address the study objectives	4
3.2	The report describes the data collection methods and analysis, the rationale for selecting them, and their limitations. Reference indicators are included where relevant.	4.5
3.3	The report describes the data sources, the rationale for their selection, and their limitations. The report includes discussion of how the mix of data sources (triangulation) was used to obtain a diversity of perspectives, ensure data accuracy and overcome data limits/bias.	4.5
3.4	The report gives a complete description of the stakeholders' consultation process, including the rationale for selecting the particular level and activities for consultation.	5
3.5	The methods employed are appropriate for the evaluation and to answer its questions.	4.5
4. Findings (score 0-5):		
4.1	Findings respond directly to the study objectives detailed in the scope and objectives section.	4
4.2	Reported findings reflect systematic and appropriate analysis and interpretation of the data.	4.5
4.3	Findings are objectively reported based on the evidence.	4.5
5. Conclusions (score 0-5):		
5.1	Conclusions present objective judgments based on findings and substantiated by evidence. These are logically connected from data collection through analysis to conclusions.	4.5
5.2	Conclusions provide insights pertinent to the object and purpose of the project presented.	4
6. Recommendations (score 0-5):		
6.1	Recommendations are pragmatic and relevant to the object and purposes of the project.	4
6.2	Recommendations clearly identify the target group for each recommendation, disaggregated by stakeholder group/ country / organisational setting/ therapeutic area, etc.	3.5
6.3	The report describes the process followed in developing the recommendations (e.g. consultation with stakeholders).	4

7. Overall comments (free text):

The draft final report for the study on a potential CBAM scope extension to other goods and transportation services presents the consolidated analytical and consultative results of the project. It builds on the analytical framework outlined in last year's inception report and proceeds to deliver quantitative and qualitative findings on the possible horizontal extension of CBAM (e.g., additional sectors), upstream inclusion of additional precursor goods, and inclusion of transportation services. The comprehensive, nearly 400-page document is structured along the seven contractual tasks, outsourcing certain information (such as data tables and documentation of stakeholder engagement) to annexes, not all of which were not included in the document provided for this review. The stated aim of the study is to identify technically and administratively feasible options for broadening CBAM coverage while maintaining proportional administrative burdens.

Overall, the report represents a mature, methodologically consistent, and evidence-rich output. Its strengths build on the approach already presented in the inception report, with breadth of data collection, a structured consultation process, and balanced treatment of technical, economic, and legal feasibility criteria. A robust set of consistently applied criteria guide the assessment, with appropriate criteria tailored to the particular features of different sectors (e.g. accounting for complexity and pursuing a "substance-centred" approach for products from the chemicals sector). Existing regulatory frameworks for emissions quantification and reporting are exhaustively surveyed. Conclusions are, overall, clear and pragmatic, identifying the specific product categories suited for CBAM scope extension based on the applied criteria. Only for refinery products, chemicals, and polymers, the conclusions are somewhat more ambiguous, owed to the attendant complexities and trade-offs.

Weaknesses are primarily editorial and structural. Spelling, syntax and grammatical errors were still fairly prevalent in the draft submitted for review. In terms of content, some chapters are excessively detailed or repetitive; cross-task synthesis occasionally remains implicit, especially where quantitative and qualitative findings intersect. Extensive signposting helps situate and bring along readers, yet the sheer density and volume of material covered renders the study inaccessible for those unfamiliar with the project. Several tables could benefit from clearer narrative explanation, and the rationale for some recommendations is under-documented. Substantively, the omission of scrap metal from the analysis is understandable, but regrettable given the political salience recently enjoyed. Despite these issues – some of which are likely to be addressed in the final version – the report succeeds in addressing the Commission's objectives and provides a solid evidence base for final policy reflection. Like the inception report, it therefore merits a high score of 4.5.

SCORING THE REPORT

The report will be evaluated by selected independent experts against the 6 evaluation criteria.

For each criterion, a score from 0 to 5 must be awarded (with a resolution of 0.5 allowed, i.e. 2.5, 3.0, 3.5, ...).

INTERPRETATION OF THE SCORES

0 — The report fails to address the criterion or cannot be assessed due to missing or incomplete information (unless the result of an 'obvious clerical error');

1 — Poor: the criterion is inadequately addressed or there are serious inherent weaknesses;

2 — Fair: the report broadly addresses the criterion but there are significant weaknesses;

3 — Good: the report addresses the criterion well but with a number of shortcomings;

4 — Very good: the report addresses the criterion very well but with a small number of shortcomings;

5 — Excellent: the report successfully addresses all relevant aspects of the criterion; any shortcomings are minor.

10.7.2 Task 5 and 6

External review of the Study on a potential CBAM scope extension to other goods and transportation services. Specific Contract No TAXUD/2024/DE/311 implementing framework contract No GROW/2021/OP/0001 Lot 2

Evaluation form

Name of Reviewer (Institution): Inge Mayeres (Transport & Mobility Leuven)		
Date of review: 21 October 2025		
Report title: Study on a potential CBAM scope extension to other goods and transportation services		Score
1 The report structure (score 0-5):		
1.1	The report is well structured, logical, clear and complete (e.g., background and objectives are presented before findings, findings are presented before conclusions and recommendations).	5
2. Study objectives and scope (score 0-5):		

2.1	The report presents a clear and full description of the 'objective' of the study.	5
2.2	Relevance of stakeholders involved in the study/ project implementation, including the consortium partners, external stakeholders, and their roles.	Not reviewed
2.3	The report identifies the implementation status of the project, including phase of the task, and any significant changes (e.g. plans, strategies, methodologies) that have occurred and explains the implications of those changes for the project.	5
3 Methodology (score 0-5):		
3.1	The report presents a transparent description of the methodology applied, explains how the study was specifically designed to address the study objectives	4
3.2	The report describes the data collection methods and analysis, the rationale for selecting them, and their limitations. Reference indicators are included where relevant.	4
3.3	The report describes the data sources, the rationale for their selection, and their limitations. The report includes discussion of how the mix of data sources (triangulation) was used to obtain a diversity of perspectives, ensure data accuracy and overcome data limits/bias.	4
3.4	The report gives a complete description of the stakeholders' consultation process, including the rationale for selecting the particular level and activities for consultation.	Not reviewed
3.5	The methods employed are appropriate for the evaluation and to answer its questions.	4
4. Findings (score 0-5):		
4.1	Findings respond directly to the study objectives detailed in the scope and objectives section.	4
4.2	Reported findings reflect systematic and appropriate analysis and interpretation of the data.	4
4.3	Findings are objectively reported based on the evidence.	4
5. Conclusions (score 0-5):		
5.1	Conclusions present objective judgments based on findings and substantiated by evidence. These are logically connected from data collection through analysis to conclusions.	4
5.2	Conclusions provide insights pertinent to the object and purpose of the project presented.	4
6. Recommendations (score 0-5):		
6.1	Recommendations are pragmatic and relevant to the object and purposes of the project.	4
6.2	Recommendations clearly identify the target group for each recommendation, disaggregated by stakeholder group/ country / organisational setting/ therapeutic area, etc.	Not applicable
6.3	The report describes the process followed in developing the recommendations (e.g. consultation with stakeholders).	Not reviewed
7. Overall comments (free text): In my review I have focused on Tasks 5 and 6 (excluding the stakeholder consultation), which are in my area of expertise. In the report itself I have indicated a number of comments, including smaller ones that can easily be addressed and suggestions for clarification for the reader. For section 7.6.1 I suggest considering the feasibility of adding a summary table for the different scenarios and their main effects, as well as more explanation on the effects. This would help the reader in understanding the main lessons from this analysis and linking them to the conclusions. For Section 8.8 (Task 6.6) I suggest considering the feasibility to have a more structured discussion, also considering the findings of the OECD on the net effective carbon tax rates.		
SCORING THE REPORT The report will be evaluated by selected independent experts against the 6 evaluation criteria. For each criterion, a score from 0 to 5 must be awarded (with a resolution of 0.5 allowed, i.e. 2.5, 3.0, 3.5, ...).		
INTERPRETATION OF THE SCORES 0 — The report fails to address the criterion or cannot be assessed due to missing or incomplete information (unless the result of an 'obvious clerical error'); 1 — Poor: the criterion is inadequately addressed or there are serious inherent weaknesses; 2 — Fair: the report broadly addresses the criterion but there are significant weaknesses; 3 — Good: the report addresses the criterion well but with a number of shortcomings; 4 — Very good: the report addresses the criterion very well but with a small number of shortcomings; 5 — Excellent: the report successfully addresses all relevant aspects of the criterion; any shortcomings are minor.		

Study team responses by task

TASKS	COMMENTS RECEIVED APPLICABLE TO THE TASK	STUDY TEAM RESPONSE. INCLUDE HOW AND WHERE IN THE REPORT THE COMMENT WAS ADDRESSED.
TASK 1, 2, 3	Some chapters are excessively detailed or repetitive; cross-task synthesis occasionally remains implicit, especially where quantitative and qualitative findings intersect. Extensive signposting helps situate and bring along readers, yet the sheer density and volume of material covered renders the study inaccessible for those unfamiliar with the project.	The length of the chapters is a consequence of the applied methodology as agreed with the Commission. Given the length and technical nature of this study, the repetition is necessary to provide the appropriate context for the reader. As noted by the reviewer, we have used extensive signposting throughout the chapters to bring the reader along and help get familiar with the study, but in the end the study is intended for the technical audience.
TASK 1, 2, 3	Several tables could benefit from clearer narrative explanation, and the rationale for some recommendations is under-documented.	The narrative explanation regarding the tables and recommendations have been reviewed and made clearer to the extent possible, noting that this study is intended for a technical audience familiar with the project.
TASK 3	The omission of scrap metal from the analysis is understandable, but regrettable given the political salience recently enjoyed	The section on scrap metal has been omitted as agreed with the Commission.
TASK 4	none	none
TASK 5	Series of clarifications. Suggestion to add a summary table for the different scenarios and their main effects, as well as more explanation on the effects. This would help the reader in understanding the main lessons from this analysis and linking them to the conclusions.	Clarifications have been added for each point highlighted by the reviewer. A comparative summary table was added, together with a synthesis of the main findings right before the presentation of the detailed analysis per scenario to facilitate the reading and understanding of the effects described.
TASK 6	Suggestion to consider the feasibility to have a more structured discussion, also considering the findings of the OECD on the net effective carbon tax rates.	The analysis of OECD net carbon price data shows that sufficient information is only available for road transport, not for all transport modes. Including subsidies and compensations would overly complicate the analysis given the project's objectives. Although EU carbon pricing instruments (CPIs) outside the EU ETS were not initially within scope, the study team included them for completeness. A description of the OECD CPIs on road transport was added below Table 52 for comparison. Overall, the World Bank's CPI data provide insights across all transport modes (directly or indirectly), whereas the OECD data mainly cover road—and to a limited extent, rail—excluding maritime and air transport.
TASK 7	none	none
GENERAL	Spelling, syntax and grammatical errors were still fairly prevalent in the draft submitted for review.	Extensive quality review, language-proofing, editing and formatting have been carried out on the current version of the study as a follow up.